Hartshorne, Chapter II: Schemes

Solutions to the exercises

Conventions. Rings are commutative with 1; unlike in Chapter I, the base field is not assumed algebraically closed unless an exercise says so. Results of the text are cited by their numbers in Hartshorne ("Prop. 2.5", "Thm. 7.6", ...), and results of other chapters as "I, ..." or "III, ...". The numbering and statements of the exercises follow the edition scanned in this repository; each exercise is restated before its solution, and starred exercises are marked "Starred in the book" at the end of the statement.

Honest notes. Several printed statements need an extra hypothesis to be true, and the solutions say what is assumed: for example $\operatorname{char} k \neq 2$ in Ex. 3.10(b), 4.12 and 7.7(b); linear independence of the forms in Ex. 7.3(b); connectedness of X in Ex. 7.9; $r \geq 1$ in Ex. 5.10 and 5.14; and the exercises 6.6(b),(c), which fail at the base point P_0. A few exercises are open-ended (Ex. 4.12(b)(3)), and some solutions use standard facts beyond the text (cohomology from Chapter III, Krull–Akizuki, Weierstrass-type theorems), which are named where used.

1. Sheaves

Throughout, "sheaf" means sheaf of abelian groups on the indicated space unless said otherwise. For a presheaf $\mathcal{F}$ we write $\mathcal{F}^+$ for the associated sheaf and $\theta\colon \mathcal{F} \to \mathcal{F}^+$ for the canonical map (Prop.–Def. 1.2); recall $\mathcal{F}^+_P = \mathcal{F}_P$, and θ is an isomorphism if $\mathcal{F}$ is already a sheaf. A morphism of presheaves from $\mathcal{F}$ to a sheaf $\mathcal{G}$ factors uniquely through θ. Two elementary facts about sheaves will be used repeatedly.

Fact A. If $\mathcal{F}$ is a sheaf and $s \in \mathcal{F}(U)$ has $s_P = 0$ for all $P \in U$, then $s = 0$. (Each $s_P = 0$ gives a neighborhood $W_P \subseteq U$ with $s|_{W_P} = 0$; the W_P cover U; apply (3).) Hence a sheaf all of whose stalks are 0 is the zero sheaf.

Fact B. Let $\mathcal{F}', \mathcal{F}''$ be subsheaves of a sheaf $\mathcal{F}$ such that $\mathcal{F}'_P = \mathcal{F}''_P$ as subgroups of $\mathcal{F}_P$ for every P. Then $\mathcal{F}' = \mathcal{F}''$. Indeed, let $s \in \mathcal{F}'(U)$. For $P \in U$ we have $s_P \in \mathcal{F}''_P$, so there is an open $W_P \ni P$, $W_P \subseteq U$, and $t \in \mathcal{F}''(W_P)$ with $t_P = s_P$; shrinking W_P we may assume $t = s|_{W_P}$. Thus $s|_{W_P} \in \mathcal{F}''(W_P)$ for all P; these sections (elements of $\mathcal{F}''(W_P) \subseteq \mathcal{F}(W_P)$) agree on overlaps because they are restrictions of s, so they glue in $\mathcal{F}''$ to some $t \in \mathcal{F}''(U)$ with $t|_{W_P} = s|_{W_P}$ for all P; by (3) for $\mathcal{F}$, $t = s$. So $\mathcal{F}'(U) \subseteq \mathcal{F}''(U)$, and by symmetry equality holds.

Exercise 1.1

Problem. Let A be an abelian group, and define the *constant presheaf* associated to A on the topological space X to be the presheaf $U \mapsto A$ for all $U \neq \emptyset$, with restriction maps the identity. Show that the constant sheaf $\mathcal{A}$ defined in the text is the sheaf associated to this presheaf.

Solution. Let $\mathcal{P}$ be the constant presheaf ($\mathcal{P}(\emptyset) = 0$, $\mathcal{P}(U) = A$ otherwise). Every open neighborhood of a point is nonempty, so all groups in the direct system defining $\mathcal{P}_P$ equal A and all maps are the identity; hence $\mathcal{P}_P = A$ and the germ of $a \in \mathcal{P}(U)$ at $P \in U$ is a.

By the construction in Prop.–Def. 1.2, $\mathcal{P}^+(U)$ is the set of functions $s\colon U \to \bigcup_{P\in U} \mathcal{P}_P = A$ (with $s(P) \in \mathcal{P}_P = A$) such that every $P \in U$ has a neighborhood $V \subseteq U$ and $t \in \mathcal{P}(V) = A$ with $s(Q) = t_Q = t$ for all $Q \in V$. That is, $\mathcal{P}^+(U)$ is exactly the set of locally constant functions $U \to A$, i.e. continuous maps $U \to A$ for the discrete topology on A; this is $\mathcal{A}(U)$ by definition (Example 1.0.3), with the group structure pointwise and restriction maps the restriction of functions on both sides. (For $U = \emptyset$ both groups are 0, being the set of the single empty function.) So $\mathcal{P}^+ = \mathcal{A}$ as sheaves, and under this identification $\theta\colon \mathcal{P} \to \mathcal{A}$ sends $a \in A = \mathcal{P}(U)$ to the constant function a. Thus $\mathcal{A}$ is the sheaf associated to $\mathcal{P}$. □

Exercise 1.2

Problem.

(a) For any morphism of sheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$, show that for each point P, $(\ker\varphi)_P = \ker(\varphi_P)$ and $(\operatorname{im}\varphi)_P = \operatorname{im}(\varphi_P)$.

(b) Show that φ is injective (respectively, surjective) if and only if the induced map φ_P on the stalks is injective (respectively, surjective) for all P.

(c) Show that a sequence $\cdots \to \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \to \dots$ of sheaves and morphisms is exact if and only if for each $P \in X$ the corresponding sequence of stalks is exact as a sequence of abelian groups.

Solution. (a) *Kernel.* $\ker\varphi$ is the subsheaf of $\mathcal{F}$ with $(\ker\varphi)(U) = \ker\varphi(U)$. The map $(\ker\varphi)_P \to \mathcal{F}_P$ is injective: if $s \in \ker\varphi(U)$ has zero germ in $\mathcal{F}_P$ then $s|_W = 0$ for some neighborhood W, which is zero in $(\ker\varphi)(W)$. Its image lies in $\ker\varphi_P$ since $\varphi_P(s_P) = \varphi(s)_P = 0$. Conversely let $g \in \ker\varphi_P$, $g = s_P$ with $s \in \mathcal{F}(U)$. Then $\varphi(s)_P = 0$, so there is a neighborhood $W \subseteq U$ of P with $0 = \varphi(s)|_W = \varphi(s|_W)$; so $s|_W \in \ker\varphi(W)$ and $g = (s|_W)_P$ is in the image of $(\ker\varphi)_P$. Hence $(\ker\varphi)_P = \ker\varphi_P$.

Image. Let $\mathcal{I}$ be the presheaf $U \mapsto \varphi(U)(\mathcal{F}(U))$, i.e. $\mathcal{I}(U) = \varphi(U)(\mathcal{F}(U)) \subseteq \mathcal{G}(U)$. Then $\operatorname{im}\varphi = \mathcal{I}^+$, and the natural map $\operatorname{im}\varphi \to \mathcal{G}$ is induced from the inclusion $\mathcal{I} \to \mathcal{G}$. As in the first paragraph, the stalk map $\mathcal{I}_P \to \mathcal{G}_P$ is injective (if $\varphi(s)$, $s \in \mathcal{F}(U)$, has zero germ then $\varphi(s)|_W = 0$ for some W, and 0 is zero in $\mathcal{I}(W)$) and its image is $\{\varphi(s)_P\} = \{\varphi_P(s_P)\} = \operatorname{im}\varphi_P$. Since $\mathcal{I}_P^+ = \mathcal{I}_P$ compatibly with the maps to $\mathcal{G}_P$, $(\operatorname{im}\varphi)_P = \operatorname{im}\varphi_P$ as subgroups of $\mathcal{G}_P$.

(b) φ is injective iff $\ker\varphi = 0$ iff all stalks $(\ker\varphi)_P = \ker\varphi_P$ vanish (Fact A for "if"), iff all φ_P are injective. φ is surjective iff $\operatorname{im}\varphi = \mathcal{G}$, i.e. iff the inclusion $\operatorname{im}\varphi \to \mathcal{G}$ of sheaves is an isomorphism; by Prop. 1.1 this holds iff it is an isomorphism on every stalk, i.e. iff $\operatorname{im}\varphi_P = \mathcal{G}_P$ for all P by (a), i.e. iff every φ_P is surjective.

(c) The sequence is exact at $\mathcal{F}^i$ iff $\operatorname{im}\varphi^{i-1} = \ker\varphi^i$ as subsheaves of $\mathcal{F}^i$. If so, taking stalks and using (a) gives $\operatorname{im}\varphi_P^{i-1} = \ker\varphi_P^i$. Conversely, if these are equal for all P, the two subsheaves have the same stalks (by (a)), so are equal by Fact B. □

Exercise 1.3

Problem.

(a) Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X. Show that φ is surjective if and only if the following condition holds: for every open set $U \subseteq X$, and for every $s \in \mathcal{G}(U)$, there is a covering $\{U_i\}$ of U, and there are elements $t_i \in \mathcal{F}(U_i)$, such that $\varphi(t_i) = s|_{U_i}$ for all i.

(b) Give an example of a surjective morphism of sheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$, and an open set U such that $\varphi(U)\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is not surjective.

Solution. (a) ($\Rightarrow$) Let φ be surjective, $s \in \mathcal{G}(U)$, $P \in U$. By Ex. 1.2(b), φ_P is surjective, so $s_P = \varphi_P(t_P)$ for some germ t_P represented by $t \in \mathcal{F}(V)$, $V \ni P$ open, $V \subseteq U$. Then $\varphi(t)_P = s_P$, so $\varphi(t)|_{V'} = s|_{V'}$ for some smaller open $V' \ni P$, and $t' := t|_{V'}$ satisfies $\varphi(t') = s|_{V'}$. Doing this for every $P \in U$ gives the covering.

($\Leftarrow$) By Ex. 1.2(b) it suffices to show each φ_P is surjective. Let $g \in \mathcal{G}_P$ be represented by $s \in \mathcal{G}(U)$. Take a covering $\{U_i\}$ and t_i as in the condition; choose i with $P \in U_i$. Then $\varphi_P((t_i)_P) = (\varphi(t_i))_P = (s|_{U_i})_P = g$.

(b) Let $X = S^1 = \{z \in \mathbf{C} : |z| = 1\}$ with its usual topology. Let $\mathcal{F}$ be the sheaf of continuous real-valued functions on X (abelian group under addition), and $\mathcal{G}$ the sheaf of continuous functions to S^1, which is an abelian group under pointwise multiplication (both are sheaves since continuity is local). Define $\varphi(U)(f) = \mathrm{e}^{2\pi i f}$; this is a homomorphism compatible with restrictions, so a morphism of sheaves.

φ is surjective: let $g \in \mathcal{G}(U)$ and $P \in U$. Put $w = g(P)$ and $V = g^{-1}(S^1 \setminus \{-w\})$, an open neighborhood of P. On the simply cut circle $S^1 \setminus \{-w\}$ there is a continuous function α with $\mathrm{e}^{2\pi i\alpha(z)} = z$ (a continuous branch of $\frac{1}{2\pi i}\log$; e.g. fix β with $\mathrm{e}^{2\pi i\beta} = w$ and let $\alpha(z)$ be the unique number in $(\beta - \frac{1}{2}, \beta + \frac{1}{2})$ with $\mathrm{e}^{2\pi i\alpha(z)} = z$; the map α is the inverse of the homeomorphism $(\beta - \frac{1}{2}, \beta + \frac{1}{2}) \to S^1 \setminus \{-w\}$ given by $\mathrm{e}^{2\pi i(\cdot)}$). Then $f = \alpha \circ g|_V \in \mathcal{F}(V)$ satisfies $\varphi(f) = g|_V$. By (a), φ is surjective.

$\varphi(X)$ is not surjective: take $g = \mathrm{id} \in \mathcal{G}(X)$. Suppose $f\colon S^1 \to \mathbf{R}$ is continuous with $\mathrm{e}^{2\pi i f(z)} = z$ for all z. Then f is injective (it determines z). Its image is compact and connected, a closed interval $[a, b]$, with $a < b$ since f is injective on an infinite set. Pick z_0 with $f(z_0) = c \in (a, b)$ (possible: only two points map to the endpoints). Then $S^1 \setminus \{z_0\}$ is connected and f maps it bijectively onto $[a, b] \setminus \{c\}$, which is disconnected, contradicting that continuous images of connected sets are connected. So id is not in the image of $\varphi(X)$. □

Exercise 1.4

Problem.

(a) Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves such that $\varphi(U)\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for each U. Show that the induced map $\varphi^+\colon \mathcal{F}^+ \to \mathcal{G}^+$ of associated sheaves is injective.

(b) Use part (a) to show that if $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves, then $\operatorname{im}\varphi$ can be naturally identified with a subsheaf of $\mathcal{G}$, as mentioned in the text.

Solution. (a) The morphism φ^+ is the unique morphism with $\varphi^+\theta_{\mathcal{F}} = \theta_{\mathcal{G}}\varphi$. Concretely, $\varphi^+(U)$ sends $s \in \mathcal{F}^+(U)$ (a function $P \mapsto s(P) \in \mathcal{F}_P$) to $P \mapsto \varphi_P(s(P))$: this function lies in $\mathcal{G}^+(U)$ (if $s = t_Q$ on a neighborhood V with $t \in \mathcal{F}(V)$, then $\varphi_Q(s(Q)) = \varphi(t)_Q$ there), the assignment is compatible with restriction, and it satisfies $\varphi^+\theta_{\mathcal{F}} = \theta_{\mathcal{G}}\varphi$; by uniqueness it is φ^+.

First, φ_P is injective for each P: if $t \in \mathcal{F}(V)$ and $\varphi_P(t_P) = \varphi(t)_P = 0$ then $\varphi(t|_W) = \varphi(t)|_W = 0$ for some neighborhood W of P, hence $t|_W = 0$ as $\varphi(W)$ is injective, so $t_P = 0$. Now if $\varphi^+(s) = 0$ then $\varphi_P(s(P)) = 0$, so $s(P) = 0$ for every P, i.e. $s = 0$. So each $\varphi^+(U)$ is injective.

(b) Let $\mathcal{I}$ be the presheaf image $U \mapsto \varphi(U)(\mathcal{F}(U)) \subseteq \mathcal{G}(U)$, so $\operatorname{im}\varphi = \mathcal{I}^+$. The inclusion $\iota\colon \mathcal{I} \to \mathcal{G}$ has $\iota(U)$ injective, so by (a) $\iota^+\colon \mathcal{I}^+ \to \mathcal{G}^+$ is injective. Composing with the isomorphism $\theta_{\mathcal{G}}^{-1}\colon \mathcal{G}^+ \to \mathcal{G}$ ($\mathcal{G}$ is a sheaf) gives an injection $\operatorname{im}\varphi \to \mathcal{G}$, which is the natural map of the text (it is the map obtained from the universal property applied to ι). Hence $\operatorname{im}\varphi$ is identified with a subsheaf of $\mathcal{G}$, whose sections over U are those $s \in \mathcal{G}(U)$ which locally (on a cover of U) lie in the image of φ; its stalks are $\operatorname{im}\varphi_P$ by Ex. 1.2(a). □

Exercise 1.5

Problem. Show that a morphism of sheaves is an isomorphism if and only if it is both injective and surjective.

Solution. If $\varphi\colon \mathcal{F} \to \mathcal{G}$ is an isomorphism, each φ_P is an isomorphism (inverse $(\varphi^{-1})_P$), hence injective and surjective, so φ is injective and surjective by Ex. 1.2(b). Conversely if φ is injective and surjective, then all φ_P are bijective by Ex. 1.2(b), hence isomorphisms, and φ is an isomorphism by Prop. 1.1. □

Exercise 1.6

Problem.

(a) Let $\mathcal{F}'$ be a subsheaf of a sheaf $\mathcal{F}$. Show that the natural map of $\mathcal{F}$ to the quotient sheaf $\mathcal{F}/\mathcal{F}'$ is surjective, and has kernel $\mathcal{F}'$. Thus there is an exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}/\mathcal{F}' \to 0$.

(b) Conversely, if $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence, show that $\mathcal{F}'$ is isomorphic to a subsheaf of $\mathcal{F}$, and that $\mathcal{F}''$ is isomorphic to the quotient of $\mathcal{F}$ by this subsheaf.

Solution. (a) The natural map $\pi\colon \mathcal{F} \to \mathcal{F}/\mathcal{F}'$ is the composite of $\mathcal{F} \to Q$, where Q is the presheaf quotient $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$, with $\theta\colon Q \to Q^+ = \mathcal{F}/\mathcal{F}'$. On stalks, $Q_P = \mathcal{F}_P/\mathcal{F}'_P = (\mathcal{F}/\mathcal{F}')_P$ (text), and π_P is the quotient map $\mathcal{F}_P \to \mathcal{F}_P/\mathcal{F}'_P$. This is surjective, so π is surjective (Ex. 1.2(b)), and $(\ker\pi)_P = \ker\pi_P = \mathcal{F}'_P$ (Ex. 1.2(a)). Thus $\ker\pi$ and $\mathcal{F}'$ are subsheaves of $\mathcal{F}$ with the same stalks, hence equal by Fact B. The inclusion $\mathcal{F}' \to \mathcal{F}$ is injective with image $\ker\pi$, so the sequence is exact.

(b) Let $\alpha\colon \mathcal{F}' \to \mathcal{F}$, $\beta\colon \mathcal{F} \to \mathcal{F}''$. Since α is injective, each $\alpha(U)$ is injective, so $\mathcal{F}_1'(U) := \alpha(U)(\mathcal{F}'(U))$ defines a subpresheaf of $\mathcal{F}$ and α induces an isomorphism of presheaves $\mathcal{F}' \to \mathcal{F}_1'$. Hence $\mathcal{F}_1'$ is a sheaf, a subsheaf of $\mathcal{F}$, isomorphic to $\mathcal{F}'$; it is the image of α (the presheaf image is already a sheaf). By exactness $\ker\beta = \operatorname{im}\alpha = \mathcal{F}_1'$. Since $\beta\alpha = 0$, $\beta(U)$ vanishes on $\mathcal{F}_1'(U)$ and induces $\bar\beta(U)\colon \mathcal{F}(U)/\mathcal{F}_1'(U) \to \mathcal{F}''(U)$, a morphism from the presheaf quotient to the sheaf $\mathcal{F}''$; it factors through a unique $\gamma\colon \mathcal{F}/\mathcal{F}_1' \to \mathcal{F}''$. On stalks, $\gamma_P\colon \mathcal{F}_P/(\mathcal{F}_1')_P \to \mathcal{F}_P''$ is induced by β_P; it is surjective since β_P is (Ex. 1.2(b)) and injective since $\ker\beta_P = (\ker\beta)_P = (\mathcal{F}_1')_P$. So γ is an isomorphism by Prop. 1.1. □

Exercise 1.7

Problem. Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves.

(a) Show that $\operatorname{im}\varphi \cong \mathcal{F}/\ker\varphi$.

(b) Show that $\operatorname{coker}\varphi \cong \mathcal{G}/\operatorname{im}\varphi$.

Solution. (a) The presheaf quotient $Q(U) = \mathcal{F}(U)/\ker\varphi(U)$ maps to the presheaf image $\mathcal{I}(U) = \varphi(U)(\mathcal{F}(U))$ by $\bar\varphi(U)$, and each $\bar\varphi(U)$ is bijective (first isomorphism theorem), compatible with restrictions. So $Q \cong \mathcal{I}$ as presheaves, whence $\mathcal{F}/\ker\varphi = Q^+ \cong \mathcal{I}^+ = \operatorname{im}\varphi$.

(b) By definition $\operatorname{coker}\varphi = C^+$ with $C(U) = \mathcal{G}(U)/\varphi(U)(\mathcal{F}(U)) = \mathcal{G}(U)/\mathcal{I}(U)$. Regard $\operatorname{im}\varphi$ as a subsheaf of $\mathcal{G}$ (Ex. 1.4(b)); then $\mathcal{I} \subseteq \operatorname{im}\varphi \subseteq \mathcal{G}$ as subpresheaves, so the identity of $\mathcal{G}(U)$ induces a morphism of presheaves $C \to D$, $D(U) = \mathcal{G}(U)/(\operatorname{im}\varphi)(U)$, and hence a morphism of sheaves $\gamma\colon \operatorname{coker}\varphi = C^+ \to D^+ = \mathcal{G}/\operatorname{im}\varphi$. On stalks (direct limits are exact, which is also the justification of $(\mathcal{F}/\mathcal{F}')_P = \mathcal{F}_P/\mathcal{F}_P'$ in the text), γ_P is the map $\mathcal{G}_P/\mathcal{I}_P \to \mathcal{G}_P/(\operatorname{im}\varphi)_P$ induced by the identity of $\mathcal{G}_P$. By the proof of Ex. 1.2(a), $\mathcal{I}_P = (\operatorname{im}\varphi)_P = \operatorname{im}\varphi_P$ inside $\mathcal{G}_P$, so γ_P is the identity of $\mathcal{G}_P/\operatorname{im}\varphi_P$. By Prop. 1.1, γ is an isomorphism. □

Exercise 1.8

Problem. For any open subset $U \subseteq X$, show that the functor $\Gamma(U, \cdot)$ from sheaves on X to abelian groups is a left exact functor, i.e., if $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ is an exact sequence of sheaves, then $0 \to \Gamma(U, \mathcal{F}') \to \Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{F}'')$ is an exact sequence of groups. The functor $\Gamma(U, \cdot)$ need not be exact; see (Ex. 1.21) below.

Solution. Let the maps be $\alpha\colon \mathcal{F}' \to \mathcal{F}$, $\beta\colon \mathcal{F} \to \mathcal{F}''$. Exactness at $\mathcal{F}'$ means $\ker\alpha = 0$, i.e. $\ker\alpha(U) = 0$: $\alpha(U)$ is injective. Exactness at $\mathcal{F}$ means $\operatorname{im}\alpha = \ker\beta$; since α is injective, the presheaf image $V \mapsto \alpha(V)(\mathcal{F}'(V))$ is already a sheaf (Ex. 1.6(b)), hence equals $\operatorname{im}\alpha$. Thus $\ker\beta(U) = (\operatorname{im}\alpha)(U) = \alpha(U)(\mathcal{F}'(U))$, which says exactly that $\ker\beta(U) = \operatorname{im}\alpha(U)$. □

Exercise 1.9

Problem. *Direct Sum.* Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Show that the presheaf $U \mapsto \mathcal{F}(U) \oplus \mathcal{G}(U)$ is a sheaf. It is called the direct sum of $\mathcal{F}$ and $\mathcal{G}$, and is denoted $\mathcal{F} \oplus \mathcal{G}$. Show that it plays the role of direct sum and of direct product in the category of sheaves of abelian groups on X.

Solution. Let $\mathcal{H}(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$ with componentwise restriction; $\mathcal{H}(\emptyset) = 0 \oplus 0 = 0$.

(3) If $(a, b) \in \mathcal{H}(U)$ restricts to 0 on all V_i of a cover, then $a|_{V_i} = 0 = b|_{V_i}$ for all i, so $a = 0 = b$.

(4) If $(a_i, b_i) \in \mathcal{H}(V_i)$ agree on overlaps, then $a_i|_{V_i \cap V_j} = a_j|_{V_i \cap V_j}$ and likewise for b, so they glue to $a \in \mathcal{F}(U)$ and $b \in \mathcal{G}(U)$, and $(a, b)|_{V_i} = (a_i, b_i)$.

Universal properties. There are morphisms of sheaves $p_1 \colon \mathcal{H} \to \mathcal{F}$, $p_2 \colon \mathcal{H} \to \mathcal{G}$ (projections) and $\iota_1 \colon \mathcal{F} \to \mathcal{H}$, $a \mapsto (a, 0)$, $\iota_2 \colon \mathcal{G} \to \mathcal{H}$, $b \mapsto (0, b)$, with $p_k \iota_k = \mathrm{id}$, $p_1 \iota_2 = 0 = p_2 \iota_1$, and $\iota_1 p_1 + \iota_2 p_2 = \mathrm{id}$ (all clearly compatible with restriction).

Product: given a sheaf $\mathcal{E}$ and morphisms $f \colon \mathcal{E} \to \mathcal{F}$, $g \colon \mathcal{E} \to \mathcal{G}$, the map $(f, g)(U) \colon e \mapsto (f(e), g(e))$ is a morphism with $p_1(f, g) = f$, $p_2(f, g) = g$; if h is any morphism with $p_1 h = f$, $p_2 h = g$, then $h(e) = (f(e), g(e))$ on every section, so $h = (f, g)$.

Coproduct: given $f \colon \mathcal{F} \to \mathcal{E}$, $g \colon \mathcal{G} \to \mathcal{E}$, the map $(a, b) \mapsto f(a) + g(b)$ is a morphism $h \colon \mathcal{H} \to \mathcal{E}$ with $h\iota_1 = f$, $h\iota_2 = g$. If h' also satisfies this then $h'(a, b) = h'(\iota_1 a + \iota_2 b) = f(a) + g(b)$, so $h' = h$. □

Exercise 1.10

> **Problem.** *Direct Limit.* Let $\{\mathcal{F}_i\}$ be a direct system of sheaves and morphisms on X. We define the direct limit of the system $\{\mathcal{F}_i\}$, denoted $\varinjlim \mathcal{F}_i$, to be the sheaf associated to the presheaf $U \mapsto \varinjlim \mathcal{F}_i(U)$. Show that this is a direct limit in the category of sheaves on X, i.e., that it has the following universal property: given a sheaf $\mathcal{G}$, and a collection of morphisms $\mathcal{F}_i \to \mathcal{G}$, compatible with the maps of the direct system, then there exists a unique map $\varinjlim \mathcal{F}_i \to \mathcal{G}$ such that for each i, the original map $\mathcal{F}_i \to \mathcal{G}$ is obtained by composing the maps $\mathcal{F}_i \to \varinjlim \mathcal{F}_i \to \mathcal{G}$.

Solution. Here the direct system is indexed by a directed set I, with maps $f_{ij} \colon \mathcal{F}_i \to \mathcal{F}_j$ for $i \leq j$. Let $\mathcal{P}(U) = \varinjlim_i \mathcal{F}_i(U)$ with the evident restriction maps (induced by those of the $\mathcal{F}_i$, which commute with the f_{ij}); $\mathcal{P}$ is a presheaf, and $\mathcal{F} := \mathcal{P}^+ = \varinjlim \mathcal{F}_i$. Let $\lambda_i \colon \mathcal{F}_i(U) \to \mathcal{P}(U)$ be the canonical maps; they are morphisms of presheaves, and $\mu_i := \theta\lambda_i \colon \mathcal{F}_i \to \mathcal{F}$ are morphisms of sheaves with $\mu_j f_{ij} = \mu_i$.

Let $\mathcal{G}$ be a sheaf with morphisms $g_i \colon \mathcal{F}_i \to \mathcal{G}$, $g_j f_{ij} = g_i$. For each U the universal property of the direct limit of groups gives a unique homomorphism $g(U) \colon \mathcal{P}(U) \to \mathcal{G}(U)$ with $g(U)\lambda_i(U) = g_i(U)$. For $V \subseteq U$, the maps $\rho_{UV} g(U)$ and $g(V)\rho_{UV}$ agree after composing with each $\lambda_i(U)$ (both equal $\rho_{UV} g_i(U) = g_i(V)\rho_{UV}$), so they are equal: $g \colon \mathcal{P} \to \mathcal{G}$ is a morphism of presheaves. Since $\mathcal{G}$ is a sheaf, $g = \bar{g}\theta$ for a unique morphism $\bar{g} \colon \mathcal{F} \to \mathcal{G}$ (Prop.–Def. 1.2). Then $\bar{g}\mu_i = \bar{g}\theta\lambda_i = g\lambda_i = g_i$.

Uniqueness: if $h \colon \mathcal{F} \to \mathcal{G}$ satisfies $h\mu_i = g_i$ for all i, then $h\theta \colon \mathcal{P} \to \mathcal{G}$ satisfies $h\theta\lambda_i = g_i = g\lambda_i$ for all i, so $h\theta = g$ on each $\mathcal{P}(U) = \varinjlim \mathcal{F}_i(U)$ (generated by the images of the $\lambda_i(U)$), and by the uniqueness in Prop.–Def. 1.2, $h = \bar{g}$. □

Exercise 1.11

> **Problem.** Let $\{\mathcal{F}_i\}$ be a direct system of sheaves on a noetherian topological space X. In this case show that the presheaf $U \mapsto \varinjlim \mathcal{F}_i(U)$ is already a sheaf. In particular, $\Gamma(X, \varinjlim \mathcal{F}_i) = \varinjlim \Gamma(X, \mathcal{F}_i)$.

Solution. Let $\mathcal{P}(U) = \varinjlim \mathcal{F}_i(U)$ (I directed). Recall that on a noetherian space every open subset is quasi-compact (every open cover of an open set has a finite subcover: if not, choose successively members of the cover not contained in the union of the previous ones to get a strictly increasing infinite chain of open sets, contradicting the ascending chain condition on opens, equivalent to the descending chain condition on closed sets). Every element of $\mathcal{P}(U)$ is the class of some $s_i \in \mathcal{F}_i(U)$; and the class of s_i is 0 iff $f_{ik}(s_i) = 0$ for some $k \geq i$.

(3) Let $U = \bigcup U_j$ and $s \in \mathcal{P}(U)$ with $s|_{U_j} = 0$ for all j. Write s as the class of $s_i \in \mathcal{F}_i(U)$. For each j there is $k_j \geq i$ with $f_{ik_j}(s_i)|_{U_j} = 0$ in $\mathcal{F}_{k_j}(U_j)$. By quasi-compactness $U = U_{j_1} \cup \cdots \cup U_{j_n}$; take $k \geq k_{j_1}, \ldots, k_{j_n}$. Then $f_{ik}(s_i)|_{U_{j_m}} = f_{k_{j_m}k}\big(f_{ik_{j_m}}(s_i)|_{U_{j_m}}\big) = 0$ for $m = 1, \ldots, n$, so $f_{ik}(s_i) = 0$ because $\mathcal{F}_k$ is a sheaf. Hence $s = 0$.

(4) Let $U = \bigcup U_j$ and $s^j \in \mathcal{P}(U_j)$ with $s^j|_{U_j \cap U_l} = s^l|_{U_j \cap U_l}$. Take a finite subcover $U_1, \ldots, U_n$ (renumbering). Represent s^m by $t_m \in \mathcal{F}_{i_m}(U_m)$ and choose $i \geq i_1, \ldots, i_n$; replacing t_m by $f_{i_m i}(t_m)$ we may assume all $t_m \in \mathcal{F}_i(U_m)$. For each pair (m, l) the equality of classes of $t_m|_{U_m \cap U_l}$ and $t_l|_{U_m \cap U_l}$ means there is $k_{ml} \geq i$ with $f_{ik_{ml}}(t_m|) = f_{ik_{ml}}(t_l|)$. Let $k \geq i$ dominate the finitely many k_{ml} and put $t'_m = f_{ik}(t_m) \in \mathcal{F}_k(U_m)$; then $t'_m|_{U_m \cap U_l} = t'_l|_{U_m \cap U_l}$ in $\mathcal{F}_k$. Since $\mathcal{F}_k$ is a sheaf, there is $t \in \mathcal{F}_k(U)$ with $t|_{U_m} = t'_m$. Let $s \in \mathcal{P}(U)$ be its class; then $s|_{U_m} = s^m$ for $m = 1, \ldots, n$. For an arbitrary j, the sections $s|_{U_j}$ and s^j of $\mathcal{P}(U_j)$ have the same restriction to each $U_j \cap U_m$: $s|_{U_j \cap U_m} = s^m|_{U_j \cap U_m} = s^j|_{U_j \cap U_m}$. As the $U_j \cap U_m$ ($m = 1..n$) cover U_j, property (3) just proved gives $s|_{U_j} = s^j$.

So $\mathcal{P}$ is a sheaf; thus $\varinjlim \mathcal{F}_i = \mathcal{P}^+ \cong \mathcal{P}$ via θ, and $\Gamma(X, \varinjlim \mathcal{F}_i) = \mathcal{P}(X) = \varinjlim \mathcal{F}_i(X) = \varinjlim \Gamma(X, \mathcal{F}_i)$. □

Exercise 1.12

> **Problem.** *Inverse Limit.* Let $\{\mathcal{F}_i\}$ be an inverse system of sheaves on X. Show that the presheaf $U \mapsto \varprojlim \mathcal{F}_i(U)$ is a sheaf. It is called the inverse limit of the system $\{\mathcal{F}_i\}$, and is denoted $\varprojlim \mathcal{F}_i$. Show that it has the universal property of an inverse limit in the category of sheaves.

Solution. Let the system be indexed by a partially ordered set I with maps $f_{ji} \colon \mathcal{F}_j \to \mathcal{F}_i$ for $i \leq j$ (with $f_{ii} = \mathrm{id}$, $f_{ji}f_{kj} = f_{ki}$). Put $\mathcal{P}(U) = \varprojlim \mathcal{F}_i(U) = \{(s_i) \in \prod_i \mathcal{F}_i(U) : f_{ji}(s_j) = s_i \ \forall i \leq j\}$, with componentwise restriction; this is a presheaf (and $\mathcal{P}(\emptyset) = 0$).

(3) If $s = (s_i) \in \mathcal{P}(U)$ and $s|_{V_a} = 0$ for a cover $\{V_a\}$, then $s_i|_{V_a} = 0$ for all a, i, so $s_i = 0$ for all i since $\mathcal{F}_i$ is a sheaf.

(4) Let $s^a = (s_i^a)_i \in \mathcal{P}(V_a)$ agree on overlaps. For each i, the $s_i^a \in \mathcal{F}_i(V_a)$ agree on overlaps, so glue to $s_i \in \mathcal{F}_i(U)$ with $s_i|_{V_a} = s_i^a$. For $i \leq j$, $f_{ji}(s_j)$ and s_i in $\mathcal{F}_i(U)$ both restrict on V_a to $f_{ji}(s_j^a) = s_i^a$, so are equal by (3) for $\mathcal{F}_i$. So $s = (s_i) \in \mathcal{P}(U)$ and $s|_{V_a} = s^a$.

Universal property. The projections $p_i \colon \mathcal{P} \to \mathcal{F}_i$ are morphisms with $f_{ji}p_j = p_i$. Let $\mathcal{G}$ be a sheaf with morphisms $g_i \colon \mathcal{G} \to \mathcal{F}_i$ such that $f_{ji}g_j = g_i$. Then $g(U) \colon \mathcal{G}(U) \to \mathcal{P}(U)$, $t \mapsto (g_i(U)(t))_i$ lands in $\mathcal{P}(U)$, is compatible with restrictions, and satisfies $p_i g = g_i$. If $h \colon \mathcal{G} \to \mathcal{P}$ is any morphism with $p_i h = g_i$ for all i, then $h(t) = (p_i h(t))_i = (g_i(t))_i = g(t)$. So g is unique. □

Exercise 1.13

Problem. *Espace Etale of a Presheaf.* (This exercise is included only to establish the connection between our definition of a sheaf and another definition often found in the literature. See for example Godement [1, Ch. II, Section 1.2].) Given a presheaf $\mathcal{F}$ on X, we define a topological space $\mathrm{Spe}(\mathcal{F})$, called the espace etale of $\mathcal{F}$, as follows. As a set, $\mathrm{Spe}(\mathcal{F}) = \bigcup_{P \in X} \mathcal{F}_P$. We define a projection map $\pi\colon \mathrm{Spe}(\mathcal{F}) \to X$ by sending $s \in \mathcal{F}_P$ to P. For each open set $U \subseteq X$ and each section $s \in \mathcal{F}(U)$, we obtain a map $\bar{s}\colon U \to \mathrm{Spe}(\mathcal{F})$ by sending $P \mapsto s_P$, its germ at P. This map has the property that $\pi \circ \bar{s} = \mathrm{id}_U$, in other words, it is a "section" of π over U. We now make $\mathrm{Spe}(\mathcal{F})$ into a topological space by giving it the strongest topology such that all the maps $\bar{s}\colon U \to \mathrm{Spe}(\mathcal{F})$ for all U, and all $s \in \mathcal{F}(U)$, are continuous. Now show that the sheaf $\mathcal{F}^+$ associated to $\mathcal{F}$ can be described as follows: for any open set $U \subseteq X$, $\mathcal{F}^+(U)$ is the set of continuous sections of $\mathrm{Spe}(\mathcal{F})$ over U. In particular, the original presheaf $\mathcal{F}$ was a sheaf if and only if for each U, $\mathcal{F}(U)$ is equal to the set of all continuous sections of $\mathrm{Spe}(\mathcal{F})$ over U.

Solution. Write $E = \mathrm{Spe}(\mathcal{F})$. The strongest topology making all $\bar{s}$ continuous is: $W \subseteq E$ is open iff $\bar{s}^{-1}(W)$ is open in U for all open U and all $s \in \mathcal{F}(U)$. (This is a topology, and any topology making all $\bar{s}$ continuous is contained in it.) A *section* of π over U is a map $\sigma\colon U \to E$ with $\pi\sigma = \mathrm{id}_U$, i.e. $\sigma(P) \in \mathcal{F}_P$; it is continuous when $\sigma^{-1}(W)$ is open for all open $W \subseteq E$.

By Prop.–Def. 1.2, $\mathcal{F}^+(U)$ is the set of sections σ of π over U such that each $P \in U$ has a neighborhood $V \subseteq U$ and $t \in \mathcal{F}(V)$ with $\sigma(Q) = t_Q$ for all $Q \in V$ (i.e. $\sigma|_V = \bar{t}$). We show this equals the set of continuous sections.

($\subseteq$) If $\sigma \in \mathcal{F}^+(U)$, then U is covered by open sets V on which $\sigma|_V = \bar{t}$ with $t \in \mathcal{F}(V)$; $\bar{t}$ is continuous by definition of the topology of E. Continuity is local, so σ is continuous.

($\supseteq$) First a claim: for $t \in \mathcal{F}(V)$, the set $\bar{t}(V)$ is open in E. Let $s \in \mathcal{F}(U')$. Then $\bar{s}^{-1}(\bar{t}(V)) = \{Q \in U' \cap V : s_Q = t_Q\}$ (if $s_Q = t_{Q'}$ then $Q = \pi(s_Q) = Q'$). This set is open: if $s_Q = t_Q$ there is an open $W \ni Q$, $W \subseteq U' \cap V$, with $s|_W = t|_W$, so $s_{Q'} = t_{Q'}$ for all $Q' \in W$. This proves the claim.

Now let σ be a continuous section over U and $P \in U$. Choose $t \in \mathcal{F}(V)$ with $t_P = \sigma(P)$. The set $N = \sigma^{-1}(\bar{t}(V))$ is an open neighborhood of P in U (by the claim and continuity), and we may assume $N \subseteq V$ because $\pi\sigma(Q) = Q$ and $\sigma(Q) \in \bar{t}(V)$ imply $Q \in V$. For $Q \in N$, $\sigma(Q) = t_{Q'}$ for some $Q' \in V$ and applying π gives $Q' = Q$. So $\sigma(Q) = t_Q = (t|_N)_Q$ for all $Q \in N$. Thus $\sigma \in \mathcal{F}^+(U)$.

Restriction maps of $\mathcal{F}^+$ are restrictions of sections, so $\mathcal{F}^+$ is the sheaf of continuous sections of $E \to X$.

Finally, $\theta\colon \mathcal{F} \to \mathcal{F}^+$ is $s \mapsto \bar{s}$. If $\mathcal{F}$ is a sheaf, θ is an isomorphism (text), so $\mathcal{F}(U) \to \{\text{continuous sections over } U\}$, $s \mapsto \bar{s}$, is bijective for each U. Conversely, if all these maps are bijective then θ is an isomorphism of presheaves onto the sheaf $\mathcal{F}^+$, so $\mathcal{F}$ is a sheaf. □

Exercise 1.14

Problem. *Support.* Let $\mathcal{F}$ be a sheaf on X, and let $s \in \mathcal{F}(U)$ be a section over an open set U. The *support* of s, denoted $\mathrm{Supp}\, s$, is defined to be $\{P \in U \mid s_P \neq 0\}$, where s_P denotes the germ of s in the stalk $\mathcal{F}_P$. Show that $\mathrm{Supp}\, s$ is a closed subset of U. We define the support of $\mathcal{F}$, $\mathrm{Supp}\, \mathcal{F}$, to be $\{P \in X \mid \mathcal{F}_P \neq 0\}$. It need not be a closed subset.

Solution. The complement of $\operatorname{Supp} s$ in U is $\{P \in U : s_P = 0\}$. If $s_P = 0$, there is an open W, $P \in W \subseteq U$, with $s|_W = 0$; then $s_Q = 0$ for all $Q \in W$. So the complement is open and $\operatorname{Supp} s$ is closed in U.

Example where $\operatorname{Supp} \mathcal{F}$ *is not closed.* Let $X = \mathbf{R}$ (usual topology) and let $\mathcal{L}$ be the constant sheaf $\mathbf{Z}$ (locally constant functions to $\mathbf{Z}$). Define $\mathcal{F}(V) = \{s \in \mathcal{L}(V) : s(0) = 0 \text{ if } 0 \in V\}$. This is a subpresheaf of $\mathcal{L}$ defined by a pointwise condition, hence a sheaf: (3) is inherited from $\mathcal{L}$ and a glued section s of $\mathcal{L}$ from sections satisfying the pointwise condition satisfies it. For $P \neq 0$, small neighborhoods avoid 0 and $\mathcal{F}_P = \mathcal{L}_P = \mathbf{Z} \neq 0$. For $P = 0$ a germ is represented by a locally constant s with $s(0) = 0$, hence $s = 0$ on a neighborhood of 0; so $\mathcal{F}_0 = 0$. Therefore $\operatorname{Supp} \mathcal{F} = \mathbf{R} \setminus \{0\}$, which is not closed. □

Exercise 1.15

Problem. *Sheaf Hom.* Let $\mathcal{F}, \mathcal{G}$ be sheaves of abelian groups on X. For any open set $U \subseteq X$, show that the set $\operatorname{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ of morphisms of the restricted sheaves is an abelian group. Show that the presheaf $U \mapsto \operatorname{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ is a sheaf. This sheaf is called the *sheaf of local morphisms* of $\mathcal{F}$ into $\mathcal{G}$, "sheaf hom" for short, and is denoted $\mathcal{H}om(\mathcal{F}, \mathcal{G})$.

Solution. A morphism $\varphi\colon \mathcal{F}|_U \to \mathcal{G}|_U$ is a family of homomorphisms $\varphi(W)\colon \mathcal{F}(W) \to \mathcal{G}(W)$, $W \subseteq U$ open, commuting with restrictions. Define $(\varphi + \psi)(W) = \varphi(W) + \psi(W)$; this is again a family of homomorphisms (as $\mathcal{G}(W)$ is abelian) commuting with restriction, with zero element the zero morphism and inverse $-\varphi$; the group axioms are inherited from $\mathcal{G}(W)$. So $\operatorname{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ is an abelian group.

For open $V \subseteq U$, the restriction map sends φ to $\varphi|_V = (\varphi(W))_{W \subseteq V}$; this is a homomorphism of groups, and $(\varphi|_V)|_{V'} = \varphi|_{V'}$, so $\mathcal{H}om$ is a presheaf (value 0 on $\emptyset$, where there is a unique morphism).

(3) Let $U = \bigcup U_i$, $\varphi \in \operatorname{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ with $\varphi|_{U_i} = 0$ for all i. For $W \subseteq U$ open and $s \in \mathcal{F}(W)$, we get $\varphi(W)(s)|_{W \cap U_i} = \varphi(W \cap U_i)(s|_{W \cap U_i}) = 0$ for all i, so $\varphi(W)(s) = 0$ as $\mathcal{G}$ is a sheaf. Thus $\varphi = 0$.

(4) Let $\varphi_i \in \operatorname{Hom}(\mathcal{F}|_{U_i}, \mathcal{G}|_{U_i})$ with $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$. For $W \subseteq U$ open and $s \in \mathcal{F}(W)$ the sections $t_i = \varphi_i(W \cap U_i)(s|_{W \cap U_i}) \in \mathcal{G}(W \cap U_i)$ satisfy $t_i|_{W \cap U_i \cap U_j} = \varphi_i(W \cap U_i \cap U_j)(s|) = \varphi_j(W \cap U_i \cap U_j)(s|) = t_j|_{W \cap U_i \cap U_j}$, so glue to a unique $t \in \mathcal{G}(W)$; put $\varphi(W)(s) = t$. This $\varphi(W)$ is a homomorphism (uniqueness of gluing), and compatible with restriction $W' \subseteq W$ (both $\varphi(W)(s)|_{W'}$ and $\varphi(W')(s|_{W'})$ restrict to $t_i|_{W' \cap U_i}$ on $W' \cap U_i$ and hence agree by (3)). For $W \subseteq U_i$ we get $\varphi(W) = \varphi_i(W)$ (the cover of W by $W \cap U_j$ and the compatibility of the φ_j with φ_i show $t|_{W \cap U_j} = \varphi_i(W \cap U_j)(s|)$; use (3)). So $\varphi|_{U_i} = \varphi_i$. □

Exercise 1.16

Problem. *Flasque Sheaves.* A sheaf $\mathcal{F}$ on a topological space X is *flasque* if for every inclusion $V \subseteq U$ of open sets, the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$ is surjective.

(a) Show that a constant sheaf on an irreducible topological space is flasque. See (I, Section 1) for irreducible topological spaces.

(b) If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ is flasque, then for any open set U, the sequence $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U) \to 0$ of abelian groups is also exact.

(c) If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ and $\mathcal{F}$ are flasque, then $\mathcal{F}''$ is flasque.

(d) If $f\colon X \to Y$ is a continuous map, and if $\mathcal{F}$ is a flasque sheaf on X, then $f_*\mathcal{F}$ is a flasque sheaf on Y.

(e) Let $\mathcal{F}$ be any sheaf on X. We define a new sheaf $\mathcal{G}$, called the sheaf of *discontinuous sections* of $\mathcal{F}$ as follows. For each open set $U \subseteq X$, $\mathcal{G}(U)$ is the set of maps $s\colon U \to \bigcup_{P\in U} \mathcal{F}_P$ such that for each $P \in U$, $s(P) \in \mathcal{F}_P$. Show that $\mathcal{G}$ is a flasque sheaf, and that there is a natural injective morphism of $\mathcal{F}$ to $\mathcal{G}$.

Solution. (a) Let $\mathcal{A}$ be the constant sheaf with group A on irreducible X. A nonempty open subset U of an irreducible space is irreducible, hence connected. $\mathcal{A}(U)$ is the set of locally constant functions $U \to A$; on a connected U such a function is constant (the fibres are open and partition U). So $\mathcal{A}(U) = A$ for $U \neq \emptyset$ and the restriction maps for nonempty $V \subseteq U$ are the identity of A, hence surjective; for $V = \emptyset$ the target is 0. So $\mathcal{A}$ is flasque.

(b) Let $\alpha\colon \mathcal{F}' \to \mathcal{F}$, $\beta\colon \mathcal{F} \to \mathcal{F}''$. By Ex. 1.8 the sequence is exact at $\mathcal{F}'(U)$ and $\mathcal{F}(U)$ (in particular $\ker \beta(U) = \alpha(U)(\mathcal{F}'(U))$), so we must show that $\beta(U)$ is surjective. Let $s'' \in \mathcal{F}''(U)$. Consider the set Σ of pairs (V,t), $V \subseteq U$ open, $t \in \mathcal{F}(V)$ with $\beta(t) = s''|_V$, ordered by $(V,t) \le (V_1,t_1)$ iff $V \subseteq V_1$ and $t_1|_V = t$. By Ex. 1.3(a) there is a cover $\{U_i\}$ of U and $t_i \in \mathcal{F}(U_i)$ with $\beta(t_i) = s''|_{U_i}$, so $\Sigma \neq \emptyset$ (also $(\emptyset,0) \in \Sigma$). If $\{(V_a,t_a)\}$ is a chain, then with $V = \bigcup V_a$ the t_a agree on overlaps (the intersection of two members is the smaller one) and glue to $t \in \mathcal{F}(V)$; $\beta(t)|_{V_a} = \beta(t_a) = s''|_{V_a}$ for all a, so $\beta(t) = s''|_V$ by (3) for $\mathcal{F}''$; thus (V,t) is an upper bound. By Zorn's lemma there is a maximal (V,t). Suppose $V \neq U$, pick $P \in U \setminus V$ and by the above an open $W \ni P$, $W \subseteq U$, and $w \in \mathcal{F}(W)$ with $\beta(w) = s''|_W$. On $V \cap W$, $\beta(t|_{V\cap W} - w|_{V\cap W}) = 0$, so $t|_{V\cap W} - w|_{V\cap W} = \alpha(r)$ with $r \in \mathcal{F}'(V \cap W)$ by Ex. 1.8. Since $\mathcal{F}'$ is flasque, $r = r'|_{V\cap W}$ with $r' \in \mathcal{F}'(W)$. Put $w' = w + \alpha(r') \in \mathcal{F}(W)$. Then $\beta(w') = s''|_W$ and $w'|_{V\cap W} = t|_{V\cap W}$. So t and w' glue to a section over $V \cup W$ lifting s'' (check as before), contradicting maximality as $P \in W \setminus V$. Hence $V = U$ and $s'' = \beta(t)$.

(c) Let $V \subseteq U$ be open and $s'' \in \mathcal{F}''(V)$. By (b) applied to V, $s'' = \beta(t)$ for some $t \in \mathcal{F}(V)$. Since $\mathcal{F}$ is flasque there is $\tilde{t} \in \mathcal{F}(U)$ with $\tilde{t}|_V = t$. Then $\beta(\tilde{t}) \in \mathcal{F}''(U)$ restricts to $\beta(t) = s''$. So $\mathcal{F}''(U) \to \mathcal{F}''(V)$ is surjective.

(d) For open $V \subseteq U \subseteq Y$, $f^{-1}V \subseteq f^{-1}U$ are open in X, and $(f_*\mathcal{F})(U) \to (f_*\mathcal{F})(V)$ is the restriction $\mathcal{F}(f^{-1}U) \to \mathcal{F}(f^{-1}V)$, surjective since $\mathcal{F}$ is flasque.

(e) $\mathcal{G}$ is a presheaf under restriction of maps, and in fact a sheaf: for (3), if $s|_{U_i} = 0$ for a cover then $s(P) = 0$ for all P, i.e. $s = 0$; for (4), compatible $s_i \in \mathcal{G}(U_i)$ define $s(P) := s_i(P)$ for any i with $P \in U_i$, which is well defined by compatibility on overlaps; clearly $s|_{U_i} = s_i$. It is flasque: for $V \subseteq U$ and $s \in \mathcal{G}(V)$, extend s by 0 on $U \setminus V$ (i.e. $\tilde{s}(P) = 0 \in \mathcal{F}_P$ for $P \notin V$); then $\tilde{s} \in \mathcal{G}(U)$ and $\tilde{s}|_V = s$.

The map $\mathcal{F} \to \mathcal{G}$, $s \mapsto \bar{s} = (P \mapsto s_P)$ is a morphism of sheaves (compatible with restriction). It is injective: if $\bar{s} = 0$ then all germs s_P vanish, so $s = 0$ by Fact A. □

Exercise 1.17

Problem. *Skyscraper Sheaves.* Let X be a topological space, let P be a point, and let A be an abelian group. Define a sheaf $i_P(A)$ on X as follows: $i_P(A)(U) = A$ if $P \in U$, 0 otherwise. Verify that the stalk of $i_P(A)$ is A at every point $Q \in \{P\}^-$, and 0 elsewhere, where $\{P\}^-$ denotes the closure of the set consisting of the point P. Hence the name "skyscraper sheaf." Show that this sheaf could also be described as $i_*(A)$, where A denotes the constant sheaf A on the closed subspace $\{P\}^-$, and $i\colon \{P\}^- \to X$ is the inclusion.

Solution. Restriction maps of $i_P(A)$ (call it $\mathcal{S}$): for $V \subseteq U$, $\rho_{UV} = \mathrm{id}_A$ if $P \in V$, and 0 otherwise. This is a presheaf: for $W \subseteq V \subseteq U$, if $P \in W$ then $P \in V, U$ and all maps are identities; if $P \notin W$ then $\rho_{VW}\rho_{UV}$ and ρ_{UW} both have target 0. Also $\mathcal{S}(\emptyset) = 0$.

Sheaf. (3) Let $U = \bigcup V_i$, $s \in \mathcal{S}(U)$ with $s|_{V_i} = 0$. If $P \notin U$, $\mathcal{S}(U) = 0$. If $P \in U$, choose i with $P \in V_i$; then $s|_{V_i} = s$ (identity), so $s = 0$. (4) Let $s_i \in \mathcal{S}(V_i)$ be compatible. If $P \notin U$, all $V_i \not\ni P$, all $s_i = 0$ and $s = 0$ works. If $P \in U$, pick i_0 with $P \in V_{i_0}$ and put $s = s_{i_0} \in A$. For i with $P \in V_i$ we have $s_i = s_i|_{V_i \cap V_{i_0}} = s_{i_0}|_{V_i \cap V_{i_0}} = s$ (identities, $P \in V_i \cap V_{i_0}$), so $s|_{V_i} = s_i$; for i with $P \notin V_i$ both sides lie in $\mathcal{S}(V_i) = 0$.

Stalks. Let $Z = \{P\}^-$. If $Q \in Z$ then every open neighborhood U of Q meets $\{P\}$, i.e. contains P, so $\mathcal{S}(U) = A$ for all $U \ni Q$ with identity restriction maps, and $\mathcal{S}_Q = A$. If $Q \notin Z$, then $U_0 = X \setminus Z$ is an open neighborhood of Q which does not contain P; every neighborhood $U \subseteq U_0$ of Q has $\mathcal{S}(U) = 0$, so $\mathcal{S}_Q = 0$.

Comparison with i_*A. Let $\mathcal{A}$ be the constant sheaf on Z; $(i_*\mathcal{A})(U) = \mathcal{A}(U \cap Z)$. The space Z is irreducible (the closure of a point), so as in Ex. 1.16(a), $\mathcal{A}(W) = A$ for every nonempty open $W \subseteq Z$ with identity restrictions, and $\mathcal{A}(\emptyset) = 0$. Moreover $U \cap Z \neq \emptyset$ iff $P \in U$ (as above: if $Q \in U \cap Z$ then $P \in U$; conversely $P \in U \cap Z$). Hence $(i_*\mathcal{A})(U) = A$ if $P \in U$ and 0 otherwise, and for $V \subseteq U$ the restriction map is the identity if $P \in V$, and zero (into 0) otherwise: this is precisely $\mathcal{S}$. So $i_P(A) = i_*\mathcal{A}$. □

Exercise 1.18

Problem. *Adjoint Property of f^{-1}.* Let $f\colon X \to Y$ be a continuous map of topological spaces. Show that for any sheaf $\mathcal{F}$ on X there is a natural map $f^{-1}f_*\mathcal{F} \to \mathcal{F}$, and for any sheaf $\mathcal{G}$ on Y there is a natural map $\mathcal{G} \to f_*f^{-1}\mathcal{G}$. Use these maps to show that there is a natural bijection of sets, for any sheaves $\mathcal{F}$ on X and $\mathcal{G}$ on Y, $\mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) = \mathrm{Hom}_Y(\mathcal{G}, f_*\mathcal{F})$. Hence we say that f^{-1} is a *left adjoint* of f_*, and that f_* is a *right adjoint* of f^{-1}.

Solution. Recall $f^{-1}\mathcal{G} = \mathcal{P}^+_{\mathcal{G}}$, where $\mathcal{P}_{\mathcal{G}}(U) = \varinjlim_{V \supseteq f(U)} \mathcal{G}(V)$, V open in Y. Write ρ for restriction maps in all sheaves.

Natural maps. (i) Let $\mathcal{F}$ be a sheaf on X. For $U \subseteq X$ open and $V \supseteq f(U)$ open, $U \subseteq f^{-1}V$, and we have $\rho\colon (f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V) \to \mathcal{F}(U)$. These are compatible with shrinking V, so they induce $\varepsilon'(U)\colon \mathcal{P}_{f_*\mathcal{F}}(U) \to \mathcal{F}(U)$, compatible with restriction in U: a morphism of presheaves $\mathcal{P}_{f_*\mathcal{F}} \to \mathcal{F}$. Since $\mathcal{F}$ is a sheaf, it factors uniquely through $f^{-1}f_*\mathcal{F} = \mathcal{P}^+_{f_*\mathcal{F}}$, giving $\varepsilon_{\mathcal{F}}\colon f^{-1}f_*\mathcal{F} \to \mathcal{F}$.

(ii) Let $\mathcal{G}$ be a sheaf on Y. For $V \subseteq Y$ open, $f(f^{-1}V) \subseteq V$, so V is one of the open sets in the limit defining $\mathcal{P}_{\mathcal{G}}(f^{-1}V)$, giving $\mathcal{G}(V) \to \mathcal{P}_{\mathcal{G}}(f^{-1}V) \xrightarrow{\theta} (f^{-1}\mathcal{G})(f^{-1}V) = (f_*f^{-1}\mathcal{G})(V)$. These

are compatible with restrictions, and define $\eta_{\mathcal{G}}\colon \mathcal{G} \to f_* f^{-1}\mathcal{G}$. Both constructions are clearly functorial in $\mathcal{F}$, resp. $\mathcal{G}$.

The bijection. By the universal property of Prop.–Def. 1.2, $\mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) = \mathrm{Hom}_{\text{presheaves}}(\mathcal{P}_{\mathcal{G}}, \mathcal{F})$ (composition with θ). By the universal property of the direct limit of groups, a morphism of presheaves $\alpha\colon \mathcal{P}_{\mathcal{G}} \to \mathcal{F}$ is the same as a family of homomorphisms $a_{U,V}\colon \mathcal{G}(V) \to \mathcal{F}(U)$, one for each pair of opens $U \subseteq X$, $V \subseteq Y$ with $f(U) \subseteq V$, such that (i) $a_{U,V'}\rho_{VV'} = a_{U,V}$ for $V' \subseteq V$ (with $f(U) \subseteq V'$), (ii) $\rho_{UU'}a_{U,V} = a_{U',V}$ for $U' \subseteq U$.

Given such a family put $b_V = a_{f^{-1}V,V}\colon \mathcal{G}(V) \to \mathcal{F}(f^{-1}V) = (f_*\mathcal{F})(V)$. For $V' \subseteq V$: $\rho_{f^{-1}V,f^{-1}V'}b_V = a_{f^{-1}V',V}$ by (ii) and this equals $a_{f^{-1}V',V'}\rho_{VV'} = b_{V'}\rho_{VV'}$ by (i); so $b\colon \mathcal{G} \to f_*\mathcal{F}$ is a morphism of sheaves. Conversely, given a morphism $b\colon \mathcal{G} \to f_*\mathcal{F}$ put $a_{U,V} = \rho_{f^{-1}V,U}\, b_V$; (ii) is clear, and (i): $a_{U,V'}\rho_{VV'} = \rho_{f^{-1}V',U}b_{V'}\rho_{VV'} = \rho_{f^{-1}V',U}\rho_{f^{-1}V,f^{-1}V'}b_V = a_{U,V}$. The two constructions are inverse: from b we get $a_{f^{-1}V,V} = b_V$; from a we get $\rho_{f^{-1}V,U}a_{f^{-1}V,V} = a_{U,V}$ by (ii). This gives the bijection $\mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_Y(\mathcal{G}, f_*\mathcal{F})$, natural in $\mathcal{F}$ and $\mathcal{G}$ (it is compatible with composing on either side, since composition just acts on the groups $\mathcal{G}(V)$, $\mathcal{F}(U)$). Unwinding, the bijection takes $\alpha \mapsto f_*(\alpha) \circ \eta_{\mathcal{G}}$, and the identity of $f_*\mathcal{F}$ corresponds to $\varepsilon_{\mathcal{F}}$, and the identity of $f^{-1}\mathcal{G}$ corresponds to $\eta_{\mathcal{G}}$; i.e. the natural maps constructed above are the unit and counit of the adjunction. □

Exercise 1.19

Problem. *Extending a Sheaf by Zero.* Let X be a topological space, let Z be a closed subset, let $i\colon Z \to X$ be the inclusion, and let $U = X - Z$ be the complementary open subset, and let $j\colon U \to X$ be its inclusion.

(a) Let $\mathcal{F}$ be a sheaf on Z. Show that the stalk $(i_*\mathcal{F})_P$ of the direct image sheaf on X is $\mathcal{F}_P$ if $P \in Z$, 0 if $P \notin Z$. Hence we call $i_*\mathcal{F}$ the sheaf obtained by *extending $\mathcal{F}$ by zero* outside Z. By abuse of notation we will sometimes write $\mathcal{F}$ instead of $i_*\mathcal{F}$, and say "consider $\mathcal{F}$ as a sheaf on X," when we mean $i_*\mathcal{F}$.

(b) Now let $\mathcal{F}$ be a sheaf on U. Let $j_!(\mathcal{F})$ be the sheaf on X associated to the presheaf $V \mapsto \mathcal{F}(V)$ if $V \subseteq U$, $V \mapsto 0$ if $V \not\subseteq U$. Show that the stalk $(j_!(\mathcal{F}))_P$ is equal to $\mathcal{F}_P$ if $P \in U$, 0 if $P \notin U$, and show that $j_!\mathcal{F}$ is the only sheaf on X which has this property, and whose restriction to U is $\mathcal{F}$. We call $j_!\mathcal{F}$ the sheaf obtained by *extending $\mathcal{F}$ by zero* outside U.

(c) Now let $\mathcal{F}$ be a sheaf on X. Show that there is an exact sequence of sheaves on X, $0 \to j_!(\mathcal{F}|_U) \to \mathcal{F} \to i_*(\mathcal{F}|_Z) \to 0$.

Solution. (a) $(i_*\mathcal{F})(W) = \mathcal{F}(W \cap Z)$. Let $P \in Z$. The map $\Phi\colon (i_*\mathcal{F})_P = \varinjlim_{W \ni P} \mathcal{F}(W \cap Z) \to \mathcal{F}_P = \varinjlim_{W' \ni P,\ W' \subseteq Z \text{ open}} \mathcal{F}(W')$, $\langle W, s\rangle \mapsto \langle W \cap Z, s\rangle$, is well defined. It is surjective, since every open $W' \subseteq Z$ is of the form $W \cap Z$ with W open in X. It is injective: if $\Phi\langle W, s\rangle = 0$ then $s|_{W''} = 0$ for some open $W'' \ni P$ of Z; write $W'' = W_1 \cap Z$ with W_1 open in X; replacing W_1 by $W_1 \cap W$ (which still satisfies $W_1 \cap Z = W''$ since $W'' \subseteq W \cap Z$) we get $W_1 \subseteq W$ and $s|_{W_1 \cap Z} = 0$, i.e. $\langle W, s\rangle = 0$ in $(i_*\mathcal{F})_P$. If $P \notin Z$, then $X \setminus Z$ is an open neighborhood of P, and for all open $W \ni P$ contained in it $(i_*\mathcal{F})(W) = \mathcal{F}(\emptyset) = 0$, so the stalk is 0.

(b) Let $\mathcal{P}$ be the presheaf in the statement; $\mathcal{P}^+ = j_!\mathcal{F}$ and $\mathcal{P}^+_P = \mathcal{P}_P$. If $P \in U$, the open neighborhoods contained in U are cofinal and $\mathcal{P}(V) = \mathcal{F}(V)$ for them, so $\mathcal{P}_P = \mathcal{F}_P$. If $P \notin U$, no neighborhood of P is contained in U, so $\mathcal{P}(V) = 0$ for all $V \ni P$ and $\mathcal{P}_P = 0$.

Restriction to U: $\mathcal{P}|_U = \mathcal{F}$ is a sheaf on U, and $\theta|_U \colon \mathcal{P}|_U = \mathcal{F} \to (j_!\mathcal{F})|_U$ is a morphism of sheaves on U which on stalks (at $Q \in U$) is $\mathcal{P}_Q \to \mathcal{P}_Q^+$, an isomorphism; by Prop. 1.1 $(j_!\mathcal{F})|_U \cong \mathcal{F}$ (compatibly with the identification of stalks).

Uniqueness: let $\mathcal{G}$ be a sheaf on X with $\mathcal{G}_P = 0$ for $P \notin U$ and an isomorphism $\psi \colon \mathcal{F} \to \mathcal{G}|_U$. Define $\kappa \colon \mathcal{P} \to \mathcal{G}$ by $\kappa(V) = \psi(V)$ for $V \subseteq U$ and $\kappa(V) = 0$ otherwise. This commutes with restrictions: for $V' \subseteq V$, if $V \subseteq U$ it is the compatibility of ψ; if $V \not\subseteq U$ then $\mathcal{P}(V) = 0$ and both composites $\mathcal{P}(V) \to \mathcal{G}(V')$ are 0. By Prop.–Def. 1.2, κ factors through $\bar{\kappa} \colon j_!\mathcal{F} = \mathcal{P}^+ \to \mathcal{G}$. On stalks at $Q \in U$, $\bar{\kappa}_Q = \kappa_Q = \psi_Q$ is an isomorphism; at $Q \notin U$, both stalks are zero. By Prop. 1.1, $\bar{\kappa}$ is an isomorphism, restricting to ψ on U (up to the identification above). So $j_!\mathcal{F}$ is the unique such sheaf.

(c) *The maps.* Let $\mathcal{P}(V) = \mathcal{F}(V)$ if $V \subseteq U$ and 0 otherwise (this is the presheaf for $j_!(\mathcal{F}|_U)$; note $(\mathcal{F}|_U)(V) = \mathcal{F}(V)$ for $V \subseteq U$). The maps $\mathcal{P}(V) \to \mathcal{F}(V)$ (identity for $V \subseteq U$, zero otherwise) are compatible with restriction, so they induce $\alpha \colon j_!(\mathcal{F}|_U) \to \mathcal{F}$. Let $\beta = \eta_{\mathcal{F}} \colon \mathcal{F} \to i_*i^{-1}\mathcal{F} = i_*(\mathcal{F}|_Z)$ be the natural map of Ex. 1.18 (for the map i).

Exactness is checked on stalks (Ex. 1.2(c)). Recall $(\mathcal{F}|_Z)_P = \mathcal{F}_P$ for $P \in Z$ (text). At $P \in U$: the stalk sequence is $0 \to (j_!\mathcal{F}|_U)_P = \mathcal{F}_P \xrightarrow{\alpha_P} \mathcal{F}_P \to (i_*\mathcal{F}|_Z)_P = 0$ (by (a), as $P \notin Z$); here α_P is the identity (computed on the presheaf $\mathcal{P}$, where it is the identity on germs of sections over $V \subseteq U$). This is exact. At $P \in Z$: the stalk sequence is $0 \to 0 \to \mathcal{F}_P \xrightarrow{\beta_P} \mathcal{F}_P \to 0$, since $(j_!\mathcal{F}|_U)_P = 0$ by (b) and $(i_*\mathcal{F}|_Z)_P = (\mathcal{F}|_Z)_P = \mathcal{F}_P$ by (a). Here β_P sends the germ of $s \in \mathcal{F}(W)$ to the germ at P of the image of s in $(i^{-1}\mathcal{F})(W \cap Z)$, which under the identification of the stalk of $i^{-1}\mathcal{F}$ with $\mathcal{F}_P$ is s_P; so $\beta_P = \mathrm{id}$ and the sequence is exact. Hence $0 \to j_!(\mathcal{F}|_U) \to \mathcal{F} \to i_*(\mathcal{F}|_Z) \to 0$ is exact (in particular the composition $\beta\alpha$ has zero stalks, hence is 0 by Fact A). □

Exercise 1.20

Problem. *Subsheaf with Supports.* Let Z be a closed subset of X, and let $\mathcal{F}$ be a sheaf on X. We define $\Gamma_Z(X, \mathcal{F})$ to be the subgroup of $\Gamma(X, \mathcal{F})$ consisting of all sections whose support (Ex. 1.14) is contained in Z.

(a) Show that the presheaf $V \mapsto \Gamma_{Z\cap V}(V, \mathcal{F}|_V)$ is a sheaf. It is called the *subsheaf of $\mathcal{F}$ with supports in Z*, and is denoted $\mathcal{H}^0_Z(\mathcal{F})$.

(b) Let $U = X - Z$, and let $j \colon U \to X$ be the inclusion. Show there is an exact sequence of sheaves on X

$$0 \to \mathcal{H}^0_Z(\mathcal{F}) \to \mathcal{F} \to j_*(\mathcal{F}|_U).$$

Furthermore, if $\mathcal{F}$ is flasque, the map $\mathcal{F} \to j_*(\mathcal{F}|_U)$ is surjective.

Solution. (a) Let $\mathcal{H}(V) = \{s \in \mathcal{F}(V) : \operatorname{Supp} s \subseteq Z \cap V\} = \{s \in \mathcal{F}(V) : s_P = 0 \ \forall P \in V \setminus Z\}$. For $W \subseteq V$ we have $(s|_W)_P = s_P$, so restriction maps $\mathcal{H}(V) \to \mathcal{H}(W)$; $\mathcal{H}$ is a subpresheaf of $\mathcal{F}$ and a subgroup in each degree. (3) is inherited from $\mathcal{F}$. (4): if $s_i \in \mathcal{H}(V_i)$ are compatible, they glue to $s \in \mathcal{F}(V)$, $V = \bigcup V_i$, and for $P \in V \setminus Z$ pick i with $P \in V_i$: $s_P = (s_i)_P = 0$. So $s \in \mathcal{H}(V)$. Thus $\mathcal{H} = \mathcal{H}^0_Z(\mathcal{F})$ is a sheaf, a subsheaf of $\mathcal{F}$.

(b) Let $\gamma \colon \mathcal{F} \to j_*(\mathcal{F}|_U)$ be given by $\gamma(V) \colon \mathcal{F}(V) \to \mathcal{F}(V \cap U) = (j_*(\mathcal{F}|_U))(V)$, the restriction; clearly a morphism of sheaves. We have $\ker\gamma(V) = \{s \in \mathcal{F}(V) : s|_{V\cap U} = 0\}$. By Fact A, $s|_{V\cap U} = 0$ iff $s_P = 0$ for all $P \in V \cap U = V \setminus Z$ (the converse implication: if $s|_{V\cap U} = 0$ then its

germs vanish), i.e. iff $s \in \mathcal{H}^0_Z(\mathcal{F})(V)$. So $\ker\gamma = \mathcal{H}^0_Z(\mathcal{F})$ as subsheaves of $\mathcal{F}$, which means that $0 \to \mathcal{H}^0_Z(\mathcal{F}) \to \mathcal{F} \to j_*(\mathcal{F}|_U)$ is exact (injectivity at the left since it is a subsheaf, exactness in the middle since the image of the inclusion is $\ker\gamma$).

If $\mathcal{F}$ is flasque, $\gamma(V)\colon \mathcal{F}(V) \to \mathcal{F}(V \cap U)$ is surjective for every open V ($V \cap U \subseteq V$), so γ is surjective as a morphism of sheaves (by Ex. 1.3(a), with the trivial cover). □

Exercise 1.21

Problem. *Some Examples of Sheaves on Varieties.* Let X be a variety over an algebraically closed field k, as in Ch. I. Let $\mathcal{O}_X$ be the sheaf of regular functions on X (1.0.1).

(a) Let Y be a closed subset of X. For each open set $U \subseteq X$, let $\mathcal{I}_Y(U)$ be the ideal in the ring $\mathcal{O}_X(U)$ consisting of those regular functions which vanish at all points of $Y \cap U$. Show that the presheaf $U \mapsto \mathcal{I}_Y(U)$ is a sheaf. It is called the *sheaf of ideals* $\mathcal{I}_Y$ of Y, and it is a subsheaf of the sheaf of rings $\mathcal{O}_X$.

(b) If Y is a subvariety, then the quotient sheaf $\mathcal{O}_X/\mathcal{I}_Y$ is isomorphic to $i_*(\mathcal{O}_Y)$, where $i\colon Y \to X$ is the inclusion, and $\mathcal{O}_Y$ is the sheaf of regular functions on Y.

(c) Now let $X = \mathbf{P}^1$, and let Y be the union of two distinct points $P, Q \in X$. Then there is an exact sequence of sheaves on X, where $\mathcal{F} = i_*\mathcal{O}_P \oplus i_*\mathcal{O}_Q$:

$$0 \to \mathcal{I}_Y \to \mathcal{O}_X \to \mathcal{F} \to 0.$$

Show that the induced map on global sections $\Gamma(X, \mathcal{O}_X) \to \Gamma(X, \mathcal{F})$ is not surjective. This shows that the global section functor $\Gamma(X, \cdot)$ is not exact (cf. Ex. 1.8 which shows that it is left exact).

(d) Again let $X = \mathbf{P}^1$, and let $\mathcal{O}$ be the sheaf of regular functions. Let $\mathcal{K}$ be the constant sheaf on X associated to the function field K of X. Show that there is a natural injection $\mathcal{O} \to \mathcal{K}$. Show that the quotient sheaf $\mathcal{K}/\mathcal{O}$ is isomorphic to the direct sum $\sum_{P \in X} i_P(I_P)$, where I_P is the group $K/\mathcal{O}_P$, and $i_P(I_P)$ denotes the skyscraper sheaf (Ex. 1.17) given by I_P at the point P.

(e) Finally show that in the case of (d) the sequence $0 \to \Gamma(X, \mathcal{O}) \to \Gamma(X, \mathcal{K}) \to \Gamma(X, \mathcal{K}/\mathcal{O}) \to 0$ is exact. (This is an analogue of what is called the "first Cousin problem" in several complex variables. See Gunning and Rossi [1, p. 248].)

Solution. Conventions: points of a variety are its closed points; $\mathcal{O}_X(U)$ is the ring of regular functions on U, and for $f \in \mathcal{O}_X(U)$, $P \in U$, $f(P) \in k$ is its value, which depends only on the germ $f_P \in \mathcal{O}_{X,P}$ and is compatible with restrictions.

(a) $\mathcal{I}_Y(U) = \{f \in \mathcal{O}_X(U) : f(P) = 0\ \forall P \in Y \cap U\}$ is an ideal (value at P is a ring homomorphism), and restriction preserves the condition, so $\mathcal{I}_Y$ is a subpresheaf of ideals of $\mathcal{O}_X$. (3) is inherited from $\mathcal{O}_X$. (4): if $f_i \in \mathcal{I}_Y(U_i)$ agree on overlaps, they glue to $f \in \mathcal{O}_X(U)$, $U = \bigcup U_i$, and for $P \in Y \cap U$, $P \in U_i$ for some i and $f(P) = f_i(P) = 0$. So $f \in \mathcal{I}_Y(U)$.

(b) Let Y be a closed subvariety (closed irreducible subset with induced structure) and $i\colon Y \to X$. For $U \subseteq X$ open, restriction of regular functions gives a ring homomorphism $r(U)\colon \mathcal{O}_X(U) \to \mathcal{O}_Y(U \cap Y) = (i_*\mathcal{O}_Y)(U)$ (a function regular on U is locally of the form g/h with g, h homogeneous polynomials of the same degree, $h \neq 0$ at the relevant points; its restriction to $Y \cap U$ is locally given by the same expression, hence is regular on $Y \cap U$ as a variety). Clearly r is a morphism of sheaves. Its kernel is $\ker r(U) = \{f : f|_{Y \cap U} = 0\} = \mathcal{I}_Y(U)$, so $\ker r = \mathcal{I}_Y$.

r is surjective (Ex. 1.3(a)): let $\varphi \in \mathcal{O}_Y(U \cap Y)$ and $P \in U$. If $P \notin Y$, then $W = U \setminus Y$ is an open neighborhood of P with $W \cap Y = \emptyset$, so $\varphi|_W \in (i_*\mathcal{O}_Y)(W) = \mathcal{O}_Y(\emptyset) = 0$ is the image of 0. If $P \in Y \cap U$, by definition of regular function on $Y \subseteq X \subseteq \mathbf{P}^n$ there is an open neighborhood $W_0 \ni P$ in $Y \cap U$ and homogeneous polynomials g, h of the same degree with h nowhere zero on W_0 and $\varphi = g/h$ on W_0. Choose W_1 open in X with $W_1 \cap Y = W_0$ (possible since W_0 is open in Y) and let $W = \{Q \in W_1 \cap U : h(Q) \neq 0\}$, an open neighborhood of P in X. Then $f = g/h \in \mathcal{O}_X(W)$ and $r(W)(f) = f|_{W\cap Y} = \varphi|_{W\cap Y}$, as $W \cap Y \subseteq W_0$. The open sets W obtained (for all $P \in U$) cover U, so r is surjective.

Therefore $\operatorname{im} r = i_*\mathcal{O}_Y$ and by Ex. 1.7(a), $\mathcal{O}_X/\mathcal{I}_Y = \mathcal{O}_X/\ker r \cong \operatorname{im} r = i_*\mathcal{O}_Y$.

(c) Here $\mathcal{O}_P$ denotes the sheaf of regular functions on the one-point variety $\{P\}$, which is the constant sheaf k on $\{P\}$; hence $i_*\mathcal{O}_P$ is the skyscraper sheaf $i_P(k)$ of Ex. 1.17 (note $\{P\}$ is closed). So $\mathcal{F}(U) = k^{\epsilon_P(U)+\epsilon_Q(U)}$ where $\epsilon_R(U) = 1$ if $R \in U$ and 0 otherwise. Define $\psi\colon \mathcal{O}_X \to \mathcal{F}$ by $\psi(U)(f) = (f(P)$ if $P \in U$, $f(Q)$ if $Q \in U)$; a morphism of sheaves (indeed (b) applied to $Y = \{P\}$ and $Y = \{Q\}$). Its kernel is $\ker\psi(U) = \{f : f(R) = 0$ for all $R \in Y \cap U\} = \mathcal{I}_Y(U)$ with $Y = \{P, Q\}$, so $\ker\psi = \mathcal{I}_Y$, giving exactness at $\mathcal{O}_X$ and at $\mathcal{I}_Y$ (inclusion). For surjectivity use stalks (Ex. 1.2(b)): $\mathcal{F}_P = k$ (from $i_*\mathcal{O}_P$; the summand for Q contributes 0 since $P \notin \{Q\}^- = \{Q\}$), $\mathcal{F}_Q = k$, and $\mathcal{F}_R = 0$ for $R \neq P, Q$ (Ex. 1.17). The map $\psi_P\colon \mathcal{O}_{X,P} \to k$ is evaluation, which is surjective (constants), similarly ψ_Q. So ψ is surjective, and $0 \to \mathcal{I}_Y \to \mathcal{O}_X \to \mathcal{F} \to 0$ is exact.

On global sections: $\Gamma(X, \mathcal{O}_X) = k$ (the regular functions on $\mathbf{P}^1$ are constants, Thm. I.3.4(a)), $\Gamma(X, \mathcal{F}) = k \oplus k$, and the map is $c \mapsto (c, c)$ (a constant function takes the same value at P and Q). Its image is the diagonal, so it is not surjective. Hence $\Gamma(X, \cdot)$ is not exact.

(d) Let $X = \mathbf{P}^1$ and $K = k(X) = k(x)$. X is irreducible, so $\mathcal{K}(U) = K$ for every nonempty open U, with identity restrictions (Ex. 1.16(a)), and $\mathcal{K}(\emptyset) = 0$. A regular function φ on a nonempty open U is locally a quotient of homogeneous polynomials of equal degree; any two such local expressions agree on the (nonempty, by irreducibility) overlap and hence define the same element of K (an element of K is an equivalence class of regular functions on nonempty open sets, two being equivalent if they agree on a common nonempty open set). So there is a well-defined map $\iota(U)\colon \mathcal{O}(U) \to K$, a ring homomorphism compatible with restrictions. It is injective: if $\iota(U)\varphi = 0$, then φ vanishes on a nonempty open subset of U, which is dense in U, and the zero set of φ is closed, so $\varphi = 0$. Thus $\iota\colon \mathcal{O} \to \mathcal{K}$ is an injective morphism of sheaves (Ex. 1.2(b) / injectivity on sections). At the stalk P, $\mathcal{O}_P \subseteq K$ is the local ring.

The sheaf $\mathcal{G}$. For U open let $\mathcal{G}(U) = \bigoplus_{P\in U} I_P = \{(g_P)_{P\in U} \in \prod_{P\in U} I_P :\ g_P = 0$ for all but finitely many $P\}$ with restriction "forget the coordinates outside the smaller set." Then $\mathcal{G}$ is a presheaf. It is a sheaf: (3) is clear coordinatewise; for (4), if $g^i \in \mathcal{G}(U_i)$ are compatible, define $g_P = g^i_P$ for any i with $P \in U_i$ (well defined by compatibility); $X = \mathbf{P}^1$ is noetherian so $U = \bigcup U_i$ is covered by finitely many $U_{i_1}, \ldots, U_{i_n}$ and the support of g is contained in the union of the finite supports of $g^{i_1}, \ldots, g^{i_n}$, hence finite. (This $\mathcal{G}$ is the subsheaf of finitely supported sections of the product $\prod_P i_P(I_P)$; it is the object denoted $\sum_P i_P(I_P)$; this is the reading of the notation used here.)

The map. A rational function $f \in K$ lies in $\mathcal{O}_P$ for all but finitely many closed points P (write $f = a/b$ with a, b homogeneous of the same degree, $b \neq 0$; $f \in \mathcal{O}_P$ whenever $b(P) \neq 0$, which holds outside the finite zero set of b). Hence for $U \neq \emptyset$ the map $\lambda(U)\colon K \to \mathcal{G}(U)$, $f \mapsto (f \bmod \mathcal{O}_P)_{P\in U}$ is well defined, a homomorphism, compatible with restriction (and $\lambda(\emptyset) = 0$). Thus $\lambda\colon \mathcal{K} \to \mathcal{G}$ is a morphism of sheaves, and $\lambda\iota = 0$ because $\varphi \in \mathcal{O}(U)$ has $\iota(\varphi) \in \mathcal{O}_P$ for

all $P \in U$. So λ factors through the presheaf quotient $U \mapsto \mathcal{K}(U)/\mathcal{O}(U)$, and hence through a morphism $\bar{\lambda}\colon \mathcal{K}/\mathcal{O} \to \mathcal{G}$.

$\bar{\lambda}$ is an isomorphism. Check stalks (Prop. 1.1). $(\mathcal{K}/\mathcal{O})_P = \mathcal{K}_P/\mathcal{O}_P = K/\mathcal{O}_P = I_P$. For $\mathcal{G}$, let $\pi_P\colon \mathcal{G}_P \to I_P$ be induced by the projection $\mathcal{G}(U) \to I_P$, $P \in U$. It is surjective: $a \in I_P$ is the image of the section supported at P with value a. It is injective: if the germ of $g \in \mathcal{G}(U)$ has $g_P = 0$ (the P-coordinate), then the support S of g is finite, so $W = U \setminus (S \setminus \{P\})$ is an open neighborhood of P (finite sets of closed points are closed) and $g|_W$ has support contained in $\{P\}$ with zero P-coordinate, i.e. $g|_W = 0$. So $\mathcal{G}_P \cong I_P$, and the composite $K/\mathcal{O}_P = (\mathcal{K}/\mathcal{O})_P \xrightarrow{\bar{\lambda}_P} \mathcal{G}_P \xrightarrow{\pi_P} I_P$ is $f \bmod \mathcal{O}_P \mapsto f \bmod \mathcal{O}_P$, the identity. Hence $\bar{\lambda}_P$ is an isomorphism for each P and $\mathcal{K}/\mathcal{O} \cong \mathcal{G} = \sum_P i_P(I_P)$.

(e) By Ex. 1.6(a) (applied to the subsheaf $\iota(\mathcal{O}) \subseteq \mathcal{K}$) $0 \to \mathcal{O} \to \mathcal{K} \to \mathcal{K}/\mathcal{O} \to 0$ is exact, so by Ex. 1.8 the sequence $0 \to \Gamma(X,\mathcal{O}) \to \Gamma(X,\mathcal{K}) \to \Gamma(X,\mathcal{K}/\mathcal{O})$ is exact. By (d), $\Gamma(X,\mathcal{K}/\mathcal{O}) \cong \mathcal{G}(X) = \bigoplus_{P\in X} I_P$ and $\Gamma(X,\mathcal{K}) = K$, with the map $K \to \bigoplus_P K/\mathcal{O}_P$, $f \mapsto (f \bmod \mathcal{O}_P)_P$. We must show it is surjective: given finitely many distinct points $P_1,\dots,P_n$ and $g_1,\dots,g_n \in K$, find $f \in K$ with $f - g_m \in \mathcal{O}_{P_m}$ for each m, and $f \in \mathcal{O}_R$ for all other points R.

It suffices to treat each m separately: find $f_m \in K$ with $f_m - g_m \in \mathcal{O}_{P_m}$ and $f_m \in \mathcal{O}_R$ for all $R \neq P_m$; then $f = \sum f_m$ works (at P_m, $f - g_m = (f_m - g_m) + \sum_{l\neq m} f_l \in \mathcal{O}_{P_m}$; at $R \notin \{P_m\}$ all $f_l \in \mathcal{O}_R$).

Fix $P = P_m$, $g = g_m$. Choose homogeneous coordinates on $\mathbf{P}^1$ (a linear change of coordinates) so that P is the point at infinity $[0:1]$ and put $x =$ the coordinate function on $\mathbf{A}^1 = \mathbf{P}^1 \setminus \{P\}$, so $K = k(x)$, $k[x] \subseteq \mathcal{O}_R$ for every $R \neq P$ (a polynomial in x is regular on $\mathbf{A}^1$), and $\mathcal{O}_P = \{a/b : a,b \in k[x], b \neq 0,\ \deg a \leq \deg b\}$. Write $g = a/b$ with $a,b \in k[x]$, $b \neq 0$, and divide: $a = qb + r$ with $q,r \in k[x]$, $\deg r < \deg b$. Then $f_m := q \in k[x]$ is regular at every $R \neq P$, and $g - f_m = r/b \in \mathcal{O}_P$. This proves surjectivity, so the sequence $0 \to \Gamma(X,\mathcal{O}) \to \Gamma(X,\mathcal{K}) \to \Gamma(X,\mathcal{K}/\mathcal{O}) \to 0$ is exact. □

Exercise 1.22

> **Problem.** *Gluing Sheaves.* Let X be a topological space, let $\mathfrak{U} = \{U_i\}$ be an open cover of X, and suppose we are given for each i a sheaf $\mathcal{F}_i$ on U_i, and for each i,j an isomorphism $\varphi_{ij}\colon \mathcal{F}_i|_{U_i\cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i\cap U_j}$, such that (1) for each i, $\varphi_{ii} = \mathrm{id}$, and (2) for each i,j,k, $\varphi_{ik} = \varphi_{jk}\circ\varphi_{ij}$ on $U_i\cap U_j\cap U_k$. Then there exists a unique sheaf $\mathcal{F}$ on X, together with isomorphisms $\psi_i\colon \mathcal{F}|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$, such that for each i,j, $\psi_j = \varphi_{ij} \circ \psi_i$ on $U_i \cap U_j$. We say loosely that $\mathcal{F}$ is obtained by *gluing* the sheaves $\mathcal{F}_i$ via the isomorphisms φ_{ij}.

Solution. *Remark on the statement.* The scan (read at high resolution) prints the compatibility as $\psi_j = \varphi_{ij}\circ\psi_i$ on $U_i\cap U_j$, which is type-correct since φ_{ij} goes from $\mathcal{F}_i$ to $\mathcal{F}_j$; this is the condition used. Uniqueness is understood up to a unique isomorphism compatible with the ψ_i.

Construction. For $V \subseteq X$ open let

$$\mathcal{F}(V) = \Big\{(s_i)_i \in \prod_i \mathcal{F}_i(V\cap U_i) :\ \varphi_{ij}\big(s_i|_{V\cap U_i\cap U_j}\big) = s_j|_{V\cap U_i\cap U_j} \text{ for all } i,j\Big\},$$

a subgroup of the product, with componentwise restrictions; clearly a presheaf with $\mathcal{F}(\emptyset) = 0$.

(3) Let $V = \bigcup V_a$ and $s = (s_i) \in \mathcal{F}(V)$ with $s|_{V_a} = 0$ for all a. Then for every i, $s_i|_{V_a \cap U_i} = 0$ for all a, and the $V_a \cap U_i$ cover $V \cap U_i$, so $s_i = 0$.

(4) Let $s^a = (s_i^a)_i \in \mathcal{F}(V_a)$ with $s^a|_{V_a \cap V_b} = s^b|_{V_a \cap V_b}$. Fix i; the sections $s_i^a \in \mathcal{F}_i(V_a \cap U_i)$ agree on overlaps, so glue to $s_i \in \mathcal{F}_i(V \cap U_i)$ with $s_i|_{V_a \cap U_i} = s_i^a$. For i, j, the sections $\varphi_{ij}(s_i|_{V \cap U_i \cap U_j})$ and $s_j|_{V \cap U_i \cap U_j}$ of $\mathcal{F}_j$ have the same restriction to each $V_a \cap U_i \cap U_j$ (namely $\varphi_{ij}(s_i^a|) = s_j^a|$), hence are equal. So $s = (s_i) \in \mathcal{F}(V)$ and $s|_{V_a} = s^a$. Thus $\mathcal{F}$ is a sheaf.

The isomorphisms ψ_i. For $V \subseteq U_i$ define $\psi_i(V)\colon \mathcal{F}(V) \to \mathcal{F}_i(V)$, $(s_l)_l \mapsto s_i$ (note $V \cap U_i = V$); a morphism of sheaves $\mathcal{F}|_{U_i} \to \mathcal{F}_i$. For $V \subseteq U_i \cap U_j$, $\psi_j(V)(s) = s_j = \varphi_{ij}(s_i) = \varphi_{ij}\psi_i(V)(s)$ by the definition of $\mathcal{F}(V)$. So $\psi_j = \varphi_{ij}\psi_i$ on $U_i \cap U_j$. $\psi_i(V)$ is bijective: given $t \in \mathcal{F}_i(V)$, $V \subseteq U_i$, put $s_l = \varphi_{il}(t|_{V \cap U_l}) \in \mathcal{F}_l(V \cap U_l)$ (here $V \cap U_l \subseteq U_i \cap U_l$); then $s_i = \varphi_{ii}(t) = t$, and by (2), $\varphi_{lm}(s_l|_{V \cap U_l \cap U_m}) = \varphi_{lm}\varphi_{il}(t|) = \varphi_{im}(t|) = s_m|_{V \cap U_l \cap U_m}$, so $(s_l) \in \mathcal{F}(V)$ and $\psi_i(V)(s) = t$: surjective. If $(s_l) \in \mathcal{F}(V)$ has $s_i = 0$, then $s_l = \varphi_{il}(s_i|) = 0$ for all l: injective. A morphism of sheaves bijective on all sections is an isomorphism, so each ψ_i is an isomorphism.

Uniqueness. Let $\mathcal{F}'$ be another sheaf on X with isomorphisms $\psi_i'\colon \mathcal{F}'|_{U_i} \to \mathcal{F}_i$ with $\psi_j' = \varphi_{ij}\psi_i'$ on $U_i \cap U_j$. For V open and $s' \in \mathcal{F}'(V)$ put $t_i = \psi_i'(s'|_{V \cap U_i}) \in \mathcal{F}_i(V \cap U_i)$. Then $\varphi_{ij}(t_i|_{V \cap U_i \cap U_j}) = \varphi_{ij}\psi_i'(s'|_{V \cap U_i \cap U_j}) = \psi_j'(s'|_{V \cap U_i \cap U_j}) = t_j|_{V \cap U_i \cap U_j}$, so $\eta(V)(s') := (t_i)_i \in \mathcal{F}(V)$. This defines a morphism $\eta\colon \mathcal{F}' \to \mathcal{F}$, and for $V \subseteq U_i$, $\psi_i\eta(V)(s') = t_i = \psi_i'(s')$, i.e. $\psi_i\eta = \psi_i'$ on U_i. It is the only morphism with this property, since the i-th component of $\eta(V)(s')$ is forced to be $\psi_i'(s'|_{V \cap U_i})$. Finally η is an isomorphism: injective by (3) for $\mathcal{F}'$, since $\eta(V)(s') = 0$ forces $s'|_{V \cap U_i} = 0$ (ψ_i' injective) for all i, hence $s' = 0$; surjective since given $(t_i) \in \mathcal{F}(V)$, the sections $s_i' = \psi_i'^{-1}(t_i) \in \mathcal{F}'(V \cap U_i)$ satisfy $\psi_j'(s_i'|_{V \cap U_i \cap U_j}) = \varphi_{ij}\psi_i'(s_i'|) = \varphi_{ij}(t_i|) = t_j| = \psi_j'(s_j'|)$, so $s_i'|_{V \cap U_i \cap U_j} = s_j'|_{V \cap U_i \cap U_j}$ (ψ_j' injective), and they glue to $s' \in \mathcal{F}'(V)$ with $\eta(V)(s') = (t_i)$. Hence $\mathcal{F}$ is unique up to a unique isomorphism compatible with the ψ_i. □

2. Schemes

Throughout this section I use the numbering of Chapter II of the scanned edition: Lemma 2.1, Prop. 2.2 (stalks and sections of $\mathcal{O}$ on $\operatorname{Spec} A$), Prop. 2.3 (morphisms of locally ringed spaces $\operatorname{Spec} B \to \operatorname{Spec} A$ come from ring homomorphisms), Lemma 2.4 and Prop. 2.5 (the scheme $\operatorname{Proj} S$), Prop. 2.6 (the functor t from varieties to schemes). For a point x of a scheme, $\mathcal{O}_x$ is the local ring, $\mathfrak{m}_x$ its maximal ideal and $k(x) = \mathcal{O}_x/\mathfrak{m}_x$ the residue field. For $f \in A$, $D(f) = \operatorname{Spec} A \setminus V(f)$. Recall that a morphism of schemes is a morphism of locally ringed spaces, and that a morphism of sheaves is an isomorphism iff it is an isomorphism on all stalks.

Exercise 2.1

> **Problem.** Let A be a ring, let $X = \operatorname{Spec} A$, let $f \in A$ and let $D(f) \subseteq X$ be the open complement of $V((f))$. Show that the locally ringed space $(D(f), \mathcal{O}_X|_{D(f)})$ is isomorphic to $\operatorname{Spec} A_f$.

Solution. Let $\varphi : A \to A_f$ be the localization map and $g = (g, g^{\#}) : Y = \operatorname{Spec} A_f \to X$ the morphism of Prop. 2.3(b). The prime ideals of A_f correspond bijectively to the primes of A not containing f via $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$; thus g is injective with image exactly $D(f)$. It is a homeomorphism onto $D(f)$: for an ideal $\mathfrak{b} \subseteq A_f$ one has $g(V(\mathfrak{b})) = V(\varphi^{-1}\mathfrak{b}) \cap D(f)$, and conversely for an ideal

$\mathfrak{a} \subseteq A$ one has $g^{-1}(V(\mathfrak{a})) = V(\mathfrak{a}A_f)$ and $g(V(\mathfrak{a}A_f)) = V(\mathfrak{a}) \cap D(f)$. So g maps the closed sets of Y exactly onto the closed sets of the subspace $D(f)$.

It remains to see that $g^\# : \mathcal{O}_X|_{D(f)} \to g_*\mathcal{O}_Y$ is an isomorphism of sheaves; we check this on stalks. Let $\mathfrak{q} \in Y$, $\mathfrak{p} = \varphi^{-1}(\mathfrak{q}) \in D(f)$. By Prop. 2.3(b) the stalk map is the localization $\varphi_\mathfrak{q} : A_\mathfrak{p} \to (A_f)_\mathfrak{q}$ of φ (and $(g_*\mathcal{O}_Y)_{g(\mathfrak{q})} = \mathcal{O}_{Y,\mathfrak{q}}$ because g is a homeomorphism onto the open set $D(f)$). Since $f \notin \mathfrak{p}$, f is a unit in $A_\mathfrak{p}$, so $A_\mathfrak{p} = (A_f)_{\mathfrak{p}A_f}$, and $\mathfrak{q} = \mathfrak{p}A_f$; hence $\varphi_\mathfrak{q}$ is the identity of $A_\mathfrak{p}$ under these identifications, in particular an isomorphism. Therefore g is an isomorphism of locally ringed spaces $\operatorname{Spec} A_f \cong (D(f), \mathcal{O}_X|_{D(f)})$. □

Exercise 2.2

> **Problem.** Let $(X, \mathcal{O}_X)$ be a scheme, and let U be any open subset of X. Show that $(U, \mathcal{O}_X|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U, and refer to $(U, \mathcal{O}_X|_U)$ as an *open subscheme* of X.

Solution. $(U, \mathcal{O}_X|_U)$ is a locally ringed space (its stalks are stalks of $\mathcal{O}_X$). Let $x \in U$. Since X is a scheme there is an open $V \ni x$ and an isomorphism of locally ringed spaces $(V, \mathcal{O}_X|_V) \cong \operatorname{Spec} A$. The set $U \cap V$ is open in V, so under this isomorphism it corresponds to an open subset of $\operatorname{Spec} A$ containing the point $\mathfrak{p}$ corresponding to x. The sets $D(h)$ form a base of the topology of $\operatorname{Spec} A$ (see the text after Lemma 2.1), so there is $h \in A$ with $\mathfrak{p} \in D(h) \subseteq U \cap V$. By Ex. 2.1, $(D(h), \mathcal{O}|_{D(h)}) \cong \operatorname{Spec} A_h$ is an affine scheme. So x has an open neighbourhood in U which, with the restricted structure sheaf, is an affine scheme. Hence $(U, \mathcal{O}_X|_U)$ is a scheme. □

Exercise 2.3

> **Problem.** *Reduced schemes.* A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subseteq X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.
>
> (a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $P \in X$, the local ring $\mathcal{O}_{X,P}$ has no nilpotent elements.
>
> (b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\mathrm{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\mathrm{red}}$, where for any ring A, we denote by A_{red} the quotient of A by its ideal of nilpotent elements. Show that $(X, (\mathcal{O}_X)_{\mathrm{red}})$ is a scheme. We call it the *reduced scheme associated to* X, and denote it by X_{red}. Show that there is a morphism of schemes $X_{\mathrm{red}} \to X$, which is a homeomorphism on the underlying topological spaces.
>
> (c) Let $f : X \to Y$ be a morphism of schemes, and assume that X is reduced. Show that there is a unique morphism $g : X \to Y_{\mathrm{red}}$ such that f is obtained by composing g with the natural map $Y_{\mathrm{red}} \to Y$.

Solution. (a) Suppose X is reduced and $\sigma \in \mathcal{O}_{X,P}$ with $\sigma^n = 0$. Represent σ by $s \in \mathcal{O}_X(U)$; then s^n has zero germ at P, so $s^n|_{U'} = 0$ for a smaller neighbourhood $U' \ni P$. Since $\mathcal{O}_X(U')$ is reduced, $s|_{U'} = 0$, so $\sigma = 0$. Conversely suppose all stalks are reduced and $s \in \mathcal{O}_X(U)$ satisfies $s^n = 0$. For each $P \in U$ the germ s_P is nilpotent, hence 0; a section with all germs zero is zero (sheaf axiom). So $s = 0$.

(b) *Stalks.* Let $\mathcal{P}$ be the presheaf $U \mapsto \mathcal{O}_X(U)_{\mathrm{red}}$; its stalk (equal to that of its sheafification) is the filtered direct limit $\varinjlim_{U \ni P} \mathcal{O}_X(U)_{\mathrm{red}}$. Since a filtered direct limit is exact, and since the

nilradical of $\varinjlim R_\alpha$ is $\varinjlim \mathrm{nil}(R_\alpha)$ (a nilpotent element of the limit is represented by an element that is nilpotent in some R_α), this equals $(\mathcal{O}_{X,P})_{\mathrm{red}}$. Thus the stalks of $(\mathcal{O}_X)_{\mathrm{red}}$ are the rings $(\mathcal{O}_{X,P})_{\mathrm{red}}$, which are local (quotient of a local ring by an ideal inside the maximal ideal) and reduced. So $(X, (\mathcal{O}_X)_{\mathrm{red}})$ is a locally ringed space, and it is reduced by (a).

Natural map. The quotient maps $\mathcal{O}_X(U) \to \mathcal{O}_X(U)_{\mathrm{red}}$ give a morphism of sheaves $\varepsilon : \mathcal{O}_X \to (\mathcal{O}_X)_{\mathrm{red}}$, which is surjective on stalks with kernel the nilradical; so on stalks it is a local homomorphism ($\varepsilon^{-1}(\mathfrak{m}/\mathrm{nil}) = \mathfrak{m}$). Hence $(\mathrm{id}_X, \varepsilon) : X_{\mathrm{red}} \to X$ is a morphism of locally ringed spaces, and it is a homeomorphism on spaces (the identity).

X_{red} *is a scheme.* The construction is compatible with restriction to open sets: $((\mathcal{O}_X)_{\mathrm{red}})|_U = (\mathcal{O}_X|_U)_{\mathrm{red}}$. So it suffices to treat $X = \operatorname{Spec} A$. Let $\pi : \operatorname{Spec} A_{\mathrm{red}} \to \operatorname{Spec} A$ be the morphism of Prop. 2.3(b) for the quotient map $A \to A_{\mathrm{red}}$. Since $\mathfrak{N}(A)$ is contained in every prime, π is a bijection, and it is a homeomorphism ($V(\mathfrak{a}) \mapsto V(\mathfrak{a} + \mathfrak{N}A)$ and the closed sets correspond; or note $\pi(V(\mathfrak{b})) = V(\varphi^{-1}\mathfrak{b})$ as in Ex. 2.18(c)). The sheaf $\pi_*\mathcal{O}_{\operatorname{Spec} A_{\mathrm{red}}}$ has reduced rings of sections: by (a) $\operatorname{Spec} A_{\mathrm{red}}$ is reduced, because its stalks $(A_{\mathrm{red}})_{\mathfrak{p}}$ are localizations of a reduced ring. Hence $\pi^\# : \mathcal{O}_{\operatorname{Spec} A}(V) \to \mathcal{O}_{\operatorname{Spec} A_{\mathrm{red}}}(V)$ kills nilpotents and factors through $\mathcal{O}_{\operatorname{Spec} A}(V)_{\mathrm{red}}$; by the universal property of sheafification we get $\mathcal{O}_{\operatorname{Spec} A,\mathrm{red}} \to \pi_*\mathcal{O}_{\operatorname{Spec} A_{\mathrm{red}}}$. On stalks at $\mathfrak{p}$ this is the map $(A_{\mathfrak{p}})_{\mathrm{red}} \to (A_{\mathrm{red}})_{\mathfrak{p}}$, an isomorphism because localization is exact and commutes with taking the nilradical ($\mathfrak{N}(A_{\mathfrak{p}}) = \mathfrak{N}(A)_{\mathfrak{p}}$). Hence $(\operatorname{Spec} A, \mathcal{O}_{\mathrm{red}}) \cong \operatorname{Spec} A_{\mathrm{red}}$ is affine, and X_{red} is a scheme in general.

(c) *Existence.* Let $i : Y_{\mathrm{red}} \to Y$ be the natural map. As i is the identity on spaces, put $g = f$ as a continuous map. For $V \subseteq Y$ open, the ring $\mathcal{O}_X(f^{-1}V)$ is reduced, so $f^\# : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}V)$ kills nilpotents and factors through $\mathcal{O}_Y(V)_{\mathrm{red}}$. This gives a morphism from the presheaf $V \mapsto \mathcal{O}_Y(V)_{\mathrm{red}}$ to the sheaf $f_*\mathcal{O}_X$, hence a morphism $g^\# : (\mathcal{O}_Y)_{\mathrm{red}} \to f_*\mathcal{O}_X$ with $g^\# \circ \varepsilon = f^\#$. On stalks at $x \in X$, $y = f(x)$, $g_x^\# : (\mathcal{O}_{Y,y})_{\mathrm{red}} \to \mathcal{O}_{X,x}$ is induced by the local homomorphism $f_x^\#$; since $f_x^\#(\mathfrak{m}_y) \subseteq \mathfrak{m}_x$, $g_x^\#$ maps the maximal ideal $\mathfrak{m}_y/\mathrm{nil}$ into $\mathfrak{m}_x$, so it is local. Thus g is a morphism of locally ringed spaces and $i \circ g = f$ (indeed $(i \circ g)^\# = i_*(g^\#) \circ \varepsilon = f^\#$ with $\varepsilon = i^\#$).

Uniqueness. Let $g' : X \to Y_{\mathrm{red}}$ satisfy $i \circ g' = f$. Since i is the identity on spaces, $g' = f = g$ as continuous maps, and $g'^\# \circ \varepsilon = f^\# = g^\# \circ \varepsilon$. As ε is surjective on stalks, $g'^\#$ and $g^\#$ agree on all stalks of $(\mathcal{O}_Y)_{\mathrm{red}}$ (as maps into the sheaf $f_*\mathcal{O}_X$ whose sections are determined by germs), hence $g'^\# = g^\#$. So $g' = g$. □

Exercise 2.4

> **Problem.** Let A be a ring and $(X, \mathcal{O}_X)$ a scheme. Given a morphism $f : X \to \operatorname{Spec} A$, we have an associated map on sheaves $f^\# : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_X$. Taking global sections we obtain a homomorphism $A \to \Gamma(X, \mathcal{O}_X)$. Thus there is a natural map
>
> $$\alpha : \mathrm{Hom}_{\mathrm{Sch}}(X, \operatorname{Spec} A) \to \mathrm{Hom}_{\mathrm{Rings}}(A, \Gamma(X, \mathcal{O}_X)).$$
>
> Show that α is bijective (cf. (I, 3.5) for an analogous statement about varieties).

Solution. Write $B = \Gamma(X, \mathcal{O}_X)$ and, for $s \in B$ and $x \in X$, $s_x \in \mathcal{O}_x$ for its germ. Recall $\Gamma(\operatorname{Spec} A, \mathcal{O}) = A$ (Prop. 2.2(c)).

Injectivity. Let $f : X \to \operatorname{Spec} A$ with $\varphi = \alpha(f) : A \to B$. For $x \in X$, $\mathfrak{p} = f(x)$, the stalk map $f_x^\# : A_{\mathfrak{p}} \to \mathcal{O}_x$ is local, and it composes with $A \to A_{\mathfrak{p}}$ to the germ map $A \xrightarrow{\varphi} B \to \mathcal{O}_x$. Locality

gives $(f_x^{\#})^{-1}(\mathfrak{m}_x) = \mathfrak{p}A_{\mathfrak{p}}$, so

$$f(x) = \{a \in A : \ \varphi(a)_x \in \mathfrak{m}_x\}. \tag{$*$}$$

Thus f is determined as a map of sets by φ. Next, $f^{\#}$ is determined on each basic open set $D(g)$: the map $f^{\#}(D(g)) : A_g \to \mathcal{O}_X(f^{-1}D(g))$ is a ring homomorphism compatible with the restrictions from $A = \Gamma(\operatorname{Spec} A, \mathcal{O})$, so it sends $a/1$ to $\varphi(a)|_{f^{-1}D(g)}$; a ring homomorphism out of A_g is determined by its values on A (the images of the elements g^{-n} are the inverses of the image of g). Since the sets $D(g)$ form a base, the morphism of sheaves $f^{\#}$ is determined (a morphism of sheaves is determined by its values on a base). So α is injective.

Surjectivity. Let $\varphi : A \to B$ be a ring homomorphism. Define $f : X \to \operatorname{Spec} A$ by $(*)$: $f(x) = \varphi_x^{-1}(\mathfrak{m}_x)$, where $\varphi_x : A \to \mathcal{O}_x$ is $a \mapsto \varphi(a)_x$; this is a prime ideal, being the preimage of the prime $\mathfrak{m}_x$. For $g \in A$,

$$f^{-1}(D(g)) = \{x : \varphi(g)_x \notin \mathfrak{m}_x\} = \{x : \ \varphi(g)_x \text{ is a unit}\} =: X_{\varphi(g)}.$$

This is open: if the germ of $s \in B$ at x is a unit, there is a neighbourhood W and $t \in \mathcal{O}_X(W)$ with $st = 1$ near x, hence on a smaller neighbourhood where $s| \cdot t = 1$, and all germs there are units. So f is continuous. Moreover $\varphi(g)|_{X_{\varphi(g)}}$ is a unit of $\mathcal{O}_X(X_{\varphi(g)})$: the local inverses exist near every point and agree on overlaps (inverses are unique), so they glue. By the universal property of localization, $A \to \mathcal{O}_X(X_{\varphi(g)})$, $a \mapsto \varphi(a)|$, extends uniquely to $\theta_g : A_g \to \mathcal{O}_X(f^{-1}D(g))$. For $D(h) \subseteq D(g)$ the diagrams with the restriction maps commute (uniqueness in the universal property). So the θ_g define a morphism of sheaves on the base $\{D(g)\}$, which extends uniquely to a morphism of sheaves $f^{\#} : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_X$. On the stalk at $\mathfrak{p} = f(x)$ it is the map $A_{\mathfrak{p}} \to \mathcal{O}_x$ induced from $\varphi_x : A \to \mathcal{O}_x$ (elements of $A \setminus \mathfrak{p}$ map to units); it is local, since $\varphi_x(\mathfrak{p}) \subseteq \mathfrak{m}_x$ by definition of $f(x)$. Hence f is a morphism of locally ringed spaces, i.e. a morphism of schemes, and $\alpha(f) = \theta_1 = \varphi$. □

Exercise 2.5

> **Problem.** Describe $\operatorname{Spec} \mathbf{Z}$, and show that it is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\operatorname{Spec} \mathbf{Z}$.

Solution. *Description.* $\mathbf{Z}$ is a PID, so its prime ideals are (0) and (p), p a prime number; the (p) are maximal. The closed subsets $V(\mathfrak{a})$ are: $V(0) = \operatorname{Spec} \mathbf{Z}$, and for $\mathfrak{a} = (n)$, $n \neq 0$, the finite set of primes dividing n (and $\emptyset$ for $n = \pm 1$). So the proper closed subsets are exactly the finite sets of closed points (p); the point (0) is a generic point, $\overline{\{(0)\}} = \operatorname{Spec} \mathbf{Z}$, and it lies in every nonempty open set. The residue field at (p) is $\mathbf{F}_p$, at (0) it is $\mathbf{Q}$; the local rings are $\mathbf{Z}_{(p)}$ and $\mathbf{Q}$. Global sections: $\Gamma(\operatorname{Spec} \mathbf{Z}, \mathcal{O}) = \mathbf{Z}$.

Final object. By Ex. 2.4, for every scheme X the set $\operatorname{Hom}(X, \operatorname{Spec} \mathbf{Z})$ is in bijection with $\operatorname{Hom}_{\text{Rings}}(\mathbf{Z}, \Gamma(X, \mathcal{O}_X))$. Since rings have 1 and homomorphisms preserve 1, there is exactly one ring homomorphism $\mathbf{Z} \to R$ for every ring R ($n \mapsto n \cdot 1_R$). Hence there is exactly one morphism $X \to \operatorname{Spec} \mathbf{Z}$. □

Exercise 2.6

Problem. Describe the spectrum of the zero ring, and show that it is an initial object for the category of schemes. (According to our conventions, all ring homomorphisms must take 1 to 1. Since $0 = 1$ in the zero ring, we see that each ring R admits a unique homomorphism to the zero ring, but that there is no homomorphism from the zero ring to R unless $0 = 1$ in R.)

Solution. The zero ring 0 has only one ideal, namely $0 = (1)$ itself, which is not a proper ideal, so it has no prime ideals: $\operatorname{Spec} 0 = \emptyset$, with the zero sheaf (the value of a sheaf on $\emptyset$ is the zero ring). It is an affine scheme (the empty space is covered by no points, so the scheme condition is vacuous).

Let X be any scheme. A morphism $\emptyset \to X$ consists of the unique (empty) continuous map and a morphism of sheaves $\mathcal{O}_X \to f_*\mathcal{O}_\emptyset$. For every open $V \subseteq X$, $(f_*\mathcal{O}_\emptyset)(V) = \mathcal{O}_\emptyset(\emptyset) = 0$, and there is exactly one ring homomorphism from any ring to the zero ring; these are compatible with restrictions. So there is exactly one morphism of sheaves, and the locality condition on stalks is vacuous (there are no points). Hence $\operatorname{Hom}(\operatorname{Spec} 0, X)$ has exactly one element for every scheme X: $\operatorname{Spec} 0$ is an initial object.

(Consistency check with Ex. 2.4: $\operatorname{Hom}(X, \operatorname{Spec} 0) = \operatorname{Hom}_{\mathrm{Rings}}(0, \Gamma(X, \mathcal{O}_X))$ is nonempty iff $\Gamma(X, \mathcal{O}_X) = 0$. If $X \neq \emptyset$ this fails, since $1 = 0$ in $\Gamma(X, \mathcal{O}_X)$ would give $1 = 0$ in a local ring $\mathcal{O}_x$, which is nonzero. So only $X = \emptyset$ maps to $\operatorname{Spec} 0$, consistently with the above.) □

Exercise 2.7

Problem. Let X be a scheme. For any point $x \in X$, let $\mathcal{O}_x$ be the local ring at x, and $\mathfrak{m}_x$ its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/\mathfrak{m}_x$. Now let K be any field. Show that to give a morphism of $\operatorname{Spec} K$ to X is equivalent to giving a point $x \in X$ and an inclusion map $k(x) \to K$.

Solution. I first prove a lemma that will also be used in Ex. 2.8.

Lemma A. Let X be a scheme and R a ring having exactly one prime ideal $\mathfrak{n}$ (e.g. a field, or $k[\varepsilon]$). Let $*$ be the unique point of $\operatorname{Spec} R$. Then $f \mapsto (f(*), f_*^{\#})$ is a bijection from $\operatorname{Hom}(\operatorname{Spec} R, X)$ onto the set of pairs (x, θ) with $x \in X$ and $\theta : \mathcal{O}_x \to R$ a local homomorphism.

Proof. The stalk map $f_*^{\#} : \mathcal{O}_{X,f(*)} \to \mathcal{O}_{\operatorname{Spec} R,*} = R_{\mathfrak{n}} = R$ is local by definition of a morphism of schemes, so the map is well defined.

Injective. Let f, f' give the same (x, θ). Choose an affine open $U = \operatorname{Spec} A \ni x$, $x \leftrightarrow \mathfrak{p}$. Both f, f' have image $\{x\} \subseteq U$, so they factor through U as morphisms of locally ringed spaces $f_U, f'_U : \operatorname{Spec} R \to \operatorname{Spec} A$. By Prop. 2.3(c) they come from ring homomorphisms $\psi, \psi' : A \to R$. The stalk map of f_U at $*$ is $A_{\mathfrak{p}} \to R$ induced by ψ, i.e. $\psi = \theta \circ (A \to A_{\mathfrak{p}})$; likewise for ψ'. So $\psi = \psi'$ and $f = f'$.

Surjective. Given (x, θ) and $U = \operatorname{Spec} A \ni x$ as above, put $\psi : A \to A_{\mathfrak{p}} = \mathcal{O}_x \xrightarrow{\theta} R$ and let f be $\operatorname{Spec} R \to \operatorname{Spec} A = U \subseteq X$ induced by ψ (Prop. 2.3(b)). Then $f(*) = \psi^{-1}(\mathfrak{n}) = (A \to A_{\mathfrak{p}})^{-1}(\theta^{-1}\mathfrak{n}) = (A \to A_{\mathfrak{p}})^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p}$ since θ is local. The stalk map $A_{\mathfrak{p}} \to R$ of f is the unique extension of ψ to $A_{\mathfrak{p}}$, namely θ. □

Now take $R = K$. A homomorphism $\theta : \mathcal{O}_x \to K$ is local iff $\theta^{-1}(0) = \mathfrak{m}_x$, i.e. iff $\ker \theta = \mathfrak{m}_x$, i.e. iff θ is the composition of $\mathcal{O}_x \to k(x)$ with a ring homomorphism $\iota : k(x) \to K$ (which

is automatically injective, $k(x)$ being a field). This identifies the pairs (x, θ) with pairs $(x, \iota : k(x) \hookrightarrow K)$. By Lemma A, morphisms $\operatorname{Spec} K \to X$ correspond bijectively to a point $x \in X$ together with an inclusion $k(x) \to K$. □

Exercise 2.8

> **Problem.** Let X be a scheme. For any point $x \in X$, we define the *Zariski tangent space* T_x to X at x to be the dual of the $k(x)$-vector space $\mathfrak{m}_x/\mathfrak{m}_x^2$. Now assume that X is a scheme over a field k, and let $k[\varepsilon]$ be the ring of dual numbers over k, i.e., $k[\varepsilon] = k[t]/(t^2)$. Show that to give a k-morphism of $\operatorname{Spec} k[\varepsilon]$ to X is equivalent to giving a point $x \in X$, rational over k (i.e., such that $k(x) = k$), and an element of T_x.

Solution. The ring $k[\varepsilon]$ has a unique prime ideal (ε) (it is local and ε is nilpotent), with residue field k. So Lemma A (Ex. 2.7) applies: morphisms $\operatorname{Spec} k[\varepsilon] \to X$ correspond to pairs (x, θ), $\theta : \mathcal{O}_x \to k[\varepsilon]$ a local homomorphism.

The k-structure. Let $\pi : X \to \operatorname{Spec} k$ be the structure morphism; by Ex. 2.4 it corresponds to a ring map $\kappa : k \to \Gamma(X, \mathcal{O}_X)$, which makes each $\mathcal{O}_x$ a k-algebra. The morphism f corresponding to (x, θ) is a k-morphism iff $\pi \circ f$ is the structure morphism of $\operatorname{Spec} k[\varepsilon]$; by Ex. 2.4 this says that $k \xrightarrow{\kappa} \Gamma(X, \mathcal{O}_X) \xrightarrow{f^\#} k[\varepsilon]$ is the inclusion. This composite factors as $k \to \mathcal{O}_x \xrightarrow{\theta} k[\varepsilon]$, and conversely it is determined by θ (we are only dealing with one point). So f is a k-morphism iff θ is a homomorphism of k-algebras.

From θ to (x, δ). Let $\theta : \mathcal{O}_x \to k[\varepsilon]$ be a local k-algebra homomorphism. Then $\theta(\mathfrak{m}_x) \subseteq (\varepsilon)$, and θ induces a k-algebra homomorphism of fields $\bar{\theta} : k(x) \to k[\varepsilon]/(\varepsilon) = k$. Both $k \to k(x)$ (a field map, hence injective) and $\bar{\theta}$ (injective) are k-algebra maps, and $\bar{\theta} \circ (k \to k(x)) = \mathrm{id}_k$. For $y \in k(x)$ choose $c \in k$ with $\bar{\theta}(y) = c = \bar{\theta}(c \cdot 1)$; injectivity of $\bar{\theta}$ gives $y = c \cdot 1$. So $k \to k(x)$ is onto: $k(x) = k$, i.e. x is rational over k. Consequently $\mathcal{O}_x = k \cdot 1 \oplus \mathfrak{m}_x$ as k-vector spaces. For $u \in \mathfrak{m}_x$ write $\theta(u) = \delta(u)\varepsilon$ with $\delta(u) \in k$; $\delta : \mathfrak{m}_x \to k$ is k-linear, and for $u, v \in \mathfrak{m}_x$, $\delta(uv)\varepsilon = \theta(u)\theta(v) = \delta(u)\delta(v)\varepsilon^2 = 0$. So δ vanishes on $\mathfrak{m}_x^2$ and defines $\delta \in \operatorname{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k) = T_x$ (recall $k(x) = k$). Moreover $\theta(a + u) = a + \delta(u)\varepsilon$ for $a \in k$, $u \in \mathfrak{m}_x$, so θ is determined by δ.

From (x, δ) to θ. Conversely let x be rational over k and $\delta \in T_x$, viewed as a k-linear map $\mathfrak{m}_x \to k$ vanishing on $\mathfrak{m}_x^2$. Define $\theta(a + u) = a + \delta(u)\varepsilon$ ($a \in k$, $u \in \mathfrak{m}_x$). It is k-linear, $\theta(1) = 1$, and

$$\theta((a+u)(b+v)) = ab + \big(b\delta(u) + a\delta(v) + \delta(uv)\big)\varepsilon = ab + (b\delta(u) + a\delta(v))\varepsilon = \theta(a+u)\theta(b+v),$$

since $\delta(uv) = 0$. It is local: $\theta^{-1}((\varepsilon)) = \mathfrak{m}_x$.

These two constructions are inverse to each other, so k-morphisms $\operatorname{Spec} k[\varepsilon] \to X$ correspond bijectively to pairs ($x \in X$ with $k(x) = k$, element of T_x). □

Exercise 2.9

> **Problem.** If X is a topological space, and Z an irreducible closed subset of X, a *generic point* for Z is a point ξ such that $Z = \overline{\{\xi\}}$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Solution. *Affine case.* Let $X = \operatorname{Spec} A$ and $Z \subseteq X$ nonempty, closed, irreducible. Write $Z = V(\mathfrak{a})$ and $I = \sqrt{\mathfrak{a}}$ (so $V(I) = Z$, by Lemma 2.1(c)). I is a proper ideal since $Z \neq \emptyset$. It is prime: if $ab \in I$ then $Z = V(I) \subseteq V(ab) = V(a) \cup V(b)$, so $Z = (Z \cap V(a)) \cup (Z \cap V(b))$ is a union of two closed subsets; by irreducibility $Z \subseteq V(a)$ or $Z \subseteq V(b)$, i.e. $a \in \sqrt{I} = I$ or $b \in I$. For a prime $\mathfrak{p}$ one has $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ (the smallest closed set $V(\mathfrak{b})$ containing $\mathfrak{p}$ has $\mathfrak{b} \subseteq \mathfrak{p}$, so $V(\mathfrak{b}) \supseteq V(\mathfrak{p})$). Hence $Z = \overline{\{I\}}$: a generic point exists. If $\overline{\{\mathfrak{p}\}} = \overline{\{\mathfrak{q}\}}$ then $V(\mathfrak{p}) = V(\mathfrak{q})$, so $\mathfrak{p} \supseteq \mathfrak{q} \supseteq \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{q}$: it is unique.

General case. Let $Z \subseteq X$ be nonempty, closed, irreducible, and pick $z \in Z$ and an affine open $U \ni z$. Then $Z \cap U$ is a nonempty open subset of the irreducible space Z, hence irreducible and dense in Z; and it is closed in U. By the affine case there is $\xi \in U$ with $Z \cap U = \overline{\{\xi\}} \cap U$ (closure in U is closure in X intersected with U). Then $\xi \in Z$, so $\overline{\{\xi\}} \subseteq Z$ (Z closed); and $Z = \overline{Z \cap U} \subseteq \overline{\{\xi\}}$. So $Z = \overline{\{\xi\}}$.

Uniqueness. If also $Z = \overline{\{\xi'\}}$, then the open set $U \cap Z$ is a nonempty open subset of $\overline{\{\xi'\}}$, so it contains ξ' (an open set meeting the closure of $\{\xi'\}$ contains ξ'). Hence $\xi' \in U$ and $\overline{\{\xi'\}} \cap U = Z \cap U = \overline{\{\xi\}} \cap U$, i.e. ξ, ξ' are generic points of the same irreducible closed subset of U. By the affine case, $\xi = \xi'$. □

Exercise 2.10

Problem. Describe $\operatorname{Spec} \mathbf{R}[x]$. How does its topological space compare to the set $\mathbf{R}$? To $\mathbf{C}$?

Solution. $\mathbf{R}[x]$ is a PID, so its primes are (0) and the ideals (p) with p irreducible monic. Over $\mathbf{R}$ the irreducible monic polynomials are $x - a$ ($a \in \mathbf{R}$) and $x^2 + bx + c$ with $b^2 - 4c < 0$ (the monic quadratics with a pair of non-real conjugate roots $z, \bar{z} \in \mathbf{C}$). Hence $\operatorname{Spec} \mathbf{R}[x]$ consists of

- the generic point $\eta = (0)$, with residue field $\mathbf{R}(x)$;
- the *real points* $(x - a)$, $a \in \mathbf{R}$, with residue field $\mathbf{R}[x]/(x - a) = \mathbf{R}$;
- the *complex points* $(x^2 + bx + c) = (x - z)(x - \bar{z})$ with $z \in \mathbf{C} \setminus \mathbf{R}$, residue field $\mathbf{R}[x]/(x^2 + bx + c) \cong \mathbf{C}$, one for each conjugate pair $\{z, \bar{z}\}$.

All the nonzero primes are maximal, i.e. closed points. Thus the set of closed points is naturally in bijection with $\mathbf{C}/(z \sim \bar{z})$, which is the closed upper half plane $\{z : \operatorname{Im} z \geq 0\}$; it contains $\mathbf{R}$ as the real points, and it is the quotient of $\mathbf{C} = \operatorname{Max} \mathbf{C}[x]$ by complex conjugation (the map $\operatorname{Spec} \mathbf{C}[x] \to \operatorname{Spec} \mathbf{R}[x]$ is $2:1$ off the real line).

Topology: every nonzero ideal is (p) with p a product of finitely many irreducibles, so $V((p))$ is a finite set of closed points. The closed sets are therefore $\emptyset$, the finite sets of closed points, and the whole space. The open sets are the cofinite sets of closed points together with η, plus $\emptyset$. So: the closed points with the induced topology form an infinite set with the cofinite topology (just as for $\mathbf{R}$ or $\mathbf{C}$ with the Zariski topology on $\mathbf{A}^1$), which is not Hausdorff and is very different from the usual topology of $\mathbf{R}$ or $\mathbf{C}$. Comparing, the set of closed points is neither $\mathbf{R}$ nor $\mathbf{C}$ but the quotient $\mathbf{C}/\text{conj}$, which contains $\mathbf{R}$ and is "half of $\mathbf{C}$"; and the whole space has in addition the generic point η whose closure is everything. Compared with $\mathbf{A}^1_{\mathbf{C}} = \operatorname{Spec} \mathbf{C}[x]$, whose closed points are $\mathbf{C}$, the space $\operatorname{Spec} \mathbf{R}[x]$ is obtained by identifying conjugate points (and the two have the same cofinite structure). □

Exercise 2.11

Problem. Let $k = \mathbf{F}_p$ be the finite field with p elements. Describe $\operatorname{Spec} k[x]$. What are the residue fields of its points? How many points are there with a given residue field?

Solution. $k[x]$ is a PID, so $\operatorname{Spec} k[x]$ consists of the generic point $\eta = (0)$ and the closed points (f), $f \in \mathbf{F}_p[x]$ monic irreducible. The topology is again: closed sets are the finite sets of closed points and the whole space. The residue fields are:

- at η: $\mathbf{F}_p(x)$, an infinite field;
- at (f), $\deg f = d$: $\mathbf{F}_p[x]/(f) \cong \mathbf{F}_{p^d}$, the field with p^d elements.

Counting. Let N_d be the number of monic irreducible polynomials of degree d in $\mathbf{F}_p[x]$. Standard fact: $x^{p^n} - x$ (which is separable, its derivative being -1) is the product of all monic irreducible polynomials whose degree divides n (a monic irreducible f of degree d divides $x^{p^n} - x$ iff the class $\bar{x}$ of x in $\mathbf{F}_p[x]/(f) = \mathbf{F}_{p^d}$ satisfies $\bar{x}^{p^n} = \bar{x}$, iff the n-th power of the Frobenius $F : y \mapsto y^p$ fixes the generator $\bar{x}$ of $\mathbf{F}_{p^d}$ over $\mathbf{F}_p$, iff $F^n = \mathrm{id}$ on $\mathbf{F}_{p^d}$, iff $d \mid n$, since F has order exactly d on $\mathbf{F}_{p^d}$; and $x^{p^n} - x$, being separable, is squarefree). Comparing degrees, $\sum_{d|n} dN_d = p^n$, and Moebius inversion gives

$$N_d = \frac{1}{d} \sum_{e|d} \mu(d/e)\, p^e.$$

Hence the number of points with residue field $\mathbf{F}_{p^d}$ is N_d (e.g. $N_1 = p$, $N_2 = (p^2 - p)/2$, $N_3 = (p^3 - p)/3$, $N_4 = (p^4 - p^2)/4$); these are ≥ 1 for every d (since $dN_d \geq p^d - \sum_{e<d} p^e > 0$). The unique point with residue field $\mathbf{F}_p(x)$ is η. For a field L not isomorphic to any $\mathbf{F}_{p^d}$ or to $\mathbf{F}_p(x)$ there is no point with residue field L. Equivalently the points with residue field $\mathbf{F}_{p^d}$ correspond to Frobenius orbits of size exactly d on $\overline{\mathbf{F}}_p$. □

Exercise 2.12

Problem. *Gluing Lemma.* Generalize the gluing procedure described in the text (2.3.5) as follows. Let $\{X_i\}$ be a family of schemes (possibly infinite). For each $i \neq j$, suppose given an open subset $U_{ij} \subseteq X_i$, and let it have the induced scheme structure (Ex. 2.2). Suppose also given for each $i \neq j$ an isomorphism of schemes $\varphi_{ij} : U_{ij} \to U_{ji}$ such that (1) for each i, j, $\varphi_{ji} = \varphi_{ij}^{-1}$, and (2) for each i, j, k, $\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}$ and $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap U_{ik}$. Then there is a scheme X, together with morphisms $\psi_i : X_i \to X$ for each i, such that (1) ψ_i is an isomorphism of X_i onto an open subscheme of X, (2) the $\psi_i(X_i)$ cover X, (3) $\psi_i(U_{ij}) = \psi_i(X_i) \cap \psi_j(X_j)$ and (4) $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij}. We say that X is obtained by *glueing* the schemes X_i along the isomorphisms φ_{ij}. An interesting special case is when the family X_i is arbitrary, but the U_{ij} are all empty. Then the scheme X is called the *disjoint union* of the X_i, and is denoted $\coprod X_i$.

Solution. Put $U_{ii} = X_i$ and $\varphi_{ii} = \mathrm{id}$; conditions (1), (2) then hold for all i, j, k (including equal indices).

The set and topology. Let X be the quotient of $\bigsqcup X_i$ by the relation $x \sim y$ ($x \in X_i$, $y \in X_j$) iff $x \in U_{ij}$ and $y = \varphi_{ij}(x)$. It is an equivalence relation: reflexivity from $\varphi_{ii} = \mathrm{id}$; symmetry

from (1). Transitivity: if $y = \varphi_{ij}(x)$ and $z = \varphi_{jk}(y)$ with $y \in U_{ji} \cap U_{jk} = \varphi_{ij}(U_{ij} \cap U_{ik})$, then $x \in U_{ij} \cap U_{ik}$ (injectivity of φ_{ij}) and by (2) $z = \varphi_{jk}\varphi_{ij}(x) = \varphi_{ik}(x)$. Let $\psi_i : X_i \to X$ be the canonical maps; give X the quotient topology: $W \subseteq X$ is open iff all $\psi_i^{-1}(W)$ are open. Two points of the same X_i are equivalent only if equal ($\varphi_{ii} = \mathrm{id}$), so each ψ_i is injective. Moreover $\psi_j^{-1}(\psi_i(X_i)) = U_{ji}$, which is open in X_j, so $\psi_i(X_i)$ is open; and for $V \subseteq X_i$ open, $\psi_j^{-1}(\psi_i(V)) = \varphi_{ij}(V \cap U_{ij})$ is open in U_{ji}, hence in X_j, so $\psi_i(V)$ is open. Thus ψ_i is a homeomorphism of X_i onto the open subset $X_i' := \psi_i(X_i)$ of X; the X_i' cover X (obvious) and $X_i' \cap X_j' = \psi_i(U_{ij}) = \psi_j(U_{ji})$; and $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij} as maps of sets (this is the definition of $\sim$).

The sheaf. For $W \subseteq X$ open put $W_i = \psi_i^{-1}(W)$, open in X_i. Since $\psi_j\varphi_{ij} = \psi_i$ on U_{ij} and $\varphi_{ij} : U_{ij} \to U_{ji}$ is bijective, $\varphi_{ij}(W_i \cap U_{ij}) = W_j \cap U_{ji}$, so the isomorphism of schemes φ_{ij} gives a ring isomorphism $\varphi_{ij}^* : \mathcal{O}_{X_j}(W_j \cap U_{ji}) \to \mathcal{O}_{X_i}(W_i \cap U_{ij})$. Define

$$\mathcal{O}_X(W) = \big\{(s_i)_i \in \textstyle\prod_i \mathcal{O}_{X_i}(W_i) : \ s_i|_{W_i \cap U_{ij}} = \varphi_{ij}^*\big(s_j|_{W_j \cap U_{ji}}\big) \text{ for all } i, j\big\},$$

with componentwise restriction. This is a subring of the product, and it is a sheaf: it is the kernel of a morphism from the sheaf $\prod_i (\psi_i)_*\mathcal{O}_{X_i}$ to a product of sheaves, namely $(s_i) \mapsto (s_i| - \varphi_{ij}^*(s_j|))_{i,j}$. Let $p_i : \mathcal{O}_X \to (\psi_i)_*\mathcal{O}_{X_i}$ be the i-th projection.

Claim: for $W \subseteq X_i'$ the map $p_i : \mathcal{O}_X(W) \to \mathcal{O}_{X_i}(W_i)$ is bijective. Note $W_j \subseteq \psi_j^{-1}(X_i') = U_{ji}$ for all j, so $W_j \cap U_{ji} = W_j$. *Injective:* the (i, j) condition says $s_i|_{W_i \cap U_{ij}} = \varphi_{ij}^*(s_j)$, which determines s_j from s_i since φ_{ij}^* is injective. *Surjective:* given s_i put $s_j = (\varphi_{ij}^*)^{-1}(s_i|_{W_i \cap U_{ij}}) \in \mathcal{O}_{X_j}(W_j)$ (s_i itself for $j = i$). The (i, j) conditions hold by construction and the (j, i) conditions by $\varphi_{ji} = \varphi_{ij}^{-1}$. For (j, k) we must show $s_j|_{W_j \cap U_{jk}} = \varphi_{jk}^*(s_k|_{W_k \cap U_{kj}})$. Put $T = W_i \cap U_{ij} \cap U_{ik}$; by (2), φ_{ij} maps T isomorphically onto $W_j \cap U_{jk}$ (recall $W_j \cap U_{ji} = W_j$). Applying the injective map $\varphi_{ij}^* : \mathcal{O}_{X_j}(W_j \cap U_{jk}) \to \mathcal{O}_{X_i}(T)$ to both sides, the left side gives $s_i|_T$ and the right side gives $(\varphi_{jk}\varphi_{ij})^*(s_k|) = \varphi_{ik}^*(s_k|_{W_k \cap U_{ki}})|_T = s_i|_T$, using $\varphi_{ik} = \varphi_{jk}\varphi_{ij}$ on $U_{ij} \cap U_{ik}$ from (2) and the definition of s_k. So the two sides agree. This proves the claim.

X is a scheme. By the claim, (ψ_i, p_i) is an isomorphism of ringed spaces $(X_i, \mathcal{O}_{X_i}) \to (X_i', \mathcal{O}_X|_{X_i'})$ (we showed ψ_i is a homeomorphism onto X_i'). Hence the stalks of $\mathcal{O}_X$ are local rings, $(X, \mathcal{O}_X)$ is a locally ringed space, and it is covered by the open sets X_i', each isomorphic to the scheme X_i; so each point has an affine open neighbourhood (Ex. 2.2 applied inside X_i'). So X is a scheme, and $\psi_i = (\psi_i, p_i) : X_i \to X$ is an isomorphism onto the open subscheme X_i'. The X_i' cover X and $\psi_i(U_{ij}) = X_i' \cap X_j'$ (shown above). Finally (4) holds as morphisms: the sheaf map of $\psi_j \circ \varphi_{ij}|_{U_{ij}}$ sends $s = (s_k) \in \mathcal{O}_X(W)$ to $\varphi_{ij}^*(s_j|_{W_j \cap U_{ji}})$, while that of $\psi_i|_{U_{ij}}$ sends it to $s_i|_{W_i \cap U_{ij}}$, and these agree by the defining condition of $\mathcal{O}_X$. This proves properties (1)–(4).

Disjoint union. If all $U_{ij} = \emptyset$ ($i \neq j$) the conditions (1), (2) are vacuous (all φ_{ij} are the empty maps), the relation $\sim$ is equality, and $X = \coprod X_i$ is the topological disjoint union with $\mathcal{O}_X|_{X_i} = \mathcal{O}_{X_i}$: a scheme in which every X_i is open and $X_i \cap X_j = \emptyset$. □

Exercise 2.13

Problem. A topological space is *quasi-compact* if every open cover has a finite subcover.

(a) Show that a topological space is noetherian (Ch. I, Section 1) if and only if every open subset is quasi-compact.

(b) If X is an affine scheme, show that $\mathrm{sp}(X)$ is quasi-compact, but not in general noetherian. We say that a scheme X is *quasi-compact* if $\mathrm{sp}(X)$ is.

(c) If A is a noetherian ring, show that $\mathrm{sp}(\operatorname{Spec} A)$ is a noetherian topological space.

(d) Give an example of a non-noetherian ring A such that $\mathrm{sp}(\operatorname{Spec} A)$ is noetherian.

Solution. Recall (Ch. I, Section 1) that a space is noetherian if it satisfies the descending chain condition for closed subsets, equivalently every nonempty family of closed subsets has a minimal element.

(a) ($\Rightarrow$) Let X be noetherian, $U \subseteq X$ open, and $\{V_\alpha\}$ an open cover of U by open subsets of X. For finite subsets F of the index set let $Z_F = X \setminus \bigcup_{\alpha\in F} V_\alpha$. This is a nonempty family of closed sets, so it has a minimal element Z_F. If $U \not\subseteq \bigcup_{\alpha\in F} V_\alpha$, pick $x \in U \setminus \bigcup_F V_\alpha$ and β with $x \in V_\beta$; then $Z_{F\cup\{\beta\}} \subsetneq Z_F$ (it misses x), contradicting minimality. So $U \subseteq \bigcup_{\alpha\in F} V_\alpha$ and U is quasi-compact. ($\Leftarrow$) Let $Z_1 \supseteq Z_2 \supseteq \cdots$ be closed. The open set $U = \bigcup_n (X \setminus Z_n)$ is the union of the increasing sequence $X \setminus Z_n$; it is quasi-compact, so $U = X \setminus Z_N$ for some N, i.e. $Z_n = Z_N$ for all $n \geq N$.

(b) Let $X = \operatorname{Spec} A$ and $X = \bigcup_\alpha U_\alpha$ an open cover. Each U_α is a union of basic sets $D(f)$, so it suffices to show that a cover $X = \bigcup_{i\in I} D(f_i)$ by basic opens has a finite subcover. Then $V(\{f_i\}) = \bigcap V(f_i) = \emptyset$, so the ideal generated by the f_i is A (otherwise it lies in a maximal, hence prime, ideal), whence $1 = \sum_{i\in F} a_i f_i$ for a finite F, so $V(f_i : i \in F) = \emptyset$, i.e. $X = \bigcup_{i\in F} D(f_i)$. So $\mathrm{sp}(X)$ is quasi-compact.

Not noetherian in general. Let $A = k[x_1, x_2, \dots]$ (infinitely many variables), $X = \operatorname{Spec} A$, and $U = X \setminus V(x_1, x_2, \dots) = \bigcup_i D(x_i)$. If U were covered by finitely many $D(x_i)$, $i \leq n$, consider $\mathfrak{p} = (x_1, \dots, x_n)$, prime because $A/\mathfrak{p} \cong k[x_{n+1}, \dots]$ is a domain. Then $\mathfrak{p} \notin D(x_i)$ for $i \leq n$, but $\mathfrak{p} \in U$ because $\mathfrak{p} \not\supseteq (x_1, x_2, \dots)$ ($x_{n+1} \notin \mathfrak{p}$). So U is an open subset which is not quasi-compact, and by (a) X is not noetherian.

(c) Let $V(\mathfrak{a}_1) \supseteq V(\mathfrak{a}_2) \supseteq \cdots$ be a descending chain of closed subsets of $\operatorname{Spec} A$. By Lemma 2.1(c), $V(\mathfrak{a}) \subseteq V(\mathfrak{b})$ iff $\sqrt{\mathfrak{a}} \supseteq \sqrt{\mathfrak{b}}$, so $\sqrt{\mathfrak{a}_1} \subseteq \sqrt{\mathfrak{a}_2} \subseteq \cdots$ is an ascending chain of ideals; as A is noetherian it stabilizes, and then so does $V(\mathfrak{a}_n) = V(\sqrt{\mathfrak{a}_n})$.

(d) Let $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, x_3^2, \dots)$. Every element of A is a polynomial without constant term plus a constant, and every non-constant-term element is nilpotent (a finite sum of nilpotents is nilpotent). So the ideal $\mathfrak{m} = (x_1, x_2, \dots)$ is the nilradical, and $A/\mathfrak{m} = k$; hence $\mathfrak{m}$ is the unique prime of A and $\operatorname{Spec} A$ is a single point, a noetherian space. But A is not noetherian: $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$ is an infinite strictly ascending chain (setting $x_1 = \cdots = x_n = 0$ gives the ring $k[x_{n+1}, \dots]/(x_j^2)$ in which $x_{n+1} \neq 0$). □

Exercise 2.14

Problem.

(a) Let S be a graded ring. Show that $\operatorname{Proj} S = \emptyset$ if and only if every element of S_+ is nilpotent.

(b) Let $\varphi : S \to T$ be a homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \operatorname{Proj} T \mid \mathfrak{p} \not\supseteq \varphi(S_+)\}$. Show that U is an open subset of $\operatorname{Proj} T$, and show that φ determines a natural morphism $f : U \to \operatorname{Proj} S$.

(c) The morphism f can be an isomorphism even when φ is not. For example, suppose $\varphi_d : S_d \to T_d$ is an isomorphism for all $d \geq d_0$, where d_0 is an integer. Then show that $U = \operatorname{Proj} T$ and the morphism $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is an isomorphism.

(d) Let V be a projective variety with homogeneous coordinate ring $S(V)$ (I, Section 2). Show that $t(V) \cong \operatorname{Proj} S(V)$.

Solution. Recall (Lemma 2.4 and its proof) that a homogeneous ideal $\mathfrak{p}$ of S is prime iff for homogeneous a, b, $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$; $\operatorname{Proj} S$ is the set of homogeneous primes not containing S_+.

(a) ($\Leftarrow$) If every element of S_+ is nilpotent and $\mathfrak{p}$ is a homogeneous prime, then $S_+ \subseteq \mathfrak{N}(S) \subseteq \mathfrak{p}$, so $\mathfrak{p} \notin \operatorname{Proj} S$. Hence $\operatorname{Proj} S = \emptyset$. ($\Rightarrow$) Suppose some element of S_+ is not nilpotent. Then (since a sum of nilpotents is nilpotent) some homogeneous $g \in S_d$, $d > 0$, is not nilpotent. Let $M = \{1, g, g^2, \dots\}$; $0 \notin M$. By Zorn's lemma the set of homogeneous ideals disjoint from M (nonempty: it contains 0) has a maximal element $\mathfrak{q}$. I claim $\mathfrak{q}$ is prime. Let a, b be homogeneous with $a, b \notin \mathfrak{q}$. By maximality $\mathfrak{q} + (a)$ and $\mathfrak{q} + (b)$ meet M; taking the homogeneous components of degrees md, nd we get $g^m = q + ar$, $g^n = q' + bs$ with $q, q' \in \mathfrak{q}$ and r, s homogeneous. Multiplying, $g^{m+n} \in \mathfrak{q} + (ab)$; so if $ab \in \mathfrak{q}$ then $g^{m+n} \in \mathfrak{q}$, a contradiction. So $ab \notin \mathfrak{q}$, and $\mathfrak{q}$ is prime. Since $g \in S_+ \setminus \mathfrak{q}$, $\mathfrak{q} \not\supseteq S_+$, so $\mathfrak{q} \in \operatorname{Proj} S$, which is nonempty.

(b) Let $\mathfrak{a} \subseteq T$ be the ideal generated by $\varphi(S_+)$; it is homogeneous, and $U = \operatorname{Proj} T \setminus V(\mathfrak{a})$ is open. For $\mathfrak{p} \in U$, $\varphi^{-1}(\mathfrak{p})$ is a homogeneous prime of S (φ preserves degrees), and it does not contain S_+ since $\mathfrak{p} \not\supseteq \varphi(S_+)$. Put $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$. If $\mathfrak{b} \subseteq S$ is homogeneous, then $f^{-1}(V(\mathfrak{b})) = U \cap V(\varphi(\mathfrak{b})T)$, so f is continuous.

Charts. Let $g \in S_+$ be homogeneous. Then $f^{-1}(D_+(g)) = D_+(\varphi(g)) \cap U = D_+(\varphi(g))$ (a prime not containing $\varphi(g)$, $g \in S_+$, certainly does not contain $\varphi(S_+)$). These sets cover U: if $\mathfrak{p} \in U$ then $\mathfrak{p} \not\supseteq \varphi(S_+)$, and since $\mathfrak{p}$ is homogeneous and S_+ is generated by homogeneous elements, there is a homogeneous $g \in S_+$ with $\varphi(g) \notin \mathfrak{p}$. The ring homomorphism φ extends to $S_g \to T_{\varphi(g)}$ and restricts to $\varphi_{(g)} : S_{(g)} \to T_{(\varphi(g))}$. By Prop. 2.5(b), $D_+(\varphi g) \cong \operatorname{Spec} T_{(\varphi g)}$ and $D_+(g) \cong \operatorname{Spec} S_{(g)}$, so $\varphi_{(g)}$ induces (Prop. 2.3(b)) a morphism of schemes $f_g : D_+(\varphi g) \to D_+(g) \subseteq \operatorname{Proj} S$. On points it is $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ (the isomorphism $D_+(g) \cong \operatorname{Spec} S_{(g)}$ sends $\mathfrak{p}$ to $\mathfrak{p}S_g \cap S_{(g)}$ and $s/g^n \in \varphi_{(g)}^{-1}(\mathfrak{p}T_{\varphi g} \cap T_{(\varphi g)})$ iff $\varphi(s) \in \mathfrak{p}$ iff $s \in \varphi^{-1}\mathfrak{p}$). For two homogeneous $g, g' \in S_+$ of degrees d, d', put $h = g'^d / g^{d'} \in S_{(g)}$; then $D_+(gg') = D_+(g) \cap D_+(g')$ corresponds to $D(h) \subseteq \operatorname{Spec} S_{(g)}$ and $S_{(gg')} = (S_{(g)})_h$ (this is part of the proof of Prop. 2.5(b)), and similarly for T with $\varphi(h) = \varphi(g')^d / \varphi(g)^{d'}$. By Ex. 2.1 (naturality of $D(h) \cong \operatorname{Spec}(S_{(g)})_h$), the restriction of f_g to $D_+(\varphi g) \cap D_+(\varphi g') = D_+(\varphi(gg'))$ is Spec of the localization $S_{(gg')} \to T_{(\varphi(gg'))}$ of $\varphi_{(g)}$ at h, which is the map $s/(gg')^n \mapsto \varphi(s)/\varphi(gg')^n$ induced by φ. The same description holds for $f_{g'}$ (the roles of g, g' are symmetric), so f_g and $f_{g'}$ agree on the overlap as morphisms into $D_+(gg')$. Morphisms of locally ringed spaces glue, so we obtain a morphism $f : U \to \operatorname{Proj} S$ with $f|_{D_+(\varphi g)} = f_g$.

(c) $U = \operatorname{Proj} T$. Let $\mathfrak{p} \in \operatorname{Proj} T$; pick a homogeneous $t \in T_e \setminus \mathfrak{p}$ with $e > 0$ (possible as $\mathfrak{p} \not\supseteq T_+$). Take m with $me \geq d_0$. Then $t^m \notin \mathfrak{p}$ and $t^m \in T_{me} = \varphi(S_{me})$, so $t^m = \varphi(s)$ with $s \in S_+$; thus $\varphi(S_+) \not\subseteq \mathfrak{p}$.

f is an isomorphism. For a homogeneous $g \in S_+$ and $m \geq 1$ we have $D_+(g) = D_+(g^m)$ and $S_{(g)} = S_{(g^m)}$, $T_{(\varphi g)} = T_{(\varphi(g)^m)}$ (as subrings of $S_g = S_{g^m}$, $T_{\varphi g} = T_{\varphi(g)^m}$); so the sets $D_+(g)$ with $g \in S_d$, $d \geq d_0$, already cover $\operatorname{Proj} S$ (each $D_+(g)$ with $\deg g = e > 0$ equals $D_+(g^m)$, $me \geq d_0$). By (b), $f^{-1}(D_+(g)) = D_+(\varphi g)$ and f restricts to the morphism induced by $\varphi_{(g)} : S_{(g)} \to T_{(\varphi g)}$. I claim $\varphi_{(g)}$ is an isomorphism when $d = \deg g \geq d_0$. *Surjective:* an element of $T_{(\varphi g)}$ is $t/\varphi(g)^n$ with $t \in T_{nd}$; multiplying numerator and denominator by a power of $\varphi(g)$ we may assume

$n \geq 1$, so $nd \geq d_0$ and we can write $t = \varphi(s)$, $s \in S_{nd}$, and $t/\varphi(g)^n = \varphi_{(g)}(s/g^n)$. *Injective:* if $\varphi(s)/\varphi(g)^n = 0$ in $T_{(\varphi g)}$ with $s \in S_{nd}$, there is $m \geq 1$ (enlarging m if necessary) with $\varphi(g)^m\varphi(s) = \varphi(g^m s) = 0$; here $g^m s \in S_{(m+n)d}$ and $\varphi_{(m+n)d}$ is injective ($(m+n)d \geq d_0$), so $g^m s = 0$, whence $s/g^n = 0$. So f is an isomorphism over each $D_+(g)$ ($\deg g \geq d_0$), these cover $\operatorname{Proj} S$, and a morphism which is an isomorphism over the members of an open cover of the target is an isomorphism (Ex. 2.17(a)). Hence $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is an isomorphism.

(d) Let $V \subseteq \mathbf{P}^n_k$ be a projective variety (k algebraically closed, V irreducible closed), $S = S(V) = k[x_0, \dots, x_n]/I(V)$, a domain since $I(V)$ is prime, and $V_i = V \cap \{x_i \neq 0\}$.

Points. By the projective Nullstellensatz (Ch. I, Ex. 2.1, 2.3) $Y \mapsto I(Y)$ is an inclusion-reversing bijection between nonempty irreducible closed subsets $Y \subseteq V$ and homogeneous prime ideals of S not containing S_+ (i.e. homogeneous primes $\mathfrak{p} \supseteq I(V)$ with $\mathfrak{p} \not\supseteq (x_0, \dots, x_n)$). So we have a bijection $\beta : t(V) \to \operatorname{Proj} S$, $Y \mapsto I(Y)$ (viewed in S). Moreover $Y \in t(V_i)$ iff $Y \cap V_i \neq \emptyset$ iff $x_i \notin I(Y)$ iff $\beta(Y) \in D_+(x_i)$. As S_+ is generated by the x_i, the $D_+(x_i)$ cover $\operatorname{Proj} S$, and the $t(V_i)$ cover $t(V)$.

Charts. The ring $S_{(x_i)} = k[x_0/x_i, \dots, x_n/x_i] \subseteq S_{x_i}$ is the image of $\mu_i : k[y_0, \dots, \widehat{y_i}, \dots, y_n] \to S_{x_i}$, $y_j \mapsto x_j/x_i$. Its kernel is $\{h : h^{\mathrm{hom}} \in I(V)\}$, where h^{hom} is the homogenization: because S is a domain, $h(x_0/x_i, \dots) = 0$ in S_{x_i} iff $x_i^N h^{\mathrm{hom}} = 0$ in S for suitable N, iff $h^{\mathrm{hom}} \in I(V)$. If $V_i \neq \emptyset$ this kernel is $I(V_i)$: a form $F \in k[x]$ vanishes on V iff it vanishes on the dense open subset V_i of the irreducible space V (if $F|_{V_i} = 0$ then $x_i F \in I(V)$ and $x_i \notin I(V)$, a prime). If $V_i = \emptyset$, both $S_{x_i} = 0$ and $D_+(x_i) = \emptyset = t(V_i)$. So μ_i induces an isomorphism of k-algebras $\bar{\mu}_i : A(V_i) \xrightarrow{\sim} S_{(x_i)}$, hence by Prop. 2.6 (where $t(V_i) \cong \operatorname{Spec} A(V_i)$) and Prop. 2.5(b)

$$\Theta_i : \ t(V_i) \cong \operatorname{Spec} A(V_i) \cong \operatorname{Spec} S_{(x_i)} \cong D_+(x_i) \subseteq \operatorname{Proj} S.$$

On points, Θ_i is $Y \mapsto I(Y)$: a prime $\mathfrak{P}$ of $A(V_i)$, being $I(Y \cap V_i)$ for $Y = \overline{Y \cap V_i} \in t(V)$, corresponds under $\bar{\mu}_i$ to the prime of $S_{(x_i)}$ given by the dehomogenizations $\{h : h^{\mathrm{hom}} \in I(Y)\}$ (same argument as above applied to the irreducible Y), i.e. $I(Y)S_{x_i} \cap S_{(x_i)}$, which is what Prop. 2.5(b) attaches to $I(Y)$. So all Θ_i are given by β on points.

Gluing. Put $W = V_i \cap V_j$ (if empty there is nothing to check). Then $t(W) = t(V_i) \cap t(V_j)$ is the principal open set $D(x_j/x_i) \subseteq \operatorname{Spec} A(V_i)$ and equals $D(x_i/x_j) \subseteq \operatorname{Spec} A(V_j)$; W is an affine variety with $A(W) = A(V_i)_{x_j/x_i} = A(V_j)_{x_i/x_j}$, and by Prop. 2.6 $\Gamma(t(W), \mathcal{O}) = \mathcal{O}_V(W) = A(W)$. Under Θ_i (resp. Θ_j) the set $t(W)$ goes isomorphically onto $D_+(x_ix_j) = D_+(x_i) \cap D_+(x_j)$, which is affine, $\cong \operatorname{Spec} S_{(x_ix_j)}$ with $S_{(x_ix_j)} = (S_{(x_i)})_{x_j/x_i} = (S_{(x_j)})_{x_i/x_j}$ (as in the proof of Prop. 2.5(b)). So $\Theta_i|_{t(W)}$ and $\Theta_j|_{t(W)}$ are two morphisms between the affine schemes $t(W) = \operatorname{Spec} A(W)$ and $\operatorname{Spec} S_{(x_ix_j)}$, hence by Prop. 2.3(c) (or Ex. 2.4) they are equal as soon as the ring maps $\lambda_i, \lambda_j : S_{(x_ix_j)} \to A(W) = \mathcal{O}_V(W)$ they induce are equal. Now $\Theta_i|_{t(W)}$ is, by Ex. 2.1 and the definition of Θ_i, Spec of the localization at x_j/x_i of $\bar{\mu}_i^{-1}$, so λ_i sends $F/(x_i^a x_j^b)$ (F a form of degree $a+b$, which equals $\big(F/x_i^{a+b}\big)/(x_j/x_i)^b\big)$ to $\big(F(y)/y_i^{a+b}\big)/(x_j/x_i)^b$, i.e. to the regular function $P \mapsto F(P)/\big(x_i(P)^a x_j(P)^b\big)$ on W (here F/x_i^{a+b} corresponds under $\bar{\mu}_i$ to the dehomogenization of F, the function $P \mapsto F(P)/x_i(P)^{a+b}$). The same computation with i, j interchanged gives the same function for λ_j. Hence $\lambda_i = \lambda_j$ and $\Theta_i = \Theta_j$ on $t(V_i) \cap t(V_j)$. Therefore the Θ_i glue to a morphism $\Theta : t(V) \to \operatorname{Proj} S$ with $\Theta^{-1}(D_+(x_i)) = t(V_i)$ (by the bijection on points above) and $\Theta|_{t(V_i)} = \Theta_i$ an isomorphism onto $D_+(x_i)$. As the $D_+(x_i)$ cover $\operatorname{Proj} S$, Θ is an isomorphism (Ex. 2.17(a)), i.e. $t(V) \cong \operatorname{Proj} S$. $\square$

Exercise 2.15

Problem.

(a) Let V be a variety over the algebraically closed field k. Show that a point $P \in t(V)$ is a closed point if and only if its residue field is k.

(b) If $f : X \to Y$ is a morphism of schemes over k, and if $P \in X$ is a point with residue field k, then $f(P) \in Y$ also has residue field k.

(c) Now show that if V, W are any two varieties over k, then the natural map

$$\mathrm{Hom}_{\mathrm{Var}(k)}(V, W) \to \mathrm{Hom}_{\mathrm{Sch}(k)}(t(V), t(W))$$

is bijective. (Injectivity is easy. The hard part is to show it is surjective.)

Solution. Recall (Prop. 2.6) the construction: $t(V)$ is the set of nonempty irreducible closed subsets of V; $\alpha : V \to t(V)$, $v \mapsto \overline{\{v\}} = \{v\}$, is a homeomorphism of V onto the set of closed points of $t(V)$ (with the induced topology); the open subsets of $t(V)$ are the sets $t(U)$, $U \subseteq V$ open, and the structure sheaf satisfies $\mathcal{O}_{t(V)}(t(U)) = \mathcal{O}_V(U)$, the ring of regular functions on U, compatibly with restriction. For $v \in U$ the value $\phi(v) \in k$ of $\phi \in \mathcal{O}_V(U)$ is the image of the section in $k(\alpha(v)) = k$. If $V_0 \subseteq V$ is an affine open subvariety then $t(V_0) \cong \operatorname{Spec} A(V_0)$, with $\alpha(v)$ corresponding to the maximal ideal $\mathfrak{m}_v$. The scheme $t(V)$ is a scheme over k (constants are regular functions).

(a) ($\Rightarrow$) If $P = \alpha(v)$ is closed, take an affine open $V_0 \ni v$; P corresponds to a maximal ideal $\mathfrak{m}$ of the finitely generated k-algebra $A = A(V_0)$, and $k(P) = A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}} = A/\mathfrak{m}$. This is a field which is finitely generated as a k-algebra, hence a finite extension of k by the Nullstellensatz (Zariski's lemma, Ch. I Thm. 1.3A), hence equal to k because k is algebraically closed. ($\Leftarrow$) Suppose $k(P) = k$, meaning that the structure map $k \to k(P)$ is an isomorphism. Choose an affine open $t(V_0) = \operatorname{Spec} A \ni P$, $P = \mathfrak{p}$. Then $k \to A \to A/\mathfrak{p} \hookrightarrow k(P) = \mathrm{Frac}(A/\mathfrak{p})$, and the image of k is already all of $k(P)$; so $A/\mathfrak{p} = k(P)$ is a field and $\mathfrak{p}$ is a maximal ideal. By the description of $t(V_0) \cong \operatorname{Spec} A(V_0)$ above, maximal ideals correspond to the points $\alpha(v)$, $v \in V_0$. So $P \in \alpha(V)$ is a closed point of $t(V)$.

(b) A morphism of schemes over k gives a local k-algebra homomorphism $f_P^{\#} : \mathcal{O}_{Y,f(P)} \to \mathcal{O}_{X,P}$, hence a k-algebra homomorphism of fields $\lambda : k(f(P)) \to k(P)$. The composite $k \to k(f(P)) \xrightarrow{\lambda} k(P)$ is the structure map of $k(P)$, an isomorphism by hypothesis. Both $k \to k(f(P))$ and λ are field homomorphisms, hence injective. Given $y \in k(f(P))$, $\lambda(y) = \lambda(c \cdot 1)$ for some $c \in k$ (surjectivity of $k \to k(P)$), so $y = c \cdot 1$ by injectivity of λ. Thus $k \to k(f(P))$ is surjective, i.e. $k(f(P)) = k$.

(c) *Injectivity.* If $h : V \to W$ is a morphism of varieties, $t(h)$ acts on closed points as $\alpha_W \circ h \circ \alpha_V^{-1}$; so $t(h) = t(h')$ gives $h = h'$ as maps, hence as morphisms of varieties.

Surjectivity. Let $g : t(V) \to t(W)$ be a morphism of schemes over k. By (a), (b), (a) again, g maps closed points to closed points. Define $h : V \to W$ by $h = \alpha_W^{-1} \circ g \circ \alpha_V$. It is continuous, being a restriction of the continuous g to the closed points, which carry the subspace topology.

(i) $g^{-1}(t(U')) = t(h^{-1}U')$ *for* $U' \subseteq W$ *open.* The open set $g^{-1}(t(U'))$ equals $t(U'')$ for some open $U'' \subseteq V$ (all open sets of $t(V)$ have this form), and $v \in U''$ iff $\alpha(v) \in g^{-1}(t(U'))$ iff $g(\alpha(v)) = \alpha_W(h(v)) \in t(U')$ iff $h(v) \in U'$. So $U'' = h^{-1}(U')$.

(ii) *h is a morphism of varieties.* Let $\phi \in \mathcal{O}_W(U') = \mathcal{O}_{t(W)}(t(U'))$ and consider $\phi' = g^\#(\phi) \in \mathcal{O}_{t(V)}(t(h^{-1}U')) = \mathcal{O}_V(h^{-1}U')$, a regular function. For $v \in h^{-1}(U')$ and $P = \alpha(v)$, the value $\phi'(v)$ is the image of ϕ'_P in $k(P) = k$. Since $g_P^\# : \mathcal{O}_{g(P)} \to \mathcal{O}_P$ is a local k-algebra map inducing the k-linear map $k(g(P)) = k \to k(P) = k$, which is the identity of k, we get $\phi'(v) = \phi(h(v))$. So $\phi \circ h = \phi'$ is regular on $h^{-1}(U')$, and h is a morphism of varieties (Ch. I, Section 3).

(iii) $t(h) = g$. On closed points they agree by construction. For a non-closed point $\eta \in t(V)$, $\eta = Y$ is an irreducible closed subset, with $\overline{\{\eta\}} = t(Y)$ whose closed points are $\alpha(Y)$, and $\alpha(Y)$ is dense in $t(Y)$ (a closed set $t(Z)$ containing $\alpha(Y)$ has $Z \supseteq Y$). Let $Z = \overline{h(Y)}$, an irreducible closed subset of W, and $\eta' = g(\eta) = Y' \in t(W)$. Since g is continuous, $g(t(Y)) \subseteq \overline{\{\eta'\}} = t(Y')$, so $h(Y) \subseteq Y'$ and $Z \subseteq Y'$. Conversely, $g(\eta) \in g(\overline{\alpha(Y)}) \subseteq \overline{g(\alpha(Y))} = \overline{\alpha_W(h(Y))} = t(Z)$, so $\eta' = Y' \in t(Z)$, i.e. $Y' \subseteq Z$. So $g(\eta) = Z = \overline{h(Y)} = t(h)(\eta)$. Hence $g = t(h)$ as continuous maps. Finally, $g^\#$ and $t(h)^\#$ both map $\mathcal{O}_{t(W)}(t(U')) = \mathcal{O}_W(U')$ into $\mathcal{O}_V(h^{-1}U')$, and both send ϕ to $\phi \circ h$ by (ii) (for $t(h)$ this is the definition); since a regular function on an open subset of a variety is determined by its values at points, $g^\# = t(h)^\#$. Hence $g = t(h)$, and the map is surjective. □

Exercise 2.16

Problem. Let X be a scheme. For any $f \in \Gamma(X, \mathcal{O}_X)$, define X_f to be the subset of points $x \in X$ such that the stalk f_x of f at x is not contained in the maximal ideal $\mathfrak{m}_x$ of the local ring $\mathcal{O}_x$.

(a) If $U = \operatorname{Spec} B$ is an open affine subscheme of X, and if $\bar{f} \in B = \Gamma(U, \mathcal{O}_X|_U)$ is the restriction of f, show that $U \cap X_f = D(\bar{f})$. Conclude that X_f is an open subset of X.

(b) Assume that X is quasi-compact. Let $A = \Gamma(X, \mathcal{O}_X)$, and let $a \in A$ be an element whose restriction to X_f is 0. Show that for some $n > 0$, $f^n a = 0$. [*Hint*: Use an open affine cover of X.]

(c) Now assume that X has a finite cover by open affines U_i such that each intersection $U_i \cap U_j$ is quasi-compact (this hypothesis is satisfied, for example, if $\operatorname{sp}(X)$ is noetherian). Let $b \in \Gamma(X_f, \mathcal{O}_{X_f})$. Show that for some $n > 0$, $f^n b$ is the restriction of an element of A.

(d) With the hypothesis of (c), conclude that $\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f$.

Solution. Here $\mathcal{O}_{X_f} = \mathcal{O}_X|_{X_f}$.

(a) Let $\mathfrak{p} \in U$. The germ $f_\mathfrak{p}$ is the image of $\bar{f}$ under $B \to B_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$, and an element $b \in B$ has image in $\mathfrak{p}B_\mathfrak{p}$ iff $b \in \mathfrak{p}$. So $\mathfrak{p} \in X_f$ iff $\bar{f} \notin \mathfrak{p}$ iff $\mathfrak{p} \in D(\bar{f})$. Thus $U \cap X_f = D(\bar{f})$ is open in U. As X is covered by affine opens, X_f is open.

(b) Cover X by finitely many affine opens $U_i = \operatorname{Spec} B_i$, $i = 1, \ldots, r$ (X quasi-compact), and let $a_i, f_i \in B_i$ be the restrictions of a, f. By (a), $X_f \cap U_i = D(f_i)$ and a restricts to 0 in $\Gamma(D(f_i), \mathcal{O}) = (B_i)_{f_i}$ (Prop. 2.2(b)); the image of a_i in $(B_i)_{f_i}$ is 0, so $f_i^{n_i} a_i = 0$ in B_i for some n_i. With $n = \max n_i$, $f^n a$ restricts to 0 on each U_i, hence $f^n a = 0$ in A.

(c) Let $U_i = \operatorname{Spec} B_i$ ($i = 1, \ldots, r$) be the given cover and $f_i = f|_{U_i}$. Then $X_f \cap U_i = D(f_i)$ and $b|_{D(f_i)} \in \Gamma(D(f_i), \mathcal{O}) = (B_i)_{f_i}$, so $b|_{D(f_i)} = b_i'/f_i^{n_i}$ with $b_i' \in B_i$. Taking $n = \max n_i$ (and multiplying numerators by powers of f_i) we may write $f^n b|_{D(f_i)} = b_i|_{D(f_i)}$ with $b_i \in B_i = \Gamma(U_i, \mathcal{O})$.

On $U_{ij} = U_i \cap U_j$ the sections $b_i|_{U_{ij}}$ and $b_j|_{U_{ij}}$ both restrict on $U_{ij} \cap X_f$ to $f^n b$ (since $U_{ij} \cap X_f \subseteq D(f_i) \cap D(f_j)$). The scheme U_{ij} is quasi-compact by hypothesis, and $(U_{ij})_f = U_{ij} \cap X_f$. By (b) applied to the scheme U_{ij}, the section $f|_{U_{ij}}$, and the section $(b_i - b_j)|_{U_{ij}}$ (which vanishes on $(U_{ij})_f$), there is m_{ij} with $f^{m_{ij}}(b_i - b_j)|_{U_{ij}} = 0$. Let m be the maximum of the finitely many m_{ij}. Then the sections $f^m b_i \in \mathcal{O}(U_i)$ agree on all overlaps, so they glue to $c \in A = \Gamma(X, \mathcal{O}_X)$ with $c|_{U_i} = f^m b_i$. On $D(f_i) = X_f \cap U_i$ we get $c = f^m b_i = f^{m+n} b$. So $c|_{X_f} = f^{m+n} b$ (sheaf axiom for the cover $\{D(f_i)\}$ of X_f).

(d) The restriction $A \to \Gamma(X_f, \mathcal{O}_X)$ sends f to a unit (the germ of f is a unit at each point of X_f, so $f|_{X_f}$ has a local inverse near each point and these glue), so it induces a ring homomorphism $\rho : A_f \to \Gamma(X_f, \mathcal{O}_X)$, $a/f^n \mapsto a|_{X_f} \cdot (f|_{X_f})^{-n}$. *Surjective:* by (c), for $b \in \Gamma(X_f, \mathcal{O}_X)$ there are N and $c \in A$ with $c|_{X_f} = f^N b$, so $b = \rho(c/f^N)$. *Injective:* if $\rho(a/f^n) = 0$ then $a|_{X_f} = 0$, so $f^m a = 0$ by (b) (X is quasi-compact, being a finite union of affine schemes), so $a/f^n = 0$ in A_f. Hence $\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f$. □

Exercise 2.17

Problem. *A Criterion for Affineness.*

(a) Let $f : X \to Y$ be a morphism of schemes, and suppose that Y can be covered by open subsets U_i, such that for each i, the induced map $f^{-1}(U_i) \to U_i$ is an isomorphism. Then f is an isomorphism.

(b) A scheme X is affine if and only if there is a finite set of elements $f_1, \ldots, f_r \in A = \Gamma(X, \mathcal{O}_X)$, such that the open subsets X_{f_i} are affine, and $f_1, \ldots, f_r$ generate the unit ideal in A. [*Hint*: Use (Ex. 2.4) and (Ex. 2.16d) above.]

Solution. (a) Let $g_i : U_i \to f^{-1}(U_i) \subseteq X$ be the inverse of the isomorphism $f_i = f|_{f^{-1}U_i} : f^{-1}(U_i) \to U_i$ (as a morphism to X). On $U_i \cap U_j$, both $g_i|_{U_{ij}}$ and $g_j|_{U_{ij}}$ are morphisms $U_{ij} \to f^{-1}(U_{ij})$ that are inverse to $f|_{f^{-1}(U_{ij})} : f^{-1}(U_{ij}) \to U_{ij}$ (the latter is an isomorphism, being the restriction of the isomorphism f_i to the open subscheme $f^{-1}(U_{ij})$). Since an inverse is unique, $g_i = g_j$ on U_{ij}, and morphisms of locally ringed spaces glue: there is $g : Y \to X$ with $g|_{U_i} = g_i$. Then $f \circ g = \mathrm{id}_Y$ (check on each U_i), and $g \circ f = \mathrm{id}_X$ (check on each $f^{-1}(U_i)$, which cover X: there $gf = g_i f_i = \mathrm{id}$). So f is an isomorphism.

(b) ($\Rightarrow$) If $X = \operatorname{Spec} A$ take $r = 1$, $f_1 = 1$: $X_1 = X$ is affine and $(1) = A$.

($\Leftarrow$) Let $f_1, \ldots, f_r \in A$ generate the unit ideal with each X_{f_i} affine. Let $\pi : X \to \operatorname{Spec} A$ be the morphism corresponding (Ex. 2.4) to $\mathrm{id}_A : A \to \Gamma(X, \mathcal{O}_X)$. By formula ($*$) in the proof of Ex. 2.4, $\pi(x) = \{a \in A : a_x \in \mathfrak{m}_x\}$, so $\pi^{-1}(D(f_i)) = \{x : (f_i)_x \notin \mathfrak{m}_x\} = X_{f_i}$. The $D(f_i)$ cover $\operatorname{Spec} A$ since the f_i generate (1). By (a) it suffices to show that $\pi|_{X_{f_i}} : X_{f_i} \to D(f_i) = \operatorname{Spec} A_{f_i}$ is an isomorphism for each i.

First, the hypotheses of Ex. 2.16(c) hold for X: the X_{f_i} are finitely many open affines covering X (they cover since $\pi^{-1}(D(f_i))$ cover X), and $X_{f_i} \cap X_{f_j}$ is the set $\{x \in X_{f_i} : (f_j)_x \notin \mathfrak{m}_x\} = D(f_j|_{X_{f_i}})$ by Ex. 2.16(a) applied to the affine scheme X_{f_i}; it is an affine scheme $\operatorname{Spec}(B_i)_{f_j}$, in particular quasi-compact (Ex. 2.13(b)). So by Ex. 2.16(d), the map $\rho_i : A_{f_i} \to \Gamma(X_{f_i}, \mathcal{O}_X) =: B_i$ is an isomorphism.

Now $\pi|_{X_{f_i}} : \operatorname{Spec} B_i = X_{f_i} \to D(f_i) \cong \operatorname{Spec} A_{f_i}$ is a morphism of affine schemes; by Prop. 2.3(c) and Ex. 2.4 it is induced by its map on global sections $\Gamma(\pi^\#) : A_{f_i} = \mathcal{O}(D(f_i)) \to \mathcal{O}_X(X_{f_i}) = B_i$.

That map is a ring homomorphism whose composite with $A \to A_{f_i}$ is the restriction $A \to B_i$ (since $\Gamma(\pi^\#) = \mathrm{id}_A$ on global sections and $\pi^\#$ commutes with restrictions); hence it equals ρ_i, an isomorphism. So $\pi|_{X_{f_i}}$ is an isomorphism of affine schemes. By (a), $\pi : X \to \operatorname{Spec} A$ is an isomorphism, i.e. X is affine. □

Exercise 2.18

Problem. In this exercise, we compare some properties of a ring homomorphism to the induced morphism of the spectra of the rings.

(a) Let A be a ring, $X = \operatorname{Spec} A$, and $f \in A$. Show that f is nilpotent if and only if $D(f)$ is empty.

(b) Let $\varphi : A \to B$ be a homomorphism of rings, and let $f : Y = \operatorname{Spec} B \to X = \operatorname{Spec} A$ be the induced morphism of affine schemes. Show that φ is injective if and only if the morphism of sheaves $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is injective. Show furthermore that in that case f is dominant, i.e., $f(Y)$ is dense in X.

(c) With the same notation, show that if φ is surjective, then f is a homeomorphism of Y onto a closed subset of X, and $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective.

(d) Prove the converse to (c), namely, if $f : Y \to X$ is a homeomorphism onto a closed subset, and if $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective, then φ is surjective. [*Hint*: Consider $X' = \operatorname{Spec} A/\operatorname{Ker}\varphi$ and use (b) and (c).]

Solution. (a) $D(f) = \emptyset$ iff f lies in every prime ideal iff $f \in \mathfrak{N}(A)$, the intersection of all primes, iff f is nilpotent (Atiyah–Macdonald, Prop. 1.8).

(b) ($\Leftarrow$) Taking global sections, $f^\#$ gives $\varphi : A = \Gamma(X, \mathcal{O}_X) \to \Gamma(X, f_*\mathcal{O}_Y) = B$ (Prop. 2.2(c)); an injective morphism of sheaves is injective on the sections over every open set (the kernel presheaf is the kernel sheaf, which is 0). ($\Rightarrow$) Assume φ injective. For $g \in A$, $f^{-1}(D(g)) = D(\varphi(g))$ and $f^\#(D(g)) : A_g \to B_{\varphi(g)}$ sends a/g^n to $\varphi(a)/\varphi(g)^n$. If this is 0, then $\varphi(g)^m\varphi(a) = \varphi(g^m a) = 0$ for some m, so $g^m a = 0$ and $a/g^n = 0$. Thus $f^\#$ is injective on sections over every basic open set $D(g)$, hence on all stalks (stalks are direct limits over the basic neighbourhoods, and direct limits of injections are injective), hence $f^\#$ is injective. *Dominance.* Let $\mathfrak{b} = \bigcap_{\mathfrak{q} \in Y} \varphi^{-1}(\mathfrak{q})$, the intersection of the primes in $f(Y)$. Then $\overline{f(Y)} = V(\mathfrak{b})$ (the closure of a set of primes Σ is $V(\bigcap \Sigma)$), and $\mathfrak{b} = \varphi^{-1}(\mathfrak{N}B) = \{a : \varphi(a) \text{ nilpotent}\} = \sqrt{\operatorname{Ker}\varphi}$, since $\varphi(a)^n = 0 \iff a^n \in \operatorname{Ker}\varphi$. Hence $\overline{f(Y)} = V(\operatorname{Ker}\varphi)$. If φ is injective this is $V(0) = X$, so f is dominant.

(c) Let φ be surjective, $\mathfrak{a} = \operatorname{Ker}\varphi$, so $B \cong A/\mathfrak{a}$ and $f(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q})$ gives a bijection of Y onto $V(\mathfrak{a})$ (the primes of A containing $\mathfrak{a}$ correspond to primes of $A/\mathfrak{a}$). For an ideal $\mathfrak{b} \subseteq B$, $f(V(\mathfrak{b})) = V(\varphi^{-1}\mathfrak{b})$ (as φ is onto, primes of A containing $\varphi^{-1}(\mathfrak{b})$ are exactly the preimages of primes containing $\mathfrak{b}$), a closed subset of X contained in $V(\mathfrak{a})$. So f is a closed continuous injection, i.e. a homeomorphism onto the closed subset $V(\mathfrak{a})$.

Surjectivity of $f^\#$. Check on stalks. For $\mathfrak{p} \in V(\mathfrak{a})$, $\mathfrak{p} = f(\mathfrak{q})$, the stalk of $f_*\mathcal{O}_Y$ at $\mathfrak{p}$ is $\mathcal{O}_{Y,\mathfrak{q}} = B_\mathfrak{q}$, because f is a homeomorphism onto the subspace $V(\mathfrak{a})$ (the sets $f^{-1}(W)$, $W \ni \mathfrak{p}$ open in X, are cofinal among the neighbourhoods of $\mathfrak{q}$). The stalk map is $A_\mathfrak{p} \to B_\mathfrak{q}$, $a/s \mapsto \varphi(a)/\varphi(s)$ which is surjective: an element of $B_\mathfrak{q}$ is b/c with $c \notin \mathfrak{q}$; write $b = \varphi(a)$, $c = \varphi(s)$ with $s \notin \mathfrak{p}$. For $\mathfrak{p} \notin V(\mathfrak{a}) = f(Y)$, since $f(Y)$ is closed there is a neighbourhood W of $\mathfrak{p}$ with $f^{-1}(W) = \emptyset$, so the stalk of $f_*\mathcal{O}_Y$ at $\mathfrak{p}$ is 0, and surjectivity is trivial. Hence $f^\#$ is surjective.

(d) Let $f : Y \to X$ be a homeomorphism onto a closed subset with $f^\#$ surjective. Let $\mathfrak{a} = \operatorname{Ker}\varphi$, $A' = A/\mathfrak{a}$, $X' = \operatorname{Spec} A'$, and factor $\varphi = \varphi' \circ q$ with $q : A \to A'$ and $\varphi' : A' \to B$ injective. Correspondingly $f = i \circ f'$, with $i : X' \to X$ induced by q and $f' : Y \to X'$ induced by φ'. By (c), i is a homeomorphism onto the closed set $V(\mathfrak{a})$ and $i^\# : \mathcal{O}_X \to i_*\mathcal{O}_{X'}$ is surjective. By (b), $f'^\# : \mathcal{O}_{X'} \to f'_*\mathcal{O}_Y$ is injective and f' is dominant.

f' is a homeomorphism. $f(Y) = i(f'(Y))$ is closed in X by hypothesis, and $f'(Y)$ is dense in X' so $f(Y) = i(f'(Y))$ is dense in $i(X') = V(\mathfrak{a})$ (i is a homeomorphism onto $V(\mathfrak{a})$). Since $f(Y)$ is closed, $f(Y) = V(\mathfrak{a}) = i(X')$, so f' is surjective. Then $f' = i^{-1} \circ f$ (where $i^{-1} : V(\mathfrak{a}) \to X'$) is a composite of homeomorphisms $Y \to V(\mathfrak{a}) \to X'$, hence a homeomorphism.

$f'^\#$ is surjective. We have $f^\# = i_*(f'^\#) \circ i^\#$. Since $f^\#$ is surjective on stalks, so is $i_*(f'^\#) : i_*\mathcal{O}_{X'} \to i_*f'_*\mathcal{O}_Y$. At a point $i(x')$ its stalk map is $f'^\#_{x'}$, since i is a homeomorphism onto a closed set (as in (c)); hence $f'^\#$ is surjective on all stalks, hence surjective.

So $f'^\#$ is an isomorphism of sheaves and f' is a homeomorphism: f' is an isomorphism of schemes (its stalk maps are isomorphisms of local rings, being local homomorphisms that are bijective). Taking global sections, $\varphi' : A' \to B$ is an isomorphism (by Prop. 2.2(c) and Prop. 2.3(c)). Thus $\varphi = \varphi' \circ q$ is surjective. □

Exercise 2.19

Problem. Let A be a ring. Show that the following conditions are equivalent:

(i) $\operatorname{Spec} A$ is disconnected;

(ii) there exist nonzero elements $e_1, e_2 \in A$ such that $e_1e_2 = 0$, $e_1^2 = e_1$, $e_2^2 = e_2$, $e_1 + e_2 = 1$ (these elements are called *orthogonal idempotents*);

(iii) A is isomorphic to a direct product $A_1 \times A_2$ of two nonzero rings.

Solution. (i)⇒(ii). Write $\operatorname{Spec} A = Y_1 \sqcup Y_2$ with Y_1, Y_2 nonempty, disjoint and closed, $Y_i = V(\mathfrak{a}_i)$. Then $V(\mathfrak{a}_1 + \mathfrak{a}_2) = Y_1 \cap Y_2 = \emptyset$, so $\mathfrak{a}_1 + \mathfrak{a}_2 = A$ (an ideal not equal to A lies in a prime), and we can write $1 = x_1 + x_2$ with $x_i \in \mathfrak{a}_i$. Also $V(\mathfrak{a}_1\mathfrak{a}_2) = Y_1 \cup Y_2 = \operatorname{Spec} A$ (Lemma 2.1(a)), so $\mathfrak{a}_1\mathfrak{a}_2$ lies in every prime, i.e. in $\mathfrak{N}(A)$; in particular $(x_1x_2)^n = 0$ for some $n \geq 1$. Expanding $1 = (x_1 + x_2)^{2n-1}$, every monomial $x_1^a x_2^b$ ($a + b = 2n - 1$) has $a \geq n$ or $b \geq n$; so $1 = u_1 + u_2$ with $u_1 \in (x_1^n)$, $u_2 \in (x_2^n)$. Say $u_1 = ux_1^n$, $u_2 = vx_2^n$. Put $e_1 = u_1$, $e_2 = u_2$. Then $e_1 + e_2 = 1$ and $e_1e_2 = uv(x_1x_2)^n = 0$, so $e_1 = e_1(e_1 + e_2) = e_1^2$ and $e_2 = e_2^2$. Finally $e_1 \neq 0$: otherwise $1 = e_2 \in (x_2) \subseteq \mathfrak{a}_2$, so $Y_2 = V(\mathfrak{a}_2) = \emptyset$, a contradiction; similarly $e_2 \neq 0$.

(ii)⇒(iii). Let $A_i = Ae_i$, a ring with identity e_i (closed under addition and multiplication, with $e_ia = a$ for $a \in A_i$), nonzero since $e_i \neq 0$. Define $\Phi : A \to A_1 \times A_2$, $a \mapsto (ae_1, ae_2)$. It is a ring homomorphism with $\Phi(1) = (e_1, e_2)$ the identity. It is injective: $ae_1 = ae_2 = 0$ implies $a = a(e_1 + e_2) = 0$. It is surjective: for $x, y \in A$, $\Phi(xe_1 + ye_2) = (xe_1^2 + ye_2e_1,\ xe_1e_2 + ye_2^2) = (xe_1, ye_2)$. So $A \cong A_1 \times A_2$.

(iii)⇒(ii). Under $A \cong A_1 \times A_2$ take $e_1 = (1, 0)$, $e_2 = (0, 1)$: they are nonzero as $A_1, A_2 \neq 0$, and satisfy the required relations.

(ii)⇒(i). $V(e_1) \cap V(e_2) = V(e_1, e_2) = V(A) = \emptyset$ since $e_1 + e_2 = 1$, and $V(e_1) \cup V(e_2) = V(e_1e_2) = V(0) = \operatorname{Spec} A$. Thus $\operatorname{Spec} A$ is the disjoint union of the two closed sets $V(e_1)$,

$V(e_2)$. Neither is empty: if $V(e_1) = \emptyset$ then e_1 lies in no prime ideal, so it is a unit; but then $e_2 = e_1^{-1} e_1 e_2 = 0$, a contradiction; similarly for e_2. Each is open (complement of the other), so $\operatorname{Spec} A$ is disconnected. □

3. First properties of schemes

Numbering follows Chapter II of the scanned edition: Prop. 3.1 (integral = reduced + irreducible), Prop. 3.2 (noetherian is local), Thm. 3.3 (fibred products). For a point x of a scheme, $\mathcal{O}_x$ is its local ring and $k(x)$ its residue field. For g in a ring B and $V = \operatorname{Spec} B$, $D_V(g) = D(g)$ is the open set where $g \notin \mathfrak{p}$; $D(g) = \operatorname{Spec} B_g$ (Ex. 2.1). If $h : T \to \operatorname{Spec} C$ is a morphism and $c \in C$, then $h^{-1}(D(c))$ is the open set of T where the section $h^{\#}(c)$ has a unit germ; if $T = \operatorname{Spec} A$ it equals $D(c \cdot 1_A) = \operatorname{Spec} A_c$, where $A_c = A \otimes_C C_c$. A quasi-compact scheme is a finite union of open affine subschemes. Two elementary algebraic remarks are used constantly:

- if A is a finitely generated C_g-algebra then it is a finitely generated C-algebra (add $1/g$ to the generators);
- if A is a finitely generated C-algebra then A_g (for $g \in A$ or $g \in C$) is a finitely generated C-algebra (add $1/g$) and $A_c = A \otimes_C C_c$ is a finitely generated C_c-algebra.

Lemma A (basic open sets).

- **(A1)** Let $V = \operatorname{Spec} B$ be affine, let $W = \operatorname{Spec} C \subseteq V$ be an open affine subscheme, let $g \in B$ with $D_V(g) \subseteq W$ and let $\bar{g} \in C$ be the restriction of g to W. Then $D_V(g) = D_W(\bar{g})$; in particular $B_g \cong C_{\bar{g}}$.
- **(A2)** Let Y be a scheme covered by open affines $V_i = \operatorname{Spec} B_i$, let $V = \operatorname{Spec} B \subseteq Y$ be open affine and $y \in V$. Then there are an index i, $h \in B_i$ and $g \in B$ with $y \in D_V(g) \subseteq D_{V_i}(h) \subseteq V \cap V_i$.

Proof. (A1) For $z \in W$, $z \in D_V(g)$ iff the germ g_z is a unit of $\mathcal{O}_z$ iff $\bar{g}_z$ is a unit, because $g_z = \bar{g}_z$. So $D_V(g) \cap W = D_W(\bar{g})$, and $D_V(g) \subseteq W$. The rings of sections are B_g and $C_{\bar{g}}$ since both sets are the same open subscheme. (A2) Pick i with $y \in V_i$. The open set $V \cap V_i$ is a neighbourhood of y in V_i, so it contains some $D_{V_i}(h) \ni y$. This is an open subset of V, so it contains some $D_V(g) \ni y$. □

Exercise 3.1

> **Problem.** Show that a morphism $f : X \to Y$ is locally of finite type if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by open affine subsets $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra.

Solution. ($\Leftarrow$) Cover Y by open affines V; the hypothesis for each of them is exactly the definition.

($\Rightarrow$) By definition there is a cover $Y = \bigcup V_i$, $V_i = \operatorname{Spec} B_i$, and covers $f^{-1}(V_i) = \bigcup_j \operatorname{Spec} A_{ij}$ with A_{ij} finitely generated B_i-algebras. Let $V = \operatorname{Spec} B \subseteq Y$ be any open affine and $y \in V$. By

(A2) choose i, $h \in B_i$, $g \in B$ with $y \in D_V(g) \subseteq W := D_{V_i}(h) \subseteq V \cap V_i$. Put $C = (B_i)_h$, so $W = \operatorname{Spec} C$.

Step 1. We have $f^{-1}(W) = \bigcup_j (f^{-1}(W) \cap \operatorname{Spec} A_{ij})$ and $f^{-1}(W) \cap \operatorname{Spec} A_{ij} = D(h \cdot 1) = \operatorname{Spec}(A_{ij})_h$. Each $(A_{ij})_h = A_{ij} \otimes_{B_i} C$ is a finitely generated C-algebra. So $f^{-1}(W)$ is covered by the $\operatorname{Spec} A'_j$, $A'_j := (A_{ij})_h$, finitely generated over $C = \mathcal{O}(W)$.

Step 2. Let $\bar{g} \in C$ be the restriction of g to W. By (A1), $D_V(g) = D_W(\bar{g}) \cong \operatorname{Spec} C_{\bar{g}}$ and $C_{\bar{g}} \cong B_g$. Applying the remark above to $f^{-1}(W) \to W = \operatorname{Spec} C$, $f^{-1}(D_V(g)) = \bigcup_j D(\bar{g}) \subseteq \operatorname{Spec} A'_j$ is covered by the $\operatorname{Spec}(A'_j)_{\bar{g}}$, and $(A'_j)_{\bar{g}}$ is a finitely generated $C_{\bar{g}} = B_g$-algebra, hence a finitely generated B-algebra (add $1/g$).

Thus every $y \in V$ has an open neighbourhood $D_V(g)$ whose preimage is covered by spectra of finitely generated B-algebras. Since the $D_V(g)$ cover V, $f^{-1}(V)$ is covered by such spectra. □

Exercise 3.2

Problem. A morphism $f : X \to Y$ of schemes is *quasi-compact* if there is a cover of Y by open affines V_i such that $f^{-1}(V_i)$ is quasi-compact for each i. Show that f is quasi-compact if and only if for *every* open affine subset $V \subseteq Y$, $f^{-1}(V)$ is quasi-compact.

Solution. ($\Leftarrow$) is clear. ($\Rightarrow$) Let $\{V_i = \operatorname{Spec} B_i\}$ be a cover with $f^{-1}(V_i)$ quasi-compact.

Claim Q. If T is a quasi-compact scheme, $h : T \to \operatorname{Spec} C$ a morphism and $c \in C$, then $h^{-1}(D(c))$ is quasi-compact. Indeed $T = U_1 \cup \cdots \cup U_n$ with $U_l = \operatorname{Spec} A_l$ affine, and $h^{-1}(D(c)) \cap U_l = \operatorname{Spec}(A_l)_c$ is quasi-compact; a finite union of quasi-compact sets is quasi-compact.

Now let $V = \operatorname{Spec} B \subseteq Y$ be open affine and $y \in V$. Choose i, h, g as in (A2): $y \in D_V(g) \subseteq W := D_{V_i}(h) \subseteq V \cap V_i$, $W = \operatorname{Spec} C$. By Claim Q applied to $f^{-1}(V_i) \to V_i$ and $h \in B_i$, $f^{-1}(W)$ is quasi-compact. By (A1), $D_V(g) = D_W(\bar{g})$, so Claim Q applied to $f^{-1}(W) \to W$ and $\bar{g}$ shows that $f^{-1}(D_V(g))$ is quasi-compact. The quasi-compact space V is covered by finitely many sets $D_V(g_y)$, so $f^{-1}(V)$ is a finite union of quasi-compact sets, hence quasi-compact. □

Exercise 3.3

Problem.

(a) Show that a morphism $f : X \to Y$ is of finite type if and only if it is locally of finite type and quasi-compact.

(b) Conclude from this that f is of finite type if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by a finite number of open affines $U_i = \operatorname{Spec} A_i$, where each A_i is a finitely generated B-algebra.

(c) Show also that if f is of finite type, then for *every* open affine subset $V = \operatorname{Spec} B \subseteq Y$, and for *every* open affine subset $U = \operatorname{Spec} A \subseteq f^{-1}(V)$, A is a finitely generated B-algebra.

Solution. (a) ($\Rightarrow$) If $Y = \bigcup V_i$ and $f^{-1}(V_i)$ is a finite union of affines $\operatorname{Spec} A_{ij}$ with A_{ij} finitely generated B_i-algebras, then f is locally of finite type, and each $f^{-1}(V_i)$ is quasi-compact (finite

union of affines), so f is quasi-compact. ($\Leftarrow$) Suppose f is locally of finite type with $Y = \bigcup V_i$, $f^{-1}(V_i) = \bigcup_j \operatorname{Spec} A_{ij}$, A_{ij} finitely generated B_i-algebras, and quasi-compact. By Ex. 3.2, $f^{-1}(V_i)$ is quasi-compact, so finitely many of the $\operatorname{Spec} A_{ij}$ already cover it. Hence f is of finite type.

(b) ($\Rightarrow$) By (a) f is locally of finite type and quasi-compact. Let $V = \operatorname{Spec} B$ be open affine. By Ex. 3.1, $f^{-1}(V)$ is covered by open affines $\operatorname{Spec} A_j$, A_j finitely generated over B, and by Ex. 3.2 $f^{-1}(V)$ is quasi-compact, so finitely many $\operatorname{Spec} A_j$ suffice. ($\Leftarrow$) Cover Y by open affines; the hypothesis gives the finite covers required in the definition of finite type.

(c) Let $V = \operatorname{Spec} B \subseteq Y$ and $U = \operatorname{Spec} A \subseteq f^{-1}(V)$ be open affines. By (b), $f^{-1}(V) = U_1 \cup \cdots \cup U_n$ with $U_j = \operatorname{Spec} A_j$, A_j finitely generated B-algebras.

Step 1: U is covered by sets $D_U(g)$ with A_g a finitely generated B-algebra. Let $u \in U$. Apply (A2) to the scheme $f^{-1}(V)$ with the cover $\{U_j\}$ and the open affine U: there are j, $h \in A_j$, $g \in A$ with $u \in D_U(g) \subseteq D_{U_j}(h) \subseteq U \cap U_j$. Put $C = (A_j)_h$; it is a finitely generated A_j-algebra, hence a finitely generated B-algebra, and $D_{U_j}(h) = \operatorname{Spec} C$. By (A1), $A_g \cong C_{\bar{g}}$, which is a finitely generated C-algebra, hence a finitely generated B-algebra.

Step 2. Since U is quasi-compact, there are $g_1, \ldots, g_m \in A$ with $D(g_1) \cup \cdots \cup D(g_m) = U$, i.e. $(g_1, \ldots, g_m) = A$, such that every A_{g_l} is a finitely generated B-algebra. We show that this forces A to be a finitely generated B-algebra. Choose $c_l \in A$ with $\sum c_l g_l = 1$. For each l choose generators of the B-algebra A_{g_l}; adding $1/g_l$ we may assume they are $z_{lk} = y_{lk}/g_l^{n_{lk}}$ with $y_{lk} \in A$ ($k = 1, \ldots, r_l$). Let $A' \subseteq A$ be the B-subalgebra generated by the g_l, the c_l and the y_{lk}; it is finitely generated. Let $x \in A$. For each l, $x/1 \in A_{g_l}$ is a polynomial $P_l(z_{l1}, \ldots, z_{lr_l})$ with coefficients in B, of degree $\le d$ say. For $N \ge d \max n_{lk}$ every monomial of $g_l^N P_l(z_l)$ is of the form $b\, g_l^{N - \sum \alpha_k n_{lk}} y_l^{\alpha} \in A'$. So $g_l^N x/1 = q_l/1$ in A_{g_l} with $q_l \in A'$, hence $g_l^m(g_l^N x - q_l) = 0$ in A for some m. Taking $M = N + m$ (uniform in l), $g_l^M x \in A'$ for all l. Now for $T \ge m'(M-1)+1$ (m' the number of g_l's) expand $1 = (\sum c_l g_l)^T$: every monomial has some g_l with exponent $\ge M$, so $1 = \sum_l e_l g_l^M$ with e_l a polynomial in the c_i, g_i, so $e_l \in A'$. Then $x = \sum e_l (g_l^M x) \in A'$. So $A = A'$ is a finitely generated B-algebra. $\square$

Exercise 3.4

> **Problem.** Show that a morphism $f : X \to Y$ is finite if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ is affine, equal to $\operatorname{Spec} A$, where A is a finite B-module.

Solution. ($\Leftarrow$) Cover Y by open affines. ($\Rightarrow$) Let $Y = \bigcup V_i$, $V_i = \operatorname{Spec} B_i$, with $f^{-1}(V_i) = \operatorname{Spec} A_i$ and A_i a finite B_i-module. Fix an open affine $V = \operatorname{Spec} B \subseteq Y$ and write $X_V = f^{-1}(V)$.

Step 1: local statement. Let $y \in V$ and choose i, $h \in B_i$, $g \in B$ as in (A2): $y \in D_V(g) \subseteq W := D_{V_i}(h) \subseteq V \cap V_i$, with $W = \operatorname{Spec} C$, $C = (B_i)_h$. Then $f^{-1}(W) = D(h) \subseteq \operatorname{Spec} A_i$, i.e. $f^{-1}(W) = \operatorname{Spec} A_W$ with $A_W = (A_i)_h = A_i \otimes_{B_i} C$ a finite C-module. By (A1), $D_V(g) = D_W(\bar{g}) \cong \operatorname{Spec} B_g$ with $C_{\bar{g}} \cong B_g$, so $f^{-1}(D_V(g)) = \operatorname{Spec}(A_W)_{\bar{g}}$, and $(A_W)_{\bar{g}} = A_W \otimes_C C_{\bar{g}}$ is a finite B_g-module.

So V is covered by sets $D(g)$ with $f^{-1}(D(g))$ affine, equal to Spec of a finite B_g-module algebra. By quasi-compactness of V we get $g_1, \ldots, g_r \in B$ generating the unit ideal with this property.

Step 2: X_V is affine. Let $A = \Gamma(X_V, \mathcal{O}_{X_V})$ and $s_l = f^{\#}(g_l) \in A$. Then $s_1, \ldots, s_r$ generate the unit ideal of A (apply $f^{\#}$ to $\sum b_l g_l = 1$), and the open set $(X_V)_{s_l} = f^{-1}(D(g_l))$ is affine. By

the affineness criterion (Ex. 2.17(b)), X_V is affine, $X_V = \operatorname{Spec} A$. Then $f^{-1}(D(g_l)) = D(s_l) = \operatorname{Spec} A_{s_l}$, i.e. $A_{s_l} = A \otimes_B B_{g_l}$ is a finite B_{g_l}-module.

Step 3: A is a finite B-module. Let $M = A$ as a B-module. Each $M_{g_l} = A_{s_l}$ is finite over B_{g_l}. Choose finitely many generators of M_{g_l} over B_{g_l}; multiplying by powers of g_l we may take them to be images of elements of M. Let $N \subseteq M$ be the submodule generated by all these (finitely many) elements. Then $N_{g_l} = M_{g_l}$ for all l, i.e. $(M/N)_{g_l} = 0$ by exactness of localization. So every element of M/N is killed by a power of each g_l, hence by the unit ideal $(g_1^n, \ldots, g_r^n) = B$ (for large n); so $M = N$ is finite over B. □

Exercise 3.5

Problem. A morphism $f : X \to Y$ is *quasi-finite* if for every point $y \in Y$, $f^{-1}(y)$ is a finite set.

(a) Show that a finite morphism is quasi-finite.

(b) Show that a finite morphism is *closed*, i.e. the image of any closed subset is closed.

(c) Show by example that a surjective, finite type, quasi-finite morphism need not be finite.

Solution. Let $f : X \to Y$ be finite, with $Y = \bigcup V_i$, $V_i = \operatorname{Spec} B_i$, $f^{-1}(V_i) = \operatorname{Spec} A_i$, A_i a finite B_i-module.

(a) Let $y \in Y$, choose i with $y \in V_i$, let $\mathfrak{p} \subset B_i$ be the prime of y and $S = B_i \smallsetminus \mathfrak{p}$. A point of $f^{-1}(y)$ is a prime $\mathfrak{q} \subset A_i$ whose contraction to B_i is $\mathfrak{p}$. Such primes correspond bijectively to the primes of $R := S^{-1}A_i/\mathfrak{p}S^{-1}A_i = A_i \otimes_{B_i} k(y)$ (primes of $S^{-1}A_i$ are the primes of A_i not meeting the image of S, i.e. with contraction $\subseteq \mathfrak{p}$, and those containing $\mathfrak{p}S^{-1}A_i$ have contraction $\supseteq \mathfrak{p}$; cf. Ex. 3.10(a)). Now R is a finite-dimensional $k(y)$-algebra, say of dimension n. Its primes are maximal (it is artinian), and if $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ are distinct maximal ideals, the Chinese remainder theorem gives a surjection $R \to \prod R/\mathfrak{m}_i$ onto a $k(y)$-vector space of dimension $\geq r$, so $r \leq n$. Hence $f^{-1}(y)$ is finite.

(b) Let $Z \subseteq X$ be closed. A subset of Y is closed iff its intersection with every V_i is closed in V_i, and $f(Z) \cap V_i = f(Z \cap \operatorname{Spec} A_i)$. Write $Z \cap \operatorname{Spec} A_i = V(\mathfrak{a})$ for an ideal $\mathfrak{a} \subseteq A_i$, and let $\mathfrak{b} = \mathfrak{a} \cap B_i$. The map $B_i/\mathfrak{b} \to A_i/\mathfrak{a}$ is injective and integral (since A_i is a finite B_i-module), so by the lying-over theorem $\operatorname{Spec}(A_i/\mathfrak{a}) \to \operatorname{Spec}(B_i/\mathfrak{b})$ is surjective. Therefore $f(V(\mathfrak{a})) = V(\mathfrak{b})$ is closed in V_i. So $f(Z)$ is closed.

(c) Let k be a field, $Y = \operatorname{Spec} k[t] = \mathbf{A}^1_k$, and $X = \operatorname{Spec} k[t, t^{-1}] \sqcup \operatorname{Spec} k$, with f the open immersion $\operatorname{Spec} k[t, t^{-1}] = D(t) \hookrightarrow Y$ on the first summand and the morphism $\operatorname{Spec} k \to Y$ onto the closed point $t = 0$ (corresponding to $k[t] \to k$, $t \mapsto 0$) on the second. Then f is surjective. It is quasi-finite: the fibre over a point $\mathfrak{p} \neq (t)$ is the single point of $D(t)$, and the fibre over (t) is the single point of $\operatorname{Spec} k$. It is of finite type: $f^{-1}(Y) = X$ is covered by two affines $\operatorname{Spec} k[t, t^{-1}]$ and $\operatorname{Spec} k$ whose rings are finitely generated $k[t]$-algebras (this is the definition of finite type, using the cover $\{Y\}$ of Y). But f is not finite: the first summand is a closed subset of X (its complement is open), and its image $D(t)$ is not closed in Y, contradicting (b). □

Exercise 3.6

Problem. Let X be an integral scheme. Show that the local ring $\mathcal{O}_\xi$ of the generic point ξ of X is a field. It is called the *function field* of X, and is denoted $K(X)$. Show also that if $U = \operatorname{Spec} A$ is any open affine subset of X, then $K(X)$ is isomorphic to the quotient field of A.

Solution. By Prop. 3.1, X is irreducible, so it has a unique generic point ξ (Ex. 2.9). Let $U = \operatorname{Spec} A$ be a nonempty open affine subset. Since $X = \overline{\{\xi\}}$, a nonempty open set contains ξ (otherwise ξ would lie in the closed set $X \smallsetminus U$, which would then contain $\overline{\{\xi\}} = X$). So $\xi \in U$, and ξ is a point of $\operatorname{Spec} A$ whose closure in U is all of U. If ξ corresponds to the prime $\mathfrak{p} \subset A$, its closure is $V(\mathfrak{p})$, so $\mathfrak{p}$ is contained in every prime of A, hence in the nilradical of A. Since X is integral, $A = \mathcal{O}_X(U)$ is a domain, so $\mathfrak{p} = (0)$. Therefore $\mathcal{O}_\xi = \mathcal{O}_{U,\xi} = A_{(0)} = \operatorname{Frac}(A)$, a field, and $K(X) \cong \operatorname{Frac} A$ for every nonempty open affine $U = \operatorname{Spec} A$. □

Exercise 3.7

Problem. A morphism $f : X \to Y$, with Y irreducible, is *generically finite* if $f^{-1}(\eta)$ is a finite set, where η is the generic point of Y. A morphism $f : X \to Y$ is *dominant* if $f(X)$ is dense in Y. Now let $f : X \to Y$ be a dominant, generically finite morphism of finite type of integral schemes. Show that there is an open dense subset $U \subseteq Y$ such that the induced morphism $f^{-1}(U) \to U$ is finite. [*Hint*: First show that the function field of X is a finite field extension of the function field of Y.]

Solution. Let ξ, η be the generic points of X, Y, $L = K(X)$, $K = K(Y)$.

Step 0: $f(\xi) = \eta$. Since f is continuous, $f(X) = f(\overline{\{\xi\}}) \subseteq \overline{\{f(\xi)\}}$, so $\overline{\{f(\xi)\}} \supseteq \overline{f(X)} = Y$, hence $f(\xi) = \eta$ by uniqueness of generic points.

Step 1: setup. Let $V = \operatorname{Spec} B \subseteq Y$ be a nonempty open affine subset (B a domain, $K = \operatorname{Frac} B$, $\eta \in V$). By Ex. 3.3(b), $f^{-1}(V) = U_1 \cup \cdots \cup U_N$ with $U_i = \operatorname{Spec} A_i$ and A_i finitely generated B-algebras; discard empty ones, so every U_i is nonempty and A_i is a domain with $\operatorname{Frac} A_i = L$ (Ex. 3.6); $N \geq 1$ since $\xi \in f^{-1}(V)$. The generic point $\xi \in U_i$ is the prime (0) of A_i and it maps to $\eta = (0)$ of B, so $B \to A_i$ is injective; regard $B \subseteq A_i \subseteq L$.

Step 2: L is a finite extension of K. With $S = B \smallsetminus \{0\}$, $f^{-1}(\eta) \cap U_1$ is the set of primes $\mathfrak{q}$ of A_1 with $\mathfrak{q} \cap B = 0$, i.e. the set of primes of $D := S^{-1}A_1$; by hypothesis it is finite. $D \subseteq L$ is a finitely generated K-algebra and a domain. By Noether normalization D is finite over a polynomial ring $K[t_1, \ldots, t_d] \subseteq D$. By lying over, $\operatorname{Spec} D \to \operatorname{Spec} K[t_1, \ldots, t_d]$ is surjective. If $d \geq 1$ the polynomial ring has infinitely many primes (the principal primes $(p(t_1))$ for the infinitely many irreducible $p \in K[t_1]$), contradicting finiteness. So $d = 0$, D is a domain finite-dimensional over K, hence a field. Since $A_1 \subseteq D \subseteq L = \operatorname{Frac} A_1$, $D = L$, so $[L : K] < \infty$.

Step 3: shrinking to make the A_i integral. Write $A_i = B[x_{i1}, \ldots, x_{in_i}] \subseteq L$. Every x_{ij} is algebraic over K; clearing denominators in a polynomial it satisfies, we get $0 \neq a_{ij} \in B$ with x_{ij} a root of a polynomial over B with leading coefficient a_{ij}. Let $a = \prod a_{ij} \in B \smallsetminus \{0\}$. Then each x_{ij} is integral over B_a, so $A_i' := (A_i)_a = B_a[x_{ij}]$ is a finite B_a-module and a domain. Moreover $U_i' := \operatorname{Spec} A_i' = U_i \cap f^{-1}(D(a))$, and $f^{-1}(D(a)) = \bigcup_i U_i'$.

Step 4: a lemma. If $R \subseteq S'$ are domains, S' integral over R, and $0 \neq r \in S'$, then $rS' \cap R \neq 0$. Take a monic equation $r^n + b_1 r^{n-1} + \cdots + b_n = 0$ ($b_i \in R$) of minimal degree n. Then $b_n \neq 0$ (otherwise cancel one factor r in the domain S'), and $b_n = -r(r^{n-1} + \cdots + b_{n-1}) \in rS' \cap R$.

Step 5: the pieces U_i' coincide over a smaller set. Let $2 \le i \le N$. The set $U_1' \cap U_i'$ is open in $U_1' = \operatorname{Spec} A_1'$ and contains ξ, so it contains a basic open set $D_{U_1'}(r)$ with $0 \ne r \in A_1'$. By Step 4 (with $R = B_a$, $S' = A_1'$) there is $0 \ne b' \in rA_1' \cap B_a$; writing $b' = b/a^m$ with $b \in B \smallsetminus \{0\}$ we have $b \in rA_1'$ and $D_{U_1'}(b) \subseteq D_{U_1'}(r) \subseteq U_1' \cap U_i'$. Note $D_{U_1'}(b) = U_1' \cap f^{-1}(D(b))$. So $U_1' \cap f^{-1}D(b) \subseteq U_i'$. Interchanging the roles of 1 and i we find $0 \ne b'' \in B$ with $U_i' \cap f^{-1}D(b'') \subseteq U_1'$. Put $c_i = bb''$. Then

$$U_1' \cap f^{-1}D(c_i) \subseteq U_i' \cap U_1' \cap f^{-1}D(c_i), \qquad U_i' \cap f^{-1}D(c_i) \subseteq U_1' \cap U_i' \cap f^{-1}D(c_i),$$

hence $U_1' \cap f^{-1}D(c_i) = U_i' \cap f^{-1}D(c_i)$.

Step 6. Let $c = a\,c_2 \cdots c_N \in B \smallsetminus \{0\}$ and $U = D(c) \subseteq V$. It is a nonempty open subset of the irreducible space Y, hence dense. Since $D(c) \subseteq D(a) \cap D(c_i)$, Step 5 gives $U_i' \cap f^{-1}(U) = U_1' \cap f^{-1}(U)$ for every i, so

$$f^{-1}(U) = \bigcup_i U_i' \cap f^{-1}(U) = U_1' \cap f^{-1}(D(c)) = \operatorname{Spec}(A_1')_c.$$

Now $(A_1')_c$ is a finite module over $(B_a)_c = B_c$ (a localization of the finite B_a-module A_1'), and $U = \operatorname{Spec} B_c$. So $f^{-1}(U) \to U$ is a finite morphism (take the cover of U by itself). □

Exercise 3.8

> **Problem.** *Normalization.* A scheme is *normal* if all of its local rings are integrally closed domains. Let X be an integral scheme. For each open affine subset $U = \operatorname{Spec} A$ of X, let $\tilde{A}$ be the integral closure of A in its quotient field, and let $\tilde{U} = \operatorname{Spec} \tilde{A}$. Show that one can glue the schemes $\tilde{U}$ to obtain a normal integral scheme $\tilde{X}$, called the *normalization* of X. Show also that there is a morphism $\tilde{X} \to X$, having the following universal property: for every normal integral scheme Z, and for every dominant morphism $f : Z \to X$, f factors uniquely through $\tilde{X}$. If X is of finite type over a field k, then the morphism $\tilde{X} \to X$ is a finite morphism. This generalizes (I, Ex. 3.17).

Solution. Let $K = K(X)$ and ξ the generic point. For a nonempty open affine $U = \operatorname{Spec} A$, A is a domain with $\operatorname{Frac} A = K$ (Ex. 3.6); let $\tilde{A}_U \subseteq K$ be its integral closure, $\tilde{U} = \operatorname{Spec} \tilde{A}_U$, $\pi_U : \tilde{U} \to U$ induced by $A \subseteq \tilde{A}_U$. This map is surjective (lying over) and $\tilde{U}$ is a normal integral scheme: $\tilde{A}_U$ is a domain, integrally closed in its quotient field K, and so are its localizations. We use that integral closure commutes with localization: for $f \in A$, $(\tilde{A_f}) = (\tilde{A})_f$ (Atiyah–Macdonald, 5.12).

Step 1: compatibility with open affine subsets. For a nonempty open affine $W = \operatorname{Spec} A_W \subseteq X$ write $\tilde{W} = \operatorname{Spec} \tilde{A}_W$ and $\pi_W : \tilde{W} \to W$. Every $A_W = \mathcal{O}_X(W)$ is a subring of $K = \mathcal{O}_{X,\xi}$ (the map to the stalk at ξ is injective), and for $W' \subseteq W$ the restriction map $A_W \to A_{W'}$ is the inclusion of subrings of K, since both commute with the maps to $\mathcal{O}_{X,\xi}$. Hence $\tilde{A}_W \subseteq \tilde{A}_{W'}$ (an element integral over A_W is integral over $A_{W'}$), and we put $\rho_{W'W} = \operatorname{Spec}(\tilde{A}_W \subseteq \tilde{A}_{W'}) : \tilde{W}' \to \tilde{W}$; it lies over the inclusion $W' \to W$, and clearly $\rho_{W''W'}$, $\rho_{W'W}$ compose to $\rho_{W''W}$ and $\rho_{WW} = \mathrm{id}$. *Claim: $\rho_{W'W}$ is an open immersion with image $\pi_W^{-1}(W')$.* Let $y \in W'$ and choose $f \in A_W$ with $y \in D_W(f) \subseteq W'$. By (A1), $D_W(f) = D_{W'}(\bar{f})$ and $(A_W)_f = (A_{W'})_{\bar{f}}$, and by (Atiyah–Macdonald, 5.12) both $(\tilde{A}_W)_f$ and $(\tilde{A}_{W'})_{\bar{f}}$ are the integral closure of this ring in K, so they are the same subring of K. The restriction of $\rho_{W'W}$ to the open set $\pi_{W'}^{-1}(D_{W'}(\bar{f})) = \operatorname{Spec}(\tilde{A}_{W'})_{\bar{f}}$ is

the spectrum of $\tilde{A}_W \to (\tilde{A}_W)_f = (\tilde{A}_{W'})_{\bar{f}}$, i.e. an isomorphism onto the open set $\mathrm{Spec}(\tilde{A}_W)_f = \pi_W^{-1}(D_W(f))$. These sets $\pi_{W'}^{-1}(D_{W'}(\bar{f}))$ cover $\tilde{W}'$, so $\rho_{W'W}$ is a local isomorphism onto open subsets covering $\pi_W^{-1}(W')$; it is injective, because two points with the same image lie over the same $y \in W'$ and so in the same piece $\pi_{W'}^{-1}(D_{W'}(\bar{f}))$ with $y \in D_{W'}(\bar{f})$, on which $\rho_{W'W}$ is injective. This proves the claim.

Step 2: gluing. Index by the family of *all* nonempty open affine subsets $U \subseteq X$, and put $\tilde{U}_{UU'} := \pi_U^{-1}(U \cap U') \subseteq \tilde{U}$. For a nonempty open affine $W \subseteq U \cap U'$, Step 1 gives $\pi_U^{-1}(W) = \rho_{WU}(\tilde{W})$, and we define $\varphi_W := \rho_{WU'} \circ \rho_{WU}^{-1} : \pi_U^{-1}(W) \to \pi_{U'}^{-1}(W)$, an isomorphism over W. If $W'' \subseteq W$ then $\varphi_{W''} = \varphi_W$ on $\pi_U^{-1}(W'')$, because $\rho_{W''U} = \rho_{WU}\rho_{W''W}$ and $\rho_{W''U'} = \rho_{WU'}\rho_{W''W}$. If $W_1, W_2 \subseteq U \cap U'$ are open affine, then $W_1 \cap W_2$ is covered by open affines $W_3 \subseteq W_1 \cap W_2$ and $\varphi_{W_1}, \varphi_{W_2}$ both agree with φ_{W_3} on $\pi_U^{-1}(W_3)$; so they agree on $\pi_U^{-1}(W_1 \cap W_2)$. Since the $W \subseteq U \cap U'$ cover $U \cap U'$, the φ_W glue to an isomorphism $\varphi_{UU'} : \tilde{U}_{UU'} \to \tilde{U}_{U'U}$ with $\pi_{U'}\varphi_{UU'} = \pi_U$, $\varphi_{U'U} = \varphi_{UU'}^{-1}$ (same formula) and $\varphi_{UU} = \mathrm{id}$. Cocycle condition: $\tilde{U}_{UU'} \cap \tilde{U}_{UU''} = \pi_U^{-1}(U \cap U' \cap U'')$ is covered by the sets $\pi_U^{-1}(W)$ with $W \subseteq U \cap U' \cap U''$ open affine, and $\varphi_{UU'}$ maps this set onto $\pi_{U'}^{-1}(U \cap U' \cap U'') = \tilde{U}_{U'U} \cap \tilde{U}_{U'U''}$ (as $\pi_{U'}\varphi_{UU'} = \pi_U$); on $\pi_U^{-1}(W)$ we have $\varphi_{U'U''}\varphi_{UU'} = \rho_{WU''}\rho_{WU'}^{-1}\rho_{WU'}\rho_{WU}^{-1} = \varphi_{UU''}$. By Ex. 2.12 we obtain a scheme $\tilde{X}$, containing every $\tilde{U}$ as an open subscheme, and a morphism $\pi : \tilde{X} \to X$ restricting to π_U on $\tilde{U}$. Moreover $\pi^{-1}(U) = \tilde{U}$ for every nonempty open affine U: a point of $\pi^{-1}(U)$ lies in some $\tilde{U}'$ over a point of $U \cap U'$, i.e. in $\tilde{U}_{U'U}$, which is identified with $\tilde{U}_{UU'} \subseteq \tilde{U}$.

$\tilde{X}$ is normal because its local rings are local rings of the $\tilde{U}_i$. It is reduced because its local rings are domains. It is irreducible: it is covered by the irreducible open sets $\tilde{U}_i$, and any two meet because π_i is surjective and $U_i \cap U_j \neq \emptyset$ (X irreducible). So $\tilde{X}$ is integral by Prop. 3.1.

Step 3: uniqueness lemma. Let T be an integral scheme with generic point ζ, let $f : T \to X$ be a morphism with $f(\zeta) = \xi$ (e.g. f dominant, Step 0 of Ex. 3.7), and let $g, g' : T \to \tilde{X}$ with $\pi g = \pi g' = f$. Then $g = g'$. Indeed, for any nonempty open affine $U = \mathrm{Spec}\, A \subseteq X$ and nonempty open affine $V = \mathrm{Spec}\, B \subseteq f^{-1}(U)$, $A \to B$ is injective (it is the restriction of $K(X) \to K(T)$, or: the prime (0) of B lies over (0) of A). Both $g|_V$ and $g'|_V$ map into $\pi^{-1}(U) = \mathrm{Spec}\, \tilde{A}_U$, so they correspond to ring homomorphisms $\tilde{A}_U \to B$ extending $A \to B$. Such an extension is unique: if $s = a/a' \in \tilde{A}_U$ ($a, a' \in A$, $a' \neq 0$) then $\varphi(s)\varphi(a') = \varphi(a)$ and $\varphi(a') \neq 0$ in the domain B. So $g|_V = g'|_V$. Such V cover T, hence $g = g'$.

Step 4: universal property. Let Z be a normal integral scheme and $f : Z \to X$ dominant. Let $U = \mathrm{Spec}\, A \subseteq X$ and $V = \mathrm{Spec}\, B \subseteq f^{-1}(U)$ be nonempty open affines; $A \subseteq B$ as in Step 3, and B is a domain with $\mathrm{Frac}\, B = K(Z) \supseteq K = \mathrm{Frac}\, A$. The domain B is integrally closed: if $s \in \mathrm{Frac}\, B$ is integral over B, it is integral over every localization $B_\mathfrak{q}$ (integrally closed since Z is normal), so $s \in \bigcap_\mathfrak{q} B_\mathfrak{q} = B$. An element of $\tilde{A}_U \subseteq K \subseteq K(Z)$ is integral over A, hence over B, so lies in B. Thus $A \to B$ extends to $\tilde{A}_U \to B$, i.e. we get $g_V : V \to \tilde{U} = \pi^{-1}(U) \subseteq \tilde{X}$ with $\pi g_V = f|_V$. For two such V, V', the morphisms g_V and $g_{V'}$ agree on $V \cap V'$ by Step 3 applied to the integral scheme $V \cap V'$ and the morphism $f|_{V \cap V'}$. So the g_V glue to $g : Z \to \tilde{X}$ with $\pi g = f$, and g is unique by Step 3.

Step 5: finiteness. Let X be of finite type over k. By Ex. 3.3(c), for every open affine $U = \mathrm{Spec}\, A \subseteq X$, A is a finitely generated k-algebra and a domain, so its integral closure $\tilde{A}$ in $\mathrm{Frac}\, A$ is a finite A-module (finiteness of integral closure for finitely generated domains over a field, as in (I, Ex. 3.17); in characteristic p one needs the general form of this theorem, e.g. Eisenbud, *Commutative Algebra*, Cor. 13.13). Hence $\pi^{-1}(U) = \mathrm{Spec}\, \tilde{A}_U$ with $\tilde{A}_U$ a finite A-module, for all

such U; so π is finite. $\square$

Exercise 3.9

Problem. *The Topological Space of a Product.* Recall that in the category of varieties, the Zariski topology on the product of two varieties is not equal to the product topology (I, Ex. 1.4). Now we see that in the category of schemes, the underlying point set of a product of schemes is not even the product set.

(a) Let k be a field, and let $\mathbf{A}^1_k = \operatorname{Spec} k[x]$ be the affine line over k. Show that $\mathbf{A}^1_k \times_{\operatorname{Spec} k} \mathbf{A}^1_k \cong \mathbf{A}^2_k$, and show that the underlying point set of the product is not the product of the underlying sets of the factors (even if k is algebraically closed).

(b) Let k be a field, let s and t be indeterminates over k. Then $\operatorname{Spec} k(s)$, $\operatorname{Spec} k(t)$, and $\operatorname{Spec} k$ are all one-point spaces. Describe the product scheme $\operatorname{Spec} k(s) \times_{\operatorname{Spec} k} \operatorname{Spec} k(t)$.

Solution. (a) By Thm. 3.3, Step 1, $\mathbf{A}^1_k \times_k \mathbf{A}^1_k = \operatorname{Spec}(k[x] \otimes_k k[y]) = \operatorname{Spec} k[x,y] = \mathbf{A}^2_k$, with projections p_1, p_2 induced by $k[x] \to k[x,y]$ and $k[y] \to k[x,y]$. Consider the two points $\mathfrak{p}_0 = (0)$ and $\mathfrak{p}_1 = (x-y)$ of $\mathbf{A}^2_k$; they are distinct primes. We have $\mathfrak{p}_0 \cap k[x] = 0$, and also $\mathfrak{p}_1 \cap k[x] = 0$: a nonzero polynomial $q(x)$ divisible by $x-y$ in $k[x,y]$ would vanish after substituting $y = x$, which gives $q(x) = 0$; similarly $\mathfrak{p}_1 \cap k[y] = 0$. So $p_1(\mathfrak{p}_0) = p_1(\mathfrak{p}_1)$ and $p_2(\mathfrak{p}_0) = p_2(\mathfrak{p}_1)$, i.e. the canonical map

$$\mathbf{A}^2_k \to \mathbf{A}^1_k \times \mathbf{A}^1_k \text{ (product of sets)}, \quad z \mapsto (p_1(z), p_2(z))$$

is not injective, a fortiori not bijective: both $\mathfrak{p}_0$ and $\mathfrak{p}_1$ (the generic points of the plane and of the diagonal) map to the pair (generic point, generic point). This holds for every field k, in particular for k algebraically closed.

(b) By Thm. 3.3, Step 1, the product is $\operatorname{Spec}(k(s) \otimes_k k(t))$. Since $k(s) = T_1^{-1}k[s]$ and $k(t) = T_2^{-1}k[t]$ with $T_1 = k[s] \smallsetminus 0$, $T_2 = k[t] \smallsetminus 0$, we have

$$k(s) \otimes_k k(t) = T^{-1}k[s,t], \qquad T = \{f(s)g(t) : f \neq 0, g \neq 0\}$$

(a multiplicative set). The primes of this ring are the primes $\mathfrak{p}$ of $k[s,t]$ with $\mathfrak{p} \cap T = \emptyset$, i.e. (as $\mathfrak{p}$ is prime) with $\mathfrak{p} \cap k[s] = 0$ and $\mathfrak{p} \cap k[t] = 0$. In the UFD $k[s,t]$ the primes are (0), the principal primes (h) with h irreducible, and the maximal ideals. A maximal ideal $\mathfrak{m}$ meets $k[s]$ nontrivially: $k[s,t]/\mathfrak{m}$ is a field finitely generated as a k-algebra, hence finite over k (Zariski's lemma), so the image of s is algebraic over k. So no maximal ideal occurs. A principal prime (h) meets $k[s]$ iff h divides a nonzero polynomial $f(s)$, iff (up to a unit) $h \in k[s]$ (all divisors of $f(s)$ lie in $k[s]$); similarly for $k[t]$. Hence:

$\operatorname{Spec} k(s) \times_{\operatorname{Spec} k} \operatorname{Spec} k(t)$ *is the spectrum of the one-dimensional noetherian domain* $T^{-1}k[s,t]$. *Its points are the generic point* (0) *and the points* (h), $h \in k[s,t]$ *irreducible and involving both* s *and* t *(i.e.* $h \notin k[s] \cup k[t]$ *up to units); the residue field at* (h) *is* $\operatorname{Frac}(k[s,t]/(h))$. For example the point $(s-t)$ lies in it. In particular the product is an infinite scheme, though each factor is a single point. $\square$

Exercise 3.10

Problem. *Fibres of a Morphism.*

(a) If $f : X \to Y$ is a morphism, and $y \in Y$ a point, show that $\mathrm{sp}(X_y)$ is homeomorphic to $f^{-1}(y)$ with the induced topology.

(b) Let $X = \operatorname{Spec} k[s,t]/(s-t^2)$, let $Y = \operatorname{Spec} k[s]$, and let $f : X \to Y$ be the morphism defined by sending $s \mapsto s$. If $y \in Y$ is the point $a \in k$ with $a \neq 0$, show that the fibre X_y consists of two points, with residue field k. If $y \in Y$ corresponds to $0 \in k$, show that the fibre X_y is a nonreduced one-point scheme. If η is the generic point of Y, show that X_η is a one-point scheme, whose residue field is an extension of degree 2 of the residue field of η. (Assume k algebraically closed.)

Solution. (a) Recall $X_y = X \times_Y \operatorname{Spec} k(y)$ and let $p : X_y \to X$ be the projection. Choose an open affine $V = \operatorname{Spec} B \ni y$, with y corresponding to the prime $\mathfrak{p} \subset B$. The morphism $\operatorname{Spec} k(y) \to Y$ factors through V, so any scheme T with maps to X and to $\operatorname{Spec} k(y)$ over Y maps into $f^{-1}(V)$; hence $X_y = f^{-1}(V) \times_V \operatorname{Spec} k(y)$, and p lands in $f^{-1}(V)$. Cover $f^{-1}(V)$ by open affines $U_j = \operatorname{Spec} A_j$. By Thm. 3.3 (Steps 1 and 4), $p^{-1}(U_j) = U_j \times_V \operatorname{Spec} k(y) = \operatorname{Spec}(A_j \otimes_B k(y))$. With $S = B \smallsetminus \mathfrak{p}$ and $\varphi : B \to A_j$ we have $k(y) = S^{-1}B/\mathfrak{p}S^{-1}B$, so

$$R_j := A_j \otimes_B k(y) = S^{-1}A_j/\mathfrak{p}S^{-1}A_j.$$

Primes of $S^{-1}A_j$ correspond to primes $\mathfrak{q}$ of A_j with $\varphi^{-1}(\mathfrak{q}) \cap S = \emptyset$, i.e. $\varphi^{-1}(\mathfrak{q}) \subseteq \mathfrak{p}$, and those containing $\mathfrak{p}S^{-1}A_j$ are the ones with $\varphi^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}$. So $\operatorname{Spec} R_j \to \operatorname{Spec} A_j$ is a bijection onto $\{\mathfrak{q} : \varphi^{-1}(\mathfrak{q}) = \mathfrak{p}\} = f^{-1}(y) \cap U_j$. It is a homeomorphism onto this subset with the induced topology: it is the composite of the closed embedding $\operatorname{Spec} R_j \hookrightarrow \operatorname{Spec} S^{-1}A_j$ and the map $\operatorname{Spec} S^{-1}A_j \to \operatorname{Spec} A_j$, which is a homeomorphism onto its image with the subspace topology (the closed sets $V(S^{-1}\mathfrak{a})$ are the traces of the closed sets $V(\mathfrak{a})$).

Since the sets $p^{-1}(U_j)$ are the preimages under p of the open sets U_j, p is injective on all of X_y (two points with the same image lie in the same $p^{-1}(U_j)$), has image $f^{-1}(y)$, and is a homeomorphism from each open set $p^{-1}(U_j)$ onto the open subset $f^{-1}(y) \cap U_j$ of $f^{-1}(y)$. These open subsets cover $f^{-1}(y)$, so p is a homeomorphism of X_y onto $f^{-1}(y)$.

(b) The ring map $k[s] \to k[s,t]/(s-t^2)$ identifies the target with $k[t]$, $s \mapsto t^2$; so $X = \operatorname{Spec} k[t]$ and f is the map $t \mapsto t^2$.

Caveat. For the statement "two points" we need $\operatorname{char} k \neq 2$, which the printed exercise does not state (it only assumes k algebraically closed); I assume it. (In characteristic 2 the fibre over $a \neq 0$ is the one-point nonreduced scheme $\operatorname{Spec} k[t]/(t-\alpha)^2$, $\alpha^2 = a$.)

Let $y = a \neq 0$, i.e. the maximal ideal $(s-a)$, with $k(y) = k[s]/(s-a) = k$. Then

$$X_y = \operatorname{Spec}\big(k[t] \otimes_{k[s]} k[s]/(s-a)\big) = \operatorname{Spec} k[t]/(t^2-a) = \operatorname{Spec} k[t]/((t-\alpha)(t+\alpha)),$$

where $\alpha^2 = a$, $\alpha \neq 0$, so $\alpha \neq -\alpha$. By the Chinese remainder theorem, $k[t]/(t^2-a) \cong k \times k$, so X_y consists of two points (reduced), each with residue field k.

For y corresponding to 0, $X_y = \operatorname{Spec} k[t]/(t^2)$, a single point (the prime (t)) with $\mathcal{O} = k[t]/(t^2)$ nonreduced ($t \neq 0$, $t^2 = 0$).

For the generic point $\eta = (0)$ of Y, $k(\eta) = k(s)$ and

$$X_\eta = \operatorname{Spec}\big(k[t] \otimes_{k[s]} k(s)\big) = \operatorname{Spec}\ k(s)[t]/(t^2-s).$$

The polynomial $t^2 - s$ is irreducible over $k(s)$: otherwise $s = (u/v)^2$ with $u, v \in k[s]$, $v \neq 0$, i.e. $sv^2 = u^2$, and the left side has odd degree while the right side has even degree. Hence

$k(s)[t]/(t^2 - s)$ is a field, of degree 2 over $k(s)$, and X_η is the one-point scheme Spec of this field. □

Exercise 3.11

Problem. *Closed Subschemes.*

(a) Closed immersions are stable under base extension: if $f : Y \to X$ is a closed immersion, and if $X' \to X$ is any morphism, then $f' : Y \times_X X' \to X'$ is also a closed immersion.

(b) If Y is a closed subscheme of an affine scheme $X = \operatorname{Spec} A$, then Y is also affine, and in fact Y is the closed subscheme determined by a suitable ideal $\mathfrak{a} \subseteq A$ as the image of the closed immersion $\operatorname{Spec} A/\mathfrak{a} \to \operatorname{Spec} A$. [*Hints*: First show that Y can be covered by a finite number of open affine subsets of the form $D(f_i) \cap Y$, with $f_i \in A$. By adding some more f_i with $D(f_i) \cap Y = \emptyset$, if necessary, show that we may assume that the $D(f_i)$ cover X. Next show that $f_1, \ldots, f_r$ generate the unit ideal of A. Then use (Ex. 2.17b) to show that Y is affine, and (Ex. 2.18d) to show that Y comes from an ideal $\mathfrak{a} \subseteq A$.] *Note*: We will give another proof of this result using sheaves of ideals later (5.10).

(c) Let Y be a closed subset of a scheme X, and give Y the reduced induced subscheme structure. If Y' is any other closed subscheme of X with the same underlying topological space, show that the closed immersion $Y \to X$ factors through Y'. We express this property by saying that the reduced induced structure is the smallest subscheme structure on a closed subset.

(d) Let $f : Z \to X$ be a morphism. Then there is a unique closed subscheme Y of X with the following property: the morphism f factors through Y, and if Y' is any other closed subscheme of X through which f factors, then $Y \to X$ factors through Y' also. We call Y the *scheme-theoretic image* of f. If Z is a reduced scheme, then Y is just the reduced induced structure on the closure of the image $f(Z)$.

(Part (b) is starred in the book.)

Solution. I prove (b) first, since (a) uses it (the proof of (b) does not use (a)). Throughout, a closed subscheme is an equivalence class of closed immersions $i : Y \to X$, and I freely identify Y with the closed subset $i(Y)$.

(b) Let $i : Y \to X = \operatorname{Spec} A$ be a closed immersion; put $\varphi = \alpha(i) : A \to C := \Gamma(Y, \mathcal{O}_Y)$ (Ex. 2.4).

Step 1: Y is covered by finitely many Y_{f_i} that are affine and $D(f_i)$ cover X. Let $y \in Y$ and let $\operatorname{Spec} C' \subseteq Y$ be an open affine neighbourhood of y. Since i is a homeomorphism onto the closed set $i(Y)$, there is an open $O \subseteq X$ with $i(\operatorname{Spec} C') = i(Y) \cap O$; choose $f \in A$ with $i(y) \in D(f) \subseteq O$. Then $i^{-1}(D(f)) \subseteq i^{-1}(O) = \operatorname{Spec} C'$, and $i^{-1}(D(f))$ is the open subset $D_{\operatorname{Spec} C'}(\bar{f})$ where $\bar{f} = i^\#(f)|_{\operatorname{Spec} C'}$ (the point z lies in it iff $f \notin \mathfrak{p}_{i(z)}$ iff the image of f in $\mathcal{O}_{Y,z}$ is a unit, by locality of $i_z^\#$). So $i^{-1}(D(f)) = \operatorname{Spec} C'_{\bar{f}}$ is affine. As Y is homeomorphic to a closed subset of the quasi-compact space X, it is quasi-compact, so finitely many of these cover Y: $Y = \bigcup_{l \le n} i^{-1} D(f_l)$. The closed set $X \smallsetminus \bigcup_{l \le n} D(f_l)$ is quasi-compact and disjoint from $i(Y)$, so it is covered by finitely many $D(g_m)$ with $D(g_m) \cap i(Y) = \emptyset$ (such sets form a base of the open set $X \smallsetminus i(Y)$). Thus, renaming, we have $f_1, \ldots, f_r \in A$ with $\bigcup D(f_j) = X$ (i.e. $(f_1, \ldots, f_r) = A$) and each $i^{-1}(D(f_j))$ affine (possibly empty).

Step 2: Y is affine. The elements $s_j = \varphi(f_j) \in C$ generate the unit ideal of C, and $Y_{s_j} = i^{-1}(D(f_j))$ is affine. By Ex. 2.17(b), Y is affine, $Y = \operatorname{Spec} C$, and $i = \operatorname{Spec} \varphi$ (Ex. 2.4).

Step 3: φ is surjective. Let $\mathfrak{p} \in \operatorname{Spec} A$. The stalk of $i_*\mathcal{O}_Y$ at $\mathfrak{p}$ is $\varinjlim_{g \notin \mathfrak{p}} \mathcal{O}_Y(i^{-1}D(g)) = \varinjlim_{g \notin \mathfrak{p}} C_{\varphi(g)} = C_{\mathfrak{p}} := C \otimes_A A_{\mathfrak{p}}$. Since $i^{\#} : \mathcal{O}_X \to i_*\mathcal{O}_Y$ is a surjective map of sheaves, it is surjective on stalks, i.e. $A_{\mathfrak{p}} \to C_{\mathfrak{p}}$ is surjective for every prime $\mathfrak{p}$. Thus the A-module $M = C/\varphi(A)$ satisfies $M_{\mathfrak{p}} = 0$ for all $\mathfrak{p}$, so $M = 0$. Hence φ is surjective, $C \cong A/\mathfrak{a}$ with $\mathfrak{a} = \operatorname{Ker} \varphi$, and i is the closed immersion $\operatorname{Spec} A/\mathfrak{a} \to \operatorname{Spec} A$.

(a) Let $f : Y \to X$ be a closed immersion and $g : X' \to X$ any morphism, $f' : Y' = Y \times_X X' \to X'$. Being a closed immersion is local on the target X' (the image of f' is closed iff its trace on each member of an open cover is closed, f' is injective and a homeomorphism onto its image iff it is so over each member, surjectivity of $\mathcal{O}_{X'} \to f'_*\mathcal{O}_{Y'}$ can be checked on stalks). Cover X' by open affines $W = \operatorname{Spec} A'$ each mapped by g into an open affine $U = \operatorname{Spec} A \subseteq X$ (cover X by affines and then each $g^{-1}(U)$ by affines). By (b), applied to the closed immersion $f^{-1}(U) \to U$, we have $f^{-1}(U) = \operatorname{Spec} A/\mathfrak{a}$ for an ideal $\mathfrak{a}$. As $W \to X$ factors through U, $f'^{-1}(W) = Y \times_X W = f^{-1}(U) \times_U W = \operatorname{Spec}(A/\mathfrak{a} \otimes_A A') = \operatorname{Spec} A'/\mathfrak{a}A'$, and $f'| : \operatorname{Spec} A'/\mathfrak{a}A' \to \operatorname{Spec} A'$ is induced by the surjection $A' \to A'/\mathfrak{a}A'$, a closed immersion (Example 3.2.3). Hence f' is a closed immersion.

(c) Let $i' : Y' \to X$ be a closed subscheme with $i'(Y') = Y$ as sets. Let $\mathcal{J}' = \operatorname{Ker}(\mathcal{O}_X \to i'_*\mathcal{O}_{Y'})$ and $\mathcal{J} = \operatorname{Ker}(\mathcal{O}_X \to i_*\mathcal{O}_Y)$ be the ideal sheaves; since i, i' are homeomorphisms onto Y and the maps $\mathcal{O}_X \to i_*\mathcal{O}$ are surjective, we may identify $\mathcal{O}_{Y'} = \mathcal{O}_X/\mathcal{J}'|_Y$ and $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{J}|_Y$ as sheaves on the space Y. Let $U = \operatorname{Spec} A$ be an open affine of X. By (b), $\mathcal{J}'(U) = \mathfrak{a}'$ where $Y' \cap U = \operatorname{Spec} A/\mathfrak{a}'$, so $V(\mathfrak{a}') = Y \cap U$. By definition of the reduced induced structure (Example 3.2.6), $\mathcal{J}(U) = \mathfrak{b} := \bigcap_{\mathfrak{p} \in Y \cap U} \mathfrak{p}$. Then $\mathfrak{a}' \subseteq \sqrt{\mathfrak{a}'} = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}'} \mathfrak{p} = \mathfrak{b}$. Thus $\mathcal{J}'(U) \subseteq \mathcal{J}(U)$ on all open affines, compatibly with restrictions; as the open affines form a base, $\mathcal{J}' \subseteq \mathcal{J}$. So the quotient maps give a surjective morphism of sheaves of rings $g^{\#} : \mathcal{O}_X/\mathcal{J}' \to \mathcal{O}_X/\mathcal{J}$ on Y. On stalks it is a surjection of local rings, hence a local homomorphism. So $(\mathrm{id}_Y, g^{\#})$ is a morphism of locally ringed spaces, i.e. of schemes, $g : Y \to Y'$, and $i' \circ g = i$ (it is the identity on spaces and the composite $\mathcal{O}_X \to \mathcal{O}_X/\mathcal{J}' \to \mathcal{O}_X/\mathcal{J}$ is the quotient map).

(d) *This part uses results of Sect. 5 (closed subschemes $\leftrightarrow$ quasi-coherent ideal sheaves, Cor. 5.10, and Prop. 5.7), as the note in the exercise indicates.* Recall that closed subschemes of X correspond bijectively to quasi-coherent ideal sheaves $\mathcal{I} \subseteq \mathcal{O}_X$, via $\mathcal{I} \mapsto Y_{\mathcal{I}} = (V(\mathcal{I}), \mathcal{O}_X/\mathcal{I})$ with $V(\mathcal{I}) = \operatorname{Supp}(\mathcal{O}_X/\mathcal{I})$. If $\mathcal{I}' \subseteq \mathcal{I}$, the quotient map $\mathcal{O}_X/\mathcal{I}' \to \mathcal{O}_X/\mathcal{I}$ gives a factorization of $Y_{\mathcal{I}} \to X$ through $Y_{\mathcal{I}'}$ (as in (c)); conversely, a factorization $g : Y_{\mathcal{I}} \to Y_{\mathcal{I}'}$ gives $\mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}' \to g_*\mathcal{O}_{Y_{\mathcal{I}}}$ equal to the quotient map by $\mathcal{I}$, which kills $\mathcal{I}'$; so $\mathcal{I}' \subseteq \mathcal{I}$.

Claim: $f : Z \to X$ factors through $Y_{\mathcal{I}}$ iff $\mathcal{I} \subseteq \mathcal{K} := \operatorname{Ker}(f^{\#} : \mathcal{O}_X \to f_*\mathcal{O}_Z)$. ($\Rightarrow$) If $f = i \circ h$ then $f^{\#}$ factors through $i^{\#} : \mathcal{O}_X \to i_*\mathcal{O}_Y$ whose kernel is $\mathcal{I}$. ($\Leftarrow$) Suppose $\mathcal{I} \subseteq \mathcal{K}$. For $z \in Z$ put $x = f(z)$. The local homomorphism $f^{\#}_z : \mathcal{O}_{X,x} \to \mathcal{O}_{Z,z}$ kills $\mathcal{I}_x$. If $\mathcal{I}_x = \mathcal{O}_{X,x}$ it would kill 1, impossible; so $\mathcal{I}_x \neq \mathcal{O}_{X,x}$, i.e. $x \in \operatorname{Supp}(\mathcal{O}_X/\mathcal{I}) = V(\mathcal{I})$. So $f(Z) \subseteq V(\mathcal{I})$, defining a continuous $h : Z \to V(\mathcal{I}) =: Y$ with $i \circ h = f$ topologically. The map $f^{\#} : \mathcal{O}_X \to f_*\mathcal{O}_Z = i_*h_*\mathcal{O}_Z$ kills $\mathcal{I}$, so it factors through $\mathcal{O}_X/\mathcal{I} = i_*\mathcal{O}_Y$ and by adjunction gives $h^{\#} : \mathcal{O}_Y \to h_*\mathcal{O}_Z$ whose stalk maps are induced by the local maps $f^{\#}_z$, hence local. So $f = i \circ h$.

Construction. Let $\mathcal{I}_f \subseteq \mathcal{O}_X$ be the sum of all quasi-coherent ideal sheaves contained in $\mathcal{K}$. It is the image of the direct sum of these sheaves, hence quasi-coherent (direct sums and images of quasi-coherent sheaves are quasi-coherent, Prop. 5.7), and obviously the largest quasi-coherent ideal contained in $\mathcal{K}$ (if f is quasi-compact then $\mathcal{K}$ itself is quasi-coherent, but this is not needed). Let $Y = Y_{\mathcal{I}_f}$. By the claim f factors through Y. If f factors through a closed subscheme $Y' = Y_{\mathcal{I}'}$ then $\mathcal{I}' \subseteq \mathcal{K}$ is quasi-coherent, so $\mathcal{I}' \subseteq \mathcal{I}_f$ and $Y \to X$ factors through Y'. Uniqueness: if Y_1 has

the same universal property then $Y \to X$ factors through Y_1 and $Y_1 \to X$ through Y, so their ideals are comparable in both directions and equal.

The reduced case. Let Z be reduced, $W = \overline{f(Z)}$ with reduced induced structure, with ideal sheaf $\mathcal{I}_W$ (on an affine open $U = \operatorname{Spec} A$ it is $\bigcap_{\mathfrak{p}\in W\cap U} \mathfrak{p}$; it is quasi-coherent). If $s \in \mathcal{I}_W(U)$, then s vanishes at every point of $W \cap U$, so $f^\#(s) \in \mathcal{O}_Z(f^{-1}U)$ vanishes at every point of $f^{-1}U$ (the value of $f^\# s$ at z is the image of the value $s(f(z)) = 0$ in $k(f(z)) \to k(z)$); on an affine open $\operatorname{Spec} R \subseteq f^{-1}U$ with R reduced such an element lies in every prime, hence in the nilradical $= 0$. So $f^\#(s) = 0$ and $\mathcal{I}_W \subseteq \mathcal{K}$, hence $\mathcal{I}_W \subseteq \mathcal{I}_f$. In particular $V(\mathcal{I}_f) \subseteq W$. Conversely $V(\mathcal{I}_f)$ is closed and contains $f(Z)$ (the claim), hence contains W. So $V(\mathcal{I}_f) = W$ as sets. Finally any section s of $\mathcal{I}_f$ has germ in $\mathfrak{m}_x$ at all $x \in W$ (as $(\mathcal{I}_f)_x \neq \mathcal{O}_x$ there), i.e. vanishes on W, so $\mathcal{I}_f \subseteq \mathcal{I}_W$. Hence $\mathcal{I}_f = \mathcal{I}_W$ and $Y = W$ with the reduced induced structure. □

Exercise 3.12

Problem. *Closed Subschemes of* $\operatorname{Proj} S$.

(a) Let $\varphi : S \to T$ be a surjective homomorphism of graded rings, preserving degrees. Show that the open set U of (Ex. 2.14) is equal to $\operatorname{Proj} T$, and the morphism $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is a closed immersion.

(b) If $I \subseteq S$ is a homogeneous ideal, take $T = S/I$ and let Y be the closed subscheme of $X = \operatorname{Proj} S$ defined as the image of the closed immersion $\operatorname{Proj} S/I \to X$. Show that different homogeneous ideals can give rise to the same closed subscheme. For example, let d_0 be an integer, and let $I' = \bigoplus_{d\geq d_0} I_d$. Show that I and I' determine the same closed subscheme.

We will see later (5.16) that every closed subscheme of X comes from a homogeneous ideal $I \subseteq S$ (at least in the case where S is a polynomial ring over S_0).

Solution. Recall (Ex. 2.14(b)) $U = \{\mathfrak{p} \in \operatorname{Proj} T : \mathfrak{p} \not\supseteq \varphi(S_+)\}$ and $f : U \to \operatorname{Proj} S$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$; for a homogeneous $s \in S_+$ we have $f^{-1}(D_+(s)) = D_+(\varphi(s))$, and f restricted to it is the morphism $\operatorname{Spec} T_{(\varphi(s))} \to \operatorname{Spec} S_{(s)}$ induced by $S_{(s)} \to T_{(\varphi(s))}$.

(a) As φ is surjective and degree preserving, $\varphi(S_d) = T_d$ for all d (apply φ to the homogeneous components of a preimage), so $\varphi(S_+) = T_+$. A point of $\operatorname{Proj} T$ is a homogeneous prime not containing $T_+ = \varphi(S_+)$; so $U = \operatorname{Proj} T$. The sets $D_+(s)$, $s \in S_+$ homogeneous, cover $\operatorname{Proj} S$, and $f^{-1}(D_+(s)) = D_+(\varphi(s)) \to D_+(s)$ is induced by $S_{(s)} \to T_{(\varphi(s))}$. This ring map is surjective: an element of $T_{(\varphi(s))}$ is $t/\varphi(s)^n$ with $t \in T_{n \deg s}$, and $t = \varphi(s')$ with $s' \in S_{n \deg s}$, so it is the image of s'/s^n. By Example 3.2.3 the map over each $D_+(s)$ is a closed immersion, and being a closed immersion is local on the target, so f is a closed immersion.

(b) Let $d_0 \in \mathbf{Z}$ and $I' = \bigoplus_{d\geq d_0} I_d$; it is a homogeneous ideal contained in I (if $x \in I'_d$, $y \in S_e$, then $xy \in I_{d+e}$ with $d + e \geq d_0$ when $e \geq 0$). Let $\psi : S/I' \to S/I$ be the natural surjection of graded rings; by (a) it gives a closed immersion $\rho : \operatorname{Proj} S/I \to \operatorname{Proj} S/I'$, and the functoriality of the construction of Ex. 2.14 shows that the composite of ρ with the closed immersion $\operatorname{Proj} S/I' \to \operatorname{Proj} S$ is the closed immersion $\operatorname{Proj} S/I \to \operatorname{Proj} S$ defined by $S \to S/I$. So it suffices to show that ρ is an isomorphism; then the two closed immersions differ by an isomorphism, i.e. I and I' define the same closed subscheme. As in (a), over $D_+(\bar{s}) \subseteq \operatorname{Proj} S/I'$ ($s \in S_+$ homogeneous of degree $e > 0$, $\bar{s}$ its class) the morphism ρ is induced by $\psi_{(s)} : (S/I')_{(\bar{s})} \to (S/I)_{(\bar{s})}$, which

is surjective. It is injective: let x/s^n ($x \in S_{ne}$) map to 0 in $(S/I)_{(\bar{s})}$. Then $s^m x \in I$ for some $m \geq 0$, and enlarging m we may assume $(n+m)e \geq d_0$. Then $s^m x \in I_{(n+m)e} = I'_{(n+m)e}$, so $\bar{s}^m \bar{x} = 0$ in S/I' and $x/s^n = 0$ in $(S/I')_{(\bar{s})}$. So $\psi_{(s)}$ is an isomorphism for all such s; these $D_+(\bar{s})$ cover $\operatorname{Proj} S/I'$, so ρ is an isomorphism.

(The point of the example: $I \neq I'$ in general, e.g. $S = k[x_0, x_1]$, $I = (x_0)$, $d_0 = 2$: $I' = (x_0^2, x_0 x_1) \neq I$, since $x_0 \in I \smallsetminus I'$; yet both define the same closed subscheme, the point $x_0 = 0$ of $\mathbf{P}^1_k$.) □

Exercise 3.13

Problem. *Properties of Morphisms of Finite Type.*

(a) A closed immersion is a morphism of finite type.

(b) A quasi-compact open immersion (Ex. 3.2) is of finite type.

(c) A composition of two morphisms of finite type is of finite type.

(d) Morphisms of finite type are stable under base extension.

(e) If X and Y are schemes of finite type over S, then $X \times_S Y$ is of finite type over S.

(f) If $X \xrightarrow{f} Y \xrightarrow{g} Z$ are two morphisms, and if f is quasi-compact, and $g \circ f$ is of finite type, then f is of finite type.

(g) If $f : X \to Y$ is a morphism of finite type, and if Y is noetherian, then X is noetherian.

Solution. (a) Let $f : Y \to X$ be a closed immersion and $U = \operatorname{Spec} A \subseteq X$ an open affine. By Ex. 3.11(b), $f^{-1}(U) \to U$ is $\operatorname{Spec} A/\mathfrak{a} \to \operatorname{Spec} A$, so $f^{-1}(U)$ is a single affine open with ring $A/\mathfrak{a}$, a finitely generated A-algebra. Covering X by such U shows f is of finite type.

(b) Let $j : U \to X$ be a quasi-compact open immersion and $V = \operatorname{Spec} A \subseteq X$ open affine. By Ex. 3.2, $j^{-1}(V) = U \cap V$ is a quasi-compact open subset of $\operatorname{Spec} A$, so it is a finite union of basic open sets $D(g_l) = \operatorname{Spec} A_{g_l}$, and A_{g_l} is a finitely generated A-algebra. Hence j is of finite type.

(c) Let $f : X \to Y$ and $g : Y \to Z$ be of finite type, $W = \operatorname{Spec} C \subseteq Z$ open affine. By Ex. 3.3(b), $g^{-1}(W) = V_1 \cup \cdots \cup V_n$ with $V_i = \operatorname{Spec} B_i$, B_i finitely generated C-algebras, and each $f^{-1}(V_i) = \bigcup_{j \leq m_i} \operatorname{Spec} A_{ij}$ with A_{ij} finitely generated B_i-algebras. Then $(gf)^{-1}(W) = \bigcup_{i,j} \operatorname{Spec} A_{ij}$ is a finite union and each A_{ij} is a finitely generated C-algebra. By Ex. 3.3(b) (applied to the cover of Z by all open affines), gf is of finite type.

(d) Let $f : X \to Y$ be of finite type and $Y' \to Y$ any morphism; $X' = X \times_Y Y'$, $f' : X' \to Y'$. Cover Y' by open affines $W = \operatorname{Spec} C'$, each mapped into an open affine $V = \operatorname{Spec} B \subseteq Y$. By Ex. 3.3(b), $f^{-1}(V) = \bigcup_{i \leq n} \operatorname{Spec} A_i$ with A_i finitely generated B-algebras. As in Ex. 3.10(a), $f'^{-1}(W) = X \times_Y W = f^{-1}(V) \times_V W$, which by Thm. 3.3 (Step 4) is covered by the $\operatorname{Spec} A_i \times_V W = \operatorname{Spec}(A_i \otimes_B C')$, $i \leq n$; each $A_i \otimes_B C'$ is a finitely generated C'-algebra. So f' is of finite type.

(e) $X \times_S Y \to Y$ is the base extension of $X \to S$ by $Y \to S$, hence of finite type by (d); the composition with $Y \to S$ is of finite type by (c).

(f) We show f is locally of finite type; then f is of finite type by Ex. 3.3(a). Let $W = \operatorname{Spec} C \subseteq Z$ and $V = \operatorname{Spec} B \subseteq g^{-1}(W)$ be open affines; such V cover Y. Let $U = \operatorname{Spec} A$ be an open affine

subset of $f^{-1}(V)$; such U cover $f^{-1}(V)$. Since $U \subseteq (gf)^{-1}(W)$ and gf is of finite type, A is a finitely generated C-algebra by Ex. 3.3(c). The C-algebra structure of A factors as $C \to B \to A$, so A is a finitely generated B-algebra (the same generators). Hence $f^{-1}(V)$ is covered by spectra of finitely generated B-algebras, and f is locally of finite type.

(g) Cover Y by finitely many $V_i = \operatorname{Spec} B_i$, B_i noetherian. By Ex. 3.3(b), each $f^{-1}(V_i)$ is a finite union of $\operatorname{Spec} A_{ij}$ with A_{ij} a finitely generated B_i-algebra, hence noetherian by the Hilbert basis theorem. So X is a finite union of spectra of noetherian rings, i.e. noetherian. □

Exercise 3.14

> **Problem.** If X is a scheme of finite type over a field k, show that the closed points of X are dense. Show by example that this is not true for arbitrary schemes.

Solution. Let $U \subseteq X$ be a nonempty open subset; we show that U contains a closed point of X. This is equivalent to the density of the set of closed points. Choose a nonempty open affine $V = \operatorname{Spec} A \subseteq U$; A is a finitely generated k-algebra by Ex. 3.3(c) (applied to $X \to \operatorname{Spec} k$), and $A \neq 0$. Let $\mathfrak{m}$ be a maximal ideal of A and $x \in V$ the corresponding point. The field $k(x) = A/\mathfrak{m}$ is finitely generated as a k-algebra, hence finite over k (Zariski's lemma / Nullstellensatz).

x is closed in every open affine $W = \operatorname{Spec} A'$ containing it. Let $\mathfrak{q} \subset A'$ be the prime of x. Then $A'/\mathfrak{q}$ is a domain which embeds in $\operatorname{Frac}(A'/\mathfrak{q}) = k(x)$, so it is a domain finite-dimensional over k, hence a field. So $\mathfrak{q}$ is maximal, and x is closed in W.

x is closed in X. Let $z \in \overline{\{x\}}$ and choose an open affine $W \ni z$. Since W is an open neighbourhood of $z \in \overline{\{x\}}$, it meets $\{x\}$, so $x \in W$. Then $z \in \overline{\{x\}} \cap W = \overline{\{x\}}^W = \{x\}$ by the previous paragraph. So $\overline{\{x\}} = \{x\}$.

Thus every nonempty open set contains a closed point of X.

Counterexample. Let R be a discrete valuation ring and $X = \operatorname{Spec} R$. It has two points, the closed point $\mathfrak{m}$ and the generic point (0); the set of closed points is $\{\mathfrak{m}\}$, whose closure is $\{\mathfrak{m}\} \neq X$. (Of course X is not of finite type over a field.) □

Exercise 3.15

> **Problem.** Let X be a scheme of finite type over a field k (not necessarily algebraically closed).
>
> (a) Show that the following three conditions are equivalent (in which case we say that X is *geometrically irreducible*): (i) $X \times_k \bar{k}$ is irreducible, where $\bar{k}$ denotes the algebraic closure of k. (By abuse of notation, we write $X \times_k \bar{k}$ to denote $X \times_{\operatorname{Spec} k} \operatorname{Spec} \bar{k}$.) (ii) $X \times_k k_s$ is irreducible, where k_s denotes the separable closure of k. (iii) $X \times_k K$ is irreducible for every extension field K of k.
>
> (b) Show that the following three conditions are equivalent (in which case we say that X is *geometrically reduced*): (i) $X \times_k \bar{k}$ is reduced. (ii) $X \times_k k_p$ is reduced, where k_p denotes the perfect closure of k. (iii) $X \times_k K$ is reduced for all extension fields K of k.
>
> (c) We say that X is *geometrically integral* if $X \times_k \bar{k}$ is integral. Give examples of integral schemes which are neither geometrically irreducible nor geometrically reduced.

Solution. Throughout, for a field extension $k \subseteq K$ and a k-scheme X write $X_K = X \times_k K$ and $p_K : X_K \to X$ for the projection; over an affine open $\operatorname{Spec} A \subseteq X$, $p_K^{-1}(\operatorname{Spec} A) = \operatorname{Spec}(A \otimes_k K)$ (Thm. 3.3, Steps 1 and 4). I use the following facts.

- **(F1)** *p_K is surjective.* The fibre of p_K over a point x with residue field $k(x)$ is $\operatorname{Spec}(k(x) \otimes_k K) \neq \emptyset$ (a tensor product of two nonzero vector spaces over a field is nonzero; cf. Ex. 3.10). Consequently, if X_K is irreducible then so is X (a continuous image of an irreducible space is irreducible).
- **(F2)** *(Tensor products of domains.)* If F is an algebraically closed field and D, E are F-algebras that are integral domains, then $D \otimes_F E$ is an integral domain (Zariski–Samuel, or Lang, *Algebra*; for finitely generated domains this is (I, Ex. 3.15), and the general case follows by taking the direct limit over finitely generated subalgebras).
- **(F3)** *(Base change and reducedness.)* If A is a reduced algebra over a field F and L/F is a separable field extension (e.g. any extension if F is perfect, or any algebraic extension of a perfect field), then $A \otimes_F L$ is reduced (Bourbaki, *Alg.* V, Sect. 15; Matsumura, *Commutative Algebra*, Sect. 26).
- **(F4)** *Purely inseparable base change.* If $F \subseteq F'$ is algebraic and purely inseparable (every element of F' has some power $\lambda^q \in F$, q a power of $p = \operatorname{char} F$; in characteristic 0 this means $F = F'$) then for every F-scheme T, $p : T_{F'} \to T$ is a homeomorphism. Indeed, on an affine piece $A \to B = A \otimes_F F'$ is injective (flatness) and integral, so $\operatorname{Spec} B \to \operatorname{Spec} A$ is surjective and closed (lying over). It is injective: for $b = \sum a_i \otimes \lambda_i$, taking q large, $b^q = \sum a_i^q \lambda_i^q \otimes 1 \in A$ (we are in characteristic p), so a prime $\mathfrak{Q}$ of B is determined by its contraction $\mathfrak{P} = \mathfrak{Q} \cap A$, namely $\mathfrak{Q} = \{b : b^q \in \mathfrak{P} \text{ for some } q\}$. A continuous closed bijection is a homeomorphism; a global statement follows because the property is local on T (the preimages $p^{-1}(\operatorname{Spec} A)$ are saturated, and closedness is local on the target).

(a) (iii)⇒(i),(ii) is trivial. (i)⇒(ii): $X_{\bar{k}} \to X_{k_s}$ is surjective by (F1) (applied to X_{k_s} over k_s), so X_{k_s} is irreducible. (ii)⇒(i): $X_{\bar{k}} = (X_{k_s})_{\bar{k}}$ and $\bar{k}/k_s$ is purely inseparable (for $\alpha \in \bar{k}$, some α^{p^e} is separable over k, hence lies in k_s). By (F4), $X_{\bar{k}}$ is homeomorphic to X_{k_s}, so it is irreducible.

(i)⇒(iii): Let K/k be an extension. The ring $K \otimes_k \bar{k} \neq 0$; let $\mathfrak{m}$ be a maximal ideal and $L = (K \otimes_k \bar{k})/\mathfrak{m}$. Then L is a field containing (copies of) K and $\bar{k}$ over k, so $X_L = (X_K)_L = (X_{\bar{k}})_L$. Put $F = \bar{k}$ and $Z = X_{\bar{k}}$, an irreducible scheme (by (i)) with a finite cover by open affines $\operatorname{Spec} D_i$. Consider $\operatorname{Spec}(D_i \otimes_F L)$. The nilradical N_i of D_i gives $N_i \otimes_F L \subseteq D_i \otimes_F L$ (flatness), an ideal consisting of nilpotent elements (a sum of nilpotents $n \otimes \lambda$), and $(D_i \otimes L)/(N_i \otimes L) = (D_i/N_i) \otimes_F L$. Hence $\operatorname{Spec}((D_i/N_i) \otimes_F L)$ and $\operatorname{Spec}(D_i \otimes_F L)$ have the same underlying space. Now Z is irreducible, so Z_{red} is integral (Prop. 3.1) and its affine pieces $E_i = D_i/N_i$ are domains (nonempty $\operatorname{Spec} D_i$); by (F2), $E_i \otimes_F L$ is a domain, so $W_i = p_L^{-1}(\operatorname{Spec} D_i) \subseteq Z_L$ is irreducible. For i, j, $W_i \cap W_j = p_L^{-1}(\operatorname{Spec} D_i \cap \operatorname{Spec} D_j)$ is nonempty since Z is irreducible (so the two nonempty opens meet) and p_L is surjective (F1). A space covered by irreducible open subsets that pairwise meet is irreducible. So X_L is irreducible, and then X_K is irreducible by (F1) applied to $(X_K)_L \to X_K$.

(b) (iii)⇒(i),(ii) is trivial. (i)⇒(ii): over an affine open $\operatorname{Spec} A$, $A \otimes_k k_p \subseteq A \otimes_k \bar{k}$ (flatness over the field k_p) and a subring of a reduced ring is reduced. (ii)⇒(i): $A \otimes_k \bar{k} = (A \otimes_k k_p) \otimes_{k_p} \bar{k}$, where $A \otimes_k k_p$ is reduced and $\bar{k}/k_p$ is algebraic over the perfect field k_p, hence separable; by

(F3) $A \otimes_k \bar{k}$ is reduced. (i)⇒(iii): let K/k be an extension and $L \supseteq K, \bar{k}$ as in (a). By (i), $A \otimes_k \bar{k}$ is reduced; as $\bar{k}$ is perfect, (F3) shows $(A \otimes_k \bar{k}) \otimes_{\bar{k}} L = A \otimes_k L$ is reduced. Finally $A \otimes_k K \to (A \otimes_k K) \otimes_K L = A \otimes_k L$ is injective (L is free over K), so $A \otimes_k K$ is reduced. As this holds on every affine open, X_K is reduced.

(c) The following integral schemes are not geometrically integral for different reasons (the first two show that each of the two properties can fail separately; the third fails both at the same time).

(1) Not geometrically irreducible. $k = \mathbf{R}$, $X = \operatorname{Spec} \mathbf{C} = \operatorname{Spec} \mathbf{R}[x]/(x^2+1)$. It is a single reduced point, hence integral. But $\mathbf{C} \otimes_{\mathbf{R}} \mathbf{C} \cong \mathbf{C} \times \mathbf{C}$ $(a \otimes b \mapsto (ab, a\bar{b}))$, so $X_{\mathbf{C}}$ consists of two points: X is not geometrically irreducible (though it is geometrically reduced).

(2) Not geometrically reduced. $k = \mathbf{F}_p(t)$, $X = \operatorname{Spec} k[x]/(x^p - t)$. The polynomial $x^p - t$ is irreducible, so $X = \operatorname{Spec} K$ with $K = k(t^{1/p})$ a field, an integral scheme. But $K \otimes_k k' = k'[x]/(x^p - t) = k'[x]/(x - t^{1/p})^p$ for $k' = k(t^{1/p})$ (or $\bar{k}$), which is not reduced. (This X is geometrically irreducible.)

(3) Neither. Let $k = \mathbf{F}_2(t)$ and

$$K = k[x,y]/(x^2 + x + t,\ y^2 + t), \qquad X = \operatorname{Spec} K.$$

First, x^2+x+t is irreducible over k: a root u/v ($u, v \in \mathbf{F}_2[t]$ coprime) would satisfy $u^2+uv+tv^2 = 0$, so $v \mid u^2$, hence $v \in \mathbf{F}_2^\times$, i.e. $v = 1$ and $u^2 + u + t = 0$ with $u \in \mathbf{F}_2[t]$; if $\deg u \geq 1$ the term u^2 has degree $2 \deg u > \max(\deg u, 1)$ and cannot cancel, and if u is constant then $u^2 + u = 0$ gives $t = 0$, a contradiction. So $k_1 = k[x]/(x^2 + x + t)$ is a field with basis $1, x$ over k. Next t is not a square in k_1: $(c + dx)^2 = c^2 + d^2x^2 = (c^2 + d^2t) + d^2x$ (char 2, $x^2 = x + t$), and this equals t only if $d = 0$ and $c^2 = t$, which is impossible in $k = \mathbf{F}_2(t)$ (degree parity). So $y^2 + t$ is irreducible over k_1, and $K = k_1[y]/(y^2 + t)$ is a field, so X is an integral scheme. Over $\bar{k}$:

$$K \otimes_k \bar{k} = \bar{k}[x,y]/(x^2 + x + t,\, y^2 + t) = \bar{k}[x,y]/\big((x + a)(x + a + 1), (y + b)^2\big),$$

where $a^2 + a + t = 0$ (so $a \neq a + 1$) and $b^2 = t$. By the Chinese remainder theorem this is $\bar{k}[y]/(y + b)^2 \times \bar{k}[y]/(y + b)^2$: two points, each nonreduced. So $X_{\bar{k}}$ is neither irreducible nor reduced: X is an integral scheme that is neither geometrically irreducible nor geometrically reduced. □

Exercise 3.16

> **Problem.** *Noetherian Induction.* Let X be a noetherian topological space, and let $\mathcal{P}$ be a property of closed subsets of X. Assume that for any closed subset Y of X, if $\mathcal{P}$ holds for every proper closed subset of Y, then $\mathcal{P}$ holds for Y. (In particular, $\mathcal{P}$ must hold for the empty set.) Then $\mathcal{P}$ holds for X.

Solution. Suppose $\mathcal{P}$ fails for X. Let $\mathcal{S}$ be the set of closed subsets of X for which $\mathcal{P}$ fails; $X \in \mathcal{S}$, so $\mathcal{S} \neq \emptyset$. Since X is noetherian, closed subsets satisfy the descending chain condition, so $\mathcal{S}$ has a minimal element Y (otherwise we could build an infinite strictly descending chain in $\mathcal{S}$). By minimality every proper closed subset of Y has $\mathcal{P}$, so by hypothesis $\mathcal{P}$ holds for Y, contradicting $Y \in \mathcal{S}$. Hence $\mathcal{P}$ holds for X. □

Exercise 3.17

Problem. *Zariski Spaces.* A topological space X is a *Zariski space* if it is noetherian and every (nonempty) closed irreducible subset has a unique generic point (Ex. 2.9). For example, let R be a discrete valuation ring, and let $T = \mathrm{sp}(\mathrm{Spec}\, R)$. Then T consists of two points $t_0 =$ the maximal ideal, $t_1 =$ the zero ideal. The open subsets are $\emptyset, \{t_1\}$, and T. This is an irreducible Zariski space with generic point t_1.

(a) Show that if X is a noetherian scheme, then $\mathrm{sp}(X)$ is a Zariski space.

(b) Show that any minimal nonempty closed subset of a Zariski space consists of one point. We call these closed points.

(c) Show that a Zariski space X satisfies the axiom T_0: given any two distinct points of X, there is an open set containing one but not the other.

(d) If X is an irreducible Zariski space, then its generic point is contained in every nonempty open subset of X.

(e) If x_0, x_1 are points of a topological space X, and if $x_0 \in \overline{\{x_1\}}$, then we say that x_1 *specializes* to x_0, written $x_1 \rightsquigarrow x_0$. We also say x_0 is a *specialization* of x_1, or that x_1 is a *generization* of x_0. Now let X be a Zariski space. Show that the minimal points, for the partial ordering determined by $x_1 > x_0$ if $x_1 \rightsquigarrow x_0$, are the closed points, and the maximal points are the generic points of the irreducible components of X. Show also that a closed subset contains every specialization of any of its points. (We say closed subsets are *stable under specialization.*) Similarly, open subsets are *stable under generization.*

(f) Let t be the functor on topological spaces introduced in the proof of (2.6). If X is a noetherian topological space, show that $t(X)$ is a Zariski space. Furthermore X itself is a Zariski space if and only if the map $\alpha : X \to t(X)$ is a homeomorphism.

Solution. (a) $\mathrm{sp}(X)$ is a noetherian topological space for a noetherian scheme X (Caution 3.1.1, from (Ex. 2.13)), and every irreducible closed subset of a scheme has a unique generic point (Ex. 2.9).

(b) Let Z be a minimal nonempty closed subset. If $Z = Z_1 \cup Z_2$ with Z_i closed and proper in Z, then by minimality $Z_1 = Z_2 = \emptyset$ and $Z = \emptyset$, a contradiction; so Z is irreducible and has a generic point z: $Z = \overline{\{z\}}$. If $z' \in Z$, then $\overline{\{z'\}} \subseteq Z$ is a nonempty closed subset, so $\overline{\{z'\}} = Z$, and z' is also a generic point of Z. By uniqueness $z' = z$. So $Z = \{z\}$. (Conversely if $\{z\}$ is closed it is clearly a minimal nonempty closed set, so "closed points" are the points z with $\{z\}$ closed.)

(c) Let $x \neq y$ and suppose every open set containing x contains y and vice versa. Then $y \in \overline{\{x\}}$ (every open neighbourhood of y meets $\{x\}$) and $x \in \overline{\{y\}}$, so $\overline{\{x\}} = \overline{\{y\}}$: this irreducible closed set has two generic points x, y, contradicting uniqueness.

(d) Let ξ be the generic point and $U \neq \emptyset$ open. If $\xi \notin U$ then $\xi \in X \smallsetminus U$, which is closed, so $X = \overline{\{\xi\}} \subseteq X \smallsetminus U$ and $U = \emptyset$, a contradiction.

(e) Define $x_1 \geq x_0$ iff $x_0 \in \overline{\{x_1\}}$. This is reflexive and transitive ($x_0 \in \overline{\{x_1\}}$, $x_1 \in \overline{\{x_2\}}$ give $\overline{\{x_0\}} \subseteq \overline{\{x_1\}} \subseteq \overline{\{x_2\}}$), and antisymmetric: $x_1 \geq x_0 \geq x_1$ gives $\overline{\{x_0\}} = \overline{\{x_1\}}$, so $x_0 = x_1$ by uniqueness of generic points. Write $x_1 > x_0$ if $x_1 \geq x_0$, $x_1 \neq x_0$.

Minimal points. z is minimal iff there is no $x_0 \neq z$ with $x_0 \in \overline{\{z\}}$, i.e. iff $\overline{\{z\}} = \{z\}$, i.e. iff z is a closed point (by (b)).

Maximal points. Let Z be an irreducible component of X (closed and irreducible, so $Z = \overline{\{z\}}$) and suppose $z \in \overline{\{x_1\}}$. The closed irreducible set $\overline{\{x_1\}}$ contains Z, so equals Z by maximality

of components; so x_1 is a generic point of Z, $x_1 = z$. Thus z is maximal. Conversely let z be maximal. $\overline{\{z\}}$ is irreducible, hence contained in an irreducible component Z of X (components exist and every irreducible subset lies in one, as X is noetherian), with generic point w. Then $z \in \overline{\{w\}}$, i.e. $w \geq z$, so $w = z$ by maximality; thus z is the generic point of the component Z.

Stability. If Z is closed, $x \in Z$ and $x_0 \in \overline{\{x\}}$ then $x_0 \in Z$ since $\overline{\{x\}} \subseteq Z$. If U is open, $x_0 \in U$ and $x_0 \in \overline{\{x\}}$ (i.e. x is a generization of x_0), then $x \in U$: otherwise $x \in X \smallsetminus U$ (closed), so $x_0 \in \overline{\{x\}} \subseteq X \smallsetminus U$, a contradiction.

(f) Recall (proof of (2.6)): $t(X)$ is the set of nonempty irreducible closed subsets of X; for $Y \subseteq X$ closed, $t(Y) = \{Z \in t(X) : Z \subseteq Y\}$; one has $t(Y_1 \cup Y_2) = t(Y_1) \cup t(Y_2)$ (an irreducible closed subset of $Y_1 \cup Y_2$ lies in one of them) and $t(\bigcap Y_i) = \bigcap t(Y_i)$, and the closed sets of $t(X)$ are the $t(Y)$. The map $Y \mapsto t(Y)$ from closed subsets of X to closed subsets of $t(X)$ is a bijection preserving inclusions in both directions: $Y = \bigcup_{y \in Y} \overline{\{y\}}$ is the union of the members of $t(Y)$, so Y can be recovered from $t(Y)$. It also preserves finite unions. Hence:

- $t(X)$ is noetherian (descending chains of closed sets correspond).
- The irreducible closed subsets of $t(X)$ are exactly the $t(Y)$ with Y irreducible closed in X (irreducibility is expressed by inclusions and finite unions of closed sets, which the bijection preserves), i.e. the sets $t(Y)$ with $Y \in t(X)$.
- For $Y \in t(X)$ (a point of $t(X)$), $\overline{\{Y\}}$ is the smallest closed set $t(Y')$ containing Y, i.e. with $Y \subseteq Y'$; this is $t(Y)$. So $t(Y)$ has the point Y as a generic point, and it is unique: if Y' is also a generic point, $t(Y') = \overline{\{Y'\}} = t(Y)$ forces $Y' = Y$.

So $t(X)$ is a Zariski space.

The map $\alpha : X \to t(X)$, $x \mapsto \overline{\{x\}}$ is continuous: $\alpha^{-1}(t(Y)) = \{x : \overline{\{x\}} \subseteq Y\} = Y$. It is injective iff distinct points have distinct closures iff X is T_0; it is surjective iff every nonempty irreducible closed subset has a generic point. If X is a Zariski space, then α is injective by (c) and surjective, hence bijective; and then $\alpha(Y) = t(Y)$ for closed Y (as $\alpha^{-1}(t(Y)) = Y$ and α is onto), so α maps closed sets onto closed sets and is a homeomorphism. Conversely, if X is noetherian and α is a homeomorphism, then $X \cong t(X)$ is a Zariski space (or directly: surjectivity gives existence of generic points and injectivity gives uniqueness).

Remark on the example: for the DVR T, $\overline{\{t_1\}} = T$ and $\overline{\{t_0\}} = \{t_0\}$, so $t_1 > t_0$, consistent with (d), (e). □

Exercise 3.18

Problem. *Constructible Sets.* Let X be a Zariski space (Ex. 3.17). A subset of X is *constructible* if it belongs to the smallest family $\mathcal{F}$ of subsets such that (1) every open subset is in $\mathcal{F}$, (2) a finite intersection of elements of $\mathcal{F}$ is in $\mathcal{F}$, and (3) the complement of an element of $\mathcal{F}$ is in $\mathcal{F}$.

(a) A subset of X is *locally closed* if it is the intersection of an open subset with a closed subset. Show that a subset of X is constructible if and only if it can be written as a finite disjoint union of locally closed subsets.

(b) Show that a constructible subset of an irreducible Zariski space X is dense if and only if it contains the generic point. Furthermore, in that case it contains a nonempty open subset.

(c) A subset S of X is closed if and only if it is constructible and stable under specialization. Similarly, a subset T of X is open if and only if it is constructible and stable under generization.

(d) If $f : X \to Y$ is a continuous map of Zariski spaces, then the inverse image of any constructible subset of Y is a constructible subset of X.

Solution. Note that $\mathcal{F}$ is closed under finite unions (A∪B= complement of the intersection of the complements) and contains X and $\emptyset$; if $B \subseteq X$ is any subset, a family of subsets of B with the analogous properties relative to B is considered below.

(a) Let $\mathcal{G}$ be the family of finite unions of locally closed sets. Closed and open sets are locally closed, so $\mathcal{G}$ contains all open sets. The intersection of two locally closed sets is locally closed, so $\mathcal{G}$ is closed under finite intersections (distributivity). The complement of a locally closed set $O \cap C$ (O open, C closed) is $(X \smallsetminus O) \cup (X \smallsetminus C)$, a union of a closed set and an open set; so the complement of a finite union of locally closed sets is a finite intersection of finite unions of locally closed sets, which lies in $\mathcal{G}$. Thus $\mathcal{G}$ satisfies (1)–(3), so $\mathcal{F} \subseteq \mathcal{G}$. Conversely a locally closed set $O \cap C = O \cap (X \smallsetminus (X \smallsetminus C))$ lies in $\mathcal{F}$, and $\mathcal{F}$ is closed under finite unions; so $\mathcal{G} \subseteq \mathcal{F}$. Hence $\mathcal{F} = \mathcal{G}$.

It remains to see that every finite union of locally closed sets is a finite *disjoint* union of locally closed sets. For locally closed $D = O \cap C$ and P locally closed, $P \smallsetminus D = (P \cap (X \smallsetminus O)) \sqcup (P \cap O \cap (X \smallsetminus C))$ is a disjoint union of two locally closed sets. Hence if P is a finite disjoint union of locally closed sets, so is $P \smallsetminus D$ for any locally closed D (do it for each piece). Now let $A = D_1 \sqcup \cdots \sqcup D_m$ be a disjoint union of locally closed sets and L locally closed. Then $A \cup L = A \sqcup (L \smallsetminus A)$ and $L \smallsetminus A = (\cdots((L \smallsetminus D_1) \smallsetminus D_2) \cdots) \smallsetminus D_m$ is a finite disjoint union of locally closed sets by the above. Induction on the number of pieces gives the claim. (The converse direction, that finite disjoint unions of locally closed sets are constructible, is the inclusion $\mathcal{G} \subseteq \mathcal{F}$.)

(b) Let ξ be the generic point and $S = L_1 \sqcup \cdots \sqcup L_n$, $L_i = O_i \cap C_i$ locally closed. Suppose $\xi \in S$, say $\xi \in L_i$. Then $\xi \in C_i$ closed, so $C_i = X$, and $L_i = O_i$ is open containing ξ (so nonempty). Thus $S \supseteq O_i \neq \emptyset$ and $S \ni \xi$ is dense since $\overline{\{\xi\}} = X$. Conversely suppose S is dense: $X = \overline{S} = \bigcup \overline{L_i}$. As X is irreducible, $X = \overline{L_i}$ for some i (and $L_i \neq \emptyset$). Since C_i is closed and contains L_i, $C_i \supseteq \overline{L_i} = X$, so $L_i = O_i$ is a nonempty open set, which contains ξ by Ex. 3.17(d). So $\xi \in S$, and S contains the nonempty open set O_i.

(c) ($\Rightarrow$) A closed set is the complement of an open one, hence constructible, and it is stable under specialization (Ex. 3.17(e)); an open set is constructible and stable under generization.

($\Leftarrow$, closed) Let S be constructible and stable under specialization, $Z = \overline{S}$. We show $Z \subseteq S$. We first note: *for a closed subset $W \subseteq X$ (with the induced topology, again a Zariski space), $S \cap W$ is constructible in W*: the family of $B \subseteq X$ with $B \cap W$ constructible in W contains the open sets, and is closed under finite intersections and complements, since $W \smallsetminus (B \cap W) = (X \smallsetminus B) \cap W$; so it contains $\mathcal{F}$. Let W be an irreducible component of Z with generic point w and $O = Z \smallsetminus$ (the other components), a nonempty open subset of Z contained in W and containing w (if w lay in another component W', then $W = \overline{\{w\}} \subseteq W'$). Since S is dense in Z, $S \cap O$ is dense in O (every nonempty open subset of O is open in Z) hence $\overline{S \cap O} \supseteq \overline{O} \supseteq \overline{\{w\}} = W$. So $S \cap W$ is a dense constructible subset of the irreducible Zariski space W, and by (b) $w \in S \cap W$.

As Z is the union of its (finitely many) irreducible components $W = \overline{\{w\}}$ and S is stable under specialization, $W \subseteq S$ for every component, so $Z \subseteq S$ and S is closed.

($\Leftarrow$, open) Let T be constructible and stable under generization. Then $S = X \smallsetminus T$ is constructible, and stable under specialization: if $x \in S$ and $x_0 \in \overline{\{x\}} \cap T$, then x is a generization of $x_0 \in T$, so $x \in T$, a contradiction. By the closed case S is closed, so T is open.

(d) The family $\{B \subseteq Y : f^{-1}(B) \in \mathcal{F}_X\}$ contains the open sets of Y (as f is continuous) and is closed under finite intersections and complements (since preimages commute with these operations). By minimality of $\mathcal{F}_Y$ it contains $\mathcal{F}_Y$. □

Exercise 3.19

Problem. The real importance of the notion of constructible subsets derives from the following theorem of Chevalley—see Cartan and Chevalley [1, exposé 7] and see also Matsumura [2, Ch. 2, Sect. 6]: let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Then the image of any constructible subset of X is a constructible subset of Y. In particular, $f(X)$, which need not be either open or closed, is a constructible subset of Y. Prove this theorem in the following steps.

(a) Reduce to showing that $f(X)$ itself is constructible, in the case where X and Y are affine, integral noetherian schemes, and f is a dominant morphism.

(b) In that case, show that $f(X)$ contains a nonempty open subset of Y by using the following result from commutative algebra: let $A \subseteq B$ be an inclusion of noetherian integral domains, such that B is a finitely generated A-algebra. Then given a nonzero element $b \in B$, there is a nonzero element $a \in A$ with the following property: if $\varphi : A \to K$ is any homomorphism of A to an algebraically closed field K, with $\varphi(a) \neq 0$, then φ extends to a homomorphism φ' of B into K, such that $\varphi'(b) \neq 0$. [*Hint*: Prove this algebraic result by induction on the number of generators of B over A. For the case of one generator, prove the result directly. In the application, take $b = 1$.]

(c) Now use noetherian induction on Y to complete the proof.

(d) Give some examples of morphisms $f : X \to Y$ of varieties over an algebraically closed field k, to show that $f(X)$ need not be either open or closed. (Part (b) is starred in the book.)

Solution. In what follows "constructible" in a noetherian scheme Y refers to the Zariski space $\mathrm{sp}(Y)$ (Ex. 3.17(a)). Basic observations:

- **(C1)** Finite unions, finite intersections, complements and differences of constructible sets are constructible.
- **(C2)** If $B \subseteq Y$ is open or closed (a Zariski space with the induced topology), every subset of B that is constructible in B is constructible in Y. Indeed the family $\{C \subseteq B : C \in \mathcal{F}_Y\}$ contains the opens of B (they are $B \cap O$, locally closed in Y, so in $\mathcal{F}_Y$) and is stable under finite intersection and complements relative to B ($B \smallsetminus C = B \cap (Y \smallsetminus C)$ and $B \in \mathcal{F}_Y$); so it contains $\mathcal{F}_B$. Consequently, if $Y = Y_1 \cup \cdots \cup Y_n$ with Y_i open and $S \cap Y_i$ is constructible in Y_i for each i, then $S = \bigcup (S \cap Y_i)$ is constructible in Y.
- **(C3)** If $W \subseteq Z \subseteq Y$ with Z closed in Y and W open in Z, then W is locally closed in Y, hence constructible.

(a) Reduction. Claim: Chevalley's theorem follows from the following statement $(\star)$*: if* $f : X \to Y$ *is a dominant morphism of finite type between integral affine noetherian schemes, then* $f(X)$ *is constructible in* Y*.* I formulate the reduction so that it can be reused for closed subsets of Y in (c). Fix an affine noetherian scheme Y and a closed subset $Z \subseteq Y$, and consider

$(\star)_Z$: for every integral closed subscheme Z' of Y with $Z' \subseteq Z$ (as sets) and every dominant finite-type morphism $f : X \to Z'$ with X integral affine noetherian, $f(X)$ is constructible in Y (it suffices to know this in Z', by (C2)).

$(T)_Z$: for every finite-type morphism $h : X \to Y$ with X noetherian and $h(X) \subseteq Z$, $h(X)$ is constructible in Y.

Claim: $(\star)_Z \Rightarrow (T)_Z$. Let $h : X \to Y$ be as in $(T)_Z$. Cover X by finitely many open affines X_j; the open immersion $X_j \to X$ is quasi-compact (X is noetherian), hence of finite type (Ex. 3.13(b)), so $X_j \to Y$ is of finite type (Ex. 3.13(c)), and $h(X) = \bigcup h(X_j)$. Each X_j is a noetherian affine scheme with finitely many irreducible components X_{jl}, each given the reduced induced structure: an integral affine noetherian scheme, with $X_{jl} \to X_j$ a closed immersion, so $X_{jl} \to Y$ is of finite type (Ex. 3.13(a),(c)), and $h(X) = \bigcup h(X_{jl})$. Let $Z' = \overline{h(X_{jl})} \subseteq Z$ (Z is closed), irreducible, with its reduced induced structure: an integral closed subscheme of the affine scheme Y, so affine. Since X_{jl} is reduced, $X_{jl} \to Y$ factors through Z' (Ex. 3.11(d), or directly: $B \to A_{jl}$ kills the prime Ker of Z' since A_{jl} is a domain and the image is dense in Z'); call the factored map f. It is dominant, and of finite type by Ex. 3.13(f) (X_{jl} is noetherian, so f is quasi-compact, and the composite with the closed immersion $Z' \to Y$ is of finite type). By $(\star)_Z$, $f(X_{jl}) = h(X_{jl})$ is constructible; so $h(X)$ is a finite union of constructible sets.

Claim: $(T)_Y$ *for all affine noetherian* Y *implies Chevalley's theorem.* First, to show that $f(C)$ is constructible for constructible $C \subseteq X$ it is enough to show that $f(X)$ is constructible for *every* finite type morphism of noetherian schemes: by Ex. 3.18(a), $C = \bigcup L_i$ with $L_i = O_i \cap C_i$ (O_i open, C_i closed), and L_i is the underlying set of the scheme $W_i = O_i \cap C_i$, an open subscheme of the closed subscheme C_i (reduced structure), which is noetherian; $W_i \to C_i \to X$ is a quasi-compact open immersion followed by a closed immersion, hence of finite type, and so is $W_i \to X \to Y$ (Ex. 3.13). Then $f(L_i)$ is the image of $W_i \to Y$. Next, for a noetherian scheme Y cover it by finitely many open affines Y_i. The set $X_i = f^{-1}(Y_i)$ is open in X, hence a noetherian scheme, and the open immersion $j_i : X_i \to X$ is quasi-compact, hence of finite type. So $f \circ j_i$ is of finite type, and it equals $(Y_i \hookrightarrow Y) \circ f_i$ with $f_i : X_i \to Y_i$ quasi-compact; by Ex. 3.13(f), f_i is of finite type. Since $f(X) \cap Y_i = f_i(X_i)$, (C2) shows that $f(X)$ is constructible once each $f_i(X_i)$ is constructible in the affine noetherian scheme Y_i, which is the case $(T)_{Y_i}$.

(b) The algebraic lemma. Let $R \subseteq S$ be domains, S a finitely generated R-algebra, $0 \neq s \in S$. *Claim (L): there is* $0 \neq r \in R$ *such that every homomorphism* $\varphi : R \to K$ *into an algebraically closed field with* $\varphi(r) \neq 0$ *extends to* $\varphi' : S \to K$ *with* $\varphi'(s) \neq 0$. (Noetherianity is not needed.)

One generator, $S = R[x]$. Let $F = \operatorname{Frac} R$.

Case 1: x *transcendental over* F. Then $S = R[x]$ is a polynomial ring; write $s = a_0x^m + \cdots + a_m$ with $a_0 \neq 0$ and take $r = a_0$. If $\varphi(a_0) \neq 0$, $\varphi(s) = \varphi(a_0)x^m + \cdots \in K[x]$ is a nonzero polynomial; K is infinite (algebraically closed), so there is $c \in K$ with $\varphi(s)(c) \neq 0$. Let φ' extend φ by $x \mapsto c$; then $\varphi'(s) = \varphi(s)(c) \neq 0$.

Case 2: x *algebraic over* F. Then $\operatorname{Frac} S = F(x) = F[x]$ is a finite extension of F, so x and $s^{-1} \in \operatorname{Frac} S$ are algebraic over F: there are relations $a_0x^n + a_1x^{n-1} + \cdots + a_n = 0$ and

$b_0 s^{-m} + b_1 s^{-m+1} + \cdots + b_m = 0$ with $a_i, b_i \in R$, $a_0 b_0 \neq 0$. Put $r = a_0 b_0$. Let $\varphi(r) \neq 0$. Then φ extends to $R_r \to K$. In $S' = S_{rs} \subseteq \operatorname{Frac} S$, which is generated over R_r by x and s^{-1}, both elements are integral over R_r (divide by the units a_0, b_0). So $R_r \subseteq S'$ is an integral extension of domains. Let $\mathfrak{q} = \operatorname{Ker}(R_r \to K)$, a prime. By lying over there is a prime $\mathfrak{Q}$ of S' with $\mathfrak{Q} \cap R_r = \mathfrak{q}$; then $R_r/\mathfrak{q} \subseteq S'/\mathfrak{Q}$ is an integral extension of domains, and $R_r/\mathfrak{q} \hookrightarrow K$. Since K is algebraically closed, the embedding $\operatorname{Frac}(R_r/\mathfrak{q}) \to K$ extends to the algebraic extension $\operatorname{Frac}(S'/\mathfrak{Q})$, giving $\varphi'' : S' \to K$ extending $\varphi|_{R_r}$. Restrict to S: $\varphi' = \varphi''|_S$ extends φ, and $\varphi'(s) \neq 0$ because s is a unit of S'.

Induction. Let $S = R[x_1, \ldots, x_n]$, $n \geq 2$, and $R_1 = R[x_1] \subseteq S$ (a domain), so $S = R_1[x_2, \ldots, x_n]$ has $n-1$ generators over R_1. By induction there is $0 \neq r_1 \in R_1$ such that every $\psi : R_1 \to K$ with $\psi(r_1) \neq 0$ extends to $\psi' : S \to K$ with $\psi'(s) \neq 0$. By the one-generator case for $R \subseteq R_1$ and the element r_1, there is $0 \neq r \in R$ such that every $\varphi : R \to K$ with $\varphi(r) \neq 0$ extends to $\varphi_1 : R_1 \to K$ with $\varphi_1(r_1) \neq 0$. Composing the two extensions proves (L) for S (the case $n = 0$ is trivial).

Application. Let $f : X = \operatorname{Spec} S \to Y = \operatorname{Spec} R$ be dominant, with X, Y integral affine noetherian and f of finite type. The kernel of $R \to S$ is a prime $\mathfrak{p}$ with $V(\mathfrak{p}) = \overline{f(X)} = Y$, so $\mathfrak{p} = 0$ and $R \subseteq S$; S is a finitely generated R-algebra (Ex. 3.3(c)). Apply (L) with $s = 1$ to get $0 \neq r \in R$. We claim $D(r) \subseteq f(X)$. Let $\mathfrak{q} \in D(r)$ and $K = \overline{k(\mathfrak{q})}$, $\varphi : R \to R/\mathfrak{q} \to k(\mathfrak{q}) \subseteq K$. Then $\varphi(r) \neq 0$ (as $r \notin \mathfrak{q}$), so φ extends to $\varphi' : S \to K$. The kernel $\mathfrak{Q}$ of φ' is a prime of S with $\mathfrak{Q} \cap R = \operatorname{Ker} \varphi = \mathfrak{q}$, i.e. $f(\mathfrak{Q}) = \mathfrak{q}$. Hence $f(X) \supseteq D(r)$, a nonempty open subset of Y since $r \neq 0$.

(c) Noetherian induction. Let Y be an affine noetherian scheme. We prove $(\star)_Z$ for all closed $Z \subseteq Y$ by noetherian induction on Z (Ex. 3.16). Assume $(\star)_{Z''}$ holds for every proper closed subset $Z'' \subsetneq Z$; by (a), $(T)_{Z''}$ holds for those. Let $Z' \subseteq Z$ be an integral closed subscheme (affine, noetherian) and $f : X \to Z'$ dominant of finite type with X integral affine noetherian. By (b) there is a nonempty open $W \subseteq Z'$ (open in Z') with $W \subseteq f(X)$. W is constructible in Y by (C3). The set $Z' \smallsetminus W$ is a closed subset of Z', hence of Y, and it is a proper subset of $Z' \subseteq Z$; so $(T)_{Z' \smallsetminus W}$ holds. Let $X'' = f^{-1}(Z' \smallsetminus W)$ with its reduced induced structure: a closed subscheme of X, hence a noetherian scheme, and $X'' \to X \to Z' \to Y$ is of finite type (Ex. 3.13(a),(c)) with image in $Z' \smallsetminus W$. By $(T)_{Z' \smallsetminus W}$, $f(X'') = f(X) \cap (Z' \smallsetminus W)$ is constructible in Y. Therefore $f(X) = W \cup (f(X) \cap (Z' \smallsetminus W))$ is constructible. This proves $(\star)_Z$. By Ex. 3.16, $(\star)_Y$ holds, hence $(T)_Y$ by (a) (the induction needs no base case since for $Z = \emptyset$ the statement is vacuous). By the last claim of (a), Chevalley's theorem holds for all finite type morphisms of noetherian schemes.

(d) Examples. Let k be algebraically closed and $f : \mathbf{A}^2_k \to \mathbf{A}^2_k$, $(x, y) \mapsto (x, xy)$ (a morphism of varieties). The image on closed points is $\{(a, b) : a \neq 0\} \cup \{(0, 0)\}$: for $a \neq 0$, $(a, b) = f(a, b/a)$, and $f(0, y) = (0, 0)$. It is constructible ($D(x) \cup \{(0, 0)\}$) but not closed: it contains the dense open set $D(x)$ but not $(0, 1)$. It is not open: let $O \ni (0, 0)$ be open, $O = \mathbf{A}^2 \smallsetminus V(J)$. Since $(0, 0) \notin V(J)$ there is $g \in J$ with $g(0, 0) \neq 0$; then $g(0, y)$ is a nonzero polynomial in y, so $V(J) \cap \{x = 0\}$ is finite and O contains infinitely many points $(0, b)$, $b \neq 0$, which are not in the image. (In scheme language the image is $D(x) \cup \{\text{the origin}\}$.) Another standard example: the projection of the closed set $V(xy - 1) \subset \mathbf{A}^2$ to the x-axis has image $\mathbf{A}^1 \smallsetminus \{0\}$, which is open but not closed. □

Exercise 3.20

Problem. *Dimension.* Let X be an integral scheme of finite type over a field k (not necessarily algebraically closed). Use appropriate results from (I, Sect. 1) to prove the following.

(a) For any closed point $P \in X$, $\dim X = \dim \mathcal{O}_P$, where for rings, we always mean the Krull dimension.

(b) Let $K(X)$ be the function field of X (Ex. 3.6). Then $\dim X = \text{tr. d. } K(X)/k$.

(c) If Y is a closed subset of X, then $\operatorname{codim}(Y, X) = \inf\{\dim \mathcal{O}_{P,X} \mid P \in Y\}$.

(d) If Y is a closed subset of X, then $\dim Y + \operatorname{codim}(Y, X) = \dim X$.

(e) If U is a nonempty open subset of X, then $\dim U = \dim X$.

(f) If $k \subseteq k'$ is a field extension, then every irreducible component of $X' = X \times_k k'$ has dimension $= \dim X$.

Solution. *Conventions.* In (c), P ranges over all (not only the closed) points of Y; over closed points alone the formula would be false (by (a) all closed points of X have $\dim \mathcal{O}_P = \dim X$). The facts from (I, Sect. 1) used: for a finitely generated k-algebra B which is a domain, (1.8A) $\dim B = \text{tr. d. Frac}(B)/k$, and $\operatorname{ht} \mathfrak{p} + \dim B/\mathfrak{p} = \dim B$ for every prime $\mathfrak{p}$; and $\dim \operatorname{Spec} B$ (as a topological space) equals the Krull dimension of B.

Topological preliminaries.

- **(D1)** Let U be an open subset of a topological space T. Then $Z \mapsto Z \cap U$ is a bijection from the irreducible closed subsets of T meeting U onto the irreducible closed subsets of U, with inverse $Z' \mapsto \overline{Z'}$, and both preserve strict inclusions (a nonempty open subset of an irreducible space is dense and irreducible). Hence $\dim U \leq \dim T$.

- **(D2)** If $T = \bigcup T_i$ is an open cover, then $\dim T = \sup \dim T_i$. Indeed "$\geq$" is (D1). For "$\leq$": let $Z_0 \subsetneq \cdots \subsetneq Z_n$ be irreducible closed subsets of T and choose i with $Z_0 \cap T_i \neq \emptyset$; then $Z_j \cap T_i$ $(j = 0, \ldots, n)$ is a strictly increasing chain of irreducible closed subsets of T_i by (D1), so $n \leq \dim T_i$.

- **(D3)** For any point P of any scheme X, $\dim \mathcal{O}_{X,P} = \operatorname{codim}(\overline{\{P\}}, X)$. Indeed let $U = \operatorname{Spec} A$ be an open affine containing P, $P \leftrightarrow \mathfrak{p}$. Every irreducible closed subset of X containing P meets U, so by (D1) the irreducible closed subsets of X containing $\overline{\{P\}}$ correspond, preserving inclusions, to those of U containing $\overline{\{P\}} \cap U = V(\mathfrak{p})$, i.e. to the primes $\mathfrak{q} \subseteq \mathfrak{p}$. So $\operatorname{codim}(\overline{\{P\}}, X) = \operatorname{ht} \mathfrak{p} = \dim A_{\mathfrak{p}} = \dim \mathcal{O}_{X,P}$.

Setup. X is covered by finitely many open affines $U_i = \operatorname{Spec} A_i$; by Ex. 3.3(c) every nonempty open affine $\operatorname{Spec} A \subseteq X$ has A a finitely generated k-algebra, and a domain because X is integral, with $\operatorname{Frac} A = K(X)$ (Ex. 3.6).

(b) For every nonempty open affine $U = \operatorname{Spec} A$, $\dim U = \dim A = \text{tr. d. } K(X)/k =: d$ by (1.8A). By (D2) applied to a cover of X by such U, $\dim X = d$.

(e) A nonempty open $U \subseteq X$ is covered by nonempty open affines $\operatorname{Spec} A$ of X, each of dimension d by the above; by (D2), $\dim U = d = \dim X$.

(a) Let P be a closed point, $U = \operatorname{Spec} A$ an open affine containing it, $P \leftrightarrow \mathfrak{p}$. As P is closed in U, $\mathfrak{p}$ is maximal, so $\dim A/\mathfrak{p} = 0$ and (1.8A) gives $\operatorname{ht} \mathfrak{p} = \dim A = \dim X$ by (b). Hence $\dim \mathcal{O}_P = \operatorname{ht} \mathfrak{p} = \dim X$.

(c) Recall $\operatorname{codim}(Y,X) = \inf \operatorname{codim}(Z,X)$ over irreducible closed $Z \subseteq Y$. Every irreducible closed $Z \subseteq Y$ is $\overline{\{P\}}$ for its generic point $P \in Y$, and conversely $\overline{\{P\}} \subseteq Y$ for each $P \in Y$. So by (D3), $\operatorname{codim}(Y,X) = \inf_{P\in Y} \operatorname{codim}(\overline{\{P\}},X) = \inf_{P\in Y} \dim \mathcal{O}_{P,X}$.

(d) *Irreducible case.* Let $Z \subseteq X$ be closed irreducible, with the reduced induced structure: an integral closed subscheme, of finite type over k (Ex. 3.13(a),(c)). Let $V = \operatorname{Spec} A \subseteq X$ be an open affine meeting Z, so $Z \cap V = V(\mathfrak{p}) = \operatorname{Spec} A/\mathfrak{p}$ for a prime $\mathfrak{p}$. By (e) for Z, $\dim Z = \dim(Z \cap V) = \dim A/\mathfrak{p}$. By (D1), the irreducible closed subsets of X containing Z correspond to those of V containing $Z \cap V$, i.e. to primes contained in $\mathfrak{p}$; so $\operatorname{codim}(Z,X) = \operatorname{ht}\mathfrak{p}$. By (1.8A) and (b), $\dim Z + \operatorname{codim}(Z,X) = \dim A/\mathfrak{p} + \operatorname{ht}\mathfrak{p} = \dim A = \dim X$.

General case. Let $Y = Y_1 \cup \cdots \cup Y_r$ be the decomposition into irreducible components. $\dim Y = \max \dim Y_j$. Every irreducible closed $Z \subseteq Y$ lies in some Y_j, and then $\operatorname{codim}(Z,X) \geq \operatorname{codim}(Y_j,X)$ (a chain starting with Y_j can be extended by $Z \subseteq Y_j$ at the beginning if $Z \neq Y_j$). So $\operatorname{codim}(Y,X) = \min_j \operatorname{codim}(Y_j,X)$, and by the irreducible case

$$\dim Y + \operatorname{codim}(Y,X) = \max_j \dim Y_j + \min_j(\dim X - \dim Y_j) = \dim X.$$

(f) By Ex. 3.13(d), X' is of finite type over k'. Let $X_i = \operatorname{Spec} A_i$ be finitely many open affines covering X; $X'_i = X_i \times_k k' = \operatorname{Spec} S_i$ with $S_i = A_i \otimes_k k'$ cover X'. Let $d = \dim X$.

Claim: if P is a minimal prime of $S = S_i$, then $\dim S/P = d$. By Noether normalization there is $P_0 = k[t_1,\dots,t_d] \subseteq A = A_i$ with A a finite P_0-module ($d = \operatorname{tr.d.Frac} A = \dim A$). Put $R = P_0 \otimes_k k' = k'[t_1,\dots,t_d]$; the inclusion $P_0 \subseteq A$ stays injective after $\otimes_k k'$ (flatness over a field), so $R \subseteq S$, and S is a finite R-module. Let $r \in R$, $r \neq 0$. We show r is a non-zero-divisor on S. Let $F_0 = k(t_1,\dots,t_d)$ and $M = \operatorname{Frac} A$; M is a finite-dimensional F_0-vector space ($A \otimes_{P_0} F_0$ is a domain finite over F_0, hence a field, equal to M). Then $S = A \otimes_k k' \subseteq M \otimes_k k' = M \otimes_{F_0} R'$ with $R' = F_0 \otimes_k k'$; here R' is the localization of R at the multiplicative set $P_0 \smallsetminus 0$, a subring of $k'(t_1,\dots,t_d)$, so a domain. As $M \otimes_{F_0} R' \cong R'^n$ is free over R' ($n = \dim_{F_0} M$), r acts injectively on it, hence on S. Now a minimal prime P of S consists of zero-divisors (in S_P, which has a unique prime, every element of PS_P is nilpotent; so for $x \in P$ there are $u \notin P$ and $n \geq 1$ minimal with $ux^n = 0$, and then $x \cdot ux^{n-1} = 0$ with $ux^{n-1} \neq 0$). Therefore $P \cap R = 0$. Then $R \hookrightarrow S/P$ is an integral extension of domains, so $\operatorname{Frac}(S/P)$ is algebraic over $k'(t_1,\dots,t_d)$ and $\operatorname{tr.d.}_{k'} \operatorname{Frac}(S/P) = d$. Since S/P is a finitely generated k'-domain, $\dim S/P = d$ by (1.8A).

Now let Z' be an irreducible component of X'. The sets $Z' \cap X'_i$ that are nonempty are irreducible components of $X'_i = \operatorname{Spec} S_i$ (the irreducible components of an open subset are the nonempty traces of those of the space), i.e. $Z' \cap X'_i = V(P)$ for a minimal prime P of S_i, of dimension $\dim S_i/P = d$ by the Claim. These sets form an open cover of Z', so by (D2) $\dim Z' = d = \dim X$. □

Exercise 3.21

> **Problem.** Let R be a discrete valuation ring containing its residue field k. Let $X = \operatorname{Spec} R[t]$ be the affine line over $\operatorname{Spec} R$. Show that statements (a), (d), (e) of (Ex. 3.20) are false for X.

Solution. Let π be a uniformizer, $K = R[1/\pi] = \operatorname{Frac} R$, and recall $R/(\pi) \cong k$.

$\dim X = 2$. There is a chain of primes $(0) \subsetneq (\pi) \subsetneq (\pi, t)$ in $R[t]$: $R[t]/(\pi) = k[t]$ is a domain, and $R[t]/(\pi, t) = k$ is a field. So $\dim R[t] \geq 2$. Since R is noetherian of dimension 1, $\dim R[t] = \dim R + 1 = 2$ (a standard fact on polynomial rings over noetherian rings, e.g. Matsumura, *Commutative Algebra*, Thm. 15.4, or Atiyah–Macdonald, Ex. 11.7).

(a) fails. Consider $\mathfrak{m} = (\pi t - 1) \subset R[t]$. Since $R_\pi = R[x]/(\pi x - 1)$ (the standard presentation of a localization), $R[t]/(\pi t - 1) \cong R_\pi = K$ via $t \mapsto 1/\pi$. Hence $R[t]/\mathfrak{m} \cong K$ is a field and $\mathfrak{m}$ is a maximal ideal, a closed point P of X. By Krull's principal ideal theorem, $\operatorname{ht} \mathfrak{m} \leq 1$, and $\mathfrak{m} \neq 0$ is prime in a domain, so $\operatorname{ht} \mathfrak{m} = 1$. Thus $\dim \mathcal{O}_P = 1 \neq 2 = \dim X$.

(d) fails. Let $Y = \{P\}$ be this closed point. Then $\dim Y = 0$ and $\operatorname{codim}(Y, X) = \operatorname{codim}(\overline{\{P\}}, X) = \operatorname{ht} \mathfrak{m} = 1$, by (D3) of Ex. 3.20 (valid on any scheme). So $\dim Y + \operatorname{codim}(Y, X) = 1 \neq 2 = \dim X$.

(e) fails. Let $U = D(\pi) = \operatorname{Spec} R[t]_\pi = \operatorname{Spec} K[t]$, a nonempty open subset of X. Then $\dim U = \dim K[t] = 1 \neq 2 = \dim X$. □

Exercise 3.22

Problem. *Dimension of the Fibres of a Morphism.* Let $f : X \to Y$ be a dominant morphism of integral schemes of finite type over a field k.

(a) Let Y' be a closed irreducible subset of Y, whose generic point η' is contained in $f(X)$. Let Z be any irreducible component of $f^{-1}(Y')$, such that $\eta' \in f(Z)$, and show that $\operatorname{codim}(Z, X) \leq \operatorname{codim}(Y', Y)$.

(b) Let $e = \dim X - \dim Y$ be the *relative dimension* of X over Y. For any point $y \in f(X)$, show that every irreducible component of the fibre X_y has dimension $\geq e$. [*Hint*: Let $Y' = \overline{\{y\}}$, and use (a) and (Ex. 3.20b).]

(c) Show that there is a dense open subset $U \subseteq X$, such that $\dim U_y = e$, for any $y \in f(U)$. [*Hint*: First reduce to the case where X and Y are affine, say $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$. Then A is a finitely generated B-algebra. Take $t_1, \ldots, t_e \in A$ which form a transcendence basis of $K(X)$ over $K(Y)$, and let $X_1 = \operatorname{Spec} B[t_1, \ldots, t_e]$. Then X_1 is isomorphic to affine e-space over Y, and the morphism $X \to X_1$ is generically finite. Now use (Ex. 3.7) above.]

(d) Going back to our original morphism $f : X \to Y$, for any integer h, let E_h be the set of points $x \in X$ such that, letting $y = f(x)$, there is an irreducible component Z of the fibre X_y, containing x, and having $\dim Z \geq h$. Show that (1) $E_e = X$ (use (b) above); (2) if $h > e$, then E_h is not dense in X (use (c) above); and (3) E_h is closed, for all h (use induction on $\dim X$).

(e) Prove the following theorem of Chevalley—see Cartan and Chevalley [1, exposé 8]. For each integer h, let C_h be the set of points $y \in Y$ such that $\dim X_y = h$. Then the subsets C_h are constructible, and C_e contains an open dense subset of Y.

(Starred in the book.)

Solution. *Conventions.* By Ex. 3.10(a) the irreducible components of the fibre X_y correspond homeomorphically to the irreducible components of the subspace $f^{-1}(y) \subseteq X$; I work with $f^{-1}(y)$ and, for dimensions, treat it as a topological space. X_y is a scheme of finite type over $k(y)$ (Ex. 3.13(d)). Note that X, Y are integral of finite type over k with generic points ξ, η, $f(\xi) = \eta$ (Step 0 of Ex. 3.7), and f is of finite type: $X \to \operatorname{Spec} k$ is of finite type, and f is

quasi-compact (X is noetherian), so Ex. 3.13(f) applies. For a point x, $\dim \mathcal{O}_x = \operatorname{codim}(\overline{\{x\}}, \cdot)$ by (D3) of Ex. 3.20.

(a) Let ζ be the generic point of Z. Since $f(Z) \subseteq \overline{\{f(\zeta)\}}$ and $\eta' \in f(Z)$, we get $\eta' \in \overline{\{f(\zeta)\}}$. Also $f(\zeta) \in Y'$ ($Z \subseteq f^{-1}(Y')$), so $\overline{\{f(\zeta)\}} \subseteq Y' = \overline{\{\eta'\}}$. Hence $\overline{\{f(\zeta)\}} = Y'$ and $f(\zeta) = \eta'$ by uniqueness of the generic point. So f induces a local homomorphism of noetherian local rings $f^{\#} : \mathcal{O}_{Y,\eta'} \to \mathcal{O}_{X,\zeta}$.

Let $d = \dim \mathcal{O}_{Y,\eta'} = \operatorname{codim}(Y', Y)$. Choose $x_1, \dots, x_d \in \mathfrak{m}_{\eta'}$ with $\mathfrak{m}_{\eta'} = \sqrt{(x_1, \dots, x_d)}$ (a system of parameters). Let $I \subseteq \mathcal{O}_{X,\zeta}$ be the ideal generated by $f^{\#}(\mathfrak{m}_{\eta'})$. We show $\sqrt{I} = \mathfrak{m}_\zeta$. Primes of $\mathcal{O}_{X,\zeta}$ correspond to points x of which ζ is a specialization ($\zeta \in \overline{\{x\}}$), and the contraction under $f^{\#}$ of the prime corresponding to x is the prime of $\mathcal{O}_{Y,\eta'}$ corresponding to $f(x)$. If a prime $\mathfrak{q}_x \supseteq I$ then its contraction contains $\mathfrak{m}_{\eta'}$, so equals it, i.e. $f(x) = \eta' \in Y'$; then $x \in f^{-1}(Y')$ and $\overline{\{x\}}$ is an irreducible subset of $f^{-1}(Y')$ containing ζ, hence containing $Z = \overline{\{\zeta\}}$; since Z is an irreducible component, $\overline{\{x\}} = Z$ and $x = \zeta$. So the only prime containing I is $\mathfrak{m}_\zeta$, i.e. $\sqrt{I} = \mathfrak{m}_\zeta = \sqrt{(f^{\#}x_1, \dots, f^{\#}x_d)}$. By Krull's height theorem, $\operatorname{ht} \mathfrak{m}_\zeta \le d$. Hence $\operatorname{codim}(Z, X) = \dim \mathcal{O}_{X,\zeta} \le d = \operatorname{codim}(Y', Y)$.

(b) Let $y \in f(X)$, $Y' = \overline{\{y\}}$, and let W' be an irreducible component of $f^{-1}(y)$ with generic point w (it exists and is unique because W' corresponds, by Ex. 3.10(a), to an irreducible closed subset of the scheme X_y; thus $W' = \overline{\{w\}} \cap f^{-1}(y)$, as $f^{-1}(y)$ has the induced topology), $Z = \overline{\{w\}} \subseteq X$ (closure in X). Let Z_0 be an irreducible component of $f^{-1}(Y')$ containing Z and ζ_0 its generic point. As in (a), $y = f(w) \in f(Z_0) \subseteq \overline{\{f(\zeta_0)\}} \subseteq Y'$ gives $f(\zeta_0) = y$. So $\zeta_0 \in f^{-1}(y)$ and $w \in \overline{\{\zeta_0\}}$; the closure of $\{\zeta_0\}$ in $f^{-1}(y)$ is an irreducible subset of $f^{-1}(y)$ containing w, hence containing W', hence equal to W' by maximality; so ζ_0 is a generic point of W', i.e. $\zeta_0 = w$, and $Z = Z_0$ is an irreducible component of $f^{-1}(Y')$ with $y \in f(Z)$. By (a), $\operatorname{codim}(Z, X) \le \operatorname{codim}(Y', Y)$. Using Ex. 3.20(d) for $Z \subseteq X$ and $Y' \subseteq Y$,

$$\dim Z = \dim X - \operatorname{codim}(Z, X) \ge \dim X - \operatorname{codim}(Y', Y) = \dim X - \dim Y + \dim Y' = e + \dim Y'.$$

Now Z with its reduced structure is an integral scheme of finite type over k with function field $k(w)$ (the residue field of X at w), so $\dim Z = \operatorname{tr.d.}_k k(w)$ by Ex. 3.20(b); likewise $\dim Y' = \operatorname{tr.d.}_k k(y)$. The component W' of $f^{-1}(y)$, i.e. of X_y (Ex. 3.10(a)), is an integral scheme of finite type over $k(y)$ (reduced structure), with function field the residue field at w of X_y, which is $k(w)$. Therefore, by Ex. 3.20(b) over the field $k(y)$ and additivity of transcendence degree,

$$\dim W' = \operatorname{tr.d.}_{k(y)} k(w) = \operatorname{tr.d.}_k k(w) - \operatorname{tr.d.}_k k(y) = \dim Z - \dim Y' \ge e.$$

(c) *Reduction to the affine case.* Let $V = \operatorname{Spec} B \subseteq Y$ be a nonempty open affine and $X_0 = \operatorname{Spec} A \subseteq f^{-1}(V)$ a nonempty open affine (nonempty since $\xi \in f^{-1}(V)$). If we find a dense open $U \subseteq X_0$ with $\dim U_y = e$ for all $y \in f(U)$, then U is dense open in X and the same holds for f (the fibre of $f|_U$ over y is the same scheme; also $\dim X_0 - \dim V = e$ by Ex. 3.20(e)). So assume $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, $B \subseteq A$ domains (injective as f is dominant, as in Ex. 3.7), finitely generated k-algebras, A a finitely generated B-algebra (Ex. 3.3(c), f being of finite type).

By Ex. 3.20(b), $\operatorname{tr.d.}_{K(Y)} K(X) = \dim X - \dim Y = e$. $K(X)$ is generated as a field over $K(Y)$ by the (finitely many) generators of A over B, so we can choose $t_1, \dots, t_e \in A$ forming a transcendence basis of $K(X)/K(Y)$. Then $B[t_1, \dots, t_e] \subseteq A$ is a polynomial ring over B, $X_1 := \operatorname{Spec} B[t_1, \dots, t_e] \to Y$ is affine e-space over Y, and $g : X \to X_1$ induced by the inclusion

is dominant, of finite type (A is a finitely generated $B[t]$-algebra), between integral schemes. It is generically finite: with $K_1 = K(Y)(t_1, \dots, t_e) = K(X_1)$ the fibre over the generic point is $\operatorname{Spec}(S^{-1}A)$, $S = B[t] \setminus 0$; $S^{-1}A \subseteq K(X)$ is a domain generated over K_1 by the generators of A, which are algebraic over K_1 ($K(X)/K_1$ is algebraic), so it is finite-dimensional over K_1, hence a field, with exactly one prime. By Ex. 3.7 there is a dense open $U_1 \subseteq X_1$ with $g^{-1}(U_1) \to U_1$ finite. Put $U = g^{-1}(U_1)$, a nonempty (it contains ξ) hence dense open subset of X. The morphism $g|_U : U \to U_1$ is finite, hence closed (Ex. 3.5(b)), and its image contains the generic point of X_1 ($g(\xi)$), so it is dense; therefore $g|_U$ is surjective.

Let $y \in f(U)$ and $p : X_1 \to Y$ the projection. The fibre $U_y = U \times_Y \operatorname{Spec} k(y) = U \times_{U_1} (U_1)_y$, and $(U_1)_y$ is an open subscheme of $(X_1)_y = \operatorname{Spec}(B[t] \otimes_B k(y)) = \mathbf{A}^e_{k(y)}$. It is nonempty (as $U_y \neq \emptyset$ and $U_y \to (U_1)_y$). The morphism $U_y \to (U_1)_y$ is the base change of $g|_U$, hence finite (Spec of finite algebras) and surjective (the fibre over z is $\operatorname{Spec}(k(z) \otimes_{k(z')} (\text{fibre over } z')) \neq \emptyset$ where $z' \in U_1$ is the image of z). Now:

Lemma E. A finite surjective morphism $h : T \to T'$ of schemes has $\dim T = \dim T'$. Indeed take an open affine cover $W = \operatorname{Spec} C$ of T' with $h^{-1}(W) = \operatorname{Spec} D$, D a finite C-algebra; $\operatorname{Spec} D \to \operatorname{Spec} C$ is surjective, so $\operatorname{Ker}(C \to D)$ lies in every prime, and $C/\operatorname{Ker} \subseteq D$ is an integral extension, so $\dim D = \dim C/\operatorname{Ker} = \dim C$ (going up and incomparability, Atiyah–Macdonald Cor. 5.9, 5.11). Then (D2) of Ex. 3.20 for the covers $\{W\}$ of T' and $\{h^{-1}W\}$ of T gives $\dim T = \sup \dim D = \sup \dim C = \dim T'$.

So $\dim U_y = \dim(U_1)_y = e$, the latter by Ex. 3.20(e) ($(U_1)_y$ is a nonempty open subset of the integral $k(y)$-scheme $\mathbf{A}^e_{k(y)}$, of dimension e by (b)).

(d) For an arbitrary finite type morphism $\phi : X \to Y$ of schemes of finite type over k (not nec. integral) define $E_h(\phi)$ as in the statement; this depends only on the topological fibres $\phi^{-1}(y)$, their irreducible components and their dimensions as topological spaces.

(1) Let $x \in X$, $y = f(x)$. The point x lies in some irreducible component Z of $f^{-1}(y)$, and $\dim Z \geq e$ by (b). So $x \in E_e$. As $E_h \supseteq E_e$ for $h \leq e$, we get $E_h = X$ for $h \leq e$.

(2) Let $h > e$ and U be as in (c). Let $x \in U$, $y = f(x)$ and Z an irreducible component of $f^{-1}(y)$ containing x. Then $Z \cap U$ is a nonempty open subset of Z, and it is a closed irreducible subset of $U_y = U \cap f^{-1}(y)$; here Z (reduced structure) is an integral scheme of finite type over $k(y)$, so by Ex. 3.20(e), $\dim Z = \dim(Z \cap U) \leq \dim U_y = e < h$. Hence $x \notin E_h$. So $E_h \cap U = \emptyset$ and E_h is not dense.

(3) We prove: for every morphism $\phi : X \to Y$ of schemes of finite type over k and every h, $E_h(\phi)$ is closed in X, by induction on $\dim X$ ($X = \emptyset$ is trivial).

Reduction to X, Y integral and ϕ dominant. Let $X_1, \dots, X_m$ be the irreducible components of X, with reduced structures (integral, of finite type over k) and $\phi_j = \phi|_{X_j}$. Then $E_h(\phi) = \bigcup_j E_h(\phi_j)$. Indeed, if $x \in E_h(\phi)$ with $y = \phi(x)$ and Z a component of $\phi^{-1}(y)$ containing x with $\dim Z \geq h$, then $Z \subseteq X_j$ for some j (Z is irreducible), and Z is then a component of $\phi_j^{-1}(y) = \phi^{-1}(y) \cap X_j$ (maximality persists in the smaller set), so $x \in E_h(\phi_j)$. Conversely a component Z_j of $\phi_j^{-1}(y)$ lies in a component Z of $\phi^{-1}(y)$ with $\dim Z \geq \dim Z_j$, so $E_h(\phi_j) \subseteq E_h(\phi)$. Also $E_h(\phi_j)$ is closed in X_j iff in X. Next, replace Y by $Y_j = \overline{\phi(X_j)}$ with the reduced structure, an integral scheme of finite type over k (this doesn't change the topological fibres), and $\phi_j : X_j \to Y_j$ is dominant. So we may assume X, Y integral and $\phi = f$ dominant (this does not increase $\dim X$; the inductive

step below then lowers it).

Inductive step. Let $f : X \to Y$ be dominant of integral finite type k-schemes, $h > e$ (for $h \le e$, $E_h = X$ by (1)). Let U be as in (c) and $X' = X \smallsetminus U$, a proper closed subset, so $\dim X' < \dim X$ by Ex. 3.20(d) (each component X'_i has codimension ≥ 1) if $X' \neq \emptyset$. By (2) $E_h \cap U = \emptyset$, so $E_h \subseteq X'$. Moreover, if $x \in E_h$ and Z is a component of $f^{-1}(y)$ through x with $\dim Z \ge h$, every point of Z lies in E_h (the same Z works), so $Z \cap U = \emptyset$, i.e. $Z \subseteq X'$. Then Z is also a component of $(f|_{X'})^{-1}(y) = f^{-1}(y) \cap X'$ and $x \in E_h(f|_{X'})$; conversely $E_h(f|_{X'}) \subseteq E_h(f)$ as above. So $E_h(f) = E_h(f|_{X'})$, where $f|_{X'} : X' \to Y$ is a morphism of finite type k-schemes (reduced structure on X') with $\dim X' < \dim X$. By the induction hypothesis $E_h(f|_{X'})$ is closed in X', hence in X. (If $X' = \emptyset$, $E_h = \emptyset$.) This completes the induction.

(e) For each integer h, let $D_h = \{y \in Y : \dim X_y \ge h\}$. A point y is in D_h iff $y \in f(X)$ and some component of $f^{-1}(y)$ has dimension $\ge h$, iff $y \in f(E_h)$ (if Z is such a component, every point of Z lies in E_h; conversely if $x \in E_h$, $\dim f^{-1}(f(x)) \ge h$). So $D_h = f(E_h)$. By (d)(3), E_h is closed; with the reduced induced structure it is a noetherian scheme and $E_h \to X \to Y$ is of finite type (Ex. 3.13(a),(c)). By Chevalley's theorem (Ex. 3.19), D_h is constructible. Hence for $h \ge 0$, $C_h = D_h \smallsetminus D_{h+1}$ is constructible. (If $h < 0$ is allowed, set $\dim \emptyset = -\infty$, so $C_h = \emptyset$.)

It remains to see that C_e contains a nonempty open set. We show $\eta \in C_e$: the fibre X_η has ξ in it, and by the argument of (b) (with $y = \eta$, $Y' = Y$) every irreducible component W' of $f^{-1}(\eta)$ has generic point w with $Z = \overline{\{w\}}$ an irreducible component of $f^{-1}(Y) = X$, so $Z = X$, $w = \xi$, and $\dim W' = \dim Z - \dim Y = \dim X - \dim Y = e$ (formula of (b)). So $f^{-1}(\eta)$ is irreducible of dimension exactly e, i.e. $\dim X_\eta = e$, $\eta \in C_e$. Since C_e is constructible and contains the generic point of the irreducible Zariski space Y, by Ex. 3.18(b) it contains a nonempty open subset of Y, which is dense. □

Exercise 3.23

> **Problem.** If V, W are two varieties over an algebraically closed field k, and if $V \times W$ is their product, as defined in (I, Ex. 3.15, 3.16), and if t is the functor of (2.6), then $t(V \times W) = t(V) \times_{\operatorname{Spec} k} t(W)$.

Solution. Recall (I, Ex. 3.15, 3.16): if $V_i \subseteq V$, $W_j \subseteq W$ are open affine subvarieties, then $V_i \times W_j$ is an affine variety with coordinate ring $A(V_i) \otimes_k A(W_j)$, the sets $V_i \times W_j = q_1^{-1}(V_i) \cap q_2^{-1}(W_j)$ are open in $V \times W$ (where q_1, q_2 are the projections) and cover it, and the isomorphism $A(V_i) \otimes_k A(W_j) \to A(V_i \times W_j)$ is $a \otimes b \mapsto q_1^* a \cdot q_2^* b$. Let $T = t(V) \times_{\operatorname{Spec} k} t(W)$ with projections p_1, p_2 (it exists by Thm. 3.3).

The projections q_1, q_2 give morphisms $t(q_1), t(q_2)$ from $t(V \times W)$ to $t(V), t(W)$ (compatible with the structure morphisms to $\operatorname{Spec} k$, since t is a functor to schemes over k, Prop. 2.6), hence a morphism $\theta : t(V \times W) \to T$ with $p_i \theta = t(q_i)$. We show θ is an isomorphism, locally on T.

By Prop. 2.6, for an affine variety V_i the scheme $t(V_i)$ is $\operatorname{Spec} A(V_i)$, and for an open subvariety t gives the corresponding open subscheme. Thus $t(V)$ is covered by the open affine subschemes $t(V_i) = \operatorname{Spec} A(V_i)$ and $t(W)$ by $t(W_j) = \operatorname{Spec} A(W_j)$. The open sets $T_{ij} = p_1^{-1}(t(V_i)) \cap p_2^{-1}(t(W_j))$ cover T, and by Thm. 3.3 (Steps 4 and 1) $T_{ij} = t(V_i) \times_{\operatorname{Spec} k} t(W_j) = \operatorname{Spec}(A(V_i) \otimes_k A(W_j))$. Also $\theta^{-1}(T_{ij}) = t(q_1)^{-1}(t(V_i)) \cap t(q_2)^{-1}(t(W_j)) = t(q_1^{-1} V_i \cap q_2^{-1} W_j) = t(V_i \times W_j) = \operatorname{Spec} A(V_i \times W_j)$, using that t takes open subvarieties to open subschemes and

preserves preimages of open sets (the homeomorphism of $V \times W$ with the closed points). The restriction $\theta : \operatorname{Spec} A(V_i \times W_j) \to \operatorname{Spec}(A(V_i) \otimes_k A(W_j))$ is, by the universal property of the product (Thm. 3.3, Step 1), the morphism corresponding to the ring homomorphism $A(V_i) \otimes_k A(W_j) \to A(V_i \times W_j)$ whose restrictions to the two factors are q_1^* and q_2^*, i.e. $a \otimes b \mapsto q_1^* a \cdot q_2^* b$. This is an isomorphism by (I, Ex. 3.15). Hence θ is an isomorphism over each T_{ij}; since the T_{ij} cover T, θ is an isomorphism, i.e. $t(V \times W) \cong t(V) \times_{\operatorname{Spec} k} t(W)$ (compatibly with the projections). □

4. Separated and proper morphisms

Exercise 4.1

Problem. Show that a finite morphism is proper.

Solution. Let $f : X \to Y$ be finite: Y is covered by open affines $V_i = \operatorname{Spec} B_i$ with $f^{-1}(V_i) = \operatorname{Spec} A_i$, A_i a finitely generated B_i-module. We check the three conditions in the definition of "proper".

Finite type. Each A_i is a finitely generated B_i-algebra, so f is of finite type.

Separated. Put $X_i = f^{-1}(V_i)$. The open sets $X_i \times_{V_i} X_i$ cover $X \times_Y X$, and $\Delta^{-1}(X_i \times_{V_i} X_i) = X_i$; the restriction of Δ is the diagonal of $X_i \to V_i$, a morphism of affine schemes, hence a closed immersion by Prop. 4.1. Being a closed immersion is local on the target, so Δ is a closed immersion.

Universally closed. (i) A finite morphism is closed. By locality on the target (if $W \subseteq X$ is closed then $f(W) \cap V_i = f(W \cap X_i)$) we may assume $X = \operatorname{Spec} A \to Y = \operatorname{Spec} B$ with A a finite B-algebra. A closed subset of X is $V(I)$; let $J = \operatorname{Ker}(B \to A/I)$. Then $B/J \hookrightarrow A/I$ is an injective integral extension, so by lying over $\operatorname{Spec} A/I \to \operatorname{Spec} B/J$ is surjective; thus $f(V(I)) = V(J)$ is closed.

(ii) Finite morphisms are stable under base extension. Let $Y' \to Y$ be any morphism and $X' = X \times_Y Y'$. Cover Y' by open affines $V' = \operatorname{Spec} B'$ each mapping into some V_i. Then the preimage of V' in X' is $\operatorname{Spec}(A_i \otimes_{B_i} B')$, and $A_i \otimes_{B_i} B'$ is a finite B'-module. So $X' \to Y'$ is finite, hence closed by (i).

Thus f is separated, of finite type, and universally closed, i.e. proper. □

Exercise 4.2

Problem. Let S be a scheme, let X be a reduced scheme over S, and let Y be a separated scheme over S. Let f and g be two S-morphisms of X to Y which agree on an open dense subset of X. Show that $f = g$. Give examples to show that this result fails if either (a) X is nonreduced, or (b) Y is nonseparated. [*Hint*: Consider the map $h : X \to Y \times_S Y$ obtained from f and g.]

Solution. Lemma. Let X be reduced, W a scheme, $Z \subseteq W$ a closed subscheme and $h : X \to W$ a morphism with $h(X) \subseteq Z$ as sets. Then h factors through Z.

Proof. The question is local on W and on X, so let $W = \operatorname{Spec} C$, $Z = \operatorname{Spec} C/I$, $X = \operatorname{Spec} A$ with A reduced, h given by $\varphi : C \to A$. For every prime $\mathfrak{q}$ of A we have $\varphi^{-1}(\mathfrak{q}) \supseteq I$ (as $h(X) \subseteq Z$), hence $\varphi(I) \subseteq \bigcap \mathfrak{q} = \mathfrak{N}(A) = 0$. So φ factors through C/I. (The local factorizations are unique, since $Z \to W$ is a monomorphism, so they glue.) $\square$

Proof of the statement. Let $U \subseteq X$ be open dense with $f|_U = g|_U$. Let $h = (f, g)_S : X \to Y \times_S Y$. Since Y is separated over S, $\Delta : Y \to Y \times_S Y$ is a closed immersion; let $Z = \Delta(Y)$ with the corresponding closed subscheme structure, so $\Delta : Y \xrightarrow{\sim} Z$. On U we have $h|_U = (f|_U, f|_U) = \Delta \circ f|_U$, so $h(U) \subseteq Z$. Now $h^{-1}(Z)$ is closed in X and contains the dense set U, hence equals X. By the Lemma (X is reduced), $h = \Delta \circ k$ for a morphism $k : X \to Y$. Then $f = p_1 \circ h = p_1 \circ \Delta \circ k = k$ and $g = p_2 \circ h = k$, so $f = g$. $\square$

Example (a): X nonreduced. Let $S = \operatorname{Spec} k$, $Y = \mathbf{A}^1_k = \operatorname{Spec} k[t]$ (affine, hence separated), and $X = \operatorname{Spec} A$ with $A = k[x, \varepsilon]/(x\varepsilon, \varepsilon^2)$. Since ε is nilpotent, $|X| = |\operatorname{Spec} k[x]|$ is irreducible, so $U = D(x)$ is a nonempty open, hence dense, subset. In A_x we have $\varepsilon = x^{-1}(x\varepsilon) = 0$, so $A_x = k[x]_x$. Define $f, g : X \to Y$ by $f^{\#}(t) = 0$ and $g^{\#}(t) = \varepsilon$. They agree on U (both give $t \mapsto 0$ in A_x) but $\varepsilon \neq 0$ in A, so $f \neq g$.

Example (b): Y nonseparated. Let $S = \operatorname{Spec} k$, $X = \mathbf{A}^1_k$ (reduced), and Y the affine line with the origin doubled (Example 2.3.6), the union of two copies Y_1, Y_2 of $\mathbf{A}^1_k$ glued along $\mathbf{A}^1_k \setminus \{0\}$. Let $f, g : X \to Y$ be the inclusions of Y_1 and Y_2. They agree on the dense open set $X \setminus \{0\}$ but $f(0) \neq g(0)$ (the two origins), so $f \neq g$. $\square$

Exercise 4.3

> **Problem.** Let X be a separated scheme over an affine scheme S. Let U and V be open affine subsets of X. Then $U \cap V$ is also affine. Give an example to show that this fails if X is not separated.

Solution. Write $S = \operatorname{Spec} A$. Since S is affine, $U \times_S V = \operatorname{Spec}(\mathcal{O}(U) \otimes_A \mathcal{O}(V))$ is affine. Let $\Delta : X \to X \times_S X$ be the diagonal; $U \times_S V = p_1^{-1}(U) \cap p_2^{-1}(V)$ is open in $X \times_S X$, and

$$\Delta^{-1}(U \times_S V) = \Delta^{-1}p_1^{-1}(U) \cap \Delta^{-1}p_2^{-1}(V) = U \cap V,$$

because $p_1\Delta = p_2\Delta = \mathrm{id}_X$. Since Δ is a closed immersion, its restriction $U \cap V \to U \times_S V$ is a closed immersion into an affine scheme. A closed subscheme of an affine scheme is affine (it is Spec of a quotient ring). Hence $U \cap V$ is affine. $\square$

Example. Let k be a field, $S = \operatorname{Spec} k$, and let X be the affine plane with the origin doubled: glue two copies $U = \operatorname{Spec} k[x, y]$ and $V = \operatorname{Spec} k[x, y]$ along $W = \mathbf{A}^2 \setminus \{0\}$ (a nonseparated scheme, as in Example 4.0.1). Both U, V are open affine in X and $U \cap V = W$. We claim W is not affine. Cover $W = D(x) \cup D(y)$. A global section of $\mathcal{O}_W$ is a pair of elements of $k[x, y]_x$ and $k[x, y]_y$ agreeing in $k[x, y]_{xy}$, i.e. an element of $k[x, y]_x \cap k[x, y]_y = k[x, y]$. So $\Gamma(W, \mathcal{O}_W) = k[x, y]$. If W were affine it would be isomorphic to $\operatorname{Spec} \Gamma(W, \mathcal{O}_W) = \mathbf{A}^2$ via the canonical map, which is the open immersion $W \hookrightarrow \mathbf{A}^2$; but that is not surjective (it misses the origin). So $W = U \cap V$ is not affine. $\square$

Exercise 4.4

Problem. Let $f : X \to Y$ be a morphism of separated schemes of finite type over a noetherian scheme S. Let Z be a closed subscheme of X which is proper over S. Show that $f(Z)$ is closed in Y, and that $f(Z)$ with its image subscheme structure (Ex. 3.11d) is proper over S. We refer to this result by saying that "the image of a proper scheme is proper." [*Hint*: Factor f into the graph morphism $\Gamma_f : X \to X \times_S Y$ followed by the second projection p_2, and show that Γ_f is a closed immersion.]

Solution. Step 1: the graph is a closed immersion. Let $f : X \to Y$ be any S-morphism with Y separated over S, and $\Gamma_f = (\mathrm{id}, f)_S : X \to X \times_S Y$. Consider the square with $\Delta_Y : Y \to Y \times_S Y$ and $f \times \mathrm{id}_Y : X \times_S Y \to Y \times_S Y$. For a scheme T, a T-point of $(X \times_S Y) \times_{Y \times_S Y} Y$ is a triple (a_1, a_2, c) with $a_1 : T \to X$, $a_2, c : T \to Y$ and $(fa_1, a_2) = (c, c)$; that is, $a_2 = c = fa_1$. Thus it is the same as a T-point a_1 of X, and the map to $X \times_S Y$ is $(a_1, fa_1) = \Gamma_f \circ a_1$. Hence $X \cong (X \times_S Y) \times_{Y \times_S Y} Y$ with Γ_f the base change of Δ_Y. Since Y is separated over S, Δ_Y is a closed immersion, so is Γ_f (stability of closed immersions under base change).

Step 2. Let $i : Z \to X$ be the closed immersion and consider $f_Z = f \circ i : Z \to Y$. By Step 1 (applied to f_Z), $\Gamma_{f_Z} : Z \to Z \times_S Y$ is a closed immersion, and $f_Z = p_2 \circ \Gamma_{f_Z}$. Now $Z \to S$ is proper, hence universally closed, so $p_2 : Z \times_S Y \to Y$ (the base change of $Z \to S$ by $Y \to S$) is a closed map. Thus $f(Z) = p_2(\Gamma_{f_Z}(Z))$ is the image of a closed set under a closed immersion followed by a closed map, hence closed in Y.

Step 3: properness. Let W be $f(Z)$ with the image subscheme structure of Ex. 3.11d: the smallest closed subscheme of Y through which f_Z factors. Since Z is noetherian, f_Z is quasi-compact, so the underlying set of W is the closure of $f(Z)$, i.e. $f(Z)$ by Step 2, and $f_Z = j \circ \varphi$ with $j : W \to Y$ the closed immersion and $\varphi : Z \to W$ surjective.

$W \to S$ is separated: j is a closed immersion, hence separated (for a closed immersion, which is a monomorphism, Δ is an isomorphism), and $Y \to S$ is separated; a composition of separated morphisms is separated (Cor. 4.6b).

Finite type: j is a closed immersion, hence of finite type; $Y \to S$ is of finite type; compose.

Universally closed: Let $S' \to S$ be any morphism, $W' = W \times_S S'$, $Z' = Z \times_S S'$, $\varphi' : Z' \to W'$. We claim φ' is surjective: $Z' = Z \times_W W'$, and for $w' \in W'$ with image $w \in W$ choose $z \in Z$ over w (surjectivity of φ); the scheme $\mathrm{Spec}(k(z) \otimes_{k(w)} k(w'))$ is nonempty (a tensor product of two nonzero vector spaces over a field is nonzero) and maps to $Z \times_W W'$, giving a point of Z' over w'. Now let $C \subseteq W'$ be closed. Then $\varphi'^{-1}(C)$ is closed in Z' and, by surjectivity, its image in S' equals the image of C in S'. But $Z' \to S'$ is closed (Z is proper over S). So $C \to S'$ has closed image. Hence $W \to S$ is universally closed.

So $W = f(Z)$ is proper over S. □

Exercise 4.5

Problem. Let X be an integral scheme of finite type over a field k, having function field K. We say that a valuation of K/k (see I, Section 6) has *center* x on X if its valuation ring R dominates the local ring $\mathcal{O}_{x,X}$.

(a) If X is separated over k, then the center of any valuation of K/k on X (if it exists) is unique.

(b) If X is proper over k, then every valuation of K/k has a unique center on X.

(c) (Starred in the book.) Prove the converses of (a) and (b). [*Hint*: While parts (a) and (b) follow quite easily from (4.3) and (4.7), their converses will require some comparison of valuations in different fields.]

(d) If X is proper over k, and if k is algebraically closed, show that $\Gamma(X, \mathcal{O}_X) = k$. This result generalizes (I, 3.4a). [*Hint*: Let $a \in \Gamma(X, \mathcal{O}_X)$, with $a \notin k$. Show that there is a valuation ring R of K/k with $a^{-1} \in \mathfrak{m}_R$. Then use (b) to get a contradiction.]

Solution. Conventions. A valuation of K/k is a valuation ring R of K with $k \subseteq R$ (the trivial one $R = K$ is allowed). Since X is integral, for every $x \in X$ the local ring $\mathcal{O}_x = \mathcal{O}_{x,X}$ is a subring of K; let η be the generic point, $\varepsilon : U_K := \operatorname{Spec} K \to X$ the canonical morphism with image η. X is noetherian, so (4.3), (4.7) apply with $Y = \operatorname{Spec} k$. For a valuation ring R put $T = \operatorname{Spec} R$, $U = \operatorname{Spec} K$, $i : U \to T$.

Centers as lifts. *Claim:* the centers of R on X correspond bijectively to the k-morphisms $h : T \to X$ with $h \circ i = \varepsilon$ (the center being $h(t_0)$, t_0 the closed point). Indeed, apply Lemma 4.4 with $x_1 = \eta$: here $Z = \{\eta\}^- = X$ (integral, so reduced), $k(\eta) = K$, and every $x \in X$ is a specialization of η. By Lemma 4.4, morphisms $h : T \to X$ with $h \circ i = \varepsilon$ correspond to points x_0 such that R dominates $\mathcal{O}_{x_0,X}$ (and the inclusion $k(\eta) = K \subseteq K$ is the identity). Such an h is automatically a k-morphism: $T \to X \to \operatorname{Spec} k$ restricted to U is $U \to \operatorname{Spec} k$, and $R \subseteq K$.

(a) Let x, x' be centers of R and h, h' the corresponding morphisms $T \to X$. Then $h \circ i = h' \circ i = \varepsilon$, and h, h' are k-morphisms. As X is separated over k, Thm 4.3 gives $h = h'$, so $x = h(t_0) = h'(t_0) = x'$.

(b) Let R be a valuation of K/k. The data $\varepsilon : U \to X$ and $T \to \operatorname{Spec} k$ form a commutative diagram as in (4.7); since X is proper over k, there is $h : T \to X$ with $h \circ i = \varepsilon$ (compatible with $T \to \operatorname{Spec} k$). By the Claim its image $h(t_0)$ is a center. Uniqueness follows from (a), since proper implies separated.

(c) We use two standard facts: (V1) (I,6.1A) every local subring of a field F is dominated by a valuation ring of F, and a valuation ring is maximal for domination among local subrings of F; (V2) if $F \subseteq L$ are fields and R'' is a valuation ring of L, then $R'' \cap F$ is a valuation ring of F with maximal ideal $\mathfrak{m}_{R''} \cap F$ (for $0 \neq x \in F$, x or x^{-1} lies in R'').

Lemma A. Let $\xi \in X$, $Z = \{\xi\}^-$ (reduced induced structure; integral with function field $F = k(\xi)$), and let R be a valuation ring of F with $k \subseteq R$. Then there is a valuation ring S of K dominating $\mathcal{O}_\xi$ and a valuation ring $\tilde{R} \subseteq S$ of K/k such that for every $x \in Z$:

$$\tilde{R} \text{ dominates } \mathcal{O}_{x,X} \iff R \text{ dominates } \mathcal{O}_{x,Z}.$$

Proof. By (V1) choose a valuation ring S of K dominating the local subring $\mathcal{O}_\xi$. Then $F = \mathcal{O}_\xi/\mathfrak{m}_\xi \hookrightarrow S/\mathfrak{m}_S =: \kappa$, so $F \subseteq \kappa$. By (V1) choose a valuation ring R'' of κ dominating the local ring $R \subseteq F \subseteq \kappa$. By (V2), $R'' \cap F$ is a valuation ring of F dominating R, so $R'' \cap F = R$ by maximality. Let $\tilde{R} = \{s \in S : \bar{s} \in R''\}$ ($\bar{s}$ the image in κ). It is a subring of S with fraction field K, and it is a valuation ring: let $x \in K^*$. If $x \notin S$ then $x^{-1} \in \mathfrak{m}_S$ so $x^{-1} \in \tilde{R}$. If $x \in S$ and $\bar{x} \in R''$ then $x \in \tilde{R}$; if $x \in S$ and $\bar{x} \notin R''$, then $\bar{x} \neq 0$, $\bar{x}^{-1} \in R''$, $x \in S^*$ and $x^{-1} \in \tilde{R}$. Its maximal ideal is $\{s \in S : \bar{s} \in \mathfrak{m}_{R''}\}$. Since $k \subseteq R \subseteq R''$ and $k \subseteq \mathcal{O}_\xi \subseteq S$ map compatibly, $k \subseteq \tilde{R}$.

Now let $x \in Z$. Then ξ is a generization of x, so $\mathcal{O}_x \subseteq \mathcal{O}_\xi \subseteq S$, and with $\mathfrak{p}_\xi = \mathfrak{m}_\xi \cap \mathcal{O}_x$ we have $\mathcal{O}_{x,Z} = \mathcal{O}_x/\mathfrak{p}_\xi$, which is the image of $\nu : \mathcal{O}_x \to \mathcal{O}_\xi \to F$; so $\mathcal{O}_{x,Z} \subseteq F$, $\mathfrak{m}_{x,Z} = \nu(\mathfrak{m}_x)$ and

$\mathfrak{m}_x = \nu^{-1}(\mathfrak{m}_{x,Z})$. For $a \in \mathcal{O}_x$ the image of a in κ is $\nu(a)$ (because $\mathcal{O}_\xi \to S \to \kappa$ factors through F). Hence, using $R'' \cap F = R$,

$$a \in \tilde{R} \iff \nu(a) \in R, \qquad a \in \mathfrak{m}_{\tilde{R}} \iff \nu(a) \in \mathfrak{m}_{R''} \cap F = \mathfrak{m}_R.$$

Therefore $\mathcal{O}_x \subseteq \tilde{R} \iff \mathcal{O}_{x,Z} \subseteq R$ and, in that case, $\mathfrak{m}_{\tilde{R}} \cap \mathcal{O}_x = \nu^{-1}(\mathfrak{m}_R)$ while $\mathfrak{m}_x = \nu^{-1}(\mathfrak{m}_{x,Z})$; as ν maps onto $\mathcal{O}_{x,Z}$, these are equal iff $\mathfrak{m}_R \cap \mathcal{O}_{x,Z} = \mathfrak{m}_{x,Z}$. This is the claim. □

Converse of (a). Assume every valuation of K/k has at most one center on X; we verify the criterion of (4.3) (for $Y = \operatorname{Spec} k$). Let K' be a field, R' a valuation ring of K', $T = \operatorname{Spec} R'$, $U = \operatorname{Spec} K'$, with a k-morphism $T \to \operatorname{Spec} k$ (so $k \subseteq R'$), a morphism $U \to X$, and two morphisms $h, h' : T \to X$ extending it. Let ξ be the image of U, so $h(t_1) = h'(t_1) = \xi$, and $F = k(\xi) \subseteq K'$ (the same inclusion for h and h', as $h|_U = h'|_U$), $Z = \{\xi\}^-$. Put $x_0 = h(t_0)$, $x_0' = h'(t_0) \in Z$. By Lemma 4.4, R' dominates $\mathcal{O}_{x_0,Z}$ and $\mathcal{O}_{x_0',Z}$. Let $R = R' \cap F$, a valuation ring of F containing k (V2); it dominates $\mathcal{O}_{x_0,Z}$ and $\mathcal{O}_{x_0',Z}$ ($\mathfrak{m}_R \cap \mathcal{O} = \mathfrak{m}_{R'} \cap \mathcal{O} = \mathfrak{m}_{\mathcal{O}}$). By Lemma A the valuation $\tilde{R}$ of K/k dominates both $\mathcal{O}_{x_0,X}$ and $\mathcal{O}_{x_0',X}$, so by hypothesis $x_0 = x_0'$. Then h and h' have the same image points $t_0 \mapsto x_0$, $t_1 \mapsto \xi$ and the same residue field inclusion, so $h = h'$ by Lemma 4.4. By (4.3), X is separated over k.

Converse of (b). Assume every valuation of K/k has a unique center. By the converse of (a), X is separated; it is of finite type and noetherian. We verify the existence part of (4.7). Let R', K', T, U be as above with a k-morphism $T \to \operatorname{Spec} k$ and $U \to X$ with image ξ, $F = k(\xi) \subseteq K'$, $Z = \{\xi\}^-$, $R = R' \cap F$. Let $S, \tilde{R}$ be as in Lemma A. As $\tilde{R}$ is a valuation of K/k, it has a center $x_0 \in X$. *Claim:* $x_0 \in Z$. Since $\mathcal{O}_{x_0} \subseteq \tilde{R} \subseteq S$, the prime $\mathfrak{q} = \mathfrak{m}_S \cap \mathcal{O}_{x_0}$ corresponds to a point y (a generization of x_0) with $\mathcal{O}_y = (\mathcal{O}_{x_0})_{\mathfrak{q}} \subseteq S$ and $\mathfrak{m}_S \cap \mathcal{O}_y = \mathfrak{m}_y$; so S (a valuation of K/k) has center y. But S also has center ξ. By uniqueness $y = \xi$, so $x_0 \in \{\xi\}^- = Z$. By Lemma A, R dominates $\mathcal{O}_{x_0,Z}$; then R' does too (R' dominates R). By Lemma 4.4 there is a morphism $h : T \to X$ with $h(t_0) = x_0$, $h(t_1) = \xi$ inducing $k(\xi) \subseteq K'$, i.e. with $h|_U$ the given morphism, and h is a k-morphism because $R' \subseteq K'$. By (4.7), X is proper over k. (c)

(d) Let k be algebraically closed, X proper over k (so integral, finite type, separated), and $a \in \Gamma(X, \mathcal{O}_X)$; as X is integral, $a \in K$ and $a \in \mathcal{O}_x$ for all $x \in X$. Suppose $a \notin k$. Since k is algebraically closed, a is transcendental over k (an algebraic a would lie in k). Put $b = a^{-1}$; $k[b]$ is a polynomial ring and $\Lambda = k[b]_{(b)}$ is a local subring of K with maximal ideal (b). By (V1) there is a valuation ring R of K dominating Λ; it contains k, so is a valuation of K/k, and $a^{-1} = b \in \mathfrak{m}_R$. By (b), R has a center $x \in X$: $\mathcal{O}_x \subseteq R$. But $a \in \mathcal{O}_x \subseteq R$, so $1 = a \cdot a^{-1} \in \mathfrak{m}_R$, a contradiction. Hence $\Gamma(X, \mathcal{O}_X) \subseteq k$, and obviously $k \subseteq \Gamma(X, \mathcal{O}_X)$. □

Exercise 4.6

> **Problem.** Let $f : X \to Y$ be a proper morphism of affine varieties over k. Then f is a finite morphism. [*Hint*: Use (4.11A).]

Solution. We regard the affine varieties as integral affine schemes of finite type over k (Ch. II, Section 2): $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$ with A, B integral domains finitely generated over k, and f is given by a k-algebra homomorphism $\varphi : B \to A$; "proper" refers to the scheme morphism. Let $K = \operatorname{Frac}(A)$ and $B_1 = \varphi(B) \subseteq A \subseteq K$.

Claim: A is integral over B_1. By Thm 4.11A applied to the subring B_1 of the field K, the integral closure of B_1 in K is the intersection of all valuation rings R of K with $B_1 \subseteq R$. So it suffices to show $A \subseteq R$ for each such R. Let $T = \operatorname{Spec} R$, $U = \operatorname{Spec} K$, $i : U \to T$. We have morphisms $U \to X$ (corresponding to $A \subseteq K$) and $T \to Y$ (corresponding to $B \to B_1 \subseteq R$), and the diagram commutes since both composites $U \to Y$ correspond to $B \to B_1 \subseteq K$. As f is proper, Thm 4.7 gives $h : T \to X$, i.e. a ring homomorphism $\psi : A \to R$ whose composite with $R \subseteq K$ is the inclusion $A \subseteq K$ (this is the commutativity of the upper triangle). Thus ψ is injective with $A \subseteq R$ as subrings of K.

Since A is finitely generated as a k-algebra, it is finitely generated as a B_1-algebra; being also integral over B_1, it is a finite B_1-module (generated by monomials of bounded degree in integral generators). So A is a finite B-module via φ, i.e. f is finite. □

Exercise 4.7

Problem. *Schemes over* $\mathbf{R}$. For any scheme X_0 over $\mathbf{R}$, let $X = X_0 \times_{\mathbf{R}} \mathbf{C}$. Let $\alpha : \mathbf{C} \to \mathbf{C}$ be complex conjugation, and let $\sigma : X \to X$ be the automorphism obtained by keeping X_0 fixed and applying α to $\mathbf{C}$. Then X is a scheme over $\mathbf{C}$, and σ is a *semi-linear* automorphism, in the sense that we have a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathbf{C} & \xrightarrow{\alpha} & \operatorname{Spec} \mathbf{C}. \end{array}$$

Since $\sigma^2 = \mathrm{id}$, we call σ an *involution*.

(a) Now let X be a separated scheme of finite type over $\mathbf{C}$, let σ be a semilinear involution on X, and assume that for any two points $x_1, x_2 \in X$, there is an open affine subset containing both. (This last condition is satisfied for example if X is quasi-projective.) Show that there is a unique separated scheme X_0 of finite type over $\mathbf{R}$, such that $X_0 \times_{\mathbf{R}} \mathbf{C} \cong X$, and such that this isomorphism identifies the given involution of X with the one on $X_0 \times_{\mathbf{R}} \mathbf{C}$ described above.

(b) For the following statements (b)–(e), X_0 will denote a separated scheme of finite type over $\mathbf{R}$, and X, σ will denote the corresponding scheme with involution over $\mathbf{C}$. Show that X_0 is affine if and only if X is.

(c) If X_0, Y_0 are two such schemes over $\mathbf{R}$, then to give a morphism $f_0 : X_0 \to Y_0$ is equivalent to giving a morphism $f : X \to Y$ which commutes with the involutions, i.e., $f \circ \sigma_X = \sigma_Y \circ f$.

(d) If $X \cong \mathbf{A}^1_{\mathbf{C}}$, then $X_0 \cong \mathbf{A}^1_{\mathbf{R}}$.

(e) If $X \cong \mathbf{P}^1_{\mathbf{C}}$, then either $X_0 \cong \mathbf{P}^1_{\mathbf{R}}$, or X_0 is isomorphic to the conic in $\mathbf{P}^2_{\mathbf{R}}$ given by the homogeneous equation $x_0^2 + x_1^2 + x_2^2 = 0$.

Solution. Notation. $\chi : \mathbf{C} \to \mathbf{C}$ is conjugation, $\alpha = \operatorname{Spec} \chi$. A *semilinear involution* of a $\mathbf{C}$-algebra A (structure map $c : \mathbf{C} \to A$, written $z \mapsto z$) is a ring automorphism s with $s^2 = \mathrm{id}$ and $s(za) = \bar{z}s(a)$; for a semilinear involution σ of a $\mathbf{C}$-scheme X we mean $\pi\sigma = \alpha\pi$ ($\pi : X \to \operatorname{Spec} \mathbf{C}$), $\sigma^2 = \mathrm{id}$. Then σ is an automorphism of the scheme X (not over $\mathbf{C}$), and for $W \subseteq X$ affine open, $\sigma(W)$ is again affine open. The check that the displayed diagram commutes for $X_0 \times_{\mathbf{R}} \mathbf{C}$: $\sigma = (p_1, \alpha p_2)$, so $p_2\sigma = \alpha p_2$.

Lemma 1 (algebra). Let A be a $\mathbf{C}$-algebra with a semilinear involution s and $A_0 = A^s$ (an $\mathbf{R}$-algebra).

(i) $A = A_0 \oplus iA_0$, and $A_0 \otimes_{\mathbf{R}} \mathbf{C} \to A$, $a \otimes z \mapsto az$, is an isomorphism carrying $\mathrm{id} \otimes \chi$ to s. Conversely, for any $\mathbf{R}$-algebra A_0 and $A = A_0 \otimes_{\mathbf{R}} \mathbf{C}$, $s = \mathrm{id} \otimes \chi$ one has $A^s = A_0$.

(ii) A is a finite A_0-module, so $\rho : \operatorname{Spec} A \to \operatorname{Spec} A_0$ is finite, surjective and closed, and its fibres are exactly the orbits of s.

(iii) For $f \in A_0$, $(A_f)^s = (A_0)_f$.

(iv) If A is a finitely generated $\mathbf{C}$-algebra, then A_0 is a finitely generated $\mathbf{R}$-algebra.

Proof. (i) For $a \in A$: $a = \frac{1}{2}(a+sa) + i \cdot \frac{1}{2i}(a-sa)$, and $s\big(\frac{1}{2i}(a-sa)\big) = \frac{1}{-2i}(sa-a) = \frac{1}{2i}(a-sa)$, so $A = A_0 + iA_0$. If $a = ib$ with $a, b \in A_0$ then $a = sa = -ib = -a$, so $a = 0$; so the sum is direct. Hence $A_0 \otimes_{\mathbf{R}} \mathbf{C} = A_0 \otimes 1 \oplus A_0 \otimes i \to A_0 \oplus iA_0 = A$ is bijective; compatibility with s is clear. The converse is the same decomposition ($s = 1$ on $A_0 \otimes 1$, -1 on $A_0 \otimes i$). (ii) $A = A_0 + A_0 i$ is finite over A_0, so ρ is finite, hence closed (Ex. 4.1, step (i)); it is surjective by lying over ($A_0 \subseteq A$ integral); fibres are orbits of the finite group $\{1, s\}$ acting on A with $A^s = A_0$ (Atiyah–Macdonald, Ex. 5.13). (iii) If $a/f^n \in A_f$ is s-invariant, then $f^m(sa - a) = 0$ for some m, so $f^m a \in A_0$ and $a/f^n = f^m a / f^{n+m} \in (A_0)_f$. (iv) Let $a_1, \dots, a_m$ generate A over $\mathbf{C}$, $a_j = u_j + iv_j$ ($u_j, v_j \in A_0$), and $A_0' \subseteq A_0$ the $\mathbf{R}$-subalgebra generated by the u_j, v_j. The image of $A_0' \otimes \mathbf{C} \to A_0 \otimes \mathbf{C} = A$ contains $\mathbf{C}$ and all a_j, so is A. As $M \otimes_{\mathbf{R}} \mathbf{C} \cong M \oplus M$ for any $\mathbf{R}$-module M, we get $(A_0/A_0') \otimes \mathbf{C} = 0$, so $A_0 = A_0'$. □

Lemma 2. Let $W = \operatorname{Spec} A$ with a semilinear involution σ (ring map s), $q : |W| \to |W|/\sigma$ the quotient map, and let $(q_*\mathcal{O}_W)^\sigma$ be the sheaf of σ-invariant sections. Then the ringed space $(|W|/\sigma, (q_*\mathcal{O}_W)^\sigma)$ is isomorphic to $\operatorname{Spec} A_0$, with q corresponding to ρ.

Proof. By Lemma 1(ii), ρ is continuous, surjective, closed with fibres the σ-orbits, so it induces a bijection $\bar\rho : |W|/\sigma \to \operatorname{Spec} A_0$, which is a homeomorphism (a closed subset of $|W|/\sigma$ is $q(C)$, C closed and σ-stable, and $\bar\rho(q(C)) = \rho(C)$ is closed). On the basic open sets $D(f) \subseteq \operatorname{Spec} A_0$, $f \in A_0$, we have $\rho^{-1}D(f) = D(f) \subseteq W$ and $\mathcal{O}_W(D(f))^\sigma = (A_f)^s = (A_0)_f = \mathcal{O}_{\operatorname{Spec} A_0}(D(f))$ by Lemma 1(iii), compatibly with restrictions. Two sheaves with a compatible isomorphism on a basis are isomorphic. □

Lemma 3. Let X_0 be any scheme over $\mathbf{R}$, $X = X_0 \times_{\mathbf{R}} \mathbf{C}$, $\sigma = \mathrm{id} \times \alpha$, $q = p_1 : X \to X_0$. Then $q\sigma = q$; q is surjective, closed, and its fibres are the σ-orbits; and $q^\# : \mathcal{O}_{X_0} \to q_*\mathcal{O}_X$ is injective with image $(q_*\mathcal{O}_X)^\sigma$.

Proof. Let $V = \operatorname{Spec} B_0 \subseteq X_0$ be open affine. Then $q^{-1}(V) = \operatorname{Spec}(B_0 \otimes_{\mathbf{R}} \mathbf{C})$ with σ corresponding to $s = \mathrm{id} \otimes \chi$, $A^s = B_0$ (Lemma 1(i)), and $q|$ is the map ρ of Lemma 1(ii). This gives the statements on q. For the sheaf statement, on a basic open $D(f) \subseteq V$ ($f \in B_0$) we have $\mathcal{O}_{X_0}(D(f)) = (B_0)_f \to \mathcal{O}_X(q^{-1}D(f)) = A_f$ injective with image $(A_f)^s$ by Lemma 1(iii); such sets form a basis of X_0. □

(a) Existence. Let (X, σ) be as in (a). Call an open set *σ-stable* if $\sigma(W) = W$.

Step 1: X is covered by σ-stable open affines. Given $x \in X$, there is an open affine $W \ni x, \sigma(x)$ by hypothesis. Then $\sigma(W)$ is open affine and contains $\sigma(x), x$. Since X is separated over the affine scheme $\operatorname{Spec} \mathbf{C}$, $W' = W \cap \sigma(W)$ is affine (Ex. 4.3); it is σ-stable ($\sigma^2 = \mathrm{id}$) and contains x.

Step 2: the ringed space. Let $q : |X| \to |X|/\sigma$ be the quotient, and $\mathcal{O}_0 = (q_*\mathcal{O}_X)^\sigma$, where σ acts on $q_*\mathcal{O}_X$ through $\sigma^\#$ (note $q^{-1}(V)$ is σ-stable for every open V). Put $X_0 = (|X|/\sigma, \mathcal{O}_0)$.

If $W = \operatorname{Spec} A$ is a σ-stable open affine, then $q(W)$ is open ($q^{-1}q(W) = W$), the semilinear involution is $s = \sigma^{\#}_W$ on A (semilinearity: $\pi|_W$ corresponds to $c : \mathbf{C} \to A$, and $\pi\sigma = \alpha\pi$ says $s(c(z)) = c(\bar{z})$), and by Lemma 2, $(q(W), \mathcal{O}_0|) \cong \operatorname{Spec} A^s$. Hence X_0 is a scheme, covered by affine schemes $W_0 = \operatorname{Spec} A^s$. Because X is quasi-compact (finite type over $\mathbf{C}$), it is covered by finitely many σ-stable affines $W_1, \dots, W_n$, and by Lemma 1(iv) each A^s_j is a finitely generated $\mathbf{R}$-algebra. So X_0 is of finite type over $\mathbf{R}$ once we give it an $\mathbf{R}$-structure: the elements of $\mathbf{R} \subseteq \mathbf{C} \to \Gamma(X, \mathcal{O}_X)$ are σ-invariant, giving $\mathbf{R} \to \Gamma(X_0, \mathcal{O}_0)$ and $\psi : X_0 \to \operatorname{Spec} \mathbf{R}$.

Step 3: q is a morphism of schemes $X \to X_0$, namely $(q, q^{\#})$ with $q^{\#} : \mathcal{O}_0 \hookrightarrow q_*\mathcal{O}_X$; on W it is $\rho_W : \operatorname{Spec} A \to \operatorname{Spec} A_0$ (Lemma 2). It satisfies $q\sigma = q$ and $\psi q = (\operatorname{Spec} \mathbf{C} \to \operatorname{Spec} \mathbf{R}) \circ \pi$.

Step 4: $X \cong X_0 \times_{\mathbf{R}} \mathbf{C}$. Let $\theta = (q, \pi) : X \to X_0 \times_{\mathbf{R}} \mathbf{C}$. The open sets $W_0 \times_{\mathbf{R}} \mathbf{C} = \operatorname{Spec}(A_0 \otimes_{\mathbf{R}} \mathbf{C})$ ($W_0 = q(W)$) cover $X_0 \times_{\mathbf{R}} \mathbf{C}$, $\theta^{-1}(W_0 \times \mathbf{C}) = q^{-1}(W_0) = W$, and $\theta|_W$ is Spec of $A_0 \otimes_{\mathbf{R}} \mathbf{C} \to A$, $a \otimes z \mapsto az$, an isomorphism by Lemma 1(i). So θ is an isomorphism of $\mathbf{C}$-schemes, and $\theta\sigma = (q\sigma, \pi\sigma) = (q, \alpha\pi) = (\mathrm{id} \times \alpha)\theta$: it identifies σ with $\mathrm{id} \times \alpha$.

Separatedness of X_0 (needed for (b)–(e)). It suffices to show $\Delta^{-1}_{X_0}(W_{i,0} \times_{\mathbf{R}} W_{j,0}) = W_{i,0} \cap W_{j,0} \to W_{i,0} \times_{\mathbf{R}} W_{j,0}$ is a closed immersion. Now $W_{i,0} \cap W_{j,0} = q(W_i \cap W_j)$, and $W_i \cap W_j = \operatorname{Spec} A_{ij}$ is affine, σ-stable, so $W_{i,0} \cap W_{j,0} = \operatorname{Spec} A^s_{ij}$ by Lemma 2. We need $A^s_i \otimes_{\mathbf{R}} A^s_j \to A^s_{ij}$ surjective. After applying $- \otimes_{\mathbf{R}} \mathbf{C}$ it becomes $A_i \otimes_{\mathbf{C}} A_j \to A_{ij}$ (by Lemma 1(i)), which is surjective because X is separated; and $M \otimes_{\mathbf{R}} \mathbf{C} = 0$ forces $M = 0$.

(a) Uniqueness. Let X_0' be another scheme over $\mathbf{R}$ with a $\mathbf{C}$-isomorphism $\kappa : X_0' \times_{\mathbf{R}} \mathbf{C} \to X$, $\kappa(\mathrm{id} \times \alpha) = \sigma\kappa$. Put $q' = p_1\kappa^{-1} : X \to X_0'$. Then $q'\sigma = q'$ and by Lemma 3 (transported by κ) q' is a surjective closed map with fibres the σ-orbits, and $\mathcal{O}_{X_0'} \cong (q'_*\mathcal{O}_X)^{\sigma}$. Since q has the same properties (Step 3), there is a unique homeomorphism $\phi : |X_0| \to |X_0'|$ with $\phi q = q'$, and an isomorphism of sheaves $\phi^{\#} : \mathcal{O}_{X_0'} \to \phi_*\mathcal{O}_{X_0}$, both being the invariants of $q_*\mathcal{O}_X$. An isomorphism of ringed spaces between schemes is an isomorphism of schemes; it is compatible with the maps to $\operatorname{Spec} \mathbf{R}$ and with the identifications with X. So $X_0' \cong X_0$ (compatibly with the isomorphisms to X).

(b) If $X_0 = \operatorname{Spec} A_0$ is affine, then $X = \operatorname{Spec}(A_0 \otimes_{\mathbf{R}} \mathbf{C})$ is affine. Conversely if $X = \operatorname{Spec} A$ is affine, then X itself is a σ-stable open affine, so by Lemma 3 and Lemma 2, $(X_0, \mathcal{O}_{X_0}) \cong (|X|/\sigma, (q_*\mathcal{O}_X)^{\sigma}) \cong \operatorname{Spec} A^s$ is affine.

(c) Given $f_0 : X_0 \to Y_0$ put $f = f_0 \times \mathrm{id}_{\mathbf{C}} : X \to Y$. It is a $\mathbf{C}$-morphism and $f\sigma_X = \sigma_Y f$ since $(f_0 \times \mathrm{id})(\mathrm{id} \times \alpha) = (\mathrm{id} \times \alpha)(f_0 \times \mathrm{id})$.

Injectivity. If $f_0 \times \mathrm{id} = f_0' \times \mathrm{id}$, then $f_0 q_X = q_Y f = f_0' q_X$. As q_X is surjective, $f_0 = f_0'$ on points; on sheaves, $f_0^{\#} : \mathcal{O}_{Y_0} \to f_{0*}\mathcal{O}_{X_0}$ composed with the injection $f_{0*}\mathcal{O}_{X_0} \hookrightarrow f_{0*}q_{X*}\mathcal{O}_X$ equals the corresponding map for f_0' (both are $(q_Y f)^{\#}$), so $f_0^{\#} = f_0'^{\#}$.

Surjectivity. Let $f : X \to Y$ be a $\mathbf{C}$-morphism commuting with the involutions. Then $q_Y f\sigma_X = q_Y\sigma_Y f = q_Y f$ is constant on σ_X-orbits, so it induces a continuous map $f_0 : |X_0| = |X|/\sigma \to |Y_0|$ with $f_0 q_X = q_Y f$. For $V_0 \subseteq Y_0$ open and $V = q_Y^{-1}V_0$, the map $f^{\#} : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}V)$ commutes with the involutions (as $f\sigma_X = \sigma_Y f$), hence maps invariants to invariants:

$$\mathcal{O}_{Y_0}(V_0) = \mathcal{O}_Y(V)^{\sigma} \to \mathcal{O}_X(f^{-1}V)^{\sigma} = \mathcal{O}_{X_0}(f_0^{-1}V_0)$$

(using $f^{-1}V = q_X^{-1}f_0^{-1}V_0$). This defines $f_0^{\#}$ and a morphism of ringed spaces. It is local: for $x \in X$, $x_0 = q_X(x)$, the stalk maps satisfy $(O_{Y_0, f_0 x_0} \to O_{X_0, x_0} \to O_{X,x}) = (O_{Y_0, f_0 x_0} \to O_{Y, fx} \to O_{X,x})$, which sends the maximal ideal into $\mathfrak{m}_x$; since $O_{X_0,x_0} \to O_{X,x}$ is a local homomorphism,

the preimage of $\mathfrak{m}_x$ is $\mathfrak{m}_{x_0}$. So f_0 is a morphism of schemes, over $\mathbf{R}$, and $(f_0 \times \mathrm{id}) = f$ (both are determined by their compositions $q_Y f = f_0 q_X$ and $\pi_Y f = \pi_X$ to Y_0 and $\operatorname{Spec} \mathbf{C}$). (c)

(The correspondence is compatible with composition, so isomorphisms correspond to isomorphisms commuting with the involutions. In particular *if* $(X, \sigma) \cong (X', \sigma')$ *then* $X_0 \cong X'_0$.)

(d) Let $X \cong \mathbf{A}^1_{\mathbf{C}} = \operatorname{Spec} \mathbf{C}[t]$; transporting σ, we may assume $X = \operatorname{Spec} \mathbf{C}[t]$ and σ is given by a semilinear involution s of $\mathbf{C}[t]$ (so $s|_{\mathbf{C}} = \chi$). Let $d = \deg s(t)$; $s(t)$ is nonconstant (else $s(\mathbf{C}[t]) \subseteq \mathbf{C}$, not surjective), and $\deg s(p) = d \deg p$ for all p (coefficients are conjugated, the degree multiplies by d). Since s is surjective some $s(p)$ has degree 1, so $d = 1$: $s(t) = at + b$, $a \neq 0$. The relation $s^2(t) = t$ gives $|a|^2 = 1$ and $\bar{a}b + \bar{b} = 0$. Choose c with $c^2 = a$ (so $|c| = 1$, $\bar{c}a = c$) and put $w = \bar{c}b$; then $w + \bar{w} = \bar{c}b + c\bar{b} = \bar{c}b - c\bar{a}b = \bar{c}b - \bar{c}b = 0$ (using $\bar{b} = -\bar{a}b$ and $c\bar{a} = \bar{c}$). So $d_0 := w/2$ satisfies $d_0 - \bar{d}_0 = w$. Let $u = ct + d_0$. Then

$$s(u) = \bar{c}(at + b) + \bar{d}_0 = ct + (\bar{c}b + \bar{d}_0) = ct + d_0 = u.$$

Since $c \neq 0$, $\mathbf{C}[t] = \mathbf{C}[u]$ with u transcendental, and $s(\sum a_k u^k) = \sum \bar{a}_k u^k$. Hence $(\mathbf{C}[t], s) \cong (\mathbf{R}[u] \otimes_{\mathbf{R}} \mathbf{C}, \mathrm{id} \otimes \chi)$, i.e. $(X, \sigma) \cong (\mathbf{A}^1_{\mathbf{R}} \times_{\mathbf{R}} \mathbf{C}, \mathrm{id} \times \alpha)$, and $X_0 \cong \mathbf{A}^1_{\mathbf{R}}$ by (c).

(e) Let $B = \mathbf{C}[x_0, x_1]$, $P = \mathbf{P}^1_{\mathbf{C}} = \operatorname{Proj} B$. By (c) we may replace (X, σ) by (P, σ) for an arbitrary $\mathbf{C}$-isomorphism $X \cong P$. A graded ring automorphism S of B (possibly semilinear: $S(zb) = \bar{z}S(b)$) induces an automorphism $S^* = \operatorname{Proj} S$ of the scheme P, with $(SS')^* = S'^* S^*$, and $\pi S^* = \alpha \pi$ if S is semilinear. We use the standard fact $\operatorname{Aut}_{\mathbf{C}} \mathbf{P}^1_{\mathbf{C}} = PGL_2(\mathbf{C})$ (II.7.1.1): every $\mathbf{C}$-automorphism is N^* for a linear graded automorphism N (substitution $x_i \mapsto \sum N_{ij} x_j$, $N \in GL_2(\mathbf{C})$), and $N^* = \mathrm{id}$ iff $N|_{B_1}$ is scalar. Let κ be the coefficient conjugation of B; $\kappa^* = \mathrm{id} \times \alpha$ on $\mathbf{P}^1_{\mathbf{R}} \times_{\mathbf{R}} \mathbf{C} = \operatorname{Proj}(\mathbf{R}[x_0, x_1] \otimes_{\mathbf{R}} \mathbf{C})$.

Step 1. If σ is a semilinear involution of P, then $g = \sigma\kappa^*$ satisfies $\pi g = \alpha\pi\kappa^* = \alpha\alpha\pi = \pi$, so $g = N^*$ is $\mathbf{C}$-linear, hence $\sigma = g\kappa^* = N^*\kappa^* = (\kappa N)^* =: S^*$ with $S = \kappa \circ N$ semilinear graded. Let $L = B_1 = \mathbf{C}x_0 \oplus \mathbf{C}x_1$ and $\varphi = S|_L$, a semilinear bijection of L. Then $(S^2)^* = \sigma^2 = \mathrm{id}$ and S^2 is linear, so $\varphi^2 = \lambda \,\mathrm{id}_L$, $\lambda \in \mathbf{C}^*$. Applying φ to $\varphi^2 = \lambda$ gives $\varphi^3 \ell = \bar{\lambda}\varphi\ell$ and $\varphi^3\ell = \lambda\varphi\ell$, so $\lambda \in \mathbf{R}$. For $c \in \mathbf{C}^*$, $S_c(f) = c^d S(f)$ for $f \in B_d$ is again a semilinear graded automorphism with $S_c^* = S^*$ and $(S_c|_L)^2 = |c|^2\lambda$. So we may assume $\lambda = \pm 1$.

Case $\lambda = 1$. φ is a semilinear involution of the $\mathbf{C}$-space L, so by Lemma 1(i) (for the module L) $L = L^\varphi \oplus iL^\varphi$; take an $\mathbf{R}$-basis y_0, y_1 of L^φ; it is a $\mathbf{C}$-basis of L, $B = \mathbf{C}[y_0, y_1]$ and $S(\sum a_I y^I) = \sum \bar{a}_I y^I$. The graded $\mathbf{C}$-isomorphism $B \to B$, $x_i \mapsto y_i$, intertwines κ and S. Hence $(P, \sigma) \cong (\mathbf{P}^1_{\mathbf{R}} \times \mathbf{C}, \kappa^*)$ and $X_0 \cong \mathbf{P}^1_{\mathbf{R}}$ by (c).

Case $\lambda = -1$. Pick $0 \neq v \in L$. Then $v, \varphi v$ are $\mathbf{C}$-independent (if $\varphi v = cv$ then $\varphi^2 v = |c|^2 v \neq -v$). Put $y_0 = v$, $y_1 = \varphi v$: $\varphi y_0 = y_1$, $\varphi y_1 = -y_0$. So (B, S) is isomorphic to $(\mathbf{C}[y_0, y_1], S)$ with S the semilinear automorphism determined by $y_0 \mapsto y_1$, $y_1 \mapsto -y_0$.

The model. Let $Q \subseteq \mathbf{P}^2_{\mathbf{R}}$ be $\operatorname{Proj} \mathbf{R}[u, v, w]/(q)$, $q = u^2 + v^2 + w^2$. Then $Q_{\mathbf{C}} = \operatorname{Proj} \mathbf{C}[u, v, w]/(q)$, with $\sigma_Q = \kappa_Q^*$ (κ_Q = conjugation of coefficients). Let $B' = \mathbf{C}[s, t]$ and $\varepsilon : \mathbf{C}[u, v, w]/(q) \to B'^{(2)}$ (even-degree part), $u \mapsto \frac{1}{2}(s^2 - t^2)$, $v \mapsto \frac{1}{2i}(s^2 + t^2)$, $w \mapsto st$. It is well defined: $\frac{1}{4}(s^2 - t^2)^2 - \frac{1}{4}(s^2 + t^2)^2 + s^2t^2 = 0$. It is surjective ($s^2 = \varepsilon(u + iv)$, $-t^2 = \varepsilon(u - iv)$, $st = \varepsilon(w)$) and injective by dimension count: in degree d, $\dim \mathbf{C}[u, v, w]_d/(q)_d = \binom{d+2}{2} - \binom{d}{2} = 2d + 1 = \dim B'_{2d}$. So ε is a graded isomorphism ($\deg d \mapsto 2d$) and $Q_{\mathbf{C}} \cong \operatorname{Proj} B'^{(2)} \cong \operatorname{Proj} B' = \mathbf{P}^1_{\mathbf{C}}$ (Ex. 5.13, naturally). Let S' be the semilinear graded automorphism of B' with $S'(s) = it$, $S'(t) = -is$. Then $S'(\frac{1}{2}(s^2 - t^2)) = \frac{1}{2}((it)^2 - (-is)^2) = \frac{1}{2}(s^2 - t^2)$, $S'(\frac{1}{2i}(s^2 + t^2)) = \frac{1}{-2i}((it)^2 + (-is)^2) = \frac{1}{2i}(s^2 + t^2)$,

$S'(st) = (it)(-is) = st$. So $S'\varepsilon = \varepsilon\kappa_Q$ (both semilinear ring maps agreeing on generators), and therefore $(Q_{\mathbf{C}}, \sigma_Q) \cong (\mathbf{P}^1_{\mathbf{C}}, S'^*)$.

Comparison. Put $y'_0 = s$, $y'_1 = it$ (a $\mathbf{C}$-basis of B'_1): $S'(y'_0) = y'_1$ and $S'(y'_1) = \bar{i}S'(t) = (-i)(-is) = -y'_0$. So the graded $\mathbf{C}$-isomorphism $\mathbf{C}[y_0, y_1] \to B'$, $y_0 \mapsto y'_0$, $y_1 \mapsto y'_1$ intertwines S and S'. Thus $(P, \sigma) \cong (Q_{\mathbf{C}}, \sigma_Q)$ and $X_0 \cong Q$ by (c): the conic $x_0^2 + x_1^2 + x_2^2 = 0$. □

(Remark: the two cases are distinct, since $Q(\mathbf{R}) = \emptyset$ while $\mathbf{P}^1_{\mathbf{R}}$ has real points.)

Exercise 4.8

Problem. Let $\mathcal{P}$ be a property of morphisms of schemes such that:

(a) a closed immersion has $\mathcal{P}$;

(b) a composition of two morphisms having $\mathcal{P}$ has $\mathcal{P}$;

(c) $\mathcal{P}$ is stable under base extension. Then show that:

(d) a product of morphisms having $\mathcal{P}$ has $\mathcal{P}$;

(e) if $f : X \to Y$ and $g : Y \to Z$ are two morphisms, and if $g \circ f$ has $\mathcal{P}$ and g is separated, then f has $\mathcal{P}$;

(f) if $f : X \to Y$ has $\mathcal{P}$, then $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$ has $\mathcal{P}$.

[*Hint*: For (e), consider the graph morphism $\Gamma_f : X \to X \times_Z Y$ and note that it is obtained by base extension from the diagonal morphism $\Delta : Y \to Y \times_Z Y$.]

Solution. (d) Let $f : X \to Y$ and $f' : X' \to Y'$ be morphisms of S-schemes with $\mathcal{P}$. Then $f \times_S f'$ is the composition

$$X \times_S X' \xrightarrow{\mathrm{id}_X \times f'} X \times_S Y' \xrightarrow{f \times \mathrm{id}_{Y'}} Y \times_S Y'.$$

The first map is the base extension of f' by $p_2 : X \times_S Y' \to Y'$ (indeed $X \times_S X' = (X \times_S Y') \times_{Y'} X'$), the second is the base extension of f by $p_1 : Y \times_S Y' \to Y$. Both have $\mathcal{P}$ by (c), so the composition has $\mathcal{P}$ by (b).

(e) As in Ex. 4.4 (Step 1, with $S = Z$), $\Gamma_f : X \to X \times_Z Y$ is the base extension of $\Delta : Y \to Y \times_Z Y$ by $f \times \mathrm{id}_Y : X \times_Z Y \to Y \times_Z Y$. Since g is separated, Δ is a closed immersion, so has $\mathcal{P}$ by (a), hence Γ_f has $\mathcal{P}$ by (c). Next, $p_2 : X \times_Z Y \to Y$ is the base extension of $g \circ f : X \to Z$ by $g : Y \to Z$; since $g \circ f$ has $\mathcal{P}$, so does p_2 by (c). Finally $f = p_2 \circ \Gamma_f$ has $\mathcal{P}$ by (b).

(f) Let $i : X_{\mathrm{red}} \to X$ and $j : Y_{\mathrm{red}} \to Y$ be the closed immersions. Then $f \circ i = j \circ f_{\mathrm{red}}$. Here $f \circ i$ has $\mathcal{P}$ by (a), (b). The closed immersion j is separated (it is a monomorphism, so its diagonal is an isomorphism). By (e) applied to f_{red} and $g = j$, f_{red} has $\mathcal{P}$. □

Exercise 4.9

Problem. Show that a composition of projective morphisms is projective. [*Hint*: Use the Segre embedding defined in (I, Ex. 2.14) and show that it gives a closed immersion $\mathbf{P}^r \times \mathbf{P}^s \to \mathbf{P}^{rs+r+s}$.] Conclude that projective morphisms have properties (a)–(f) of (Ex. 4.8) above.

Solution. Recall $\mathbf{P}^n_Y = \mathbf{P}^n_{\operatorname{Spec}\mathbf{Z}} \times_{\operatorname{Spec}\mathbf{Z}} Y$; $f : X \to Y$ is projective if it is a closed immersion $X \to \mathbf{P}^n_Y$ followed by the projection.

Step 1: Segre. Let $N = rs + r + s$. We construct a closed immersion $\mathbf{P}^r_{\mathbf{Z}} \times_{\mathbf{Z}} \mathbf{P}^s_{\mathbf{Z}} \to \mathbf{P}^N_{\mathbf{Z}}$, where $\mathbf{P}^r = \operatorname{Proj}\mathbf{Z}[x_0, \ldots, x_r]$, $\mathbf{P}^s = \operatorname{Proj}\mathbf{Z}[y_0, \ldots, y_s]$, $\mathbf{P}^N = \operatorname{Proj}\mathbf{Z}[z_{kl}]$. For $0 \le i \le r$, $0 \le j \le s$, let $V_{ij} = D_+(x_i) \times D_+(y_j) = \operatorname{Spec}\mathbf{Z}[x_k/x_i,\ y_l/y_j]$ and $W_{ij} = D_+(z_{ij}) = \operatorname{Spec}\mathbf{Z}[z_{kl}/z_{ij}]$, and define $\psi_{ij} : V_{ij} \to W_{ij}$ by the ring map

$$z_{kl}/z_{ij} \longmapsto (x_k/x_i)(y_l/y_j).$$

It is surjective on rings (it hits x_k/x_i via z_{kj}/z_{ij} and y_l/y_j via z_{il}/z_{ij}, which generate). On V_{ij} one has $\psi_{ij}^{-1} D(z_{i'j'}/z_{ij}) = D\big((x_{i'}/x_i)(y_{j'}/y_j)\big) = V_{ij} \cap V_{i'j'}$, and on $V_{ij} \cap V_{i'j'}$ both ψ_{ij} and $\psi_{i'j'}$ send $z_{kl}/z_{i'j'} = (z_{kl}/z_{ij})/(z_{i'j'}/z_{ij})$ to $x_k y_l/(x_{i'} y_{j'})$. So the ψ_{ij} glue to a morphism $\psi : \mathbf{P}^r \times \mathbf{P}^s \to \mathbf{P}^N$ with $\psi^{-1}(W_{ij}) = V_{ij}$. Since the W_{ij} cover $\mathbf{P}^N$ and each ψ_{ij} is a closed immersion (surjective ring map), ψ is a closed immersion (this is local on the target). For an arbitrary base scheme S, base change along $S \to \operatorname{Spec}\mathbf{Z}$ gives a closed immersion $\mathbf{P}^r_S \times_S \mathbf{P}^s_S \to \mathbf{P}^N_S$.

Step 2: composition. Let $f : X \to Y$ and $g : Y \to Z$ be projective, with closed immersions $X \hookrightarrow \mathbf{P}^r_Y$ and $Y \hookrightarrow \mathbf{P}^s_Z$. Then

$$X \hookrightarrow \mathbf{P}^r_Y = \mathbf{P}^r_{\mathbf{Z}} \times Y \hookrightarrow \mathbf{P}^r_{\mathbf{Z}} \times \mathbf{P}^s_Z = \mathbf{P}^r_Z \times_Z \mathbf{P}^s_Z \hookrightarrow \mathbf{P}^N_Z,$$

where the second map is the base extension of $Y \hookrightarrow \mathbf{P}^s_Z$ and the third is Step 1. A composition of closed immersions is a closed immersion, and the composite $X \to \mathbf{P}^N_Z \to Z$ equals $g \circ f$. So $g \circ f$ is projective.

Step 3: properties (a)–(c). (a) A closed immersion $X \to Y = \mathbf{P}^0_Y$ is projective. (b) is Step 2. (c) If $X \hookrightarrow \mathbf{P}^n_Y$ and $Y' \to Y$ is any morphism, then $X \times_Y Y' \hookrightarrow \mathbf{P}^n_Y \times_Y Y' = \mathbf{P}^n_{Y'}$ is a closed immersion (base change), compatible with the projections to Y'. So projective morphisms satisfy (a), (b), (c) of Ex. 4.8, and therefore, by Ex. 4.8, also (d), (e), (f) there. □

Exercise 4.10

Problem. (Starred in the book.) *Chow's Lemma.* This result says that proper morphisms are fairly close to projective morphisms. Let X be proper over a noetherian scheme S. Then there is a scheme X' and a morphism $g : X' \to X$ such that X' is projective over S, and there is an open dense subset $U \subseteq X$ such that g induces an isomorphism of $g^{-1}(U)$ to U. Prove this result in the following steps.

(a) Reduce to the case X irreducible.

(b) Show that X can be covered by a finite number of open subsets U_i, $i = 1, \ldots, n$, each of which is quasi-projective over S. Let $U_i \to P_i$ be an open immersion of U_i into a scheme P_i which is projective over S.

(c) Let $U = \bigcap U_i$, and consider the map

$$f : U \to X \times_S P_1 \times_S \cdots \times_S P_n$$

deduced from the given maps $U \to X$ and $U \to P_i$. Let X' be the closed image subscheme structure (Ex. 3.11d) $\overline{f(U)}$. Let $g : X' \to X$ be the projection onto the first factor, and let $h : X' \to P = P_1 \times_S \cdots \times_S P_n$ be the projection onto the product of the remaining factors. Show that h is a closed immersion, hence X' is projective over S.

(d) Show that $g^{-1}(U) \to U$ is an isomorphism, thus completing the proof.

Solution. Conventions. X is of finite type over the noetherian S, hence noetherian, and separated. "Projective" and "quasi-projective" are as in the text: quasi-projective means an open immersion followed by a projective morphism. By Ex. 4.9 and Ex. 4.8, projective morphisms are stable under composition and base change, closed immersions are projective, and proper schemes are stable under products etc. (Cor. 4.8). All morphisms $U \to Y$ of noetherian schemes are quasi-compact, so for a morphism $f : U \to Y$ the sheaf $\ker(\mathcal{O}_Y \to f_*\mathcal{O}_U)$ is quasi-coherent and defines the image subscheme structure of Ex. 3.11d: the smallest closed subscheme through which f factors. This construction is compatible with restriction to open subsets of Y.

(a) Reduction to X irreducible. Let $X_1, \dots, X_m$ be the irreducible components of $|X|$ and $W_i = X \setminus \bigcup_{j \neq i} X_j$, a nonempty open subset of X, irreducible. Let $\bar{X}_i \subseteq X$ be the closed subscheme which is the scheme-theoretic closure of the open subscheme W_i (image structure of $W_i \hookrightarrow X$). Then $|\bar{X}_i| = X_i$, $\bar{X}_i|_{W_i} = X|_{W_i}$ (the ideal sheaf restricted to W_i vanishes), $\bar{X}_i$ is irreducible, and proper over S (closed immersion followed by $X \to S$, Cor. 4.8). Suppose the theorem holds for irreducible schemes, giving $g_i : X_i' \to \bar{X}_i$ projective over S and $U_i \subseteq \bar{X}_i$ open dense with $g_i^{-1}(U_i) \cong U_i$. Replacing U_i by $U_i \cap W_i$ (still open, dense in $\bar{X}_i$, since $W_i \neq \emptyset$; and the isomorphism persists over a smaller open set), we may assume $U_i \subseteq W_i$. Then U_i is also an open subscheme of X (as $\bar{X}_i|_{W_i} = X|_{W_i}$), the U_i are pairwise disjoint (since $W_i \cap W_j = \emptyset$), and $U = \bigcup U_i \cong \bigsqcup U_i$ is open and dense in X ($\overline{U_i} = X_i$). Let $X' = \bigsqcup X_i'$ and $g : X' \to X$ the morphism with $g|_{X_i'} = (\bar{X}_i \hookrightarrow X) \circ g_i$. Then $g^{-1}(U) \cap X_i' = g_i^{-1}(U \cap X_i) = g_i^{-1}(U_i) \cong U_i$ (points of U_j, $j \neq i$, are not in X_i), so $g^{-1}(U) \to U$ is an isomorphism.

It remains to see X' is projective over S. If $X_i' \hookrightarrow \mathbf{P}_S^{n_i}$ are closed immersions, let $N + 1 = \sum(n_i + 1)$ and embed $\mathbf{P}_S^{n_i}$ in $\mathbf{P}_S^N$ as the linear subspace Λ_i where all coordinates outside the i-th block of $n_i + 1$ coordinates vanish. The Λ_i are closed, pairwise disjoint, so $\bigsqcup \Lambda_i \to \mathbf{P}_S^N$ is a homeomorphism onto a closed set and is surjective on stalks of structure sheaves (near a point of Λ_i, the other Λ_j are absent), i.e. a closed immersion. Hence $\bigsqcup X_i' \hookrightarrow \mathbf{P}_S^N$ is a closed immersion and X' is projective over S.

So from now on X is irreducible (and nonempty).

(b) Finite cover by quasi-projective opens. Cover S by finitely many open affines $S_k = \operatorname{Spec} A_k$ and X by finitely many open affines $V = \operatorname{Spec} C$ lying over some S_k (possible as X is quasi-compact, $f^{-1}(S_k)$ noetherian). Here C is a finitely generated A_k-algebra, $C = A_k[t_1, \dots, t_n]/J$. So $V \hookrightarrow \mathbf{A}_{S_k}^n = D_+(x_0) \subseteq \mathbf{P}_{S_k}^n \subseteq \mathbf{P}_S^n$ is an immersion (closed in an open subscheme of $\mathbf{P}_S^n$; here $\mathbf{P}_{S_k}^n$ is open in $\mathbf{P}_S^n$). Let $P_V \subseteq \mathbf{P}_S^n$ be the scheme-theoretic closure of V (image structure of $V \to \mathbf{P}_S^n$). Since V is closed in an open subscheme $G \subseteq \mathbf{P}_S^n$ (namely $G = \mathbf{A}_{S_k}^n$), $P_V \cap G = V$ and so $V \to P_V$ is an open immersion; $P_V \to \mathbf{P}_S^n \to S$ is a closed immersion followed by the projection, i.e. P_V is projective over S. Hence each V is quasi-projective over S. Take these finitely many V as $U_1, \dots, U_n$ (nonempty ones) with open immersions $\iota_i : U_i \to P_i$, P_i projective over S.

(c) h is a closed immersion. Let $U = \bigcap U_i$, nonempty open, so dense in the irreducible X. Put $P = P_1 \times_S \cdots \times_S P_n$, $Y = X \times_S P$. Each P_i is projective, hence proper, hence separated, over S; so P is projective over S by Ex. 4.9, $P \to S$ is separated, and $Y \to S$ and $Y \to X$ are proper (Cor. 4.8). Let $j = (\iota_i|_U)_i : U \to P$ and $f = (\mathrm{incl}, j)_S : U \to Y$. Then f is the composition of the graph $\Gamma_j : U \to U \times_S P$, a closed immersion since P is separated (Ex. 4.4,

Step 1), with the open immersion $U \times_S P = W_0 \hookrightarrow Y$ ($W_0 = p_X^{-1}(U)$). Let $X' \subseteq Y$ be the image subscheme of f, $g = p_X|_{X'}$ and $h = p_P|_{X'}$.

Facts. (F1) $X' \cap W_0 = f(U)$ as subschemes: restricting the image subscheme structure to the open set W_0 gives that of $f : U \to W_0$, which is a closed immersion, so its image subscheme is $f(U)$ itself. Hence $f : U \to X'$ is an open immersion. (F2) X' is irreducible, being the closure of $f(U)$ with U irreducible; its generic point η' lies in the open set $f(U)$. (F3) X' is proper over S (closed in Y), so h is proper by Cor. 4.8(e) (here $P \to S$ is separated), in particular h is closed.

Local structure over Q_i. Let $a_i : X' \to P_i$ be the i-th component of h, and $Q_i = p_{P_i}^{-1}(U_i) \subseteq P$, open (we identify U_i with its image in P_i). We claim

$$(*) \qquad h^{-1}(Q_i) = g^{-1}(U_i) \quad \text{and} \quad h| : g^{-1}(U_i) \to Q_i \text{ is a closed immersion.}$$

First the closed immersion. Let $Y_i = p_X^{-1}(U_i) = U_i \times_S P$ (open in Y) and write $P = P_i \times_S P''$, $P'' = \prod_{l \neq i} P_l$ (empty product $= S$), so $Y_i = (U_i \times_S P'') \times_S P_i$. The two morphisms $Y_i \to P_i$, namely the projection pr to P_i and $\iota_i \circ p_{U_i}$, agree on $f(U)$ (indeed $\operatorname{pr} f = \iota_i \circ \mathrm{incl}$). Since P_i is separated over S, the equalizer $E_i \subseteq Y_i$ (the pullback of Δ_{P_i}) is a closed subscheme through which f factors. Moreover E_i is the graph of the morphism $\iota_i p_{U_i} : U_i \times_S P'' \to P_i$, so projecting gives an isomorphism $E_i \cong U_i \times_S P''$, and under it $h|_{E_i}$ is $\iota_i \times \mathrm{id} : U_i \times_S P'' \to P_i \times_S P'' = P$, an open immersion with image Q_i; so $E_i \cong Q_i$. As $g^{-1}(U_i) = X' \cap Y_i$ is the image subscheme of $f : U \to Y_i$, the minimality of the image structure shows that $g^{-1}(U_i)$ is a closed subscheme of $E_i \cong Q_i$, and $h|$ is the resulting closed immersion into Q_i.

Now the equality of sets. Clearly $g^{-1}(U_i) \subseteq h^{-1}(Q_i)$ (by what we just saw). Conversely let $x' \in X'$ with $a_i(x') \in U_i$. Let $K' = k(\eta')$ and $Z' = X'_{\mathrm{red}}$ (integral, function field K', η' its generic point, and $x' \in Z'$ is a specialization of η'). Let Λ be the local ring of x' on Z', a local subring of K', and R a valuation ring of K' dominating Λ (I,6.1A). By Lemma 4.4 we get $\tau : T = \operatorname{Spec} R \to X'$ with $\tau(t_0) = x'$, $\tau(t_1) = \eta'$, inducing the identity on K', so $\tau|_{\operatorname{Spec} K'}$ is the generic point of the open subscheme $f(U) \cong U$. The morphism $a_i\tau : T \to P_i$ sends the closed point t_0 into the open set U_i; every point of T specializes to t_0 and open sets are stable under generization, so $a_i\tau$ factors through U_i: $a_i\tau = \iota_i c$ with $c : T \to U_i$. Now T has two morphisms to X extending $\operatorname{Spec} K' \to X$, namely $g\tau$ and $(U_i \subseteq X) \circ c$: on $\operatorname{Spec} K'$, $g\tau$ is the generic point $\operatorname{Spec} K' \to U \subseteq X$, and $a_i\tau|_{\operatorname{Spec} K'} = \iota_i \circ (\operatorname{Spec} K' \to U)$ (since $a_i f = \iota_i \circ \mathrm{incl}$), so $c|_{\operatorname{Spec} K'} = (\operatorname{Spec} K' \to U \subseteq U_i)$. Both lifts lie over the same morphism $T \to S$ (all maps involved are S-morphisms). As X is separated over the noetherian S, Thm 4.3 gives $g\tau = (U_i \subseteq X) \circ c$. Hence $g(x') = g\tau(t_0) \in U_i$, i.e. $x' \in g^{-1}(U_i)$. This proves $(*)$.

Conclusion. The sets $g^{-1}(U_i)$ cover X' (the U_i cover X), so $h(X') \subseteq Q := \bigcup Q_i$, open in P, and by $(*)$ $h : X' \to Q$ is a closed immersion locally on Q (indeed $h^{-1}(Q_i) \to Q_i$ is a closed immersion for each i), hence a closed immersion $X' \to Q$. As h is proper, $h(X')$ is closed in P; so $h : X' \to P$ is a homeomorphism onto a closed subset Z_0, and $\mathcal{O}_P \to h_*\mathcal{O}_{X'}$ is surjective on stalks at points of $Z_0 \subseteq Q$ (as it is for Q) and the target stalks vanish off Z_0. So h is a closed immersion. Since P is projective over S and h is a closed immersion, $X' \to P \to S$ is projective by Ex. 4.9.

(d) $g^{-1}(U) \to U$ is an isomorphism. By (F1), $g^{-1}(U) = X' \cap p_X^{-1}(U) = X' \cap W_0 = f(U)$ as open subschemes of X'. The composition $U \xrightarrow{f} f(U) = g^{-1}(U) \xrightarrow{g} U$ is the identity (as $p_X f = \mathrm{incl}$), and $f : U \to f(U)$ is an isomorphism (closed immersion onto its image subscheme). Therefore $g| : g^{-1}(U) \to U$ is an isomorphism. This completes the proof, with X' projective

over S and U open dense in X. □

Exercise 4.11

Problem. If you are willing to do some harder commutative algebra, and stick to noetherian schemes, then we can express the valuative criteria of separatedness and properness using only *discrete* valuation rings.

(a) If $\mathcal{O}, \mathfrak{m}$ is a noetherian local domain with quotient field K, and if L is a finitely generated field extension of K, then there exists a discrete valuation ring R of L dominating $\mathcal{O}$. Prove this in the following steps. By taking a polynomial ring over $\mathcal{O}$, reduce to the case where L is a *finite* extension field of K. Then show that for a suitable choice of generators $x_1, \ldots, x_n$ of $\mathfrak{m}$, the ideal $\mathfrak{a} = (x_1)$ in $\mathcal{O}' = \mathcal{O}[x_2/x_1, \ldots, x_n/x_1]$ is not equal to the unit ideal. Then let $\mathfrak{p}$ be a minimal prime ideal of $\mathfrak{a}$, and let $\mathcal{O}'_{\mathfrak{p}}$ be the localization of $\mathcal{O}'$ at $\mathfrak{p}$. This is a noetherian local domain of dimension 1 dominating $\mathcal{O}$. Let $\tilde{\mathcal{O}}'_{\mathfrak{p}}$ be the integral closure of $\mathcal{O}'_{\mathfrak{p}}$ in L. Use the theorem of Krull–Akizuki (see Nagata [7, p. 115]) to show that $\tilde{\mathcal{O}}'_{\mathfrak{p}}$ is a noetherian domain of dimension 1. Finally, take R to be a localization of $\tilde{\mathcal{O}}'_{\mathfrak{p}}$ at one of its maximal ideals.

(b) Let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Show that f is separated (respectively, proper) if and only if the criterion of (4.3) (respectively, (4.7)) holds for all *discrete* valuation rings.

Solution. (a) *Caveat.* If $\mathcal{O}$ is a field and L is algebraic over $K = \mathcal{O}$, then the only valuation ring of L containing $\mathcal{O}$ is L itself (an element algebraic over a field is integral over it, hence lies in every valuation ring containing it, as does its inverse), which is not a discrete valuation ring. So we assume (as is intended) that $\mathfrak{m} \neq 0$ or L is not algebraic over K.

Reduction to L/K finite. Choose a transcendence basis $t_1, \ldots, t_r$ of L/K (L is finitely generated); then L is finite over $K(t) = K(t_1, \ldots, t_r)$. Let $\mathcal{O}_1 = \mathcal{O}[t_1, \ldots, t_r]_{\mathfrak{n}}$ with $\mathfrak{n} = (\mathfrak{m}, t_1, \ldots, t_r)$; $\mathfrak{n}$ is a maximal ideal, since $\mathcal{O}[t]/\mathfrak{n} = \mathcal{O}/\mathfrak{m}$. So $\mathcal{O}_1$ is a noetherian local domain with quotient field $K(t)$, $\mathfrak{n} \cap \mathcal{O} = \mathfrak{m}$ so $\mathcal{O}_1$ dominates $\mathcal{O}$, and its maximal ideal is nonzero. A discrete valuation ring of L dominating $\mathcal{O}_1$ also dominates $\mathcal{O}$. So we may replace $(\mathcal{O}, K)$ by $(\mathcal{O}_1, K(t))$: assume L/K finite and $\mathfrak{m} \neq 0$.

A good chart. Let $x_1, \ldots, x_n$ be nonzero generators of $\mathfrak{m}$ (a minimal generating set), and $\mathcal{O}_i = \mathcal{O}[x_j/x_i : j \neq i] \subseteq K$; note $\mathfrak{m}\mathcal{O}_i = x_i\mathcal{O}_i$. *Claim: $x_i\mathcal{O}_i \neq \mathcal{O}_i$ for some i.* Suppose $x_i\mathcal{O}_i = \mathcal{O}_i$ for all i. Then $1 = x_i \sum_{|\beta| \leq N} c_\beta \prod_j (x_j/x_i)^{\beta_j}$ for some common N and $c_\beta \in \mathcal{O}$; multiplying by x_i^N gives $x_i^N = \sum c_\beta x_i^{N+1-|\beta|} \prod x_j^{\beta_j} \in \mathfrak{m}^{N+1}$. Let $M = n(N-1)+1$. Every monomial of degree M in $x_1, \ldots, x_n$ contains some x_i with exponent $\geq N$, so lies in $x_i^N \mathfrak{m}^{M-N} \subseteq \mathfrak{m}^{M+1}$. So $\mathfrak{m}^M = \mathfrak{m}^{M+1}$, and Nakayama's lemma gives $\mathfrak{m}^M = 0$, so $\mathfrak{m} = 0$ in the domain $\mathcal{O}$, a contradiction. Renumber so that $i = 1$, and put $\mathcal{O}' = \mathcal{O}_1$, $\mathfrak{a} = x_1\mathcal{O}' \neq \mathcal{O}'$.

A one-dimensional local ring. $\mathcal{O}' \subseteq K$ is a finitely generated $\mathcal{O}$-algebra, a noetherian domain with quotient field K. Let $\mathfrak{p}$ be a minimal prime over the nonzero principal ideal $\mathfrak{a}$. By Krull's principal ideal theorem $\operatorname{ht}\mathfrak{p} \leq 1$, and $\mathfrak{p} \neq 0$, so $\mathcal{O}'_{\mathfrak{p}}$ is a noetherian local domain of dimension 1 with quotient field K. It dominates $\mathcal{O}$: all $x_j = x_1 \cdot (x_j/x_1) \in \mathfrak{a} \subseteq \mathfrak{p}$, so $\mathfrak{m} \subseteq \mathfrak{p} \cap \mathcal{O}$, and $\mathfrak{p} \cap \mathcal{O}$ is a proper prime, so equals $\mathfrak{m}$.

Normalization. Let B be the integral closure of $\mathcal{O}'_{\mathfrak{p}}$ in the finite extension L of K. By Krull–Akizuki, B (a ring between $\mathcal{O}'_{\mathfrak{p}}$ and L) is a noetherian domain of dimension ≤ 1; as it is integral

over $\mathcal{O}'_\mathfrak{p}$ it has dimension exactly 1. Take a maximal ideal $\mathfrak{P}$ of B; it lies over the maximal ideal of $\mathcal{O}'_\mathfrak{p}$ (integral extension), so $\mathfrak{P} \neq 0$. Then $R = B_\mathfrak{P}$ is a noetherian local integrally closed domain of dimension 1 with quotient field L, hence a discrete valuation ring (I,6.2A). Since $\mathfrak{P} \cap \mathcal{O}'_\mathfrak{p} = \mathfrak{p}\mathcal{O}'_\mathfrak{p}$, R dominates $\mathcal{O}'_\mathfrak{p}$, hence $\mathcal{O}$. (a)

(b) One direction is trivial: a discrete valuation ring is a valuation ring, so if f is separated (proper), the criterion holds in particular for discrete valuation rings (note that the criterion in (4.7) includes uniqueness, so by (4.3) it implies the criterion of (4.3) for DVRs).

Separated. Assume the criterion of (4.3) holds for all discrete valuation rings. X and $X \times_Y X$ are noetherian (the latter is of finite type over Y), so Δ is quasi-compact, and by Lemma 4.5 and Cor. 4.2 it suffices to show that $\Delta(X)$ is stable under specialization. Let $\xi_1 \in \Delta(X)$ and $\xi_1 \rightsquigarrow \xi_0$, $\xi_0 \neq \xi_1$ (otherwise nothing to prove). Put $K = k(\xi_1)$ and let $\mathcal{O}$ be the local ring of ξ_0 on $\{\xi_1\}^-$ (reduced structure): a noetherian local domain with quotient field K and $\mathfrak{m}_\mathcal{O} \neq 0$. By (a) with $L = K$ there is a discrete valuation ring R of K dominating $\mathcal{O}$. Exactly as in the proof of (4.3), Lemma 4.4 gives $h'' : T = \operatorname{Spec} R \to X \times_Y X$ with $h''(t_0) = \xi_0$, $h''(t_1) = \xi_1$ inducing K; $h_1 = p_1 h''$, $h_2 = p_2 h''$ are two lifts in the diagram $U = \operatorname{Spec} K \to X$, $T \to Y$ (the restrictions to U agree because $\xi_1 \in \Delta(X)$, so $U \to X \times_Y X$ factors through Δ). By hypothesis $h_1 = h_2$, i.e. $h'' = \Delta \circ h_1$, so $\xi_0 = h''(t_0) \in \Delta(X)$.

Proper: reduction to noetherian bases. Assume the criterion of (4.7) holds for all DVRs. Then f is separated by the previous paragraph, and finite type. We show f is universally closed.

Step 1: closed after base change to noetherian Y'. Let $Y' \to Y$ be a morphism with Y' noetherian, $X' = X \times_Y Y'$ (noetherian, as $X' \to Y'$ is of finite type), $f' : X' \to Y'$, $Z \subseteq X'$ a closed subset with the reduced induced structure. $f'|_Z : Z \to Y'$ is quasi-compact, so by Lemma 4.5 it suffices to show $f'(Z)$ is stable under specialization. Let $z_1 \in Z$, $y_1 = f'(z_1) \rightsquigarrow y_0$, $y_0 \neq y_1$. Let $\mathcal{O}$ be the local ring of y_0 on $\{y_1\}^- \subseteq Y'$ (reduced): a noetherian local domain with quotient field $k(y_1)$, $\mathfrak{m}_\mathcal{O} \neq 0$. Put $K = k(z_1)$; as z_1 is a point of a scheme of finite type over $k(y_1)$ (the fibre), K is a finitely generated field extension of $k(y_1)$. By (a) there is a DVR R of K dominating $\mathcal{O}$. By Lemma 4.4 we get $U = \operatorname{Spec} K \to Z$ (the point z_1) and $T = \operatorname{Spec} R \to Y'$ forming a commutative square with $i : U \to T$ and $U \to Y'$. Compose with $Z \to X' \to X$ and $Y' \to Y$: these form a diagram as in (4.7) for the DVR R, so there is $h : T \to X$ making it commute. Together with $T \to Y'$ it gives $T \to X' = X \times_Y Y'$ (restricting to $U \to Z \subseteq X'$ on U); the generic point t_1 goes to $z_1 \in Z$, so $T = \overline{\{t_1\}}$ maps into the closed set Z, and as T is reduced, $T \to X'$ factors through Z (Lemma in Ex. 4.2). The image z_0 of t_0 satisfies $f'(z_0) = y_0$, so $y_0 \in f'(Z)$.

Step 2: arbitrary base change. We show: if f' is closed for every base change $Y' \to Y$ with Y' noetherian, then f is universally closed. Let $g : Y' \to Y$ be arbitrary and $Z \subseteq X' = X \times_Y Y'$ closed. Closedness of $f'(Z)$ is local on Y' and we may work over an open affine $\operatorname{Spec} B \subseteq Y$ (B noetherian), so assume $Y = \operatorname{Spec} B$, $Y' = \operatorname{Spec} C$. Let $\{C_\lambda\}$ be the directed family of finitely generated B-subalgebras of C; each $\operatorname{Spec} C_\lambda$ is noetherian, so $f_\lambda : X_\lambda = X \times_B C_\lambda \to \operatorname{Spec} C_\lambda$ is closed (Step 1). Let $\pi_\lambda : X' \to X_\lambda$, $\rho_\lambda : \operatorname{Spec} C \to \operatorname{Spec} C_\lambda$, and $\bar{Z}_\lambda = \overline{\pi_\lambda(Z)}$.

(i) $Z = \bigcap_\lambda \pi_\lambda^{-1}(\bar{Z}_\lambda)$. Take $x \notin Z$. If $X = \bigcup \operatorname{Spec} A_k$ (finitely many affines, A_k finitely generated B-algebras), then $X' = \bigcup \operatorname{Spec}(A_k \otimes_B C)$ and $A_k \otimes_B C = \bigcup_\lambda \operatorname{Im}(A_k \otimes_B C_\lambda)$, so every basic open $D(a)$ in $\operatorname{Spec}(A_k \otimes C)$ is π_λ^{-1} of a basic open of X_λ for some λ (and for all larger ones). Hence there are λ and $V \subseteq X_\lambda$ open with $x \in \pi_\lambda^{-1}(V) \subseteq X' \setminus Z$. Then $V \cap \pi_\lambda(Z) = \emptyset$, so $V \cap \bar{Z}_\lambda = \emptyset$ and $x \notin \pi_\lambda^{-1}(\bar{Z}_\lambda)$. (The other inclusion is trivial.)

(ii) Let $y \in \bigcap_\lambda \rho_\lambda^{-1}(f_\lambda(\bar{Z}_\lambda))$; we show $y \in f'(Z)$. For each λ let $F_\lambda = \pi_\lambda^{-1}(\bar{Z}_\lambda) \cap f'^{-1}(y)$, a closed subset of the fibre X'_y. It is nonempty: there is $\zeta \in \bar{Z}_\lambda$ with $f_\lambda(\zeta) = \rho_\lambda(y)$, and since $X' = X_\lambda \times_{\operatorname{Spec} C_\lambda} \operatorname{Spec} C$, the scheme $\operatorname{Spec}(k(\zeta) \otimes_{k(\rho_\lambda y)} k(y)) \neq \emptyset$ gives a point of X' over ζ and y. If $\mu \geq \lambda$ then $\pi_\mu^{-1}(\bar{Z}_\mu) \subseteq \pi_\lambda^{-1}(\bar{Z}_\lambda)$ (as $\pi_{\mu\lambda}(\bar{Z}_\mu) \subseteq \bar{Z}_\lambda$), so the F_λ form a directed family of nonempty closed subsets of the quasi-compact space X'_y (finite type over a field), and $\bigcap F_\lambda \neq \emptyset$. By (i) it lies in $Z \cap f'^{-1}(y)$, so $y \in f'(Z)$.

(iii) Since $f'(Z) \subseteq \rho_\lambda^{-1} f_\lambda(\pi_\lambda Z) \subseteq \rho_\lambda^{-1} f_\lambda(\bar{Z}_\lambda)$, (ii) gives $f'(Z) = \bigcap_\lambda \rho_\lambda^{-1}(f_\lambda(\bar{Z}_\lambda))$, an intersection of closed sets (f_λ is closed). Hence $f'(Z)$ is closed. So f is universally closed, hence proper. □

Exercise 4.12

Problem. *Examples of Valuation Rings.* Let k be an algebraically closed field.

(a) If K is a function field of dimension 1 over k (I, Section 6), then every valuation ring of K/k (except for K itself) is discrete. Thus the set of all of them is just the abstract nonsingular curve C_K of (I, Section 6).

(b) If K/k is a function field of dimension two, there are several different kinds of valuations. Suppose that X is a complete nonsingular surface with function field K.
(1) If Y is an irreducible curve on X, with generic point x_1, then the local ring $R = \mathcal{O}_{x_1,X}$ is a discrete valuation ring of K/k with center at the (nonclosed) point x_1 on X.
(2) If $f : X' \to X$ is a birational morphism, and if Y' is an irreducible curve in X' whose image in X is a single closed point x_0, then the local ring R of the generic point of Y' on X' is a discrete valuation ring of K/k with center at the closed point x_0 on X.
(3) Let $x_0 \in X$ be a closed point. Let $f : X_1 \to X$ be the blowing-up of x_0 (I, Section 4) and let $E_1 = f^{-1}(x_0)$ be the exceptional curve. Choose a closed point $x_1 \in E_1$, let $f_2 : X_2 \to X_1$ be the blowing-up of x_1, and let $E_2 = f_2^{-1}(x_1)$ be the exceptional curve. Repeat. In this manner we obtain a sequence of varieties X_i, with closed points x_i chosen on them, and for each i, the local ring $\mathcal{O}_{x_{i+1},X_{i+1}}$ dominates $\mathcal{O}_{x_i,X_i}$. Let $R_0 = \bigcup_{i=0}^{\infty} \mathcal{O}_{x_i,X_i}$. Then R_0 is a local ring, so it is dominated by some valuation ring R of K/k (I, 6.1A). Show that R is a valuation ring of K/k, and that it has center x_0 on X. When is R a discrete valuation ring?

Note. We will see later (V, Ex. 5.6) that in fact the R_0 of (3) is already a valuation ring itself, so $R_0 = R$. Furthermore, every valuation ring of K/k (except K itself) is one of the three kinds just described.

Solution. (a) Let $R \neq K$ be a valuation ring of K/k. Then $\mathfrak{m}_R \neq 0$; choose $0 \neq a \in \mathfrak{m}_R$. Since $a \in \mathfrak{m}_R$, a is not a unit, so $a \notin k^*$; as k is algebraically closed, a is transcendental over k. As K has dimension 1 and is finitely generated over k, K is a finite extension of $k(a)$. Let B_0 be the integral closure of $k[a]$ in K: it is a finite $k[a]$-module (I,3.9A), hence a noetherian integrally closed domain of dimension 1 (a Dedekind domain). Since $R \supseteq k[a]$ and R is integrally closed (a valuation ring), $B_0 \subseteq R$. Let $\mathfrak{p} = \mathfrak{m}_R \cap B_0$, a prime ideal containing $a \neq 0$, hence maximal. Then $(B_0)_\mathfrak{p}$ is a noetherian integrally closed local domain of dimension 1, so a discrete valuation ring (I,6.2A), and $(B_0)_\mathfrak{p} \subseteq R \subsetneq K$ (units of $B_0 \setminus \mathfrak{p}$ are units in R). A ring R with $S = (B_0)_\mathfrak{p} \subseteq R \subseteq K$ equals S or K: if $x \in R \setminus S$, write $x = u\pi^{-m}$ with $m > 0$, u a unit of S, π a uniformizer; then $\pi^{-1} \in R$, so $R \supseteq S[\pi^{-1}] = K$. Since $R \neq K$, $R = (B_0)_\mathfrak{p}$ is a discrete valuation ring.

Conversely every discrete valuation ring of K/k is a valuation ring of K/k. So the valuation rings $\neq K$ of K/k are exactly the DVRs of K/k, i.e. the points of the abstract curve C_K of (I,Section 6). (a)

(b)(1) Let $\mathcal{O}_{x_1} = \mathcal{O}_{x_1,X}$; it is a subring of K with quotient field K containing k. Let P be a closed point of Y. As X is nonsingular, $\mathcal{O}_{P,X}$ is a regular local ring (I,5.2A/II.8), and $\mathcal{O}_{x_1,X}$ is its localization at the prime corresponding to Y; a localization of a regular local ring is regular (Serre). Its dimension equals $\operatorname{codim}(Y,X) = 1$ (II, Ex. 3.20). So $\mathcal{O}_{x_1}$ is a one-dimensional regular local ring, i.e. a discrete valuation ring (I,6.2A). It contains k, has fraction field K, so it is a discrete valuation ring of K/k, and it dominates itself: its center on X is x_1.

(2) Here (as the exercise intends) X' is also a nonsingular surface (otherwise the local ring at the generic point of Y' need not be regular). Since f is birational, f induces an isomorphism $K(X') \cong K(X) = K$ compatible with $f^{\#}$, and we regard $\mathcal{O}_{x',X'} \subseteq K$ for points $x' \in X'$. Let η' be the generic point of Y'. By (1) applied to X', $R = \mathcal{O}_{\eta',X'}$ is a discrete valuation ring of K/k. Since $f(Y') = \{x_0\}$, $f(\eta') = x_0$, and $f^{\#} : \mathcal{O}_{x_0,X} \to \mathcal{O}_{\eta',X'} = R$ is a local homomorphism which, under the identification of function fields, is the inclusion $\mathcal{O}_{x_0,X} \subseteq R$. A local homomorphism of local rings (the preimage of $\mathfrak{m}_R$ is $\mathfrak{m}_{x_0}$) is exactly the statement that R dominates $\mathcal{O}_{x_0,X}$. So R has center x_0 on X.

(3) All X_i are integral, and the blowing-up maps $X_{i+1} \to X_i$ are birational, so all function fields are identified with K and all the local rings $\mathcal{O}_i := \mathcal{O}_{x_i,X_i}$ are subrings of K. Since $x_{i+1} \mapsto x_i$, the map $\mathcal{O}_i \to \mathcal{O}_{i+1}$ is the inclusion and it is local: $\mathfrak{m}_{i+1} \cap \mathcal{O}_i = \mathfrak{m}_i$. So $\mathcal{O}_0 \subseteq \mathcal{O}_1 \subseteq \cdots$ and $\mathfrak{m}_0 \subseteq \mathfrak{m}_1 \subseteq \cdots$.

R_0 *is local* with maximal ideal $M = \bigcup \mathfrak{m}_i$: an element of R_0 not in M lies in some $\mathcal{O}_i \setminus \mathfrak{m}_i$, hence is a unit there. Elements of M are non-units of R_0: if $a \in \mathfrak{m}_i$ and $a^{-1} \in R_0$, then $a^{-1} \in \mathcal{O}_l$ for some $l \geq i$ and $1 = aa^{-1} \in \mathfrak{m}_l$, absurd. Also $M \cap \mathcal{O}_i = \mathfrak{m}_i$ for every i.

Let R be a valuation ring of K dominating R_0 (I,6.1A). Then R is a valuation ring of K with $k \subseteq \mathcal{O}_0 \subseteq R_0 \subseteq R$, so it is a valuation ring of K/k, and $\mathfrak{m}_R \cap \mathcal{O}_0 = \mathfrak{m}_R \cap R_0 \cap \mathcal{O}_0 = M \cap \mathcal{O}_0 = \mathfrak{m}_0$. Hence $R \supseteq \mathcal{O}_{x_0,X}$ and $\mathfrak{m}_R \cap \mathcal{O}_{x_0,X} = \mathfrak{m}_{x_0}$: R has center x_0 on X.

When is R discrete? This depends on the sequence of points x_i (the answer is governed by Zariski's theory of infinitely near points; I only prove the following two facts, which show that both cases occur, and do not attempt a complete classification). Take $X = \mathbf{P}^2_k$, $x_0 = (0,0)$ in the affine chart with coordinates x, y. Blowing up x_0, the chart of X_1 with coordinates (x, y_1), $y = xy_1$ contains $E_1 = \{x = 0\}$. Inductively, at each stage the chart of X_n has coordinates (x, y_n) with $E_n = \{x = 0\}$, and the blow-up of a point $x_n = (0, a_n)$ has chart (x, y_{n+1}) with $y_n - a_n = xy_{n+1}$.

(i) A non-discrete case. Choose $x_n = (x = 0, y_n = 0)$ for all n (all $a_n = 0$), so $y_n = y/x^n$ and x_n lies on the strict transform of the curve $y = 0$. Then $x,\ y_n \in \mathfrak{m}_n \subseteq M \subseteq \mathfrak{m}_R$ for all n. If R had a discrete valuation $v : K^* \to \mathbf{Z}$ with $R = \{v \geq 0\}$, then $v(x) \geq 1$ and $v(y_n) \geq 1$ so $v(y) = v(y_n) + nv(x) \geq n + 1$ for every n, which is impossible. So here R is *not* a discrete valuation ring of rank one (one valuation ring dominating R_0 is the composite of $\operatorname{ord}_{\{y=0\}}$ with the valuation of $k(x)$ at 0, with value group $\mathbf{Z}^2$ ordered lexicographically; by V, Ex. 5.6 it is R_0 itself).

(ii) A discrete case. Let $\operatorname{char} k = 0$ and let $\varphi = e^x - 1 = \sum_{m \geq 1} a_m x^m \in k[[x]]$, which is transcendental over $k(x)$ (any $\varphi \in xk[[x]]$ transcendental over $k(x)$ would do; here is a proof for e^x: let $\sum_{i=0}^{d} p_i(x)E^i = 0$, $E = e^x$, $p_i \in k[x]$, $p_d \neq 0$, be a relation with $d \geq 1$ minimal. Applying d/dx ($E' = E$) and subtracting $\frac{p'_d + dp_d}{p_d}$ times the original gives a relation of degree $< d$, so all its coefficients vanish; for $i = 0$ this reads $p_d p'_0 - p'_d p_0 = dp_d p_0$, and comparing

degrees (char 0) forces $p_0 = 0$; dividing by E gives a relation of degree $d-1$ with $p_d \neq 0$, contradicting minimality). Choose $x_n = (x = 0, y_n = a_n)$, where $y_n = (y - p_{n-1}(x))/x^n$, $p_{n-1} = \sum_{m<n} a_m x^m$ (the points of the branch $y = \varphi(x)$). Since x, φ are algebraically independent, $y \mapsto \varphi$ extends to an embedding $K = k(x,y) \hookrightarrow k((x))$, giving the discrete valuation $v = \mathrm{ord}_x$ and the DVR $R_v = K \cap k[[x]]$. For each n, $\mathcal{O}_n = k[x, y_n]_{(x, y_n - a_n)} \subseteq R_v$ and $v(y_n - a_n) > 0$, $v(x) > 0$, so R_v dominates every $\mathcal{O}_n$, hence $R_0 \subseteq R_v$ with domination. In fact $R_0 = R_v$: let $f = P/Q \in R_v$ with $P, Q \in k[x,y]$ and put $c = \mathrm{ord}_x Q(x, \varphi)$, so $\mathrm{ord}_x P(x,\varphi) \geq c$. For $n > c$ write $y = p_{n-1} + x^n y_n$: then $Q = Q(x, p_{n-1}) + x^n G$, $P = P(x, p_{n-1}) + x^n H$ with $G, H \in k[x, y_n]$, and $p_{n-1} \equiv \varphi \pmod{x^n}$ gives $Q(x, p_{n-1}) = x^c u(x)$ with $u(0) \neq 0$ and $P(x, p_{n-1}) = x^c P_1(x)$. Hence $f = (P_1 + x^{n-c} H)/(u + x^{n-c} G) \in \mathcal{O}_n$ (the denominator is a unit of $\mathcal{O}_n$ since $u(0) \neq 0$). So $R_0 = R_v$ is a valuation ring, and since R dominates the valuation ring R_0 (maximal for domination), $R = R_0 = R_v$ is a discrete valuation ring, with center x_0.

In general (Zariski) R can also be of rank one with non-discrete value group, or of rank two, according to the behaviour of the infinitely near points x_i; this classification belongs to the later chapters (V, Ex. 5.6) and is not proved here. □

5. Sheaves of modules

Exercise 5.1

Problem. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of finite rank. We define the *dual* of $\mathcal{E}$, denoted $\mathcal{E}^\vee$, to be the sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$.

(a) Show that $\mathcal{E}^{\vee\vee} \cong \mathcal{E}$.

(b) For any $\mathcal{O}_X$-module $\mathcal{F}$, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$.

(c) For any $\mathcal{O}_X$-modules $\mathcal{F}, \mathcal{G}$, $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G}))$.

(d) (*Projection formula*) If $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, if $\mathcal{F}$ is an $\mathcal{O}_X$-module, and if $\mathcal{E}$ is a locally free $\mathcal{O}_Y$-module of finite rank, then there is a natural isomorphism $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}) \cong f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E}$.

Solution. Throughout, $\mathcal{H}om(\mathcal{A}, \mathcal{B})|_U = \mathcal{H}om(\mathcal{A}|_U, \mathcal{B}|_U)$, and if $\mathcal{A} = \mathcal{O}_U^n$ with basis $e_1, \ldots, e_n$ then $\varphi \mapsto (\varphi(e_1), \ldots, \varphi(e_n))$ is an isomorphism $\mathcal{H}om(\mathcal{O}_U^n, \mathcal{B}) \cong \mathcal{B}^n$ (a morphism out of a free module is determined by the images of a basis, which can be chosen arbitrarily). In particular $\mathcal{E}^\vee|_U \cong \mathcal{O}_U^n$ with the dual basis $e_1^*, \ldots, e_n^*$, $e_i^*(e_j) = \delta_{ij}$.

(a) For an open set U define $\theta_U : \mathcal{E}(U) \to \mathcal{E}^{\vee\vee}(U) = \mathrm{Hom}(\mathcal{E}^\vee|_U, \mathcal{O}_U)$ by $\theta_U(s)(\xi) = \xi_V(s|_V) \in \mathcal{O}(V)$ for $\xi \in \mathcal{E}^\vee(V)$, $V \subseteq U$. This is $\mathcal{O}(U)$-linear, compatible with restrictions, hence a morphism $\theta : \mathcal{E} \to \mathcal{E}^{\vee\vee}$ of $\mathcal{O}_X$-modules. To show it is an isomorphism it suffices to do so on the open sets U over which $\mathcal{E}|_U \cong \mathcal{O}_U^n$ (a morphism of sheaves that is an isomorphism locally is an isomorphism). The construction is compatible with restriction to U and natural with respect to isomorphisms of $\mathcal{E}|_U$, so we may take $\mathcal{E}|_U = \mathcal{O}_U^n$ with basis e_i. Then $\mathcal{E}^\vee|_U$ has basis e_i^*, and $\mathcal{E}^{\vee\vee}|_U$ has the dual basis e_i^{**}; by definition $\theta(e_i)(e_j^*) = e_j^*(e_i) = \delta_{ij}$, i.e. $\theta(e_i) = e_i^{**}$. So θ maps a basis to a basis and is an isomorphism.

(b) For $\xi \in \mathcal{E}^\vee(U)$ and $t \in \mathcal{F}(U)$ let $\mu(\xi \otimes t) \in \mathcal{H}om(\mathcal{E}, \mathcal{F})(U)$ be the morphism $s \mapsto \xi(s)\, t|_V$ ($s \in \mathcal{E}(V)$, $V \subseteq U$). This is $\mathcal{O}(U)$-bilinear in (ξ, t) and compatible with restrictions, so it defines

a morphism from the presheaf $U \mapsto \mathcal{E}^\vee(U) \otimes_{\mathcal{O}(U)} \mathcal{F}(U)$ to the sheaf $\mathcal{H}om(\mathcal{E}, \mathcal{F})$, hence (universal property of sheafification) a morphism $\mu : \mathcal{E}^\vee \otimes \mathcal{F} \to \mathcal{H}om(\mathcal{E}, \mathcal{F})$. It is an isomorphism locally: if $\mathcal{E}|_U = \mathcal{O}_U^n$, then $\mathcal{E}^\vee \otimes \mathcal{F}|_U$ is the sheaf $\mathcal{F}^n$ (the presheaf $U' \mapsto \mathcal{O}(U')^n \otimes \mathcal{F}(U') = \mathcal{F}(U')^n$ is already a sheaf), and $\mu(\sum e_i^* \otimes t_i)(e_j) = t_j$, so μ corresponds to the identification $\mathcal{F}^n \cong \mathcal{H}om(\mathcal{O}_U^n, \mathcal{F})$ above.

(c) (This holds for every $\mathcal{O}_X$-module $\mathcal{E}$.) By the definition of $\mathcal{E} \otimes \mathcal{F}$ as a sheafification, a morphism $\psi : \mathcal{E} \otimes \mathcal{F} \to \mathcal{G}$ is the same as a compatible family of $\mathcal{O}(U)$-linear maps $\psi_U : \mathcal{E}(U) \otimes_{\mathcal{O}(U)} \mathcal{F}(U) \to \mathcal{G}(U)$, i.e. of $\mathcal{O}(U)$-bilinear maps $B_U : \mathcal{E}(U) \times \mathcal{F}(U) \to \mathcal{G}(U)$ commuting with restrictions. Given such B, define $\Phi(B) : \mathcal{F} \to \mathcal{H}om(\mathcal{E}, \mathcal{G})$ by $t \in \mathcal{F}(U) \mapsto \big(s \in \mathcal{E}(V) \mapsto B_V(s, t|_V)\big)_{V \subseteq U}$; this is a morphism of $\mathcal{O}_X$-modules by bilinearity and compatibility with restrictions. Conversely, a morphism $\chi : \mathcal{F} \to \mathcal{H}om(\mathcal{E}, \mathcal{G})$ gives $B_U(s,t) = \chi_U(t)_U(s)$. These two constructions are inverse to each other and additive, which gives the isomorphism of groups. (Applying this on every open set U to the restrictions gives the isomorphism of sheaves $\mathcal{H}om(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om(\mathcal{F}, \mathcal{H}om(\mathcal{E}, \mathcal{G}))$.)

(d) Recall $f^*\mathcal{E} = f^{-1}\mathcal{E} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$. For $e \in \mathcal{E}(V)$, $V \subseteq Y$ open, let $f^*e \in (f^*\mathcal{E})(f^{-1}V)$ be the image of e under $\mathcal{E}(V) \to (f^{-1}\mathcal{E})(f^{-1}V) \to (f^*\mathcal{E})(f^{-1}V)$, $e \mapsto 1 \otimes e$. Define

$$B_V : \ (f_*\mathcal{F})(V) \times \mathcal{E}(V) = \mathcal{F}(f^{-1}V) \times \mathcal{E}(V) \to (f_*(\mathcal{F} \otimes f^*\mathcal{E}))(V), \qquad (t, e) \mapsto t \otimes f^*e.$$

It is biadditive, and $\mathcal{O}_Y(V)$-bilinear: for $a \in \mathcal{O}_Y(V)$ we have $f^*(ae) = f^\#(a) \cdot f^*e$ and the $\mathcal{O}_Y$-module structure of $f_*\mathcal{F}$ is $a \cdot t = f^\#(a)t$, so $(at) \otimes f^*e = t \otimes f^\#(a) f^*e = t \otimes f^*(ae)$. The B_V commute with restrictions, so by the argument of (c) they give a morphism $\lambda : f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E} \to f_*(\mathcal{F} \otimes f^*\mathcal{E})$. It is natural in $\mathcal{F}$ and in $\mathcal{E}$ (a morphism $\mathcal{E} \to \mathcal{E}'$ induces compatible f^*).

To see that λ is an isomorphism we may work locally on Y: for $V \subseteq Y$ open both sides restrict to the corresponding sheaves for $f|_{f^{-1}V}$, and λ is compatible with this. So assume $\mathcal{E} \cong \mathcal{O}_Y^n$. By naturality in $\mathcal{E}$ it suffices to treat $\mathcal{E} = \mathcal{O}_Y^n$. Then $f^*\mathcal{E} = \mathcal{O}_X^n$, $\mathcal{F} \otimes f^*\mathcal{E} = \mathcal{F}^n$, $f_*(\mathcal{F}^n) = (f_*\mathcal{F})^n = f_*\mathcal{F} \otimes \mathcal{O}_Y^n$, and $\lambda(t \otimes (a_1, \dots, a_n)) = (f^\#(a_k)t)_k = (a_k t)_k$, i.e. λ is the identity under these identifications. □

Exercise 5.2

Problem. Let R be a discrete valuation ring with quotient field K, and let $X = \operatorname{Spec} R$.

(a) To give an $\mathcal{O}_X$-module is equivalent to giving an R-module M, a K-vector space L, and a homomorphism $\rho : M \otimes_R K \to L$.

(b) That $\mathcal{O}_X$-module is quasi-coherent if and only if ρ is an isomorphism.

Solution. Let π be a uniformizer. $X = \{\eta, s\}$ with $\eta = (0)$ the generic point and $s = (\pi)$ the closed point. The open sets are $\emptyset$, $\{\eta\} = D(\pi)$, and X; $\mathcal{O}_X(X) = R$ and $\mathcal{O}_X(\{\eta\}) = R_\pi = K$.

(a) Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Put $M = \mathcal{F}(X)$ (an R-module), $L = \mathcal{F}(\{\eta\})$ (a K-module) and let $r : M \to L$ be the restriction map. Compatibility of restriction with the module structures means $r(am) = \bar{a}\, r(m)$, where $\bar{a}$ is the image of $a \in R$ in K; so r is R-linear for the R-module structure of L coming from $R \subseteq K$, and by the universal property of $\otimes_R K$ it factors uniquely as $r = \rho \circ (m \mapsto m \otimes 1)$ with $\rho : M \otimes_R K \to L$ K-linear, $\rho(m \otimes c) = c\, r(m)$.

Conversely, given (M, L, ρ), put $\mathcal{F}(\emptyset) = 0$, $\mathcal{F}(X) = M$, $\mathcal{F}(\{\eta\}) = L$ with restriction $m \mapsto \rho(m \otimes 1)$. This is a sheaf: any open covering of X contains X itself (the only open set containing s is X), any covering of $\{\eta\}$ contains $\{\eta\}$, and $\mathcal{F}(\emptyset) = 0$; so the sheaf axioms are trivial. It is an $\mathcal{O}_X$-module, since r is semilinear as required. A morphism $\mathcal{F} \to \mathcal{F}'$ is a pair $(\varphi_X : M \to M', \varphi_\eta : L \to L')$ with φ_X R-linear, φ_η K-linear and $\varphi_\eta r = r' \varphi_X$; equivalently $(\varphi_X, \varphi_\eta)$ is a morphism of triples, i.e. $\varphi_\eta \rho = \rho'(\varphi_X \otimes 1)$. The two constructions are mutually inverse, so we have an equivalence of categories between $\mathcal{O}_X$-modules and triples (M, L, ρ).

(b) If ρ is an isomorphism, then $\mathcal{F} \cong \widetilde{M}$: indeed $\widetilde{M}(X) = M$ and $\widetilde{M}(\{\eta\}) = \widetilde{M}(D(\pi)) = M_\pi = M \otimes_R K$ (5.1(c)), with restriction the canonical map $M \to M \otimes K$, i.e. $\widetilde{M}$ corresponds to the triple $(M, M \otimes K, \mathrm{id})$. So $\mathcal{F}$ is quasi-coherent. Conversely, if $\mathcal{F}$ is quasi-coherent, then since X is affine $\mathcal{F} \cong \widetilde{N}$ for some R-module N (the definition together with 5.4, or 5.5), and $N = \Gamma(X, \mathcal{F}) = M$ by 5.1(d). An isomorphism $\mathcal{F} \cong \widetilde{M}$ is an isomorphism of triples $(M, L, \rho) \cong (M, M \otimes K, \mathrm{id})$ and hence ρ is an isomorphism. □

Exercise 5.3

> **Problem.** Let $X = \operatorname{Spec} A$ be an affine scheme. Show that the functors $\widetilde{\ }$ and Γ are adjoint, in the following sense: for any A-module M, and for any sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, there is a natural isomorphism
> $$\operatorname{Hom}_A(M, \Gamma(X, \mathcal{F})) \cong \operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathcal{F}).$$

Solution. Given $\varphi : \widetilde{M} \to \mathcal{F}$, let $\Phi(\varphi) = \Gamma(X, \varphi) : M = \Gamma(X, \widetilde{M}) \to \Gamma(X, \mathcal{F})$ (5.1(d)); it is A-linear.

Conversely let $\psi : M \to \Gamma(X, \mathcal{F})$ be A-linear. For $f \in A$ define $\psi_f : M_f = \widetilde{M}(D(f)) \to \mathcal{F}(D(f))$ by $\psi_f(m/f^n) = f^{-n} \cdot \psi(m)|_{D(f)}$, where f is a unit in $\mathcal{O}_X(D(f))$. This is well defined: if $m/f^n = m'/f^k$ in M_f, then $f^N(f^k m - f^n m') = 0$ for some N, so $f^N(f^k \psi(m) - f^n \psi(m'))|_{D(f)} = 0$, and multiplying by the unit f^{-N-n-k} gives $f^{-n}\psi(m)| = f^{-k}\psi(m')|$. It is A_f-linear, and for $D(g) \subseteq D(f)$ it is compatible with restriction (both composites send m/f^n to the restriction of $f^{-n}\psi(m)|_{D(f)}$ to $D(g)$). Now a section $\sigma \in \widetilde{M}(U)$ is, after covering U by basic open sets $D(f_i)$, given by $\sigma_i \in M_{f_i}$ that agree on $D(f_i f_j)$; the elements $\psi_{f_i}(\sigma_i) \in \mathcal{F}(D(f_i))$ agree on the overlaps $D(f_i f_j)$ by the compatibility just shown, so they glue to a unique element of $\mathcal{F}(U)$ (independent of the covering). This defines $\widetilde{\psi} : \widetilde{M} \to \mathcal{F}$, a morphism of $\mathcal{O}_X$-modules, with $\Gamma(\widetilde{\psi}) = \psi$ (take $f = 1$).

Then $\Phi(\widetilde{\psi}) = \psi$. Conversely, given φ with $\psi = \Phi(\varphi)$: on $D(f)$, $m/f^n = f^{-n} \cdot (m|_{D(f)})$ in $\widetilde{M}(D(f))$ and φ is $\mathcal{O}_X$-linear and commutes with restriction, so $\varphi(m/f^n) = f^{-n}\varphi(m)|_{D(f)} = \psi_f(m/f^n)$. Hence φ and $\widetilde{\psi}$ agree on all $D(f)$, hence everywhere (sections are locally of that form), so $\widetilde{\Phi(\varphi)} = \varphi$. Thus Φ is bijective. It is clearly additive, and natural in M and in $\mathcal{F}$ (the formulas above are natural). □

Exercise 5.4

> **Problem.** Show that a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ on a scheme X is quasi-coherent if and only if every point of X has a neighborhood U, such that $\mathcal{F}|_U$ is isomorphic to a cokernel of

a morphism of free sheaves on U. If X is noetherian, then $\mathcal{F}$ is coherent if and only if it is locally a cokernel of a morphism of free sheaves of finite rank. (These properties were originally the definition of quasi-coherent and coherent sheaves.)

Solution. Recall: $\mathcal{O}_X$ is quasi-coherent (Ex. 5.2.1 of the text), $\widetilde{\ }$ is exact and commutes with direct sums (5.2(a),(c)), and $\widetilde{A} = \mathcal{O}_{\operatorname{Spec} A}$; hence for a free module $A^{(I)}$ we get $\widetilde{A^{(I)}} = \mathcal{O}^{(I)}$ on $\operatorname{Spec} A$. Kernels, cokernels of morphisms of quasi-coherent sheaves are quasi-coherent (5.7).

($\Rightarrow$) Let $\mathcal{F}$ be quasi-coherent. Every point has an open affine neighborhood $U = \operatorname{Spec} A$ with $\mathcal{F}|_U \cong \widetilde{M}$. Choose a surjection $A^{(I)} \to M$, let N be its kernel and choose a surjection $A^{(J)} \to N$; this gives an exact sequence $A^{(J)} \to A^{(I)} \to M \to 0$. Applying $\widetilde{\ }$ gives an exact sequence $\mathcal{O}_U^{(J)} \to \mathcal{O}_U^{(I)} \to \mathcal{F}|_U \to 0$. If X is noetherian and $\mathcal{F}$ is coherent, we may take $U = \operatorname{Spec} A$ with A noetherian and M finitely generated (5.4 of the text), so I can be taken finite; then $N \subseteq A^{|I|}$ is finitely generated since A is noetherian, so J can be taken finite too.

($\Leftarrow$) Suppose $x \in U$ and $\mathcal{O}_U^{(J)} \xrightarrow{\varphi} \mathcal{O}_U^{(I)} \to \mathcal{F}|_U \to 0$ is exact. Free sheaves are quasi-coherent on the scheme U, so $\mathcal{F}|_U = \operatorname{coker} \varphi$ is quasi-coherent by 5.7. Being quasi-coherent means covered by open affines on which the sheaf is some $\widetilde{M}$; covering each such U by affines $\operatorname{Spec} A \subseteq U$ on which $\mathcal{F} \cong \widetilde{M}$, we cover X by such affines, so $\mathcal{F}$ is quasi-coherent. If X is noetherian and I, J finite, take an open affine $V = \operatorname{Spec} A \subseteq U$, A noetherian. Then $\mathcal{F}|_V = \widetilde{M}$ with $M = \Gamma(V, \mathcal{F})$ the cokernel of $A^{|J|} \to A^{|I|}$ (by exactness of $\widetilde{\ }$ and 5.5, Γ is the inverse equivalence), which is finitely generated. Hence $\mathcal{F}$ is coherent. □

Exercise 5.5

Problem. Let $f : X \to Y$ be a morphism of schemes.

(a) Show by example that if $\mathcal{F}$ is coherent on X, then $f_*\mathcal{F}$ need not be coherent on Y, even if X and Y are varieties over a field k.

(b) Show that a closed immersion is a finite morphism (Sect. 3).

(c) If f is a finite morphism of noetherian schemes, and if $\mathcal{F}$ is coherent on X, then $f_*\mathcal{F}$ is coherent on Y.

Solution. (a) Let k be a field (algebraically closed if one wishes varieties in the sense of Ch. I), $X = \mathbf{A}^1_k = \operatorname{Spec} k[t]$, $Y = \operatorname{Spec} k$, and f the structure morphism; both are varieties over k. Then $\mathcal{F} = \mathcal{O}_X = \widetilde{k[t]}$ is coherent, but $f_*\mathcal{F} = \widetilde{(k[t])}$ with $k[t]$ viewed as a k-module (5.2(d)), i.e. the module $k[t]$, which has infinite dimension over k. Since coherent on $\operatorname{Spec} k$ means finitely generated, $f_*\mathcal{F}$ is quasi-coherent but not coherent. (One can also take $f : \mathbf{A}^2_k \to \mathbf{A}^1_k$ the projection: $f_*\mathcal{O}_X = \widetilde{k[x,y]}$ and $k[x,y]$ is not a finitely generated $k[x]$-module.)

(b) Let $i : Z \to X$ be a closed immersion. Cover X by affine opens $V = \operatorname{Spec} B$. Then $i^{-1}(V) \to V$ is a closed immersion, so $i^{-1}(V)$ is a closed subscheme of V, and by Cor. 5.10 it is isomorphic over V to $\operatorname{Spec} B/\mathfrak{a}$ for an ideal $\mathfrak{a}$ of B. Thus $i^{-1}(V)$ is affine and $B/\mathfrak{a}$ is a finitely generated B-module (generated by 1). So i is finite.

(c) The question is local on Y ($f_*\mathcal{F}|_V = (f|_{f^{-1}V})_*(\mathcal{F}|_{f^{-1}V})$). By definition of finiteness we can cover Y by open affines $V = \operatorname{Spec} A$ such that $f^{-1}V = \operatorname{Spec} B$ with B a finitely generated

A-module. Since Y is noetherian, A is noetherian, and B (a finitely generated module, hence finitely generated algebra, over A) is noetherian. As $\mathcal{F}|_{f^{-1}V}$ is coherent on $\operatorname{Spec} B$, $\mathcal{F}|_{f^{-1}V} = \widetilde{N}$ with N a finitely generated B-module. By 5.2(d), $f_*\widetilde{N} = ({}_AN)^\sim$, and ${}_AN$ is a finitely generated A-module since N is finitely generated over B and B over A. So $f_*\mathcal{F}|_V$ is coherent. □

Exercise 5.6

Problem. *Support.* Recall the notions of support of a section of a sheaf, support of a sheaf, and subsheaf with supports from (Ex. 1.14) and (Ex. 1.20).

(a) Let A be a ring, let M be an A-module, let $X = \operatorname{Spec} A$, and let $\mathcal{F} = \widetilde{M}$. For any $m \in M = \Gamma(X, \mathcal{F})$, show that $\operatorname{Supp} m = V(\operatorname{Ann} m)$, where $\operatorname{Ann} m = \{a \in A \mid am = 0\}$.

(b) Now suppose that A is noetherian, and M finitely generated. Show that $\operatorname{Supp} \mathcal{F} = V(\operatorname{Ann} M)$.

(c) The support of a coherent sheaf on a noetherian scheme is closed.

(d) For any ideal $\mathfrak{a} \subseteq A$, we define a submodule $\Gamma_\mathfrak{a}(M)$ of M by $\Gamma_\mathfrak{a}(M) = \{m \in M \mid \mathfrak{a}^n m = 0 \text{ for some } n > 0\}$. Assume that A is noetherian, and M any A-module. Show that $\Gamma_\mathfrak{a}(M)^\sim \cong \mathcal{H}^0_Z(\mathcal{F})$, where $Z = V(\mathfrak{a})$ and $\mathcal{F} = \widetilde{M}$. [*Hint*: Use (Ex. 1.20) and (5.8) to show a priori that $\mathcal{H}^0_Z(\mathcal{F})$ is quasi-coherent. Then show that $\Gamma_\mathfrak{a}(M) \cong \Gamma_Z(\mathcal{F})$.]

(e) Let X be a noetherian scheme, and let Z be a closed subset. If $\mathcal{F}$ is a quasi-coherent (respectively, coherent) $\mathcal{O}_X$-module, then $\mathcal{H}^0_Z(\mathcal{F})$ is also quasi-coherent (respectively, coherent).

Solution. Recall $\operatorname{Supp} s = \{x : s_x \neq 0\}$ for a section s, $\operatorname{Supp} \mathcal{F} = \{x : \mathcal{F}_x \neq 0\}$, and $\mathcal{H}^0_Z(\mathcal{F})(U) = \{s \in \mathcal{F}(U) : \operatorname{Supp} s \subseteq Z\}$ (a subsheaf of $\mathcal{F}$).

(a) The germ of m at $\mathfrak{p}$ is $m/1 \in M_\mathfrak{p} = (\widetilde{M})_\mathfrak{p}$. It is zero iff there is $a \notin \mathfrak{p}$ with $am = 0$, i.e. iff $\operatorname{Ann} m \not\subseteq \mathfrak{p}$. Hence $\operatorname{Supp} m = \{\mathfrak{p} : \operatorname{Ann} m \subseteq \mathfrak{p}\} = V(\operatorname{Ann} m)$.

(b) Let $m_1, \ldots, m_r$ generate M. Then $M_\mathfrak{p} \neq 0$ iff some $m_i/1 \neq 0$ in $M_\mathfrak{p}$, so $\operatorname{Supp} \mathcal{F} = \bigcup \operatorname{Supp} m_i = \bigcup V(\operatorname{Ann} m_i) = V(\bigcap \operatorname{Ann} m_i)$ (finite union), and $\bigcap_i \operatorname{Ann} m_i = \operatorname{Ann} M$ since M is generated by the m_i.

(c) Let $\mathcal{F}$ be coherent on the noetherian scheme X. Cover X by finitely many open affines $U_i = \operatorname{Spec} A_i$ with $\mathcal{F}|_{U_i} = \widetilde{M_i}$, M_i finitely generated. By (b), $\operatorname{Supp} \mathcal{F} \cap U_i = \operatorname{Supp}(\mathcal{F}|_{U_i}) = V(\operatorname{Ann} M_i)$ is closed in U_i (stalks are local). A subset whose intersection with each U_i is closed in U_i is closed in X (its complement is open in each U_i).

(d) *Step 1: $\mathcal{H}^0_Z(\mathcal{F})$ is the kernel of $\mathcal{F} \to j_*(\mathcal{F}|_W)$,* where $W = X \setminus Z$, $j : W \to X$ is the inclusion and the map is the restriction $\mathcal{F}(V) \to \mathcal{F}(V \cap W)$. Indeed for $s \in \mathcal{F}(V)$, $\operatorname{Supp} s \subseteq Z$ iff all germs of s at points of $V \cap W$ vanish iff $s|_{V \cap W} = 0$.

Now let X be any noetherian scheme (in particular $X = \operatorname{Spec} A$ here) and $\mathcal{F}$ quasi-coherent. The scheme W is noetherian and $\mathcal{F}|_W$ is quasi-coherent, so $j_*(\mathcal{F}|_W)$ is quasi-coherent by 5.8(c) (hypothesis: the source W is noetherian). Hence $\mathcal{H}^0_Z(\mathcal{F})$, the kernel of a morphism of quasi-coherent sheaves, is quasi-coherent by 5.7. (This proves the quasi-coherent part of (e) too.)

Step 2. Now $X = \operatorname{Spec} A$, $\mathcal{F} = \widetilde{M}$, $Z = V(\mathfrak{a})$. Since $\mathcal{H}^0_Z(\mathcal{F})$ is quasi-coherent on the affine scheme X, it equals $\widetilde{N}$ with $N = \Gamma(X, \mathcal{H}^0_Z(\mathcal{F})) = \{m \in M : \operatorname{Supp} m \subseteq Z\}$ (5.5), as a submodule

of M (and $\widetilde{N} \to \widetilde{M}$ is the inclusion). By (a), $\operatorname{Supp} m \subseteq V(\mathfrak{a})$ iff $V(\operatorname{Ann} m) \subseteq V(\mathfrak{a})$ iff $\mathfrak{a} \subseteq \sqrt{\operatorname{Ann} m}$. Since A is noetherian, $\mathfrak{a} = (a_1, \ldots, a_s)$ is finitely generated; $\mathfrak{a} \subseteq \sqrt{\operatorname{Ann} m}$ means each $a_i^{n_i} \in \operatorname{Ann} m$, which holds iff $\mathfrak{a}^n m = 0$ for some n (if $a_i^{n_i} m = 0$ then $\mathfrak{a}^n m = 0$ for $n \geq \sum(n_i - 1) + 1$, since $\mathfrak{a}^n$ is generated by monomials of degree n in the a_i, each containing some $a_i^{n_i}$; the converse is clear). Thus $N = \Gamma_{\mathfrak{a}}(M)$ and $\mathcal{H}^0_Z(\widetilde{M}) = \Gamma_{\mathfrak{a}}(M)^{\sim}$.

(e) Quasi-coherence was shown in Step 1. If $\mathcal{F}$ is coherent, let $U = \operatorname{Spec} A$ be an open affine of X with A noetherian, $\mathcal{F}|_U = \widetilde{M}$, M finitely generated. Then $\mathcal{H}^0_Z(\mathcal{F})|_U = \mathcal{H}^0_{Z \cap U}(\mathcal{F}|_U) = \Gamma_{\mathfrak{a}}(M)^{\sim}$ by (d), where $Z \cap U = V(\mathfrak{a})$. Now $\Gamma_{\mathfrak{a}}(M)$ is a submodule of the finitely generated module M over the noetherian ring A, hence finitely generated. So $\mathcal{H}^0_Z(\mathcal{F})$ is coherent. □

Exercise 5.7

Problem. Let X be a noetherian scheme, and let $\mathcal{F}$ be a coherent sheaf.

(a) If the stalk $\mathcal{F}_x$ is a free $\mathcal{O}_x$-module for some point $x \in X$, then there is a neighborhood U of x such that $\mathcal{F}|_U$ is free.

(b) $\mathcal{F}$ is locally free if and only if its stalks $\mathcal{F}_x$ are free $\mathcal{O}_x$-modules for all $x \in X$.

(c) $\mathcal{F}$ is invertible (i.e., locally free of rank 1) if and only if there is a coherent sheaf $\mathcal{G}$ such that $\mathcal{F} \otimes \mathcal{G} \cong \mathcal{O}_X$. (This justifies the terminology *invertible*: it means that $\mathcal{F}$ is an invertible element of the monoid of coherent sheaves under the operation $\otimes$.)

Solution. (a) Let $\mathcal{F}_x \cong \mathcal{O}_x^n$ (n is finite as $\mathcal{F}_x$ is finitely generated). Take an affine open neighborhood $\operatorname{Spec} A$ of x, A noetherian, with $\mathcal{F} = \widetilde{M}$, M finitely generated, and $x = \mathfrak{p}$. Then $M_{\mathfrak{p}} \cong A_{\mathfrak{p}}^n$. Pick a basis of $M_{\mathfrak{p}}$; each basis element has the form m_i/u_i with $m_i \in M$, $u_i \notin \mathfrak{p}$, and u_i is a unit in $A_{\mathfrak{p}}$, so the images of $m_1, \ldots, m_n$ also form a basis. Let $\varphi : A^n \to M$, $e_i \mapsto m_i$. Then $\varphi_{\mathfrak{p}}$ is an isomorphism. The modules $K = \ker \varphi$ and $C = \operatorname{coker} \varphi$ are finitely generated (A noetherian) and $K_{\mathfrak{p}} = C_{\mathfrak{p}} = 0$ (localization is exact). A finitely generated module with zero localization at $\mathfrak{p}$ is killed by some element not in $\mathfrak{p}$ (take the product of annihilating elements for the generators). So there is $s \notin \mathfrak{p}$ with $K_s = C_s = 0$, i.e. $\varphi_s : A_s^n \to M_s$ is an isomorphism, and $\mathcal{F}|_{D(s)} = \widetilde{M_s} \cong \mathcal{O}^n_{D(s)}$.

(b) If $\mathcal{F}|_U \cong \mathcal{O}_U^{(I)}$ then $\mathcal{F}_x \cong \mathcal{O}_x^{(I)}$ for $x \in U$, so stalks are free. The converse is (a).

(c) ($\Rightarrow$) Let $\mathcal{F}$ be invertible and $\mathcal{G} = \mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$, which is locally $\cong \mathcal{O}$, hence coherent. The evaluation morphism $\mathcal{F} \otimes \mathcal{F}^\vee \to \mathcal{O}_X$, $s \otimes \varphi \mapsto \varphi(s)$ is an isomorphism: on U with $\mathcal{F}|_U = \mathcal{O}_U e$, $\mathcal{F}^\vee|_U = \mathcal{O}_U e^*$ and $e \otimes e^* \mapsto 1$.

($\Leftarrow$) Suppose $\mathcal{F} \otimes \mathcal{G} \cong \mathcal{O}_X$ with $\mathcal{F}, \mathcal{G}$ coherent. Fix x, $R = \mathcal{O}_x$ with maximal ideal $\mathfrak{m}$, $k = R/\mathfrak{m}$. Then $F = \mathcal{F}_x$, $G = \mathcal{G}_x$ are finitely generated and $F \otimes_R G \cong R$ (stalk of a tensor product). Tensoring with k: $(F/\mathfrak{m}F) \otimes_k (G/\mathfrak{m}G) \cong k$, so both have dimension 1. By Nakayama F and G are cyclic: $F \cong R/I$, $G \cong R/J$, so $R \cong F \otimes G \cong R/(I+J)$, which forces $I + J = 0$ (annihilators), hence $I = 0$ and $F \cong R$. By (b), $\mathcal{F}$ is locally free, of rank 1. □

Exercise 5.8

Problem. Again let X be a noetherian scheme and $\mathcal{F}$ a coherent sheaf on X. We will consider the function

$$\varphi(x) = \dim_{k(x)} \mathcal{F}_x \otimes_{\mathcal{O}_x} k(x),$$

where $k(x) = \mathcal{O}_x/\mathfrak{m}_x$ is the residue field at the point x. Use Nakayama's lemma to prove the following results.

(a) The function φ is *upper semi-continuous*, i.e., for any $n \in \mathbf{Z}$, the set $\{x \in X \mid \varphi(x) \geq n\}$ is closed.

(b) If $\mathcal{F}$ is locally free, and X is connected, then φ is a constant function.

(c) Conversely, if X is reduced, and φ is constant, then $\mathcal{F}$ is locally free.

Solution. (a) We show: every x has a neighborhood on which $\varphi \leq \varphi(x)$. Then $\{\varphi < n\}$ is open for every n (if $\varphi(x) < n$ the neighborhood lies in $\{\varphi < n\}$), i.e. $\{\varphi \geq n\}$ is closed.

Let $\varphi(x) = r$. Take an affine neighborhood $\operatorname{Spec} A$ of x, A noetherian, $\mathcal{F} = \widetilde{M}$, M finitely generated, $x = \mathfrak{p}$. The $k(\mathfrak{p})$-vector space $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p}$ has dimension r; choose $m_1, \ldots, m_r \in M$ whose images span it. By Nakayama ($M_\mathfrak{p}$ is a finitely generated $A_\mathfrak{p}$-module) the m_i generate $M_\mathfrak{p}$. Then $N = M/\sum Am_i$ is finitely generated with $N_\mathfrak{p} = 0$, so $N_s = 0$ for some $s \notin \mathfrak{p}$. Thus on $D(s)$ the module M_s is generated by the m_i, so for every $\mathfrak{q} \in D(s)$, $M_\mathfrak{q}$ is generated by r elements, whence $\varphi(\mathfrak{q}) = \dim M_\mathfrak{q}/\mathfrak{q}M_\mathfrak{q} \leq r$.

(b) If $\mathcal{F}|_U \cong \mathcal{O}_U^{(I)}$, then $\mathcal{F}_y \otimes k(y) = k(y)^{(I)}$ for $y \in U$; as $\mathcal{F}_y$ is finitely generated I is finite and $\varphi = |I|$ on U. So φ is locally constant, hence constant on the connected space X.

(c) Let $\varphi \equiv r$. Fix x and, as in (a), an affine neighborhood $\operatorname{Spec} A$ (A noetherian and reduced since X is reduced), $\mathcal{F} = \widetilde{M}$, $x = \mathfrak{p}$, and $m_1, \ldots, m_r \in M$ generating $M_\mathfrak{p}$ and, after shrinking to some $D(s)$ and replacing A by A_s, M by M_s, generating M (see (a)). So we have a surjection $\pi : A^r \to M$ with kernel $K \subseteq A^r$. Let $\mathfrak{q}$ be any prime of A. Tensoring $A_\mathfrak{q}^r \to M_\mathfrak{q} \to 0$ with $k(\mathfrak{q})$ gives a surjection $k(\mathfrak{q})^r \to M_\mathfrak{q} \otimes k(\mathfrak{q})$ between vector spaces of the same dimension r (because $\varphi(\mathfrak{q}) = r$), which is therefore an isomorphism. From the exact sequence $K_\mathfrak{q} \to A_\mathfrak{q}^r \to M_\mathfrak{q} \to 0$ and right exactness of $\otimes k(\mathfrak{q})$, the image of $K_\mathfrak{q} \otimes k(\mathfrak{q}) \to k(\mathfrak{q})^r$ is the kernel of this isomorphism, i.e. zero; this says $K_\mathfrak{q} \subseteq \mathfrak{q}A_\mathfrak{q}^r$.

Let $\kappa = (a_1, \ldots, a_r) \in K \subseteq A^r$. For every prime $\mathfrak{q}$ the image of κ in $A_\mathfrak{q}^r$ lies in $\mathfrak{q}A_\mathfrak{q}^r$, so $a_i/1 \in \mathfrak{q}A_\mathfrak{q}$, i.e. $a_i \in \mathfrak{q}$. Hence $a_i \in \bigcap \mathfrak{q} = \mathfrak{N}(A) = 0$ (A reduced). So $K = 0$, $M \cong A^r$ is free, and $\mathcal{F}$ is free in a neighborhood of x. □

Exercise 5.9

Problem. Let S be a graded ring, generated by S_1 as an S_0-algebra, let M be a graded S-module, and let $X = \operatorname{Proj} S$.

(a) Show that there is a natural homomorphism $\alpha : M \to \Gamma_*(\widetilde{M})$.

(b) Assume now that $S_0 = A$ is a finitely generated k-algebra for some field k, that S_1 is a finitely generated A-module, and that M is a finitely generated S-module. Show that the map α is an isomorphism in all large enough degrees, i.e., there is a $d_0 \in \mathbf{Z}$ such that for all $d \geq d_0$, $\alpha_d : M_d \to \Gamma(X, \widetilde{M}(d))$ is an isomorphism. [*Hint*: Use the methods of the proof of (5.19).]

(c) With the same hypotheses, we define an equivalence relation $\approx$ on graded S-modules by saying $M \approx M'$ if there is an integer d such that $M_{\geq d} \cong M'_{\geq d}$. Here $M_{\geq d} = \bigoplus_{n \geq d} M_n$. We will say that a graded S-module M is *quasi-finitely generated* if it is equivalent to a finitely generated module. Now show that the functors $\widetilde{\ }$ and Γ_* induce an equivalence of categories between the category of quasi-finitely generated graded S-modules modulo the equivalence relation $\approx$, and the category of coherent $\mathcal{O}_X$-modules.

Solution. (a) By 5.12(b), $\widetilde{M}(n) = \widetilde{M} \otimes \mathcal{O}_X(n) \cong \widetilde{M(n)}$. For $m \in M_n = M(n)_0$ let $\alpha(m)$ be the global section of $\widetilde{M(n)}$ whose value at $\mathfrak{p}$ is $m/1 \in M(n)_{(\mathfrak{p})}$; it is a section (locally of the form m/f with $f = 1$). This defines $\alpha_n : M_n \to \Gamma(X, \widetilde{M}(n))$, hence $\alpha : M \to \Gamma_*(\widetilde{M}) = \bigoplus_{n \in \mathbf{Z}} \Gamma(X, \widetilde{M}(n))$, additive. It is S-linear: $s \in S_d$ acts on Γ_* through the section $s/1 \in \Gamma(\mathcal{O}_X(d))$ and the maps $\mathcal{O}_X(d) \otimes \widetilde{M}(n) \to \widetilde{M}(n+d)$, which on stalks at $\mathfrak{p}$ send $(s/1) \otimes (m/1)$ to $sm/1$; so $\alpha(sm) = s\alpha(m)$. It is natural in M: a graded homomorphism $\varphi : M \to N$ induces $\Gamma_*(\widetilde{\varphi}) \circ \alpha_M = \alpha_N \circ \varphi$. Also, if $M \to M'$ is an inclusion of a graded submodule $M_{\geq d}$ etc. the corresponding α's are compatible by naturality.

A lemma (the method of 5.19). Let T be a graded ring which is an integral domain, with T_0 a finitely generated algebra over a field k, T generated over T_0 by T_1, T_1 a finitely generated T_0-module, $T_1 \neq 0$. Choose nonzero generators $x_0, \ldots, x_m$ of the T_0-module T_1, put $X_T = \operatorname{Proj} T$ (covered by the $D_+(x_i)$ since the x_i generate T_+), $K = \operatorname{Frac} T$. Then T is a finitely generated k-algebra, hence noetherian. Let $L \subseteq K$ be the localization of T at the nonzero homogeneous elements; it is a graded ring and each T_{x_i} is a graded subring of L. For $n \geq 0$ let $T'_n = \{s \in L_n : s \in T_{x_i} \text{ for all } i\}$ and $T' = \bigoplus_{n \geq 0} T'_n$, a graded subring of $L \subseteq K$ containing T.

(i) $\Gamma(X_T, \widetilde{T}(n)) = T'_n$ for $n \geq 0$ (multiplication matching the product in Γ_*), and $\alpha_n : T_n \to T'_n$ is the inclusion.

(ii) For $s \in T'_d$ ($d \geq 0$) there is N such that $y\,s \in T$ for all $y \in T_{\geq N}$.

(iii) T' is integral over T and is a finitely generated T-module.

(iv) $T'_n = T_n$ for all $n \gg 0$.

Proof. (i) By 5.12(b) and 5.11(b), the sections of $\widetilde{T}(n) = \widetilde{T(n)}$ over $D_+(x_i)$ are $T(n)_{(x_i)} = \{a/x_i^k : a \in T_{n+k}\} = (T_{x_i})_n$, and over $D_+(x_i x_j)$ they are $(T_{x_i x_j})_n$. All rings $T_{x_i} \to T_{x_i x_j}$ are subrings of K (T is a domain and $x_i \neq 0$), so by the sheaf axiom a global section is a tuple (t_i) with $t_i \in (T_{x_i})_n$ and $t_i = t_j$ in K, i.e. an element of $\bigcap_i (T_{x_i})_n = T'_n$. The map α sends $t \in T_n$ to the tuple $(t, \ldots, t)$.

(ii) Write $s = a_i/x_i^{k_i}$ with $a_i \in T$; so $x_i^{n_0} s \in T$ for $n_0 = \max k_i$ and all i. Let $N = (m+1)(n_0 - 1) + 1$. Any monomial of degree $e \geq N$ in $x_0, \ldots, x_m$ is divisible by some $x_i^{n_0}$ (pigeonhole), and T_e is spanned over T_0 by such monomials. Hence $ys \in T$ for $y \in T_e$, $e \geq N$.

(iii) Let $s \in T'_d$ ($d \geq 0$) and N as in (ii). For $y \in T_{\geq N}$, $ys \in T$ and $\deg(ys) \geq N$, so by induction $ys^q \in T$ for all $q \geq 1$. With $y = x_0^N \neq 0$ we get $T[s] \subseteq y^{-1}T$, a finitely generated T-module (T is noetherian), so s is integral over T. As T' is generated by such homogeneous elements it is integral over T, hence contained in the integral closure $\overline{T}$ of T in K, which is a finite T-module (I, 3.9A, since T is a finitely generated k-algebra). So T', a submodule, is finitely generated.

(iv) Let $u_1, \ldots, u_t$ be homogeneous generators of T' over T, $D = \max \deg u_j$, and by (ii) N' with $T_{\geq N'} u_j \subseteq T$ for all j. For $n \geq D + N'$: $T'_n = \sum_j T_{n - \deg u_j} u_j \subseteq T$, so $T'_n = T_n$. $\square$

(b) S is a finitely generated A-algebra, hence a finitely generated k-algebra and noetherian. Let $P(M)$ be the assertion "α_d is bijective for all $d \gg 0$".

Domain case. Let $\mathfrak{p} \subseteq S$ be a homogeneous prime and $T = S/\mathfrak{p}$. Then T satisfies the hypotheses of the lemma, except possibly $T_1 \neq 0$. If $T_1 = 0$, then $T = T_0$, $\operatorname{Proj} T = \emptyset$, $\widetilde{T} = 0$ and $T_d = 0$ for

$d \geq 1$, so $P(T)$ holds. If $T_1 \neq 0$, then by the lemma (i), (iv) $\alpha_d : T_d \to \Gamma(\widetilde{T}(d))$ is bijective for $d \gg 0$. (Here $\operatorname{Proj} T$ is the closed subscheme $j : \operatorname{Proj} T \hookrightarrow X$ defined by $\mathfrak{p}$; for $f \in S_1$ we have $(S/\mathfrak{p})_{(f)} = T_{(\bar{f})}$ and $j^{-1}D_+(f) = D_+(\bar{f})$, so the $\mathcal{O}_X$-module $\widetilde{S/\mathfrak{p}}$ is j_* of the sheaf $\widetilde{T}$ on $\operatorname{Proj} T$, and likewise for all twists; hence the global sections of $(\widetilde{S/\mathfrak{p}})(d)$ over X are those of $\widetilde{T}(d)$ over $\operatorname{Proj} T$, and under this identification α_d for $S/\mathfrak{p}$ is α_d for T, since both send m to the section with value $m/1$.) For the twist $T(l)$, α_d for $T(l)$ is α_{d+l} for T, so $P(T(l))$ holds too.

Extensions. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of graded modules with $P(M')$ and $P(M'')$. Twisting the exact sequence $0 \to \widetilde{M'} \to \widetilde{M} \to \widetilde{M''} \to 0$ by the invertible sheaf $\mathcal{O}(d)$ (exactness of $\widetilde{\ }$ follows from exactness of localization on stalks $M_{(\mathfrak{p})}$) and applying the left exact functor Γ gives a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \to & M'_d & \to & M_d & \to & M''_d & \to & 0 \\ & & \downarrow \alpha' & & \downarrow \alpha & & \downarrow \alpha'' & & \\ 0 & \to & \Gamma(\widetilde{M'}(d)) & \to & \Gamma(\widetilde{M}(d)) & \to & \Gamma(\widetilde{M''}(d)) & & \end{array}$$

For $d \gg 0$, α' and α'' are isomorphisms. Then α is injective (if $\alpha(m) = 0$ then $\alpha''(\bar{m}) = 0$, so $\bar{m} = 0$, $m \in M'_d$ and $\alpha'(m) = 0$, so $m = 0$). It is surjective: given $t \in \Gamma(\widetilde{M}(d))$, its image in $\Gamma(\widetilde{M''}(d))$ equals $\alpha''(\bar{m})$ for some $m \in M_d$, so $t - \alpha(m)$ maps to 0, hence comes from $\Gamma(\widetilde{M'}(d)) = \alpha'(M'_d)$, say $t - \alpha(m) = \alpha(m')$ with $m' \in M'_d \subseteq M_d$, and $t = \alpha(m + m')$. Thus $P(M)$ holds.

General case. For M finitely generated over the noetherian graded ring S, there is a filtration $0 = M^0 \subseteq \cdots \subseteq M^r = M$ by graded submodules with $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(l_i)$, $\mathfrak{p}_i$ homogeneous primes (I, 7.4). By the domain case and induction on r (using the extension step), $P(M)$ holds.

(c) Interpretation: the category $\mathcal{C}$ has as objects the quasi-finitely generated graded S-modules, and $\operatorname{Hom}_{\mathcal{C}}(M, N) = \varinjlim_d \operatorname{Hom}_{S\text{-gr}}(M_{\geq d}, N_{\geq d})$ (graded homomorphisms of degree 0, transition maps by restriction). Two objects are isomorphic in $\mathcal{C}$ iff $M \approx N$.

Step 1. For any graded M and d, $\widetilde{M_{\geq d}} \to \widetilde{M}$ is an isomorphism: for $\mathfrak{p} \in X$ choose $x \in S_1 \setminus \mathfrak{p}$ (exists as S_+ is generated by S_1); a degree zero element m/f of $(M/M_{\geq d})_{(\mathfrak{p})}$ equals $x^d m/(x^d f) = 0$ since $x^d m \in M_{\geq d}$. Hence if $M \approx N$ with N finitely generated, then $\widetilde{M} \cong \widetilde{N}$ is coherent (5.11(c), S noetherian), and by naturality of α and (b), $\alpha_d(M)$ is an isomorphism for all $d \geq d_1(M)$.

Step 2: the functor. A morphism in $\mathcal{C}$ represented by $\varphi : M_{\geq d} \to N_{\geq d}$ gives $\widetilde{\varphi} : \widetilde{M} = \widetilde{M_{\geq d}} \to \widetilde{N}$, independent of the representative. So $M \mapsto \widetilde{M}$ is a functor $\mathcal{C} \to \operatorname{Coh}(X)$.

Step 3: fully faithful. Let $\Theta : \operatorname{Hom}_{\mathcal{C}}(M, N) \to \operatorname{Hom}(\widetilde{M}, \widetilde{N})$ be the map. Let d_1 be such that $\alpha_n(M)$, $\alpha_n(N)$ are isomorphisms for $n \geq d_1$. *Injective:* if $\widetilde{\varphi} = 0$ for $\varphi : M_{\geq d} \to N_{\geq d}$, then $\alpha_N \varphi = \Gamma_*(\widetilde{\varphi})\alpha_M = 0$, so $\varphi = 0$ in degrees $\geq d_1$. *Surjective:* given $\psi : \widetilde{M} \to \widetilde{N}$, put $\varphi = \alpha_N^{-1} \circ \Gamma_*(\psi) \circ \alpha_M : M_{\geq d_1} \to N_{\geq d_1}$, a graded S-homomorphism. We check $\widetilde{\varphi} = \psi$ on $D_+(f)$, $f \in S_1$ (these cover X): $\widetilde{M}(D_+(f)) = M_{(f)} \ni m/f^k$, and we may assume $m \in M_k$, $k \geq d_1$. The section f of $\mathcal{O}(1)$ trivializes $\mathcal{O}(1)$ on $D_+(f)$, and under the resulting identification $\widetilde{M}(k)|_{D_+(f)} \cong \widetilde{M}|_{D_+(f)}$ the section $\alpha(m)$ corresponds to m/f^k; ψ commutes with these identifications, so $\psi(m/f^k)$ corresponds to $\Gamma_*(\psi)(\alpha(m)) = \alpha(\varphi(m))$, which corresponds to $\varphi(m)/f^k = \widetilde{\varphi}(m/f^k)$.

Step 4: essentially surjective. Let $\mathcal{F}$ be coherent on X. As S_1 is a finitely generated A-module, X is a closed subscheme of $\mathbf{P}^r_A$ and $\mathcal{O}_X(1)$ is the restriction of $\mathcal{O}_{\mathbf{P}^r}(1)$ (5.12(c)), so it is very

ample; by Serre's theorem (5.17) there are n and finitely many $s_1, \dots, s_t \in \Gamma(X, \mathcal{F}(n))$ generating $\mathcal{F}(n)$. Let $M = \Gamma_*(\mathcal{F})$ and let $N \subseteq M$ be the graded submodule generated by the $s_j \in M_n$; N is finitely generated. By 5.15, $\widetilde{M} \cong \mathcal{F}$. The image of $\widetilde{N}(n) \to \widetilde{M}(n) \cong \mathcal{F}(n)$ is an $\mathcal{O}_X$-submodule that contains all the s_j, hence is all of $\mathcal{F}(n)$; since $\widetilde{N} \to \widetilde{M}$ is injective, $\widetilde{N}(n) \cong \mathcal{F}(n)$, and $\widetilde{N} \cong \mathcal{F}$. So $\mathcal{F} \cong \widetilde{N}$ with N finitely generated, and $N \in \mathcal{C}$. (Moreover $M = \Gamma_*(\mathcal{F}) \approx N$: since $\widetilde{N} = \widetilde{M}$, the map $\alpha_d(N) : N_d \to \Gamma(X, \widetilde{N}(d)) = \Gamma(X, \mathcal{F}(d)) = M_d$ is the inclusion $N_d \subseteq M_d$, and it is bijective for $d \gg 0$ by (b) applied to the finitely generated module N. So $\Gamma_*(\mathcal{F})$ is quasi-finitely generated and, by Step 3, Γ_* is a quasi-inverse to $\widetilde{\ }$.)

Hence $\widetilde{\ } : \mathcal{C} \to \mathrm{Coh}(X)$ is fully faithful and essentially surjective, i.e. an equivalence, with quasi-inverse Γ_*. □

Exercise 5.10

Problem. Let A be a ring, let $S = A[x_0, \dots, x_r]$ and let $X = \operatorname{Proj} S$. We have seen that a homogeneous ideal I in S defines a closed subscheme of X (Ex. 3.12), and that conversely every closed subscheme of X arises in this way (5.16).

(a) For any homogeneous ideal $I \subseteq S$, we define the *saturation* $\bar{I}$ of I to be $\{s \in S \mid$ for each $i = 0, \dots, r$, there is an n such that $x_i^n s \in I\}$. We say that I is *saturated* if $I = \bar{I}$. Show that $\bar{I}$ is a homogeneous ideal of S.

(b) Two homogeneous ideals I_1 and I_2 of S define the same closed subscheme of X if and only if they have the same saturation.

(c) If Y is any closed subscheme of X, then the ideal $\Gamma_*(\mathcal{I}_Y)$ is saturated. Hence it is the largest homogeneous ideal defining the subscheme Y.

(d) There is a 1-1 correspondence between saturated ideals of S and closed subschemes of X.

Solution. (Throughout $r \geq 1$, as in 5.13 and 5.16; this is needed for $\Gamma_*(\mathcal{O}_X) = S$. For $r = 0$ one has $\Gamma_*(\mathcal{O}_X) = A[x_0, x_0^{-1}] \neq S$, so (c) is not meaningful as stated; we do not treat that case.)

(a) Clearly $I \subseteq \bar{I}$. If $s, t \in \bar{I}$ with $x_i^n s \in I$, $x_i^m t \in I$, then $x_i^{\max(n,m)}(s+t) \in I$; if $u \in S$ then $x_i^n(us) \in I$. So $\bar{I}$ is an ideal. If $s = \sum s_d$ (homogeneous components) lies in $\bar{I}$ and $x_i^n s \in I$, then all components $x_i^n s_d$ lie in I because I is homogeneous; taking the maximum of the finitely many exponents needed (for each i, over the components s_d), we get $s_d \in \bar{I}$. So $\bar{I}$ is homogeneous.

Two facts. For a homogeneous ideal I let $Y_I \subseteq X$ be the closed subscheme it defines, with ideal sheaf $\widetilde{I} \subseteq \widetilde{S} = \mathcal{O}_X$ (proof of 5.16(a)); a closed subscheme is determined by its ideal sheaf (5.9).

Fact 1: $\widetilde{I} = \widetilde{\bar{I}}$. The module $\bar{I}/I$ has $(\bar{I}/I)_{(\mathfrak{p})} = 0$ for every $\mathfrak{p} \in X$: choose i with $x_i \notin \mathfrak{p}$ ($\mathfrak{p} \not\supseteq S_+$); a degree-0 element of $T^{-1}(\bar{I}/I)$ is $\bar{s}/f$ with $s \in \bar{I}$ homogeneous, and $x_i^n s \in I$ for some n, so $\bar{s}/f = \overline{x_i^n s}/(x_i^n f) = 0$. As $\widetilde{\ }$ is exact, $\widetilde{\bar{I}}/\widetilde{I} = 0$.

Fact 2: $\Gamma_*(\widetilde{I}) = \bar{I}$ (as submodules of $\Gamma_*(\mathcal{O}_X) = S$, 5.13). Since $\widetilde{I(n)} \subseteq \widetilde{S(n)}$, a homogeneous $s \in S_n$ gives a section of $\widetilde{I}(n)$ iff its restriction to each $D_+(x_i)$ lies in $I(n)_{(x_i)} \subseteq S(n)_{(x_i)}$ (5.11(b)), i.e. iff $s \in I_{x_i}$ for all i, i.e. iff for each i there is n_i with $x_i^{n_i} s \in I$, i.e. iff $s \in \bar{I}$. Both sides are graded, so they agree.

(b) If I_1, I_2 have the same saturation, then $\widetilde{I_1} = \widetilde{\bar{I}_1} = \widetilde{\bar{I}_2} = \widetilde{I_2}$ (Fact 1), so they define the same closed subscheme. Conversely, if $Y_{I_1} = Y_{I_2}$, then $\widetilde{I_1} = \widetilde{I_2}$ and by Fact 2, $\bar{I}_1 = \Gamma_*(\widetilde{I_1}) = \Gamma_*(\widetilde{I_2}) = \bar{I}_2$.

(c) Let $Y \subseteq X$ be closed and $I = \Gamma_*(\mathcal{I}_Y)$, a homogeneous ideal of S. By 5.9 $\mathcal{I}_Y$ is quasi-coherent, and by 5.15 $\widetilde{I} = \widetilde{\Gamma_*(\mathcal{I}_Y)} \cong \mathcal{I}_Y$ (the isomorphism being compatible with the inclusion into $\mathcal{O}_X$), so I defines Y. By Fact 2, $\bar{I} = \Gamma_*(\widetilde{I}) = \Gamma_*(\mathcal{I}_Y) = I$: I is saturated. If J is any homogeneous ideal defining Y (i.e. $\widetilde{J} = \mathcal{I}_Y$), then $J \subseteq \bar{J} = \Gamma_*(\widetilde{J}) = \Gamma_*(\mathcal{I}_Y) = I$. So I is the largest.

(d) Map $I \mapsto Y_I$ from saturated homogeneous ideals to closed subschemes. Injective by (b): saturated I_1, I_2 defining the same subscheme satisfy $I_1 = \bar{I}_1 = \bar{I}_2 = I_2$. Surjective by (c): $Y = Y_I$ with $I = \Gamma_*(\mathcal{I}_Y)$ saturated. The inverse is $Y \mapsto \Gamma_*(\mathcal{I}_Y)$. □

Exercise 5.11

> **Problem.** Let S and T be two graded rings with $S_0 = T_0 = A$. We define the *Cartesian product* $S \times_A T$ to be the graded ring $\bigoplus_{d \geq 0} S_d \otimes_A T_d$. If $X = \operatorname{Proj} S$ and $Y = \operatorname{Proj} T$, show that $\operatorname{Proj}(S \times_A T) \cong X \times_A Y$, and show that the sheaf $\mathcal{O}(1)$ on $\operatorname{Proj}(S \times_A T)$ is isomorphic to $p_1^*(\mathcal{O}_X(1)) \otimes p_2^*(\mathcal{O}_Y(1))$ on $X \times Y$.
>
> The Cartesian product of rings is related to the Segre embedding of projective spaces (I, Ex. 2.14) in the following way. If $x_0, \ldots, x_r$ is a set of generators for S_1 over A, corresponding to a projective embedding $X \hookrightarrow \mathbf{P}^r_A$, and if $y_0, \ldots, y_s$ is a set of generators for T_1, corresponding to a projective embedding $Y \hookrightarrow \mathbf{P}^s_A$, then $\{x_i \otimes y_j\}$ is a set of generators for $(S \times_A T)_1$, and hence defines a projective embedding $\operatorname{Proj}(S \times_A T) \hookrightarrow \mathbf{P}^N_A$, with $N = rs + r + s$. This is just the image of $X \times Y \subseteq \mathbf{P}^r \times \mathbf{P}^s$ in $\mathbf{P}^N$ under its Segre embedding.

Solution. *Convention.* As in 5.12 (which is needed for $\mathcal{O}(1)$ to be invertible) and in the discussion of the Segre embedding, we assume that S and T are generated over A by S_1 and T_1 respectively. (The statement about $\mathcal{O}(1)$ is meaningless otherwise; we make no claim about the first statement without this hypothesis.) Put $R = S \times_A T$, so $R_d = S_d \otimes_A T_d$ and $R_1 = S_1 \otimes_A T_1$. R is generated by R_1 over A because R_d is spanned by $(x_{i_1} \cdots x_{i_d}) \otimes (y_{j_1} \cdots y_{j_d}) = \prod_k (x_{i_k} \otimes y_{j_k})$.

Step 1: affine pieces. Let $x \in S_a$, $y \in T_a$ $(a \geq 1)$, so $x \otimes y \in R_a$. Then

$$R_{(x \otimes y)} \;\cong\; S_{(x)} \otimes_A T_{(y)}, \qquad \frac{s \otimes t}{(x \otimes y)^k} \leftrightarrow \frac{s}{x^k} \otimes \frac{t}{y^k} \quad (s \in S_{ak},\, t \in T_{ak}).$$

Indeed, $S_{(x)} = (S_x)_0 = \varinjlim_k (S_{ak}, \cdot x)$, $T_{(y)} = \varinjlim_k (T_{ak}, \cdot y)$, $R_{(x \otimes y)} = \varinjlim_k (R_{ak}, \cdot x \otimes y)$, and tensor products commute with filtered colimits, so $S_{(x)} \otimes_A T_{(y)} = \varinjlim_k S_{ak} \otimes_A T_{ak} = \varinjlim_k R_{ak}$; the identification respects the ring structures. Similarly for $M = S(1)$, $N = T(1)$ one gets (for $a = 1$) an isomorphism of modules over this ring

$$R(1)_{(x \otimes y)} = \varinjlim_k (S_{k+1} \otimes T_{k+1}) \;\cong\; S(1)_{(x)} \otimes_A T(1)_{(y)}.$$

Step 2: the isomorphism of schemes. Choose generating sets $\{x_\alpha\}$ of S_1 and $\{y_\beta\}$ of T_1 over A. Then $X = \bigcup D_+(x_\alpha)$, $Y = \bigcup D_+(y_\beta)$, $X \times_A Y = \bigcup U_{\alpha\beta}$ with $U_{\alpha\beta} = D_+(x_\alpha) \times_A D_+(y_\beta) = \operatorname{Spec}(S_{(x_\alpha)} \otimes T_{(y_\beta)})$ (3.3), and the $x_\alpha \otimes y_\beta$ generate the A-module R_1, hence the ideal R_+, so $Z = \operatorname{Proj} R = \bigcup V_{\alpha\beta}$, $V_{\alpha\beta} = D_+(x_\alpha \otimes y_\beta) = \operatorname{Spec} R_{(x_\alpha \otimes y_\beta)}$. By Step 1 we have isomorphisms

$\phi_{\alpha\beta} : V_{\alpha\beta} \to U_{\alpha\beta}$. On overlaps, $V_{\alpha\beta} \cap V_{\alpha'\beta'} = D_+(x_\alpha x_{\alpha'} \otimes y_\beta y_{\beta'}) = \operatorname{Spec} R_{(x_\alpha x_{\alpha'} \otimes y_\beta y_{\beta'})}$ and $U_{\alpha\beta} \cap U_{\alpha'\beta'} = D_+(x_\alpha x_{\alpha'}) \times D_+(y_\beta y_{\beta'}) = \operatorname{Spec} S_{(x_\alpha x_{\alpha'})} \otimes T_{(y_\beta y_{\beta'})}$; the isomorphism of Step 1 (with $a = 2$) for this pair restricts from both $\phi_{\alpha\beta}$ and $\phi_{\alpha'\beta'}$, because all these isomorphisms are given by the same formula $\frac{s\otimes t}{(x\otimes y)^k} \mapsto \frac{s}{x^k} \otimes \frac{t}{y^k}$ and the localization maps are induced by the identity on $S \times_A T$. (Explicitly, write $x = x_\alpha, x' = x_{\alpha'}$, $y = y_\beta, y' = y_{\beta'}$. The ring map $S_{(x)} \otimes T_{(y)} \to S_{(xx')} \otimes T_{(yy')} \cong R_{(xx'\otimes yy')}$ coming from $U_{\alpha'\beta'} \cap U_{\alpha\beta} \subseteq U_{\alpha\beta}$ sends $\frac{s}{x^k} \otimes \frac{t}{y^k}$ to $\frac{sx'^k}{(xx')^k} \otimes \frac{ty'^k}{(yy')^k} \leftrightarrow \frac{(s\otimes t)(x'\otimes y')^k}{((x\otimes y)(x'\otimes y'))^k} = \frac{s\otimes t}{(x\otimes y)^k}$, which is the localization map $R_{(x\otimes y)} \to R_{(xx'\otimes yy')}$ composed with the isomorphism of Step 1 for $(x \otimes y)$. The same holds with the two indices exchanged.) Hence the $\phi_{\alpha\beta}$ glue to an isomorphism $\Phi : Z \to X \times_A Y$.

Step 3: the sheaf $\mathcal{O}(1)$. On $V = V_{\alpha\beta}$, $\mathcal{O}_Z(1)|_V = \widetilde{R(1)_{(x\otimes y)}}$ (5.11(b), 5.12(b)). On $U = U_{\alpha\beta} = \operatorname{Spec} B$, $B = S_{(x)} \otimes_A T_{(y)}$, put $M_1 = S(1)_{(x)}$, $N_1 = T(1)_{(y)}$; then $\mathcal{O}_X(1)|_{D_+(x)} = \widetilde{M_1}$, $\mathcal{O}_Y(1)|_{D_+(y)} = \widetilde{N_1}$ and by 5.2(e), $p_1^*\widetilde{M_1} = (M_1 \otimes_{S_{(x)}} B)^\sim = (M_1 \otimes_A T_{(y)})^\sim$, $p_2^*\widetilde{N_1} = (S_{(x)} \otimes_A N_1)^\sim$. By 5.2(b), $p_1^*\mathcal{O}_X(1) \otimes p_2^*\mathcal{O}_Y(1)|_U = \big((M_1 \otimes_A T_{(y)}) \otimes_B (S_{(x)} \otimes_A N_1)\big)^\sim = (M_1 \otimes_A N_1)^\sim$. By Step 1, $M_1 \otimes_A N_1 \cong R(1)_{(x\otimes y)}$ as $B \cong R_{(x\otimes y)}$-modules. So $\Phi^*(p_1^*\mathcal{O}_X(1) \otimes p_2^*\mathcal{O}_Y(1))|_V \cong \mathcal{O}_Z(1)|_V$, via the formula $(\frac{s}{x^k}) \otimes (\frac{t}{y^k}) \mapsto \frac{s\otimes t}{(x\otimes y)^k}$ (with $s \in S_{k+1}, t \in T_{k+1}$); these local isomorphisms agree on overlaps for the same reason as in Step 2, hence glue to a global isomorphism $\mathcal{O}_Z(1) \cong \Phi^*\big(p_1^*\mathcal{O}_X(1) \otimes p_2^*\mathcal{O}_Y(1)\big)$.

The Segre embedding. By the above, R is generated over A by the elements $x_i \otimes y_j$ of degree 1, so $A[z_{ij}] \to R$, $z_{ij} \mapsto x_i \otimes y_j$, is a surjective graded homomorphism ($N + 1 = (r + 1)(s + 1)$ variables), and $\operatorname{Proj} R \hookrightarrow \mathbf{P}^N_A$ is a closed immersion (Ex. 3.12). For $S = A[x_0..x_r]$, $T = A[y_0..y_s]$ this is the Segre embedding $\sigma : \mathbf{P}^r \times \mathbf{P}^s \to \mathbf{P}^N$, and $\sigma^*\mathcal{O}(1) \cong p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)$ by the above. For general $S = A[x]/I$, $T = A[y]/J$, $S \times_A T$ is a quotient of $A[x] \times_A A[y]$, so $\operatorname{Proj}(S \times_A T)$ is a closed subscheme of $\mathbf{P}^r \times \mathbf{P}^s \subseteq \mathbf{P}^N$; by naturality of Φ with respect to the surjections $A[x] \to S$, $A[y] \to T$ it is $X \times_A Y \subseteq \mathbf{P}^r \times \mathbf{P}^s$, embedded by σ. □

Exercise 5.12

Problem.

(a) Let X be a scheme over a scheme Y, and let $\mathcal{L}, \mathcal{M}$ be two very ample invertible sheaves on X. Show that $\mathcal{L} \otimes \mathcal{M}$ is also very ample. [*Hint*: Use a Segre embedding.]

(b) Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of schemes. Let $\mathcal{L}$ be a very ample invertible sheaf on X relative to Y, and let $\mathcal{M}$ be a very ample invertible sheaf on Y relative to Z. Show that $\mathcal{L} \otimes f^*\mathcal{M}$ is a very ample invertible sheaf on X relative to Z.

Solution. In (a) "very ample" means very ample relative to Y: there is an immersion $i : X \to \mathbf{P}^r_Y$ of Y-schemes with $i^*\mathcal{O}(1) \cong \mathcal{L}$ (the definition after 5.16).

The Segre embedding over a base. Apply Ex. 5.11 with $A = \mathbf{Z}$, $S = \mathbf{Z}[x_0, \dots, x_r]$, $T = \mathbf{Z}[y_0, \dots, y_s]$: there is a closed immersion $\sigma_{\mathbf{Z}} : \mathbf{P}^r_{\mathbf{Z}} \times_{\mathbf{Z}} \mathbf{P}^s_{\mathbf{Z}} \to \mathbf{P}^N_{\mathbf{Z}}$, $N = rs + r + s$, with $\sigma_{\mathbf{Z}}^*\mathcal{O}(1) \cong p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)$. For any scheme W put $\mathbf{P}^n_W = \mathbf{P}^n_{\mathbf{Z}} \times W$ and $\mathcal{O}_{\mathbf{P}^n_W}(1) = h^*\mathcal{O}_{\mathbf{P}^n_{\mathbf{Z}}}(1)$ with h the projection. Base changing by W, $\sigma_W : \mathbf{P}^r_W \times_W \mathbf{P}^s_W \to \mathbf{P}^N_W$ is a closed immersion (base change of a closed immersion), and since $h \circ \sigma_W = \sigma_{\mathbf{Z}} \circ h'$ for the projection h' and $p_k \circ h' = h \circ q_k$ (q_k the projections of $\mathbf{P}^r_W \times_W \mathbf{P}^s_W$),

$$\sigma_W^*\mathcal{O}(1) \cong q_1^*\mathcal{O}_{\mathbf{P}^r_W}(1) \otimes q_2^*\mathcal{O}_{\mathbf{P}^s_W}(1),$$

using that u^* commutes with $\otimes$ and $(uv)^* = v^*u^*$.

We will use the standard facts: open and closed immersions, hence immersions, are stable under composition and base change; the diagonal $\Delta : X \to X \times_Y X$ is an immersion (Sect. 4).

(a) Let $i : X \to \mathbf{P}^r_Y$, $j : X \to \mathbf{P}^s_Y$ be immersions with $i^*\mathcal{O}(1) = \mathcal{L}$, $j^*\mathcal{O}(1) = \mathcal{M}$. Let $\kappa = (i \times_Y j) \circ \Delta : X \to X \times_Y X \to \mathbf{P}^r_Y \times_Y \mathbf{P}^s_Y$; here $i \times_Y j = (i \times \mathrm{id}) \circ (\mathrm{id} \times j)$ is a composition of base changes of immersions, so an immersion, and Δ is an immersion; hence κ is an immersion. Then $k = \sigma_Y \circ \kappa : X \to \mathbf{P}^N_Y$ is an immersion of Y-schemes, and since $q_1\kappa = i$, $q_2\kappa = j$,

$$k^*\mathcal{O}(1) = \kappa^*\big(q_1^*\mathcal{O}(1) \otimes q_2^*\mathcal{O}(1)\big) = i^*\mathcal{O}(1) \otimes j^*\mathcal{O}(1) \cong \mathcal{L} \otimes \mathcal{M}.$$

So $\mathcal{L} \otimes \mathcal{M}$ is very ample relative to Y (it is invertible as a tensor product of invertible sheaves).

(b) Let $i : X \to \mathbf{P}^r_Y$ (Y-immersion) with $i^*\mathcal{O}(1) = \mathcal{L}$ and $j : Y \to \mathbf{P}^s_Z$ (Z-immersion) with $j^*\mathcal{O}(1) = \mathcal{M}$. Write $\mathbf{P}^r_Y = \mathbf{P}^r_Z \times_Z Y$ with projections $q : \mathbf{P}^r_Y \to \mathbf{P}^r_Z$, $\pi : \mathbf{P}^r_Y \to Y$; by definition $\mathcal{O}_{\mathbf{P}^r_Y}(1) = q^*\mathcal{O}_{\mathbf{P}^r_Z}(1)$. Let $\tau = \mathrm{id} \times j : \mathbf{P}^r_Z \times_Z Y \to \mathbf{P}^r_Z \times_Z \mathbf{P}^s_Z$, an immersion (base change of j), and $k = \sigma_Z \circ \tau \circ i : X \to \mathbf{P}^N_Z$, an immersion of Z-schemes (i is an immersion; composites of immersions are immersions). With $p_1\tau = q$, $p_2\tau = j\pi$:

$$\tau^*\sigma_Z^*\mathcal{O}(1) = \tau^*(p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)) = q^*\mathcal{O}(1) \otimes \pi^*j^*\mathcal{O}(1) = \mathcal{O}_{\mathbf{P}^r_Y}(1) \otimes \pi^*\mathcal{M},$$

and pulling back by i, using $\pi \circ i = f$ (i is a morphism over Y): $k^*\mathcal{O}(1) = i^*\mathcal{O}(1) \otimes i^*\pi^*\mathcal{M} = \mathcal{L} \otimes f^*\mathcal{M}$. □

Exercise 5.13

> **Problem.** Let S be a graded ring, generated by S_1 as an S_0-algebra. For any integer $d > 0$, let $S^{(d)}$ be the graded ring $\bigoplus_{n \geq 0} S^{(d)}_n$ where $S^{(d)}_n = S_{nd}$. Let $X = \operatorname{Proj} S$. Show that $\operatorname{Proj} S^{(d)} \cong X$, and that the sheaf $\mathcal{O}(1)$ on $\operatorname{Proj} S^{(d)}$ corresponds via this isomorphism to $\mathcal{O}_X(d)$.
>
> This construction is related to the d-uple embedding (I, Ex. 2.12) in the following way. If $x_0, \ldots, x_r$ is a set of generators for S_1, corresponding to an embedding $X \hookrightarrow \mathbf{P}^r_A$, then the set of monomials of degree d in the x_i is a set of generators for $S^{(d)}_1 = S_d$. These define a projective embedding of $\operatorname{Proj} S^{(d)}$ which is none other than the image of X under the d-uple embedding of $\mathbf{P}^r_A$.

Solution. Write $A = S_0$. Note that $S^{(d)}$ is generated over A by $S^{(d)}_1 = S_d$: S_{nd} is spanned over A by products of nd elements of S_1, i.e. by products of n elements of S_d.

The schemes. For $h \in S_d$ let $D_+(h) \subseteq X$ and $D'_+(h) \subseteq X' = \operatorname{Proj} S^{(d)}$ be the corresponding basic open sets (h has degree d in S and degree 1 in $S^{(d)}$). Then

$$S^{(d)}_{(h)} = \left\{ \tfrac{s}{h^k} : s \in S_{dk} \right\} = S_{(h)}$$

as subrings of S_h: indeed $(S_h)_0$ consists of the elements s/h^k with $s \in S_{dk}$ (the denominators h^k have degree dk), and both rings are by definition this set. So $D_+(h) = \operatorname{Spec} S_{(h)} = D'_+(h)$ as schemes. Now $X = \bigcup_{h \in S_d} D_+(h)$: if $\mathfrak{p} \in X$, some $f \in S_1$ is not in $\mathfrak{p}$ (S_+ generated by S_1), and $f^d \in S_d \setminus \mathfrak{p}$. Likewise $X' = \bigcup_{h \in S_d} D'_+(h)$ because $S^{(d)}$ is generated by $S^{(d)}_1 = S_d$. For $h, h' \in S_d$,

$D_+(h) \cap D_+(h') = D_+(hh') = \operatorname{Spec} S_{(hh')}$ and $D'_+(h) \cap D'_+(h') = D'_+(hh') = \operatorname{Spec} S^{(d)}_{(hh')} = \operatorname{Spec} S_{(hh')}$ (same computation with hh' of degree $2d$), and the inclusions are given by the same localization maps. Thus X and X' are glued from the same affine schemes $\operatorname{Spec} S_{(h)}$ ($h \in S_d$) along the same open subschemes by the same maps, so there is a canonical isomorphism $X' \cong X$ restricting to the identity of $\operatorname{Spec} S_{(h)}$ for each h.

The sheaf. On $D'_+(h)$, $\mathcal{O}_{X'}(1) = \widetilde{S^{(d)}(1)}$ has sections $S^{(d)}(1)_{(h)} = \{s/h^k : s \in S^{(d)}_{k+1} = S_{d(k+1)}\}$, and on $D_+(h)$, $\mathcal{O}_X(d) = \widetilde{S(d)}$ has sections $S(d)_{(h)} = \{s/h^k : s \in S(d)_{dk} = S_{dk+d}\}$. These are the same $S_{(h)}$-submodules of S_h (the degree d part of S_h), compatible with restriction to $D_+(hh')$. So $\mathcal{O}_{X'}(1) \cong \mathcal{O}_X(d)$ under $X' \cong X$ (5.11(b), 5.12(b)).

The d-uple embedding. Let $x_0, \dots, x_r$ generate S_1 and $M_0, \dots, M_N$ be the monomials of degree d in them; they generate S_d over A. The surjection $A[z_0, \dots, z_N] \to S^{(d)}$, $z_j \mapsto M_j$, gives a closed immersion $X' = \operatorname{Proj} S^{(d)} \hookrightarrow \mathbf{P}^N_A$ (Ex. 3.12). In particular for $S = A[x_0, \dots, x_r]$ this gives $\mathbf{P}^r_A \cong \operatorname{Proj} S^{(d)} \hookrightarrow \mathbf{P}^N_A$, which is the d-uple embedding ν_d (the map defined by the monomials of degree d, with $\nu_d^* \mathcal{O}(1) = \mathcal{O}(d)$ by the above). For general $S = A[x]/I$, the surjection $A[x] \to S$ induces a surjection $A[x]^{(d)} \to S^{(d)}$, so X' is a closed subscheme of $\operatorname{Proj} A[x]^{(d)} \cong \mathbf{P}^r_A$, and the isomorphisms of the first part are compatible with these maps (they are identities on each $\operatorname{Spec} S_{(h)}$). Hence the embedding of $X' \cong X$ in $\mathbf{P}^N_A$ is the composite $X \hookrightarrow \mathbf{P}^r_A \xrightarrow{\nu_d} \mathbf{P}^N_A$, the image of X under the d-uple embedding. □

Exercise 5.14

Problem. Let A be a ring, and let X be a closed subscheme of $\mathbf{P}^r_A$. We define the *homogeneous coordinate ring* $S(X)$ of X for the given embedding to be $A[x_0, \dots, x_r]/I$, where I is the ideal $\Gamma_*(\mathcal{I}_X)$ constructed in the proof of (5.16). (Of course if A is a field and X a variety, this coincides with the definition given in (I, Sect. 2)!) Recall that a scheme X is *normal* if its local rings are integrally closed domains. A closed subscheme $X \subseteq \mathbf{P}^r_k$ is *projectively normal* for the given embedding, if its homogeneous coordinate ring $S(X)$ is an integrally closed domain (cf. (I, Ex. 3.18)). Now assume that k is an algebraically closed field, and that X is a connected, normal closed subscheme of $\mathbf{P}^r_k$. Show that for some $d > 0$, the d-uple embedding of X is projectively normal, as follows.

(a) Let S be the homogeneous coordinate ring of X, and let $S' = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n))$. Show that S is a domain, and that S' is its integral closure. [*Hint*: First show that X is integral. Then regard S' as the global sections of the sheaf of rings $\mathcal{S} = \bigoplus_{n \geq 0} \mathcal{O}_X(n)$ on X, and show that $\mathcal{S}$ is a sheaf of integrally closed domains.]

(b) Use (Ex. 5.9) to show that $S_d = S'_d$ for all sufficiently large d.

(c) Show that $S^{(d)}$ is integrally closed for sufficiently large d, and hence conclude that the d-uple embedding of X is projectively normal.

(d) As a corollary of (a), show that a closed subscheme $X \subseteq \mathbf{P}^r_k$ is projectively normal if and only if it is normal, and for every $n \geq 0$ the natural map $\Gamma(\mathbf{P}^r, \mathcal{O}_{\mathbf{P}^r}(n)) \to \Gamma(X, \mathcal{O}_X(n))$ is surjective.

Solution. Throughout $X \neq \emptyset$ (connected spaces are nonempty), $P = k[x_0, \dots, x_r]$, $I = \Gamma_*(\mathcal{I}_X)$ (a saturated homogeneous ideal with $\widetilde{I} = \mathcal{I}_X$, by 5.10(c)), $S = P/I$, so $X = \operatorname{Proj} S$ as a closed subscheme (Ex. 3.12) and $\mathcal{O}_X(n) = i^* \mathcal{O}_{\mathbf{P}^r}(n) = \widetilde{S}(n)$ (5.12(c)). S is generated by S_1 over $S_0 = k$ and S_1 is finite dimensional. Also $S_1 \neq 0$, since $S_1 = 0$ would give $\operatorname{Proj} S = \emptyset$. We assume $r \geq 1$

(needed for 5.10 and 5.13); if $r = 0$ then X is the point $\mathbf{P}^0_k$, $S = k[x_0]$ and $S^{(d)} = k[x_0^d]$ are polynomial rings in one variable, hence integrally closed, and everything is trivial.

(a) *X is integral.* X is noetherian and its local rings are domains. If a point lay on two irreducible components, its local ring would have two minimal primes. So the (finitely many) irreducible components are pairwise disjoint closed sets, hence each is open; as X is connected, X is irreducible. Local rings are domains, so X is reduced. Hence X is integral; with K its function field. (It is a variety.)

S injects into S'. The exact sequence $0 \to \mathcal{I}_X \to \mathcal{O}_{\mathbf{P}} \to i_*\mathcal{O}_X \to 0$ twisted by $\mathcal{O}(n)$ and the left exactness of Γ give $0 \to \Gamma(\mathcal{I}_X(n)) = I_n \to P_n \to \Gamma(X, \mathcal{O}_X(n))$, so $S_n = P_n/I_n \hookrightarrow \Gamma(X, \mathcal{O}_X(n))$ for $n \geq 0$; this is compatible with products, so $S \subseteq S'$ is a subring.

$\mathcal{S}$ is a sheaf of integrally closed domains. Let ξ be the generic point. Each $\mathcal{O}_X(n)$ is invertible, and $\mathcal{S} = \bigoplus_{n\geq 0} \mathcal{O}_X(n)$ is a sheaf of graded rings. Let $U = \operatorname{Spec} B$ be an open affine on which $\mathcal{O}_X(1)|_U = \tilde{\mathcal{O}}_U e$ is free; then $\mathcal{O}(n)|_U = \mathcal{O}_U e^n$ and $\mathcal{S}(U) = B[e]$ is a polynomial ring, with B an integrally closed domain (the local rings of B are integrally closed and $B = \bigcap B_{\mathfrak{m}}$). A polynomial ring over an integrally closed domain is integrally closed. The stalk $\mathcal{S}_\xi = K[e]$. For every nonempty open V the restriction map $\mathcal{S}(V) \to \mathcal{S}_\xi$ is injective (check on trivializing affine opens, where it is $B[e] \subseteq K[e]$). Since X is quasi-compact, $S' = \Gamma(X, \mathcal{S})$, and $S' \hookrightarrow \mathcal{S}_\xi = K[e]$ is injective, so S' is a domain, and so is $S \subseteq S'$.

S' is integrally closed in $\operatorname{Frac} S'$. Let $\theta \in \operatorname{Frac}(S')$ be integral over S'. Embed $\operatorname{Frac} S' \subseteq \Omega = K(e)$. For a trivializing open affine U as above, $S' \to \mathcal{S}(U) \subseteq \mathcal{S}_\xi$, so θ is integral over $\mathcal{S}(U) = B[e]$ and lies in its fraction field $K(e) = \Omega$, hence $\theta \in \mathcal{S}(U)$ (integrally closed). Call it θ_U. For two such U, U', $\theta_U|_{U\cap U'}$ and $\theta_{U'}|_{U\cap U'}$ have the same germ θ at ξ, so they are equal; the θ_U glue to $\tilde{\theta} \in \mathcal{S}(X) = S'$ with germ θ. So $\theta \in S'$.

S' is integral over S and $\operatorname{Frac} S' = \operatorname{Frac} S$. Apply the Lemma in the solution of 5.9 to $T = S$ (a graded domain with $T_0 = k$): it shows S' (which equals the T' of the lemma by part (i) of the lemma) is integral over S and every $s' \in S'$ satisfies $x_i^n s' \in S$ with some $x_i \neq 0$, so $s' \in \operatorname{Frac} S$. Hence $\operatorname{Frac} S' = \operatorname{Frac} S$. Thus S' is integral over S and integrally closed in $\operatorname{Frac} S$, so S' is the integral closure of S.

(b) Apply Ex. 5.9(b) to $M = S$ ($S_0 = k$, S_1 finite-dimensional; $\widetilde{S}(d) = \mathcal{O}_X(d)$): $\alpha_d : S_d \to \Gamma(X, \mathcal{O}_X(d)) = S'_d$, which is the inclusion $S \subseteq S'$, is an isomorphism for $d \geq d_0$ (this is also (iv) of the Lemma).

(c) Let $d \geq d_0$. First, $S'_0 = \Gamma(X, \mathcal{O}_X)$ is a domain, and integral over $S_0 = k$: an element $a \in S'_0$ is integral over S by the lemma's (iii), and taking degree-0 components in an equation of integral dependence $a^q + c_1 a^{q-1} + \cdots + c_q = 0$ ($c_i \in S$) gives one with coefficients in $S_0 = k$. So every element of S'_0 is algebraic over the algebraically closed field k, hence in k: $S'_0 = k = S_0$. By (b), $S'_{nd} = S_{nd}$ for $n \geq 1$. Hence $S^{(d)} = S'^{(d)}$.

$S'^{(d)}$ is integrally closed. Let $\theta \in \operatorname{Frac}(S'^{(d)})$ be integral over $S'^{(d)}$. Then $\theta \in \operatorname{Frac} S'$ is integral over S', so $\theta \in S'$ by (a). Write $\theta = \sum \theta_m$ in homogeneous components, and $\theta = a/b$ with $a, b \in S'^{(d)}$, $b \neq 0$. We claim all θ_m with $\theta_m \neq 0$ have $d \mid m$. Let m_0 be the least m with $\theta_m \neq 0$ and c_0 the least degree of a nonzero component of b (so $d \mid c_0$). The component of $a = \theta b$ of degree $m_0 + c_0$ is $\theta_{m_0} b_{c_0} \neq 0$ (S' is a domain), and $a \in S'^{(d)}$, so $d \mid m_0 + c_0$, so $d \mid m_0$. Then $\theta_{m_0} \in S'^{(d)}$ and $\theta - \theta_{m_0} = (a - \theta_{m_0} b)/b$ again has the same form with fewer nonzero components; by induction every component has degree divisible by d, i.e. $\theta \in S'^{(d)}$. So $S^{(d)} = S'^{(d)}$ is an

integrally closed domain.

Conclusion. The d-uple embedding of X is $X \cong \operatorname{Proj} S^{(d)} \hookrightarrow \mathbf{P}_k^N$ (Ex. 5.13), given by the surjection $P_N = k[z_0, \ldots, z_N] \to S^{(d)}$, $z_j \mapsto M_j$, with kernel J. J is saturated: if $z_i^n s \in J$ for all i, then $M_i^n \bar{s} = 0$ in the domain $S^{(d)} \subseteq S$; as $S_d \neq 0$ some $M_i \neq 0$, so $\bar{s} = 0$, $s \in J$. The closed subscheme $X_d = \operatorname{Proj}(P_N/J)$ of $\mathbf{P}_k^N$ has ideal sheaf $\widetilde{J}$, so by Fact 2 in the solution of 5.10, $\Gamma_*(\mathcal{I}_{X_d}) = \Gamma_*(\widetilde{J}) = \bar{J} = J$. So the homogeneous coordinate ring of the d-uple embedded X is $P_N/J \cong S^{(d)}$, an integrally closed domain: the d-uple embedding is projectively normal.

(d) ($\Rightarrow$) Let $S = S(X)$ be an integrally closed domain (so $S \neq 0$ and $X \neq \emptyset$). $X = \operatorname{Proj} S$ is irreducible (generic point (0)), hence connected. For $\mathfrak{p} \in X$ let Σ be the set of homogeneous elements not in $\mathfrak{p}$; $\mathcal{O}_{X,\mathfrak{p}} = S_{(\mathfrak{p})} = (\Sigma^{-1}S)_0$ is a domain. Let $\theta \in \operatorname{Frac} S_{(\mathfrak{p})}$ be integral over $S_{(\mathfrak{p})}$. As $\Sigma^{-1}S$ is an integrally closed domain (localization of S), $\theta \in \Sigma^{-1}S$. Write $\theta = a/b$ with $a, b \in S_{(\mathfrak{p})}$, $b \neq 0$ homogeneous of degree 0; then $\theta b = a$ has degree 0, so $\theta_m b = 0$ for the components θ_m, $m \neq 0$, hence $\theta_m = 0$ and $\theta \in S_{(\mathfrak{p})}$. So X is normal. Now X is a connected normal closed subscheme, so by (a) S' is the integral closure of S, hence $S' = S$, i.e. $P_n \to S_n = \Gamma(X, \mathcal{O}_X(n))$ is surjective for all $n \geq 0$.

($\Leftarrow$) Suppose X is normal and $P_n = \Gamma(\mathbf{P}^r, \mathcal{O}(n)) \to \Gamma(X, \mathcal{O}_X(n))$ is surjective for all $n \geq 0$ (and $X \neq \emptyset$). For $n = 0$ this says $\Gamma(X, \mathcal{O}_X) = k$; if X were a disjoint union of two nonempty open-closed subsets, $\Gamma(X, \mathcal{O}_X)$ would contain $k \times k$, of dimension ≥ 2. So X is connected, and (a) applies: S is a domain and S' its integral closure. Surjectivity means $S_n = S'_n$ for all $n \geq 0$, i.e. $S = S'$ is integrally closed. So X is projectively normal. □

Exercise 5.15

Problem. *Extension of Coherent Sheaves.* We will prove the following theorem in several steps: Let X be a noetherian scheme, let U be an open subset, and let $\mathcal{F}$ be a coherent sheaf on U. Then there is a coherent sheaf $\mathcal{F}'$ on X such that $\mathcal{F}'|_U \cong \mathcal{F}$.

(a) On a noetherian affine scheme, every quasi-coherent sheaf is the union of its coherent subsheaves. We say a sheaf $\mathcal{F}$ is the *union* of its subsheaves $\mathcal{F}_\alpha$ if for every open set U, the group $\mathcal{F}(U)$ is the union of the subgroups $\mathcal{F}_\alpha(U)$.

(b) Let X be an affine noetherian scheme, U an open subset, and $\mathcal{F}$ coherent on U. Then there exists a coherent sheaf $\mathcal{F}'$ on X with $\mathcal{F}'|_U \cong \mathcal{F}$. [*Hint*: Let $i : U \to X$ be the inclusion map. Show that $i_*\mathcal{F}$ is quasi-coherent, then use (a).]

(c) With $X, U, \mathcal{F}$ as in (b), suppose furthermore we are given a quasi-coherent sheaf $\mathcal{G}$ on X such that $\mathcal{F} \subseteq \mathcal{G}|_U$. Show that we can find $\mathcal{F}'$ a coherent subsheaf of $\mathcal{G}$, with $\mathcal{F}'|_U \cong \mathcal{F}$. [*Hint*: Use the same method, but replace $i_*\mathcal{F}$ by $\rho^{-1}(i_*\mathcal{F})$, where ρ is the natural map $\mathcal{G} \to i_*(\mathcal{G}|_U)$.]

(d) Now let X be a noetherian scheme, U an open subset, $\mathcal{F}$ a coherent sheaf on U, and $\mathcal{G}$ a quasi-coherent sheaf on X such that $\mathcal{F} \subseteq \mathcal{G}|_U$. Show that there is a coherent subsheaf $\mathcal{F}' \subseteq \mathcal{G}$ on X with $\mathcal{F}'|_U \cong \mathcal{F}$. Taking $\mathcal{G} = i_*\mathcal{F}$ proves the result announced at the beginning. [*Hint*: Cover X with open affines, and extend over one of them at a time.]

(e) As an extra corollary, show that on a noetherian scheme X, every quasi-coherent sheaf $\mathcal{F}$ is the union of its coherent subsheaves. [*Hint*: If s is a section of $\mathcal{F}$ over an open set U, apply (d) to the subsheaf of $\mathcal{F}|_U$ generated by s.]

Solution. (a) Let $X = \operatorname{Spec} A$, A noetherian, $\mathcal{F} = \widetilde{M}$ (5.4/5.5). M is the union of its finitely generated submodules N; $\widetilde{N} \subseteq \widetilde{M}$ is a subsheaf (exactness of $\widetilde{\ }$) and is coherent. Let $U \subseteq X$

be open and $s \in \mathcal{F}(U)$. U is quasi-compact (X noetherian), so $U = D(f_1) \cup \cdots \cup D(f_n)$. Then $s|_{D(f_i)} \in M_{f_i}$ has the form $m_i/f_i^{k_i}$ with $m_i \in M$. Let $N = \sum Am_i$, finitely generated. Then $s|_{D(f_i)} \in (N_{f_i}) = \widetilde{N}(D(f_i))$ for all i; as $\widetilde{N}$ is a subsheaf of $\mathcal{F}$ these sections glue (the restrictions to overlaps agree in $\mathcal{F}$, hence in $\widetilde{N}$) to a section of $\widetilde{N}(U)$ mapping to s. So $s \in \widetilde{N}(U)$.

We note: if $\mathcal{G}_1, \mathcal{G}_2 \subseteq \mathcal{F}$ are coherent subsheaves on a noetherian scheme, so is $\mathcal{G}_1 + \mathcal{G}_2$ (image of $\mathcal{G}_1 \oplus \mathcal{G}_2 \to \mathcal{F}$, coherent by 5.7), so coherent subsheaves form a directed family.

(b) Let $i : U \to X$ be the inclusion. Since U is noetherian and $\mathcal{F}$ is quasi-coherent, $i_*\mathcal{F}$ is quasi-coherent by 5.8(c). Moreover $(i_*\mathcal{F})|_U = \mathcal{F}$. Part (b) is therefore the special case $\mathcal{G} = i_*\mathcal{F}$ of (c): (c) gives a coherent $\mathcal{F}' \subseteq i_*\mathcal{F}$ with $\mathcal{F}'|_U = \mathcal{F}$. (The proof of (c) below is exactly the method of the hint, applied to $\mathcal{H} = \rho^{-1}(i_*\mathcal{F})$, which for $\mathcal{G} = i_*\mathcal{F}$ is $i_*\mathcal{F}$ itself.)

(c) Let $\mathcal{H} = \rho^{-1}(i_*\mathcal{F})$, i.e. $\mathcal{H}(W) = \{g \in \mathcal{G}(W) : g|_{W\cap U} \in \mathcal{F}(W \cap U)\}$; this is the kernel of the composite $\mathcal{G} \xrightarrow{\rho} i_*(\mathcal{G}|_U) \to i_*(\mathcal{G}|_U)/i_*\mathcal{F}$. Now $i_*(\mathcal{G}|_U)$ is quasi-coherent by 5.8(c) (U noetherian), and so is $i_*\mathcal{F}$ (as $\mathcal{F}$ is quasi-coherent); $i_*\mathcal{F} \to i_*(\mathcal{G}|_U)$ is a morphism of quasi-coherent sheaves, so the quotient is quasi-coherent (5.7), and $\mathcal{H}$, the kernel of a morphism of quasi-coherent sheaves, is quasi-coherent (5.7). Moreover, for $W \subseteq U$, $\mathcal{H}(W) = \mathcal{F}(W)$, i.e. $\mathcal{H}|_U = \mathcal{F}$ (equal as subsheaves of $\mathcal{G}|_U$).

By (a), $\mathcal{H}$ is the union of its coherent subsheaves. Since U is noetherian, cover it by finitely many open affines $V_j = \operatorname{Spec} B_j$ with $\mathcal{F}|_{V_j} = \widetilde{N_j}$, N_j finitely generated by $n_{j1}, \ldots, n_{jt_j} \in \mathcal{F}(V_j) = \mathcal{H}(V_j)$. Each of these finitely many sections lies in $\mathcal{G}_\alpha(V_j)$ for some coherent subsheaf $\mathcal{G}_\alpha \subseteq \mathcal{H}$; as the family of coherent subsheaves is directed, a single coherent $\mathcal{F}' \subseteq \mathcal{H} \subseteq \mathcal{G}$ contains all of them. Then $\mathcal{F}'|_U \subseteq \mathcal{H}|_U = \mathcal{F}$, and $\mathcal{F}'|_{V_j}$ contains the sections n_{jk}, which generate $\mathcal{F}|_{V_j}$ (their germs generate the stalks), so $\mathcal{F}'|_{V_j} = \mathcal{F}|_{V_j}$. Hence $\mathcal{F}'|_U = \mathcal{F}$ as subsheaves.

(d) Let $X = V_1 \cup \cdots \cup V_n$ with V_k open affine, and $U_k = U \cup V_1 \cup \cdots \cup V_k$ ($U_0 = U$). We construct, by induction on k, coherent subsheaves $\mathcal{F}_k \subseteq \mathcal{G}|_{U_k}$ with $\mathcal{F}_k|_{U_{k-1}} = \mathcal{F}_{k-1}$ and $\mathcal{F}_0 = \mathcal{F}$. Given $\mathcal{F}_{k-1}$, let $V = V_k$ (affine noetherian) and $V' = V \cap U_{k-1}$ open in V. Then $\mathcal{F}_{k-1}|_{V'}$ is a coherent subsheaf of $(\mathcal{G}|_V)|_{V'}$, with $\mathcal{G}|_V$ quasi-coherent. By (c) there is a coherent $\mathcal{F}'' \subseteq \mathcal{G}|_V$ with $\mathcal{F}''|_{V'} = \mathcal{F}_{k-1}|_{V'}$ (equality of subsheaves, as shown in (c)). Define $\mathcal{F}_k \subseteq \mathcal{G}|_{U_k}$ by $\mathcal{F}_k(W) = \{g \in \mathcal{G}(W) : g|_{W\cap U_{k-1}} \in \mathcal{F}_{k-1},\ g|_{W\cap V} \in \mathcal{F}''\}$; this is a subsheaf with $\mathcal{F}_k|_{U_{k-1}} = \mathcal{F}_{k-1}$ and $\mathcal{F}_k|_V = \mathcal{F}''$ (the two descriptions agree on V'), and it is coherent since coherence is local. Then $\mathcal{F}' = \mathcal{F}_n \subseteq \mathcal{G}$ is coherent on $X = U_n$ with $\mathcal{F}'|_U = \mathcal{F}$.

For the announced theorem, let $\mathcal{F}$ be coherent on U and $\mathcal{G} = i_*\mathcal{F}$, quasi-coherent by 5.8(c), so that $(i_*\mathcal{F})|_U = \mathcal{F}$ contains $\mathcal{F}$. This yields a coherent $\mathcal{F}'$ on X with $\mathcal{F}'|_U = \mathcal{F}$.

(e) Let $\mathcal{F}$ be quasi-coherent on X, $U \subseteq X$ open and $s \in \mathcal{F}(U)$. Let $\mathcal{F}_s \subseteq \mathcal{F}|_U$ be the subsheaf generated by s, the image of $\mathcal{O}_U \to \mathcal{F}|_U$, $1 \mapsto s$. On an open affine $V = \operatorname{Spec} A \subseteq U$ with $\mathcal{F}|_V = \widetilde{M}$ we get $\mathcal{F}_s|_V = \widetilde{As|_V}$, which is coherent ($A$ noetherian, $As|_V$ cyclic). So $\mathcal{F}_s$ is a coherent subsheaf of $\mathcal{F}|_U$. By (d) (with $\mathcal{G} = \mathcal{F}$) there is a coherent $\mathcal{F}' \subseteq \mathcal{F}$ with $\mathcal{F}'|_U = \mathcal{F}_s$. Then $s \in \mathcal{F}_s(U) = \mathcal{F}'(U)$. □

Exercise 5.16

Problem. *Tensor Operations on Sheaves.* First we recall the definitions of various tensor operations on a module. Let A be a ring, and let M be an A-module. Let $T^n(M)$ be the tensor product $M \otimes \cdots \otimes M$ of M with itself n times, for $n \geq 1$. For $n = 0$ we put

$T^0(M) = A$. Then $T(M) = \bigoplus_{n\geq 0} T^n(M)$ is a (noncommutative) A-algebra, which we call the *tensor algebra* of M. We define the *symmetric algebra* $S(M) = \bigoplus_{n\geq 0} S^n(M)$ of M to be the quotient of $T(M)$ by the two-sided ideal generated by all expressions $x\otimes y - y\otimes x$, for all $x, y \in M$. Then $S(M)$ is a commutative A-algebra. Its component in degree n is called the nth *symmetric product* of M. We denote the image of $x \otimes y$ in $S(M)$ by xy, $x, y \in M$. As an example, note that if M is a free A-module of rank r, then $S(M) \cong A[x_1, \ldots, x_r]$.

We define the *exterior algebra* $\bigwedge(M)$ of M to be the quotient of $T(M)$ by the two-sided ideal generated by all expressions $x \otimes x$ for $x \in M$. Note that this expression contains all expressions of the form $x\otimes y+y\otimes x$, so that $\bigwedge(M)$ is a skew commutative graded A-algebra. This means that if $u \in \bigwedge^r(M)$ and $v \in \bigwedge^s(M)$, then $u \wedge v = (-1)^{rs} v \wedge u$ (here we denote the multiplication in this algebra by $\wedge$, i.e. the image of $x\otimes y$ in $\bigwedge^2(M)$ is denoted by $x\wedge y$). The nth component $\bigwedge^n(M)$ is called the nth *exterior power* of M.

Now let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. We define the *tensor algebra, symmetric algebra, and exterior algebra* of $\mathcal{F}$ by taking the sheaves associated to the presheaf which to an open set U assigns the corresponding tensor operation applied to $\mathcal{F}(U)$ as an $\mathcal{O}_X(U)$-module. The results are $\mathcal{O}_X$-algebras, and their components are $\mathcal{O}_X$-modules.

(a) Suppose $\mathcal{F}$ is locally free of rank n. Then $T^r(\mathcal{F})$, $S^r(\mathcal{F})$, and $\bigwedge^r(\mathcal{F})$ are also locally free, of ranks n^r, $\binom{n+r-1}{n-1}$, and $\binom{n}{r}$ respectively.

(b) Again let $\mathcal{F}$ be locally free of rank n. Then the multiplication map $\bigwedge^r \mathcal{F} \otimes \bigwedge^{n-r} \mathcal{F} \to \bigwedge^n \mathcal{F}$ is a perfect pairing for any r, i.e., it induces an isomorphism of $\bigwedge^r \mathcal{F}$ with $(\bigwedge^{n-r} \mathcal{F})^\vee \otimes \bigwedge^n \mathcal{F}$. As a special case, note that if $\mathcal{F}$ has rank 2, then $\mathcal{F} \cong \mathcal{F}^\vee \otimes \bigwedge^2 \mathcal{F}$.

(c) Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be an exact sequence of locally free sheaves. Then for any r there is a finite filtration of $S^r(\mathcal{F})$,

$$S^r(\mathcal{F}) = F^0 \supseteq F^1 \supseteq \cdots \supseteq F^r \supseteq F^{r+1} = 0$$

with quotients $F^p/F^{p+1} \cong S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F}'')$ for each p.

(d) Same statement as (c), with exterior powers instead of symmetric powers. In particular, if $\mathcal{F}', \mathcal{F}, \mathcal{F}''$ have ranks n', n, n'' respectively, there is an isomorphism $\bigwedge^n \mathcal{F} \cong \bigwedge^{n'} \mathcal{F}' \otimes \bigwedge^{n''} \mathcal{F}''$.

(e) Let $f : X \to Y$ be a morphism of ringed spaces, and let $\mathcal{F}$ be an $\mathcal{O}_Y$-module. Then f^* commutes with all the tensor operations on $\mathcal{F}$: $f^*(S^n(\mathcal{F})) \cong S^n(f^*\mathcal{F})$, etc.

Solution. (The binomial coefficient in (a) is printed as $\binom{n+r-1}{n-1}$; this equals $\binom{n+r-1}{r}$, the rank of S^r of a free module of rank n.) We use that the construction commutes with restriction to open subsets, and it is natural for isomorphisms of $\mathcal{F}$.

(a) Let U be open with $\mathcal{F}|_U \cong \mathcal{O}_U^n$, basis $e_1, \ldots, e_n$. For every open $U' \subseteq U$, $\mathcal{F}(U') = \mathcal{O}(U')^n$ is free over $B = \mathcal{O}(U')$ with basis $e_i|_{U'}$, so $T^r(\mathcal{F}(U'))$ is free with basis $e_{i_1} \otimes \cdots \otimes e_{i_r}$ (n^r elements), $S^r(\mathcal{F}(U'))$ is free with basis the monomials $e_1^{a_1} \cdots e_n^{a_n}$, $\sum a_i = r$ ($\binom{n+r-1}{r}$ elements, as $S(B^n) = B[e_1, \ldots, e_n]$), and $\bigwedge^r(\mathcal{F}(U'))$ is free with basis $e_{i_1} \wedge \cdots \wedge e_{i_r}$, $i_1 < \cdots < i_r$ ($\binom{n}{r}$ elements). The resulting presheaf, e.g. $U' \mapsto \mathcal{O}(U')^N$, is already a sheaf ($\mathcal{O}_U^N$) compatible with the restriction maps (which send basis elements to basis elements), so its sheafification is $\mathcal{O}_U^N$ with $N = n^r, \binom{n+r-1}{r}, \binom{n}{r}$ respectively.

(b) The wedge product on presheaves induces a morphism $\mathcal{F}_r \otimes \mathcal{F}_{n-r} \to \mathcal{F}_n$ where $\mathcal{F}_j = \bigwedge^j \mathcal{F}$; by 5.1(c) (applied with $\mathcal{E} = \mathcal{F}_{n-r}$, which is locally free of finite rank by (a)) it corresponds to $\phi : \mathcal{F}_r \to \mathcal{H}om(\mathcal{F}_{n-r}, \mathcal{F}_n)$, $\phi(s)(t) = s \wedge t$, and by 5.1(b) $\mathcal{H}om(\mathcal{F}_{n-r}, \mathcal{F}_n) \cong (\bigwedge^{n-r} \mathcal{F})^\vee \otimes \bigwedge^n \mathcal{F}$.

To show ϕ is an isomorphism we may work on U with $\mathcal{F}|_U$ free with basis $e_1, \dots, e_n$. Then $\mathcal{F}_r$, $\mathcal{F}_{n-r}$, $\mathcal{F}_n$ have bases e_I ($|I| = r$), e_J ($|J| = n - r$), $e_{[n]} = e_1 \wedge \cdots \wedge e_n$, and $e_I \wedge e_J = \pm e_{[n]}$ if $J = I^c$ and 0 otherwise. Hence $\phi(e_I) = \pm\, e_{I^c}^* \otimes e_{[n]}$, i.e. ϕ maps a basis to $\pm$ a basis: an isomorphism. For $n = 2, r = 1$: $\mathcal{F} \cong \mathcal{F}^\vee \otimes \bigwedge^2 \mathcal{F}$.

(c) Put $\mathcal{F}^p$ = the image of $S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F}) \to S^r(\mathcal{F})$ (induced by the multiplication $S^p(\mathcal{F}) \otimes S^{r-p}(\mathcal{F}) \to S^r(\mathcal{F})$ and $\mathcal{F}' \to \mathcal{F}$), for $0 \le p \le r$, and $F^{r+1} = 0$. Clearly $F^0 = S^r(\mathcal{F})$ and $F^p \supseteq F^{p+1}$.

Local structure. Let $x \in X$. Since $\mathcal{F}', \mathcal{F}''$ are locally free of finite rank we can find an open neighborhood U of x with $\mathcal{F}'|_U$, $\mathcal{F}''|_U$ free, with bases $e'_1, \dots, e'_{n'}$, $g''_1, \dots, g''_{n''}$. As $\mathcal{F} \to \mathcal{F}''$ is surjective, each g''_j lifts to a section of $\mathcal{F}$ locally; shrinking U we may assume lifts $g_j \in \mathcal{F}(U)$ exist. Then $\mathcal{F}'|_U \oplus \mathcal{O}_U^{n''} \to \mathcal{F}|_U$, $(e'_i, \text{basis}) \mapsto (e_i, g_j)$ is a morphism of short exact sequences (identity on the ends), hence an isomorphism (5-lemma on stalks). So $\mathcal{F}|_U$ is free with basis $e_1, \dots, e_{n'}, g_1, \dots, g_{n''}$ (e_i the image of e'_i), and $S^r(\mathcal{F})|_U$ is free with basis the monomials $e^a g^b$, $|a| + |b| = r$. The image of $S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F})$ is spanned by the products of p elements e_i with a monomial of degree $r - p$, i.e. by the monomials with $|a| \ge p$. So $F^p|_U$ is free on $\{e^a g^b : |a| \ge p\}$, F^p/F^{p+1} is free on $\{e^a g^b : |a| = p\}$.

Global isomorphism. The composite $\Psi : S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F}) \to F^p \to F^p/F^{p+1}$ vanishes on $S^p(\mathcal{F}') \otimes K$ where $K = \mathcal{F}' \cdot S^{r-p-1}(\mathcal{F})$ = the kernel of $S^{r-p}(\mathcal{F}) \to S^{r-p}(\mathcal{F}'')$ (locally: the span of the monomials with $|a| \ge 1$, which is the kernel of the map killing e_i): indeed $x \cdot (f'y) = (xf')y \in F^{p+1}$. By right exactness of $\otimes$, Ψ induces $\bar\Psi : S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F}'') \to F^p/F^{p+1}$; locally it sends the basis element $e^a \otimes g^b$ ($|a| = p$, $|b| = r - p$) to the class of $e^a g^b$, which is a basis of F^p/F^{p+1}. Hence $\bar\Psi$ is an isomorphism.

(d) Same proof: F^p = image of $\bigwedge^p \mathcal{F}' \otimes \bigwedge^{r-p} \mathcal{F} \to \bigwedge^r \mathcal{F}$, $F^{r+1} = 0$. Locally $\bigwedge^r \mathcal{F}$ has basis $e_I \wedge g_J$ ($|I| + |J| = r$), F^p is spanned by the $e_I \wedge g_J$ with $|I| \ge p$, so F^p/F^{p+1} is free on those with $|I| = p$. The map $\bigwedge^p \mathcal{F}' \otimes \bigwedge^{r-p} \mathcal{F} \to F^p/F^{p+1}$ kills $\bigwedge^p \mathcal{F}' \otimes (\mathcal{F}' \wedge \bigwedge^{r-p-1} \mathcal{F})$ (as $x \wedge (f' \wedge y) = \pm(x \wedge f') \wedge y \in F^{p+1}$), which is the kernel of $\bigwedge^{r-p} \mathcal{F} \to \bigwedge^{r-p} \mathcal{F}''$ (locally: spanned by the $e_I \wedge g_J$ with $I \ne \emptyset$), so we get $\bar\Psi : \bigwedge^p \mathcal{F}' \otimes \bigwedge^{r-p} \mathcal{F}'' \to F^p/F^{p+1}$ sending $e_I \otimes g_J$ to $e_I \wedge g_J$, an isomorphism.

If the ranks are n', n, n'' ($n = n' + n''$) take $r = n$: $F^p/F^{p+1} \cong \bigwedge^p \mathcal{F}' \otimes \bigwedge^{n-p} \mathcal{F}''$ is 0 unless $p \le n'$ and $n - p \le n''$, i.e. unless $p = n'$. So $F^0 = \cdots = F^{n'}$ and $F^{n'+1} = \cdots = F^{n+1} = 0$, giving $\bigwedge^n \mathcal{F} = F^{n'} = F^{n'}/F^{n'+1} \cong \bigwedge^{n'} \mathcal{F}' \otimes \bigwedge^{n''} \mathcal{F}''$.

(e) Let Λ^n denote one of $T^n, S^n, \bigwedge^n$. For $\mathcal{O}_Y$-modules $\mathcal{A}, \mathcal{B}$ let $\mathrm{Mult}^n_\Lambda(\mathcal{A}, \mathcal{B})$ be the set of compatible families of n-linear maps $\mathcal{A}(V)^n \to \mathcal{B}(V)$ over $\mathcal{O}(V)$ (all V), which are, for $\Lambda = S$, symmetric, and for $\Lambda = \bigwedge$, alternating (the value is 0 whenever two adjacent arguments coincide; this is exactly the condition for the map $T^n \to \mathcal{B}$ to kill the two-sided ideal generated by the $x \otimes x$). By the universal property of the tensor power of modules and of sheafification, $\mathrm{Hom}(\Lambda^n \mathcal{A}, \mathcal{B}) = \mathrm{Mult}^n_\Lambda(\mathcal{A}, \mathcal{B})$, naturally in $\mathcal{B}$. Now fix an $\mathcal{O}_X$-module $\mathcal{H}$. We claim there is a natural bijection

$$\mathrm{Mult}^n_\Lambda(\mathcal{F}, f_*\mathcal{H}) \;\cong\; \mathrm{Mult}^n_\Lambda(f^*\mathcal{F}, \mathcal{H}), \qquad g \mapsto g^\flat, \quad g^\flat(f^*x_1, \dots, f^*x_n) = g(x_1, \dots, x_n),$$

where $x_i \in \mathcal{F}(V)$ and f^*x denotes the image of x in $(f^*\mathcal{F})(f^{-1}V)$. Granting this, by the adjunction $\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{A}, \mathcal{B}) = \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{A}, f_*\mathcal{B})$ we get natural bijections $\mathrm{Hom}(f^*\Lambda^n\mathcal{F}, \mathcal{H}) = \mathrm{Hom}(\Lambda^n\mathcal{F}, f_*\mathcal{H}) = \mathrm{Mult}^n_\Lambda(\mathcal{F}, f_*\mathcal{H}) = \mathrm{Mult}^n_\Lambda(f^*\mathcal{F}, \mathcal{H}) = \mathrm{Hom}(\Lambda^n(f^*\mathcal{F}), \mathcal{H})$, natural in $\mathcal{H}$, so $f^*\Lambda^n\mathcal{F} \cong \Lambda^n(f^*\mathcal{F})$ by Yoneda.

Proof of the claim. For ordinary n-linear maps, a family g corresponds to a morphism $\mathcal{F} \to \mathcal{H}om(\mathcal{F}, \mathcal{H}om(\dots, \mathcal{H}om(\mathcal{F}, f_*\mathcal{H})))$ (5.1(c), $n-1$ internal Homs; it holds for all $\mathcal{E}$), and for $\mathcal{O}_Y$-modules $\mathcal{A}$ there is the natural isomorphism $\mathcal{H}om(\mathcal{A}, f_*\mathcal{H}) \cong f_*\mathcal{H}om(f^*\mathcal{A}, \mathcal{H})$ (the adjunction on each open set V, applied to $\mathcal{A}|_V$, $f^{-1}V \to V$). Applying these and the adjunction repeatedly gives a bijection $g \mapsto g^\flat$ between n-linear maps $\mathcal{F}^n \to f_*\mathcal{H}$ and n-linear maps $(f^*\mathcal{F})^n \to \mathcal{H}$, with $g^\flat(f^*x_1, \dots, f^*x_n) = g(x_1, \dots, x_n)$; since every local section of $f^*\mathcal{F}$ is locally a finite sum $\sum c_k f^*x_k$ with $c_k \in \mathcal{O}_X$, $g^\flat$ is determined by this formula. Symmetry of g clearly gives symmetry of $g^\flat$ (expand multilinearly) and conversely. For alternating: if $g^\flat$ is alternating then so is g (restrict to the f^*x_i). Conversely if g is alternating, then g is skew-symmetric ($g(\dots, x, z, \dots) + g(\dots, z, x, \dots) = g(\dots, x+z, x+z, \dots) - g(\dots, x, x, \dots) - g(\dots, z, z, \dots) = 0$), and for a local section $y = \sum_k c_k f^*x_k$ in two adjacent slots,

$$g^\flat(\dots, y, y, \dots) = \sum_k c_k^2 g(\dots, x_k, x_k, \dots) + \sum_{k<l} c_k c_l \big(g(\dots, x_k, x_l, \dots) + g(\dots, x_l, x_k, \dots)\big) = 0.$$

□

Exercise 5.17

Problem. *Affine Morphisms.* A morphism $f : X \to Y$ of schemes is *affine* if there is an open affine cover $\{V_i\}$ of Y such that $f^{-1}(V_i)$ is affine for each i.

(a) Show that $f : X \to Y$ is an affine morphism if and only if for *every* open affine $V \subseteq Y$, $f^{-1}(V)$ is affine. [*Hint*: Reduce to the case Y affine and use (Ex. 2.17).]

(b) An affine morphism is quasi-compact and separated. Any finite morphism is affine.

(c) Let Y be a scheme, and let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras (i.e., a sheaf of rings which is at the same time a quasi-coherent $\mathcal{O}_Y$-module). Show that there is a unique scheme X, and a morphism $f : X \to Y$, such that for every open affine $V \subseteq Y$, $f^{-1}(V) \cong \operatorname{Spec} \mathcal{A}(V)$, and for every inclusion $U \hookrightarrow V$ of open affines of Y, the morphism $f^{-1}(U) \hookrightarrow f^{-1}(V)$ corresponds to the restriction homomorphism $\mathcal{A}(V) \to \mathcal{A}(U)$. The scheme X is called $\mathbf{Spec}\, \mathcal{A}$. [*Hint*: Construct X by glueing the schemes $\operatorname{Spec} \mathcal{A}(V)$, for V open affine in Y.]

(d) If $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_Y$-algebra, then $f : X = \mathbf{Spec}\, \mathcal{A} \to Y$ is an affine morphism, and $\mathcal{A} \cong f_*\mathcal{O}_X$. Conversely, if $f : X \to Y$ is an affine morphism, then $\mathcal{A} = f_*\mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras, and $X \cong \mathbf{Spec}\, \mathcal{A}$ (over Y).

(e) Let $f : X \to Y$ be an affine morphism, and let $\mathcal{A} = f_*\mathcal{O}_X$. Show that f_* induces an equivalence of categories from the category of quasi-coherent $\mathcal{O}_X$-modules to the category of quasi-coherent $\mathcal{A}$-modules (i.e., quasi-coherent $\mathcal{O}_Y$-modules having a structure of $\mathcal{A}$-module). [*Hint*: For any quasi-coherent $\mathcal{A}$-module $\mathcal{M}$, construct a quasi-coherent $\mathcal{O}_X$-module $\widetilde{\mathcal{M}}$, and show that the functors f_* and $\widetilde{\ }$ are inverse to each other.]

Solution. (a) ($\Leftarrow$) is trivial. ($\Rightarrow$) Let $\{V_i\}$ be an affine cover of Y with $f^{-1}V_i$ affine, and let $V = \operatorname{Spec} R$ be any open affine of Y. Let $y \in V \cap V_i$. Choose $h \in \mathcal{O}(V_i)$ with $y \in D_{V_i}(h) \subseteq V \cap V_i$; then $f^{-1}(D_{V_i}(h)) = D(f^\# h)$ is a basic open subset of the affine scheme $f^{-1}V_i$, hence affine. Now choose $g \in R = \mathcal{O}(V)$ with $y \in D_V(g) \subseteq D_{V_i}(h)$; $D_V(g) = D(g|_{D_{V_i}(h)})$ is a basic open subset of the affine scheme $D_{V_i}(h)$, so $f^{-1}(D_V(g)) = D(f^\# g)$ is a basic open of the affine scheme $f^{-1}(D_{V_i}(h))$, hence affine. Thus V is covered by basic opens $D(g)$ with $f^{-1}(D(g))$ affine. As V is quasi-compact, $V = D(g_1) \cup \dots \cup D(g_m)$, so $(g_1, \dots, g_m) = R$. Put $W = f^{-1}V$ and

$a_k = f^\#(g_k) \in \Gamma(W, \mathcal{O}_W)$. Then $a_1, \dots, a_m$ generate the unit ideal of $\Gamma(W, \mathcal{O}_W)$ (the same relation $\sum c_k g_k = 1$ maps to $\sum f^\#(c_k) a_k = 1$), and $W_{a_k} = f^{-1}(D(g_k))$ is affine. By Ex. 2.17(b), $W = f^{-1}(V)$ is affine.

(b) *Quasi-compact:* f affine; the affine cover $\{V_i\}$ of Y has $f^{-1}V_i$ affine, hence quasi-compact (Ex. 3.2). *Separated:* the preimage of V_i in $X \times_Y X$ is $f^{-1}V_i \times_{V_i} f^{-1}V_i = \operatorname{Spec}(B_i \otimes_{A_i} B_i)$ ($V_i = \operatorname{Spec} A_i$, $f^{-1}V_i = \operatorname{Spec} B_i$), these cover $X \times_Y X$, and the restriction of Δ over each is the morphism $\operatorname{Spec} B_i \to \operatorname{Spec}(B_i \otimes_{A_i} B_i)$ induced by the multiplication $B_i \otimes_{A_i} B_i \to B_i$, which is surjective, hence a closed immersion. Being a closed immersion is local on the target, so Δ is a closed immersion. *Finite $\Rightarrow$ affine:* by definition, Y has an open affine cover $V_i = \operatorname{Spec} A_i$ with $f^{-1}(V_i) = \operatorname{Spec} B_i$ affine (B_i a finite A_i-module); so f is affine.

(c) For an open affine $V \subseteq Y$ let $X_V = \operatorname{Spec} \mathcal{A}(V)$ with $f_V : X_V \to V = \operatorname{Spec} \mathcal{O}_Y(V)$ induced by $\mathcal{O}_Y(V) \to \mathcal{A}(V)$. If $U \subseteq V$ are open affines, then since $\mathcal{A}|_V = \widetilde{\mathcal{A}(V)}$ (quasi-coherent on the affine V, 5.4/5.5) and $\widetilde{M}|_U = (M \otimes_{\mathcal{O}(V)} \mathcal{O}(U))^\sim$ (5.2(e) applied to the open immersion $U \to V$; see also 5.1(c)), we get $\mathcal{A}(U) = \mathcal{A}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$, i.e. $X_U = X_V \times_V U = f_V^{-1}(U)$, an open subscheme of X_V, and the corresponding morphism $X_U \to X_V$ is the open immersion given by the restriction map $\mathcal{A}(V) \to \mathcal{A}(U)$.

Existence. Glue the schemes X_V (V ranging over open affines of Y): for V, V' let $X_{V,V'} = f_V^{-1}(V \cap V') \subseteq X_V$, open. For every open affine $W \subseteq V \cap V'$ the open subscheme $f_V^{-1}(W)$ of X_V and $f_{V'}^{-1}(W) \subseteq X_{V'}$ are both canonically identified with X_W (by the previous paragraph). These identifications are compatible for $W_1 \subseteq W_2$ (they come from the restriction maps of the sheaf $\mathcal{A}$, which compose), so they glue to an isomorphism $\varphi_{V'V} : f_V^{-1}(V \cap V') \to f_{V'}^{-1}(V \cap V')$ of Y-schemes. Again by compatibility with restriction, $\varphi_{V''V} = \varphi_{V''V'} \circ \varphi_{V'V}$ on triple overlaps (check on $f_V^{-1}(W)$ for affine $W \subseteq V \cap V' \cap V''$, where all three are the identity of X_W). So we can glue the X_V to a scheme X with a morphism $f : X \to Y$ (the f_V agree on overlaps), in which each X_V is an open subscheme. Moreover $f^{-1}(V) = X_V$: a point of X over V lies in some $X_{V'}$, hence in $f_{V'}^{-1}(V \cap V')$, which was identified with $f_V^{-1}(V \cap V') \subseteq X_V$. So $f^{-1}(V) = X_V \cong \operatorname{Spec} \mathcal{A}(V)$ for every open affine V, and for open affines $U \subseteq V$ the inclusion $f^{-1}U \subseteq f^{-1}V$ is $X_U \to X_V$, which corresponds to the restriction $\mathcal{A}(V) \to \mathcal{A}(U)$ by construction.

Uniqueness. If (X', f') is another such pair, the isomorphisms $f'^{-1}(V) \cong \operatorname{Spec} \mathcal{A}(V) = f^{-1}(V)$ for open affine V (over V) agree on $f'^{-1}(U)$ for affine $U \subseteq V$, by the compatibility with the inclusions $U \subseteq V$; so they glue to an isomorphism $X' \cong X$ over Y.

(d) For $X = \mathbf{Spec}\mathcal{A}$, $f^{-1}(V) = \operatorname{Spec} \mathcal{A}(V)$ is affine for each open affine V, so f is affine. Also $(f_*\mathcal{O}_X)(V) = \mathcal{O}_X(f^{-1}V) = \mathcal{A}(V)$, compatible with the restriction maps (by the compatibility in (c)); as open affines form a basis, this gives an isomorphism of sheaves of $\mathcal{O}_Y$-algebras $f_*\mathcal{O}_X \cong \mathcal{A}$.

Conversely, let f be affine. By (b) it is quasi-compact and separated, and $\mathcal{O}_X$ is quasi-coherent, so $\mathcal{A} = f_*\mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_Y$-module (5.8(c)) and a sheaf of $\mathcal{O}_Y$-algebras. For $V \subseteq Y$ open affine, $f^{-1}(V)$ is affine by (a), with ring $\mathcal{O}_X(f^{-1}V) = \mathcal{A}(V)$, so $f^{-1}(V) \cong \operatorname{Spec} \mathcal{A}(V)$ over V; for $U \subseteq V$ open affine, $f^{-1}U \subseteq f^{-1}V$ corresponds to the restriction $\mathcal{A}(V) \to \mathcal{A}(U)$ (a morphism of affine schemes is determined by the ring homomorphism). By the uniqueness in (c), $X \cong \mathbf{Spec}\mathcal{A}$ over Y.

(e) *The functor f_*.* For $\mathcal{F}$ quasi-coherent on X, $f_*\mathcal{F}$ is quasi-coherent (5.8(c) with (b)) and a module over $f_*\mathcal{O}_X = \mathcal{A}$.

The functor $\widetilde{\ }$. Let $\mathcal{M}$ be a quasi-coherent $\mathcal{A}$-module. For $V \subseteq Y$ open affine, $f^{-1}(V) = \operatorname{Spec} \mathcal{A}(V)$ and $\mathcal{M}(V)$ is an $\mathcal{A}(V)$-module; let $\widetilde{\mathcal{M}}_V = \widetilde{\mathcal{M}(V)}$ be the associated quasi-coherent sheaf on $f^{-1}(V)$ (relative to the ring $\mathcal{A}(V)$). For affine $U \subseteq V$: $\mathcal{M}(U) = \mathcal{M}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$ (quasi-coherence of $\mathcal{M}$) and $\mathcal{A}(U) = \mathcal{A}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$, so $\mathcal{M}(V) \otimes_{\mathcal{A}(V)} \mathcal{A}(U) = \mathcal{M}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) = \mathcal{M}(U)$. By 5.2(e) (for $\operatorname{Spec} \mathcal{A}(U) = f^{-1}U \hookrightarrow f^{-1}V$ an open immersion) $\widetilde{\mathcal{M}}_V|_{f^{-1}U} \cong \widetilde{\mathcal{M}}_U$, canonically and compatibly with further restrictions. Hence the $\widetilde{\mathcal{M}}_V$ glue to a quasi-coherent $\mathcal{O}_X$-module $\widetilde{\mathcal{M}}$ with $\widetilde{\mathcal{M}}|_{f^{-1}V} = \widetilde{\mathcal{M}(V)}$; the construction is functorial in $\mathcal{M}$.

Inverse functors. $f_*(\widetilde{\mathcal{M}})(V) = \widetilde{\mathcal{M}}(f^{-1}V) = \widetilde{\mathcal{M}(V)}(\operatorname{Spec} \mathcal{A}(V)) = \mathcal{M}(V)$ (5.1(d)), compatibly with restrictions and with the $\mathcal{A}$-module structures, for open affine V; as the open affines form a basis, $f_*\widetilde{\mathcal{M}} \cong \mathcal{M}$ functorially. Conversely for $\mathcal{F}$ quasi-coherent on X and V open affine in Y, $\mathcal{F}|_{f^{-1}V} = \widetilde{P}$ with $P = \mathcal{F}(f^{-1}V) = (f_*\mathcal{F})(V)$ an $\mathcal{A}(V)$-module (5.4/5.5), so $\widetilde{f_*\mathcal{F}}|_{f^{-1}V} = \widetilde{(f_*\mathcal{F})(V)} = \mathcal{F}|_{f^{-1}V}$; these identifications are compatible on overlaps and natural, so $\widetilde{f_*\mathcal{F}} \cong \mathcal{F}$ functorially. Therefore f_* is an equivalence with quasi-inverse $\widetilde{\ }$. □

Exercise 5.18

Problem. *Vector Bundles.* Let Y be a scheme. A (geometric) *vector bundle* of rank n over Y is a scheme X and a morphism $f : X \to Y$, together with additional data consisting of an open covering $\{U_i\}$ of Y, and isomorphisms $\psi_i : f^{-1}(U_i) \to \mathbf{A}^n_{U_i}$, such that for any i, j, and for any open affine subset $V = \operatorname{Spec} A \subseteq U_i \cap U_j$, the automorphism $\psi = \psi_j \circ \psi_i^{-1}$ of $\mathbf{A}^n_V = \operatorname{Spec} A[x_1, \ldots, x_n]$ is given by a *linear* automorphism θ of $A[x_1, \ldots, x_n]$, i.e., $\theta(a) = a$ for any $a \in A$, and $\theta(x_i) = \sum a_{ij} x_j$ for suitable $a_{ij} \in A$.

An *isomorphism* $g : (X, f, \{U_i\}, \{\psi_i\}) \to (X', f', \{U_i'\}, \{\psi_i'\})$ of one vector bundle of rank n to another is an isomorphism $g : X \to X'$ of the underlying schemes, such that $f = f' \circ g$, and such that X, f, together with the covering of Y consisting of all the U_i and U_i', and the isomorphisms ψ_i and $\psi_i' \circ g$, is also a vector bundle structure on X.

(a) Let $\mathcal{E}$ be a locally free sheaf of rank n on a scheme Y. Let $S(\mathcal{E})$ be the symmetric algebra on $\mathcal{E}$, and let $X = \mathbf{Spec}\, S(\mathcal{E})$, with the projection morphism $f : X \to Y$. For each open affine subset $U \subseteq Y$ for which $\mathcal{E}|_U$ is free, choose a basis of $\mathcal{E}|_U$, and let $\psi : f^{-1}(U) \to \mathbf{A}^n_U$ be the isomorphism resulting from the identification of $S(\mathcal{E}(U))$ with $\mathcal{O}(U)[x_1, \ldots, x_n]$. Then $(X, f, \{U\}, \{\psi\})$ is a vector bundle of rank n over Y, which (up to isomorphism) does not depend on the bases of the $\mathcal{E}|_U$ chosen. We call it the *geometric vector bundle associated to* $\mathcal{E}$, and denote it by $\mathbf{V}(\mathcal{E})$.

(b) For any morphism $f : X \to Y$, a *section of* f *over an open set* $U \subseteq Y$ is a morphism $s : U \to X$ such that $f \circ s = \mathrm{id}_U$. It is clear how to restrict sections to smaller open sets, or how to glue them together, so we have a sheaf of sets on Y, which we denote $\mathcal{S}(X/Y)$. If $f : X \to Y$ is a vector bundle of rank n, show that the sheaf of sections $\mathcal{S}(X/Y)$ has a natural structure of an $\mathcal{O}_Y$-module, which is locally free of rank n. [*Hint*: It is enough to define the module structure locally, so we can assume $Y = \operatorname{Spec} A$ affine and $X = \mathbf{A}^n_Y$. Then a section $s : Y \to X$ comes from an A-algebra homomorphism $\theta : A[x_1, \ldots, x_n] \to A$, which in turn determines an ordered n-tuple $\langle \theta(x_1), \ldots, \theta(x_n) \rangle$ of elements of A. Use this correspondence between sections s and n-tuples of elements of A to define the module structure.]

(c) Again let $\mathcal{E}$ be a locally free sheaf of rank n on Y, let $X = \mathbf{V}(\mathcal{E})$, and let $\mathcal{S} = \mathcal{S}(X/Y)$ be the sheaf of sections of X over Y. Show that $\mathcal{S} \cong \mathcal{E}^\vee$, as follows. Given a section $s \in \Gamma(V, \mathcal{E}^\vee)$ over an open set V, we think of s as an element of $\operatorname{Hom}(\mathcal{E}|_V, \mathcal{O}_V)$. So s

determines an $\mathcal{O}_V$-algebra homomorphism $S(\mathcal{E}|_V) \to \mathcal{O}_V$. This determines a morphism of spectra $V = \mathbf{Spec}\,\mathcal{O}_V \to \mathbf{Spec}\,S(\mathcal{E}|_V) = f^{-1}(V)$, which is a section of X/Y. Show that this construction gives an isomorphism of $\mathcal{E}^\vee$ to $\mathcal{S}$.

(d) Summing up, show that we have established a one-to-one correspondence between isomorphism classes of locally free sheaves of rank n on Y, and isomorphism classes of vector bundles of rank n over Y. Because of this, we sometimes use the terms "locally free sheaf" and "vector bundle" interchangeably, if no confusion seems likely to result.

Solution. We use freely: morphisms $Z \to \mathbf{A}^n_W = W \times \operatorname{Spec}\mathbf{Z}[x_1, \dots, x_n]$ over W correspond bijectively to n-tuples of elements of $\Gamma(Z, \mathcal{O}_Z)$ (Ex. 2.4 applied to $\operatorname{Spec}\mathbf{Z}[x]$).

(a) $S(\mathcal{E})$ is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras: for $U = \operatorname{Spec} R$ affine and $\mathcal{E}|_U = \widetilde{N}$, we have for each $g \in R$ that $S(\mathcal{E}(D(g))) = S_{R_g}(N_g) = S_R(N)_g$, and the stalks are $S(\mathcal{E})_\mathfrak{p} = S(N_\mathfrak{p}) = S_R(N)_\mathfrak{p}$; hence $S(\mathcal{E})|_U = \widetilde{S_R(N)}$ and $S(\mathcal{E})(U) = S(\mathcal{E}(U))$. So $X = \mathbf{Spec}\,S(\mathcal{E})$ exists (5.17(c)) and for U affine with $\mathcal{E}|_U$ free on $e_1, \dots, e_n$, $f^{-1}(U) = \operatorname{Spec} S(\mathcal{E}(U)) = \operatorname{Spec}\mathcal{O}(U)[e_1, \dots, e_n] = \mathbf{A}^n_U$; call this ψ_U. These U form an open cover of Y. Let $V = \operatorname{Spec} A \subseteq U \cap U'$ be affine. The restriction of $\mathcal{E}$ to V is free with the two bases $\{e_i|_V\}$ and $\{e'_i|_V\}$ coming from U, U'; so $e'_j = \sum_i a_{ji} e_i$ with $(a_{ji}) \in GL_n(A)$, and $S(\mathcal{E}(V)) = A[e_1, \dots, e_n] = A[e'_1, \dots, e'_n]$. The automorphism $\psi_{U'}\psi_U^{-1}$ of $\mathbf{A}^n_V$ corresponds to the A-algebra automorphism θ of $A[x_1..x_n]$ with $\theta(x_j) = \sum a_{ji} x_i$ (identifying $x_i \leftrightarrow e_i$, $x_j \leftrightarrow e'_j$), which is linear. Thus we have a vector bundle structure.

Independence. If $\{U'\}$ with bases e' is another choice, then $g = \mathrm{id}_X$ with the union of the two coverings and the maps ψ_U, $\psi'_{U'}$ is again a vector bundle structure by the same computation (for $V \subseteq U \cap U'$ the transition is given by the change of bases of $\mathcal{E}(V)$, a linear automorphism). So id_X is an isomorphism between the two bundles. (An isomorphism $\mathcal{E} \cong \mathcal{E}'$ likewise induces $S(\mathcal{E}) \cong S(\mathcal{E}')$ and an isomorphism of the bundles, by the same argument; so $\mathbf{V}(\mathcal{E})$ depends up to isomorphism only on the isomorphism class of $\mathcal{E}$.)

(b) Let $(X, f, \{U_i\}, \{\psi_i\})$ be a vector bundle. For $W \subseteq U_i$ open, a section $s : W \to f^{-1}(U_i) \cong \mathbf{A}^n_{U_i}$ with $fs = \mathrm{incl}$ is the same as a morphism $W \to \mathbf{A}^n_{U_i}$ over U_i, i.e. an n-tuple $(s^\#(x_1), \dots, s^\#(x_n)) \in \mathcal{O}(W)^n$. Thus $\mathcal{S}|_{U_i} \cong \mathcal{O}^n_{U_i}$ as sheaves of sets and we transport the $\mathcal{O}_{U_i}$-module structure (componentwise addition and scalar multiplication). We must check that the structures from U_i and U_j agree on $U_i \cap U_j$. Both module structures are determined (compatibility with restriction) by their values on affine $V = \operatorname{Spec} A \subseteq U_i \cap U_j$ and sections over open subsets $W \subseteq V$. On such W the coordinates for the two charts are related by the linear automorphism θ of $A[x_1, \dots, x_n]$ with $\theta(x_j) = \sum_i a_{ji} x_i$: $s^\#(x'_j) = s^\#(\theta(x_j)) = \sum_i a_{ji}\, s^\#(x_i)$ with $a_{ji} \in A$. So the identification $\mathcal{O}(W)^n \to \mathcal{O}(W)^n$ between the two coordinatizations is given by a matrix with entries in A, hence is $\mathcal{O}(W)$-linear, and the two structures coincide. So $\mathcal{S}(X/Y)$ is an $\mathcal{O}_Y$-module, locally isomorphic to $\mathcal{O}^n$: locally free of rank n.

(c) Let $\mathcal{E}$ be locally free of rank n, $X = \mathbf{V}(\mathcal{E})$. For $s \in \mathcal{E}^\vee(V) = \mathrm{Hom}(\mathcal{E}|_V, \mathcal{O}_V)$ let $\varphi_s : S(\mathcal{E}|_V) \to \mathcal{O}_V$ be the $\mathcal{O}_V$-algebra homomorphism extending s. For every open affine $W \subseteq V$, φ_s gives $S(\mathcal{E})(W) = S(\mathcal{E}(W)) \to \mathcal{O}(W)$, compatible with restrictions, hence a morphism $W \to f^{-1}(W)$ over W (a section) and these glue to a section $\sigma_s : V \to f^{-1}(V) \subseteq X$ of f. The map $\Phi : \mathcal{E}^\vee \to \mathcal{S}$, $s \mapsto \sigma_s$, is compatible with restriction. It is an isomorphism of $\mathcal{O}_Y$-modules, which can be checked locally. Let $U = \operatorname{Spec} R$ be an open affine with $\mathcal{E}|_U$ free on $e_1, \dots, e_n$, $X|_U = \operatorname{Spec} R[e_1, \dots, e_n]$ with coordinates $x_i = e_i$, and let $W \subseteq U$ be open. Then $\mathcal{E}^\vee|_U$ is free on the dual basis e_i^*, and for $s = \sum c_i e_i^* \in \mathcal{E}^\vee(W) = \mathcal{O}(W)^n$ with $c_i = s(e_i) \in \mathcal{O}(W)$,

$\varphi_s : \mathcal{O}(W)[e_1..e_n] \to \mathcal{O}(W)$ is $e_i \mapsto c_i$, so $\sigma_s^\#(x_i) = c_i$. That is, under the chart ψ_U the section σ_s is the n-tuple $(c_1, \dots, c_n)$, which is exactly the identification $\mathcal{S}|_U \cong \mathcal{O}_U^n$ used to define the module structure in (b). Hence $\Phi|_U : \mathcal{E}^\vee|_U = \mathcal{O}_U^n \to \mathcal{S}|_U \cong \mathcal{O}_U^n$ is the identity, an $\mathcal{O}$-linear isomorphism. So $\Phi : \mathcal{E}^\vee \cong \mathcal{S}(\mathbf{V}(\mathcal{E})/Y)$.

(d) Let $\mathrm{Loc}_n(Y)$ be the set of isomorphism classes of locally free sheaves of rank n and $\mathrm{Vec}_n(Y)$ the set of isomorphism classes of vector bundles of rank n. Define $\alpha : \mathrm{Loc}_n \to \mathrm{Vec}_n$, $[\mathcal{E}] \mapsto [\mathbf{V}(\mathcal{E})]$ (well defined by (a)) and $\beta : \mathrm{Vec}_n \to \mathrm{Loc}_n$, $[X] \mapsto [\mathcal{S}(X/Y)^\vee]$. Here β is well defined: an isomorphism $g : X \to X'$ of vector bundles induces the bijections $s \mapsto g \circ s$ of sections, which are $\mathcal{O}$-linear because in the charts ψ_i, $\psi_i' \circ g$ (both bundle structures on X) the transition maps are linear, as in (b); the dual of a locally free sheaf of rank n is locally free of rank n (5.1).

$\beta\alpha = \mathrm{id}$*:* by (c) and 5.1(a), $\mathcal{S}(\mathbf{V}(\mathcal{E})/Y)^\vee \cong \mathcal{E}^{\vee\vee} \cong \mathcal{E}$. In particular α is injective and β surjective.

$\alpha\beta = \mathrm{id}$*:* let X be a vector bundle with charts $\psi_i : f^{-1}(U_i) \to \mathbf{A}^n_{U_i}$, let $\mathcal{S} = \mathcal{S}(X/Y)$ and $\mathcal{E} = \mathcal{S}^\vee$ (locally free of rank n). We construct an isomorphism $X \cong \mathbf{V}(\mathcal{E})$ of vector bundles.

Linear functions. Over U_i let $\varepsilon_1, \dots, \varepsilon_n \in \mathcal{S}(U_i)$ be the sections with coordinates $(1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$ in the chart ψ_i; they form a basis of $\mathcal{S}|_{U_i} \cong \mathcal{O}^n$, and $x_1, \dots, x_n \in \mathcal{O}_X(f^{-1}U_i)$ denote the coordinate functions of ψ_i. For $V \subseteq U_i$ open and $\lambda \in \mathcal{E}(V) = \mathrm{Hom}(\mathcal{S}|_V, \mathcal{O}_V)$ put $c_i(\lambda) = \sum_k \lambda(\varepsilon_k)\, x_k \in \mathcal{O}_X(f^{-1}V)$; it is $\mathcal{O}(V)$-linear in λ and compatible with restrictions. It is independent of the chart: let $V = \operatorname{Spec} A \subseteq U_i \cap U_j$ be affine and $\psi_j\psi_i^{-1}$ given by θ, $\theta(x_j') = \sum_i a_{ji}x_i$. The matrix (a_{ji}) lies in $GL_n(A)$: θ is an automorphism preserving degrees, so it is surjective on the degree-1 part, i.e. (a_{ji}) is a surjective endomorphism of A^n, hence invertible. A section s with i-coordinates $u_i = s^\#(x_i)$ has j-coordinates $u_j' = s^\#(x_j') = \sum_i a_{ji}u_i$. So the unit section ε_l' of the chart j satisfies $\sum_i a_{ji}u_i = \delta_{jl}$, i.e. $u_i = (a^{-1})_{il}$ and $\varepsilon_l' = \sum_i (a^{-1})_{il}\varepsilon_i$. Therefore

$$\sum_l \lambda(\varepsilon_l')x_l' = \sum_{l,i,j} (a^{-1})_{il}\, \lambda(\varepsilon_i)\, a_{lj}\, x_j = \sum_i \lambda(\varepsilon_i)x_i.$$

So the c_i glue (they agree on affine opens, hence on all opens) to an $\mathcal{O}_Y$-linear morphism $c : \mathcal{E} \to f_*\mathcal{O}_X$. By the universal property of the symmetric algebra, c extends to a homomorphism of $\mathcal{O}_Y$-algebras $\gamma : S(\mathcal{E}) \to f_*\mathcal{O}_X$.

The isomorphism. The morphism f is affine (for every open affine $V \subseteq U_i$, $f^{-1}(V) \cong \mathbf{A}^n_V$ is affine, and these V cover Y), so $f_*\mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_Y$-algebra and $X \cong \mathbf{Spec} f_*\mathcal{O}_X$ (5.17(d)). Hence γ induces a morphism $g : X \to \mathbf{Spec}\, S(\mathcal{E}) = \mathbf{V}(\mathcal{E})$ over Y. Over an open affine $U \subseteq U_i$, $\mathcal{E}|_U$ is free with the basis ε_k^* dual to ε_k; use it to define the chart $\psi_U^{\mathcal{E}} : \mathbf{V}(\mathcal{E})|_U \cong \mathbf{A}^n_U$ of (a), so that the coordinate functions are $e_k = \varepsilon_k^*$. Then $\gamma : \mathcal{O}(U)[e_1, \dots, e_n] \to \mathcal{O}_X(f^{-1}U) = \mathcal{O}(U)[x_1, \dots, x_n]$ sends $e_k \mapsto c(\varepsilon_k^*) = \sum_l \varepsilon_k^*(\varepsilon_l)x_l = x_k$, an isomorphism; i.e. over U, g is an isomorphism and $\psi_U^{\mathcal{E}} \circ g = \psi_i|_{f^{-1}U}$. Thus g is an isomorphism of schemes over Y, and the vector bundle structure of X together with the charts $\psi_{U'}^{\mathcal{E}} \circ g$ (U' affine with $\mathcal{E}|_{U'}$ free, with arbitrary bases) is again a vector bundle structure: for ψ_i vs. $\psi_{U'}^{\mathcal{E}} \circ g$ over an affine $V \subseteq U_i \cap U'$ the transition is determined by the change of basis of the free module $\mathcal{E}(V)$ from ε^* to the chosen basis of $\mathcal{E}|_{U'}$, hence linear, as in (a). So g is an isomorphism of vector bundles $X \cong \mathbf{V}(\mathcal{E})$, and $\alpha\beta[X] = [X]$.

Thus α and β are inverse bijections. (Note that the sheaf of sections of $\mathbf{V}(\mathcal{E})$ is $\mathcal{E}^\vee$, not $\mathcal{E}$: in Hartshorne's convention $\mathbf{V}(\mathcal{E})$ corresponds to the *dual* of its sheaf of sections.) $\square$

6. Divisors

Throughout, results of Hartshorne are cited by number ("(6.5)", "Ex. 6.2", "Thm. I.7.7", ...). Condition $(*)$ is the standing hypothesis of the section: X is a noetherian integral separated scheme which is regular in codimension one. For a prime divisor Y of such an X we write η_Y for its generic point, v_Y for the valuation of the discrete valuation ring $\mathcal{O}_{X,\eta_Y}$, and $K = K(X)$ for the function field. In the exercises that deal with varieties over a field k (6.2, 6.3, 6.5–6.7, 6.9, 6.11, 6.12) k is algebraically closed, as in the text; where an extra hypothesis (characteristic, $\sqrt{-1} \in k$) is needed it is stated. For a linear form or homogeneous polynomial h I write $\bar{h}$ for its image in a homogeneous coordinate ring.

Exercise 6.1

> **Problem.** Let X be a scheme satisfying $(*)$. Then $X \times \mathbf{P}^n$ also satisfies $(*)$, and $\mathrm{Cl}(X \times \mathbf{P}^n) \cong (\mathrm{Cl}\, X) \times \mathbf{Z}$.

Solution. Write $P := X \times_{\mathrm{Spec}\,\mathbf{Z}} \mathbf{P}^n_{\mathbf{Z}} = \mathbf{P}^n_X$, with projection $p : P \to X$ and homogeneous coordinates $x_0, \dots, x_n$. The open sets $U_i = D_+(x_i)$ cover P and $U_i \cong X \times \mathbf{A}^n = \mathbf{A}^n_X$.

Step 1: P satisfies $(*)$. By (6.6), applied n times, $X \times \mathbf{A}^n$ satisfies $(*)$, in particular it is integral, noetherian and regular in codimension one. Any two U_i, U_j meet (their intersection contains $X \times \mathbf{G}_m^n \neq \emptyset$ over any open of X), so P is covered by integral open subschemes that pairwise meet; hence P is irreducible and reduced, i.e. integral. It is noetherian (finite type over X) and separated ($P \to X$ is separated and X is separated). Regularity in codimension one is a statement about local rings, hence can be checked on the cover by the U_i. So $(*)$ holds.

Step 2: an exact sequence. Let $H = V_+(x_0) \cong \mathbf{P}^{n-1}_X$. By Step 1 (for $n-1$; for $n = 1$ it is $\cong X$) H is integral, and over an affine open $\mathrm{Spec}\, A \subseteq X$, in the chart $\mathrm{Spec}\, A[t_0, t_2, \dots, t_n] \subseteq U_1$ ($t_j = x_j/x_1$), it is cut out by the prime element t_0 (the quotient is a polynomial ring over the domain A). By Krull's Hauptidealsatz it has codimension 1; so H is a prime divisor, and $P - H = U_0 \cong X \times \mathbf{A}^n$. By (6.5)(c) there is an exact sequence

$$\mathbf{Z} \xrightarrow{1 \mapsto [H]} \mathrm{Cl}\, P \longrightarrow \mathrm{Cl}(X \times \mathbf{A}^n) \to 0,$$

and by (6.6), applied n times, $\mathrm{Cl}(X \times \mathbf{A}^n) \cong \mathrm{Cl}\, X$.

Step 3: a retraction $\rho : \mathrm{Cl}\, P \to \mathbf{Z}$ with $\rho[H] = 1$. Let $K = K(X) = \mathcal{O}_{X,\eta}$, η the generic point, and let $P_K = \mathbf{P}^n_K = P \times_X \mathrm{Spec}\, K$ be the generic fibre. The points of P_K are the points $\xi \in P$ with $p(\xi) = \eta$, and for such ξ, $\mathcal{O}_{P_K,\xi} = \mathcal{O}_{P,\xi}$ (localization). Hence the prime divisors of P_K correspond bijectively to the prime divisors W of P which *dominate* X (generic point over η), $W \mapsto W_K$, with the same valuation: $v_{W_K} = v_W$ on $K(P) = K(P_K)$. Define $\rho_0 : \mathrm{Div}\, P \to \mathrm{Div}\, P_K$ by $W \mapsto W_K$ if W dominates X and $W \mapsto 0$ otherwise. For $f \in K(P)^*$ we get $\rho_0((f)) = (f)_{P_K}$, so ρ_0 induces $\mathrm{Cl}\, P \to \mathrm{Cl}\, P_K \cong \mathbf{Z}$, the last isomorphism being the degree of (6.4) (valid over the field K). The hyperplane H dominates X and H_K is a hyperplane in $\mathbf{P}^n_K$, of degree 1; so $\rho[H] = 1$.

Conclusion. Since $\rho \circ (1 \mapsto [H]) = \mathrm{id}_{\mathbf{Z}}$, the map $\mathbf{Z} \to \mathrm{Cl}\, P$ of Step 2 is injective and the short exact sequence $0 \to \mathbf{Z} \to \mathrm{Cl}\, P \to \mathrm{Cl}\, X \to 0$ splits: $\mathrm{Cl}\, P = \mathbf{Z}[H] \oplus \ker \rho \cong \mathbf{Z} \oplus \mathrm{Cl}\, X$. □

Exercise 6.2

Problem. *Varieties in projective space.* Let k be an algebraically closed field and let X be a closed subvariety of $\mathbf{P}^n_k$ which is nonsingular in codimension one (hence satisfies $(*)$). For any divisor $D = \sum n_i Y_i$ on X, we define the *degree* of D to be $\sum n_i \deg Y_i$, where $\deg Y_i$ is the degree of Y_i as a projective variety itself (I, Section 7). (Starred in the book.)

(a) Let V be an irreducible hypersurface in $\mathbf{P}^n$ which does not contain X, and let Y_i be the irreducible components of $V \cap X$. They all have codimension 1 in X (I, Ex. 1.8). For each i, let f_i be a local equation for V on some open set U_i on which $Y_i \cap U_i \neq \emptyset$, and let $n_i = v_{Y_i}(\bar{f}_i)$, where $\bar{f}_i$ is the restriction of f_i to $U_i \cap X$. Then we define the *divisor* $V.X$ to be $\sum n_i Y_i$. Extend by linearity, and show that this gives a well-defined homomorphism from the subgroup of $\operatorname{Div} \mathbf{P}^n$ consisting of divisors none of whose components contain X, to $\operatorname{Div} X$.

(b) If D is a principal divisor on $\mathbf{P}^n$, for which $D.X$ is defined as in (a), show that $D.X$ is principal on X. Thus we get a homomorphism $\operatorname{Cl} \mathbf{P}^n \to \operatorname{Cl} X$.

(c) Show that the integer n_i defined in (a) is the same as the intersection multiplicity $i(X, V; Y_i)$ defined in (I, Section 7). Then use the generalized Bezout theorem (I, 7.7) to show that for any divisor D on $\mathbf{P}^n$, none of whose components contain X, $\deg(D.X) = (\deg D) \cdot (\deg X)$.

(d) If D is a principal divisor on X, show that there is a rational function f on $\mathbf{P}^n$ such that $D = (f).X$. Conclude that $\deg D = 0$. Thus the degree function defines a homomorphism $\deg : \operatorname{Cl} X \to \mathbf{Z}$. (This gives another proof of (6.10), since any complete nonsingular curve is projective.) Finally, there is a commutative diagram with top row $\operatorname{Cl} \mathbf{P}^n \to \operatorname{Cl} X$, left vertical map $\deg : \operatorname{Cl} \mathbf{P}^n \xrightarrow{\cong} \mathbf{Z}$, right vertical map $\deg : \operatorname{Cl} X \to \mathbf{Z}$, and bottom row $\cdot(\deg X) : \mathbf{Z} \to \mathbf{Z}$, and in particular we see that the map $\operatorname{Cl} \mathbf{P}^n \to \operatorname{Cl} X$ is injective.

Solution. Notation: $S = k[x_0, \ldots, x_n]$, $I_X \subset S$ the (prime, homogeneous) ideal of X, $S_X = S/I_X$ the homogeneous coordinate ring; we assume $\dim X \geq 1$ (otherwise $\operatorname{Div} X = 0$ and there is nothing to prove). For a prime divisor Y of X let $\mathfrak{p}_Y \subset S_X$ be its homogeneous prime. Then $Y \cap D_+(x_i) \neq \emptyset$ iff $\bar{x}_i \notin \mathfrak{p}_Y$ iff $\eta_Y \in D_+(x_i)$. Let $\mathcal{G} \subseteq \operatorname{Div} \mathbf{P}^n$ be the subgroup of divisors none of whose components contain X; it is free on the irreducible hypersurfaces $V \not\supseteq X$.

(a) An irreducible hypersurface $V = V_+(h)$, h irreducible of degree d. On $D_+(x_i)$ the ideal sheaf of V is generated by h/x_i^d, a nonzerodivisor, so locally V is given by one equation (this is also clear since $\mathbf{P}^n$ is locally factorial, (6.11)). Since $V \not\supseteq X$, $h \notin I_X$, so $\bar{h} \neq 0$ in the domain S_X and every $\bar{f}_i \neq 0$.

Components. $V \cap X \subsetneq X$ is a proper closed subset of the variety X, so each of its components has dimension $\leq \dim X - 1$; by I.Ex.1.8 it is $\geq \dim X - 1$. So the Y_i are prime divisors of X.

Independence of choices. Let f and f' be two local equations of V defined on open sets U, U' both containing $\eta = \eta_{Y_i}$. Then f, f' both generate the stalk of the ideal sheaf $\mathcal{I}_V$ at η, so $f = uf'$ with $u \in \mathcal{O}^*_{\mathbf{P}^n,\eta}$. Restricting to X gives $\bar{f} = \bar{u}\bar{f}'$ with $\bar{u} \in \mathcal{O}^*_{X,\eta}$, so $v_{Y_i}(\bar{f}) = v_{Y_i}(\bar{f}')$. Moreover $\eta \in V$ gives $f \in \mathfrak{m}_{\mathbf{P}^n,\eta}$, so $n_i = v_{Y_i}(\bar{f}_i) \geq 1$. Thus n_i depends only on V and Y_i, and $V.X := \sum n_i Y_i$ is a well-defined effective divisor. Defining $D.X = \sum m_j (V_j.X)$ for $D = \sum m_j V_j \in \mathcal{G}$ is then a homomorphism $\mathcal{G} \to \operatorname{Div} X$, because $\mathcal{G}$ is free on the V_j. Note that if Y is a prime divisor of X and $\eta_Y \notin V$, then $\bar{f}$ is a unit at η_Y, so the *coefficient of Y in $V.X$ is $v_Y(\overline{h/x_i^d})$ for every i with*

$\eta_Y \in D_+(x_i)$, whether or not $Y \subseteq V$ (it is 0 in the latter case). I record this:

$$\operatorname{coeff}_Y(V.X) = v_Y\big(\bar{h}/\bar{x}_i^d\big) \quad \text{for all } i \text{ with } \bar{x}_i \notin \mathfrak{p}_Y. \qquad (1)$$

(b) *Lemma.* Let $A, B \in S$ be homogeneous of the same degree d with $A, B \notin I_X$, and $f = A/B \in K(\mathbf{P}^n)^*$. Factor $A = c\prod h_j^{a_j}$, $B = c'\prod h_j^{b_j}$ into irreducibles and set $(A) = \sum a_j V_+(h_j)$, $(B) = \sum b_j V_+(h_j)$. Then $(f) = (A)-(B) \in \mathcal{G}$ (no h_j lies in I_X, since I_X is prime and $A, B \notin I_X$), and

$$(f).X = \big(\bar{A}/\bar{B}\big) \quad \text{as divisors on } X,$$

where $\bar{A}/\bar{B} \in K(X)^*$. Indeed, let Y be a prime divisor of X and choose i with $\bar{x}_i \notin \mathfrak{p}_Y$. By (1),

$$\operatorname{coeff}_Y((A).X) = \sum_j a_j v_Y(\bar{h}_j/\bar{x}_i^{d_j}) = v_Y\Big(\prod(\bar{h}_j/\bar{x}_i^{d_j})^{a_j}\Big) = v_Y(\bar{A}/\bar{x}_i^d),$$

(using $\sum a_j d_j = \deg A = d$ and that c is a unit), and similarly for B. Subtracting, $\operatorname{coeff}_Y((f).X) = v_Y(\bar{A}/\bar{B})$.

Now let $D \in \mathcal{G}$ be principal, $D = (f)$. Write $f = A/B$ with A, B coprime forms of the same degree. Then (A) and (B) are effective without common components and $D = (A) - (B)$, so the components of $(A), (B)$ are those of D; none contains X, so $A, B \notin I_X$, and the Lemma gives $D.X = (\bar{A}/\bar{B})$, principal. So $\mathcal{G} \cap \{\text{principal}\} \to \{\text{principal divisors of } X\}$, and we get $\mathcal{G}/(\mathcal{G} \cap \operatorname{Prin}) \to \operatorname{Cl} X$. By (6.4)(a), every class of $\operatorname{Cl}\mathbf{P}^n$ contains $d \cdot H'$ with H' any hyperplane, and we can choose $H' = V_+(x_i)$ with $x_i \notin I_X$ (possible since $X \neq \emptyset$); so $\mathcal{G} \to \operatorname{Cl}\mathbf{P}^n$ is surjective with kernel $\mathcal{G} \cap \operatorname{Prin}$, i.e. $\operatorname{Cl}\mathbf{P}^n = \mathcal{G}/(\mathcal{G} \cap \operatorname{Prin})$. We obtain the homomorphism $\varphi : \operatorname{Cl}\mathbf{P}^n \to \operatorname{Cl} X$, $[D] \mapsto [D.X]$ for $D \in \mathcal{G}$.

(c) *Local computation.* Let Y be a prime divisor of X, $\mathfrak{p} = \mathfrak{p}_Y$ (homogeneous prime of S_X, of height one), $x \in S_1$ with $\bar{x} \notin \mathfrak{p}$, $A = (S_X)_{(x)}$ (degree-0 part of $(S_X)_x$, the coordinate ring of $X \cap D_+(x)$), $R = \mathcal{O}_{X,\eta_Y} = A_{\mathfrak{q}}$ where $\mathfrak{q} = \mathfrak{p}(S_X)_x \cap A$. Since $\bar{x}$ is homogeneous of degree one and a unit, $(S_X)_x = A[x, x^{-1}]$ is a Laurent polynomial ring (every homogeneous element of degree m is $(s/x^m)x^m$ with $s/x^m \in A$, and x is algebraically independent over A by degree reasons). As $\mathfrak{p}$ is homogeneous, so is the ideal $\mathfrak{p}(S_X)_x$, whence $\mathfrak{p}(S_X)_x = \bigoplus_m (\mathfrak{p}(S_X)_x)_m = \bigoplus_m \mathfrak{q}x^m = \mathfrak{q}A[x, x^{-1}]$. Let π be a uniformizer of the DVR R and $\kappa = R/\pi R$. Localizing, $T := (S_X)_{\mathfrak{p}} = R[x, x^{-1}]_{\pi R[x,x^{-1}]}$. Here $\pi R[x, x^{-1}]$ is prime (quotient $\kappa[x, x^{-1}]$), so T is a noetherian local domain whose maximal ideal πT is principal and nonzero, hence T *is a DVR with uniformizer* π, and $v_T|_R = v_R$. Consequently, for $h \in S_d$ with $\bar{h} \neq 0$ in S_X,

$$\operatorname{length}_{S_{\mathfrak{p}}}\big(S/(I_X + (h))\big)_{\mathfrak{p}} = \operatorname{length}_T\big(T/\bar{h}T\big) = v_T(\bar{h}) = v_R(\bar{h}/\bar{x}^d), \qquad (2)$$

since $(S/(I_X + (h)))_{\mathfrak{p}} = T/\bar{h}T$ is a module over the quotient $S_{\mathfrak{p}} \to T$ of $S_{\mathfrak{p}}$ (so lengths over $S_{\mathfrak{p}}$ and T agree), $T/\pi^n T$ has length n, and $\bar{x}$ is a unit of T.

Now take $V = V_+(h)$, h irreducible of degree d, $V \not\supseteq X$, and $Y = Y_i$ a component of $V \cap X$. The intersection multiplicity of (I, Section 7) is $i(X, V; Y) = \mu_{\mathfrak{p}}(S/(I_X + I_V)) = \operatorname{length}_{S_{\mathfrak{p}}}(S/(I_X + (h)))_{\mathfrak{p}}$ ($I_V = (h)$ as h is irreducible). By (2) and (1) this equals $v_Y(\bar{h}/\bar{x}^d) = n_i$. So $n_i = i(X, V; Y_i)$.

Bezout. Let V be an irreducible hypersurface not containing X; by Thm. I.7.7, $\sum_i i(X, V; Y_i) \deg Y_i = (\deg X)(\deg V)$, where $\deg V = \deg h$ by (I, 7.6(d)). The left side is $\deg(V.X)$ by the first part.

For $D = \sum m_j V_j \in \mathcal{G}$, linearity gives $\deg(D.X) = \sum m_j \deg(V_j.X) = \deg X \sum m_j \deg V_j = (\deg X)(\deg D)$, where $\deg D$ is as in (6.4).

(d) Let $D = (g)$ with $g \in K(X)^*$. Write $g = \bar{A}/\bar{B}$ with A, B forms of the same degree, $A, B \notin I_X$ (this is the description of $K(X)$ for a projective variety). Put $f = A/B \in K(\mathbf{P}^n)^*$. By the Lemma in (b), $(f) \in \mathcal{G}$ and $(f).X = (g) = D$. By (c) and (6.4)(b), $\deg D = \deg((f).X) = \deg(f) \cdot \deg X = 0$. So deg vanishes on principal divisors and defines $\deg : \operatorname{Cl} X \to \mathbf{Z}$.

Diagram. For $D \in \mathcal{G}$, (c) says $\deg(\varphi[D]) = \deg(D.X) = (\deg X) \deg D$, i.e. $\deg_X \circ \varphi = (\deg X) \cdot \deg_{\mathbf{P}^n}$, which is the commutativity. Since $\deg_{\mathbf{P}^n}$ is an isomorphism (6.4)(c) and $\deg X > 0$ (I, 7.6(a)), multiplication by $\deg X$ is injective, hence φ is injective. □

Exercise 6.3

Problem. *Cones.* In this exercise we compare the class group of a projective variety V and the class group of its cone (I, Ex. 2.10). So let V be a projective variety in $\mathbf{P}^n$, which is of dimension ≥ 1 and nonsingular in codimension 1. Let $X = C(V)$ be the affine cone over V in $\mathbf{A}^{n+1}$, and let $\bar{X}$ be its projective closure in $\mathbf{P}^{n+1}$. Let $P \in X$ be the vertex of the cone. (Starred in the book.)

(a) Let $\pi : \bar{X} - P \to V$ be the projection map. Show that V can be covered by open subsets U_i such that $\pi^{-1}(U_i) \cong U_i \times \mathbf{A}^1$ for each i, and then show as in (6.6) that $\pi^* : \operatorname{Cl} V \to \operatorname{Cl}(\bar{X} - P)$ is an isomorphism. Since $\operatorname{Cl} \bar{X} \to \operatorname{Cl}(\bar{X} - P)$ is an isomorphism, we have also $\operatorname{Cl} V \cong \operatorname{Cl} \bar{X}$.

(b) We have $V \subseteq \bar{X}$ as the hyperplane section at infinity. Show that the class of the divisor V in $\operatorname{Cl} \bar{X}$ is equal to π^*(class of $V.H$), where H is any hyperplane of $\mathbf{P}^n$ not containing V. Thus conclude using (6.5) that there is an exact sequence

$$0 \to \mathbf{Z} \to \operatorname{Cl} V \to \operatorname{Cl} X \to 0,$$

where the first map sends $1 \mapsto V.H$, and the second is π^* followed by the restriction to $X - P$ and inclusion in X. (The injectivity of the first arrow follows from the previous exercise.)

(c) Let $S(V)$ be the homogeneous coordinate ring of V (which is also the affine coordinate ring of X). Show that $S(V)$ is a unique factorization domain if and only if (1) V is projectively normal (Ex. 5.14), and (2) $\operatorname{Cl} V \cong \mathbf{Z}$ and is generated by the class of $V.H$.

(d) Let $\mathcal{O}_P$ be the local ring of P on X. Show that the natural restriction map induces an isomorphism $\operatorname{Cl} X \to \operatorname{Cl}(\operatorname{Spec} \mathcal{O}_P)$.

Solution. *Setup.* Let $S = S(V) = k[x_0, \ldots, x_n]/I(V)$, a graded domain generated by S_1 with $S_0 = k$, $\mathfrak{m} = S_+$, $\dim S = \dim V + 1 \geq 2$. Then $X = \operatorname{Spec} S$ and P is the closed point $\mathfrak{m}$. Let $\mathbf{P}^{n+1}$ have coordinates $x_0, \ldots, x_n, w$, so that $\mathbf{A}^{n+1} = \{w \neq 0\}$ and the hyperplane at infinity $\{w = 0\}$ is a copy of $\mathbf{P}^n \supseteq V$. The homogeneous coordinate ring $S[w]$ of $\bar{X} = V_+(I(V))$ is a domain, $\bar{X} \cap \{w \neq 0\} = X$ and $\bar{X} \cap \{w = 0\} = V$; $P = (0 : \cdots : 0 : 1)$ and $\bar{X} - P = \bigcup_i D_+(x_i) \cap \bar{X}$. Since P is the only point at which all x_i vanish, $\pi(x : w) = (x)$ is a morphism $\bar{X} - P \to V$.

(a) Let $U_i = V \cap D_+(x_i) = \operatorname{Spec} S_{(x_i)}$. The ring of $\bar{X} \cap D_+(x_i)$ is $(S[w]_{x_i})_0 = S_{(x_i)}[t_i]$, $t_i = w/x_i$, a polynomial ring (w is algebraically independent over S). Hence $\pi^{-1}(U_i) \cong U_i \times \mathbf{A}^1$, π being the projection. Put $L = \bar{X} - P$; it is a noetherian integral separated scheme, regular in codimension one (locally $U_i \times \mathbf{A}^1$, and V is regular in codimension one, so (6.6) applies to the

local rings), i.e. L satisfies $(*)$. Also $\bar{X}$ satisfies $(*)$: all codimension one points lie in L since $\operatorname{codim}(P, \bar{X}) = \dim V + 1 \geq 2$. So by (6.5)(b), $\operatorname{Cl} \bar{X} \to \operatorname{Cl} L$ is an isomorphism.

The map π^*. For a prime divisor $Y \subset V$, $\pi^{-1}(Y) \cap \pi^{-1}(U_i) = (Y \cap U_i) \times \mathbf{A}^1$ is integral of codimension one in $U_i \times \mathbf{A}^1$ whenever $Y \cap U_i \neq \emptyset$; any two such open pieces meet, so $\pi^{-1}(Y)$ is a prime divisor of L ("type 1"), with $v_{\pi^{-1}Y}|_{K(V)} = v_Y$ (as in (6.6): $\mathcal{O}_{L,\pi^{-1}Y} = \mathcal{O}_{V,Y}[t]_{(\pi_Y)}$, a DVR with the same uniformizer π_Y). Define $\pi^* : \operatorname{Div} V \to \operatorname{Div} L$ by $Y \mapsto \pi^{-1}(Y)$. The prime divisors of L are of two kinds: type 1, $\pi^{-1}(Y)$, and type 2, those whose generic point maps to the generic point of V (if Z is a prime divisor, $\overline{\pi(Z)}$ is V or a prime divisor of V by a dimension count, $\dim Z = \dim V$ and $Z \subseteq \pi^{-1}(\overline{\pi(Z)})$; in the latter case $Z = \pi^{-1}(\overline{\pi(Z)})$). For $f \in K(V)^*$, $\pi^* f \in K(L)^*$ has $(\pi^* f) = \pi^*((f))$: at type 1 primes the valuation is $v_Y(f)$; at type 2 primes the local ring contains $K(V)$, so $\pi^* f$ is a unit. Thus $\pi^* : \operatorname{Cl} V \to \operatorname{Cl} L$ is a well-defined homomorphism.

Injective. Suppose $\pi^* D = (g)$, $g \in K(L)^* = K(V)(t_i)$. On $\pi^{-1}(U_i) = U_i \times \mathbf{A}^1$ the type 2 primes correspond to the irreducible polynomials of $K(V)[t_i]$ (the closed points of the generic fibre $\operatorname{Spec} K(V)[t_i]$), and for such a prime $Z_q = (q)$, $v_{Z_q}(g)$ is the exponent of q in $g = a/b$ ($a, b \in K(V)[t_i]$ coprime). Since $(g) = \pi^* D$ has no type 2 component, a, b have no irreducible factor of positive degree, so $g \in K(V)^*$. Then $\pi^*(D - (g)) = 0$, so $D = (g)$ because π^* is injective on Div (distinct Y give distinct $\pi^{-1}Y$).

Surjective. Type 1 divisors are in the image. If Z is of type 2, in the chart $U_i \times \mathbf{A}^1$ it is Z_q for an irreducible $q \in K(V)[t_i]$; viewing $q \in K(L)^*$, $v_{Z_q}(q) = 1$, $v_{Z'}(q) = 0$ for every other type 2 prime Z', so $Z - (q)$ is a combination of type 1 primes. Hence π^* is onto. Therefore $\pi^* : \operatorname{Cl} V \xrightarrow{\cong} \operatorname{Cl} L \xleftarrow{\cong} \operatorname{Cl} \bar{X}$.

(b) Choose a linear form $\ell \in k[x_0, \ldots, x_n]$ with $\ell \notin I(V)$, $H = V_+(\ell)$. In $K(\bar{X})$ consider $f = \ell/w$. On $\pi^{-1}(U_i) = U_i \times \mathbf{A}^1$ we have $f = \pi^*(\ell/x_i) \cdot t_i^{-1}$. The divisor of t_i is $\{t_i = 0\} = V \cap \pi^{-1}(U_i)$ with multiplicity one (t_i is a coordinate), and $(\pi^*(\ell/x_i)) = \pi^*((\ell/x_i))$ with $(\ell/x_i)|_{U_i} = (V.H)|_{U_i}$ by the definition of $V.H$ in Ex. 6.2(a). So $(f) = \pi^*(V.H) - V$ on every $\pi^{-1}(U_i)$, hence on L. Therefore $[V] = \pi^*[V.H]$ in $\operatorname{Cl} L \cong \operatorname{Cl} \bar{X}$.

Now V is an irreducible closed subset of codimension one of $\bar{X}$ with $\bar{X} - V = X$. By (6.5)(c),

$$\mathbf{Z} \xrightarrow{1 \mapsto [V]} \operatorname{Cl} \bar{X} \longrightarrow \operatorname{Cl} X \to 0$$

is exact. Composing with the isomorphism $\pi^* : \operatorname{Cl} V \cong \operatorname{Cl} \bar{X}$ of (a) the first map becomes $1 \mapsto [V.H]$, and the second map $\operatorname{Cl} \bar{X} \to \operatorname{Cl} X$ (restriction) is, after restricting further to $X - P$, the map described in the statement (note $\operatorname{Cl} X \cong \operatorname{Cl}(X - P)$ by (6.5)(b) since $\operatorname{codim}(P, X) \geq 2$). Finally $1 \mapsto [V.H]$ is injective: by Ex. 6.2(c),(d) applied to $V \subseteq \mathbf{P}^n$ and the hyperplane $H \not\supseteq V$ (of degree one), $\deg(V.H) = \deg V > 0$, so $m[V.H] = 0$ forces $m = 0$. This gives the exact sequence $0 \to \mathbf{Z} \to \operatorname{Cl} V \to \operatorname{Cl} X \to 0$.

(c) By (6.2), the noetherian domain S is a UFD iff $X = \operatorname{Spec} S$ is normal and $\operatorname{Cl} X = 0$. X is normal iff S is integrally closed, i.e. iff V is projectively normal (Ex. 5.14: V is projectively normal iff $S(V)$ is an integrally closed domain). By (b), $\operatorname{Cl} X = 0$ iff $\operatorname{Cl} V$ is generated by the image of $\mathbf{Z}$, i.e. by $[V.H]$; since $\mathbf{Z} \to \operatorname{Cl} V$ is injective, this holds iff $\operatorname{Cl} V \cong \mathbf{Z}$ with generator $[V.H]$. This is the claim.

(d) Let $\mathcal{O}_P = S_{\mathfrak{m}}$. The scheme $\operatorname{Spec} \mathcal{O}_P$ is noetherian, integral, separated and regular in codimension one (its local rings are local rings of X), so satisfies $(*)$; its prime divisors are

$Y \cap \operatorname{Spec} \mathcal{O}_P$ for prime divisors $Y \ni P$ of X, i.e. correspond to height-one primes $\mathfrak{p} \subseteq \mathfrak{m}$ of S, and for these $\mathcal{O}_{X,Y}$ is also the local ring of $\operatorname{Spec} \mathcal{O}_P$ at the corresponding point. Define $\operatorname{res} : \operatorname{Div} X \to \operatorname{Div} \operatorname{Spec} \mathcal{O}_P$ by $Y \mapsto Y \cap \operatorname{Spec} \mathcal{O}_P$ (which is 0 if $P \notin Y$). It is surjective (take closures), and $\operatorname{res}((f)) = (f)$ for $f \in K^*$ since valuations are unchanged. So res induces a surjection $r : \operatorname{Cl} X \to \operatorname{Cl} \operatorname{Spec} \mathcal{O}_P$. The kernel of r: if $\operatorname{res} D = (f)$ then $D - (f)$ lies in $\ker(\operatorname{res})$, the group generated by prime divisors *not* through P. So it suffices to prove:

Claim. Every prime divisor $Y \not\ni P$ of X is principal.

Granting the Claim, $D - (f)$ is principal, so $[D] = 0$, and r is injective.

Proof of the Claim. Let $\mathfrak{p} \subset S$ be the height-one prime of Y; $P \notin Y$ means $\mathfrak{p} \not\subseteq \mathfrak{m}$. Let $\mathcal{H}$ be the set of nonzero homogeneous elements of S. We use two standard facts: a minimal prime over a homogeneous ideal of a graded ring is homogeneous, and a homogeneous proper ideal of S is contained in $\mathfrak{m}$ (its degree-0 part is a proper ideal of $S_0 = k$).

(i) $\mathfrak{p} \cap \mathcal{H} = \emptyset$. If $h \in \mathfrak{p} \cap \mathcal{H}$, then $\mathfrak{p}$ (of height one in the domain S) is minimal over (h), hence homogeneous, hence $\subseteq \mathfrak{m}$; contradiction. Likewise, $\mathfrak{p} \not\subseteq \mathfrak{m}$ gives an element $g = g_0 + g_1 + \cdots + g_D \in \mathfrak{p}$ ($g_j \in S_j$) with $g_0 \neq 0$; we normalize $g_0 = 1$.

(ii) *The element G.* Let $S_h = \mathcal{H}^{-1}S$ and fix $0 \neq x \in S_1$. As in Ex. 6.2(c), $S_h = K_0[x, x^{-1}]$ where $K_0 = (S_h)_0$ is a field (and x is transcendental over it). The prime $\mathfrak{p}S_h$ is proper (by (i)) and nonzero, hence a maximal ideal of the PID $K_0[x, x^{-1}]$. Now $g = \sum a_j x^j \in K_0[x]$ with $a_j = g_j/x^j \in A := S_{(x)} \subseteq K_0$, $a_0 = 1$, and $g \in \mathfrak{p}S_h$. Let $G \in K_0[x]$ be a generator of $\mathfrak{p}S_h$ which is a polynomial with $G(0) = 1$: take an irreducible factor of g in $K_0[x]$ lying in $\mathfrak{p}S_h$ (it is not associate to x as $g(0) \neq 0$), multiplied by a constant. It is unique: two generators differ by a unit λx^m of $K_0[x, x^{-1}]$, and both being polynomials with constant term 1 forces $m = 0, \lambda = 1$. By this uniqueness, G does not depend on the choice of x (replacing x by $x' = ux$, $u \in K_0^*$, the polynomial $G(x'/u)$ is again a polynomial in x' with constant term 1 generating $\mathfrak{p}S_h$). Moreover $G \mid g$ in $K_0[x]$.

(iii) $(G) = Y$ *as divisors on* X. Let $\mathfrak{q} \subset S$ be any height-one prime ($Z = V(\mathfrak{q})$).

Case $\mathfrak{q} \cap \mathcal{H} = \emptyset$. Then $S_h \subseteq S_\mathfrak{q}$ and $S_\mathfrak{q} = (S_h)_{\mathfrak{q}S_h}$, with $\mathfrak{q}S_h$ a maximal ideal of $K_0[x, x^{-1}]$. If $\mathfrak{q} = \mathfrak{p}$, $S_\mathfrak{p}$ is the localization of the PID at the maximal ideal (G), so $v_Y(G) = 1$. If $\mathfrak{q} \neq \mathfrak{p}$, then $\mathfrak{q}S_h \neq \mathfrak{p}S_h$ (they contract to $\mathfrak{q}, \mathfrak{p}$), so $G \notin \mathfrak{q}S_h$ and $v_Z(G) = 0$.

Case $\mathfrak{q} \cap \mathcal{H} \neq \emptyset$. Then $\mathfrak{q}$ is homogeneous (minimal over a homogeneous element) and $\mathfrak{q} \neq \mathfrak{m}$ (as $\operatorname{ht} \mathfrak{m} = \dim S \geq 2$), so some $x' \in S_1$ is not in $\mathfrak{q}$ (else $\mathfrak{m} \subseteq \mathfrak{q}$). Use this x' in place of x: $G = \sum_{j=0}^{e} c_j {x'}^j$ with $c_j \in K_0$, $c_0 = 1$, and $G \mid g$ in $K_0[x']$, $g = \sum a_j {x'}^j$, $a_j = g_j/{x'}^j \in A' = S_{(x')}$. With τ an indeterminate, $g^*(\tau) = \tau^D + a_1\tau^{D-1} + \cdots + a_D$ is monic over A' and divisible by $G^*(\tau) = \tau^e + c_1\tau^{e-1} + \cdots + c_e \in K_0[\tau]$ (reverse the polynomials). Hence every root of G^* in an algebraic closure of K_0 is integral over A', and so are the coefficients c_j (symmetric functions of the roots). As in Ex. 6.2(c) we have $\mathfrak{q}(S_{x'}) = \mathfrak{q}_0 S_{x'}$ with $\mathfrak{q}_0 = \mathfrak{q}S_{x'} \cap A'$ prime (of height one, since $S_{x'} = A'[x', {x'}^{-1}]$ and $\mathfrak{q}S_{x'} = \mathfrak{q}_0 A'[x', {x'}^{-1}]$), $R = A'_{\mathfrak{q}_0}$ is the local ring of V along the prime divisor $V_+(\mathfrak{q})$ of V, hence a DVR (as V is nonsingular in codimension one), with uniformizer π, and $T := S_\mathfrak{q} = \mathcal{O}_{X,Z} = R[x', {x'}^{-1}]_{(\pi)}$ is a DVR with uniformizer π and residue field $\operatorname{Frac}(\kappa[x', {x'}^{-1}])$, $\kappa = R/\pi R$. The $c_j \in K_0$ are integral over $A' \subseteq T$ and T is integrally closed with fraction field $K \supseteq K_0$, so $c_j \in T \cap K_0$. Now $R \to T$ is a flat local homomorphism ($R[x', {x'}^{-1}]$ is free over R and T is a localization of it), hence faithfully flat, so $aT \cap R = aR$ for $a \in R$; therefore $T \cap K_0 = R$ (write $c \in T \cap K_0$ as a/b, $a, b \in R$; then $a \in bT \cap R = bR$). So

$c_j \in R$ for all j, and $G \in T$ has image $\sum \bar{c}_j x'^j$ in $T/\pi T$ with $\bar{c}_0 = 1$, i.e. $G \notin \pi T$. Hence G is a unit of T and $v_Z(G) = 0$.

Thus $(G) = 1 \cdot Y$ and Y is principal. This proves the Claim and (d). □

Exercise 6.4

> **Problem.** Let k be a field of characteristic $\neq 2$. Let $f \in k[x_1, \ldots, x_n]$ be a *square-free* nonconstant polynomial, i.e. in the unique factorization of f into irreducible polynomials, there are no repeated factors. Let $A = k[x_1, \ldots, x_n, z]/(z^2 - f)$. Show that A is an integrally closed ring. [*Hint*: The quotient field K of A is just $k(x_1, \ldots, x_n)[z]/(z^2 - f)$. It is a Galois extension of $k(x_1, \ldots, x_n)$ with Galois group $\mathbf{Z}/2\mathbf{Z}$ generated by $z \mapsto -z$. If $\alpha = g + hz \in K$, where $g, h \in k(x_1, \ldots, x_n)$, then the minimal polynomial of α is $X^2 - 2gX + (g^2 - h^2 f)$. Now show that α is integral over $k[x_1, \ldots, x_n]$ if and only if $g, h \in k[x_1, \ldots, x_n]$. Conclude that A is the integral closure of $k[x_1, \ldots, x_n]$ in K.]

Solution. Put $R = k[x_1, \ldots, x_n]$ (a UFD, hence integrally closed) and $F = \operatorname{Frac} R = k(x_1, \ldots, x_n)$.

A is a domain, $A = R \oplus Rz$. Since f is nonconstant and square-free, it is not a square in F (some irreducible factor occurs to the first power, and valuations of squares are even); so $z^2 - f$ is irreducible over F, hence (Gauss) irreducible in $R[z]$ (it is monic). Thus $A = R[z]/(z^2 - f)$ is a domain, free over R with basis $1, z$, so $A \subset K := F[z]/(z^2 - f)$ and $K = \operatorname{Frac} A$, $[K : F] = 2$. As $\operatorname{char} k \neq 2$, $\sigma : z \mapsto -z$ is a nontrivial F-automorphism of K, so K/F is Galois with group $\{1, \sigma\}$.

Integrality criterion. Let $\alpha = g + hz \in K$ $(g, h \in F)$. Then $\sigma\alpha = g - hz$, $\alpha + \sigma\alpha = 2g$, $\alpha\sigma\alpha = g^2 - h^2 f$, and α is a root of $X^2 - 2gX + (g^2 - h^2 f) \in F[X]$. Since A is integral over R $(z^2 = f)$, an element of K is integral over A iff it is integral over R. If α is integral over R then so is $\sigma\alpha$ (apply σ to a monic equation with coefficients in $R \subset F$), hence so are $2g = \alpha + \sigma\alpha$ and $g^2 - h^2 f = \alpha\sigma\alpha$; these lie in F, and R is integrally closed, so $2g \in R$ and $g^2 - h^2 f \in R$. Conversely if $2g$, $g^2 - h^2 f \in R$ then α is a root of a monic polynomial over R.

Let α be integral over A. Then $2g \in R$, so $g \in R$ (2 is a unit), hence $h^2 f = g^2 - (g^2 - h^2 f) \in R$. Write $h = u/v$ with $u, v \in R$ coprime. Then $v^2 \mid u^2 f$, hence $v^2 \mid f$. If v had an irreducible factor q, then $q^2 \mid f$, contradicting square-freeness. So v is a unit and $h \in R$, i.e. $\alpha = g + hz \in A$. Hence A is integrally closed in $K = \operatorname{Frac} A$, and, being integral over R with the same fraction field extension, it is the integral closure of R in K (an element of K integral over R is integral over A, hence in A). □

Exercise 6.5

> **Problem.** *Quadric Hypersurfaces.* Let $\operatorname{char} k \neq 2$, and let X be the affine quadric hypersurface $\operatorname{Spec} k[x_0, \ldots, x_n]/(x_0^2 + x_1^2 + \cdots + x_r^2)$ (cf. I, Ex. 5.12). (Starred in the book.)
>
> (a) Show that X is normal if $r \geq 2$ (use Ex. 6.4).
>
> (b) Show by a suitable linear change of coordinates that the equation of X could be written as $x_0 x_1 = x_2^2 + \cdots + x_r^2$. Now imitate the method of (6.5.2) to show that: (1) If $r = 2$, then $\operatorname{Cl} X \cong \mathbf{Z}/2\mathbf{Z}$; (2) If $r = 3$, then $\operatorname{Cl} X \cong \mathbf{Z}$ (use (6.6.1) and Ex. 6.3 above); (3) If $r \geq 4$ then $\operatorname{Cl} X = 0$.

(c) Now let Q be the projective quadric hypersurface in $\mathbf{P}^n$ defined by the same equation. Show that: (1) If $r = 2$, $\operatorname{Cl} Q \cong \mathbf{Z}$, and the class of a hyperplane section $Q.H$ is twice the generator; (2) If $r = 3$, $\operatorname{Cl} Q \cong \mathbf{Z} \oplus \mathbf{Z}$; (3) If $r \geq 4$, $\operatorname{Cl} Q \cong \mathbf{Z}$, generated by $Q.H$.

(d) Prove Klein's theorem, which says that if $r \geq 4$, and if Y is an irreducible subvariety of codimension 1 on Q, then there is an irreducible hypersurface $V \subseteq \mathbf{P}^n$ such that $V \cap Q = Y$, with multiplicity one. In other words, Y is a complete intersection. (First show that for $r \geq 4$, the homogeneous coordinate ring $S(Q) = k[x_0, \ldots, x_n]/(x_0^2 + \cdots + x_r^2)$ is a UFD.)

Solution. *Reading of the statement.* Here $0 \leq r \leq n$. Part (a) holds over any field of characteristic $\neq 2$. Parts (b)–(d) need $\sqrt{-1} \in k$ (for the coordinate change) and, for $r = 3$, the fact $Q_3 \cong \mathbf{P}^1 \times \mathbf{P}^1$ with its classes over an algebraically closed field (6.6.1), and the geometric Ex. 6.2, 6.3 are for k algebraically closed; so from (b) on I assume, as in the neighbouring exercises, that k is algebraically closed of characteristic $\neq 2$. (For $k = \mathbf{R}$ and $r = 2$, for instance, $x_0^2 + x_1^2 + x_2^2$ cannot be brought to the form $x_0 x_1 - x_2^2$.)

(a) Let $r \geq 2$ and $q = x_1^2 + \cdots + x_r^2 \in k[x_1, \ldots, x_n]$. Then $x_0^2 + q = 0$ defines $A = k[x_1, \ldots, x_n, z]/(z^2 - f)$ with $z = x_0$ and $f = -q$. By Ex. 6.4 it suffices to see that f (equivalently q) is nonconstant and square-free. It is nonconstant. Suppose an irreducible a with $a^2 \mid q$, $q = a^2 b$. Then $\partial q/\partial x_i = 2x_i = a(2\partial_i a \cdot b + a\partial_i b)$ for $i = 1, \ldots, r$, so $a \mid x_1$ and $a \mid x_2$ (here $r \geq 2$ and 2 is a unit); the irreducible a would then be associate to both x_1 and x_2, absurd. So q is square-free, A is an integrally closed domain, and X is normal. In particular X satisfies $(*)$.

(b) Since $\sqrt{-1} =: i \in k$, $x_0^2 + x_1^2 = (x_0 + ix_1)(x_0 - ix_1)$ and $x_j^2 = -(ix_j)^2$. The linear change $y_0 = x_0 + ix_1$, $y_1 = x_0 - ix_1$, $y_j = ix_j$ $(2 \leq j \leq r)$, $y_j = x_j$ $(j > r)$ is invertible (determinant $-2i \cdot i^{r-1} \neq 0$, as $\operatorname{char} k \neq 2$), and $x_0^2 + \cdots + x_r^2 = y_0 y_1 - (y_2^2 + \cdots + y_r^2)$ because $x_j^2 = -y_j^2$ for $2 \leq j \leq r$. So after renaming, $X = \operatorname{Spec} A$ with $A = k[x_0, \ldots, x_n]/(x_0 x_1 - q)$, $q = x_2^2 + \cdots + x_r^2$. Let $m = (x_0, \ldots, x_n)$. Since $x_0 x_1 - q \in m^2$, m/m^2 has basis $\bar{x}_0, \ldots, \bar{x}_n$. Observe also

$$A_{x_1} = k[x_0, \ldots, x_n, x_1^{-1}]/(x_0 x_1 - q) \cong k[x_1, x_1^{-1}, x_2, \ldots, x_n] \qquad (x_0 = q/x_1). \qquad (1)$$

This is a Laurent polynomial ring over a polynomial ring, hence a UFD, so $\operatorname{Cl}\operatorname{Spec} A_{x_1} = 0$ by (6.2). Note X is normal by (a) (the change of coordinates is an isomorphism), so (6.2), (6.5), (6.6) apply.

(1) $r = 2$. $X : x_0 x_1 = x_2^2$ in $\mathbf{A}^{n+1}$. Let $Y = V(x_1, x_2)$, $\mathfrak{p} = (x_1, x_2)$; $A/\mathfrak{p} = k[x_0, x_3, \ldots, x_n]$ is a domain of dimension $n - 1 = \dim A - 1$, so Y is a prime divisor. Set-theoretically $V(x_1) = Y$ ($x_1 = 0 \Rightarrow x_2^2 = 0$). Since $x_0 \notin \mathfrak{p}$, x_0 is a unit in $A_\mathfrak{p}$ and $x_1 = x_2^2/x_0$, so $\mathfrak{p}A_\mathfrak{p} = (x_2)$ and $v_Y(x_1) = 2$. So $(x_1) = 2Y$, i.e. $2[Y] = 0$. By (6.5)(c) applied to the irreducible Y and $X - Y = D(x_1) = \operatorname{Spec} A_{x_1}$ we have an exact sequence $\mathbf{Z} \to \operatorname{Cl} X \to \operatorname{Cl}(X - Y) = 0$, so $\operatorname{Cl} X$ is generated by $[Y]$, of order ≤ 2. It remains to see $[Y] \neq 0$. If $Y = (g)$, $g \in K$, then $g \in A_{\mathfrak{p}'}$ for every height-one prime $\mathfrak{p}'$, so $g \in A$ by (6.3A), and every $a \in \mathfrak{p}$ has $a/g \in A_{\mathfrak{p}'}$ for all $\mathfrak{p}'$, so $a \in gA$; also $g \in \mathfrak{p}A_\mathfrak{p} \cap A = \mathfrak{p}$. Hence $\mathfrak{p} = gA$ is principal (this is the argument of (6.2)). But then the image of $\mathfrak{p}$ in m/m^2 would be spanned by the class of g, of dimension ≤ 1, whereas it contains the linearly independent classes $\bar{x}_1, \bar{x}_2$. Contradiction. Hence $\operatorname{Cl} X = \mathbf{Z}/2\mathbf{Z}$, generated by $[Y]$.

(2) $r = 3$. Put $u = x_2 + ix_3$, $v = x_2 - ix_3$; then $q = uv$ and $X : x_0 x_1 = uv$, i.e. $X \cong X_3 \times \mathbf{A}^{n-3}$ where $X_3 = \operatorname{Spec} k[x_0, x_1, u, v]/(x_0 x_1 - uv)$. By (6.6) (iterated; X_3 satisfies $(*)$ being normal by (a)), $\operatorname{Cl} X \cong \operatorname{Cl} X_3$. Now $X_3 = C(Q_3)$ is the affine cone over the quadric surface $Q_3 =$

$V_+(x_0x_1 - uv) \subset \mathbf{P}^3$, which is nonsingular, of dimension 2. By (6.6.1) $Q_3 \cong \mathbf{P}^1 \times \mathbf{P}^1$ and $\operatorname{Cl} Q_3 = \mathbf{Z} \oplus \mathbf{Z}$, and by (6.6.2) a hyperplane section $Q_3.H$ is of type $(1,1)$. Ex. 6.3(b) gives $\operatorname{Cl} X_3 \cong (\mathbf{Z} \oplus \mathbf{Z})/\mathbf{Z}(1,1) \cong \mathbf{Z}$ (via $(a,b) \mapsto a - b$). So $\operatorname{Cl} X \cong \mathbf{Z}$.

(3) $r \geq 4$. Now $\mathfrak{p} = (x_1) \subset A$ has quotient $A/(x_1) = k[x_0, x_2, \ldots, x_n]/(q)$ with $q = x_2^2 + \cdots + x_r^2$ a quadratic form of rank $r - 1 \geq 3$, hence irreducible (a reducible quadratic form is a product of two linear forms, so has rank ≤ 2). So $\mathfrak{p} = (x_1)$ is a principal prime, $Y = V(x_1)$ is a prime divisor, and $(x_1) = Y$ (as $\mathfrak{p}A_{\mathfrak{p}} = x_1 A_{\mathfrak{p}}$, $v_Y(x_1) = 1$, and x_1 is a unit elsewhere), so $[Y] = 0$. By (6.5)(c), $\mathbf{Z} \xrightarrow{1 \mapsto [Y]} \operatorname{Cl} X \to \operatorname{Cl}(X - Y) \to 0$ and $\operatorname{Cl}(X - Y) = \operatorname{Cl} \operatorname{Spec} A_{x_1} = 0$ by (1); hence $\operatorname{Cl} X = 0$.

(c) For $r \geq 2$, $Q \subset \mathbf{P}^n$ is a projective variety of dimension $n - 1 \geq 1$ (irreducible by (a)), normal (its homogeneous coordinate ring is integrally closed by (a), so the local rings of Q, which are localizations of degree-0 parts, are integrally closed), hence nonsingular in codimension one; and $X = \operatorname{Spec} S(Q)$ is the affine cone $C(Q)$. So Ex. 6.3(b) gives an exact sequence

$$0 \to \mathbf{Z} \xrightarrow{1 \mapsto Q.H} \operatorname{Cl} Q \to \operatorname{Cl} X \to 0. \qquad (2)$$

(3) $r \geq 4$: $\operatorname{Cl} X = 0$ by (b)(3), so $\mathbf{Z} \cong \operatorname{Cl} Q$, generated by $Q.H$.

(2) $r = 3$: $\operatorname{Cl} X \cong \mathbf{Z}$ by (b)(2), which is free; so (2) splits and $\operatorname{Cl} Q \cong \mathbf{Z} \oplus \mathbf{Z}$.

(1) $r = 2$: $\bar{Y} := V_+(x_1, x_2) \subset Q$ is a linear space $\mathbf{P}^{n-2}$, an irreducible subvariety of codimension 1 in Q and of degree 1; the cone over it is Y, and the map $\operatorname{Cl} Q \to \operatorname{Cl} X$ of Ex. 6.3(b) sends $[\bar{Y}] \mapsto [Y]$, a generator of $\operatorname{Cl} X$ by (b)(1). By exactness of (2), $\operatorname{Cl} Q$ is generated by $[\bar{Y}]$ and $[Q.H]$. Take $H = V_+(x_1) \not\supseteq Q$. In the chart $x_0 = 1$, Q is $\operatorname{Spec} k[x_2', x_3', \ldots, x_n'] \cong \mathbf{A}^{n-1}$ ($x_1' = x_2'^2$), $\bar{Y}$ is $\{x_2' = 0\}$ and x_1' has valuation 2 along $\bar{Y}$; moreover $V(x_1) \cap Q = \bar{Y}$ as a set. So $Q.H = 2\bar{Y}$ and $\operatorname{Cl} Q$ is cyclic, generated by $[\bar{Y}]$. By Ex. 6.2(d) there is a degree homomorphism $\operatorname{Cl} Q \to \mathbf{Z}$ with $\deg \bar{Y} = 1$; so $\operatorname{Cl} Q \cong \mathbf{Z}$ generated by $[\bar{Y}]$, and $[Q.H] = 2[\bar{Y}]$ is twice the generator. (Consistently, $\deg(Q.H) = \deg Q = 2$.)

(d) Let $r \geq 4$. By (b)(3) $\operatorname{Cl} X = 0$ and by (a) X is normal, so $S := S(Q) = k[x_0, \ldots, x_n]/(x_0^2 + \cdots + x_r^2)$ is a UFD by (6.2). Let $Y \subset Q$ be an irreducible closed subvariety of codimension one. Its ideal $\bar{\mathfrak{p}} \subset S$ is a homogeneous prime with $\dim S/\bar{\mathfrak{p}} = \dim Y + 1 = \dim S - 1$, hence of height one. In a UFD a height-one prime is principal: pick a nonzero homogeneous $a \in \bar{\mathfrak{p}}$ (possible, $\bar{\mathfrak{p}}$ being homogeneous and nonzero); some prime factor ϖ of a lies in $\bar{\mathfrak{p}}$, and $(\varpi) \subseteq \bar{\mathfrak{p}}$ forces $\bar{\mathfrak{p}} = (\varpi)$ by height. The factors of a homogeneous element of a graded domain are homogeneous (compare lowest and highest degree components of a factorization), so $\varpi =: \bar{g}$ is homogeneous, of degree $d \geq 1$.

Lift $\bar{g}$ to a homogeneous $g \in k[x_0, \ldots, x_n]$ of degree d. *g is irreducible:* if $g = g_1 g_2$ with homogeneous g_i of positive degrees $d_i < d$, then $\bar{g} = \bar{g}_1 \bar{g}_2$ and, $\bar{g}$ being prime, $\bar{g} \mid \bar{g}_1$ say; $\bar{g}_1 = \bar{g}\bar{w}$ with $\bar{w}$ homogeneous of degree $d_1 - d < 0$ forces $\bar{g}_1 = 0$, hence $\bar{g} = 0$, absurd. So $V = V_+(g)$ is an irreducible hypersurface; $Q \not\subseteq V$ since $\bar{g} \neq 0$. The ideal (f_Q, g), $f_Q = x_0^2 + \cdots + x_r^2$, has quotient $S/(\bar{g}) = S/\bar{\mathfrak{p}}$, a domain; so it is prime and $V \cap Q = Y$ as sets, with the single component Y. Finally the multiplicity: by Ex. 6.2(c), the coefficient of Y in $Q.V$ is $i(Q, V; Y) = \operatorname{length}_{S'_{\mathfrak{P}}}(S'/(f_Q, g))_{\mathfrak{P}}$ with $S' = k[x_0, \ldots, x_n]$, $\mathfrak{P}$ the preimage of $\bar{\mathfrak{p}}$; this length is that of $S_{\bar{\mathfrak{p}}}/\bar{g}S_{\bar{\mathfrak{p}}} = S_{\bar{\mathfrak{p}}}/\bar{\mathfrak{p}}S_{\bar{\mathfrak{p}}}$, a field, so it is 1. Therefore $Q.V = Y$: $V \cap Q = Y$ with multiplicity one. $\square$

Exercise 6.6

Problem. Let X be the nonsingular plane cubic curve $y^2z = x^3 - xz^2$ of (6.10.2).

(a) Show that three points P, Q, R of X are collinear if and only if $P + Q + R = 0$ in the group law on X. (Note that the point $P_0 = (0, 1, 0)$ is the zero element in the group structure on X.)

(b) A point $P \in X$ has order 2 in the group law on X if and only if the tangent line at P passes through P_0.

(c) A point $P \in X$ has order 3 in the group law on X if and only if P is an inflection point. (An *inflection point* of a plane curve is a nonsingular point P of the curve, whose tangent line (I, Ex. 7.3) has intersection multiplicity ≥ 3 with the curve at P.)

(d) Let $k = \mathbf{C}$. Show that the points of X with coordinates in $\mathbf{Q}$ form a subgroup of the group of all points. Can you determine the structure of this subgroup explicitly?

Solution. *Setting.* k is algebraically closed with $\operatorname{char} k \neq 2$, as in (6.10.2) (for (d), $k = \mathbf{C}$). $X \subset \mathbf{P}^2$ is a nonsingular cubic with $P_0 = (0, 1, 0)$ and group law defined as in (6.10.2): $P \mapsto [P - P_0]$ is a bijection from the closed points to $\ker(\deg : \operatorname{Cl} X \to \mathbf{Z})$, and the sum $S = P + Q$ in the group is the point with $[S - P_0] = [P - P_0] + [Q - P_0]$ in $\operatorname{Cl} X$, i.e. $S \sim P + Q - P_0$. The injectivity of $P \mapsto [P - P_0]$ was shown in (6.10.2): X is not rational, so two distinct points are not linearly equivalent (6.10.1). For a line $L \subset \mathbf{P}^2$ ($L \not\subseteq X$ since X is irreducible), $L.X$ (Ex. 6.2) is an effective divisor of degree $\deg L \cdot \deg X = 3$, whose coefficient at P is the intersection multiplicity $i(X, L; P)$ (Ex. 6.2(c)). "P, Q, R are collinear" means: there is a line L with $L.X = P + Q + R$ (so points are counted with multiplicity; e.g. for $P = Q$ the line is tangent). For a nonsingular point P of X and a line $L \ni P$, $i(X, L; P) \geq 2$ iff L is the tangent line T_P (the order of vanishing at P of the equation of X restricted to L is ≥ 2 iff the gradient of the equation kills the direction of L). All lines are linearly equivalent: $L.X \sim L_0.X$ with $L_0 = \{z = 0\}$, since $(\ell/z).X = L.X - L_0.X$ is principal by Ex. 6.2(b); and $L_0.X = 3P_0$ (put $z = 0$ in the equation: $x^3 = 0$). Hence for every line L,

$$L.X \sim 3P_0. \qquad (3)$$

(This is the relation $P + Q + R \sim 3P_0$ of (6.10.2).) Finally, given $P, Q \in X$, let L be the line PQ if $P \neq Q$ and T_P if $P = Q$; then $L.X - P - Q \geq 0$, say $L.X = P + Q + R'$ (for $P = Q$, $i(X, T_P; P) \geq 2$), which defines the "third point" R'.

(a) If P, Q, R are collinear, $P+Q+R = L.X \sim 3P_0$ by (3), i.e. $(P-P_0)+(Q-P_0)+(R-P_0) \sim 0$, which says $P + Q + R = 0$ in the group. Conversely let $P + Q + R = 0$ in the group, i.e. $(P - P_0) + (Q - P_0) + (R - P_0) \sim 0$, so $P + Q + R \sim 3P_0$. Let L be the line PQ (tangent line if $P = Q$) and $L.X = P + Q + R'$. By (3), $P + Q + R' \sim 3P_0$, so $R \sim R'$, hence $R = R'$ by injectivity of $P \mapsto [P - P_0]$ (two points linearly equivalent are equal). So $L.X = P + Q + R$.

(b) Reading: the statement is about $P \neq P_0$ (P_0 is the zero element, of order 1, though its tangent $z = 0$ passes through P_0). P has order 2 iff $P \neq P_0$ and $2P = 0$, iff (since P_0 is the zero) $P + P + P_0 = 0$, iff by (a) there is a line L with $L.X = 2P + P_0$. If such L exists, $L \ni P, P_0$ and $i(X, L; P) \geq 2$, so $L = T_P$ and $T_P \ni P_0$. Conversely if $P \neq P_0$ and $P_0 \in T_P$ then $T_P.X \geq 2P+P_0$ (the multiplicities at P and at $P_0 \neq P$), and has degree 3, so $T_P.X = 2P + P_0$.

(c) Again P_0 itself satisfies the equivalence in the form "$3P = 0$ iff P is an inflection point" (so "order 3" means $P \neq P_0$). Now $3P = 0$ iff $P+P+P = 0$ iff (a) there is a line L with $L.X = 3P$.

If so, $i(X, L; P) = 3 \geq 2$ so $L = T_P$ and $i(X, T_P; P) = 3$: P is an inflection point. Conversely if $i(X, T_P; P) \geq 3$ then $T_P.X = 3P$ (degree 3).

(d) *Subgroup.* Let $X(\mathbf{Q})$ be the set of points with coordinates in $\mathbf{Q}$ (the equation has rational coefficients). $P_0 \in X(\mathbf{Q})$. Let $P, Q \in X(\mathbf{Q})$. The line $L = PQ$ has rational coefficients; if $P = Q$, T_P is given by the gradient of the defining polynomial at P, again rational. Choose a rational point $Q' \in L$ with $Q' \neq P$ (take $Q' = Q$ if $Q \neq P$; if $P = Q$, $L = T_P$ is a line with rational coefficients and has other rational points). Parametrizing L by $(s, t) \mapsto sP + tQ'$, the cubic form $F|_L$ in (s, t) has rational coefficients and vanishes at the rational root $(1 : 0)$ corresponding to P, to order ≥ 2 if $P = Q$ and $L = T_P$, and also at $(0 : 1)$ if $Q' = Q$; dividing out the rational linear factors corresponding to the known roots P, Q (counted with multiplicity), the remaining linear factor is rational, so the third point R' of (a) is rational. Hence the third point of the line through two rational points is rational. By (a), $-P$ is the third point of the line PP_0 (since $P + P_0 + (-P) = 0$ means $P, P_0, -P$ collinear), so $-P \in X(\mathbf{Q})$; and $P + Q = -R'$ with R' the third point of PQ, so $P + Q \in X(\mathbf{Q})$. Hence $X(\mathbf{Q})$ is a subgroup.

Structure. Claim: $X(\mathbf{Q}) = \{P_0,\ (0, 0, 1),\ (1, 0, 1),\ (-1, 0, 1)\} \cong \mathbf{Z}/2 \times \mathbf{Z}/2$.

The four points are on X ($y = 0$ and $x \in \{0, \pm 1\}$ gives $0 = x^3 - x$), rational. Each $(e, 0, 1)$ has tangent line $x = ez$ (the equation is $F = y^2z - x^3 + xz^2$, $F_y = 2yz = 0$ at such a point, so the tangent $F_x x + F_y y + F_z z = 0$ is a line through $(0, 1, 0) = P_0$), so by (b) each of the three has order 2. A group of order 4 in which three elements have order 2 is $\mathbf{Z}/2 \times \mathbf{Z}/2$. It remains to show there are no other rational points, i.e. that $y^2 = x^3 - x$ has no rational solution with $y \neq 0$ (the only point at infinity is P_0: $z = 0$ forces $x = 0$).

Suppose $(x, y) \in \mathbf{Q}^2$, $y \neq 0$, $y^2 = x^3 - x = x(x - 1)(x + 1) > 0$. Then $x \in (-1, 0) \cup (1, \infty)$. If $x \in (-1, 0)$, then $(x', y') = (-1/x,\ y/x^2)$ is a rational point with $y' \neq 0$ and $x' > 1$ (check: $y'^2 = (x^3 - x)/x^4 = (x^2 - 1)/x^3 = x'^3 - x'$). So it suffices to exclude $x > 1$. Write $x = a/d^2$ with $a, d \in \mathbf{Z}$, $d \geq 1$, $\gcd(a, d) = 1$ (possible: if $x = a/b_0$ in lowest terms, $b_0 \geq 1$, then $x^3 - x = a(a^2 - b_0^2)/b_0^3$ is again in lowest terms, because $\gcd(a, b_0) = 1$; it equals the square y^2 of a rational number, whose reduced denominator is a perfect square; so b_0^3, hence b_0, is a perfect square d^2). Then $(yd^3)^2 = a(a^2 - d^4) = a(a - d^2)(a + d^2) \in \mathbf{Z}$, so $b := yd^3 \in \mathbf{Z}$ and $b^2 = a(a - d^2)(a + d^2)$, with $a > d^2 > 0$ since $x > 1$. Put $B = a - d^2$, $C = a + d^2$. Then $\gcd(a, B) = \gcd(a, C) = \gcd(a, d^2) = 1$ and $\gcd(B, C) \mid \gcd(2a, 2d^2) = 2$.

- If a is even, or d is even (a odd): B, C are odd, so a, B, C are pairwise coprime positive integers with product a square; hence $a = s^2$, $B = t^2$, $C = u^2$. If a even then d is odd (coprime); in both cases $s^2 - d^2 = t^2$, $s^2 + d^2 = u^2$, whence $s^4 - d^4 = (tu)^2$ with s, d, tu positive integers.

- If a and d are both odd: B, C are even and $\gcd(B, C) = 2$. As a is coprime to BC and $a \cdot BC = b^2$, $a = s^2$ and BC is a square. Put $B = 2B'$, $C = 2C'$, $\gcd(B', C') = 1$, $B'C' = BC/4$ a square, so $B' = p^2$, $C' = q^2$ with p, q positive integers. Then $s^2 = a = B' + C' = p^2 + q^2$ and $d^2 = C' - B' = q^2 - p^2$, so $q^4 - p^4 = (sd)^2$ with q, p, sd positive integers.

Both cases contradict Fermat's theorem (by infinite descent) that $X^4 - Y^4 = Z^2$ has no solution in positive integers. Hence $X(\mathbf{Q})$ consists of the four points listed. So $X(\mathbf{Q}) \cong \mathbf{Z}/2 \times \mathbf{Z}/2$: the group of rational points is finite. $\square$

Exercise 6.7

> **Problem.** Let X be the nodal cubic curve $y^2z = x^3 + x^2z$ in $\mathbf{P}^2$. Imitate (6.11.4) and show that the group of Cartier divisors of degree 0, $\mathrm{CaCl}^0 X$, is naturally isomorphic to the multiplicative group $\mathbf{G}_m$. (Starred in the book.)

Solution. *Setting.* k is algebraically closed, $\mathrm{char}\, k \neq 2$ (so that $y^2 = x^2(1+x)$ is really a node). $\mathrm{CaCl}\, X$ is the group of Cartier divisors modulo principal ones, $\mathrm{CaCl}^0 X$ the kernel of the degree $\mathrm{CaCl}\, X \to \mathbf{Z}$ defined as in (6.11.4); I recall and verify that definition below. Let $Z = (0,0,1)$ and $P_0 = (0,1,0)$.

Step 1: the normalization. Define $\varphi : \mathbf{P}^1 \to X$, $(T : S) \mapsto (S(T^2 - S^2) : T(T^2 - S^2) : S^3)$. In the coordinate $t = T/S$ it is $t \mapsto (t^2 - 1,\ t(t^2 - 1))$ on the affine part $z = 1$ (the line $y = tx$ through the node meets the curve again at $x = t^2 - 1$), and $\varphi(\infty) = P_0$. It lands in X: $y^2z = T^2(T^2 - S^2)^2S^3 = x^3 + x^2z$ (both equal $S^3(T^2 - S^2)^2T^2$ after the substitution, using $(T^2 - S^2) + S^2 = T^2$). $\varphi^{-1}(Z) = \{t = \pm 1\}$. The only singular point of X is Z: with $F = y^2z - x^3 - x^2z$, $F_x = -3x^2 - 2xz$, $F_y = 2yz$, $F_z = y^2 - x^2$; $F_y = 0$ means $y = 0$ or $z = 0$; $z = 0$ forces $x = 0$ on X, and then $F_z = y^2 \neq 0$ at P_0; $y = 0$ forces $x^2(x + z) = 0$ and $F_z = -x^2 = 0$, so $x = 0$, the point Z. The map φ is a nonconstant morphism of complete curves, hence finite (6.8), and birational ($t = y/x$). Therefore $\varphi^{-1}(X - Z) \to X - Z$ is a finite birational morphism onto a nonsingular, hence normal, curve, hence an isomorphism. So φ is the normalization, $X - Z \cong \mathbf{P}^1 - \{1, -1\}$, and with the coordinate

$$u = \frac{t+1}{t-1} = \frac{y+x}{y-x}$$

(which sends $t = -1,\ 1,\ \infty$ to $0,\ \infty,\ 1$) we get an isomorphism $u : X - Z \to \mathbf{G}_m = \mathbf{P}^1 - \{0, \infty\}$ with $u(P_0) = 1$.

Step 2: the local ring at Z. Let $A = k[x,y]/(y^2 - x^3 - x^2)$ (the affine part $z = 1$). Via φ, $A \to k[t]$ is $x \mapsto t^2 - 1$, $y \mapsto t(t^2 - 1)$. Its image is $B = \{g \in k[t] : g(1) = g(-1)\}$: indeed with $w = t^2 - 1$ one has $k[t] = k[w] \oplus t\,k[w]$ and $B = k + (t^2 - 1)k[t] = k[w] \oplus tw\,k[w]$, which is generated as an algebra by w and $v = tw$ (note $v^2 = w^2(w + 1)$), i.e. equals the image of A; and $A \to B$ is injective: the kernel of $k[x,y] \to k[t]$ is a prime of height one (the image has dimension one) containing the prime $(y^2 - x^3 - x^2)$ of height one (irreducible, as $x^2(x+1)$ is not a square), hence equal to it. The local ring is $\mathcal{O}_{X,Z} = B_{\mathfrak{m}}$, $\mathfrak{m} = \{g \in B : g(1) = g(-1) = 0\}$. Claim:

$$\mathcal{O}_{X,Z} = \{f \in k(t) : f \text{ regular at } t = \pm 1,\ f(1) = f(-1)\}.$$

"$\subseteq$" is clear. Conversely, take such $f = a/b$ with $a, b \in k[t]$, $b(1)b(-1) \neq 0$ (the semilocal ring of $\mathbf{P}^1$ at $\{\pm 1\}$). Choose a polynomial $\tilde{b}$ of degree ≤ 1 with $\tilde{b}(1) = b(-1)$, $\tilde{b}(-1) = b(1)$ (both nonzero). Then $h = b\tilde{b} \in B$, $h(1) = h(-1) \neq 0$, and $g = f\,h = a\tilde{b} \in k[t]$ has $g(1) = f(1)h(1) = f(-1)h(-1) = g(-1)$, so $g \in B$ and $f = g/h \in B_{\mathfrak{m}}$. Consequently $\mathcal{O}^*_{X,Z} = \{f$ regular at $t = \pm 1,\ f(1) = f(-1) \neq 0\}$, i.e. in the coordinate u: f regular at $u = 0, \infty$ with $f(0) = f(\infty) \neq 0$. (This is just the statement that the node is obtained from $\mathbf{P}^1$ by gluing $t = 1$ and $t = -1$.)

Step 3: Cartier divisors and degree. Let $\mathcal{C}$ be the group of Cartier divisors of X which are trivial in a neighbourhood of Z. Every Cartier divisor D is linearly equivalent to one in $\mathcal{C}$: if $f_Z \in K^*$ is a local equation of D near Z, then $D - (f_Z)$ is given by 1 near Z. A Cartier divisor in $\mathcal{C}$ is

the same as a Cartier divisor on the nonsingular curve $X - Z$ with finite support (its support is a finite set $T \subset X - Z$ and it is trivial on $X - T \ni Z$); by (6.11) $\mathcal{C} \cong \mathrm{Div}(X - Z) = \bigoplus_{P \in X-Z} \mathbf{Z}$. A principal Cartier divisor (f), $f \in K^*$, lies in $\mathcal{C}$ iff f is a unit in a neighbourhood of Z, iff $f \in \mathcal{O}^*_{X,Z}$. Hence

$$\mathrm{CaCl}\, X = \mathrm{Div}(X - Z)/\{(f)|_{X-Z} : f \in \mathcal{O}^*_{X,Z}\}.$$

The degree of $D = \sum n_i P_i \in \mathrm{Div}(X - Z)$ is $\sum n_i$; it vanishes on the principal divisors in this quotient (as in (6.11.4)): for $f \in \mathcal{O}^*_{X,Z}$ viewed as a rational function on $\mathbf{P}^1$, the divisor $(f)_{\mathbf{P}^1}$ has degree 0 (6.10 for $\mathbf{P}^1$) and no support at $t = \pm 1$, so equals $(f)|_{X-Z}$. This is the degree of (6.11.4). Thus $\mathrm{CaCl}^0 X = \mathrm{Div}^0(X - Z)/\{(f)|_{X-Z} : f \in \mathcal{O}^*_{X,Z}\}$.

Step 4: the isomorphism. Transport by u: $\mathrm{Div}(X - Z) = \mathrm{Div}(\mathbf{G}_m)$, points $a \in k^*$ (note $P_0 \leftrightarrow a = 1$). Define

$$\Psi : \mathrm{Div}^0(\mathbf{G}_m) \to \mathbf{G}_m, \qquad \sum n_i[a_i] \mapsto \prod a_i^{n_i} \quad (\textstyle\sum n_i = 0).$$

It is a surjective homomorphism ($[a] - [1] \mapsto a$). We show $\ker \Psi = \{(f)|_{\mathbf{G}_m} : f \in \mathcal{O}^*_{X,Z}\}$.

Let $f \in \mathcal{O}^*_{X,Z}$. Write $f = c\, u^m \prod (u - a_i)^{n_i}$ with $c \in k^*$, $a_i \in k^*$ distinct. Then $v_0(f) = m$ and $v_\infty(f) = -(m + \sum n_i)$; f being a unit at $u = 0$ and $u = \infty$ forces $m = 0$ and $\sum n_i = 0$. So $(f)|_{\mathbf{G}_m} = \sum n_i[a_i] \in \mathrm{Div}^0$, $f(\infty) = c$ and $f(0) = c \prod (-a_i)^{n_i} = c \prod a_i^{n_i}$ (as $\sum n_i = 0$). The condition $f(0) = f(\infty)$ is $\prod a_i^{n_i} = 1$, i.e. $\Psi((f)) = 1$. Conversely if $\Psi(D) = 1$, $D = \sum n_i[a_i]$, then $f = \prod (u - a_i)^{n_i}$ satisfies $f(\infty) = 1 = f(0)$, so $f \in \mathcal{O}^*_{X,Z}$ with $(f)|_{\mathbf{G}_m} = D$. Hence Ψ induces an isomorphism

$$\mathrm{CaCl}^0 X \xrightarrow{\cong} \mathbf{G}_m, \qquad [P - P_0] \mapsto u(P) = \frac{y + x}{y - x}(P) \quad (P \neq Z).$$

It is natural: it is the identification of the group of nonsingular points (the group variety $X - Z \cong \mathbf{G}_m$, with identity P_0) with $\mathrm{CaCl}^0 X$, exactly as in (6.10.2). □

Exercise 6.8

Problem.

(a) Let $f : X \to Y$ be a morphism of schemes. Show that $\mathcal{L} \mapsto f^*\mathcal{L}$ induces a homomorphism of Picard groups, $f^* : \mathrm{Pic}\, Y \to \mathrm{Pic}\, X$.

(b) If f is a finite morphism of nonsingular curves, show that this homomorphism corresponds to the homomorphism $f^* : \mathrm{Cl}\, Y \to \mathrm{Cl}\, X$ defined in the text, via the isomorphisms of (6.16).

(c) If X is a locally factorial integral closed subscheme of $\mathbf{P}^n_k$, and if $f : X \to \mathbf{P}^n$ is the inclusion map, then f^* on Pic agrees with the homomorphism on divisor class groups defined in Ex. 6.2, via the isomorphisms of (6.16).

Solution. (a) If $\mathcal{L}$ is invertible on Y, choose an open cover $\{U_i\}$ with $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$. Then $(f^*\mathcal{L})|_{f^{-1}U_i} = (f|_{f^{-1}U_i})^*(\mathcal{L}|_{U_i}) \cong f^*\mathcal{O}_{U_i} = \mathcal{O}_{f^{-1}U_i}$, so $f^*\mathcal{L}$ is invertible. Isomorphic sheaves have isomorphic pull-backs, $f^*\mathcal{O}_Y = \mathcal{O}_X$, and $f^*(\mathcal{L} \otimes \mathcal{M}) \cong f^*\mathcal{L} \otimes f^*\mathcal{M}$ (Prop. II.5.2(e)). Hence $[\mathcal{L}] \mapsto [f^*\mathcal{L}]$ is a well-defined homomorphism $\mathrm{Pic}\, Y \to \mathrm{Pic}\, X$ (unit and inverse are preserved).

(b) Let $f : X \to Y$ be a finite morphism of nonsingular curves; f is finite, hence has finite fibres, so it is not constant; thus $f(X)$ is dense in Y, f is dominant and $K(Y) \subseteq K(X)$ via f. Both X, Y are integral, separated, noetherian and locally factorial (regular local rings are UFD's, (6.11A)); by (6.11), (6.15), (6.16) we have isomorphisms $\mathrm{Cl} \cong \mathrm{CaCl} \cong \mathrm{Pic}$, $D \mapsto \mathcal{L}(D)$. So we must show $f^*\mathcal{L}(D) \cong \mathcal{L}(f^*D)$ for every divisor D on Y, where f^*D is the divisor of the text.

Let D be the Cartier divisor $\{(U_i, g_i)\}$ on Y, $g_i \in K(Y)^*$. Define the Cartier divisor $f^\# D = \{(f^{-1}U_i, g_i)\}$ on X ($g_i \in K(Y)^* \subseteq K(X)^*$; $g_i/g_j \in \mathcal{O}^*(U_i \cap U_j)$ pulls back to a unit). The sheaf $\mathcal{L}(D) \subseteq \mathcal{K}_Y$ is generated by g_i^{-1} on U_i. The natural map $f^*\mathcal{L}(D) = f^{-1}\mathcal{L}(D) \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X \to \mathcal{K}_X$, $s \otimes a \mapsto sa$ is, on $f^{-1}U_i$, a map from the free module generated by $g_i^{-1} \otimes 1$ to $\mathcal{K}_X$, sending the generator to the nonzero element g_i^{-1}; since $\mathcal{O}_X$ is a sheaf of domains this is injective, with image $\mathcal{O}_{f^{-1}U_i} g_i^{-1} = \mathcal{L}(f^\# D)|_{f^{-1}U_i}$. Hence $f^*\mathcal{L}(D) \cong \mathcal{L}(f^\# D)$.

It remains to see that the Weil divisor of $f^\# D$ is the f^*D of the text. Both sides are additive in D, so it suffices to take $D = Q$ a closed point. Choose an open neighbourhood U of Q in Y and $t \in K(Y)$ a local parameter at Q with t a unit on $U - \{Q\}$ (shrink U), and use the cover $\{U,\ Y - \{Q\}\}$ with local equations t and 1. For $P \in X$ with $f(P) = Q$, the coefficient of P in the Weil divisor of $f^\# Q$ is $v_P(t)$, since $P \in f^{-1}U$ where the local equation is t (coefficient v_P of the local equation, by the definition in the proof of (6.11)); for $f(P) \neq Q$ it is $v_P(1) = 0$. This is $f^*Q = \sum_{f(P)=Q} v_P(t) P$ by definition. Hence the diagram $\mathrm{Cl}\, Y \to \mathrm{Pic}\, Y$, $\mathrm{Cl}\, X \to \mathrm{Pic}\, X$ with vertical maps f^* commutes.

(c) Here k is algebraically closed, X is a closed integral subscheme of $\mathbf{P}^n$ and locally factorial; hence normal, so regular in codimension one, and X is a closed subvariety of $\mathbf{P}^n$ nonsingular in codimension one: the homomorphism $\varphi : \mathrm{Cl}\, \mathbf{P}^n \to \mathrm{Cl}\, X$ of Ex. 6.2 is defined. The scheme $\mathbf{P}^n$ is locally factorial, so $\mathrm{Cl}\, \mathbf{P}^n \cong \mathrm{CaCl}\, \mathbf{P}^n \cong \mathrm{Pic}\, \mathbf{P}^n$ by (6.16) (and (6.17)). Let $D = \sum m_j V_j \in \mathcal{G}$ (Ex. 6.2): no $V_j = V_+(h_j)$ contains X; every class of $\mathrm{Cl}\, \mathbf{P}^n$ has such a representative (Ex. 6.2(b)). With $U_i = D_+(x_i)$, the Cartier divisor of D is $\{(U_i, g_i)\}$, $g_i = \prod_j (h_j / x_i^{d_j})^{m_j}$, $d_j = \deg h_j$. Since no h_j vanishes identically on X, restricting gives the Cartier divisor $f^\# D = \{(X \cap U_i, \bar{g}_i)\}$ on X (with $\bar{g}_i \in K(X)^*$; when $X \cap U_i = \emptyset$ the i-th term is omitted). The sheaf $\mathcal{L}(D)$ is invertible with transition functions $u_{ij} = g_j/g_i \in \mathcal{O}^*(U_i \cap U_j)$ with respect to the generators g_i^{-1}, and $f^*\mathcal{L}(D)$ is the invertible sheaf with the restricted transition functions $\bar{u}_{ij} = \bar{g}_j/\bar{g}_i$ with respect to the generators $g_i^{-1} \otimes 1$; these are also the transition functions of $\mathcal{L}(f^\# D)$ with respect to $\bar{g}_i^{-1}$. Hence $f^*\mathcal{L}(D) \cong \mathcal{L}(f^\# D)$ (isomorphic invertible sheaves, as they are trivialized on the same cover with the same cocycle). Finally, the Weil divisor of $f^\# D$ has, for a prime divisor Y of X, coefficient $v_Y(\bar{g}_i)$ (i with $\bar{x}_i \notin \mathfrak{p}_Y$), which by (1) of Ex. 6.2(a) and linearity is $\sum_j m_j \operatorname{coeff}_Y(V_j.X) = \operatorname{coeff}_Y(D.X)$. So the Weil divisor of $f^\# D$ is $D.X$, and $f^*\mathcal{L}(D) \cong \mathcal{L}(D.X)$: under $\mathrm{Cl} \cong \mathrm{Pic}$, f^* corresponds to φ. □

Exercise 6.9

Problem. *Singular Curves.* Here we give another method of calculating the Picard group of a singular curve. Let X be a projective curve over k, let $\tilde{X}$ be its normalization, let $\pi : \tilde{X} \to X$ be the projection map (Ex. 3.8). For each point $P \in X$, let $\mathcal{O}_P$ be its local ring, and let $\tilde{\mathcal{O}}_P$ be the integral closure of $\mathcal{O}_P$. We use $*$ to denote the group of units in a ring. (Starred in the book.)

(a) Show there is an exact sequence

$$0 \to \bigoplus_{P \in X} \tilde{\mathcal{O}}_P^*/\mathcal{O}_P^* \to \operatorname{Pic} X \xrightarrow{\pi^*} \operatorname{Pic} \tilde{X} \to 0.$$

[*Hint*: Represent $\operatorname{Pic} X$ and $\operatorname{Pic} \tilde{X}$ as the groups of Cartier divisors modulo principal divisors, and use the exact sequence of sheaves $0 \to \pi_* \tilde{\mathcal{O}}_{\tilde{X}}^*/\mathcal{O}_X^* \to \mathcal{K}^*/\mathcal{O}_X^* \to \mathcal{K}^*/\pi_* \tilde{\mathcal{O}}_{\tilde{X}}^* \to 0$.]

(b) Use (a) to give another proof of the fact that if X is a plane cuspidal cubic curve, then there is an exact sequence $0 \to \mathbf{G}_a \to \operatorname{Pic} X \to \mathbf{Z} \to 0$, and if X is a plane nodal cubic curve, there is an exact sequence $0 \to \mathbf{G}_m \to \operatorname{Pic} X \to \mathbf{Z} \to 0$.

Solution. *Setting.* k is algebraically closed, a curve is integral, $K = K(X) = K(\tilde{X})$, $\tilde{X}$ is a nonsingular projective curve and π is finite and birational (Ex. 3.8). For $P \in X$, $\tilde{\mathcal{O}}_P = (\pi_* \mathcal{O}_{\tilde{X}})_P = \bigcap_{Q \in \pi^{-1}(P)} \mathcal{O}_{\tilde{X},Q}$, a semilocal Dedekind domain, hence a PID. Since π is an isomorphism over the (dense, open) nonsingular locus, the set Σ of P with $\mathcal{O}_P \neq \tilde{\mathcal{O}}_P$ is finite; so the sum in (a) is finite. As X and $\tilde{X}$ are integral, (6.13)–(6.15) identify Pic with CaCl = Cart / Prin, where $\operatorname{Cart} X = \Gamma(X, \mathcal{K}^*/\mathcal{O}_X^*)$ and Prin is the image of K^*.

(a) Define $\pi^\# : \operatorname{Cart} X \to \operatorname{Cart} \tilde{X}$ by $\{(U_i, f_i)\} \mapsto \{(\pi^{-1} U_i, f_i)\}$ (well defined since units pull back to units; $\pi^\#$ is induced by $\mathcal{K}^*/\mathcal{O}_X^* \to \pi_*(\mathcal{K}^*/\mathcal{O}_{\tilde{X}}^*)$). It maps principal to principal, $\pi^\#(g) = (g)$, and, as in Ex. 6.8(b) (π is dominant), $\mathcal{L}(\pi^\# D) \cong \pi^* \mathcal{L}(D)$, so on Pic = Cart / Prin the map induced by $\pi^\#$ is π^*. I show:

$$0 \to N \to \operatorname{Cart} X \xrightarrow{\pi^\#} \operatorname{Cart} \tilde{X} \to 0 \qquad \text{is exact, with } N \cong \bigoplus_P \tilde{\mathcal{O}}_P^*/\mathcal{O}_P^*. \tag{4}$$

(Recall that this is just the long exact sequence of the hint, with $\mathcal{N} = \pi_* \tilde{\mathcal{O}}_{\tilde{X}}^*/\mathcal{O}_X^*$ a skyscraper sheaf supported on Σ, so flasque; I give the elementary argument.)

Kernel. A Cartier divisor D is determined by its germs $f_P \in K^*/\mathcal{O}_P^*$, and $\pi^\# D = 0$ iff $f_P \in \mathcal{O}_{\tilde{X},Q}^*$ for all $Q \mapsto P$, i.e. $f_P \in \bigcap_Q \mathcal{O}_{\tilde{X},Q}^* = \tilde{\mathcal{O}}_P^*$ (if $f, f^{-1} \in \bigcap \mathcal{O}_Q = \tilde{\mathcal{O}}_P$). So $D \mapsto (f_P \bmod \mathcal{O}_P^*)_P$ is an injection of $N := \ker \pi^\#$ into $\bigoplus_{P \in \Sigma} \tilde{\mathcal{O}}_P^*/\mathcal{O}_P^*$ (the germs vanish at $P \notin \Sigma$). It is onto: given $u_P \in \tilde{\mathcal{O}}_P^*$ ($P \in \Sigma$), $u_P^{\pm 1}$ are regular on a neighbourhood of $\pi^{-1}(P)$, hence u_P is a unit of $\tilde{\mathcal{O}}_{P'}$ for P' in an open neighbourhood W_P of P which we may take with $W_P \cap \Sigma = \{P\}$. The cover $\{W_P\}_{P \in \Sigma} \cup \{W_0 = X - \Sigma\}$ with local equations u_P and 1 is a Cartier divisor: on overlaps we have only points $P' \notin \Sigma$, where $\mathcal{O}_{P'} = \tilde{\mathcal{O}}_{P'}$ and u_P is a unit. It has germs u_P at P.

Surjectivity of $\pi^\#$. By (6.11) ($\tilde{X}$ nonsingular), $\operatorname{Cart} \tilde{X} = \operatorname{Div} \tilde{X}$. Let $\tilde{D} = \sum n_Q Q \in \operatorname{Div} \tilde{X}$. The finite set $T_\Sigma = \pi^{-1}(\Sigma)$ lies in an affine open $\operatorname{Spec} C$ of $\tilde{X}$ (complement of a hyperplane section of the projective curve $\tilde{X}$ missing T_Σ). Let $B = \bigcap_{Q \in T_\Sigma} \mathcal{O}_{\tilde{X},Q}$, the localization of C at the complement of $\bigcup_Q \mathfrak{m}_Q \cap C$; it is a semilocal Dedekind domain, hence a PID, with maximal ideals $\mathfrak{m}_Q B$ and $B_{\mathfrak{m}_Q B} = \mathcal{O}_{\tilde{X},Q}$. The fractional ideal $\prod_Q (\mathfrak{m}_Q B)^{n_Q}$ is principal, $= gB$ with $g \in K^*$, and $v_Q(g) = n_Q$ for all $Q \in T_\Sigma$ (compare in each localization $B_{\mathfrak{m}_Q B}$). Then $\tilde{D}' = \tilde{D} - (g)$ has no component over Σ; its (finite) support T lies in $\pi^{-1}(X - \Sigma) \cong X - \Sigma$. Let D' be the Cartier divisor on X which on the open set $X - \Sigma \cong \pi^{-1}(X - \Sigma)$ is $\tilde{D}'$, and which is 0 on the open set $X - \pi(T) \supseteq \Sigma$; they agree on the overlap. Then $\pi^\# D' = \tilde{D}'$, so $\tilde{D} = \pi^\# D' + (g) = \pi^\#(D' + (g))$.

Passing to Pic. We now pass to the quotient by principal divisors. The map $\operatorname{Pic} X \to \operatorname{Pic} \tilde{X}$ is surjective by (4). If $\pi^\# D = (g)$ is principal, $g \in K^*$, then $D - (g) \in N$, so $\ker(\operatorname{Pic} X \to \operatorname{Pic} \tilde{X})$

is the image of N. Finally $N \to \operatorname{Pic} X$ is injective: if $n \in N$ is principal, $n = (g)$ in $\operatorname{Cart} X$, then $0 = \pi^{\#} n = (g)$ in $\operatorname{Cart} \tilde{X}$, so $g \in \Gamma(\tilde{X}, \mathcal{O}^*) = k^*$ ($\tilde{X}$ is a complete integral variety over algebraically closed k), and then $(g) = 0$ in $\operatorname{Cart} X$, i.e. $n = 0$. This proves the exact sequence of (a).

(b) Both curves have exactly one singular point Z, and $\tilde{X} = \mathbf{P}^1$, so $\operatorname{Pic} \tilde{X} = \mathbf{Z}$ (6.17) and the sum in (a) reduces to $\tilde{\mathcal{O}}_Z^*/\mathcal{O}_Z^*$. So it suffices to show that this group is $\mathbf{G}_a$ (cusp), resp. $\mathbf{G}_m$ (node). We use the normal forms $y^2z = x^3$ and $y^2z = x^3 + x^2z$ of (6.11.4) and Ex. 6.7 (the cuspidal, resp. nodal, cubic of the statement).

Node ($y^2z = x^3 + x^2z$, $\operatorname{char} k \neq 2$): by Ex. 6.7, $\tilde{\mathcal{O}}_Z = k[t]_S$ (S = polynomials not vanishing at ± 1), and $\mathcal{O}_Z = \{f \in \tilde{\mathcal{O}}_Z : f(1) = f(-1)\}$. The map $\tilde{\mathcal{O}}_Z^* \to k^*$, $f \mapsto f(1)/f(-1)$, is a homomorphism, surjective (for $c \in k^*$ the polynomial $f = \frac{c+1}{2} + \frac{c-1}{2}t$ has $f(1) = c$, $f(-1) = 1$), with kernel $\{f : f(1) = f(-1)\} \cap \tilde{\mathcal{O}}_Z^* = \mathcal{O}_Z^*$ (if $f \in \mathcal{O}_Z$ is a unit of $\tilde{\mathcal{O}}_Z$, then $f(\pm 1) \neq 0$ and $1/f \in \mathcal{O}_Z$). So $\tilde{\mathcal{O}}_Z^*/\mathcal{O}_Z^* \cong \mathbf{G}_m$.

Cusp ($y^2z = x^3$): the parametrization $t \mapsto (t^2, t^3)$, i.e. $\varphi(T : S) = (T^2S : T^3 : S^3)$ ($\varphi(\infty) = (0 : 1 : 0) = P_0$), is the normalization $\mathbf{P}^1 \to X$ (finite, birational with $t = y/x$; over $x \neq 0$ it is an iso, as $k[t^2, t^3][t^{-2}] = k[t, t^{-1}]$, and near P_0, in the chart $y = 1$ the curve is $z = x^3$, nonsingular); the singular point is $Z = (0, 0, 1)$ with the single preimage $t = 0$. Here $A = k[x, y]/(y^2 - x^3) \cong k[t^2, t^3] = \{g \in k[t] : g'(0) = 0\}$ (the coefficient of t vanishes), and $\tilde{\mathcal{O}}_Z = k[t]_{(t)}$, $\mathcal{O}_Z = \{f \in \tilde{\mathcal{O}}_Z : f'(0) = 0\}$, where $f'(0)$ is the coefficient of t in the expansion of f at 0. ("$\subseteq$" is clear; conversely if $f = a/b$ with $a, b \in k[t]$, $b(0) \neq 0$, $f'(0) = 0$, put $\tilde{b} = b(0) - b'(0)t$; then $b\tilde{b}$ and $a\tilde{b}$ have no t-term: the coefficient is $b(0)b'(0) - b'(0)b(0) = 0$, resp. $a'(0)b(0) - a(0)b'(0) = b(0)^2 f'(0) = 0$. So $f = (a\tilde{b})/(b\tilde{b})$ with numerator and denominator in A, $b\tilde{b}(0) = b(0)^2 \neq 0$.) The map $\tilde{\mathcal{O}}_Z^* \to (k, +)$, $f \mapsto f'(0)/f(0)$ is a homomorphism (logarithmic derivative), surjective ($f = 1 + ct \mapsto c$), with kernel $\{f : f'(0) = 0\} \cap \tilde{\mathcal{O}}_Z^* = \mathcal{O}_Z^*$ (if $f \in \mathcal{O}_Z$ is a unit of $\tilde{\mathcal{O}}_Z$ then $(1/f)'(0) = -f'(0)/f(0)^2 = 0$). So $\tilde{\mathcal{O}}_Z^*/\mathcal{O}_Z^* \cong \mathbf{G}_a$.

In both cases (a) gives the stated exact sequences $0 \to \mathbf{G}_a \to \operatorname{Pic} X \to \mathbf{Z} \to 0$, $0 \to \mathbf{G}_m \to \operatorname{Pic} X \to \mathbf{Z} \to 0$. □

Exercise 6.10

Problem. *The Grothendieck Group $K(X)$.* Let X be a noetherian scheme. We define $K(X)$ to be the quotient of the free abelian group generated by all the coherent sheaves on X, by the subgroup generated by all expressions $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$, whenever there is an exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ of coherent sheaves on X. If $\mathcal{F}$ is a coherent sheaf, we denote by $\gamma(\mathcal{F})$ its image in $K(X)$.

(a) If $X = \mathbf{A}_k^1$, then $K(X) \cong \mathbf{Z}$.

(b) If X is any integral scheme, define the *rank* of a coherent sheaf $\mathcal{F}$ to be $\dim_K \mathcal{F}_\xi$, where ξ is the generic point of X, and $K = \mathcal{O}_\xi$ is the function field of X. Show that the rank function defines a surjective homomorphism $\operatorname{rank} : K(X) \to \mathbf{Z}$.

(c) If Y is a closed subscheme of X, there is an exact sequence $K(Y) \to K(X) \to K(X - Y) \to 0$, where the first map is extension by zero, and the second is restriction. [*Hint*: For exactness in the middle, show that if $\mathcal{F}$ is a coherent sheaf on X, whose support is contained in Y, then there is a finite filtration $\mathcal{F} = \mathcal{F}_0 \supseteq \mathcal{F}_1 \supseteq \cdots \supseteq \mathcal{F}_n = 0$ such that each $\mathcal{F}_i/\mathcal{F}_{i+1}$ is an $\mathcal{O}_Y$-module. To show surjectivity on the right use (Ex. 5.15).]

Solution. (b) The stalk functor $\mathcal{F} \mapsto \mathcal{F}_\xi$ is exact, and $\mathcal{F}_\xi$ is a finite-dimensional K-vector space ($\mathcal{F}$ coherent), so $\dim_K$ is additive on short exact sequences. Hence rank defined on generators $\mathcal{F}$ kills the relations $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$ and descends to a homomorphism $K(X) \to \mathbf{Z}$. It is surjective since $\operatorname{rank} \mathcal{O}_X = 1$.

(a) Let $X = \operatorname{Spec} A$, $A = k[t]$, a PID. Coherent sheaves on X are the $\widetilde{M}$ with M a finitely generated A-module, and exact sequences of such sheaves correspond to exact sequences of modules (II.5.4, 5.5). By the structure theorem, $M \cong A^r \oplus \bigoplus_i A/(f_i)$ with $f_i \neq 0$. For $f \neq 0$, the exact sequence $0 \to A \xrightarrow{f} A \to A/(f) \to 0$ gives $\gamma(\widetilde{A/(f)}) = 0$ in $K(X)$, and $\gamma(\widetilde{M}) = \sum \gamma(\text{summands})$ by the exact sequences $0 \to M' \to M' \oplus M'' \to M'' \to 0$. So $\gamma(\widetilde{M}) = r\gamma(\mathcal{O}_X)$, $K(X)$ is generated by $\gamma(\mathcal{O}_X)$, and by (b) rank : $K(X) \to \mathbf{Z}$ is onto with $\operatorname{rank} \gamma(\mathcal{O}_X) = 1$. Thus $\mathbf{Z} \to K(X)$, $n \mapsto n\gamma(\mathcal{O}_X)$ is surjective and splits by rank; hence it is an isomorphism and $K(X) \cong \mathbf{Z}$.

(c) Let $i : Y \to X$ be the closed immersion, $\mathcal{J}$ its ideal sheaf, $U = X - Y$, $j : U \to X$.

The maps. For $\mathcal{G}$ coherent on Y, $i_*\mathcal{G}$ is coherent on X (on an affine open $\operatorname{Spec} B$ of X with $Y \cap \operatorname{Spec} B = \operatorname{Spec} B/I$, $i_*\widetilde{N} = \widetilde{N}$ with N a finitely generated B/I-module, hence a finitely generated B-module), and i_* is exact, so $i_* : K(Y) \to K(X)$ is a homomorphism; restriction to U is exact on coherent sheaves, so $\rho : K(X) \to K(U)$ is a homomorphism; $\rho \circ i_* = 0$ since $(i_*\mathcal{G})|_U = 0$.

Surjectivity of ρ. By Ex. 5.15, every coherent $\mathcal{G}$ on U extends to a coherent $\bar{\mathcal{G}}$ on X with $\bar{\mathcal{G}}|_U \cong \mathcal{G}$; so $\gamma(\mathcal{G}) = \rho(\gamma(\bar{\mathcal{G}}))$, and these generate $K(U)$.

Supports in Y. Let $\mathcal{F}$ be coherent with $\operatorname{Supp} \mathcal{F} \subseteq Y$. Then $\mathcal{J}^N\mathcal{F} = 0$ for some N: on an affine $\operatorname{Spec} B$, $\mathcal{F} = \widetilde{M}$ with M finitely generated and $V(\operatorname{Ann} M) = \operatorname{Supp} M \subseteq V(I)$, $I = \mathcal{J}(B)$ finitely generated, so $I \subseteq \sqrt{\operatorname{Ann} M}$ and $I^N \subseteq \operatorname{Ann} M$; take the maximum over a finite affine cover. The filtration $\mathcal{F} \supseteq \mathcal{J}\mathcal{F} \supseteq \cdots \supseteq \mathcal{J}^N\mathcal{F} = 0$ has coherent quotients $\mathcal{F}_m/\mathcal{F}_{m+1}$ annihilated by $\mathcal{J}$, i.e. of the form $i_*\mathcal{G}_m$ with $\mathcal{G}_m$ coherent on Y (a coherent $\mathcal{O}_X$-module $\mathcal{M}$ with $\mathcal{J}\mathcal{M} = 0$ is locally $\widetilde{M}$ with M a finitely generated module over B/I, and by II.5.2 this is the direct image of $\widetilde{M}$ on $\operatorname{Spec} B/I$; so $\mathcal{M} = i_*\mathcal{G}$ with $\mathcal{G} = i^{-1}\mathcal{M}$ coherent on Y). Hence $\gamma(\mathcal{F}) = \sum_m i_*\gamma(\mathcal{G}_m) \in \operatorname{im}(i_*)$.

Exactness in the middle. Let $Q = K(X)/\operatorname{im}(i_*)$. We construct $\theta : K(U) \to Q$ with $\theta \circ \rho =$ the quotient map $K(X) \to Q$; then $\rho(x) = 0 \Rightarrow \bar{x} = \theta(\rho x) = 0$, i.e. $\ker \rho \subseteq \operatorname{im}(i_*)$, as required.

(i) *Two extensions agree in Q.* Let $\mathcal{F}_1, \mathcal{F}_2$ be coherent on X and $\alpha : \mathcal{F}_1|_U \xrightarrow{\sim} \mathcal{F}_2|_U$. The graph $\Gamma \subseteq (\mathcal{F}_1 \oplus \mathcal{F}_2)|_U$ of α is a coherent subsheaf, so by Ex. 5.15 there is a coherent subsheaf $\bar{\Gamma} \subseteq \mathcal{F}_1 \oplus \mathcal{F}_2$ with $\bar{\Gamma}|_U = \Gamma$. The projections $p_\nu : \bar{\Gamma} \to \mathcal{F}_\nu$ restrict to isomorphisms over U (they restrict to $\Gamma \to \mathcal{F}_\nu|_U$, which are isomorphisms). So $\ker p_\nu$ and $\operatorname{coker} p_\nu$ are coherent with support in Y, hence in $\operatorname{im}(i_*)$ by the above. From $0 \to \ker p_\nu \to \bar{\Gamma} \to \operatorname{im} p_\nu \to 0$ and $0 \to \operatorname{im} p_\nu \to \mathcal{F}_\nu \to \operatorname{coker} p_\nu \to 0$ we get $\gamma(\bar{\Gamma}) - \gamma(\mathcal{F}_\nu) = \gamma(\ker p_\nu) - \gamma(\operatorname{coker} p_\nu) \in \operatorname{im}(i_*)$ for $\nu = 1, 2$. So $\gamma(\mathcal{F}_1) \equiv \gamma(\mathcal{F}_2)$ in Q.

(ii) For $\mathcal{G}$ coherent on U put $\theta(\gamma(\mathcal{G})) = \overline{\gamma(\bar{\mathcal{G}})}$ for any coherent extension $\bar{\mathcal{G}}$ (exists by Ex. 5.15; independent of the choice by (i)); this defines θ on the free abelian group on coherent sheaves of U. It kills the relations: for $0 \to \mathcal{G}' \to \mathcal{G} \to \mathcal{G}'' \to 0$ on U, extend $\mathcal{G}$ to $\bar{\mathcal{G}}$, then (Ex. 5.15(d) applied to $\mathcal{G}' \subseteq \bar{\mathcal{G}}|_U$) choose a coherent $\bar{\mathcal{G}}' \subseteq \bar{\mathcal{G}}$ with $\bar{\mathcal{G}}'|_U = \mathcal{G}'$, and put $\bar{\mathcal{G}}'' = \bar{\mathcal{G}}/\bar{\mathcal{G}}'$, a coherent extension of $\mathcal{G}''$. From $0 \to \bar{\mathcal{G}}' \to \bar{\mathcal{G}} \to \bar{\mathcal{G}}'' \to 0$ we get $\theta\gamma(\mathcal{G}) = \theta\gamma(\mathcal{G}') + \theta\gamma(\mathcal{G}'')$. So $\theta : K(U) \to Q$ is a homomorphism, and $\theta(\rho\gamma(\mathcal{F})) = \theta(\gamma(\mathcal{F}|_U)) = \overline{\gamma(\mathcal{F})}$ (take $\bar{\mathcal{G}} = \mathcal{F}$), so $\theta \circ \rho$ is the quotient

map. Hence $K(Y) \to K(X) \to K(X-Y) \to 0$ is exact. $\square$

Exercise 6.11

Problem. *The Grothendieck Group of a Nonsingular Curve.* Let X be a nonsingular curve over an algebraically closed field k. We will show that $K(X) \cong \operatorname{Pic} X \oplus \mathbf{Z}$, in several steps. (Starred in the book.)

(a) For any divisor $D = \sum n_i P_i$ on X, let $\psi(D) = \sum n_i \gamma(k(P_i)) \in K(X)$, where $k(P_i)$ is the skyscraper sheaf k at P_i and 0 elsewhere. If D is an effective divisor, let $\mathcal{O}_D$ be the structure sheaf of the associated subscheme of codimension 1, and show that $\psi(D) = \gamma(\mathcal{O}_D)$. Then use (6.18) to show that for any D, $\psi(D)$ depends only on the linear equivalence class of D, so ψ defines a homomorphism $\psi : \operatorname{Cl} X \to K(X)$.

(b) For any coherent sheaf $\mathcal{F}$ on X, show that there exist locally free sheaves $\mathcal{E}_0, \mathcal{E}_1$ and an exact sequence $0 \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathcal{F} \to 0$. Let $r_0 = \operatorname{rank} \mathcal{E}_0$, $r_1 = \operatorname{rank} \mathcal{E}_1$, and define $\det \mathcal{F} = (\bigwedge^{r_0} \mathcal{E}_0) \otimes (\bigwedge^{r_1} \mathcal{E}_1)^{-1} \in \operatorname{Pic} X$. Here $\bigwedge$ denotes the exterior power (Ex. 5.16). Show that $\det \mathcal{F}$ is independent of the resolution chosen, and that it gives a homomorphism $\det : K(X) \to \operatorname{Pic} X$. Finally show that if D is a divisor, then $\det(\psi(D)) = \mathcal{L}(D)$.

(c) If $\mathcal{F}$ is any coherent sheaf of rank r, show that there is a divisor D on X and an exact sequence $0 \to \mathcal{L}(D)^{\oplus r} \to \mathcal{F} \to \mathcal{T} \to 0$, where $\mathcal{T}$ is a torsion sheaf. Conclude that if $\mathcal{F}$ is a coherent sheaf of rank r, then $\gamma(\mathcal{F}) - r\gamma(\mathcal{O}_X) \in \operatorname{Im} \psi$.

(d) Using the maps ψ, det, rank, and $1 \mapsto \gamma(\mathcal{O}_X)$ from $\mathbf{Z} \to K(X)$, show that $K(X) \cong \operatorname{Pic} X \oplus \mathbf{Z}$.

Solution. *Preliminaries.* X is integral, noetherian, separated, and all local rings $\mathcal{O}_P$ (P closed) are DVR's, so X is locally factorial and (6.16) gives $\operatorname{Cl} X \cong \operatorname{Pic} X$, $D \mapsto \mathcal{L}(D)$. Prime divisors are closed points. X is quasi-projective (it is an open subscheme of a complete nonsingular, hence projective, curve (I.6.7, II.6.7)). A coherent sheaf $\mathcal{F}$ is a *torsion sheaf* if $\mathcal{F}_\xi = 0$, equivalently its support is a finite set of closed points; then $\mathcal{F} = \bigoplus_P \mathcal{F}_P$ (skyscraper) and each $\mathcal{F}_P$ is a module of finite length ℓ_P over $\mathcal{O}_P$.

(a) Let $D = \sum n_i P_i$ be effective; list its points with multiplicity as $Q_1, \dots, Q_m$ and put $D_j = Q_1 + \cdots + Q_j$, so $D_0 = 0$, $D_m = D$, $D_{j+1} = D_j + Q_{j+1}$. By (6.18), $\mathcal{I}_{D_j} = \mathcal{L}(-D_j) \subseteq \mathcal{O}_X$ is the ideal sheaf of the subscheme D_j, so $\mathcal{O}_{D_j} = \mathcal{O}_X / \mathcal{L}(-D_j)$. Put $P = Q_{j+1}$. Since $\mathcal{L}(-D_{j+1}) = \mathcal{L}(-D_j) \otimes \mathcal{L}(-P) = \mathcal{L}(-D_j) \cdot \mathcal{I}_P$ (6.13(b), (6.18)), the kernel of the surjection $\mathcal{O}_{D_{j+1}} \to \mathcal{O}_{D_j}$ is $\mathcal{L}(-D_j)/\mathcal{L}(-D_j)\mathcal{I}_P \cong \mathcal{L}(-D_j) \otimes k(P) \cong k(P)$ (an invertible sheaf is trivial near P). So $\gamma(\mathcal{O}_{D_{j+1}}) = \gamma(\mathcal{O}_{D_j}) + \gamma(k(Q_{j+1}))$, and by induction $\gamma(\mathcal{O}_D) = \sum_j \gamma(k(Q_j)) = \sum n_i \gamma(k(P_i)) = \psi(D)$.

Let D, D' be effective with $D \sim D'$. By (6.13)(c), $\mathcal{L}(-D) \cong \mathcal{L}(-D')$ as abstract sheaves. From $0 \to \mathcal{L}(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$ we get $\psi(D) = \gamma(\mathcal{O}_D) = \gamma(\mathcal{O}_X) - \gamma(\mathcal{L}(-D))$, which depends only on the isomorphism class of $\mathcal{L}(-D)$; so $\psi(D) = \psi(D')$. For arbitrary $D \sim D'$, $D - D' = (f) = E_1 - E_2$ with E_1, E_2 effective (zeros and poles of f), and $E_1 \sim E_2$, so $\psi(E_1) = \psi(E_2)$; since ψ is additive on $\operatorname{Div} X$ by definition, $\psi(D) - \psi(D') = \psi(E_1) - \psi(E_2) = 0$. So $\psi : \operatorname{Cl} X \to K(X)$ is a well-defined homomorphism.

(b) *Existence of a resolution.* Embed X as an open subscheme of a projective curve $\bar{X} \subseteq \mathbf{P}^N$. By Ex. 5.15, $\mathcal{F}$ extends to a coherent sheaf $\bar{\mathcal{F}}$ on $\bar{X}$; by (5.18), $\bar{\mathcal{F}}(m)$ is generated by finitely

many global sections for $m \gg 0$, i.e. there is a surjection $\mathcal{O}_{\bar{X}}(-m)^{\oplus s} \to \bar{\mathcal{F}}$; restricting to X gives a surjection $\mathcal{E}_0 \to \mathcal{F}$ with $\mathcal{E}_0$ locally free of finite rank. Let $\mathcal{E}_1$ be its kernel; it is coherent, and each stalk is a finitely generated submodule of a free module over a DVR (or a field), hence free; so $\mathcal{E}_1$ is locally free.

We use Ex. 5.16(d): for an exact sequence $0 \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to 0$ of locally free sheaves, $\det \mathcal{B} := \bigwedge^{\mathrm{rk}} \mathcal{B} \cong \det \mathcal{A} \otimes \det \mathcal{C}$ (write $\det \mathcal{E}$ for the top exterior power of a locally free $\mathcal{E}$). Thus $\det \mathcal{F} = \det \mathcal{E}_0 \otimes (\det \mathcal{E}_1)^{-1}$.

Independence. Let $0 \to \mathcal{E}_1 \to \mathcal{E}_0 \xrightarrow{\epsilon} \mathcal{F} \to 0$ and $0 \to \mathcal{E}_1' \to \mathcal{E}_0' \xrightarrow{\epsilon'} \mathcal{F} \to 0$ be two resolutions. Let $\mathcal{E}_0'' = \ker(\mathcal{E}_0 \oplus \mathcal{E}_0' \xrightarrow{\epsilon - \epsilon'} \mathcal{F})$, locally free (a subsheaf of a locally free sheaf on X, by the argument above). The projection $\mathcal{E}_0'' \to \mathcal{E}_0$ is surjective (given a local section a of $\mathcal{E}_0$, since ϵ' is onto locally there is b with $\epsilon'(b) = \epsilon(a)$) with kernel $\{(0, b) : \epsilon'(b) = 0\} \cong \mathcal{E}_1'$; symmetrically $\mathcal{E}_0'' \to \mathcal{E}_0'$ is onto with kernel $\mathcal{E}_1$. So there are exact sequences of locally free sheaves $0 \to \mathcal{E}_1' \to \mathcal{E}_0'' \to \mathcal{E}_0 \to 0$ and $0 \to \mathcal{E}_1 \to \mathcal{E}_0'' \to \mathcal{E}_0' \to 0$, whence $\det \mathcal{E}_0'' \cong \det \mathcal{E}_1' \otimes \det \mathcal{E}_0 \cong \det \mathcal{E}_1 \otimes \det \mathcal{E}_0'$. Rearranging in the group $\operatorname{Pic} X$: $\det \mathcal{E}_0 \otimes (\det \mathcal{E}_1)^{-1} \cong \det \mathcal{E}_0' \otimes (\det \mathcal{E}_1')^{-1}$. So $\det \mathcal{F} \in \operatorname{Pic} X$ is well defined.

Additivity. Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be exact. Take a resolution $0 \to \mathcal{E}_1 \to \mathcal{E}_0 \xrightarrow{\epsilon} \mathcal{F} \to 0$ of $\mathcal{F}$ and let $\mathcal{K}'' = \ker(\mathcal{E}_0 \to \mathcal{F} \to \mathcal{F}'')$, locally free. Then ϵ maps $\mathcal{K}''$ onto $\mathcal{F}'$ with kernel $\mathcal{E}_1$, so $0 \to \mathcal{E}_1 \to \mathcal{K}'' \to \mathcal{F}' \to 0$ is a resolution of $\mathcal{F}'$, and $0 \to \mathcal{K}'' \to \mathcal{E}_0 \to \mathcal{F}'' \to 0$ is a resolution of $\mathcal{F}''$. Hence

$$\det \mathcal{F}' \otimes \det \mathcal{F}'' = \left(\det \mathcal{K}'' \otimes \det \mathcal{E}_1^{-1}\right) \otimes \left(\det \mathcal{E}_0 \otimes \det \mathcal{K}''^{-1}\right) = \det \mathcal{E}_0 \otimes \det \mathcal{E}_1^{-1} = \det \mathcal{F}.$$

By independence of the resolution, det is additive on exact sequences, so det defined on generators extends to a homomorphism $\det : K(X) \to \operatorname{Pic} X$.

$\det \psi(D) = \mathcal{L}(D)$. For a point P, $0 \to \mathcal{L}(-P) \to \mathcal{O}_X \to k(P) \to 0$ is a resolution of $k(P)$ ((6.18): $\mathcal{I}_P = \mathcal{L}(-P)$), so $\det k(P) = \det \mathcal{O}_X \otimes \mathcal{L}(-P)^{-1} \cong \mathcal{L}(P)$ (6.13(b)). Since det is a homomorphism and $D \mapsto \mathcal{L}(D)$ is one, $\det \psi(\sum n_i P_i) = \bigotimes \mathcal{L}(P_i)^{n_i} \cong \mathcal{L}(D)$.

(c) First a lemma: for every divisor D, $\gamma(\mathcal{L}(D)) = \gamma(\mathcal{O}_X) + \psi(D)$. (*Proof.*) For a point P, $0 \to \mathcal{L}(D - P) \to \mathcal{L}(D) \to \mathcal{L}(D) \otimes k(P) \to 0$ is exact (tensor $0 \to \mathcal{I}_P \to \mathcal{O}_X \to k(P) \to 0$ with the invertible $\mathcal{L}(D)$, using $\mathcal{L}(D) \otimes \mathcal{I}_P = \mathcal{L}(D - P)$) and $\mathcal{L}(D) \otimes k(P) \cong k(P)$. So $\gamma(\mathcal{L}(D - P)) = \gamma(\mathcal{L}(D)) - \gamma(k(P))$. The function $\phi(D) = \gamma(\mathcal{L}(D)) - \gamma(\mathcal{O}_X) - \psi(D)$ therefore satisfies $\phi(D - P) = \phi(D)$ for all D, P, hence also $\phi(D + P) = \phi(D)$; as every divisor is reached from 0 by finitely many such steps, $\phi(D) = \phi(0) = 0$.

Now let $\mathcal{F}$ be coherent of rank r. Let $\mathcal{T}_0 \subseteq \mathcal{F}$ be the torsion subsheaf (sections killed by nonzero elements locally); it is coherent with $(\mathcal{T}_0)_\xi = 0$ (the generic stalk $\mathcal{F}_\xi$ is a K-vector space, torsion-free), so it is a torsion sheaf, supported on a finite set Σ. The quotient $\mathcal{F}_0 = \mathcal{F}/\mathcal{T}_0$ is torsion-free, hence locally free of rank r (stalks are finitely generated torsion-free over a DVR). Choose a basis $e_1, \dots, e_r$ of the K-vector space $(\mathcal{F}_0)_\xi$; since $\mathcal{F}_0$ is torsion-free, $\mathcal{F}_0 \subseteq$ the constant sheaf $(\mathcal{F}_0)_\xi = \bigoplus K e_\nu$. Let $\mathcal{G}_\nu = \mathcal{F}_0 \cap K e_\nu$; this is a coherent subsheaf of $\mathcal{F}_0$ of rank one, torsion-free, hence invertible, so $\mathcal{G}_\nu \cong \mathcal{L}(D_\nu)$ for some divisor D_ν (6.16). Let S be the (finite) union of the supports of $D_1, \dots, D_r$, let N be larger than the absolute values of all coefficients of all D_ν, and put $D = -N \sum_{P \in S} P$, so that $D \le D_\nu$ for all ν (coefficientwise). Then $\mathcal{L}(D) \subseteq \mathcal{L}(D_\nu)$ as sub-$\mathcal{O}_X$-modules of $\mathcal{K}$, and composing with the isomorphisms $\mathcal{L}(D_\nu) \cong \mathcal{G}_\nu$ we get an injection $\mathcal{L}(D)^{\oplus r} \hookrightarrow \bigoplus_\nu \mathcal{G}_\nu \hookrightarrow \mathcal{F}_0$. At ξ it is an isomorphism ($\mathcal{L}(D)_\xi = K \to \mathcal{G}_{\nu,\xi} = K e_\nu$ is a nonzero map of one-dimensional spaces and $\bigoplus K e_\nu = (\mathcal{F}_0)_\xi$), so its cokernel $\mathcal{T}'$ is a torsion sheaf. (Why

$\mathcal{G}_\nu$ is coherent: it is the kernel of $\mathcal{F}_0 \to \underline{K}^r/Ke_\nu$, a map from a coherent to a quasi-coherent sheaf on a noetherian scheme.)

Lift to $\mathcal{F}$. Let $\iota : \mathcal{L}(D)^{\oplus r} \to \mathcal{F}_0$ be this injection. We lift it to $\tilde{\iota} : \mathcal{L}(D)^{\oplus r} \to \mathcal{F}$. On $V = X - \Sigma$, $\mathcal{T}_0|_V = 0$ so $\mathcal{F}|_V = \mathcal{F}_0|_V$ and we take $\iota|_V$. For $P \in \Sigma$ choose an open neighbourhood W_P of P with $W_P \cap \Sigma = \{P\}$ and $\mathcal{L}(D)|_{W_P}$ trivial; then $\iota|_{W_P}$ is given by r sections of $\mathcal{F}_0(W_P)$, and since $\mathcal{F} \to \mathcal{F}_0$ is surjective, after shrinking W_P these lift to sections of $\mathcal{F}(W_P)$. The liftings on $W_P \cap V = W_P - \{P\}$ agree with $\iota|_V$ because $\mathcal{F}|_V = \mathcal{F}_0|_V$. So the local lifts glue to $\tilde{\iota}$. The composite $\mathcal{L}(D)^{\oplus r} \to \mathcal{F} \to \mathcal{F}_0$ is ι, injective, so $\tilde{\iota}$ is injective; its cokernel $\mathcal{T}$ is an extension of $\mathcal{T}'$ by $\mathcal{T}_0$ ($0 \to \mathcal{T}_0 \to \mathcal{T} \to \mathcal{T}' \to 0$ from the snake lemma), so a torsion sheaf. This is the exact sequence $0 \to \mathcal{L}(D)^{\oplus r} \to \mathcal{F} \to \mathcal{T} \to 0$.

Consequence. A torsion sheaf $\mathcal{T} = \bigoplus_P \mathcal{T}_P$ with $\mathcal{T}_P$ of finite length ℓ_P has a filtration with quotients $k(P)$ (a composition series of $\mathcal{T}_P$ over $\mathcal{O}_P$, whose simple module is k), so $\gamma(\mathcal{T}) = \sum \ell_P \gamma(k(P)) = \psi(\sum \ell_P P)$. Hence $\gamma(\mathcal{F}) = r\gamma(\mathcal{L}(D)) + \gamma(\mathcal{T}) = r\gamma(\mathcal{O}_X) + \psi(rD) + \psi(\sum \ell_P P)$, i.e. $\gamma(\mathcal{F}) - r\gamma(\mathcal{O}_X) \in \operatorname{Im}\psi$.

(d) Define $\Theta : \operatorname{Pic} X \oplus \mathbf{Z} \to K(X)$, $\Theta(\mathcal{L}(D), m) = \psi([D]) + m\gamma(\mathcal{O}_X)$ (well defined by (6.16) and (a)), and $\Phi = (\det, \operatorname{rank}) : K(X) \to \operatorname{Pic} X \oplus \mathbf{Z}$. Both are homomorphisms.

$\Phi\Theta = \mathrm{id}$: $\det \psi(D) = \mathcal{L}(D)$ by (b), $\operatorname{rank} \psi(D) = 0$ (torsion sheaves have rank 0); $\det \gamma(\mathcal{O}_X) = \mathcal{O}_X$ (the resolution $0 \to 0 \to \mathcal{O}_X \to \mathcal{O}_X \to 0$) and $\operatorname{rank} \gamma(\mathcal{O}_X) = 1$. So $\Phi\Theta(\mathcal{L}(D), m) = (\mathcal{L}(D), m)$.

$\Theta\Phi = \mathrm{id}$: it suffices to check on the generators $\gamma(\mathcal{F})$. Let $r = \operatorname{rank} \mathcal{F}$. By (c) there is a divisor D'' with $\gamma(\mathcal{F}) - r\gamma(\mathcal{O}_X) = \psi(D'')$. Applying det (and using (b), $\det \gamma(\mathcal{O}_X) = \mathcal{O}_X$): $\det \mathcal{F} = \mathcal{L}(D'')$. So the class $[D''] \in \operatorname{Cl} X$ is the one corresponding to $\det \mathcal{F}$ under $\operatorname{Cl} \cong \operatorname{Pic}$, and

$$\Theta\Phi(\gamma\mathcal{F}) = \Theta(\det \mathcal{F}, r) = \psi(D'') + r\gamma(\mathcal{O}_X) = \gamma(\mathcal{F}).$$

Thus Θ and Φ are inverse isomorphisms and $K(X) \cong \operatorname{Pic} X \oplus \mathbf{Z}$. □

Exercise 6.12

> **Problem.** Let X be a complete nonsingular curve. Show that there is a unique way to define the *degree* of any coherent sheaf on X, $\deg \mathcal{F} \in \mathbf{Z}$, such that:
>
> (1) If D is a divisor, $\deg \mathcal{L}(D) = \deg D$;
>
> (2) If $\mathcal{F}$ is a *torsion sheaf* (meaning a sheaf whose stalk at the generic point is zero), then $\deg \mathcal{F} = \sum_{P \in X} \operatorname{length}(\mathcal{F}_P)$; and
>
> (3) If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of coherent sheaves, then $\deg \mathcal{F} = \deg \mathcal{F}' + \deg \mathcal{F}''$.

Solution. The curve X is projective (Prop. 6.7) and all the previous results of Ex. 6.10, 6.11 apply; $\deg D = \sum n_i$ for $D = \sum n_i P_i$, and $\deg : \operatorname{Cl} X \to \mathbf{Z}$ is a well-defined homomorphism (6.10), which with $\operatorname{Cl} X \cong \operatorname{Pic} X$ (6.16) gives $\deg : \operatorname{Pic} X \to \mathbf{Z}$ with $\deg \mathcal{L}(D) = \deg D$.

Existence. For coherent $\mathcal{F}$ put $\deg \mathcal{F} := \deg(\det \mathcal{F})$, with $\det : K(X) \to \operatorname{Pic} X$ of Ex. 6.11(b) (the determinant is defined for all coherent $\mathcal{F}$). Property (3) holds since $\det \mathcal{F} \cong \det \mathcal{F}' \otimes \det \mathcal{F}''$ and deg is additive. For (1): $\mathcal{L}(D)$ is locally free of rank one, with resolution $0 \to 0 \to$

$\mathcal{L}(D) \to \mathcal{L}(D) \to 0$, so $\det \mathcal{L}(D) = \mathcal{L}(D)$ and $\deg \mathcal{L}(D) = \deg D$. For (2): a torsion sheaf $\mathcal{T}$ has $\gamma(\mathcal{T}) = \sum_P \ell_P \gamma(k(P))$, $\ell_P = \operatorname{length} \mathcal{T}_P$ (a composition series, as in Ex. 6.11(c)), so $\det \mathcal{T} = \bigotimes \mathcal{L}(P)^{\ell_P}$ by Ex. 6.11(b), and $\deg \mathcal{T} = \sum \ell_P$.

Uniqueness. Let $\deg'$ be any function satisfying (1)–(3). By (3) it induces a homomorphism $K(X) \to \mathbf{Z}$. Let $\mathcal{F}$ be coherent of rank r. By Ex. 6.11(c) there is a divisor $D = \sum n_i P_i$ with $\gamma(\mathcal{F}) = r\gamma(\mathcal{O}_X) + \sum n_i \gamma(k(P_i))$. Now $\deg'(\mathcal{O}_X) = \deg'(\mathcal{L}(0)) = \deg 0 = 0$ by (1), and $\deg'(k(P_i)) = \operatorname{length} k(P_i) = 1$ by (2). Hence $\deg'(\mathcal{F}) = \sum n_i = \deg D$. In particular this number is the same for every divisor D with $\gamma(\mathcal{F}) - r\gamma(\mathcal{O}_X) = \psi(D)$, and equals $\deg' \mathcal{F}$; since the same computation applies to any other function $\deg''$ with (1)–(3), $\deg''(\mathcal{F}) = \deg D = \deg'(\mathcal{F})$. So the degree is unique. □

7. Projective morphisms

Throughout this section I use the numbering of Hartshorne's Chapter II: Thm. 7.1 is the correspondence between morphisms to $\mathbf{P}^n_A$ and generated invertible sheaves, Prop. 7.2/7.3 are the criteria for closed immersions, Thm. 7.6 says $\mathcal{L}$ is ample iff some power is very ample, Prop. 7.10 and 7.11 concern $\operatorname{Proj} \mathcal{S}$ and $\mathbf{P}(\mathcal{E})$, Prop. 7.12 is the universal property of $\mathbf{P}(\mathcal{E})$, and Prop. 7.13–7.15 concern blowing up. "Very ample" always means very ample relative to the base in the sense of Section 5 of the text. Where a printed statement needs an extra hypothesis (this happens in Ex. 7.3(b), 7.7(b) and 7.9), I say so at that point.

Exercise 7.1

> **Problem.** Let $(X, \mathcal{O}_X)$ be a locally ringed space, and let $f : \mathcal{L} \to \mathcal{M}$ be a surjective map of invertible sheaves on X. Show that f is an isomorphism. [*Hint:* Reduce to a question of modules over a local ring by looking at the stalks.]

Solution. A morphism of sheaves is an isomorphism if and only if it is an isomorphism on every stalk, and surjectivity of f means that every f_x is surjective. Fix $x \in X$. Since $\mathcal{L}, \mathcal{M}$ are invertible, $\mathcal{L}_x \cong \mathcal{O}_x \cong \mathcal{M}_x$, so after choosing generators f_x becomes an $\mathcal{O}_x$-linear surjection $g : \mathcal{O}_x \to \mathcal{O}_x$, which is multiplication by $a = g(1)$. Surjectivity gives $b \in \mathcal{O}_x$ with $g(b) = ab = 1$, so a is a unit. Hence g, and so f_x, is bijective. (Only that $\mathcal{O}_x$ is a commutative ring is used; locality is not even needed.) Therefore f is an isomorphism. □

Exercise 7.2

> **Problem.** Let X be a scheme over a field k, and let $\mathcal{L}$ be an invertible sheaf on X. Let $\{s_0, \dots, s_n\}$ and $\{t_0, \dots, t_m\}$ be two sets of sections of $\mathcal{L}$, which generate the same subspace $V \subseteq \Gamma(X, \mathcal{L})$, and which generate the sheaf $\mathcal{L}$ at every point. Suppose $n \le m$. Show that the corresponding morphisms $\varphi : X \to \mathbf{P}^n_k$ and $\psi : X \to \mathbf{P}^m_k$ differ by a suitable linear projection $\mathbf{P}^m - L \to \mathbf{P}^n$ and an automorphism of $\mathbf{P}^m$, where L is a linear subspace of $\mathbf{P}^m$ of dimension $m - n - 1$.

Solution. Let $\sigma : k^{n+1} \to \Gamma(X, \mathcal{L})$, $e_i \mapsto s_i$ and $\tau : k^{m+1} \to \Gamma(X, \mathcal{L})$, $e'_j \mapsto t_j$; both have image V. Put $d = \dim V$, $K = \operatorname{Ker} \sigma$ ($\dim K = n+1-d$) and $K' = \operatorname{Ker} \tau$ ($\dim K' = m+1-d \ge \dim K$).

Claim. There is an injective linear map $\iota : k^{n+1} \to k^{m+1}$ with $\tau \circ \iota = \sigma$.

Choose a complement U of K in k^{n+1}, so $\sigma|_U : U \to V$ is an isomorphism, and a linear section $\varsigma : V \to k^{m+1}$ of the surjection $\tau : k^{m+1} \to V$. Choose an injective linear map $j : K \to K'$ (possible since $\dim K \leq \dim K'$). Define $\iota|_U = \varsigma \circ \sigma|_U$ and $\iota|_K = j$. Then $\tau\iota = \sigma$ on U (as $\tau\varsigma = \mathrm{id}$) and on K (both sides are 0). If $\iota(u+\kappa) = 0$ with $u \in U, \kappa \in K$, then $\iota(u) = -j(\kappa) \in K'$, so $\sigma(u) = \tau\iota(u) = 0$, hence $u = 0$, and then $j(\kappa) = 0$ gives $\kappa = 0$. So ι is injective.

Extend the linearly independent vectors $\iota(e_0), \ldots, \iota(e_n)$ to a basis $b_0 = \iota(e_0), \ldots, b_n = \iota(e_n), b_{n+1}, \ldots, b_m$ of k^{m+1}, and let $g \in GL_{m+1}(k)$ be defined by $g(e'_j) = b_j = \sum_l g_{lj} e'_l$. Put $t'_j = \tau(b_j) = \sum_l g_{lj} t_l$. Then:

- $t'_i = \tau(\iota(e_i)) = \sigma(e_i) = s_i$ for $0 \leq i \leq n$;
- $t'_0, \ldots, t'_m$ span V (as g is invertible), so they generate $\mathcal{L}$ at every point and define a morphism $\psi' : X \to \mathbf{P}^m_k$.

Let α be the automorphism of $\mathbf{P}^m_k$ induced by the graded automorphism $x_j \mapsto \sum_l g_{lj} x_l$ of $k[x_0, \ldots, x_m]$ (Example 7.1.1); then $\alpha^*\mathcal{O}(1) \cong \mathcal{O}(1)$ with $\alpha^* x_j = \sum_l g_{lj} x_l$. Hence $(\alpha\psi)^*\mathcal{O}(1) \cong \mathcal{L}$ and $(\alpha\psi)^* x_j = \sum_l g_{lj} t_l = t'_j$, so by the uniqueness in Thm. 7.1(b), $\psi' = \alpha \circ \psi$.

Now let $L = \{x_0 = \cdots = x_n = 0\} \subseteq \mathbf{P}^m$, a linear subspace of dimension $m - n - 1$ (empty if $m = n$), and let $p : \mathbf{P}^m - L \to \mathbf{P}^n$, $(x_0 : \cdots : x_m) \mapsto (x_0 : \cdots : x_n)$ be the linear projection from L; by 7.1 it is the morphism defined by the sections $x_0, \ldots, x_n$ of $\mathcal{O}(1)$, which generate $\mathcal{O}(1)$ on $\mathbf{P}^m - L$, so $p^*\mathcal{O}(1) \cong \mathcal{O}(1)|_{\mathbf{P}^m - L}$ and $p^* x_i = x_i$. Since $\psi'^* x_i = s_i$ $(i \leq n)$ generate $\mathcal{L}$ at every point $P \in X$, for each P some x_i $(i \leq n)$ does not vanish at $\psi'(P)$, i.e. $\psi'(X) \subseteq \mathbf{P}^m - L$. Thus $p \circ \psi' : X \to \mathbf{P}^n$ is defined, $(p\psi')^*\mathcal{O}(1) \cong \psi'^*\mathcal{O}(1) \cong \mathcal{L}$ and $(p\psi')^* x_i = \psi'^* x_i = s_i$. By uniqueness in 7.1(b), $p \circ \psi' = \varphi$. Altogether

$$\varphi = p \circ \alpha \circ \psi,$$

a linear projection composed with an automorphism of $\mathbf{P}^m$ and ψ. □

Exercise 7.3

Problem. Let $\varphi : \mathbf{P}^n_k \to \mathbf{P}^m_k$ be a morphism. Then:

(a) either $\varphi(\mathbf{P}^n) = \mathrm{pt}$ or $m \geq n$ and $\dim \varphi(\mathbf{P}^n) = n$;

(b) in the second case, φ can be obtained as the composition of (1) a d-uple embedding $\mathbf{P}^n \to \mathbf{P}^N$ for a uniquely determined $d \geq 1$, (2) a linear projection $\mathbf{P}^N - L \to \mathbf{P}^m$, and (3) an automorphism of $\mathbf{P}^m$. Also, φ has finite fibres.

Solution. Here k is any field and φ is a k-morphism; for $n = 0$ everything is trivial, so let $n \geq 1$. Let $S = k[x_0, \ldots, x_n]$. By Prop. 6.4/(6.17), $\mathrm{Pic}\, \mathbf{P}^n_k = \mathbf{Z} \cdot [\mathcal{O}(1)]$, so $\varphi^*\mathcal{O}(1) \cong \mathcal{O}(d)$ for a unique $d \in \mathbf{Z}$, and by Thm. 7.1(a) it is generated by the global sections $s_i = \varphi^* x_i$ $(i = 0, \ldots, m)$. Since $\Gamma(\mathbf{P}^n, \mathcal{O}(d)) = 0$ for $d < 0$ (Thm. 5.13) and a sheaf generated by zero sections is zero, $d \geq 0$; so $s_i = F_i \in S_d$, forms of degree d.

Case $d = 0$**.** Then $F_i = c_i \in \Gamma(\mathbf{P}^n, \mathcal{O}) = k$, not all zero (the s_i generate $\mathcal{O}$). The constant morphism $\mathbf{P}^n \to \mathrm{Spec}\, k \xrightarrow{(c_0 : \cdots : c_m)} \mathbf{P}^m$ has the same $(\mathcal{L}, s_i)$, so by uniqueness in 7.1(b) it equals φ. So $\varphi(\mathbf{P}^n)$ is a point.

Case $d > 0$**.** The F_i have no common zero on the scheme $\mathbf{P}^n$ (they generate $\mathcal{O}(d)$ everywhere).

A lemma. Let K be a field, $Z \subseteq \mathbf{P}^n_K$ a nonempty closed subset of dimension ≥ 1, and $F \in K[x_0, \dots, x_n]$ homogeneous of positive degree. Then $Z \cap V_+(F) \neq \emptyset$. Indeed, replace Z by an irreducible component Z_1 of dimension ≥ 1; $Z_1 = V_+(\mathfrak{p})$ for a homogeneous prime $\mathfrak{p} \not\supseteq S_+$, $B = S_K/\mathfrak{p}$ is a finitely generated graded K-domain with $\dim B = \dim Z_1 + 1 \geq 2$, and $\mathfrak{m}_+ = B_+$ is a maximal ideal (with residue field K), hence of height $\dim B \geq 2$ (finitely generated domains over a field are catenary and equidimensional). If $\bar{F} = 0$ in B, then $V_+(F) \supseteq Z_1$. Otherwise suppose $Z_1 \cap V_+(F) = \emptyset$; then the only homogeneous prime of B containing $\bar{F}$ is $\mathfrak{m}_+$ (a homogeneous prime not containing B_+ would be a point of $Z_1 \cap V_+(F)$). A minimal prime of $(\bar{F})$ is homogeneous, so it equals $\mathfrak{m}_+$, and Krull's Hauptidealsatz gives $\operatorname{ht} \mathfrak{m}_+ \leq 1$, a contradiction.

Finite fibres. Let $Q \in \mathbf{P}^m$ be any point, $K = \kappa(Q)$, and choose j with $Q \in D_+(x_j)$. The fibre $X_Q = \mathbf{P}^n_k \times_{\mathbf{P}^m_k} \operatorname{Spec} K$ is homeomorphic to $\varphi^{-1}(Q)$ (Ex. 3.10), and, after base change to K, is a closed subscheme of $\mathbf{P}^n_K$ (the K-point Q' of $\mathbf{P}^m_K$ is a closed immersion since $\mathbf{P}^m_K$ is separated) contained in $\varphi_K^{-1}(D_+(x_j)) = D_+(F_j)$. If $X_Q \neq \emptyset$ then $F_j \neq 0$, and $X_Q \cap V_+(F_j) = \emptyset$, so by the Lemma $\dim X_Q = 0$; a noetherian space of dimension 0 is finite. Thus all fibres of φ are finite.

The image. $\mathbf{P}^n_k$ is proper over k and $\mathbf{P}^m_k$ is separated, so φ is proper (4.8(e)) and closed. Let η be the generic point of $\mathbf{P}^n$, $\xi = \varphi(\eta)$. Then $\varphi(\mathbf{P}^n) = \overline{\{\xi\}} =: Z$ is an irreducible closed subset with $\dim Z = \operatorname{trdeg}_k \kappa(\xi)$. The fibre over ξ is a finite-type $\kappa(\xi)$-scheme with finitely many points, hence of dimension 0, and η is one of its points, so $\kappa(\eta) = K(\mathbf{P}^n)$ is a finite extension of $\kappa(\xi)$. Therefore $\dim Z = \operatorname{trdeg}_k K(\mathbf{P}^n) = n$, and $n = \dim Z \leq m$. This proves (a), and the finiteness of the fibres.

(b) We are in the case $d > 0$; d is unique since $\varphi^*\mathcal{O}(1) \cong \mathcal{O}(d)$ determines it. Let $M_0, \dots, M_N$ be the monomials of degree d ($N + 1 = \binom{n+d}{d}$) and $\rho_d : \mathbf{P}^n \to \mathbf{P}^N$ the d-uple embedding, i.e. the morphism defined by the sections M_l of $\mathcal{O}(d)$ (7.1); it is a closed immersion (Ex. 5.13). Write $F_i = \sum_l c_{il} M_l$ and let $\ell_i = \sum_l c_{il} y_l$ ($i = 0, \dots, m$) be linear forms on $\mathbf{P}^N$. Then $\rho_d^* \ell_i = F_i$. Let $L' = V_+(\ell_0, \dots, \ell_m) \subseteq \mathbf{P}^N$; since the F_i have no common zero, $\rho_d(\mathbf{P}^n) \cap L' = \emptyset$. The morphism $\lambda = (\ell_0 : \cdots : \ell_m) : \mathbf{P}^N - L' \to \mathbf{P}^m$ satisfies $(\lambda\rho_d)^*\mathcal{O}(1) \cong \mathcal{O}(d)$, $(\lambda\rho_d)^* x_i = F_i$, so by uniqueness

$$\varphi = \lambda \circ \rho_d.$$

Now assume (as is implicit in the statement) that the forms $F_0, \dots, F_m$ are linearly independent, i.e. that $\varphi(\mathbf{P}^n)$ is not contained in a hyperplane. Then $\ell_0, \dots, \ell_m$ are linearly independent, $m \leq N$, L' is a linear subspace of dimension $N - m - 1$, and λ is the linear projection from L' (a projection from L' is defined by a basis of the linear forms vanishing on L'; two choices differ by an element of PGL_{m+1}, which is the automorphism in (3)). So φ is the composite of the d-uple embedding, a linear projection and an automorphism of $\mathbf{P}^m$, as claimed.

Caveat (the statement needs the independence assumption). If the F_i are linearly dependent, the printed claim is false: for $\varphi : \mathbf{P}^1 \to \mathbf{P}^2$, $(x_0 : x_1) \mapsto (x_0^2 : x_1^2 : x_0^2)$, we have $d = 2$, $N = m = 2$, so $L = \emptyset$ and (2)–(3) would make φ an automorphism of $\mathbf{P}^2$ composed with ρ_2, whose image is a smooth conic, whereas $\varphi(\mathbf{P}^1)$ is the line $\{y_0 = y_2\}$. In general one proves: let $r + 1 = \dim \operatorname{span}(F_i)$; then $\varphi = \iota \circ (\text{linear projection } \mathbf{P}^N - L \to \mathbf{P}^r) \circ \rho_d$ with $\iota : \mathbf{P}^r \hookrightarrow \mathbf{P}^m$ a linear embedding (choose a basis of the span, and re-express the F_i in it); the previous argument applies verbatim to r in place of m. □

Exercise 7.4

Problem.

(a) Use (7.6) to show that if X is a scheme of finite type over a noetherian ring A, and if X admits an ample invertible sheaf, then X is separated.

(b) Let X be the affine line over a field k with the origin doubled (4.0.1). Calculate $\operatorname{Pic} X$, determine which invertible sheaves are generated by global sections, and then show directly (without using (a)) that there is no ample invertible sheaf on X.

Solution. (a) Let $\mathcal{L}$ be ample on X. By Thm. 7.6, $\mathcal{L}^m$ is very ample relative to $\operatorname{Spec} A$ for some $m > 0$, i.e. there is an immersion $i : X \to \mathbf{P}^r_A$ over A. An immersion is a composite of a closed immersion and an open immersion, hence separated (Cor. 4.6(a),(b)), and $\mathbf{P}^r_A \to \operatorname{Spec} A$ is separated (it is projective, hence proper, Thm. 4.9). A composite of separated morphisms is separated, so $X \to \operatorname{Spec} A$ is separated.

(b) Write $X = U_1 \cup U_2$ with $U_i \cong \operatorname{Spec} k[t]$ glued along $U_{12} = U_1 \cap U_2 = \operatorname{Spec} k[t, t^{-1}]$ by the identity.

Pic. Since $k[t]$ is a PID, every invertible sheaf $\mathcal{L}$ is trivial on U_1 and U_2; choose trivializations $\alpha_i : \mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$. Then $\alpha_1\alpha_2^{-1}$ is multiplication by a unit of $k[t, t^{-1}]$, i.e. by ct^n, $c \in k^*$, $n \in \mathbf{Z}$. Replacing α_2 by $c\alpha_2$ we may take $c = 1$. So $\mathcal{L} \cong \mathcal{L}_n$, where $\mathcal{L}_n$ is obtained by gluing $\mathcal{O}_{U_1}$ and $\mathcal{O}_{U_2}$ so that a section over X is a pair $(f_1, f_2) \in k[t] \times k[t]$ with $f_1 = t^n f_2$ in $k[t, t^{-1}]$. If $\mathcal{L}_n \cong \mathcal{L}_{n'}$, then the two trivializations on each U_i differ by units of $k[t]$, i.e. by constants $c_1, c_2 \in k^*$, so $t^n = (c_1/c_2)t^{n'}$ and $n = n'$. Clearly $\mathcal{L}_n \otimes \mathcal{L}_{n'} \cong \mathcal{L}_{n+n'}$. Hence $\operatorname{Pic} X \cong \mathbf{Z}$, generated by $\mathcal{L}_1$.

Global sections. For $n \geq 0$ the sections of $\mathcal{L}_n$ are $(t^n f, f)$, $f \in k[t]$; for $n < 0$ they are $(f, t^{|n|} f)$, $f \in k[t]$. In both cases $\Gamma(X, \mathcal{L}_n)$ is generated by one section s_n, namely $s_n = (t^n, 1)$ for $n \geq 0$, $s_n = (1, t^{|n|})$ for $n < 0$. For $n > 0$ every global section has first component divisible by t, so vanishes at the origin $0_1 \in U_1$; for $n < 0$ every global section vanishes at the origin $0_2 \in U_2$. Thus $\mathcal{L}_n$ is generated by global sections only for $n = 0$: *the only invertible sheaf generated by its global sections is (up to isomorphism) $\mathcal{O}_X$.*

No ample sheaf. Suppose $\mathcal{L}$ is ample. Applying the definition to the coherent sheaf $\mathcal{O}_X$, $\mathcal{L}^n$ is generated by global sections for all $n \geq n_0$, so $\mathcal{L}^{n_0} \cong \mathcal{O}_X \cong \mathcal{L}^{n_0+1}$, whence $\mathcal{L} \cong \mathcal{O}_X$. Then for $\mathcal{F} = \mathcal{L}_1$ (coherent) the sheaf $\mathcal{F} \otimes \mathcal{L}^n \cong \mathcal{L}_1$ is never generated by global sections, contradicting ampleness. So X has no ample invertible sheaf (consistently with (a), since X is not separated). □

Exercise 7.5

Problem. Establish the following properties of ample and very ample invertible sheaves on a noetherian scheme X. $\mathcal{L}, \mathcal{M}$ will denote invertible sheaves, and for (d), (e) we assume furthermore that X is of finite type over a noetherian ring A.

(a) If $\mathcal{L}$ is ample and $\mathcal{M}$ is generated by global sections, then $\mathcal{L} \otimes \mathcal{M}$ is ample.

(b) If $\mathcal{L}$ is ample and $\mathcal{M}$ is arbitrary, then $\mathcal{M} \otimes \mathcal{L}^n$ is ample for sufficiently large n.

(c) If $\mathcal{L}, \mathcal{M}$ are both ample, so is $\mathcal{L} \otimes \mathcal{M}$.

(d) If $\mathcal{L}$ is very ample and $\mathcal{M}$ is generated by global sections, then $\mathcal{L} \otimes \mathcal{M}$ is very ample.

(e) If $\mathcal{L}$ is ample, then there is an $n_0 > 0$ such that $\mathcal{L}^n$ is very ample for all $n \geq n_0$.

Solution. I use: if $\mathcal{F}_1, \mathcal{F}_2$ are generated by global sections, so is $\mathcal{F}_1 \otimes \mathcal{F}_2$ (if s_i generate $\mathcal{F}_1$ and t_j generate $\mathcal{F}_2$, the $s_i \otimes t_j$ generate the stalks of $\mathcal{F}_1 \otimes \mathcal{F}_2$).

(a) Let $\mathcal{F}$ be coherent. There is n_0 with $\mathcal{F} \otimes \mathcal{L}^n$ generated by global sections for $n \geq n_0$. Then $\mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^n = (\mathcal{F} \otimes \mathcal{L}^n) \otimes \mathcal{M}^n$ is generated by global sections for $n \geq n_0$, as $\mathcal{M}^n$ is.

(b) Since $\mathcal{M}$ is coherent and $\mathcal{L}$ ample, $\mathcal{M} \otimes \mathcal{L}^j$ is generated by global sections for $j \geq j_0$. For $n \geq j_0 + 1$ write $\mathcal{M} \otimes \mathcal{L}^n = \mathcal{L} \otimes (\mathcal{M} \otimes \mathcal{L}^{n-1})$ and apply (a).

(c) Apply the definition of ample to $\mathcal{O}_X$: $\mathcal{M}^m$ is generated by global sections for some $m > 0$. By Prop. 7.5, $\mathcal{L}^m$ is ample, so by (a) $\mathcal{L}^m \otimes \mathcal{M}^m = (\mathcal{L} \otimes \mathcal{M})^m$ is ample, and by Prop. 7.5 again $\mathcal{L} \otimes \mathcal{M}$ is ample.

(d) Let $i : X \to \mathbf{P}^r_A$ be an immersion with $\mathcal{L} \cong i^*\mathcal{O}(1)$, $s_a = i^*x_a$. Since X is quasi-compact, $\mathcal{M}$ is generated by finitely many global sections $t_0, \ldots, t_q$; let $\psi : X \to \mathbf{P}^q_A$ be the corresponding morphism (7.1), so $\psi^*\mathcal{O}(1) \cong \mathcal{M}$, $\psi^* y_b = t_b$. Consider $(i, \psi) : X \to \mathbf{P}^r_A \times_A \mathbf{P}^q_A$. It is an immersion: it is the composite of the graph $X \to X \times_A \mathbf{P}^q_A$ of ψ, a closed immersion because $\mathbf{P}^q_A$ is separated over A, and $i \times \mathrm{id} : X \times_A \mathbf{P}^q_A \to \mathbf{P}^r_A \times_A \mathbf{P}^q_A$, an immersion as a base change of i. Let $\sigma : \mathbf{P}^r_A \times_A \mathbf{P}^q_A \to \mathbf{P}^{(r+1)(q+1)-1}_A$ be the Segre closed immersion (Ex. 5.11), $\sigma^* w_{ab} = x_a y_b$, $\sigma^*\mathcal{O}(1) = p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)$. Then $\sigma \circ (i, \psi)$ is an immersion with $(\sigma(i, \psi))^*\mathcal{O}(1) \cong \mathcal{L} \otimes \mathcal{M}$, so $\mathcal{L} \otimes \mathcal{M}$ is very ample.

(e) By Thm. 7.6 there is $m_1 > 0$ with $\mathcal{L}^{m_1}$ very ample over $\operatorname{Spec} A$; X is noetherian. Since $\mathcal{L}$ is ample, there is j_0 with $\mathcal{L}^j$ generated by global sections for all $j \geq j_0$. Let $n \geq n_0 := j_0 + 2m_1$. Choose r' with $j_0 \leq r' \leq j_0 + m_1 - 1$ and $r' \equiv n \pmod{m_1}$, and write $n = qm_1 + r'$; then $q \geq 1$. The sheaf $\mathcal{L}^{m_1}$ is very ample, hence generated by global sections (7.1(a)); by (d) and induction, $(\mathcal{L}^{m_1})^q$ is very ample for all $q \geq 1$. As $\mathcal{L}^{r'}$ is generated by global sections, (d) shows that $\mathcal{L}^n = (\mathcal{L}^{m_1})^q \otimes \mathcal{L}^{r'}$ is very ample. □

Exercise 7.6

Problem. *The Riemann–Roch Problem.* Let X be a nonsingular projective variety over an algebraically closed field, and let D be a divisor on X. For any $n > 0$ we consider the complete linear system $|nD|$. Then the Riemann–Roch problem is to determine $\dim |nD|$ as a function of n, and, in particular, its behavior for large n. If $\mathcal{L}$ is the corresponding invertible sheaf, then $\dim |nD| = \dim \Gamma(X, \mathcal{L}^n) - 1$, so an equivalent problem is to determine $\dim \Gamma(X, \mathcal{L}^n)$ as a function of n.

(a) Show that if D is very ample, and if $X \hookrightarrow \mathbf{P}^N$ is the corresponding embedding in projective space, then for all n sufficiently large, $\dim |nD| = P_X(n) - 1$, where P_X is the Hilbert polynomial of X (I, Section 7). Thus in this case $\dim |nD|$ is a polynomial function of n, for $n \gg 0$.

(b) If D corresponds to a torsion element of $\operatorname{Pic} X$, of order r, then $\dim |nD| = 0$ if $r \mid n$, -1 otherwise. In this case the function is periodic of period r.

It follows from the general Riemann–Roch theorem that $\dim |nD|$ is a polynomial function in n for n large, whenever D is an *ample* divisor. See (IV, 1.3.2), (V, 1.6), and Appendix A. In the case of algebraic surfaces, Zariski [7] has shown for any effective divisor D, that there is a finite set of polynomials $P_1, \ldots, P_r$, such that for all n sufficiently large, $\dim |nD| = P_{i(n)}(n)$, where $i(n) \in \{1, 2, \ldots, r\}$ is a function of n.

Solution. Since X is nonsingular, $\operatorname{Cl} X \cong \operatorname{Pic} X$ (6.16), and $|nD|$ is the projective space of the nonzero sections of $\mathcal{L}^n = \mathcal{L}(nD)$ modulo k^* (Prop. 7.7), so $\dim |nD| = \dim_k \Gamma(X, \mathcal{L}^n) - 1$ as in the statement.

(a) D very ample means $\mathcal{L} \cong i^*\mathcal{O}(1)$ for an immersion $i : X \to \mathbf{P}^N$, which is a closed immersion since X is projective (complete); identify X with the closed subvariety $i(X)$ (integral, so with the reduced structure). Then $\mathcal{L}^n \cong \mathcal{O}_X(n)$. Let $S(X) = k[x_0, \ldots, x_N]/I(X)$ be the homogeneous coordinate ring. It is generated by $S(X)_1$ over k, $X = \operatorname{Proj} S(X)$, and $\widetilde{S(X)} = \mathcal{O}_X$. By Ex. 5.9(b) (with $M = S = S(X)$), $S(X)_n \to \Gamma(X, \mathcal{O}_X(n))$ is an isomorphism for $n \geq n_0$. By definition of the Hilbert polynomial (I, Section 7), $\dim_k S(X)_n = P_X(n)$ for $n \gg 0$. Hence for $n \gg 0$,

$$\dim |nD| = \dim_k \Gamma(X, \mathcal{L}^n) - 1 = P_X(n) - 1.$$

(b) Let $r \geq 1$ be minimal with $\mathcal{L}^r \cong \mathcal{O}_X$; then $\mathcal{L}^n \cong \mathcal{O}_X$ iff $r \mid n$ (the set of such n is the subgroup of $\mathbf{Z}$ generated by r). If $r \mid n$, then $\Gamma(X, \mathcal{L}^n) \cong \Gamma(X, \mathcal{O}_X) = k$ (I, Thm. 3.4, X a projective variety), so $\dim |nD| = 0$. Let $r \nmid n$ and suppose, for contradiction, that $0 \neq s \in \Gamma(X, \mathcal{L}^n)$. Since X is integral, the germ of s at the generic point is nonzero in the one-dimensional $K(X)$-vector space $(\mathcal{L}^n)_\eta$, so $s^r \in \Gamma(X, \mathcal{L}^{nr}) \cong \Gamma(X, \mathcal{O}_X) = k$ is nonzero, i.e. a nonzero constant, hence it generates $\mathcal{L}^{nr}$ at every point. If $s_P \in \mathfrak{m}_P(\mathcal{L}^n)_P$ for some P, then $s_P^r \in \mathfrak{m}_P(\mathcal{L}^{nr})_P$, a contradiction; so s generates $\mathcal{L}^n$ everywhere, i.e. $\mathcal{O}_X \to \mathcal{L}^n$, $1 \mapsto s$, is surjective, hence an isomorphism by Ex. 7.1. Then $r \mid n$, a contradiction. Hence $\Gamma(X, \mathcal{L}^n) = 0$ and $|nD| = \emptyset$ has dimension -1 for $r \nmid n$. The function is periodic of period r. □

Exercise 7.7

Problem. *Some Rational Surfaces.* Let $X = \mathbf{P}_k^2$, and let $|D|$ be the complete linear system of all divisors of degree 2 on X (conics). D corresponds to the invertible sheaf $\mathcal{O}(2)$, whose space of global sections has a basis $x^2, y^2, z^2, xy, xz, yz$, where x, y, z are the homogeneous coordinates of X.

(a) The complete linear system $|D|$ gives an embedding of $\mathbf{P}^2$ in $\mathbf{P}^5$, whose image is the Veronese surface (I, Ex. 2.13).

(b) Show that the subsystem defined by $x^2, y^2, z^2, x(y-z), (x-y)z$ gives a closed immersion of X into $\mathbf{P}^4$. The image is called the Veronese surface in $\mathbf{P}^4$. Cf. (IV, Ex. 3.11).

(c) Let $\mathfrak{d} \subseteq |D|$ be the linear system of all conics passing through a fixed point P. Then $\mathfrak{d}$ gives an immersion of $U = X - P$ into $\mathbf{P}^4$. Furthermore, if we blow up P, to get a surface $\tilde{X}$, then this map extends to a closed immersion of $\tilde{X}$ in $\mathbf{P}^4$. Show that $\tilde{X}$ is a surface of degree 3 in $\mathbf{P}^4$, and that the lines in X through P are transformed into straight lines in $\tilde{X}$ which do not meet. $\tilde{X}$ is the union of all these lines, so we say $\tilde{X}$ is a ruled surface (V, 2.19.1).

Solution. Throughout k is algebraically closed (as needed for linear systems; the closed-immersion verifications below work over any field except where char $\neq 2$ is flagged). Let $S = k[x, y, z]$. For a form s of positive degree, $X_s = D_+(s) = \operatorname{Spec} S_{(s)}$ is affine, so condition (1) of Prop. 7.2 holds automatically for any family of forms of the same degree; only condition (2) must be checked. For a family $s_0, \ldots, s_N \in S_2$ with no common zero, φ is the morphism of 7.1, and it is a closed immersion iff for each i the ring $S_{(s_i)}$ is generated as k-algebra by the elements s_j/s_i.

(a) The six monomials $s_{ab} = x_a x_b$ are the basis of $\Gamma(\mathcal{O}(2))$, with no common zero. The 2-uple embedding is by definition the morphism they define (7.1). For $s = x_a x_b$, $D_+(x_a x_b) = D_+(x_a) \cap D_+(x_b)$ has ring $k[x_c/x_a \ (c = 0, 1, 2),\ x_a/x_b]$ and each of these generators is a ratio s'/s: $x_c/x_a = x_b x_c/(x_a x_b)$ and $x_a/x_b = x_a^2/(x_a x_b)$. So (2) holds, and φ is a closed immersion of $\mathbf{P}^2$ in $\mathbf{P}^5$, with image the Veronese surface (the image of the 2-uple embedding, I, Ex. 2.13).

(b) *Hypothesis on the characteristic.* The statement needs $\operatorname{char} k \neq 2$ (which I assume): in characteristic 2 the morphism is not an immersion at $(1 : 1 : 0)$. Indeed, in the chart $x \neq 0$ with $u = y/x, v = z/x$ the five coordinates are $u^2, v^2, u - v, (1 - u)v$ and the differentials in characteristic 2 are $0, 0, du + dv, (1 + u)dv + v\,du$, which at $(u, v) = (1, 0)$ span only a line.

So let $\operatorname{char} k \neq 2$ and $s_0 = x^2, s_1 = y^2, s_2 = z^2, s_3 = x(y - z), s_4 = (x - y)z$. These have no common zero ($x^2 = y^2 = z^2 = 0$ is impossible). For each i let $R_i \subseteq S_{(s_i)}$ be the k-subalgebra generated by the s_j/s_i; we show $R_i = S_{(s_i)}$.

$i = 0$: $S_{(s_0)} = k[u, v]$ with $u = y/x$, $v = z/x$. R_0 contains $u^2, v^2, w = u - v, b = (1 - u)v$. From $w^2 = u^2 - 2uv + v^2$ we get $uv \in R_0$ (2 is invertible), so $v = b + uv \in R_0$ and $u = w + v \in R_0$.

$i = 1$: $S_{(s_1)} = k[p, q]$, $p = x/y, q = z/y$; $R_1 \ni p^2, q^2, p(1 - q), (p - 1)q$. The sum of the last two is $p - q$; $(p - q)^2$ gives $pq \in R_1$; then $p = p(1 - q) + pq$ and $q = pq - (p - 1)q$ lie in R_1.

$i = 2$: $S_{(s_2)} = k[a, b]$, $a = x/z$, $b = y/z$; $R_2 \ni a^2, b^2, a(b-1), a-b$. From $(a-b)^2$ we get $ab \in R_2$, then $a = ab - a(b - 1)$ and $b = a - (a - b)$.

$i = 3$: Put $X' = x$, $T = y - z$, $Z = z$ (a linear change of coordinates), so $s_3 = X'T$ and $S_{(s_3)} = k[\alpha, \beta, \gamma]/(\beta\gamma - 1)$ with $\alpha = Z/X'$, $\beta = T/X'$, $\gamma = \beta^{-1}$. Then

$$\tfrac{s_0}{s_3} = \gamma, \quad \tfrac{s_2}{s_3} = \alpha^2\gamma, \quad \tfrac{s_1}{s_3} = \tfrac{(T+Z)^2}{X'T} = \beta + 2\alpha + \alpha^2\gamma, \quad \tfrac{s_4}{s_3} = \tfrac{(X'-T-Z)Z}{X'T} = \alpha\gamma - \alpha - \alpha^2\gamma.$$

So $R_3 \ni \gamma, \alpha^2\gamma$, hence $m = \beta + 2\alpha \in R_3$; then $m\gamma = 1 + 2\alpha\gamma$ gives $\alpha\gamma \in R_3$, then $\alpha = \alpha\gamma - \alpha^2\gamma - s_4/s_3 \in R_3$ and $\beta = m - 2\alpha \in R_3$.

$i = 4$: Put $r = x - y$, so $s_4 = rz$ and $S_{(s_4)} = k[\alpha, \beta, \gamma]/(\beta\gamma - 1)$ with $\alpha = y/z$, $\beta = r/z$, $\gamma = \beta^{-1}$, $x/z = \alpha + \beta$. Then

$$\tfrac{s_2}{s_4} = \gamma, \quad \tfrac{s_1}{s_4} = \alpha^2\gamma, \quad \tfrac{s_0}{s_4} = (\alpha+\beta)^2\gamma = \alpha^2\gamma + 2\alpha + \beta, \quad \tfrac{s_3}{s_4} = (\alpha+\beta)(\alpha-1)\gamma = \alpha^2\gamma - \alpha\gamma + \alpha - 1.$$

So $m = 2\alpha + \beta \in R_4$, $m\gamma = 2\alpha\gamma + 1$ gives $\alpha\gamma \in R_4$, then $\alpha \in R_4$ (from s_3/s_4) and $\beta = m - 2\alpha \in R_4$.

By Prop. 7.2, $\varphi : X \to \mathbf{P}^4$ is a closed immersion; its image is the Veronese surface in $\mathbf{P}^4$ (the projection of the Veronese surface of (a) from a point not on its secant variety).

(c) Choose coordinates with $P = (0 : 0 : 1)$. The conics through P are the forms in the 5-dimensional space $V = \langle x^2, xy, y^2, xz, yz \rangle$ $(= \Gamma(X, \mathcal{I}_P(2)))$, so $\dim \mathfrak{d} = 4$.

The blow-up. Let $B \subseteq \mathbf{P}^2 \times \mathbf{P}^1$ be the closed subscheme $x\eta = y\xi$, with $(\xi : \eta)$ homogeneous coordinates on $\mathbf{P}^1$, and $\pi : B \to \mathbf{P}^2$ the projection. I claim $B \to X$ is the blow-up of P (ideal $\mathcal{I}_P$). Over $D_+(z) = \operatorname{Spec} k[a, b]$ ($a = x/z$, $b = y/z$) with $\mathcal{I}_P = (a, b)$, the blow-up is the closed subscheme $a\eta = b\xi$ of $\mathbf{P}^1 \times D_+(z)$ (Example 7.12.1), which is $B|_{D_+(z)}$. Over $X - P$ the blow-up is the identity (7.13(b)), and so is $B \to X$ there (a point of B over $(x : y : z)$ with $(x, y) \neq 0$ has $(\xi : \eta) = (x : y)$). On the overlap $D_+(z) - P$ both identifications are the unique X-isomorphism (both schemes map isomorphically to $D_+(z) - P$). So they glue: $\tilde{X} \cong B$ over X.

The map. On B consider the five sections $w_0 = x\xi,\ w_1 = x\eta (= y\xi),\ w_2 = y\eta,\ w_3 = z\xi,\ w_4 = z\eta$ of $\mathcal{O}(1,1)|_B$. They have no common zero on B: if $z \neq 0$ then $\xi = \eta = 0$ is impossible; so $z = 0$; then if $x \neq 0$ again $\xi = \eta = 0$; so $x = 0$, then $y \neq 0$, and $w_2 = w_1 = 0$ force $\eta = 0 = \xi$, impossible. Let $\Phi : B \to \mathbf{P}^4$ be the corresponding morphism (7.1). On $B - \pi^{-1}(P) \cong U$ we have $(\xi : \eta) = (x : y)$, so $\mathcal{O}(1,1)|_U \cong \mathcal{O}(2)|_U$ with $\xi \leftrightarrow x, \eta \leftrightarrow y$, and the w_i become x^2, xy, y^2, xz, yz: thus $\Phi|_U$ is the morphism defined by $\mathfrak{d}$ on U, and Φ extends it to $B = \tilde{X}$.

Φ is a closed immersion (Prop. 7.2). The sets B_{w_i} are affine and the charts are as follows (coordinates in each chart; all B_{w_i} are open in B):

- $w_0 \neq 0$: $x \neq 0, \xi \neq 0$; set $x = \xi = 1$, $u = y, v = z$, then $\eta = u$: $B_{w_0} = \operatorname{Spec} k[u,v]$; the ratios w_j/w_0 are u, u^2, v, uv, which generate $k[u,v]$.
- $w_2 \neq 0$: symmetric: $B_{w_2} = \operatorname{Spec} k[p,q]$ ($p = x/y, q = z/y$), ratios p^2, p, pq, q.
- $w_1 \neq 0$: $x \neq 0, \eta \neq 0$; with $x = 1$, $u = y, v = z$, $\eta = u\xi$, so u must be invertible: $B_{w_1} = \operatorname{Spec} k[u, u^{-1}, v]$; the ratios w_j/w_1 are $u^{-1}, u, v/u, v$, which generate.
- $w_3 \neq 0$: $z \neq 0, \xi \neq 0$; set $z = \xi = 1$, $a = x$, η free, and then $y = a\eta$: $B_{w_3} = \operatorname{Spec} k[a,\eta]$; the ratios are $w_0/w_3 = a$, $w_1/w_3 = a\eta$, $w_2/w_3 = a\eta^2$, $w_4/w_3 = \eta$, which generate.
- $w_4 \neq 0$: $z \neq 0, \eta \neq 0$; set $z = \eta = 1$, $b = y$, ξ free, and then $x = b\xi$: $B_{w_4} = \operatorname{Spec} k[b,\xi]$; the ratios are $w_0/w_4 = b\xi^2$, $w_1/w_4 = b\xi$, $w_2/w_4 = b$, $w_3/w_4 = \xi$, which generate.

So Φ is a closed immersion: $\tilde{X} = B$ is identified with the closed subscheme $\Phi(B) \subseteq \mathbf{P}^4$; in particular $\Phi|_U$ is an immersion of U (an open subscheme of $\tilde{X}$).

Degree 3. $\tilde{X} = \Phi(B)$ is an integral surface. In the chart $B_{w_0} \cong \mathbf{A}^2$, $\Phi = (1 : u : u^2 : v : uv)$. Since k is infinite and B_{w_0} is dense in B, a form $G(w_0, \dots, w_4)$ of degree n lies in $I(\tilde{X})$ iff $G(1, u, u^2, v, uv) = 0$ in $k[u,v]$. Hence $S(\tilde{X})_n$ is isomorphic to the k-span of the monomials $u^a v^b$ that are products of n elements of $\{1, u, u^2, v, uv\}$. Such a product has v-exponent $b = n_3 + n_4 \in [0, n]$, and for fixed b the u-exponent takes exactly the values $0, 1, \dots, 2(n-b) + b$ (the $n - b$ factors from $\{1, u, u^2\}$ contribute any value in $[0, 2(n-b)]$, the $n_4 \in [0, b]$ factors uv contribute n_4). So the Hilbert function is

$$\sum_{b=0}^{n} (2n - b + 1) = \frac{(n+1)(3n+2)}{2} = \frac{3}{2}n^2 + \frac{5}{2}n + 1,$$

the Hilbert polynomial of $\tilde{X}$. It has degree 2 and leading coefficient $3/2$, so $\dim \tilde{X} = 2$ and $\deg \tilde{X} = 2! \cdot \frac{3}{2} = 3$. (Check: $\tilde{X}$ lies on the three 2×2 minors $w_0 w_2 - w_1^2$, $w_0 w_4 - w_1 w_3$, $w_1 w_4 - w_2 w_3$ of $\left(\begin{smallmatrix} w_0 & w_1 & w_3 \\ w_1 & w_2 & w_4 \end{smallmatrix}\right)$, the cubic scroll.)

The lines. The lines of X through P are $\ell_{(\xi_0:\eta_0)} = \{x\eta_0 = y\xi_0\}$, $(\xi_0 : \eta_0) \in \mathbf{P}^1$. The strict transform of $\ell_{(\xi_0:\eta_0)}$ is the closure of $\pi^{-1}(\ell - P)$, namely the fibre $F_{(\xi_0:\eta_0)}$ of $B \to \mathbf{P}^1$ over $(\xi_0 : \eta_0)$, i.e. $\{((\lambda\xi_0 : \lambda\eta_0 : z), (\xi_0 : \eta_0))\} \cong \mathbf{P}^1$ with coordinates $(\lambda : z)$. Its image under Φ is

$$(\lambda\xi_0^2 : \lambda\xi_0\eta_0 : \lambda\eta_0^2 : z\xi_0 : z\eta_0) = \lambda\,(\xi_0^2, \xi_0\eta_0, \eta_0^2, 0, 0) + z\,(0, 0, 0, \xi_0, \eta_0),$$

the straight line through the points $(\xi_0^2 : \xi_0\eta_0 : \eta_0^2 : 0 : 0)$ and $(0 : 0 : 0 : \xi_0 : \eta_0)$. Distinct fibres of $B \to \mathbf{P}^1$ are disjoint and Φ is injective, so these lines are pairwise disjoint. Every point of B lies in exactly one fibre, so $\tilde{X} = \bigcup_{(\xi_0:\eta_0)} \Phi(F_{(\xi_0:\eta_0)})$ is the union of these lines: a ruled surface.$\square$

Exercise 7.8

Problem. Let X be a noetherian scheme, let $\mathcal{E}$ be a coherent locally free sheaf on X, and let $\pi : \mathbf{P}(\mathcal{E}) \to X$ be the corresponding projective space bundle. Show that there is a natural 1-1 correspondence between sections of π (i.e., morphisms $\sigma : X \to \mathbf{P}(\mathcal{E})$ such that $\pi \circ \sigma = \mathrm{id}_X$) and quotient invertible sheaves $\mathcal{E} \to \mathcal{L} \to 0$ of $\mathcal{E}$.

Solution. Apply Prop. 7.12 with $Y = X$ and $g = \mathrm{id}_X$: there $\mathbf{P}(\mathcal{E})$ comes with the surjection $\pi^*\mathcal{E} \to \mathcal{O}(1)$ (7.11(b)), and a morphism $f : X \to \mathbf{P}(\mathcal{E})$ over X gives the pair $(f^*\mathcal{O}(1),\ f^*(\pi^*\mathcal{E} \to \mathcal{O}(1)))$ with $f^*\pi^*\mathcal{E} = (\pi f)^*\mathcal{E} = \mathcal{E}$. A morphism over X from X is exactly a section σ of π (the condition $\pi\sigma = \mathrm{id}$ *is* being over X).

Define $\Theta(\sigma) = \big(\mathcal{E} \to \sigma^*\mathcal{O}(1)\big)$; it is a surjection onto an invertible sheaf, i.e. a quotient invertible sheaf. Two quotients $\mathcal{E} \to \mathcal{L}$, $\mathcal{E} \to \mathcal{L}'$ are identified if there is an isomorphism $\mathcal{L} \cong \mathcal{L}'$ compatible with the maps from $\mathcal{E}$ (equivalently, if they have the same kernel).

Surjective. Given $\mathcal{E} \to \mathcal{L} \to 0$ with $\mathcal{L}$ invertible, Prop. 7.12 (proof) gives a morphism $f : X \to \mathbf{P}(\mathcal{E})$ over X with $\mathcal{L} \cong f^*\mathcal{O}(1)$ such that the given surjection is f^* of $\pi^*\mathcal{E} \to \mathcal{O}(1)$; being over X it is a section, and $\Theta(f)$ is the given quotient.

Injective. The uniqueness statement in the proof of 7.12 says that for a given pair $(\mathcal{L},\ g^*\mathcal{E} \to \mathcal{L})$ there is a unique f with $\mathcal{L} \cong f^*\mathcal{O}(1)$ and with the map obtained from $\pi^*\mathcal{E} \to \mathcal{O}(1)$ by f^* (it is deduced from the uniqueness in 7.1 after localizing so that $\mathcal{E}$ is free). So if $\Theta(\sigma)$ and $\Theta(\sigma')$ are isomorphic quotients, then $\sigma = \sigma'$.

Thus $\sigma \mapsto \Theta(\sigma)$ is a bijection between sections of π and isomorphism classes of quotient invertible sheaves of $\mathcal{E}$; its inverse sends $\mathcal{E} \to \mathcal{L}$ to the section determined by $\mathcal{L}$ and the images of local bases (locally, if $\mathcal{E} = \mathcal{O}^{n+1}$, the section $(\mathcal{L};\ s_0, \dots, s_n)$ of 7.1). Naturality: it is compatible with restriction to open subsets of X and with isomorphisms of $\mathcal{E}$. □

Exercise 7.9

Problem. Let X be a regular noetherian scheme, and $\mathcal{E}$ a locally free coherent sheaf of rank ≥ 2 on X.

(a) Show that $\operatorname{Pic}\mathbf{P}(\mathcal{E}) \cong \operatorname{Pic} X \times \mathbf{Z}$.

(b) If $\mathcal{E}'$ is another locally free coherent sheaf on X, show that $\mathbf{P}(\mathcal{E}) \cong \mathbf{P}(\mathcal{E}')$ (over X) if and only if there is an invertible sheaf $\mathcal{L}$ on X such that $\mathcal{E}' \cong \mathcal{E} \otimes \mathcal{L}$.

Solution. *Conventions.* A regular noetherian scheme is a finite disjoint union of connected, hence integral, regular schemes, so as usual I assume X connected (hence integral); for disconnected X the statement (a) holds componentwise, i.e. with $\mathbf{Z}$ replaced by the group of locally constant functions $X \to \mathbf{Z}$ (e.g. for X two points, $\operatorname{Pic} X = 0$ but $\operatorname{Pic} P = \mathbf{Z}^2$, so connectedness is really needed in (a)). Part (b) needs no such assumption: $\mathbf{P}(\mathcal{E})$ and $\mathbf{P}(\mathcal{E}')$ are compatible with the decomposition of X into its finitely many connected components, and invertible sheaves on X are glued from invertible sheaves on the components. Let $\operatorname{rank}\mathcal{E} = n+1$, $n \geq 1$, $P = \mathbf{P}(\mathcal{E})$, $\pi : P \to X$ the projection. Recall $\pi_*\mathcal{O}_P(l) = S^l(\mathcal{E})$ for $l \geq 0$ (Prop. 7.11(a)), in particular $\pi_*\mathcal{O}_P = \mathcal{O}_X$ and $\pi_*\mathcal{O}_P(1) = \mathcal{E}$, and π is proper (7.10(a)).

Lemma A (local structure of line bundles on $\mathbf{P}^n$-bundles). Let $\pi : P \to X$ be a morphism such that X is covered by opens U with $\pi^{-1}(U) \cong \mathbf{P}^n_U$ ($n \geq 1$), X regular noetherian, and let $\mathcal{N} \in \operatorname{Pic} P$. For $x \in X$ the fibre $P_x \cong \mathbf{P}^n_{\kappa(x)}$ carries $\mathcal{N}|_{P_x} \cong \mathcal{O}(d(x))$ for a unique $d(x) \in \mathbf{Z}$ (independent of the identification of P_x with $\mathbf{P}^n_{\kappa(x)}$, since an automorphism of $\mathbf{P}^n_\kappa$ pulls $\mathcal{O}(1)$ back to a generator of $\operatorname{Pic} = \mathbf{Z}$ having sections, i.e. to $\mathcal{O}(1)$). Then every $x \in X$ has an open neighbourhood U with a trivialization $\psi : \pi^{-1}(U) \cong \mathbf{P}^n_U$ such that $\mathcal{N}|_{\pi^{-1}U} \cong \psi^*\mathcal{O}(d(x))$. In particular $d : X \to \mathbf{Z}$ is locally constant.

Proof. Let $R = \mathcal{O}_{X,x}$, a regular local ring, hence a UFD (6.11A / Auslander–Buchsbaum). The scheme $\mathbf{P}^n_R$ is noetherian, integral, separated, and regular (it is covered by $\operatorname{Spec} R[y_1, \ldots, y_n]$, and polynomial rings over regular rings are regular), hence locally factorial, so $\operatorname{Pic} \mathbf{P}^n_R = \operatorname{Cl} \mathbf{P}^n_R$ (6.16). The hyperplane $H = V_+(x_0)$ is a prime divisor with $\mathbf{P}^n_R - H = \mathbf{A}^n_R = \operatorname{Spec} R[y_1, \ldots, y_n]$, and $R[y_1, \ldots, y_n]$ is a UFD, so $\operatorname{Cl} \mathbf{A}^n_R = 0$ (6.2). By (6.5), $\mathbf{Z} \to \operatorname{Cl} \mathbf{P}^n_R \to \operatorname{Cl} \mathbf{A}^n_R = 0$ is exact, so $\operatorname{Cl} \mathbf{P}^n_R$ is generated by $[H]$, i.e. by $\mathcal{O}(1)$. Hence $\mathcal{N}|_{P_R} \cong \mathcal{O}(d)$ for some d (here $P_R = P \times_X \operatorname{Spec} R$, trivialized over an affine neighbourhood $U_0 = \operatorname{Spec} A$ of x), and restricting to the closed fibre gives $d = d(x)$.

The isomorphism is a nowhere vanishing section σ of the invertible sheaf $\mathcal{G} = \mathcal{N} \otimes \mathcal{O}(-d)$ over $P_R = \mathbf{P}^n_R$. Since $\Gamma(\mathbf{P}^n_R, \mathcal{G}) = \Gamma(\mathbf{P}^n_A, \mathcal{G}) \otimes_A R$ (flat base change; $\mathbf{P}^n_A$ is quasi-compact and separated), σ extends to a section σ_U over $\pi^{-1}(U)$ for some smaller affine neighbourhood $U = D(a)$ of x. The zero set Z of σ_U is closed in $\pi^{-1}(U)$ and misses $\pi^{-1}(x)$; as π is proper (locally on X it is a projection $\mathbf{P}^n_U \to U$, and properness is local on the base, 4.8(f)), $\pi(Z)$ is closed in U and does not contain x. Replacing U by $U - \pi(Z)$, σ_U is nowhere vanishing, so $\mathcal{O} \to \mathcal{G}|_{\pi^{-1}U}$, $1 \mapsto \sigma_U$, is a surjection of invertible sheaves, hence an isomorphism (Ex. 7.1), i.e. $\mathcal{N} \cong \mathcal{O}(d)$ there. Then for $x' \in U$, $d(x') = d$. □

(a) Define $\Psi : \operatorname{Pic} X \times \mathbf{Z} \to \operatorname{Pic} P$, $(\mathcal{L}, d) \mapsto \pi^*\mathcal{L} \otimes \mathcal{O}_P(d)$, a homomorphism.

Injective. Suppose $\pi^*\mathcal{L} \otimes \mathcal{O}_P(d) \cong \mathcal{O}_P$. Restrict to a fibre $P_x \cong \mathbf{P}^n_{\kappa(x)}$: $\pi^*\mathcal{L}$ is trivial there, so $\mathcal{O}(d) \cong \mathcal{O}$ on $\mathbf{P}^n_\kappa$, whence $d = 0$ ($n \geq 1$). Then $\pi^*\mathcal{L} \cong \mathcal{O}_P$ and applying π_* and the projection formula, $\mathcal{L} \cong \mathcal{L} \otimes \pi_*\mathcal{O}_P = \pi_*\pi^*\mathcal{L} \cong \pi_*\mathcal{O}_P = \mathcal{O}_X$.

Surjective. Let $\mathcal{N} \in \operatorname{Pic} P$. By Lemma A (applied to the local trivializations coming from free neighbourhoods of $\mathcal{E}$), $d(x)$ is locally constant, hence equal to a constant d on the connected space X, and each point has a neighbourhood U with $\mathcal{N}(-d)|_{\pi^{-1}U} \cong \mathcal{O}$. Put $\mathcal{L} = \pi_*(\mathcal{N}(-d))$. On such a U, $\mathcal{L}|_U \cong \pi_*\mathcal{O} = \mathcal{O}_U$, so $\mathcal{L}$ is invertible, and the natural map $\pi^*\pi_*\mathcal{N}(-d) \to \mathcal{N}(-d)$ is an isomorphism over U (it is the identity of $\mathcal{O}$ after the trivialization). Hence $\mathcal{N} \cong \pi^*\mathcal{L} \otimes \mathcal{O}_P(d) = \Psi(\mathcal{L}, d)$. So Ψ is an isomorphism.

(b) ($\Leftarrow$) Let $\mathcal{E}' \cong \mathcal{E} \otimes \mathcal{L}$. Put $\mathcal{S} = S(\mathcal{E})$ and consider $\mathcal{S} * \mathcal{L} = \bigoplus_d \mathcal{S}_d \otimes \mathcal{L}^d$ as in the Lemma numbered 7.9 of the text. The natural map $\mathcal{E} \otimes \mathcal{L} \to (\mathcal{S} * \mathcal{L})_1$ induces an algebra homomorphism $S(\mathcal{E} \otimes \mathcal{L}) \to \mathcal{S} * \mathcal{L}$, which is an isomorphism (check locally where $\mathcal{L}$ and $\mathcal{E}$ are free: $S^d(\mathcal{E} \otimes \mathcal{L}) \cong S^d(\mathcal{E}) \otimes \mathcal{L}^d$). The Lemma 7.9 gives $\operatorname{Proj}(\mathcal{S} * \mathcal{L}) \cong \operatorname{Proj} \mathcal{S}$ over X, i.e. $\mathbf{P}(\mathcal{E} \otimes \mathcal{L}) \cong \mathbf{P}(\mathcal{E})$ (and $\mathcal{O}_{\mathbf{P}(\mathcal{E} \otimes \mathcal{L})}(1) \cong \mathcal{O}_{\mathbf{P}(\mathcal{E})}(1) \otimes \pi^*\mathcal{L}$).

($\Rightarrow$) Let $f : \mathbf{P}(\mathcal{E}') \to \mathbf{P}(\mathcal{E})$ be an isomorphism over X, with projections π', π ($\pi f = \pi'$). Put $\mathcal{N} = f^*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1) \in \operatorname{Pic} \mathbf{P}(\mathcal{E}')$. By (a) applied to $\mathcal{E}'$ (rank ≥ 2), $\mathcal{N} \cong \pi'^*\mathcal{L} \otimes \mathcal{O}_{\mathbf{P}(\mathcal{E}')}(d)$ for some $\mathcal{L}, d$. On a fibre over x, f_x is an isomorphism of copies of $\mathbf{P}^n_{\kappa(x)}$, so $f_x^*\mathcal{O}(1) \cong \mathcal{O}(1)$ (it is a generator of $\operatorname{Pic} = \mathbf{Z}$ with nonzero sections), whereas $\mathcal{N}|_{P'_x} \cong \mathcal{O}(d)$; thus $d = 1$. Apply π'_*:

$$\pi'_*\mathcal{N} = \pi_* f_* f^*\mathcal{O}(1) = \pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1) = \mathcal{E}, \qquad \pi'_*(\pi'^*\mathcal{L} \otimes \mathcal{O}(1)) = \mathcal{L} \otimes \pi'_*\mathcal{O}(1) = \mathcal{L} \otimes \mathcal{E}'$$

(projection formula, 7.11(a)). So $\mathcal{E} \cong \mathcal{E}' \otimes \mathcal{L}$, i.e. $\mathcal{E}' \cong \mathcal{E} \otimes \mathcal{L}^{-1}$. □

Exercise 7.10

Problem. $\mathbf{P}^n$*-Bundles Over a Scheme.* Let X be a noetherian scheme.

(a) By analogy with the definition of a vector bundle (Ex. 5.18), define the notion of a *projective n-space bundle* over X, as a scheme P with a morphism $\pi : P \to X$ such that P is locally isomorphic to $U \times \mathbf{P}^n$, $U \subseteq X$ open, and the transition automorphisms on $\operatorname{Spec} A \times \mathbf{P}^n$ are given by A-linear automorphisms of the homogeneous coordinate ring $A[x_0, \ldots, x_n]$ (e.g., $x_i' = \sum a_{ij} x_j$, $a_{ij} \in A$).

(b) If $\mathcal{E}$ is a locally free sheaf of rank $n+1$ on X, then $\mathbf{P}(\mathcal{E})$ is a $\mathbf{P}^n$-bundle over X.

(c) Assume that X is regular, and show that every $\mathbf{P}^n$-bundle P over X is isomorphic to $\mathbf{P}(\mathcal{E})$ for some locally free $\mathcal{E}$ on X. [*Hint:* Let $U \subseteq X$ be an open set such that $\pi^{-1}(U) \cong U \times \mathbf{P}^n$, and let $\mathcal{L}_0$ be the invertible sheaf $\mathcal{O}(1)$ on $U \times \mathbf{P}^n$. Show that $\mathcal{L}_0$ extends to an invertible sheaf $\mathcal{L}'$ on P. Then show that $\pi_* \mathcal{L}' = \mathcal{E}$ is a locally free sheaf on X and that $P \cong \mathbf{P}(\mathcal{E})$.] Can you weaken the hypothesis "X regular"?

(d) Conclude (in the case X regular) that we have a 1-1 correspondence between $\mathbf{P}^n$-bundles over X, and equivalence classes of locally free sheaves of rank $n+1$ under the equivalence relation $\mathcal{E} \sim \mathcal{E}'$ if and only if $\mathcal{E}' \cong \mathcal{E} \otimes \mathcal{M}$ for some invertible sheaf $\mathcal{M}$ on X.

(Part (c) is starred in the book.)

Solution. (a) Definition. A *projective n-space bundle* over X is a scheme P with a morphism $\pi : P \to X$, together with an open cover $\{U_i\}$ of X and isomorphisms $\psi_i : \pi^{-1}(U_i) \to \mathbf{P}^n_{U_i} = U_i \times \mathbf{P}^n$ over U_i (local trivializations), such that for all i, j the transition automorphism $\psi_i \psi_j^{-1}$ of $\mathbf{P}^n_{U_i \cap U_j}$ over $U_i \cap U_j$ is *linear*: over every affine open $\operatorname{Spec} A \subseteq U_i \cap U_j$ it is Proj of an A-linear graded automorphism of $A[x_0, \ldots, x_n]$, $x_a \mapsto \sum_b a_{ab} x_b$ with $(a_{ab}) \in GL_{n+1}(A)$. (This is stable under restriction to smaller affines, so it suffices to ask it on an affine cover.) An isomorphism of $\mathbf{P}^n$-bundles over X is an isomorphism of X-schemes. Exactly as for vector bundles, the transition data are cocycles with values in PGL_{n+1} in place of GL_n.

(b) Let $\mathcal{E}$ be locally free of rank $n+1$. Choose an open cover $\{U_i\}$ of X and isomorphisms $\varepsilon_i : \mathcal{E}|_{U_i} \cong \mathcal{O}_{U_i}^{n+1}$. Then $S(\mathcal{E})|_{U_i} \cong \mathcal{O}_{U_i}[T_0, \ldots, T_n]$ (the symmetric algebra of a free module of rank $n+1$ is a polynomial ring, see before Ex. 5.16), so $\pi^{-1}(U_i) = \operatorname{Proj} S(\mathcal{E}|_{U_i}) \cong \operatorname{Proj} \mathcal{O}_{U_i}[T_0, \ldots, T_n] = \mathbf{P}^n_{U_i}$ (Example 7.8.7); call this ψ_i. Over an affine $\operatorname{Spec} A \subseteq U_i \cap U_j$, $\varepsilon_i \varepsilon_j^{-1}$ is an automorphism of A^{n+1}, i.e. a matrix $(a_{ab}) \in GL_{n+1}(A)$, and the induced automorphism of $S(A^{n+1}) = A[T_0, \ldots, T_n]$ is $T_a \mapsto \sum_b a_{ab} T_b$. Hence $\psi_i \psi_j^{-1} = \operatorname{Proj}$ of this A-linear graded automorphism. So $\mathbf{P}(\mathcal{E})$ is a $\mathbf{P}^n$-bundle.

(c) Let $\pi : P \to X$ be a $\mathbf{P}^n$-bundle over the regular noetherian scheme X. If $n = 0$ then π is an isomorphism (P is locally $U \times \mathbf{P}^0 = U$) and $P = \mathbf{P}(\mathcal{O}_X)$; so let $n \geq 1$. Treating the connected components separately, assume X connected, hence integral (for disconnected X apply this to each of the finitely many components and take the disjoint union of the resulting sheaves).

Step 1: P is a regular integral noetherian scheme. P is locally $\mathbf{P}^n_U$ with U regular, hence regular and noetherian ($\mathbf{P}^n_U$ is covered by spectra of polynomial rings over regular rings), so every local ring of P is a UFD (6.11A). It is connected: π is open (locally the projection $\mathbf{P}^n_U \to U$, flat of finite presentation), and a decomposition $P = P_1 \sqcup P_2$ into open-closed sets would give

$X = \pi(P_1) \sqcup \pi(P_2)$ with both open (each fibre $\mathbf{P}^n_{\kappa(x)}$ is connected, hence lies in one P_i), forcing one of them to be empty. A connected regular noetherian scheme is integral.

Step 2: extension of $\mathcal{L}_0$. Let $U \neq \emptyset$ be one of the U_i, $V = \pi^{-1}(U)$, $\psi : V \cong \mathbf{P}^n_U$, $\mathcal{L}_0 = \psi^* p_2^* \mathcal{O}(1) \in \operatorname{Pic} V$. (This is the argument of 6.5/6.16, done directly so that no separatedness of X is needed.) Choose a nonzero rational section s of $\mathcal{L}_0$ (possible since V is integral). For each prime divisor $Y \subset V$ the local ring $\mathcal{O}_{V,Y}$ is a DVR, and let $n_Y = v_Y(s)$ (valuation with respect to a generator of $(\mathcal{L}_0)_Y$); $n_Y = 0$ for all but finitely many Y since V is noetherian, and $\mathcal{L}_0 \cong \mathcal{O}_V(D_0)$, $D_0 = \sum n_Y Y$, via $f \mapsto fs$. Let $D = \sum n_Y \overline{Y}$ on P ($\overline{Y}$ the closure, a prime divisor of P with $\overline{Y} \cap V = Y$). Every local ring of P is a UFD, so every prime divisor of P is locally principal (a height-one prime of a UFD is principal), hence D is a Cartier divisor, $\mathcal{L}' = \mathcal{O}_P(D)$ is invertible, and $\mathcal{L}'|_V = \mathcal{O}_V(D_0) \cong \mathcal{L}_0$.

Step 3: $\mathcal{L}'$ has degree 1 on all fibres. Let η be the generic point of X; $P_\eta \cong \mathbf{P}^n_{K(X)}$ and $\mathcal{L}'|_{P_\eta} = \mathcal{L}_0|_{P_\eta} \cong \mathcal{O}(1)$, so $d(\eta) = 1$ in the notation of Lemma A (which was proved in Ex. 7.9 and applies to any $\mathbf{P}^n$-bundle). By Lemma A, d is locally constant, hence $d \equiv 1$ on the connected X, and every $x \in X$ has an open neighbourhood U_x with a trivialization $\psi_x : \pi^{-1}(U_x) \cong \mathbf{P}^n_{U_x}$ and $\mathcal{L}'|_{\pi^{-1}U_x} \cong \psi_x^* \mathcal{O}(1)$.

Step 4: $\mathcal{E} = \pi_* \mathcal{L}'$ is locally free of rank $n+1$ and $P \cong \mathbf{P}(\mathcal{E})$. Over U_x, $\pi_* \mathcal{L}' \cong \pi_* \mathcal{O}_{\mathbf{P}^n_{U_x}}(1) \cong \mathcal{O}_{U_x} x_0 \oplus \cdots \oplus \mathcal{O}_{U_x} x_n$ (Thm. 5.13 / 7.11(a)), so $\mathcal{E}$ is locally free of rank $n+1$ (in particular coherent). The natural map $\pi^* \mathcal{E} = \pi^* \pi_* \mathcal{L}' \to \mathcal{L}'$ is, over U_x, the map $\mathcal{O}^{n+1} \to \mathcal{O}(1)$, $e_a \mapsto x_a$, which is surjective since $x_0, \ldots, x_n$ generate $\mathcal{O}(1)$ (7.11(b)). By Prop. 7.12 (with $Y = P$, $g = \pi$) there is a unique morphism $f : P \to \mathbf{P}(\mathcal{E})$ over X with $f^* \mathcal{O}(1) \cong \mathcal{L}'$ compatibly with the surjections from $\pi^* \mathcal{E}$. Over U_x, after identifying $\mathbf{P}(\mathcal{E})|_{U_x}$ with $\mathbf{P}^n_{U_x}$ via the basis $x_0, \ldots, x_n$ of $\mathcal{E}|_{U_x}$, f is the morphism determined (7.1) by the line bundle $\psi_x^* \mathcal{O}(1)$ and its sections $\psi_x^* x_a$, which by uniqueness in 7.1 is ψ_x itself. So f is an isomorphism over each U_x, hence an isomorphism: $P \cong \mathbf{P}(\mathcal{E})$.

Weakening the hypothesis. Regularity was used only to know that (i) the local rings of X are UFD's, so that line bundles on $\mathbf{P}^n$ over them are $\mathcal{O}(d)$, and (ii) every local ring of P is a UFD, so that line bundles extend from open sets (Step 2). Both hold if X is merely a noetherian *locally factorial* scheme (all local rings UFD's) (then $\mathbf{P}^n_R$ and P are again locally factorial). So the statements of (c) and (d), and of Ex. 7.9, hold for locally factorial X. (In general the obstruction to P being some $\mathbf{P}(\mathcal{E})$ is a class in $H^2(X, \mathcal{O}_X^*)$, which vanishes in the locally factorial case; I do not claim a counterexample beyond that case.)

(d) Let X be regular. By (b), $\mathcal{E} \mapsto \mathbf{P}(\mathcal{E})$ sends rank $n+1$ locally free sheaves to $\mathbf{P}^n$-bundles; by Ex. 7.9(b)($\Leftarrow$) it is constant on the equivalence classes of $\sim$ ($\mathbf{P}(\mathcal{E} \otimes \mathcal{M}) \cong \mathbf{P}(\mathcal{E})$); by 7.9(b)($\Rightarrow$) (needs $n \geq 1$) two sheaves with isomorphic projective bundles are equivalent, so the induced map $\{\text{classes}\} \to \{\mathbf{P}^n\text{-bundles}\}/\cong$ is injective; and by (c) it is surjective. (For $n = 0$ all line bundles are equivalent and $\mathbf{P}^0$-bundles are X itself.) This gives the 1-1 correspondence. □

Exercise 7.11

Problem. On a noetherian scheme X, different sheaves of ideals can give rise to isomorphic blown up schemes.

(a) If $\mathcal{I}$ is any coherent sheaf of ideals on X, show that blowing up $\mathcal{I}^d$ for any $d \geq 1$ gives a scheme isomorphic to the blowing up of $\mathcal{I}$ (cf. Ex. 5.13).

(b) If $\mathcal{I}$ is any coherent sheaf of ideals, and if $\mathcal{J}$ is an invertible sheaf of ideals, then $\mathcal{I}$ and $\mathcal{I} \cdot \mathcal{J}$ give isomorphic blowings-up.

(c) If X is regular, show that (7.17) can be strengthened as follows. Let $U \subseteq X$ be the largest open set such that $f : f^{-1}U \to U$ is an isomorphism. Then $\mathcal{I}$ can be chosen such that the corresponding closed subscheme Y has support equal to $X - U$.

Solution. Write $\mathcal{S} = \bigoplus_{n \geq 0} \mathcal{I}^n$, so $\tilde{X} = \operatorname{Proj} \mathcal{S}$.

(a) The sheaf of graded algebras of the blow-up of $\mathcal{I}^d$ is $\mathcal{S}' = \bigoplus_n (\mathcal{I}^d)^n = \bigoplus_n \mathcal{I}^{dn} = \mathcal{S}^{(d)}$, the d-th Veronese subalgebra. For each open affine U, $\mathcal{S}(U) = \bigoplus I^n$ is generated by its degree-1 part over $\mathcal{S}_0(U) = \mathcal{O}(U)$, so by Ex. 5.13, $\operatorname{Proj} \mathcal{S}(U)^{(d)} \cong \operatorname{Proj} \mathcal{S}(U)$ over U (with $\mathcal{O}(1)$ corresponding to $\mathcal{O}(d)$). This isomorphism is canonical: for $f \in S_e$ ($e \geq 1$) it identifies $D_+(f) \subseteq \operatorname{Proj} S$ with $D_+(f^d) \subseteq \operatorname{Proj} S^{(d)}$ via the equality of rings $S_{(f)} = (S^{(d)})_{(f^d)}$ (an element g/f^k, $g \in S_{ek}$, equals $gf^{(d-1)k}/f^{dk}$ with numerator of degree divisible by d). It is therefore compatible with passing to smaller affine opens $U' \subseteq U$, and the isomorphisms glue to an isomorphism of the blow-ups over X.

(b) Since $\mathcal{J} \subseteq \mathcal{O}_X$ is invertible, locally $\mathcal{J} = (h)$ with $\mathcal{O} \to \mathcal{J}$, $1 \mapsto h$ an isomorphism, so h is a nonzerodivisor. Hence multiplication by h^n gives isomorphisms $\mathcal{I}^n \cong h^n \mathcal{I}^n = (\mathcal{I}\mathcal{J})^n$ locally, and globally the multiplication map $\mathcal{I}^n \otimes \mathcal{J}^{\otimes n} \to \mathcal{I}^n \mathcal{J}^n = (\mathcal{I}\mathcal{J})^n$ is an isomorphism (check locally). These are compatible with the products, so $\mathcal{S}' := \bigoplus_n (\mathcal{I}\mathcal{J})^n \cong \mathcal{S} * \mathcal{J}$ (notation of the Lemma numbered 7.9 in the text, with $\mathcal{S}'_d = \mathcal{S}_d \otimes \mathcal{J}^d$). Both satisfy (†) (Lemma 7.9). By that Lemma, $\operatorname{Proj} \mathcal{S}' \cong \operatorname{Proj} \mathcal{S}$ over X: the blow-ups of $\mathcal{I}\mathcal{J}$ and of $\mathcal{I}$ are isomorphic (and $\mathcal{O}_{\tilde{X}}(1)$ changes by $\pi^* \mathcal{J}$).

(c) Let X be a regular noetherian integral scheme (a nonsingular variety), $f : Z \to X$ as in (7.17), and $\mathcal{I}_0$ a coherent sheaf of ideals with $Z \cong \operatorname{Bl}_{\mathcal{I}_0} X$ over X; then $\mathcal{I}_0 \neq 0$. Let U be the largest open subset over which f is an isomorphism (the union of all such opens). I produce $\mathcal{I}$ with $\operatorname{Bl}_{\mathcal{I}} X \cong Z$ over X and $\operatorname{Supp}(\mathcal{O}_X/\mathcal{I}) = X - U$.

Step 1: removing the divisorial part. For each prime divisor $Y \subset X$ (closed integral of codimension 1) the ring $\mathcal{O}_{X,Y}$ is a DVR with valuation v_Y; let $m_Y = \min\{v_Y(g) : g \in (\mathcal{I}_0)_Y\}$. Since $\mathcal{I}_0 \neq 0$ is coherent, $m_Y = 0$ except for finitely many Y. Let $D = \sum m_Y Y$, an effective Cartier divisor (X is locally factorial), and $\mathcal{J} = \mathcal{O}_X(-D) \subseteq \mathcal{O}_X$, an invertible sheaf of ideals, locally generated at x by $h = \prod \pi_Y^{m_Y}$ ($Y \ni x$, π_Y a generator of the prime ideal of Y in the UFD $\mathcal{O}_{X,x}$). If $g \in (\mathcal{I}_0)_x$ then $v_Y(g) \geq m_Y$ for all $Y \ni x$, so $h \mid g$ in the UFD $\mathcal{O}_{X,x}$. Thus $\mathcal{I}_0 \subseteq \mathcal{J}$, and $\mathcal{I} := \mathcal{J}^{-1}\mathcal{I}_0 \subseteq \mathcal{O}_X$ is a coherent ideal sheaf with $\mathcal{I}_0 = \mathcal{J}\mathcal{I}$. By (b), $\operatorname{Bl}_{\mathcal{I}} X \cong \operatorname{Bl}_{\mathcal{I}_0} X \cong Z$ over X. Moreover $\mathcal{I}_Y = \mathcal{O}_{X,Y}$ for every prime divisor Y (its valuation minimum is $m_Y - m_Y = 0$), so *no irreducible component of $V(\mathcal{I})$ has codimension* ≤ 1 (and $\mathcal{I} \neq 0$).

Step 2: $\operatorname{Supp}(\mathcal{O}_X/\mathcal{I}) = X - U$. Let $\pi : \tilde{X} = Z \to X$ be the blow-up of $\mathcal{I}$. By (7.13(b)), π is an isomorphism over $X - V(\mathcal{I})$, so $X - V(\mathcal{I}) \subseteq U$, i.e. $X - U \subseteq V(\mathcal{I})$. Conversely let $x \in V(\mathcal{I})$ and suppose $x \in U$; let $W \ni x$ be an open set with $\pi^{-1}W \to W$ an isomorphism. By (7.13(a)) $\mathcal{I} \cdot \mathcal{O}_{\tilde{X}}$ is invertible, and over W it corresponds under the isomorphism to $\mathcal{I}|_W$; so $\mathcal{I}_x = (g)$ is a principal ideal of $\mathcal{O}_{X,x}$, with $g \neq 0$ (as $\mathcal{I} \neq 0$) and g a non-unit (as $x \in V(\mathcal{I})$). By Krull's Hauptidealsatz a minimal prime over (g) has height 1, i.e. $V(\mathcal{I})$ has an irreducible component of codimension 1 through x, contradicting Step 1. Hence $V(\mathcal{I}) \subseteq X - U$. So $V(\mathcal{I}) = X - U$, i.e. the closed subscheme Y defined by $\mathcal{I}$ has support $X - U$. □

Exercise 7.12

Problem. Let X be a noetherian scheme, and let Y, Z be two closed subschemes, neither one containing the other. Let $\tilde{X}$ be obtained by blowing up $Y \cap Z$ (defined by the ideal sheaf $\mathcal{I}_Y + \mathcal{I}_Z$). Show that the strict transforms $\tilde{Y}$ and $\tilde{Z}$ of Y and Z in $\tilde{X}$ do not meet.

Solution. The question is local on X (the sets $\pi^{-1}(U)$, $U \subseteq X$ affine open, cover $\tilde{X}$), so let $X = \operatorname{Spec} A$, $\mathcal{I}_Y = \mathfrak{a}$, $\mathcal{I}_Z = \mathfrak{b}$, $I = \mathfrak{a} + \mathfrak{b}$, $S = \bigoplus_{d \geq 0} I^d$ and $\tilde{X} = \operatorname{Proj} S$.

By Cor. 7.15 and its proof, $\tilde{Y}$ is the blow-up of Y along the ideal $I\mathcal{O}_Y = (I + \mathfrak{a})/\mathfrak{a}$ (the ideal of $Y \cap Z$ in Y), namely $\operatorname{Proj} S'$ with $S' = \bigoplus_d (I\mathcal{O}_Y)^d = \bigoplus_d (I^d + \mathfrak{a})/\mathfrak{a} \cong \bigoplus_d I^d/(I^d \cap \mathfrak{a})$, embedded in $\operatorname{Proj} S$ via the surjection $S \to S'$ whose kernel is the graded ideal $\mathfrak{A} = \bigoplus_d (I^d \cap \mathfrak{a})$. So $\tilde{Y} = V_+(\mathfrak{A})$ and likewise $\tilde{Z} = V_+(\mathfrak{B})$, $\mathfrak{B} = \bigoplus_d (I^d \cap \mathfrak{b})$. Hence $\tilde{Y} \cap \tilde{Z} = V_+(\mathfrak{A} + \mathfrak{B})$.

Claim: $(\mathfrak{A} + \mathfrak{B})_d = I^d$ for every $d \geq 1$. We have $\subseteq$ trivially. Conversely $I^d = \sum_{i=0}^{d} \mathfrak{a}^i \mathfrak{b}^{d-i}$. For $i \geq 1$, $\mathfrak{a}^i \mathfrak{b}^{d-i} \subseteq \mathfrak{a} \cap I^d$, and $\mathfrak{b}^d \subseteq \mathfrak{b} \cap I^d$. So $I^d \subseteq (I^d \cap \mathfrak{a}) + (I^d \cap \mathfrak{b})$.

Hence $\mathfrak{A} + \mathfrak{B} \supseteq S_+ = \bigoplus_{d \geq 1} S_d$, so $V_+(\mathfrak{A} + \mathfrak{B}) = \emptyset$ (a homogeneous prime of S containing $\mathfrak{A} + \mathfrak{B}$ contains S_+, so it is not a point of $\operatorname{Proj} S$). Thus $\tilde{Y} \cap \tilde{Z} = \emptyset$. (The hypothesis that neither of Y, Z contains the other is not needed for the disjointness; it only guarantees that the strict transforms are the honest, proper, transforms.) □

Exercise 7.13

Problem. *A Complete Nonprojective Variety.* Let k be an algebraically closed field of char $\neq 2$. Let $C \subseteq \mathbf{P}^2_k$ be the nodal cubic curve $y^2 z = x^3 + x^2 z$. If $P_0 = (0, 0, 1)$ is the singular point, then $C - P_0$ is isomorphic to the multiplicative group $\mathbf{G}_m = \operatorname{Spec} k[t, t^{-1}]$ (Ex. 6.7). For each $a \in k$, $a \neq 0$, consider the translation of $\mathbf{G}_m$ given by $t \mapsto at$. This induces an automorphism of C which we denote by φ_a.

Now consider $C \times (\mathbf{P}^1 - \{0\})$ and $C \times (\mathbf{P}^1 - \{\infty\})$. We glue their open subsets $C \times (\mathbf{P}^1 - \{0, \infty\})$ by the isomorphism $\varphi : \langle P, u \rangle \mapsto \langle \varphi_u(P), u \rangle$ for $P \in C$, $u \in \mathbf{G}_m = \mathbf{P}^1 - \{0, \infty\}$. Thus we obtain a scheme X, which is our example. The projections to the second factor are compatible with φ, so there is a natural morphism $\pi : X \to \mathbf{P}^1$.

(a) Show that π is a proper morphism, and hence that X is a complete variety over k.

(b) Use the method of (Ex. 6.9) to show that $\operatorname{Pic}(C \times \mathbf{A}^1) \cong \mathbf{G}_m \times \mathbf{Z}$ and $\operatorname{Pic}(C \times (\mathbf{A}^1 - \{0\})) \cong \mathbf{G}_m \times \mathbf{Z} \times \mathbf{Z}$. [*Hint:* If A is a domain and $*$ denotes the group of units, then $(A[u])^* \cong A^*$ and $(A[u, u^{-1}])^* \cong A^* \times u^{\mathbf{Z}}$.]

(c) Now show that the restriction map $\operatorname{Pic}(C \times \mathbf{A}^1) \to \operatorname{Pic}(C \times (\mathbf{A}^1 - \{0\}))$ is of the form $\langle t, n \rangle \mapsto \langle t, 0, n \rangle$, and that the automorphism φ of $C \times (\mathbf{A}^1 - \{0\})$ induces a map of the form $\langle t, d, n \rangle \mapsto \langle t, d + n, n \rangle$ on its Picard group.

(d) Conclude that the image of the restriction map $\operatorname{Pic} X \to \operatorname{Pic}(C \times \{0\})$ consists entirely of divisors of degree 0 on C. Hence X is not projective over k and π is not a projective morphism.

(Starred in the book.)

Solution. 0. Structure of C. Use homogeneous coordinates $(\tau_0 : \tau_1)$ on $\mathbf{P}^1$, $0 = (1 : 0)$, $\infty = (0 : 1)$, and $t = \tau_1/\tau_0$ (the parameter of $\mathbf{G}_m = \mathbf{P}^1 - \{0, \infty\}$). The morphism

$$\nu : \mathbf{P}^1 \to C, \qquad (\tau_0 : \tau_1) \mapsto \big(4\tau_0\tau_1(\tau_1 - \tau_0) : 4\tau_0\tau_1(\tau_1 + \tau_0) : (\tau_1 - \tau_0)^3\big)$$

lands on C (a direct check: $y^2z = 16\tau_0^2\tau_1^2(\tau_1+\tau_0)^2(\tau_1-\tau_0)^3 = x^2(x+z)$), and I use the following standard facts about the nodal cubic.

(F1) ν is finite and birational, $\nu^{-1}(P_0) = \{0,\infty\}$, and ν restricts to an isomorphism $\mathbf{G}_m \cong C - P_0 =: C^\circ$ (this is the isomorphism of Ex. 6.7).

(F2) There is an exact sequence of sheaves on C: $0 \to \mathcal{O}_C \to \nu_*\mathcal{O}_{\mathbf{P}^1} \to k_{P_0} \to 0$, the last map being $F \mapsto F(0) - F(\infty)$ (the difference of the values at the two points over P_0).

Verification: on the chart $z \neq 0$, C is $y^2 = x^2(x+1)$; with the slope $s = y/x$ we get $x = s^2-1$, $y = s(s^2-1)$, $s = (t+1)/(t-1)$, the chart is $\nu(\mathbf{A}^1_s)$ ($z \neq 0 \iff t \neq 1$), and the points $t = \infty, 0$ correspond to $s = 1, -1$. Since $k[s] = k[x] \oplus s\,k[x]$ ($s^2 = x+1$), one finds $k[x,y]/(y^2 - x^2(x+1)) = k[x] \oplus xs\,k[x] = k + (s^2-1)k[s] = \{f \in k[s] : f(1) = f(-1)\}$ (here $\operatorname{char} k \neq 2$ is used: $1 \neq -1$). The remaining point $(0:1:0)$ ($t = 1$) is a smooth point of C (local coordinate $x/y = 1/s$) over which ν is an isomorphism, and $C - P_0$ is the union of $D(x)$ in the chart and this point.

Let W be $\mathbf{A}^1 = \operatorname{Spec} k[u]$ or $\mathbf{G}_m = \operatorname{Spec} k[u,u^{-1}]$, $A = \mathcal{O}(W)$, $Y_W = C \times W$, $\tilde{Y}_W = \mathbf{P}^1 \times W$, $\nu_W = \nu \times \mathrm{id}$ (finite, birational), $D_W = P_0 \times W \cong W \subseteq Y_W$. Tensoring (F2) with A over k (flat base change) gives the exact sequence

$$0 \to \mathcal{O}_{Y_W} \to \nu_{W*}\mathcal{O}_{\tilde{Y}_W} \to i_*\mathcal{O}_{D_W} \to 0, \qquad F \mapsto F|_{\{0\}\times W} - F|_{\{\infty\}\times W}, \tag{$*$}$$

where $\{0\} \times W$ and $\{\infty\} \times W$ are both identified with D_W.

The gluing. The automorphism $\tilde{\varphi} : \mathbf{P}^1 \times \mathbf{G}_m \to \mathbf{P}^1 \times \mathbf{G}_m$, $((\tau_0:\tau_1), u) \mapsto ((\tau_0 : u\tau_1), u)$ (i.e. $t \mapsto ut$) is the identity on $\{0\} \times \mathbf{G}_m$ and on $\{\infty\} \times \mathbf{G}_m$, so $\tilde{\varphi}^*$ preserves the subsheaf $\mathcal{O}_Y \subseteq \nu_*\mathcal{O}_{\tilde{Y}}$ in $(*)$ and induces an automorphism φ of $Y_{\mathbf{G}_m} = C \times \mathbf{G}_m$ over $\mathbf{G}_m$ with $\nu\tilde{\varphi} = \varphi\nu$; on $C^\circ \times \mathbf{G}_m$ it is $(t,u) \mapsto (ut,u)$, and φ_u is its fibre over u (the translation of C extending $t \mapsto ut$, fixing P_0). This is the isomorphism used in the gluing. Write $X_1 = C \times (\mathbf{P}^1 - \{\infty\}) = C \times \mathbf{A}^1_u$ and $X_2 = C \times (\mathbf{P}^1 - \{0\}) = C \times \mathbf{A}^1_{u'}$ ($u' = u^{-1}$), glued along $X_{12} = C \times \mathbf{G}_m$ by identifying $(P,u) \in X_1$ with $\varphi(P,u) \in X_2$.

(a) $\pi^{-1}(\mathbf{P}^1 - \{\infty\}) = X_1 \to \mathbf{P}^1 - \{\infty\}$ and $\pi^{-1}(\mathbf{P}^1 - \{0\}) = X_2 \to \mathbf{P}^1 - \{0\}$ are the projections $C \times \mathbf{A}^1 \to \mathbf{A}^1$, base changes of $C \to \operatorname{Spec} k$, which is proper ($C$ is projective, Thm. 4.9). Properness is local on the base (4.8(f)), so π is proper. Since $\mathbf{P}^1 \to \operatorname{Spec} k$ is proper, so is the composite $X \to \operatorname{Spec} k$ (4.8(b)); in particular X is separated and of finite type. X is integral: X_1, X_2 are integral (C is an integral variety over an algebraically closed field), and they overlap in the nonempty open set X_{12}. Hence X is a complete variety over k.

(b) *Claim.* For W, A as above let $\mathcal{L}_W(\theta, n)$, for $\theta \in A^*$, $n \in \mathbf{Z}$, be the following subsheaf of $\nu_{W*}\mathcal{O}_{\tilde{Y}}(n)$ ($\mathcal{O}(n) = p_1^*\mathcal{O}_{\mathbf{P}^1}(n)$): near $\{0\} \times W$ write a section as $F = f\tau_0^n$, near $\{\infty\} \times W$ as $F = g\tau_1^n$ (the sections τ_0^n, τ_1^n generate $\mathcal{O}(n)$ near 0, resp. ∞); then $F \in \mathcal{L}_W(\theta,n)$ iff $g|_{\{\infty\}\times W} = \theta \cdot f|_{\{0\}\times W}$. Then:

(i) $\mathcal{L}_W(\theta,n)$ is an invertible $\mathcal{O}_{Y_W}$-module, $\nu_W^*\mathcal{L}_W(\theta,n) \cong \mathcal{O}_{\tilde{Y}_W}(n)$, and $\mathcal{L}_W(\theta,n) \otimes \mathcal{L}_W(\theta',n') \cong \mathcal{L}_W(\theta\theta', n+n')$ (by multiplication of sections);

(ii) every invertible sheaf on Y_W is isomorphic to $\mathcal{L}_W(\theta,n)$ for exactly one pair $(\theta,n) \in A^* \times \mathbf{Z}$; so $\operatorname{Pic} Y_W \cong A^* \times \mathbf{Z}$.

Away from D_W the sheaf is $\nu_*\mathcal{O}(n)$, invertible since ν_W is an isomorphism there. Near D_W, on $\nu^{-1}(C_{z\neq 0})\times W = \mathbf{A}^1_s \times W$ the section $\rho = (\tau_1 - \tau_0)^n$ trivializes $\mathcal{O}(n)$, and writing $F = h\rho$ the condition becomes $h(s{=}1) = \theta' h(s{=}-1)$ with $\theta' = (-1)^n\theta$ (recall $t=\infty \leftrightarrow s = 1$, $t = 0 \leftrightarrow s = -1$), while $\mathcal{O}_Y = \{h : h(1) = h(-1)\}$ by $(*)$. Let $h_0 = \frac{1-s}{2} + \frac{1+s}{2}\theta' \in A[s]$, so $h_0(-1) = 1$, $h_0(1) = \theta'$ are units. The zero set G of h_0 misses $\{s = \pm 1\}\times W$, so $V = (C_{z\neq 0}\times W) - \nu_W(G)$ (ν_W is closed) is an open neighbourhood of D_W, on which h_0 is a unit of $\nu_*\mathcal{O}$ and $\mathcal{L}_W(\theta,n)|_V = h_0\rho\cdot\mathcal{O}_V$ (if $h\in\mathcal{L}$ then h/h_0 takes the same value at $s=\pm1$; conversely $h_0\mathcal{O}_V \subseteq \mathcal{L}$). So $\mathcal{L}$ is invertible and $\nu^*\mathcal{L} = h_0\rho\cdot\mathcal{O}_{\tilde{Y}} = \mathcal{O}(n)$ on $\nu^{-1}V$. The tensor-product statement is checked on these local generators. This proves (i).

For (ii) use the method of Ex. 6.9. By $(*)$ there is an exact sequence of sheaves of abelian groups

$$1 \to \mathcal{O}_Y^* \to \nu_*\mathcal{O}_{\tilde{Y}}^* \to i_*\mathcal{O}_D^* \to 1, \qquad F \mapsto F|_{\{0\}\times W}/F|_{\{\infty\}\times W}$$

(kernel: units F with equal values at $0,\infty$ lie in $\mathcal{O}_Y$, together with F^{-1}; surjectivity: near a point (P_0,w), given a unit ϑ of $\mathcal{O}_{W,w}$, the function $F = \frac{1-s}{2} + \frac{1+s}{2}\vartheta$ on $\mathbf{A}^1_s\times W'$ has $F|_{s=-1} = 1$, $F|_{s=1} = \vartheta$, so it is a unit on a neighbourhood of $\{s=\pm1\}\times W'$, which contains ν^{-1} of a neighbourhood of (P_0,w) (ν is closed); its ratio of values at $t=0$ and $t=\infty$ is ϑ^{-1}, and ϑ was arbitrary). Take cohomology on $Y = Y_W$ (Zariski). We have $H^1(Y,\mathcal{O}_Y^*) = \operatorname{Pic} Y$ (Ex. III.4.5: line bundles are classified by $H^1(\mathcal{O}^*)$); $H^0(Y,\nu_*\mathcal{O}_{\tilde{Y}}^*) = \mathcal{O}(\tilde{Y})^* = A^*$ since $\Gamma(\mathbf{P}^1\times W,\mathcal{O}) = A$; $H^0(Y,\mathcal{O}_Y) = \{F\in A : F|_0 = F|_\infty\} = A$ by $(*)$, so $H^0(Y,\mathcal{O}_Y^*) = A^*$ too; $H^0(Y,i_*\mathcal{O}_D^*) = A^*$; $H^1(Y,i_*\mathcal{O}_D^*) = H^1(D,\mathcal{O}_D^*) = \operatorname{Pic} W = 0$ (A is a PID); and $H^1(Y,\nu_*\mathcal{O}_{\tilde{Y}}^*) = \operatorname{Pic}\tilde{Y}$, because $R^1\nu_*\mathcal{O}_{\tilde{Y}}^* = 0$: its stalk at y is the direct limit of $\operatorname{Pic}(\nu^{-1}V)$ over neighbourhoods V of y, and a line bundle on the quasi-projective scheme $\tilde{Y}$ is trivial on a neighbourhood of the finite set $\nu^{-1}(y)$ (semilocal rings have trivial Picard group), which contains some $\nu^{-1}(V')$ as ν is closed. So we get the exact sequence

$$1 \to A^* \xrightarrow{\mathrm{id}} A^* \xrightarrow{0} A^* \xrightarrow{\delta} \operatorname{Pic} Y \xrightarrow{\nu^*} \operatorname{Pic}\tilde{Y} \to 0,$$

where the middle map is zero because $F\in\mathcal{O}(\tilde{Y})^* = A^*$ is constant along $\mathbf{P}^1$ so $F|_0/F|_\infty = 1$. Thus $\delta : A^* \to \operatorname{Pic} Y$ is injective and $\operatorname{Pic} Y/\delta(A^*) \cong \operatorname{Pic}\tilde{Y}$. Unwinding δ: $\theta\in A^* = H^0(i_*\mathcal{O}_D^*)$ is lifted locally to units g_V of $\nu_*\mathcal{O}_{\tilde{Y}}$ with $g_V|_0/g_V|_\infty = \theta$, and $\delta(\theta)$ is (up to the convention $\delta(\theta)^{\pm1}$, which is immaterial below) the class of the invertible sheaf $\{g_V\mathcal{O}_V\} = \{F\in\nu_*\mathcal{O}_{\tilde{Y}} : F|_\infty = \theta^{-1}F|_0\} = \mathcal{L}_W(\theta^{-1},0)$. Hence $\delta(A^*) = \{[\mathcal{L}_W(\theta,0)] : \theta\in A^*\}$, and $\theta\mapsto[\mathcal{L}_W(\theta,0)]$ is injective. Finally $\operatorname{Pic}\tilde{Y} = \operatorname{Pic}(\mathbf{P}^1\times W) = \mathbf{Z}\cdot[\mathcal{O}(1)]$: by Ex. 6.1, $\operatorname{Cl}(\mathbf{P}^1\times W)\cong\operatorname{Cl} W\times\mathbf{Z} = \mathbf{Z}$ with generator $V_+(\tau_0)\times W$, i.e. $\mathcal{O}(1)$, and Pic = Cl since $\mathbf{P}^1\times W$ is regular; $\mathcal{O}(n)$ for $n\neq 0$ is nontrivial. By (i), $\nu^*\mathcal{L}_W(1,n) = \mathcal{O}(n)$, so $\mathcal{L}_W(1,n)$ lifts $[\mathcal{O}(n)]$. Therefore every invertible sheaf is $\mathcal{L}_W(\theta,0)\otimes\mathcal{L}_W(1,n)\cong\mathcal{L}_W(\theta,n)$, and (θ,n) is unique (n is read off from ν^*, then θ by injectivity of δ). This proves (ii).

For $W = \mathbf{A}^1$: $A^* = k[u]^* = k^* = \mathbf{G}_m$, so $\operatorname{Pic}(C\times\mathbf{A}^1)\cong\mathbf{G}_m\times\mathbf{Z}$, $\langle c,n\rangle\leftrightarrow\mathcal{L}(c,n)$. For $W = \mathbf{G}_m$: $A^* = k[u,u^{-1}]^* = k^*\times u^{\mathbf{Z}}$, so $\operatorname{Pic}(C\times\mathbf{G}_m)\cong\mathbf{G}_m\times\mathbf{Z}\times\mathbf{Z}$, $\langle c,d,n\rangle\leftrightarrow\mathcal{L}(cu^d,n)$. (I write c for the first coordinate to avoid a clash with the parameter t.)

(c) The sheaf $\mathcal{L}_{\mathbf{A}^1}(c,n)$ restricted to $C\times\mathbf{G}_m$ is, by definition, $\mathcal{L}_{\mathbf{G}_m}(c,n)$ (the same description, the constant $c\in k^*\subseteq A^*$). So restriction is $\langle c,n\rangle\mapsto\langle c,0,n\rangle$.

For φ^*: $\varphi^*\mathcal{L}_{\mathbf{G}_m}(\theta,n)$ consists of the pull-backs $\tilde{\varphi}^*F$ of sections F of $\mathcal{L}(\theta,n)$, viewed in $\nu_*\mathcal{O}(n)$ through the identification $\tilde{\varphi}^*\mathcal{O}(n)\cong\mathcal{O}(n)$ for which $\tilde{\varphi}^*\tau_0^n = \tau_0^n$ and $\tilde{\varphi}^*\tau_1^n = u^n\tau_1^n$ (the morphism

$\tilde\varphi$ is defined by the sections $\tau_0, u\tau_1$ of $\mathcal{O}(1)$, 7.1). If $F = f\tau_0^n = g\tau_1^n$ near 0, resp. ∞, then $\tilde\varphi^* F = f'\tau_0^n = g'\tau_1^n$ with $f' = \tilde\varphi^* f$, $g' = u^n \tilde\varphi^* g$; since $\tilde\varphi$ is the identity on $\{0,\infty\} \times \mathbf{G}_m$, $f'|_0 = f|_0$ and $g'|_\infty = u^n g|_\infty$. So $g|_\infty = \theta f|_0$ becomes $g'|_\infty = u^n \theta f'|_0$:

$$\varphi^* \mathcal{L}_{\mathbf{G}_m}(\theta, n) = \mathcal{L}_{\mathbf{G}_m}(u^n\theta, n), \qquad \text{i.e.} \qquad \varphi^*\langle c, d, n\rangle = \langle c, d+n, n\rangle.$$

(d) Let $\mathcal{M} \in \operatorname{Pic} X$. By (b), $\mathcal{M}|_{X_1} \cong \mathcal{L}_{\mathbf{A}^1_u}(c,n) = \langle c, n\rangle$ and $\mathcal{M}|_{X_2} \cong \langle c', n'\rangle$ (with respect to the coordinate u'; the constants $k^* \subseteq A^*$ are the same for u and $u' = u^{-1}$). Restricted to X_{12} these are $\langle c, 0, n\rangle$ and $\langle c', 0, n'\rangle$ by (c). The gluing identifies $X_{12} \subseteq X_1$ with $X_{12} \subseteq X_2$ via φ, so $\mathcal{M}|_{X_{12}\subseteq X_1} \cong \varphi^*(\mathcal{M}|_{X_{12}\subseteq X_2})$, i.e. $\langle c, 0, n\rangle = \varphi^*\langle c', 0, n'\rangle = \langle c', n', n'\rangle$ in $\operatorname{Pic}(C \times \mathbf{G}_m)$. Comparing coordinates: $c = c'$, $0 = n'$, $n = n'$. So $n = n' = 0$ and $\mathcal{M}|_{X_1} \cong \mathcal{L}_{\mathbf{A}^1}(c, 0)$.

The closed subscheme $C \times \{0\} \subseteq X_1$ is $\cong C$, and $\mathcal{L}_{\mathbf{A}^1}(c,0)|_{C\times\{0\}}$ is the sheaf $\{F \in \nu_* \mathcal{O}_{\mathbf{P}^1} : F(\infty) = cF(0)\}$ on C, whose pull-back to $\mathbf{P}^1$ is $\mathcal{O}_{\mathbf{P}^1}$. So every element of the image of $\operatorname{Pic} X \to \operatorname{Pic}(C \times \{0\}) = \operatorname{Pic} C$ has degree 0, where $\deg \mathcal{N} := \deg \nu^* \mathcal{N} \in \mathbf{Z} = \operatorname{Pic} \mathbf{P}^1$ is the degree of the corresponding Cartier divisor class (these form $\operatorname{CaCl}^0 C \cong \mathbf{G}_m$ of Ex. 6.7).

X is not projective. Suppose X were projective over k. Then there is a very ample invertible sheaf $\mathcal{M}$ on X, i.e. a closed immersion $X \hookrightarrow \mathbf{P}^N$ with $\mathcal{M} \cong \mathcal{O}(1)|_X$. Restricting to the closed subscheme $C \times \{0\} \cong C$ gives a closed immersion $j : C \hookrightarrow \mathbf{P}^N$ with $\mathcal{M}|_C \cong j^*\mathcal{O}(1)$. Then $(j\nu)^*\mathcal{O}(1) \cong \nu^*(\mathcal{M}|_C)$ has degree 0 by what we have shown, hence is $\mathcal{O}_{\mathbf{P}^1}$, which is generated by the constant sections $(j\nu)^* x_i$; so $j\nu$ is constant. But ν is nonconstant and j injective, a contradiction. So X is not projective over k.

π is not projective. If $\pi : X \to \mathbf{P}^1$ were projective, X would be a closed subscheme of $\mathbf{P}^r_{\mathbf{P}^1} = \mathbf{P}^r_k \times_k \mathbf{P}^1_k$, which is a closed subscheme of a projective space over k by the Segre embedding; then X would be projective over k, contradicting the above. Hence X is a complete variety over k that is not projective, and $\pi : X \to \mathbf{P}^1$ is proper but not projective. □

Exercise 7.14

Problem.

(a) Give an example of a noetherian scheme X and a locally free coherent sheaf $\mathcal{E}$ on X, such that the invertible sheaf $\mathcal{O}(1)$ on $\mathbf{P}(\mathcal{E})$ is not very ample relative to X.

(b) Let $f : X \to Y$ be a morphism of finite type, let $\mathcal{L}$ be an ample invertible sheaf on X, and let $\mathcal{S}$ be a sheaf of graded $\mathcal{O}_X$-algebras satisfying (†). Let $P = \operatorname{Proj} \mathcal{S}$, let $\pi : P \to X$ be the projection, and let $\mathcal{O}_P(1)$ be the associated invertible sheaf. Show that for all $n \gg 0$, the sheaf $\mathcal{O}_P(1) \otimes \pi^* \mathcal{L}^n$ is very ample on P relative to Y. [*Hint:* Use (7.10) and (Ex. 5.12).]

Solution. (a) Let k be a field, $X = \mathbf{P}^1_k$, $\mathcal{E} = \mathcal{O}_X(-1) \oplus \mathcal{O}_X(-1)$ (locally free of rank 2), $P = \mathbf{P}(\mathcal{E})$. By Prop. 7.11(a) (rank ≥ 2), $\pi_* \mathcal{O}_P(1) = \mathcal{E}$, so $\Gamma(P, \mathcal{O}_P(1)) = \Gamma(X, \mathcal{E}) = \Gamma(\mathbf{P}^1, \mathcal{O}(-1))^2 = 0$. Suppose $\mathcal{O}_P(1)$ were very ample relative to X: there would be an immersion $i : P \to \mathbf{P}^N_X$ over X with $i^*\mathcal{O}(1) \cong \mathcal{O}_P(1)$. Then $\mathcal{O}_P(1)$ would be generated by the global sections $i^*x_0, \ldots, i^*x_N$ (7.1(a)), all zero, so $\mathcal{O}_P(1) = 0$, absurd since $P \neq \emptyset$. Hence $\mathcal{O}_P(1)$ is not very ample relative to X. (More generally, if $\mathcal{O}_P(1)$ is very ample relative to X then $\mathcal{E} = \pi_* \mathcal{O}_P(1)$ is generated by finitely many global sections, since the $i^*x_a \in \Gamma(P, \mathcal{O}_P(1)) = \Gamma(X, \mathcal{E})$ generate $\mathcal{O}_P(1)$ and hence,

restricted to each fibre $\mathbf{P}(\mathcal{E}\otimes\kappa(x))$, span $\mathcal{E}\otimes\kappa(x)$. Conversely if $\mathcal{O}^{N+1}\to\mathcal{E}$ is surjective then $\mathbf{P}(\mathcal{E})\hookrightarrow\mathbf{P}^N_X$ is a closed immersion with $\mathcal{O}(1)$ restricting to $\mathcal{O}_P(1)$. So for affine X the sheaf $\mathcal{O}_P(1)$ is always very ample, and a non-affine X with $\mathcal{E}$ not globally generated is the natural place to look for examples.)

(b) *Setup.* All schemes are noetherian (Y noetherian, X of finite type over Y). Let n_0 be such that $\mathcal{S}_1\otimes\mathcal{L}^n$ is generated by global sections for all $n\geq n_0$ ($\mathcal{L}$ ample, $\mathcal{S}_1$ coherent). The proof of Prop. 7.10(b) uses only that $\mathcal{S}_1\otimes\mathcal{L}^n$ is generated by finitely many global sections; hence, for every $n\geq n_0$, $\mathcal{O}_P(1)\otimes\pi^*\mathcal{L}^n$ is very ample on P relative to X (there is an immersion, indeed closed, $P\hookrightarrow\mathbf{P}^M_X$ over X pulling $\mathcal{O}(1)$ back to it).

Lemma (ample $\Rightarrow$ some power very ample relative to Y). Let $f:X\to Y$ be of finite type, Y noetherian, and $\mathcal{L}$ ample on X. Then $\mathcal{L}^{m_1}$ is very ample on X relative to Y for some $m_1>0$. (For $Y=\operatorname{Spec}A$ this is Thm. 7.6; the proof of 7.6 adapts as follows.) Cover Y by finitely many affine opens $U_j=\operatorname{Spec}A_j$. As in the proof of Thm. 7.6 (which uses only that $\mathcal{L}$ is ample on the noetherian scheme X), for every $P\in X$ and every affine open neighbourhood U of P on which $\mathcal{L}$ is trivial there are $n>0$ and $s\in\Gamma(X,\mathcal{L}^n)$ with $P\in X_s\subseteq U$ and X_s affine; taking $U\subseteq f^{-1}(U_j)$ for some $j\ni f(P)$ and using quasi-compactness of X, we get $s_1,\dots,s_k$, all in $\Gamma(X,\mathcal{L}^n)$ for one n (replace s_i by powers), with $X_i=X_{s_i}$ affine, $\bigcup X_i=X$ and $X_i\subseteq f^{-1}(U_{j(i)})$. Since $f^{-1}(U_{j(i)})$ is of finite type over $A_{j(i)}$, $B_i=\Gamma(X_i,\mathcal{O}_X)$ is a finitely generated $A_{j(i)}$-algebra, say by $b_{i1},\dots,b_{il_i}$. By (5.14) there is N such that all $s_i^Nb_{il}$ extend to sections $c_{il}\in\Gamma(X,\mathcal{L}^{nN})$. The sections s_i^N generate $\mathcal{L}^{nN}$, so $\{s_i^N,c_{il}\}$ define (7.1) a morphism $g:X\to\mathbf{P}^M_{\mathbf{Z}}$ and $h=(g,f):X\to\mathbf{P}^M_{\mathbf{Z}}\times Y=\mathbf{P}^M_Y$ over Y, with $h^*\mathcal{O}(1)=\mathcal{L}^{nN}$. Let $V_i\subseteq\mathbf{P}^M_Y$ be the open set where the coordinate of s_i^N is nonzero and the image in Y lies in $U_{j(i)}$; $V_i\cong\mathbf{A}^M_{A_{j(i)}}=\operatorname{Spec}A_{j(i)}[y_l]$, $h^{-1}(V_i)=X_i$, and $h:X_i\to V_i$ corresponds to $y_l\mapsto c_{il}/s_i^N=b_{il}$ (and the y's attached to the other sections go to elements of B_i), which is surjective onto B_i. So h restricts to closed immersions $X_i\to V_i$; since the V_i cover the open set $W=\bigcup V_i$, h is a closed immersion $X\to W$ and so an immersion $X\to\mathbf{P}^M_Y$. Thus $\mathcal{L}^{m_1}$, $m_1=nN$, is very ample relative to Y.

Conclusion. Apply Ex. 5.12(b) to $P\xrightarrow{\pi}X\xrightarrow{f}Y$: for $a\geq n_0$ the sheaf $\mathcal{O}_P(1)\otimes\pi^*\mathcal{L}^a$ is very ample on P relative to X, and $\mathcal{L}^{m_1}$ is very ample on X relative to Y, so $\big(\mathcal{O}_P(1)\otimes\pi^*\mathcal{L}^a\big)\otimes\pi^*\mathcal{L}^{m_1}=\mathcal{O}_P(1)\otimes\pi^*\mathcal{L}^{a+m_1}$ is very ample on P relative to Y. Taking $a=n-m_1$, this holds for every $n\geq n_0+m_1$.

(Recall the proof of Ex. 5.12(b) for completeness: if $P\hookrightarrow\mathbf{P}^M_X$ and $X\hookrightarrow\mathbf{P}^r_Y$ are immersions, then $P\hookrightarrow\mathbf{P}^M_X=\mathbf{P}^M_Y\times_Y X\hookrightarrow\mathbf{P}^M_Y\times_Y\mathbf{P}^r_Y\hookrightarrow\mathbf{P}^{(M+1)(r+1)-1}_Y$, the last map being the Segre embedding, and the pull-back of $\mathcal{O}(1)$ is the tensor product of the two sheaves.) □

8. Differentials

Conventions for this section. "Variety" and "nonsingular variety" are in the sense of Chapter II, so k is algebraically closed whenever the text speaks of varieties (this is the case in Ex. 8.2–8.8; in Ex. 8.1 only a perfect field is needed). Results of the text are cited by number (Prop. 8.7, Thm. 8.17, …), $\mathcal{N}_{Y/X}=\mathcal{H}om(\mathcal{I}/\mathcal{I}^2,\mathcal{O}_Y)$ is the normal sheaf, and $\mathcal{O}(D)$ is the invertible sheaf of a divisor (written $\mathcal{L}(D)$ in the book).

Exercise 8.1

Problem. Here we will strengthen the results of the text to include information about the sheaf of differentials at a not necessarily closed point of a scheme X.

(a) Generalize (8.7) as follows. Let B be a local ring containing a field k, and assume that the residue field $k(B) = B/\mathfrak{m}$ of B is a separably generated extension of k. Then the exact sequence of (8.4A),

$$0 \to \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\delta} \Omega_{B/k} \otimes_B k(B) \to \Omega_{k(B)/k} \to 0$$

is exact on the left also. [*Hint*: In copying the proof of (8.7), first pass to $B/\mathfrak{m}^2$, which is a complete local ring, and then use (8.25A) to choose a field of representatives for $B/\mathfrak{m}^2$.]

(b) Generalize (8.8) as follows. With B, k as above, assume furthermore that k is perfect, and that B is a localization of an algebra of finite type over k. Then show that B is a regular local ring if and only if $\Omega_{B/k}$ is free of rank $= \dim B + \text{tr. d. } k(B)/k$.

(c) Strengthen (8.15) as follows. Let X be an irreducible scheme of finite type over a perfect field k, and let $\dim X = n$. Show that the local ring $\mathcal{O}_{X,x}$ at any point $x \in X$, not necessarily closed, is a regular local ring if and only if the stalk $(\Omega_{X/k})_x$ of the sheaf of differentials at x is free of rank n.

(d) Strengthen (8.16) as follows. If X is a variety over an algebraically closed field k, then $U = \{x \in X \mid \mathcal{O}_x \text{ is a regular local ring}\}$ is an open dense subset of X.

Solution. Throughout, $\kappa = k(B) = B/\mathfrak{m}$.

(a) By (8.4A) (applied to the k-algebra B and its quotient $B/\mathfrak{m} = \kappa$) the sequence is exact except possibly on the left, so we must show that δ is injective. Since δ is a κ-linear map of κ-vector spaces, it suffices to show that the transposed map

$$\delta' : \mathrm{Hom}_\kappa(\Omega_{B/k} \otimes_B \kappa, \kappa) \to \mathrm{Hom}_\kappa(\mathfrak{m}/\mathfrak{m}^2, \kappa)$$

is surjective: if $\delta(w) = 0$ with $w \neq 0$ we could choose a linear form h with $h(w) \neq 0$ and write $h = \delta'(\varphi) = \varphi \circ \delta$, a contradiction. By the universal property of $\Omega_{B/k}$, $\mathrm{Hom}_\kappa(\Omega_{B/k} \otimes_B \kappa, \kappa) = \mathrm{Hom}_B(\Omega_{B/k}, \kappa) = \mathrm{Der}_k(B, \kappa)$, where κ is a B-module via $B \to \kappa$. If d is such a derivation then $d(ab) = a\,db + b\,da = 0$ for $a, b \in \mathfrak{m}$, so $d(\mathfrak{m}^2) = 0$; and $\delta'(d)$ is the restriction $d|_{\mathfrak{m}} : \mathfrak{m}/\mathfrak{m}^2 \to \kappa$ (indeed $\delta(\bar{x}) = dx \otimes 1$ and the linear map attached to d sends $dx \otimes 1$ to $d(x)$). Thus we must show: every κ-linear $h : \mathfrak{m}/\mathfrak{m}^2 \to \kappa$ is the restriction of a k-derivation $d : B \to \kappa$.

Put $\bar{B} = B/\mathfrak{m}^2$, a local ring with maximal ideal $\bar{\mathfrak{m}} = \mathfrak{m}/\mathfrak{m}^2$ satisfying $\bar{\mathfrak{m}}^2 = 0$. It is complete (its $\bar{\mathfrak{m}}$-adic topology is discrete), it contains k (the map $k \to B \to \bar{B}$ is injective because k is a field and $\bar{B} \neq 0$), and its residue field is κ, separably generated over k. By Cohen's theorem (8.25A) there is a subfield $K \subseteq \bar{B}$ containing k such that $K \to \bar{B}/\bar{\mathfrak{m}} = \kappa$ is an isomorphism. Every element of $\bar{B}$ is then uniquely $\lambda + c$ with $\lambda \in K$, $c \in \bar{\mathfrak{m}}$. Define $\bar{d} : \bar{B} \to \kappa$ by $\bar{d}(\lambda + c) = h(c)$. It is additive and vanishes on $K \supseteq k$. For $b = \lambda + c$, $b' = \lambda' + c'$ we have $bb' = \lambda\lambda' + (\lambda c' + \lambda' c)$ because $cc' = 0$; here $\lambda c'$ is the κ-vector space action of $\bar{\lambda} \in \kappa$ (the residue class of λ) on $\bar{\mathfrak{m}} = \mathfrak{m}/\mathfrak{m}^2$. Hence

$$\bar{d}(bb') = h(\lambda c' + \lambda' c) = \bar{\lambda}\, h(c') + \bar{\lambda}'\, h(c) = \bar{b}\, \bar{d}(b') + \bar{b}'\, \bar{d}(b),$$

where $\bar{b}, \bar{b}' \in \kappa$ are residues. So $\bar{d}$ is a k-derivation $\bar{B} \to \kappa$ with $\bar{d}|_{\bar{\mathfrak{m}}} = h$. The composite $d = \bar{d} \circ (B \to \bar{B})$ is a k-derivation $B \to \kappa$ whose restriction to $\mathfrak{m}$ is h. Hence δ' is surjective and δ is injective.

(b) Put $s = \text{tr. d. } \kappa/k$ and $r = \dim B$. Write $B = A_{\mathfrak{p}}$ with A a finitely generated k-algebra. Then κ is a finitely generated field extension of k (the fraction field of a quotient of A), hence separably generated because k is perfect (I, 4.8A). So (a) applies, and by (8.6A) $\dim_\kappa \Omega_{\kappa/k} = s$. The exact sequence of (a) gives

$$\dim_\kappa(\Omega_{B/k} \otimes_B \kappa) = \dim_\kappa \mathfrak{m}/\mathfrak{m}^2 + s. \qquad (*)$$

Also $\Omega_{B/k}$ is a finitely generated B-module by (8.5).

"If". Suppose $\Omega_{B/k}$ is free of rank $r+s$. Then the left side of $(*)$ is $r+s$, so $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2 = r = \dim B$, i.e. B is regular.

"Only if". Suppose B is regular, so $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2 = r$ and, by $(*)$, $\dim_\kappa(\Omega_{B/k} \otimes_B \kappa) = r+s$. A regular local ring is a domain; let L be its quotient field. By (8.2A), $\Omega_{B/k} \otimes_B L = \Omega_{L/k}$, and L is a finitely generated, hence (as k is perfect) separably generated, extension of k, so $\dim_L \Omega_{L/k} = \text{tr. d. } L/k$ by (8.6A). We claim $\text{tr. d. } L/k = r+s$. Let $\mathfrak{q} = \ker(A \to B)$, a prime of A because B is a domain; then $B = (A/\mathfrak{q})_{\mathfrak{p}/\mathfrak{q}}$, so we may assume A is a domain with $L = \operatorname{Frac} A$, $\kappa = \operatorname{Frac}(A/\mathfrak{p})$. Then $r = \operatorname{ht}\mathfrak{p}$ and $s = \dim A/\mathfrak{p}$ by (I, 1.8A(a)), and $\operatorname{ht}\mathfrak{p} + \dim A/\mathfrak{p} = \dim A = \text{tr. d. } L/k$ by (I, 1.8A(b),(a)). So $\dim_L(\Omega_{B/k} \otimes L) = \dim_\kappa(\Omega_{B/k} \otimes \kappa) = r+s$, and Lemma 8.9 (a finitely generated module over a noetherian local domain whose two fibre dimensions agree) shows $\Omega_{B/k}$ is free of rank $r+s$.

(c) Let $x \in X$ and $B = \mathcal{O}_{X,x}$. Taking an affine neighbourhood $\operatorname{Spec} A$ of x, B is a localization of a finitely generated k-algebra, and $(\Omega_{X/k})_x = \Omega_{B/k}$ by Remark 8.9.2/(8.2A). Also $k(B) = k(x)$. By (b), B is regular iff $\Omega_{B/k}$ is free of rank $\dim B + \text{tr. d. } k(x)/k$. So it suffices to show $\dim \mathcal{O}_{X,x} + \text{tr. d. } k(x)/k = n$. Take $x \in \operatorname{Spec} A = U$ affine open, $x = \mathfrak{p}$. U is a nonempty open subset of the irreducible X, so A_{red} is a domain; heights and dimensions are topological, so we may replace A by A_{red} (a finitely generated domain). Then $\dim \mathcal{O}_{X,x} = \operatorname{ht}\mathfrak{p}$, $\text{tr. d. } k(x)/k = \dim A/\mathfrak{p}$, and $\operatorname{ht}\mathfrak{p} + \dim A/\mathfrak{p} = \dim A = \text{tr. d. } K(X_{\text{red}})/k$ by (I, 1.8A). Finally $\dim X = \sup_i \dim U_i$ over an affine open cover (a chain of irreducible closed subsets of X meeting U_i restricts to a chain of the same length in U_i), and each $\dim U_i = \text{tr. d. } K/k$, so this common value is n. Hence $\dim \mathcal{O}_{X,x} + \text{tr. d. } k(x)/k = n$, and (c) follows.

(d) X is irreducible of finite type over the perfect field k, $n = \dim X$. By (c), $x \in U$ iff $(\Omega_{X/k})_x$ is free of rank n. The sheaf $\Omega_{X/k}$ is coherent (Remark 8.9.1). If $x \in U$, then by (Ex. 5.7) there is an open neighbourhood V of x on which $\Omega_{X/k}$ is free (of rank n); every point of V then lies in U by (c). So U is open. It contains the generic point η, since $\mathcal{O}_{X,\eta}$ is a field and fields are regular local rings; so U is a nonempty open subset of the irreducible space X, hence dense. □

Exercise 8.2

Problem. Let X be a variety of dimension n over k. Let $\mathcal{E}$ be a locally free sheaf of rank $> n$ on X, and let $V \subseteq \Gamma(X, \mathcal{E})$ be a vector space of global sections which generate $\mathcal{E}$. Then show that there is an element $s \in V$, such that for each $x \in X$, we have $s_x \notin \mathfrak{m}_x \mathcal{E}_x$. Conclude that there is a morphism $\mathcal{O}_X \to \mathcal{E}$ giving rise to an exact sequence

$$0 \to \mathcal{O}_X \to \mathcal{E} \to \mathcal{E}' \to 0$$

where $\mathcal{E}'$ is also locally free. [*Hint*: Use a method similar to the proof of Bertini's theorem (8.18).]

Solution. Let $r = \operatorname{rk}\mathcal{E} > n$. Write $\mathcal{E}(x) = \mathcal{E}_x/\mathfrak{m}_x\mathcal{E}_x$ for the fibre ($\cong k(x)^r$); for a section s we say $s(x) = 0$ if $s_x \in \mathfrak{m}_x\mathcal{E}_x$.

Step 0: reductions. X is noetherian, hence quasi-compact. Since V generates $\mathcal{E}$, every x has finitely many elements of V generating $\mathcal{E}_x$, hence (the cokernel of the corresponding map $\mathcal{O}^m \to \mathcal{E}$ being coherent with zero stalk at x) generating $\mathcal{E}$ on a neighbourhood of x. Finitely many neighbourhoods cover X, so a finite-dimensional subspace $V_0 \subseteq V$ generates $\mathcal{E}$; it suffices to find $s \in V_0$. So assume $\dim V = N < \infty$ and fix a basis $s_1, \dots, s_N$. Next, the set $\{x : s(x) = 0\}$ is closed (locally, in a trivialization $\mathcal{E}|_U \cong \mathcal{O}_U^r$, $s = (f_1, \dots, f_r)$ and the set is $V(f_1, \dots, f_r)$). A nonempty closed subset of the noetherian scheme X contains a closed point (a minimal nonempty closed subset W satisfies $\overline{\{w\}} = W$ for all $w \in W$, so W is one point because schemes are T_0). Hence it is enough to find $s \in V$ with $s(x) \neq 0$ for all *closed* points x.

Step 1: the incidence scheme. Let $\mathbf{A}^N = \operatorname{Spec} k[v_1, \dots, v_N]$; its k-points are identified with V via $a \mapsto s_a = \sum a_i s_i$. Cover X by finitely many affine opens U with $\mathcal{E}|_U \cong \mathcal{O}_U^r$; on U write $s_j = (m_{1j}, \dots, m_{rj})$, giving an $r \times N$ matrix M over $\mathcal{O}(U)$. Since the s_j generate $\mathcal{E}$, $M(x)$ has rank r at every $x \in U$, so some $r \times r$ minor Δ (using columns J, $|J| = r$) is nonzero at x, hence invertible on a neighbourhood. Thus X is covered by finitely many affine opens $W_1, \dots, W_m$ on each of which $\mathcal{E}$ is trivial and the matrix M has an invertible $r \times r$ minor M_J (columns J). Let $Z_a \subseteq W_a \times \mathbf{A}^N$ be the closed subscheme defined by the r equations $M \cdot v = 0$. As M_J is invertible we can solve $v_J = -M_J^{-1} M_{J'} v_{J'}$, so $Z_a \cong W_a \times \mathbf{A}^{N-r}$ with coordinates $(x, v_{J'})$; Z_a is integral (its ring is a polynomial ring over the domain $\mathcal{O}(W_a)$) with function field of transcendence degree $n + N - r$ over k (W_a is a nonempty open subset of the variety X, so $\operatorname{tr.d.} k(W_a)/k = n$). Let $q : Z_a \to \mathbf{A}^N$ be the projection and T_a the closure of its image, an integral closed subscheme. The coordinate ring of T_a embeds in $\mathcal{O}(Z_a)$, so $k(T_a) \subseteq k(Z_a)$ and $\dim T_a = \operatorname{tr.d.} k(T_a)/k \leq n + N - r \leq N - 1$, because $r \geq n + 1$. So T_a is a proper closed subset of $\mathbf{A}^N$, and so is $T_1 \cup \cdots \cup T_m$ ($\mathbf{A}^N$ is irreducible). Since k is algebraically closed, the nonempty open complement contains a k-point a (Nullstellensatz). Put $s = s_a \in V$.

Step 2: s works. Let x be a closed point of X, say $x \in W_b$; it is k-rational. If $s(x) = 0$, then $M(x)a = 0$, i.e. $(x, a) \in Z_b$, hence $a \in q(Z_b) \subseteq T_b$, contradicting the choice of a. So $s(x) \neq 0$ for all closed x, hence for all x by Step 0.

The exact sequence. Define $\sigma : \mathcal{O}_X \to \mathcal{E}$ by $1 \mapsto s$. Let $x \in X$ and trivialize $\mathcal{E}$ near x, $s = (f_1, \dots, f_r)$. Since $s(x) \neq 0$, some $f_i \notin \mathfrak{m}_x$, so f_i is a unit on a neighbourhood U' of x. On U' the sections s, e_j ($j \neq i$) form a basis of $\mathcal{O}_{U'}^r$ (the transition matrix to the standard basis is, up to column order, triangular with the unit f_i on the diagonal). Therefore on U', σ is injective and $\mathcal{E}' = \operatorname{coker}\sigma$ is free of rank $r - 1$ (generated by the classes of the e_j, $j \neq i$). Thus $0 \to \mathcal{O}_X \xrightarrow{\sigma} \mathcal{E} \to \mathcal{E}' \to 0$ is exact with $\mathcal{E}'$ locally free of rank $r - 1$. □

Exercise 8.3

Problem. *Product Schemes.*

(a) Let X and Y be schemes over another scheme S. Use (8.10) and (8.11) to show that $\Omega_{X \times_S Y/S} \cong p_1^*\Omega_{X/S} \oplus p_2^*\Omega_{Y/S}$.

(b) If X and Y are nonsingular varieties over a field k, show that $\omega_{X \times Y} \cong p_1^*\omega_X \otimes p_2^*\omega_Y$.

(c) Let Y be a nonsingular plane cubic curve, and let X be the surface $Y \times Y$. Show that $p_g(X) = 1$ but $p_a(X) = -1$ (I, Ex. 7.2). This shows that the arithmetic genus and the geometric genus of a nonsingular projective variety may be different.

Solution. (a) Let $Z = X \times_S Y$ with projections p_1, p_2. The morphism $Z \to Y$ is the base extension of $X \to S$ by $Y \to S$, so by (8.10)

$$\Omega_{Z/Y} \cong p_1^*\Omega_{X/S}, \qquad \text{and symmetrically} \qquad \Omega_{Z/X} \cong p_2^*\Omega_{Y/S}.$$

By (8.11) applied to $Z \to Y \to S$ and $Z \to X \to S$ we have natural maps $p_2^*\Omega_{Y/S} \to \Omega_{Z/S} \to \Omega_{Z/Y} \cong p_1^*\Omega_{X/S}$ and $p_1^*\Omega_{X/S} \to \Omega_{Z/S} \to \Omega_{Z/X} \cong p_2^*\Omega_{Y/S}$. Let

$$\varphi : p_1^*\Omega_{X/S} \oplus p_2^*\Omega_{Y/S} \to \Omega_{Z/S}, \qquad \psi : \Omega_{Z/S} \to \Omega_{Z/Y} \oplus \Omega_{Z/X} \cong p_1^*\Omega_{X/S} \oplus p_2^*\Omega_{Y/S}$$

be the maps built from these. We show φ, ψ are inverse isomorphisms; this is local, so let $S = \operatorname{Spec} R$, $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, $C = A \otimes_R B$. Then $\Omega_{C/R}$ is generated as a C-module by the elements $d(a \otimes b) = (1 \otimes b)\, d(a \otimes 1) + (a \otimes 1)\, d(1 \otimes b)$. Under the identifications of (8.10), $\Omega_{C/B} = \Omega_{A/R} \otimes_A C$ and $\Omega_{C/A} = \Omega_{B/R} \otimes_B C$, the maps are

$$\varphi(da \otimes c) = c\, d(a \otimes 1), \quad \varphi(db \otimes c) = c\, d(1 \otimes b), \qquad \psi(d\gamma) = (d_{C/B}\gamma,\ d_{C/A}\gamma).$$

Now $\psi\varphi(da \otimes c) = c\,(d_{C/B}(a \otimes 1), d_{C/A}(a \otimes 1)) = (da \otimes c,\, 0)$ since $a \otimes 1$ is a constant for $d_{C/A}$; similarly for $db \otimes c$. So $\psi\varphi = \mathrm{id}$. Conversely $\psi(d(a \otimes b)) = (da \otimes (1 \otimes b),\ db \otimes (a \otimes 1))$, so $\varphi\psi(d(a \otimes b)) = (1 \otimes b)d(a \otimes 1) + (a \otimes 1)d(1 \otimes b) = d(a \otimes b)$; as the $d(a \otimes b)$ generate $\Omega_{C/R}$ and $\varphi\psi$ is C-linear, $\varphi\psi = \mathrm{id}$. Since φ and ψ are globally defined, they give the isomorphism $\Omega_{X \times_S Y/S} \cong p_1^*\Omega_{X/S} \oplus p_2^*\Omega_{Y/S}$.

(b) Let $n = \dim X$, $m = \dim Y$. $Z = X \times_k Y$ is integral (II, Ex. 3.15; k algebraically closed), separated and of finite type, so a variety. By (a), $\Omega_{Z/k} \cong p_1^*\Omega_{X/k} \oplus p_2^*\Omega_{Y/k}$ is locally free of rank $n+m$, since X, Y nonsingular (8.15). At the generic point, $\operatorname{rk} \Omega_{Z/k} = \dim_K \Omega_{K/k} = \text{tr. d. } K/k = \dim Z$ (K the function field, separably generated as k is perfect; (8.6A)). Hence $\dim Z = n+m$, and Z is nonsingular by (8.15). Then

$$\omega_Z = \textstyle\bigwedge^{n+m}(p_1^*\Omega_X \oplus p_2^*\Omega_Y) = \bigoplus_{i+j=n+m} \bigwedge^i p_1^*\Omega_X \otimes \bigwedge^j p_2^*\Omega_Y = \bigwedge^n p_1^*\Omega_X \otimes \bigwedge^m p_2^*\Omega_Y,$$

because $\bigwedge^i$ of a locally free sheaf of rank n vanishes for $i > n$ (and likewise for $j > m$), so only $(i, j) = (n, m)$ survives (check locally on free modules). As pull-back commutes with exterior powers and tensor products, this is $p_1^*\omega_X \otimes p_2^*\omega_Y$.

(c) $Y \subseteq \mathbf{P}^2$ is a nonsingular curve of degree 3, so by (8.20.3) $\omega_Y \cong \mathcal{O}_Y(3 - 2 - 1) = \mathcal{O}_Y$. By (b), $\omega_X \cong p_1^*\mathcal{O}_Y \otimes p_2^*\mathcal{O}_Y = \mathcal{O}_X$, so $p_g(X) = \dim_k \Gamma(X, \mathcal{O}_X)$. X is a projective variety (Segre embedding $Y \times Y \subseteq \mathbf{P}^2 \times \mathbf{P}^2 \subseteq \mathbf{P}^8$), so $\Gamma(X, \mathcal{O}_X)$ is a finite-dimensional k-algebra (II, Thm. 5.19) which is a domain (X integral), hence a finite field extension of k, hence k. So $p_g(X) = 1$.

For p_a we use the Hilbert polynomial of $X \subseteq \mathbf{P}^8$ (the arithmetic genus is independent of the projective embedding, III, Ex. 5.3; for the definition (I, Ex. 7.2) we simply use this embedding). Let $S_Y = k[x_0, x_1, x_2]/(f)$ with f the cubic; $\dim(S_Y)_m = \binom{m+2}{2} - \binom{m-1}{2} = 3m$ for $m \geq 1$. The homogeneous coordinate ring of $X \subseteq \mathbf{P}^8$ is the Segre product $R = \bigoplus_m (S_Y)_m \otimes_k (S_Y)_m$: indeed R is a quotient of the Segre ring $\bigoplus_m k[x]_m \otimes k[y]_m = k[z_{ij}]/(\text{Segre relations})$, it is a domain (a subring of $S_Y \otimes_k S_Y$, a domain since k is algebraically closed), so $R = k[z]/J$ with J prime, and the zero set of J is $Y \times Y$ (every element of J vanishes at all points $x \otimes y = (x_i y_j)$ with x, y in the affine cone of Y, so $Y \times Y \subseteq V(J)$; conversely J contains the Segre relations, so a point of $V(J)$ has the form $x \otimes y$, and for each j the element $f(z_{0j}, z_{1j}, z_{2j})$ lies in J, because it maps to $y_j^3 f(x_0, x_1, x_2) = 0$ in $S_Y \otimes S_Y$; hence $f(x) = 0$, and symmetrically $f(y) = 0$); since

J is a homogeneous prime defining X, the Nullstellensatz gives $J = I(X)$. Hence $\dim R_m = (3m)^2 = 9m^2$ for $m \geq 1$, so the Hilbert polynomial is $P_X(m) = 9m^2$ and $P_X(0) = 0$. As $\dim X = 2$, $p_a(X) = (-1)^2(P_X(0) - 1) = -1$. (Alternatively, by III, Ex. 5.3 and Kunneth, $\chi(\mathcal{O}_X) = \chi(\mathcal{O}_Y)^2 = 0$.) So $p_g(X) = 1 \neq -1 = p_a(X)$. □

Exercise 8.4

Problem. *Complete Intersections in* $\mathbf{P}^n$. A closed subscheme Y of $\mathbf{P}^n_k$ is called a (strict, global) *complete intersection* if the homogeneous ideal I of Y in $S = k[x_0, \ldots, x_n]$ can be generated by $r = \operatorname{codim}(Y, \mathbf{P}^n)$ elements (I, Ex. 2.17).

(a) Let Y be a closed subscheme of codimension r in $\mathbf{P}^n$. Then Y is a complete intersection if and only if there are hypersurfaces (i.e., locally principal subschemes of codimension 1) $H_1, \ldots, H_r$, such that $Y = H_1 \cap \cdots \cap H_r$ as schemes, i.e., $\mathcal{I}_Y = \mathcal{I}_{H_1} + \cdots + \mathcal{I}_{H_r}$. [*Hint*: Use the fact that the unmixedness theorem holds in S (Matsumura [2, p. 107]).]

(b) If Y is a complete intersection of dimension ≥ 1 in $\mathbf{P}^n$, and if Y is normal, then Y is projectively normal (Ex. 5.14). [*Hint*: Apply (8.23) to the affine cone over Y.]

(c) With the same hypotheses as (b), conclude that for all $l \geq 0$, the natural map $\Gamma(\mathbf{P}^n, \mathcal{O}_{\mathbf{P}^n}(l)) \to \Gamma(Y, \mathcal{O}_Y(l))$ is surjective. In particular, taking $l = 0$, show that Y is connected.

(d) Now suppose given integers $d_1, \ldots, d_r \geq 1$, with $r < n$. Use Bertini's theorem (8.18) to show that there exist nonsingular hypersurfaces $H_1, \ldots, H_r$ in $\mathbf{P}^n$, with $\deg H_i = d_i$, such that the scheme $Y = H_1 \cap \cdots \cap H_r$ is irreducible and nonsingular of codimension r in $\mathbf{P}^n$.

(e) If Y is a nonsingular complete intersection as in (d), show that $\omega_Y \cong \mathcal{O}_Y(\sum d_i - n - 1)$.

(f) If Y is a nonsingular hypersurface of degree d in $\mathbf{P}^n$, use (c) and (e) above to show that $p_g(Y) = \binom{d-1}{n}$. Thus $p_g(Y) = p_a(Y)$ (I, Ex. 7.2). In particular, if Y is a nonsingular plane curve of degree d, then $p_g(Y) = \frac{1}{2}(d-1)(d-2)$.

(g) If Y is a nonsingular curve in $\mathbf{P}^3$, which is a complete intersection of nonsingular surfaces of degrees d, e, then $p_g(Y) = \frac{1}{2}de(d+e-4)+1$. Again the geometric genus is the same as the arithmetic genus (I, Ex. 7.2).

Solution. *Conventions.* k is algebraically closed, $S = k[x_0, \ldots, x_n]$, $\mathfrak{n} = (x_0, \ldots, x_n)$. The "homogeneous ideal" of a closed subscheme Y is $I_Y = \Gamma_*(\mathcal{I}_Y) = \bigoplus_l \Gamma(\mathbf{P}^n, \mathcal{I}_Y(l))$ (II, Ex. 5.9, 5.10): the saturated ideal, with $\widetilde{I_Y} = \mathcal{I}_Y$ and $Y = \operatorname{Proj} S/I_Y$. We use: *if J is any homogeneous ideal with $\widetilde{J} = \mathcal{I}_Y$, then $I_Y = \bigcup_m (J : \mathfrak{n}^m)$; in particular $J = I_Y$ if $\mathfrak{n} \notin \operatorname{Ass}(S/J)$.* (By Ex. 5.9, $J_l = (I_Y)_l$ for $l \gg 0$; so for $f \in I_Y$ homogeneous, $\mathfrak{n}^m f \subseteq J$ for some m, whence $f \in J$ if $\mathfrak{n}$ is not an associated prime of S/J.) A hypersurface H has invertible ideal sheaf, $\mathcal{I}_H \cong \mathcal{O}(-d)$ for some $d > 0$ (Pic $\mathbf{P}^n = \mathbf{Z}$), and the inclusion $\mathcal{O}(-d) \to \mathcal{O}$ is multiplication by a nonzero form $F \in S_d = \Gamma(\mathcal{O}(d))$; thus $H = \operatorname{Proj} S/(F)$, $\mathcal{I}_H = \widetilde{(F)}$, and conversely every nonconstant form defines a hypersurface (codimension 1 by Krull). For an irreducible closed $Z \subseteq \mathbf{P}^n$, the homogeneous prime $\mathfrak{p}_Z$ has $\operatorname{ht} \mathfrak{p}_Z = n - \dim Z = \operatorname{codim}(Z, \mathbf{P}^n)$, so $\operatorname{ht} I_Y = \operatorname{codim}(Y, \mathbf{P}^n)$ for $Y \neq \emptyset$.

(a) ($\Rightarrow$) Let $I = I_Y$ be generated by r elements. The k-vector space $I/\mathfrak{n}I$ is generated by the images of these, so by the graded Nakayama lemma I is generated by $r' \leq r$ homogeneous elements $F_1, \ldots, F_{r'}$ (lifts of a basis of $I/\mathfrak{n}I$). By Krull's height theorem $r = \operatorname{ht} I \leq r'$, so $r' = r$. No F_i is 0 (else $r-1$ generators) or a nonzero constant (else $Y = \emptyset$). So $H_i = \operatorname{Proj} S/(F_i)$ are hypersurfaces, and $\mathcal{I}_Y = \widetilde{I} = \sum \widetilde{(F_i)} = \sum \mathcal{I}_{H_i}$.

($\Leftarrow$) Suppose $\mathcal{I}_Y = \sum \mathcal{I}_{H_i}$, $H_i = \operatorname{Proj} S/(F_i)$ with $\deg F_i = d_i \geq 1$, and $\operatorname{codim} Y = r$. Put $J = (F_1, \ldots, F_r)$; then $J = \mathcal{I}_Y$. As $Y \neq \emptyset$ we have $r \leq n$. By Krull every minimal prime of J has height $\leq r$. The zero set of J in $\mathbf{A}^{n+1}$ is the cone over Y together with the origin; every component of $V(J)$ has dimension $\geq n+1-r \geq 1$, so the origin is not a component, and the components are the cones over the components of Y; the latter have dim $\leq n-r$ (as $\operatorname{codim} Y = r$) and $\geq n-r$ (Krull). Hence all minimal primes of J have height exactly r, and $\operatorname{ht} J = r$. Now S is Cohen–Macaulay (regular, (8.21Aa)) and J is generated by $\operatorname{ht} J$ elements, so by the unmixedness theorem every associated prime of S/J has height $r \leq n < n+1 = \operatorname{ht} \mathfrak{n}$. Hence $\mathfrak{n} \notin \operatorname{Ass}(S/J)$ and $J = I_Y$ is generated by r elements: Y is a complete intersection. (This also shows that, for a complete intersection, $I_Y = (F_1, \ldots, F_r)$ with $F_1, \ldots, F_r$ forming a regular sequence.)

(b) Let $I = I_Y = (F_1, \ldots, F_r)$ (by (a), with homogeneous F_i), $S' = S/I$, and let $C = \operatorname{Spec} S' \subseteq \mathbf{A}^{n+1}$ be the affine cone, $\mathfrak{n}' = \mathfrak{n}/I$ the vertex. By the proof of (a), C has pure dimension $n+1-r = \dim Y + 1 \geq 2$, i.e. $\operatorname{codim}(C, \mathbf{A}^{n+1}) = r$ and I is generated by r elements. So C is a locally complete intersection in the nonsingular variety $\mathbf{A}^{n+1}$ and (8.23) applies: C is Cohen–Macaulay, and C is normal iff it is regular in codimension 1. Moreover $\dim S'_{\mathfrak{n}'} = n+1-r \geq 2$.

$C \setminus \{\mathfrak{n}'\}$ is normal. It is covered by $D(x_j)$, with ring $S'_{x_j} \cong A_j[t, t^{-1}]$, $A_j = (S'_{x_j})_0 = \mathcal{O}(Y \cap D_+(x_j))$, $t = x_j$ (every element of S'_{x_j} is a finite sum of homogeneous parts, and x_j is transcendental over the degree-0 part). Now Y is normal, so A_j is a (finite product of) integrally closed domain(s), and $A_j[t, t^{-1}]$ is again a normal ring. Hence $C \setminus \{\mathfrak{n}'\}$ is normal; in particular it is regular in codimension 1 (normal noetherian local rings of dimension 1 are DVRs). Points of codimension 1 of C lie in $C \setminus \{\mathfrak{n}'\}$ because $\dim S'_{\mathfrak{n}'} \geq 2$. Therefore C is regular in codimension 1, hence normal by (8.23). So S' is a normal ring; $\operatorname{Spec} S'$ is connected (every minimal prime of the homogeneous ideal I is homogeneous, hence contained in $\mathfrak{n}'$), and a normal noetherian ring is a finite product of normal domains, so S' is a domain. Thus $S' = S/I_Y$ is an integrally closed domain: Y is projectively normal.

(c) By (b), S' is a normal noetherian domain, and $U = C \setminus \{\mathfrak{n}'\} = \bigcup D(x_j)$ contains all height-one primes. By (6.3A), $S' = \bigcap_{\operatorname{ht}\mathfrak{p}=1} S'_{\mathfrak{p}}$ inside $K = \operatorname{Frac} S'$. A section of $\mathcal{O}_U$ lies in each $S'_{\mathfrak{p}}$ ($\operatorname{ht}\mathfrak{p} = 1$), so $\Gamma(U, \mathcal{O}_U) = S'$. On the other hand, $\Gamma(U, \mathcal{O}_U) = \bigoplus_{l \in \mathbf{Z}} \Gamma(Y, \mathcal{O}_Y(l))$: a section is a family (s_j), $s_j \in S'_{x_j}$, compatible on overlaps; all localization maps are graded, so the homogeneous components of degree l of the s_j form again a compatible family, i.e. a section of $\mathcal{O}_Y(l)$ (a degree-l element of S'_{x_j} is exactly a section of $\mathcal{O}_Y(l)$ over $D_+(x_j) \cap Y$; here $Y = \operatorname{Proj} S'$ and $\mathcal{O}_Y(l) = \mathcal{O}_{\mathbf{P}^n}(l)|_Y$), and only finitely many degrees occur since there are finitely many j. The isomorphism $S' \to \Gamma(U, \mathcal{O}_U)$ is graded, so $S'_l \to \Gamma(Y, \mathcal{O}_Y(l))$ is bijective for all $l \geq 0$. As $S_l = \Gamma(\mathbf{P}^n, \mathcal{O}(l)) \to S'_l$ is surjective, so is $\Gamma(\mathbf{P}^n, \mathcal{O}(l)) \to \Gamma(Y, \mathcal{O}_Y(l))$.

For $l = 0$: $\Gamma(Y, \mathcal{O}_Y) = S'_0$ is a quotient of $S_0 = k$, so it equals k ($Y \neq \emptyset$). If $Y = Y_1 \sqcup Y_2$ with Y_1, Y_2 nonempty and closed, then $\Gamma(Y, \mathcal{O}_Y) = \Gamma(Y_1, \mathcal{O}) \times \Gamma(Y_2, \mathcal{O})$, a product of two nonzero k-algebras, of k-dimension ≥ 2, contradiction. So Y is connected.

(d) *Setup.* Let $v_d : \mathbf{P}^n \to \mathbf{P}^N$ be the d-uple embedding. Hyperplanes $H' \subset \mathbf{P}^N$ correspond bijectively (linear isomorphism of projective spaces) to degree-d forms F up to scalars, via $v_d^{-1}(H') = \operatorname{Proj} S/(F)$ (the coordinates of $\mathbf{P}^N$ are the monomials of degree d). If $X \subseteq \mathbf{P}^n$ is a nonsingular closed subvariety, Bertini (8.18) applied to $v_d(X) \subseteq \mathbf{P}^N$ gives a nonempty open set $\mathcal{U}_d(X)$ of forms F of degree d such that $X \not\subseteq V(F)$ and the scheme $X \cap \operatorname{Proj} S/(F)$ is regular at every point (v_d is an isomorphism onto its image, so scheme-theoretic intersections correspond).

Induction. We construct $H_1, \dots, H_r$ with $H_i = \operatorname{Proj} S/(F_i)$, $\deg F_i = d_i$, such that, with $Y_i = H_1 \cap \cdots \cap H_i$ ($Y_0 = \mathbf{P}^n$): H_i is nonsingular, and Y_i is a nonsingular irreducible closed subscheme of codimension i in $\mathbf{P}^n$. Suppose Y_{i-1} is constructed ($i \le r < n$; $Y_0 = \mathbf{P}^n$). The open sets $\mathcal{U}_{d_i}(\mathbf{P}^n)$ and $\mathcal{U}_{d_i}(Y_{i-1})$ of the irreducible space $\mathbf{P}(S_{d_i})$ are nonempty, hence meet; pick F_i in the intersection. Then H_i is nonsingular and $Y_i = Y_{i-1} \cap H_i$ is regular. Since Y_{i-1} is an irreducible projective variety of dimension $n - i + 1 \ge 1$ not contained in H_i, every component of Y_i has dimension exactly $n - i$ (Krull's Hauptidealsatz and (I, Thm. 7.2)), and $Y_i \ne \emptyset$ (a hypersurface meets every projective variety of positive dimension, (I, Thm. 7.2)). Thus $Y_i = H_1 \cap \cdots \cap H_i$ has codimension i, and by (a) it is a complete intersection. Being regular, it is normal, and $\dim Y_i = n - i \ge n - r \ge 1$; so by (c) Y_i is connected. A connected regular scheme is irreducible (its local rings are domains, so distinct irreducible components cannot meet). This completes the induction; $Y = Y_r$ is as required.

(e) Let $Y = H_1 \cap \cdots \cap H_r$, $H_i = \operatorname{Proj} S/(F_i)$, $\deg F_i = d_i$, be a nonsingular complete intersection of codimension r (nonsingular subvariety of $\mathbf{P}^n$). The maps $\mathcal{O}(-d_i) \to \mathcal{I}_Y$, $1 \mapsto F_i$, give a map $\bigoplus \mathcal{O}_Y(-d_i) \to \mathcal{I}_Y/\mathcal{I}_Y^2$, surjective because $\mathcal{I}_Y = \sum F_i \mathcal{O}$. By (8.17) the target is locally free of rank r, and so is the source; a surjection of locally free sheaves of the same finite rank is an isomorphism (on stalks it is a surjective homomorphism $\mathcal{O}_y^r \to \mathcal{O}_y^r$ of finitely generated modules, hence bijective). Hence $\mathcal{N}_{Y/\mathbf{P}^n} = \mathcal{H}om(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y) \cong \bigoplus \mathcal{O}_Y(d_i)$ and $\bigwedge^r \mathcal{N}_{Y/\mathbf{P}^n} \cong \mathcal{O}_Y(\sum d_i)$. By (8.20) and (8.20.1),

$$\omega_Y \cong \omega_{\mathbf{P}^n} \otimes \textstyle\bigwedge^r \mathcal{N}_{Y/\mathbf{P}^n} \cong \mathcal{O}_Y(-n-1) \otimes \mathcal{O}_Y(\sum d_i) = \mathcal{O}_Y(\sum d_i - n - 1).$$

(f) Let $n \ge 2$ and $Y = \operatorname{Proj} S/(F)$ nonsingular of degree d (a complete intersection with $r = 1$; $\dim Y = n - 1 \ge 1$; Y regular hence normal). By (e), $\omega_Y \cong \mathcal{O}_Y(d - n - 1)$, so $p_g(Y) = \dim \Gamma(Y, \mathcal{O}_Y(l))$ with $l = d - n - 1$. By the proof of (c), $\Gamma(Y, \mathcal{O}_Y(l)) = S'_l$ for all $l \in \mathbf{Z}$, where $S' = S/(F)$ (this is 0 for $l < 0$); so $\dim \Gamma(Y, \mathcal{O}_Y(l)) = \dim S_l - \dim S_{l-d}$. Here $l - d = -n - 1 < 0$, so $S_{l-d} = 0$, and

$$p_g(Y) = \dim S_{d-n-1} = \tbinom{d-1}{n}$$

(for $d - n - 1 \ge 0$ this is $\binom{d-1}{n}$; for $d \le n$ both sides are 0). On the other hand the Hilbert polynomial of Y is $P_Y(m) = \binom{m+n}{n} - \binom{m-d+n}{n}$, so $P_Y(0) = 1 - \binom{n-d}{n}$ (polynomial binomial), and since $\binom{n-d}{n} = (-1)^n \binom{d-1}{n}$,

$$p_a(Y) = (-1)^{n-1}(P_Y(0) - 1) = (-1)^n \tbinom{n-d}{n} = \tbinom{d-1}{n}.$$

So $p_g(Y) = p_a(Y)$. For $n = 2$ this is $\frac{1}{2}(d-1)(d-2)$.

(g) Let $Y = H_1 \cap H_2$, $H_1 = V(F)$, $H_2 = V(G)$, $\deg F = d$, $\deg G = e$, a nonsingular curve in $\mathbf{P}^3$. By (a), $I_Y = (F, G)$. By (e), $\omega_Y \cong \mathcal{O}_Y(d + e - 4)$, and by the proof of (c) (here Y is normal of dimension 1) $\Gamma(Y, \mathcal{O}_Y(l)) = (S/I_Y)_l$ for all $l \in \mathbf{Z}$; hence, with $l = d + e - 4$, $p_g(Y) = \dim(S/(F, G))_l$. The forms F, G are coprime (a common nonconstant factor π would give $V(\pi) \subseteq Y$ with $\dim V(\pi) = 2$), so G is a nonzerodivisor on $S/(F)$ (S is a UFD: if $F \mid GH$ and $F \nmid H$ some prime power $\pi^a \parallel F$ does not divide H, so $\pi \mid G$). Hence $0 \to S(-d-e) \xrightarrow{(G,-F)} S(-d) \oplus S(-e) \xrightarrow{(F,G)} S \to S/(F, G) \to 0$ is exact, and

$$\dim(S/(F,G))_l = \sigma(l) - \sigma(l-d) - \sigma(l-e) + \sigma(l-d-e), \qquad \sigma(m) = \dim S_m = \tbinom{m+3}{3}\ (m \ge 0),\ 0\ (m < 0).$$

At $l = d+e-4$ the last term is $\sigma(-4) = 0$, and the sum is $\sigma(d+e-4) - \sigma(e-4) - \sigma(d-4)$ (also correct if $l < 0$). Since $d, e \geq 1$ we have $\sigma(m) = \frac{(m+3)(m+2)(m+1)}{6}$ for all $m \geq -3$ (the polynomial vanishes at $-1, -2, -3$), so with $s = d+e$, $p = de$:

$$p_g(Y) = \frac{(s-1)(s-2)(s-3) - (e-1)(e-2)(e-3) - (d-1)(d-2)(d-3)}{6} = \frac{3p(s-4)+6}{6} = \tfrac{1}{2}de(d+e-4)+1,$$

where we used $(s-1)(s-2)(s-3) = s^3 - 6s^2 + 11s - 6$, $d^3 + e^3 = s^3 - 3ps$, $d^2 + e^2 = s^2 - 2p$. Finally, for $m \gg 0$, $P_Y(m) = \sigma(m) - \sigma(m-d) - \sigma(m-e) + \sigma(m-d-e)$ as polynomials, so $P_Y(0) = 1 + \frac{(d-1)(d-2)(d-3)+(e-1)(e-2)(e-3)-(s-1)(s-2)(s-3)}{6} = -\frac{1}{2}p(s-4)$ (use $\binom{3-d}{3} = -\frac{(d-1)(d-2)(d-3)}{6}$ etc.), and $p_a(Y) = (-1)^1(P_Y(0) - 1) = 1 + \frac{1}{2}de(d+e-4) = p_g(Y)$. □

Exercise 8.5

Problem. *Blowing up a Nonsingular Subvariety.* As in (8.24), let X be a nonsingular variety. Let Y be a nonsingular subvariety of codimension $r \geq 2$, let $\pi : \widetilde{X} \to X$ be the blow-up of X along Y, and let $Y' = \pi^{-1}(Y)$.

(a) Show that the maps $\pi^* : \operatorname{Pic} X \to \operatorname{Pic} \widetilde{X}$, and $\mathbf{Z} \to \operatorname{Pic} \widetilde{X}$ defined by $n \mapsto$ class of nY', give rise to an isomorphism $\operatorname{Pic} \widetilde{X} \cong \operatorname{Pic} X \oplus \mathbf{Z}$.

(b) Show that $\omega_{\widetilde{X}} \cong f^*\omega_X \otimes \mathcal{L}((r-1)Y')$. [*Hint*: By (a) we can write in any case $\omega_{\widetilde{X}} \cong f^*\mathcal{M} \otimes \mathcal{L}(qY')$ for some invertible sheaf $\mathcal{M}$ on X, and some integer q. By restricting to $\widetilde{X} - Y' \cong X - Y$, show that $\mathcal{M} \cong \omega_X$. To determine q, proceed as follows. First show that $\omega_{Y'} \cong f^*\omega_X \otimes \mathcal{O}_{Y'}(-q-1)$. Then take a closed point $y \in Y$ and let Z be the fibre of Y' over y. Then show that $\omega_Z \cong \mathcal{O}_Z(-q-1)$. But since $Z \cong \mathbf{P}^{r-1}$, we have $\omega_Z \cong \mathcal{O}_Z(-r)$, so $q = r-1$.]

Solution. (The book writes f for π in (b); $\mathcal{L}(D) = \mathcal{O}(D)$.) By (8.24), $\widetilde{X}$ is a nonsingular variety (of dimension $n = \dim X$), $Y' \cong \mathbf{P}(\mathcal{I}/\mathcal{I}^2)$ is a $\mathbf{P}^{r-1}$-bundle over Y, locally $Y \times \mathbf{P}^{r-1}$, hence a nonsingular (reduced) subvariety; it is irreducible (it is covered by the opens $Y_i \times \mathbf{P}^{r-1}$, which are integral, and any two of them meet since Y is irreducible and $Y' \to Y$ is surjective). Its ideal sheaf $\mathcal{I}_{Y'} = \pi^{-1}\mathcal{I} \cdot \mathcal{O}_{\widetilde{X}}$ is invertible, so Y' is a prime divisor with $\mathcal{I}_{Y'} = \mathcal{O}(-Y')$; and by (8.24c) the normal sheaf $\mathcal{N}_{Y'/\widetilde{X}} = \mathcal{O}(Y')|_{Y'}$ corresponds to $\mathcal{O}_{\mathbf{P}(\mathcal{I}/\mathcal{I}^2)}(-1)$. Hence

$$\mathcal{O}(nY')|_{Y'} \cong \mathcal{O}_{Y'}(-n), \tag{1}$$

and its restriction to a fibre $Z \cong \mathbf{P}^{r-1}$ is $\mathcal{O}_{\mathbf{P}^{r-1}}(-n)$. Put $U = \widetilde{X} \setminus Y' \cong X \setminus Y$.

(a) All of $X, \widetilde{X}, U$ are nonsingular, so Pic = Cl (6.16), compatibly with restriction to open sets. Since $\operatorname{codim}(Y, X) \geq 2$, restriction $\operatorname{Pic} X \to \operatorname{Pic}(X \setminus Y)$ is an isomorphism (6.5(b)). Since Y' is an irreducible closed subset of codimension 1, (6.5(c)) gives an exact sequence

$$\mathbf{Z} \xrightarrow{n \mapsto nY'} \operatorname{Pic} \widetilde{X} \xrightarrow{\rho} \operatorname{Pic} U \to 0.$$

The composite $\rho \circ \pi^* : \operatorname{Pic} X \to \operatorname{Pic} U = \operatorname{Pic}(X \setminus Y)$ is the restriction map, because $\pi|_U : U \to X \setminus Y$ is an isomorphism; so it is an isomorphism. Consider $\Phi : \operatorname{Pic} X \oplus \mathbf{Z} \to \operatorname{Pic} \widetilde{X}$, $(L, n) \mapsto \pi^*L \otimes \mathcal{O}(nY')$, a homomorphism.

Injective: if $\pi^*L \otimes \mathcal{O}(nY') \cong \mathcal{O}$, restrict to U: $\rho\pi^*L = L|_{X \setminus Y}$ is trivial (as $\mathcal{O}(nY')|_U$ is trivial), so L is trivial, so $\mathcal{O}(nY')$ is trivial. Restricting to Y' and a fibre $Z \cong \mathbf{P}^{r-1}$ and using (1),

$\mathcal{O}_{\mathbf{P}^{r-1}}(-n)$ is trivial; since $r-1 \geq 1$ this forces $n = 0$ (for $n > 0$, $\Gamma(\mathcal{O}(-n)) = 0 \neq \Gamma(\mathcal{O})$; for $n < 0$, $\dim \Gamma(\mathcal{O}(-n)) > 1$).

Surjective: let $\mathcal{N} \in \operatorname{Pic} \widetilde{X}$. Choose $L \in \operatorname{Pic} X$ with $L|_{X\setminus Y} \cong \mathcal{N}|_U$ (possible since $\rho\pi^*$ is onto). Then $\mathcal{N} \otimes \pi^* L^{-1} \in \ker \rho$, which is generated by $\mathcal{O}(Y')$, so it is $\mathcal{O}(nY')$ for some n, and $\mathcal{N} = \Phi(L, n)$. Hence Φ is an isomorphism.

(b) By (a), $\omega_{\widetilde{X}} \cong \pi^*\mathcal{M} \otimes \mathcal{O}(qY')$ for some $\mathcal{M} \in \operatorname{Pic} X$, $q \in \mathbf{Z}$. Restrict to $U \cong X \setminus Y$: $\omega_{\widetilde{X}}|_U = \omega_U$ is identified with $\omega_X|_{X\setminus Y}$ under $\pi|_U$, while $\pi^*\mathcal{M}|_U = \mathcal{M}|_{X\setminus Y}$ and $\mathcal{O}(qY')|_U$ is trivial. So $\mathcal{M}|_{X\setminus Y} \cong \omega_X|_{X\setminus Y}$ and, since $\operatorname{Pic} X \to \operatorname{Pic}(X \setminus Y)$ is injective (by (a)), $\mathcal{M} \cong \omega_X$. Therefore $\omega_{\widetilde{X}} \cong \pi^*\omega_X \otimes \mathcal{O}(qY')$.

Determination of q. By (8.20) with codimension 1 and by (1),

$$\omega_{Y'} \cong \omega_{\widetilde{X}}|_{Y'} \otimes \mathcal{O}(Y')|_{Y'} \cong (\pi^*\omega_X)|_{Y'} \otimes \mathcal{O}_{Y'}(-q) \otimes \mathcal{O}_{Y'}(-1) = (\pi^*\omega_X)|_{Y'} \otimes \mathcal{O}_{Y'}(-q-1).$$

Let $y \in Y$ be a closed point, $g : Y' \to Y$ the projection and $Z = g^{-1}(y) \cong \mathbf{P}^{r-1}$ the fibre. Over a neighbourhood W of y in Y we have $g^{-1}W \cong W \times \mathbf{P}^{r-1}$ and $Z = \{y\} \times \mathbf{P}^{r-1}$. If $t_1, \ldots, t_s$ ($s = \dim Y$) generate the maximal ideal of y on W, then $\mathcal{I}_Z = (t_1, \ldots, t_s)\mathcal{O}_{g^{-1}W}$; Z is a nonsingular subvariety of Y' of codimension s, so $\mathcal{I}_Z/\mathcal{I}_Z^2$ is locally free of rank s (8.17), and it is generated by the s global sections $\bar{t}_i$, hence free on them. So $\mathcal{N}_{Z/Y'} \cong \mathcal{O}_Z^s$ and, by (8.20), $\omega_Z \cong \omega_{Y'}|_Z \otimes \bigwedge^s \mathcal{N}_{Z/Y'} \cong \omega_{Y'}|_Z$. Moreover Z is mapped by π to the point y, so $(\pi^*\omega_X)|_Z$ is the pull-back of a line bundle on $\operatorname{Spec} k(y)$, i.e. trivial. Hence

$$\omega_Z \cong \mathcal{O}_Z(-q-1).$$

But $Z \cong \mathbf{P}^{r-1}$, so $\omega_Z \cong \mathcal{O}_Z(-r)$ (8.20.1). As $\operatorname{Pic} \mathbf{P}^{r-1} = \mathbf{Z}$ (generated by $\mathcal{O}(1)$) and $r - 1 \geq 1$, $q + 1 = r$, i.e. $q = r - 1$. Thus $\omega_{\widetilde{X}} \cong \pi^*\omega_X \otimes \mathcal{O}((r-1)Y')$. □

Exercise 8.6

Problem. *The Infinitesimal Lifting Property.* The following result is very important in studying deformations of nonsingular varieties. Let k be an algebraically closed field, let A be a finitely generated k-algebra such that $\operatorname{Spec} A$ is a nonsingular variety over k. Let $0 \to I \to B' \to B \to 0$ be an exact sequence, where B' is a k-algebra, and I is an ideal with $I^2 = 0$. Finally suppose given a k-algebra homomorphism $f : A \to B$. Then there exists a k-algebra homomorphism $g : A \to B'$ making a commutative diagram

$$\begin{array}{ccc} & & 0 \\ & & \downarrow \\ & & I \\ & & \downarrow \\ & & B' \\ & \overset{g}{\nearrow} & \downarrow \\ A & \xrightarrow{f} & B \\ & & \downarrow \\ & & 0 \end{array}$$

We call this result the *infinitesimal lifting property* for A. We prove this result in several steps.

(a) First suppose that $g : A \to B'$ is a given homomorphism lifting f. If $g' : A \to B'$ is another such homomorphism, show that $\theta = g - g'$ is a k-derivation of A into I, which we can consider as an element of $\mathrm{Hom}_A(\Omega_{A/k}, I)$. Note that since $I^2 = 0$, I has a natural structure of B-module, and hence of A-module. Conversely, for any $\theta \in \mathrm{Hom}_A(\Omega_{A/k}, I)$, $g' = g + \theta$ is another homomorphism lifting f. (For this step, you do not need the hypothesis about $\mathrm{Spec}\, A$ being nonsingular.)

(b) Now let $P = k[x_1, \ldots, x_n]$ be a polynomial ring over k of which A is a quotient, and let J be the kernel. Show that there does exist a homomorphism $h : P \to B'$ making a commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & J & \to & P & \to & A & \to & 0 \\ & & \downarrow & & \downarrow h & & \downarrow f & & \\ 0 & \to & I & \to & B' & \to & B & \to & 0 \end{array}$$

and that h induces an A-linear map $\bar{h} : J/J^2 \to I$.

(c) Now use the hypothesis $\mathrm{Spec}\, A$ nonsingular and (8.17) to obtain an exact sequence

$$0 \to J/J^2 \to \Omega_{P/k} \otimes_P A \to \Omega_{A/k} \to 0.$$

Show furthermore that applying the functor $\mathrm{Hom}_A(\,\cdot\,, I)$ gives an exact sequence

$$0 \to \mathrm{Hom}_A(\Omega_{A/k}, I) \to \mathrm{Hom}_P(\Omega_{P/k}, I) \to \mathrm{Hom}_A(J/J^2, I) \to 0.$$

Let $\theta \in \mathrm{Hom}_P(\Omega_{P/k}, I)$ be an element whose image gives $\bar{h} \in \mathrm{Hom}_A(J/J^2, I)$. Consider θ as a derivation of P into B'. Let $h' = h - \theta$. Then show that h' is a homomorphism $P \to B'$ such that $h'(J) = 0$. Thus h' induces the desired homomorphism $g : A \to B'$.

Solution. Throughout, I is annihilated by I, so multiplication by $b' \in B'$ on I depends only on the class of b' in $B = B'/I$; this makes I a B-module, an A-module via f, and a P-module via $P \to A \to B$.

(a) Let $g, g' : A \to B'$ both lift f. Then $\theta = g - g'$ takes values in $\ker(B' \to B) = I$, is additive and k-linear, and for $a, a' \in A$,

$$\theta(aa') = g(a)g(a') - g'(a)g'(a') = g(a)\theta(a') + \theta(a)g'(a') = a\,\theta(a') + a'\,\theta(a),$$

since $g(a)$ and $g'(a')$ act on I as multiplication by $f(a)$ and $f(a')$, i.e. by a and a' in the A-module structure. So $\theta \in \mathrm{Der}_k(A, I) = \mathrm{Hom}_A(\Omega_{A/k}, I)$. Conversely let $\theta \in \mathrm{Der}_k(A, I)$ and $g' = g + \theta$. Then g' is additive, k-linear, $g'(1) = 1$ (a derivation kills 1), lifts f (as $\theta(A) \subseteq I$), and

$$g'(a)g'(a') = g(a)g(a') + g(a)\theta(a') + \theta(a)g(a') + \theta(a)\theta(a') = g(aa') + a\theta(a') + a'\theta(a) = g'(aa')$$

because $\theta(a)\theta(a') \in I^2 = 0$. So g' is a k-algebra homomorphism lifting f.

(b) Let $\bar{x}_i$ be the image of x_i in A and choose $b_i' \in B'$ lifting $f(\bar{x}_i) \in B$. Let $h : P \to B'$ be the k-algebra homomorphism with $h(x_i) = b_i'$. The composite $P \to B' \to B$ and $P \to A \xrightarrow{f} B$ are k-algebra homomorphisms agreeing on the x_i, hence equal: the right square commutes. Consequently $h(J) \subseteq I$, giving the left vertical map $J \to I$. As $h(J^2) \subseteq I^2 = 0$, h induces $\bar{h} : J/J^2 \to I$. It is A-linear: for $p \in P$, $j \in J$, $h(pj) = h(p)h(j) = \bar{p} \cdot h(j)$, where $\bar{p}$ is the image of p in A (acting on I via f).

(c) $X = \mathbf{A}^n_k = \mathrm{Spec}\, P$ is a nonsingular variety and $Y = \mathrm{Spec}\, A$ is a nonsingular closed subvariety with ideal $\widetilde{J}$. By (8.17) (taking global sections on the affine Y) we get the exact sequence

$0 \to J/J^2 \xrightarrow{\delta} \Omega_{P/k} \otimes_P A \to \Omega_{A/k} \to 0$ ($\delta(\bar{j}) = dj \otimes 1$), and $\Omega_{A/k}$ is locally free of finite rank (8.15), hence a projective A-module. Therefore this sequence splits, and applying $\mathrm{Hom}_A(\,\cdot\,, I)$ gives the exact sequence

$$0 \to \mathrm{Hom}_A(\Omega_{A/k}, I) \to \mathrm{Hom}_A(\Omega_{P/k} \otimes_P A, I) \to \mathrm{Hom}_A(J/J^2, I) \to 0,$$

where $\mathrm{Hom}_A(\Omega_{P/k} \otimes_P A, I) = \mathrm{Hom}_P(\Omega_{P/k}, I)$, and the last map is $\theta \mapsto \theta \circ \delta$, i.e. $j \mapsto \theta(dj)$ for $j \in J$.

Choose $\theta \in \mathrm{Hom}_P(\Omega_{P/k}, I)$ mapping to $\bar{h}$: $\theta(dj) = h(j)$ for all $j \in J$. Let $D = \theta \circ d : P \to I$, a k-derivation of P into I (the P-module structure on I is the one above). Apply the converse part of (a) with P in place of A, the map $P \to B$ in place of f, and the lift h: since $-D \in \mathrm{Der}_k(P, I)$, $h' = h - D$ is a k-algebra homomorphism $P \to B'$ lifting $P \to B$. For $j \in J$,

$$h'(j) = h(j) - \theta(dj) = h(j) - h(j) = 0.$$

So $h'(J) = 0$ and h' factors through a k-algebra homomorphism $g : A = P/J \to B'$. It lifts f: $g(\bar{p}) = h'(p) \equiv h(p) \equiv f(\bar{p}) \pmod{I}$. This proves the infinitesimal lifting property. □

Exercise 8.7

> **Problem.** As an application of the infinitesimal lifting property, we consider the following general problem. Let X be a scheme of finite type over k, and let $\mathcal{F}$ be a coherent sheaf on X. We seek to classify schemes X' over k, which have a sheaf of ideals $\mathcal{I}$ such that $\mathcal{I}^2 = 0$ and $(X', \mathcal{O}_{X'}/\mathcal{I}) \cong (X, \mathcal{O}_X)$, and such that $\mathcal{I}$ with its resulting structure of $\mathcal{O}_X$-module is isomorphic to the given sheaf $\mathcal{F}$. We call such an X', together with $\mathcal{I}$, an *infinitesimal extension* of the scheme X by the sheaf $\mathcal{F}$. One such extension, the *trivial* one, is obtained as follows. Take $\mathcal{O}_{X'} = \mathcal{O}_X \oplus \mathcal{F}$ as sheaves of abelian groups, and define multiplication by $(a \oplus f) \cdot (a' \oplus f') = aa' \oplus (af' + a'f)$. Then the topological space X with the sheaf of rings $\mathcal{O}_{X'}$ is an infinitesimal extension of X by $\mathcal{F}$.
>
> The general problem of classifying extensions of X by $\mathcal{F}$ can be quite complicated. So for now, just prove the following special case: if X is affine and nonsingular, then any extension of X by a coherent sheaf $\mathcal{F}$ is isomorphic to the trivial one. (See (III, Ex. 4.10) for another case.)

Solution. (Here, as in Ex. 8.6, k is algebraically closed and "nonsingular" means that X is a nonsingular variety.) Let $X = \mathrm{Spec}\, A$ with A a finitely generated k-algebra, and let $(X', \mathcal{I})$ be an extension of X by $\mathcal{F}$. The underlying space of X' is $|X|$, and we have an exact sequence of sheaves of k-algebras

$$0 \to \mathcal{F} \to \mathcal{O}_{X'} \xrightarrow{\pi} \mathcal{O}_X \to 0,$$

with $\mathcal{F} = \mathcal{I}$, $\mathcal{F}^2 = 0$, and the $\mathcal{O}_X$-module structure on $\mathcal{F}$ given by $b' \cdot f = \pi(b')f$.

Step 1: a ring-theoretic section. Put $B' = \Gamma(X, \mathcal{O}_{X'})$ and $I = \Gamma(X, \mathcal{F})$. Since X is affine and $\mathcal{F}$ is quasi-coherent, $H^1(X, \mathcal{F}) = 0$ (III, Thm. 3.5; cohomology of a sheaf of abelian groups on the space X is the same whether computed in $\mathfrak{Ab}(X)$ or in $\mathcal{O}_X$-modules, III, Prop. 2.6). Thus the long exact cohomology sequence gives an exact sequence $0 \to I \to B' \to A \to 0$ of k-algebras, with $I^2 = 0$. Apply Ex. 8.6 to the finitely generated k-algebra A (with $\mathrm{Spec}\, A$ a nonsingular variety), the quotient $B' \to B := A$ and $f = \mathrm{id}_A$: there is a k-algebra homomorphism $g : A \to B'$ with $\pi \circ g = \mathrm{id}_A$.

Step 2: a section of sheaves of rings. For $a \in A$ the element $g(a)|_{D(a)} \in \mathcal{O}_{X'}(D(a))$ maps to the unit $a \in \mathcal{O}_X(D(a)) = A_a$. An element $x \in \mathcal{O}_{X'}(U)$ whose image in $\mathcal{O}_X(U)$ is a unit is itself a unit: at each stalk, the image of x_z is a unit, so (surjectivity of $\mathcal{O}_{X',z} \to \mathcal{O}_{X,z}$) there is y_z with $x_z y_z = 1 + \varepsilon$, $\varepsilon \in \mathcal{F}_z$, $\varepsilon^2 = 0$, and then $x_z \cdot y_z(1 - \varepsilon) = 1$; so x_z is a unit for all $z \in U$, and an element of a sheaf of rings which is a unit in every stalk is a unit (the local inverses are unique, hence glue). Hence by the universal property of localization g extends uniquely to k-algebra maps $\sigma_a : A_a \to \mathcal{O}_{X'}(D(a))$ with $\pi\sigma_a = \mathrm{id}$; by uniqueness they are compatible with restriction between basic open sets, and so define a morphism of sheaves of k-algebras $\sigma : \mathcal{O}_X \to \mathcal{O}_{X'}$ with $\pi\sigma = \mathrm{id}_{\mathcal{O}_X}$.

Step 3: the isomorphism with the trivial extension. Define $\Phi : \mathcal{O}_X \oplus \mathcal{F} \to \mathcal{O}_{X'}$, $\Phi(a \oplus f) = \sigma(a) + f$. It is additive, and

$$\Phi(a \oplus f)\Phi(a' \oplus f') = \sigma(aa') + \sigma(a)f' + \sigma(a')f + ff' = \sigma(aa') + af' + a'f = \Phi\big((a \oplus f)(a' \oplus f')\big),$$

because $ff' \in \mathcal{I}^2 = 0$ and $\sigma(a)f' = \pi(\sigma(a))f' = af'$. It is bijective on every open set: $x = \sigma(\pi x) + (x - \sigma\pi x)$ with $x - \sigma\pi x \in \ker \pi = \mathcal{F}$, uniquely. So Φ is an isomorphism of sheaves of rings; it restricts to the identity on $\mathcal{F}$ and induces the identity modulo $\mathcal{F}$. Thus $(X', \mathcal{I})$ is isomorphic to the trivial extension. □

Exercise 8.8

> **Problem.** Let X be a projective nonsingular variety over k. For any $n > 0$ we define the nth *plurigenus* of X to be $P_n = \dim_k \Gamma(X, \omega_X^{\otimes n})$. Thus in particular $P_1 = p_g$. Also, for any q, $0 \le q \le \dim X$ we define an integer $h^{q,0} = \dim_k \Gamma(X, \Omega^q_{X/k})$ where $\Omega^q_{X/k} = \bigwedge^q \Omega_{X/k}$ is the sheaf of regular q-forms on X. In particular, for $q = \dim X$, we recover the geometric genus again. The integers $h^{q,0}$ are called *Hodge numbers*.
>
> Using the method of (8.19), show that P_n and $h^{q,0}$ are *birational invariants*, i.e., if X and X' are birationally equivalent nonsingular projective varieties, then $P_n(X) = P_n(X')$ and $h^{q,0}(X) = h^{q,0}(X')$.

Solution. Let X, X' be birationally equivalent nonsingular projective varieties; they have the same function field, hence the same dimension d. Let $\mathcal{G}$ denote either $\Omega^q_{X/k}$ or $\omega_X^{\otimes n}$, and $\mathcal{G}'$ the corresponding sheaf on X'; all of these are locally free. Let $V \subseteq X$ be the largest open set on which the given birational map is a morphism $f : V \to X'$.

Step 1: $\operatorname{codim}(X \setminus V, X) \ge 2$. As in the proof of (8.19): if $P \in X$ has codimension 1, then $\mathcal{O}_{X,P}$ is a discrete valuation ring (X nonsingular), and X' is proper over k, so by the valuative criterion (4.7) the map $\operatorname{Spec} K \to X'$ extends to $\operatorname{Spec} \mathcal{O}_{X,P} \to X'$, hence the rational map is defined at P and $P \in V$ (a morphism $\operatorname{Spec} \mathcal{O}_{X,P} \to X'$ of finite type over k extends to a morphism on an open neighbourhood of P, which agrees with the given rational map at the generic point, hence on a dense open set, hence by separatedness of X' is the rational map). The generic point is in V as well, so every irreducible component of $X \setminus V$ has codimension ≥ 2.

Step 2: $\Gamma(X, \mathcal{G}) \to \Gamma(V, \mathcal{G}|_V)$ *is bijective.* Let $U \subseteq X$ be an affine open with $\mathcal{G}|_U$ free. Then $\Gamma(U, \mathcal{G}) \to \Gamma(U \cap V, \mathcal{G})$ is, componentwise, the map $\Gamma(U, \mathcal{O}_U) \to \Gamma(U \cap V, \mathcal{O})$, which is bijective: X is nonsingular, hence normal, and $U \setminus U \cap V$ has codimension ≥ 2, so this is a consequence of (6.3A), exactly as in the proof of (8.19). Gluing, the claim follows.

Step 3: a map $f^* : \Gamma(X', \mathcal{G}') \to \Gamma(V, \mathcal{G}|_V)$. By (8.11) there is a map $f^*\Omega_{X'/k} \to \Omega_{V/k}$ of locally free sheaves of the same rank d; taking qth exterior powers, resp. the top exterior power followed by the nth tensor power, we get maps $f^*\mathcal{G}' \to \mathcal{G}|_V$, hence $f^* : \Gamma(X', \mathcal{G}') \to \Gamma(V, f^*\mathcal{G}') \to \Gamma(V, \mathcal{G}|_V)$.

Step 4: f^* is injective. Since f is birational there is a nonempty open $W \subseteq V$ such that $f(W)$ is open in X' and $f : W \to f(W)$ is an isomorphism (I, 4.5 / its scheme version). An isomorphism induces an isomorphism on differentials, hence $f^*\mathcal{G}' \to \mathcal{G}$ is an isomorphism over W. Let $s' \in \Gamma(X', \mathcal{G}')$ with $f^*s' = 0$. Then $s'|_{f(W)} = 0$ (pull-back by the isomorphism $W \cong f(W)$ is bijective on sections). Now $\mathcal{G}'$ is locally free on the integral scheme X' and $f(W)$ is a nonempty open set, so restriction $\Gamma(X', \mathcal{G}') \to \Gamma(f(W), \mathcal{G}')$ is injective; hence $s' = 0$.

Combining Steps 2–4, $\dim \Gamma(X', \mathcal{G}') \le \dim \Gamma(V, \mathcal{G}) = \dim \Gamma(X, \mathcal{G})$. By symmetry (exchange X and X') the reverse inequality holds. Therefore $P_n(X) = P_n(X')$ for all $n > 0$ (taking $\mathcal{G} = \omega_X^{\otimes n}$) and $h^{q,0}(X) = h^{q,0}(X')$ for all q (taking $\mathcal{G} = \Omega^q_{X/k}$). □

9. Formal schemes

Conventions. $\widehat{X}$ denotes the completion of X along Y, $\mathfrak{X}$ a noetherian formal scheme, $\mathfrak{J}$ an ideal of definition, $Y_n = (\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathfrak{J}^n)$. For a finitely generated module M over a noetherian ring A with ideal I, $M^\Delta = \widetilde{\widehat{M}}$ on $\widehat{X} = \operatorname{Spec} A^\wedge$. We use the following facts established in the text: $M \mapsto M^\Delta$ is exact (9.4c) and $M^\Delta \cong \widetilde{M} \otimes_{\mathcal{O}_X} \mathcal{O}_{\widehat{X}}$ (9.4b); if A is I-adically complete then $\Gamma(\widehat{X}, M^\Delta) = M$ and $M \mapsto M^\Delta$ is an equivalence of finitely generated A-modules with coherent $\mathcal{O}_{\widehat{X}}$-modules (9.7); for a coherent $\mathfrak{F}$, $\mathfrak{F}/\mathfrak{J}^n\mathfrak{F}$ is a coherent $\mathcal{O}_{Y_n}$-module and $\mathfrak{F} \cong \varprojlim \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}$ (9.6a). As in the proofs in the text, since the questions are local we may (and do) work on an open affine formal subscheme $\mathfrak{U} = \widehat{U}$, $U = \operatorname{Spec} A$ with A noetherian and I-adically complete (completing A does not change $\widehat{U}$), with $\mathfrak{J}|_{\mathfrak{U}} = I^\Delta$, so that $Y_n \cap \mathfrak{U} = \operatorname{Spec} A/I^n$. (By (9.5a) any two ideals of definition are cofinal, so nothing depends on the choice of $\mathfrak{J}$.) Coherence of a sheaf on $\mathfrak{X}$ is a local property (as in (9.8), (9.9)), and by (9.7) a coherent sheaf restricted to such a $\mathfrak{U}$ is of the form M^Δ with M finitely generated.

Exercise 9.1

Problem. Let X be a noetherian scheme, Y a closed subscheme, and $\widehat{X}$ the completion of X along Y. We call the ring $\Gamma(\widehat{X}, \mathcal{O}_{\widehat{X}})$ the ring of *formal-regular functions* on X along Y. In this exercise we show that if Y is a connected, nonsingular, positive-dimensional subvariety of $X = \mathbf{P}^n$ over an algebraically closed field k, then $\Gamma(\widehat{X}, \mathcal{O}_{\widehat{X}}) = k$.

(a) Let $\mathcal{I}$ be the ideal sheaf of Y. Use (8.13) and (8.17) to show that there is an inclusion of sheaves on Y, $\mathcal{I}/\mathcal{I}^2 \hookrightarrow \mathcal{O}_Y(-1)^{n+1}$.

(b) Show that for any $r \ge 1$, $\Gamma(Y, \mathcal{I}^r/\mathcal{I}^{r+1}) = 0$.

(c) Use the exact sequences

$$0 \to \mathcal{I}^r/\mathcal{I}^{r+1} \to \mathcal{O}_X/\mathcal{I}^{r+1} \to \mathcal{O}_X/\mathcal{I}^r \to 0$$

and induction on r to show that $\Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = k$ for all $r \ge 1$. (Use (8.21Ae).)

(d) Conclude that $\Gamma(\widehat{X}, \mathcal{O}_{\widehat{X}}) = k$. (Actually, the same result holds without the hypothesis Y nonsingular, but the proof is more difficult—see Hartshorne [3, (7.3)].)

Solution. Y is connected and nonsingular (all local rings regular, hence domains), so it is irreducible, i.e. a nonsingular closed subvariety of the nonsingular variety $\mathbf{P}^n$; let $q = \operatorname{codim}(Y, \mathbf{P}^n)$ and $s = \dim Y \geq 1$. Then Y is a projective variety, so $\Gamma(Y, \mathcal{O}_Y)$ is a finite-dimensional k-algebra (II, Thm. 5.19) and a domain, hence a finite field extension of k, hence equal to k.

(a) By (8.17) there is an exact sequence $0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbf{P}^n/k} \otimes \mathcal{O}_Y \to \Omega_{Y/k} \to 0$, with $\Omega_{Y/k}$ locally free. Tensoring the Euler sequence (8.13) $0 \to \Omega_{\mathbf{P}^n} \to \mathcal{O}(-1)^{n+1} \to \mathcal{O} \to 0$ with $\mathcal{O}_Y$ stays exact (the third term is locally free), giving $0 \to \Omega_{\mathbf{P}^n}|_Y \to \mathcal{O}_Y(-1)^{n+1} \to \mathcal{O}_Y \to 0$. The composite

$$\mathcal{I}/\mathcal{I}^2 \hookrightarrow \Omega_{\mathbf{P}^n}|_Y \hookrightarrow \mathcal{O}_Y(-1)^{n+1}$$

is injective. Both inclusions have locally free cokernels ($\Omega_{Y/k}$ and $\mathcal{O}_Y$), hence are locally split; so the composite is locally split, with cokernel locally free (extension of locally free sheaves by locally free).

(b) For $y \in Y$, $\mathcal{O}_{X,y}$ is regular (so Cohen–Macaulay, (8.21Aa)) and $\mathcal{O}_{X,y}/\mathcal{I}_y = \mathcal{O}_{Y,y}$ is regular of dimension $\dim \mathcal{O}_{X,y} - q$, so $\mathcal{I}_y$ is generated by q elements forming part of a regular system of parameters, which is a regular sequence (8.21Ac), exactly as in the proof of (8.24); by (8.21Ae) the natural map $S^r(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}^r/\mathcal{I}^{r+1}$ is an isomorphism on every stalk, hence an isomorphism of sheaves on Y (as in the proof of (8.24)). By (a), $\mathcal{I}/\mathcal{I}^2$ is locally a direct summand of $\mathcal{O}_Y(-1)^{n+1}$, so S^r of the inclusion is also an inclusion (locally a direct summand):

$$\mathcal{I}^r/\mathcal{I}^{r+1} \cong S^r(\mathcal{I}/\mathcal{I}^2) \hookrightarrow S^r(\mathcal{O}_Y(-1)^{n+1}) \cong \mathcal{O}_Y(-r)^N, \qquad N = \tbinom{n+r}{r}.$$

So $\Gamma(Y, \mathcal{I}^r/\mathcal{I}^{r+1}) \hookrightarrow \Gamma(Y, \mathcal{O}_Y(-r))^N$, and it suffices to prove $\Gamma(Y, \mathcal{O}_Y(-r)) = 0$ for $r \geq 1$. Suppose $0 \neq t \in \Gamma(Y, \mathcal{O}_Y(-r))$. For each monomial $u_\alpha = x^\alpha$ of degree r (a section of $\mathcal{O}_Y(r)$) we get $tu_\alpha \in \Gamma(Y, \mathcal{O}_Y) = k$, say $tu_\alpha = c_\alpha$. The u_α generate $\mathcal{O}_Y(r)$ and $t \neq 0$ at the generic point of the integral scheme Y, so some $c_\alpha \neq 0$. Then t and u_α are nowhere vanishing, and for every β, $u_\beta = (c_\beta/c_\alpha)u_\alpha$ (multiply $tu_\beta = c_\beta$ and $tu_\alpha = c_\alpha$ and cancel the unit t). So all monomials of degree r are proportional on Y, i.e. the r-uple Veronese embedding maps Y to a single point; as it is injective, Y is a point, contradicting $\dim Y \geq 1$. So $\Gamma(Y, \mathcal{O}_Y(-r)) = 0$.

(c) For $r = 1$: $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Y$ and $\Gamma(Y, \mathcal{O}_Y) = k$. Suppose $\Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = k$ (the constants, i.e. the image of k; the map $k \to \Gamma$ is injective as $Y \neq \emptyset$). Left exactness of Γ and (b) give an exact sequence $0 = \Gamma(\mathcal{I}^r/\mathcal{I}^{r+1}) \to \Gamma(Y, \mathcal{O}_X/\mathcal{I}^{r+1}) \to \Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = k$. So $\Gamma(Y, \mathcal{O}_X/\mathcal{I}^{r+1})$ injects into k and contains k; hence it equals k, and the map to $\Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = k$ is the identity of k.

(d) By definition $\mathcal{O}_{\widehat{X}} = \varprojlim \mathcal{O}_X/\mathcal{I}^r$, so by (9.2) $\Gamma(\widehat{X}, \mathcal{O}_{\widehat{X}}) = \varprojlim_r \Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = \varprojlim(k \xleftarrow{\text{id}} k \leftarrow \cdots) = k$. □

Exercise 9.2

Problem. Use the result of (Ex. 9.1) to prove the following geometric result. Let $Y \subseteq X = \mathbf{P}^n_k$ be as above, and let $f : X \to Z$ be a morphism of k-varieties. Suppose that $f(Y)$ is a single closed point $P \in Z$. Then $f(X) = P$ also.

Solution. Here Y is connected, nonsingular, of positive dimension (so an irreducible nonsingular subvariety), as in Ex. 9.1, and $\mathcal{I}$ is its ideal sheaf. Let $W = \operatorname{Spec} C$ be an affine open neighbourhood of P in Z, $\mathfrak{m} \subseteq C$ the maximal ideal of P, and $X_0 = f^{-1}(W)$, an open subset of X containing Y. The morphism $f|_{X_0}$ corresponds to a k-algebra homomorphism $f^{\#} : C \to \Gamma(X_0, \mathcal{O}_X)$.

Claim 1: $f^{\#}(\mathfrak{m}^m) \subseteq \Gamma(X_0, \mathcal{I}^m)$ for all $m \geq 1$. Let $c \in \mathfrak{m}$ and $g = f^{\#}(c)$. For every point $y \in Y$ we have $f(y) = P$ and $c \in \mathfrak{m}_P$, so $g(y) = c(P) = 0$ in $k(y)$ (the residue field map $k(P) \to k(y)$ is a ring map). So $g|_Y \in \Gamma(Y, \mathcal{O}_Y)$ vanishes at every point of the reduced scheme Y, hence is 0, i.e. $g \in \Gamma(X_0, \mathcal{I})$. As $f^{\#}$ is a ring homomorphism and $\mathcal{I}^m$ is the mth power, $f^{\#}(\mathfrak{m}^m) \subseteq \Gamma(X_0, \mathcal{I}^m)$.

Claim 2: for every $c \in C$ there is $\lambda(c) \in k$ with $f^{\#}(c) - \lambda(c) \cdot 1 = 0$. Fix c. For each m, the image of $f^{\#}(c)$ in $\Gamma(X_0, \mathcal{O}_X/\mathcal{I}^m) = \Gamma(Y, \mathcal{O}_X/\mathcal{I}^m)$ (the sheaf $\mathcal{O}_X/\mathcal{I}^m$ is supported on $Y \subseteq X_0$) lies in k by Ex. 9.1(c): it equals $\lambda_m \cdot 1$ with $\lambda_m \in k$; since the transition maps $\Gamma(\mathcal{O}_X/\mathcal{I}^{m+1}) \to \Gamma(\mathcal{O}_X/\mathcal{I}^m)$ are the identity of k (proof of 9.1(c)), $\lambda_m = \lambda$ is independent of m. Let $e = f^{\#}(c) - \lambda \cdot 1 \in \Gamma(X_0, \mathcal{O}_X)$. Its image in $\Gamma(X_0, \mathcal{O}_X/\mathcal{I}^m)$ is 0 for all m, so $e_y \in \mathcal{I}_y^m$ for every $y \in Y$ and all m. By Krull's intersection theorem in the noetherian local ring $\mathcal{O}_{X,y}$ (with $\mathcal{I}_y \subseteq \mathfrak{m}_y$), $\bigcap_m \mathcal{I}_y^m = 0$, so $e_y = 0$. Thus e vanishes near each point of Y; in particular at the generic point η of X (any open neighbourhood of a point of Y contains η): $e_\eta = 0$. As X_0 is a nonempty open subset of the integral scheme X, $\Gamma(X_0, \mathcal{O}_X) \to \mathcal{O}_{X,\eta}$ is injective, so $e = 0$.

Hence $f^{\#}(C) \subseteq k \cdot 1$. This means that $f|_{X_0} : X_0 \to \operatorname{Spec} C$ factors through $\operatorname{Spec} k \to \operatorname{Spec} C$, so $f(X_0)$ is a single point; it contains $f(Y) = P$ (as $Y \neq \emptyset$), so $f(X_0) = \{P\}$. Since X is irreducible, X_0 is dense, and $f(X) \subseteq \overline{f(X_0)} = \overline{\{P\}} = \{P\}$ (P is closed in Z). Thus $f(X) = P$. □

Exercise 9.3

Problem. Prove the analogue of (5.6) for formal schemes, which says, if $\mathfrak{X}$ is an affine formal scheme, and if

$$0 \to \mathfrak{F}' \to \mathfrak{F} \to \mathfrak{F}'' \to 0$$

is an exact sequence of $\mathcal{O}_{\mathfrak{X}}$-modules, and if $\mathfrak{F}'$ is coherent, then the sequence of global sections

$$0 \to \Gamma(\mathfrak{X}, \mathfrak{F}') \to \Gamma(\mathfrak{X}, \mathfrak{F}) \to \Gamma(\mathfrak{X}, \mathfrak{F}'') \to 0$$

is exact. For the proof, proceed in the following steps.

(a) Let $\mathfrak{J}$ be an ideal of definition for $\mathfrak{X}$, and for each $n > 0$ consider the exact sequence

$$0 \to \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}' \to \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}' \to \mathfrak{F}'' \to 0.$$

Use (5.6), slightly modified, to show that for every open affine subset $\mathfrak{U} \subseteq \mathfrak{X}$, the sequence

$$0 \to \Gamma(\mathfrak{U}, \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}') \to \Gamma(\mathfrak{U}, \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}') \to \Gamma(\mathfrak{U}, \mathfrak{F}'') \to 0$$

is exact.

(b) Now pass to the limit, using (9.1), (9.2), and (9.6). Conclude that $\mathfrak{F} \cong \varprojlim \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}'$ and that the sequence of global sections above is exact.

Solution. (The quotient in the middle is $\mathfrak{F}/\mathfrak{J}^n\mathfrak{F}'$, where $\mathfrak{J}^n\mathfrak{F}' \subseteq \mathfrak{F}' \subseteq \mathfrak{F}$; this is what makes the first sequence exact, and it is the reading used.) Write $\mathcal{G}'_n = \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}'$ and $\mathcal{G}_n = \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}'$; the transition maps $\mathcal{G}_{n+1} \to \mathcal{G}_n$, $\mathcal{G}'_{n+1} \to \mathcal{G}'_n$ are the obvious surjections, and the sequences

$$0 \to \mathcal{G}'_n \to \mathcal{G}_n \to \mathfrak{F}'' \to 0 \tag{1}$$

are exact (the kernel of $\mathfrak{F}/\mathfrak{J}^n\mathfrak{F}' \to \mathfrak{F}''$ is $\mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}'$) and compatible with the transition maps (the maps on $\mathfrak{F}''$ being identities).

(a) Let $\mathfrak{U} = \widehat{U}$ be an open affine formal subscheme, $U = \operatorname{Spec} B$; then $(\mathfrak{U}, \mathcal{O}_{Y_n}|_{\mathfrak{U}})$ is an affine noetherian scheme: by (9.5a) $\mathfrak{J}^n \supseteq (I^\Delta)^N$ for some N on $\mathfrak{U}$, so it is a closed subscheme of $(\mathfrak{U}, \mathcal{O}/(I^\Delta)^N) = \operatorname{Spec} B/I^N$ (9.4a), hence affine. (The same holds for any open $\mathfrak{U}$ for which $(\mathfrak{U}, \mathcal{O}_{Y_1}|_{\mathfrak{U}})$ is affine, since a scheme whose reduction is affine is affine, III, Ex. 3.1.) By (9.6a), $\mathcal{G}'_n = \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}'$ is a coherent, in particular quasi-coherent, sheaf on this affine scheme, so $H^1(\mathfrak{U}, \mathcal{G}'_n) = 0$ (III, Thm. 3.5). Cohomology of a sheaf of abelian groups is the same computed in abelian groups or in modules over a sheaf of rings (III, Prop. 2.6), so the long exact sequence of cohomology of (1) on $\mathfrak{U}$ (in the category of sheaves of abelian groups) gives

$$0 \to \Gamma(\mathfrak{U}, \mathcal{G}'_n) \to \Gamma(\mathfrak{U}, \mathcal{G}_n) \to \Gamma(\mathfrak{U}, \mathfrak{F}'') \to H^1(\mathfrak{U}, \mathcal{G}'_n) = 0.$$

(This is exactly (5.6): the proof of (5.6) only needs that the subsheaf is quasi-coherent on an affine scheme with quasi-compact base of principal opens $D(f)$, and here the sheaves $\mathcal{G}_n, \mathfrak{F}''$ are merely sheaves of abelian groups on the underlying space of the affine scheme $\operatorname{Spec} B/I^nB$; that is the "slight modification".)

(b) Fix $\mathfrak{U}$ as above. The transition map $\Gamma(\mathfrak{U}, \mathcal{G}'_{n+1}) \to \Gamma(\mathfrak{U}, \mathcal{G}'_n)$ is surjective: its kernel sheaf is $\mathfrak{J}^n\mathfrak{F}'/\mathfrak{J}^{n+1}\mathfrak{F}'$, a coherent (kernel of a surjection of coherent $\mathcal{O}_{Y_{n+1}}$-modules, (9.6a)), in particular quasi-coherent, sheaf on the affine noetherian scheme $(\mathfrak{U}, \mathcal{O}_{Y_{n+1}})$, so its H^1 vanishes (III, Thm. 3.5, 2.6). Hence the inverse system $(\Gamma(\mathfrak{U}, \mathcal{G}'_n))_n$ has surjective transition maps and satisfies (ML) (Example 9.1.1). Take the inverse limit of the exact sequences of (a); by (9.1b) we obtain an exact sequence

$$0 \to \varprojlim \Gamma(\mathfrak{U}, \mathcal{G}'_n) \to \varprojlim \Gamma(\mathfrak{U}, \mathcal{G}_n) \to \varprojlim \Gamma(\mathfrak{U}, \mathfrak{F}'') \to 0.$$

The last system is constant, with limit $\Gamma(\mathfrak{U}, \mathfrak{F}'')$; by (9.2), $\varprojlim \Gamma(\mathfrak{U}, \mathcal{G}_n) = \Gamma(\mathfrak{U}, \varprojlim \mathcal{G}_n)$ and $\varprojlim \Gamma(\mathfrak{U}, \mathcal{G}'_n) = \Gamma(\mathfrak{U}, \varprojlim \mathcal{G}'_n)$. So for all such $\mathfrak{U}$ (which form a base of the topology),

$$0 \to \Gamma(\mathfrak{U}, \varprojlim \mathcal{G}'_n) \to \Gamma(\mathfrak{U}, \varprojlim \mathcal{G}_n) \to \Gamma(\mathfrak{U}, \mathfrak{F}'') \to 0$$

is exact; exactness on sections over a base of opens implies exactness of the sequence of sheaves (stalks are direct limits over the base). Now $\varprojlim \mathcal{G}'_n = \varprojlim \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}' \cong \mathfrak{F}'$ by (9.6a) ($\mathfrak{F}'$ is coherent), and the natural maps $\mathfrak{F} \to \mathcal{G}_n$ are compatible, giving a morphism $\mathfrak{F} \to \varprojlim \mathcal{G}_n$ (9.2) and a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \to & \mathfrak{F}' & \to & \mathfrak{F} & \to & \mathfrak{F}'' & \to & 0 \\ & & \downarrow\cong & & \downarrow & & \| & & \\ 0 & \to & \varprojlim \mathcal{G}'_n & \to & \varprojlim \mathcal{G}_n & \to & \mathfrak{F}'' & \to & 0 \end{array}$$

By the five lemma, $\mathfrak{F} \cong \varprojlim \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}'$. Finally, take $\mathfrak{U} = \mathfrak{X}$ (which is affine): $\Gamma(\mathfrak{X}, \mathfrak{F}) = \Gamma(\mathfrak{X}, \varprojlim \mathcal{G}_n)$ and $\Gamma(\mathfrak{X}, \mathfrak{F}') = \Gamma(\mathfrak{X}, \varprojlim \mathcal{G}'_n)$, so the middle row above on global sections is $0 \to \Gamma(\mathfrak{X}, \mathfrak{F}') \to \Gamma(\mathfrak{X}, \mathfrak{F}) \to \Gamma(\mathfrak{X}, \mathfrak{F}'') \to 0$, which is therefore exact. □

Exercise 9.4

Problem. Use (Ex. 9.3) to prove that if

$$0 \to \mathfrak{F}' \to \mathfrak{F} \to \mathfrak{F}'' \to 0$$

is an exact sequence of $\mathcal{O}_{\mathfrak{X}}$-modules on a noetherian formal scheme $\mathfrak{X}$, and if $\mathfrak{F}', \mathfrak{F}''$ are coherent, then $\mathfrak{F}$ is coherent also.

Solution. Coherence is local. Cover $\mathfrak{X}$ by finitely many open affine formal subschemes $\mathfrak{U} = \widehat{U}$, $U = \operatorname{Spec} A$, A complete, on which $\mathfrak{F}'$ and $\mathfrak{F}''$ are both coherent (refine the covers on which each is coherent by smaller affine formal opens), i.e. by (9.7) $\mathfrak{F}'|_{\mathfrak{U}} \cong M'^{\Delta}$, $\mathfrak{F}''|_{\mathfrak{U}} \cong M''^{\Delta}$ with M', M'' finitely generated A-modules. It suffices to show $\mathfrak{F}|_{\mathfrak{U}} \cong M^{\Delta}$ for a finitely generated A-module M.

Let $M = \Gamma(\mathfrak{U}, \mathfrak{F})$, an A-module ($A = \Gamma(\mathfrak{U}, \mathcal{O}_{\mathfrak{U}})$ by (9.3.1)). By (9.7), $M' = \Gamma(\mathfrak{U}, \mathfrak{F}')$ and $M'' = \Gamma(\mathfrak{U}, \mathfrak{F}'')$; by Ex. 9.3 applied to $\mathfrak{U}$ (affine) the sequence

$$0 \to M' \to M \to M'' \to 0$$

is exact. Hence M is finitely generated over the noetherian ring A. For any $\mathcal{O}_{\mathfrak{U}}$-module $\mathfrak{G}$ with $N = \Gamma(\mathfrak{U}, \mathfrak{G})$ there is a natural map $\psi_{\mathfrak{G}} : N \otimes_A \mathcal{O}_{\mathfrak{U}} \to \mathfrak{G}$, $m \otimes g \mapsto g \cdot m|_V$ on a small open set V, natural in $\mathfrak{G}$. If N is finitely generated, $N \otimes_A \mathcal{O}_{\mathfrak{U}} = N^{\Delta}$ (both are right exact in N and agree for $N = A$, by (9.4b)). For $\mathfrak{G} = \mathfrak{F}', \mathfrak{F}''$, $\psi_{\mathfrak{G}}$ is the isomorphism of (9.7) (the counit of the equivalence $M \leftrightarrow M^{\Delta}$). We obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & M'^{\Delta} & \to & M^{\Delta} & \to & M''^{\Delta} & \to & 0 \\ & & \downarrow\cong & & \downarrow \psi_{\mathfrak{F}} & & \downarrow\cong & & \\ 0 & \to & \mathfrak{F}' & \to & \mathfrak{F} & \to & \mathfrak{F}'' & \to & 0 \end{array}$$

whose top row is exact by (9.4c) (exactness of Δ) and whose bottom row is exact by hypothesis. By the five lemma $\psi_{\mathfrak{F}}$ is an isomorphism, so $\mathfrak{F}|_{\mathfrak{U}} \cong M^{\Delta}$ is coherent. Since $\mathfrak{X}$ is covered by finitely many such $\mathfrak{U}$, $\mathfrak{F}$ is coherent. □

Exercise 9.5

Problem. If $\mathfrak{F}$ is a coherent sheaf on a noetherian formal scheme $\mathfrak{X}$, which can be generated by global sections, show in fact that it can be generated by a finite number of its global sections.

Solution. Cover $\mathfrak{X}$ by finitely many open affine formal subschemes $\mathfrak{U}_i = \widehat{U}_i$, $U_i = \operatorname{Spec} A_i$ with A_i complete, on which $\mathfrak{F}|_{\mathfrak{U}_i} = M_i^{\Delta}$, $M_i = \Gamma(\mathfrak{U}_i, \mathfrak{F})$ finitely generated over A_i (9.7). Fix i and let $M_i' \subseteq M_i$ be the A_i-submodule generated by the images of $\Gamma(\mathfrak{X}, \mathfrak{F}) \to \Gamma(\mathfrak{U}_i, \mathfrak{F}) = M_i$. Since A_i is noetherian, M_i' is finitely generated, so it is generated by the images of finitely many global sections $s_{i1}, \ldots, s_{ir_i} \in \Gamma(\mathfrak{X}, \mathfrak{F})$ (each of finitely many generators is a finite A_i-combination of images of global sections).

Claim: $M_i' = M_i$. The natural map $M_i'^{\Delta} \to M_i^{\Delta} = \mathfrak{F}|_{\mathfrak{U}_i}$ is injective (exactness of Δ) and its image contains $(s|_{\mathfrak{U}_i})_x$ for every global section s and every $x \in \mathfrak{U}_i$; since the global sections generate $\mathfrak{F}$, these generate every stalk, so the map is surjective. Hence $(M_i/M_i')^{\Delta} = \operatorname{coker} = 0$, and since A_i is complete, $M_i/M_i' = \Gamma(\mathfrak{U}_i, (M_i/M_i')^{\Delta}) = 0$ by (9.7). So $M_i' = M_i$.

Therefore the sections s_{ij} ($j \le r_i$) restricted to $\mathfrak{U}_i$ generate M_i, hence generate $M_i \otimes_{A_i} \mathcal{O}_{\mathfrak{U}_i} = \mathfrak{F}|_{\mathfrak{U}_i}$ (a surjection $\mathcal{O}^{r_i} \to M_i^{\Delta}$ is obtained from generators of M_i). The finitely many global sections $\{s_{ij}\}_{i,j}$ then generate $\mathfrak{F}$ on every $\mathfrak{U}_i$, i.e. on $\mathfrak{X}$. □

Exercise 9.6

Problem. Let $\mathfrak{X}$ be a noetherian formal scheme, let $\mathfrak{J}$ be an ideal of definition, and for each n, let Y_n be the scheme $(\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathfrak{J}^n)$. Assume that the inverse system of groups $(\Gamma(Y_n, \mathcal{O}_{Y_n}))$ satisfies the Mittag-Leffler condition. Then prove that $\operatorname{Pic}\mathfrak{X} = \varprojlim \operatorname{Pic} Y_n$. As in the case of a scheme, we define $\operatorname{Pic}\mathfrak{X}$ to be the group of locally free $\mathcal{O}_{\mathfrak{X}}$-modules of rank 1 under the operation $\otimes$. Proceed in the following steps.

(a) Use the fact that $\ker(\Gamma(Y_{n+1}, \mathcal{O}_{Y_{n+1}}) \to \Gamma(Y_n, \mathcal{O}_{Y_n}))$ is a nilpotent ideal to show that the inverse system $(\Gamma(Y_n, \mathcal{O}_{Y_n}^*))$ of units also satisfies (ML).

(b) Let $\mathfrak{F}$ be a coherent sheaf of $\mathcal{O}_{\mathfrak{X}}$-modules, and assume that for each n, there is some isomorphism $\varphi_n : \mathfrak{F}/\mathfrak{J}^n\mathfrak{F} \cong \mathcal{O}_{Y_n}$. Then show that there is an isomorphism $\mathfrak{F} \cong \mathcal{O}_{\mathfrak{X}}$. Be careful, because the φ_n may not be compatible with the maps of the two inverse systems $(\mathfrak{F}/\mathfrak{J}^n\mathfrak{F})$ and $(\mathcal{O}_{Y_n})$! Conclude that the natural map $\operatorname{Pic}\mathfrak{X} \to \varprojlim \operatorname{Pic} Y_n$ is injective.

(c) Given an invertible sheaf $\mathcal{L}_n$ on Y_n for each n, and given isomorphisms $\mathcal{L}_{n+1} \otimes \mathcal{O}_{Y_n} \cong \mathcal{L}_n$, construct maps $\mathcal{L}_{n'} \to \mathcal{L}_n$ for each $n' \geq n$ so as to make an inverse system, and show that $\mathfrak{L} = \varprojlim \mathcal{L}_n$ is a coherent sheaf on $\mathfrak{X}$. Then show that $\mathfrak{L}$ is locally free of rank 1, and thus conclude that the map $\operatorname{Pic}\mathfrak{X} \to \varprojlim \operatorname{Pic} Y_n$ is surjective. Again, be careful, because even though each $\mathcal{L}_n$ is locally free of rank 1, the open sets needed to make them free might get smaller and smaller with n.

(d) Show that the hypothesis "$(\Gamma(Y_n, \mathcal{O}_{Y_n}))$ satisfies (ML)" is satisfied if either $\mathfrak{X}$ is affine, or each Y_n is projective over a field k.

Note: See (III, Ex. 11.5–11.7) for further examples and applications.

Solution. Put $R_n = \Gamma(Y_n, \mathcal{O}_{Y_n})$. The map $\operatorname{Pic}\mathfrak{X} \to \varprojlim \operatorname{Pic} Y_n$ is $\mathfrak{L} \mapsto (\mathfrak{L}/\mathfrak{J}^n\mathfrak{L})_n$; it is a homomorphism since $(\mathfrak{L} \otimes \mathfrak{L}')/\mathfrak{J}^n = \mathfrak{L}/\mathfrak{J}^n \otimes \mathfrak{L}'/\mathfrak{J}^n$, and an invertible $\mathfrak{L}$ is coherent ($\mathcal{O}_{\mathfrak{X}}$ is) with $\mathfrak{L}/\mathfrak{J}^n\mathfrak{L}$ invertible on Y_n. Note that $\mathfrak{F}_{n+1} \otimes \mathcal{O}_{Y_n} = \mathfrak{F}/\mathfrak{J}^n\mathfrak{F} = \mathfrak{F}_n$ for $\mathfrak{F}_n := \mathfrak{F}/\mathfrak{J}^n\mathfrak{F}$.

(a) The kernel of $\mathcal{O}_{Y_{n'}} \to \mathcal{O}_{Y_n}$ is $\mathfrak{J}^n/\mathfrak{J}^{n'}$, and $(\mathfrak{J}^n/\mathfrak{J}^{n'})^m = 0$ for $nm \geq n'$; by left exactness of Γ, $K_{n'n} := \ker(R_{n'} \to R_n) = \Gamma(Y_{n'}, \mathfrak{J}^n/\mathfrak{J}^{n'})$ is a nilpotent ideal of $R_{n'}$. Let $S_n(n') = \operatorname{im}(R_{n'} \to R_n) \cong R_{n'}/K_{n'n}$, a subring of R_n. By (ML) for (R_n), for each n there is $n_0 = n_0(n) \geq n$ such that $S_n(n') = S_n$ is the same subring for all $n' \geq n_0$. Since units lift modulo a nilpotent ideal ($y \in R_{n'}$ is a unit iff its image in $R_{n'}/K_{n'n}$ is a unit), we get

$$\operatorname{im}(R_{n'}^* \to R_n^*) = \{\text{ units of the ring } S_n(n')\} = S_n^* \quad (n' \geq n_0).$$

Here a unit of $S_n(n') = R_{n'}/K_{n'n}$ means an element with an inverse *in* $S_n(n')$, and is the image of a unit of $R_{n'}$. In particular the images $\operatorname{im}(R_{n'}^* \to R_n^*)$ are constant for $n' \geq n_0(n)$, so (R_n^*) satisfies (ML). Call this stable subgroup $U_n = S_n^* \subseteq R_n^*$.

(b) Let $\mathfrak{F}$ be coherent with $\mathfrak{F}_n := \mathfrak{F}/\mathfrak{J}^n\mathfrak{F} \cong \mathcal{O}_{Y_n}$ for all n. Let $E_n = \operatorname{Isom}_{\mathcal{O}_{Y_n}}(\mathfrak{F}_n, \mathcal{O}_{Y_n}) \neq \emptyset$. The group $R_n^* = \operatorname{Aut}(\mathcal{O}_{Y_n})$ acts simply transitively on E_n (two isomorphisms differ by an automorphism of $\mathcal{O}_{Y_n}$, i.e. multiplication by a unit of R_n). Reduction $\psi \mapsto \psi \otimes \mathcal{O}_{Y_n}$ gives maps $E_{n+1} \to E_n$ (as $\mathfrak{F}_{n+1} \otimes \mathcal{O}_{Y_n} = \mathfrak{F}_n$), compatible with the actions via $R_{n+1}^* \to R_n^*$. So (E_n) is an inverse system of nonempty sets.

(E_n) satisfies (ML). Fix n and $n' \geq n_0(n)$, and choose any $e \in E_{n'}$ with image $\bar{e} \in E_n$. Then $\operatorname{im}(E_{n'} \to E_n) = \{\bar{u}\bar{e} : u \in R_{n'}^*\} = U_n \cdot \bar{e}$ by (a), a coset of the fixed subgroup U_n of R_n^* acting on E_n. For $n'' \geq n' \geq n_0$ we have $\operatorname{im}(E_{n''} \to E_n) \subseteq \operatorname{im}(E_{n'} \to E_n)$, and both are cosets of the

same subgroup U_n; a coset contained in another coset of the same subgroup equals it. So the images stabilize.

$\varprojlim E_n \neq \emptyset$. Let E_n' be the stable image of $(E_{n'})_{n' \geq n}$ in E_n; these are nonempty, form an inverse system with surjective transition maps, so we choose successively $e_1 \in E_1'$, $e_2 \in E_2'$ mapping to e_1, etc. (the argument of (9.1b)). A compatible family $(\psi_n) \in \varprojlim E_n$ is an isomorphism of inverse systems $(\mathfrak{F}_n) \to (\mathcal{O}_{Y_n})$, hence induces an isomorphism on the limits. By (9.6a) $\varprojlim \mathfrak{F}_n \cong \mathfrak{F}$ ($\mathfrak{F}$ coherent) and $\varprojlim \mathcal{O}_{Y_n} = \mathcal{O}_{\mathfrak{X}}$. Thus $\mathfrak{F} \cong \mathcal{O}_{\mathfrak{X}}$.

Injectivity. If $\mathfrak{L} \in \operatorname{Pic} \mathfrak{X}$ maps to the identity of $\varprojlim \operatorname{Pic} Y_n$, then $\mathfrak{L}/\mathfrak{J}^n\mathfrak{L} \cong \mathcal{O}_{Y_n}$ for each n, so $\mathfrak{L} \cong \mathcal{O}_{\mathfrak{X}}$ by the above. So the kernel of the homomorphism is trivial.

(c) An element of $\varprojlim \operatorname{Pic} Y_n$ is a sequence of classes $[\mathcal{L}_n]$ with $[\mathcal{L}_{n+1} \otimes \mathcal{O}_{Y_n}] = [\mathcal{L}_n]$; choose invertible sheaves $\mathcal{L}_n$ and isomorphisms $\theta_n : \mathcal{L}_{n+1} \otimes \mathcal{O}_{Y_n} \to \mathcal{L}_n$. Let $\rho_{n+1,n} : \mathcal{L}_{n+1} \to \mathcal{L}_{n+1} \otimes \mathcal{O}_{Y_n} = \mathcal{L}_{n+1}/\mathfrak{J}^n\mathcal{L}_{n+1} \xrightarrow{\theta_n} \mathcal{L}_n$, a surjection with kernel $\mathfrak{J}^n\mathcal{L}_{n+1}$, and let $\rho_{n'n} = \rho_{n+1,n} \circ \cdots \circ \rho_{n',n'-1}$ for $n' > n$ (and $\rho_{nn} = \mathrm{id}$). This makes $(\mathcal{L}_n)$ an inverse system with surjective maps; and $\ker \rho_{n'n} = \mathfrak{J}^n\mathcal{L}_{n'}$ (induction on $n' - n$: $\ker \rho_{n'n}$ is the preimage under $\rho_{n',n+1}$ of $\ker \rho_{n+1,n} = \mathfrak{J}^n\mathcal{L}_{n+1}$; since $\rho_{n',n+1}$ is onto with kernel $\mathfrak{J}^{n+1}\mathcal{L}_{n'} \subseteq \mathfrak{J}^n\mathcal{L}_{n'}$, this preimage is $\mathfrak{J}^n\mathcal{L}_{n'}$). Each $\mathcal{L}_n$ is a coherent $\mathcal{O}_{Y_n}$-module. By (9.6b), $\mathfrak{L} = \varprojlim \mathcal{L}_n$ is a coherent $\mathcal{O}_{\mathfrak{X}}$-module and $\mathfrak{L}/\mathfrak{J}^n\mathfrak{L} \cong \mathcal{L}_n$ for all n, compatibly with the maps.

$\mathfrak{L}$ is locally free of rank 1. Let $x \in \mathfrak{X}$; take an open affine formal subscheme $\mathfrak{U} = \widehat{U} \ni x$, $U = \operatorname{Spec} A$, $\mathfrak{J} = I^{\Delta}$ on $\mathfrak{U}$ and $Y_n \cap \mathfrak{U} = \operatorname{Spec} A/I^n$ as in the conventions. Since $\mathcal{L}_1$ is invertible on Y_1, there is $g \in A$ with $x \in D = D(g) \cap Y_1$ and $\mathcal{L}_1|_D \cong \mathcal{O}_{Y_1}|_D$, via a generator $m_1 \in \Gamma(D, \mathcal{L}_1)$. The key point: D is the same open set for all n; it is an affine open $D_n = \operatorname{Spec}(A/I^n)_g$ of Y_n. Because D_{n+1} is affine and $\mathcal{L}_{n+1} \to \mathcal{L}_n$ is a surjection of quasi-coherent sheaves on D_{n+1} ($\mathcal{L}_n$ viewed on Y_{n+1}), sections lift: choose successively $m_{n+1} \in \Gamma(D, \mathcal{L}_{n+1})$ with $\rho_{n+1,n}(m_{n+1}) = m_n$. Consider $\mu_n : \mathcal{O}_{Y_n}|_D \to \mathcal{L}_n|_D$, $1 \mapsto m_n$. At each $y \in D$ the stalk $\mathcal{L}_{n,y}$ modulo $\mathfrak{J}_y$ is $\mathcal{L}_{1,y}$, generated by $m_{1,y}$, and $\mathfrak{J}_y$ is nilpotent in $\mathcal{O}_{Y_n,y}$, so by Nakayama $m_{n,y}$ generates $\mathcal{L}_{n,y}$. Thus μ_n is a surjection from $\mathcal{O}_{Y_n}|_D$ onto an invertible sheaf, hence an isomorphism (stalkwise a surjective map $\mathcal{O}_{Y_n,y} \to \mathcal{O}_{Y_n,y}$ of finitely generated modules is bijective). The μ_n are compatible with the transition maps, because $\rho_{n+1,n}(m_{n+1}) = m_n$. Passing to the inverse limit, $\mathcal{O}_{\mathfrak{X}}|_D = \varprojlim \mathcal{O}_{Y_n}|_D \xrightarrow{\cong} \varprojlim \mathcal{L}_n|_D = \mathfrak{L}|_D$. Thus $\mathfrak{L}$ is free of rank 1 on the neighbourhood D of x (open in $\mathfrak{X}$, equal to the formal scheme $\widehat{D(g)}$). So $\mathfrak{L} \in \operatorname{Pic} \mathfrak{X}$ maps to $([\mathfrak{L}/\mathfrak{J}^n\mathfrak{L}]) = ([\mathcal{L}_n])$. Hence $\operatorname{Pic} \mathfrak{X} \to \varprojlim \operatorname{Pic} Y_n$ is surjective, and by (b) it is an isomorphism: $\operatorname{Pic} \mathfrak{X} = \varprojlim \operatorname{Pic} Y_n$ (under (ML) for (R_n), which was used in (a) and (b)).

(d) *$\mathfrak{X}$ affine.* Then $\mathfrak{X} = \widehat{X}$, $X = \operatorname{Spec} A$, and with $\mathfrak{J} = I^{\Delta}$ we have $Y_n = \operatorname{Spec} A/I^n$ and $R_n = A/I^n$ (this is (9.4a) and its proof). The transition maps $A/I^{n+1} \to A/I^n$ are surjective, so (ML) holds (Example 9.1.1). (For another ideal of definition $\mathfrak{J}'$ the inverse systems (Y_n) and (Y_n') are cofinal by (9.5a), so the systems (R_n) and (R_n') are cofinal and (ML) for one gives (ML) for the other; alternatively $\Gamma(Y_n', \mathcal{O}) = \Gamma(\mathfrak{X}, \mathcal{O})/\Gamma(\mathfrak{X}, \mathfrak{J}'^n)$ is a quotient by Ex. 9.3, again with surjective transition maps.)

Each Y_n projective over a field k (compatibly, i.e. the closed immersions $Y_n \hookrightarrow Y_{n+1}$ are k-morphisms, so the maps $R_{n+1} \to R_n$ are k-algebra homomorphisms). By (II, Thm. 5.19), $R_n = \Gamma(Y_n, \mathcal{O}_{Y_n})$ is a finite-dimensional k-vector space. By Example 9.1.2, an inverse system of finite-dimensional vector spaces satisfies (ML): for fixed n the decreasing chain of subspaces $\operatorname{im}(R_{n'} \to R_n)$ of the finite-dimensional space R_n is stationary. □